

Polytopes at Surreal Scales

Lifting compression, truncation histories, affine residues, lattice counts, invisible extension complexity, mixed-volume tomography and realization, contact depth, omnific integer hulls, and infinite presentations

AI-assisted research manuscripts of 30 September 2026, merged in the batch-58 and batch-59 write phases

Batch 58 of the ProveIt collection: manuscripts 02–07, pinned to 6996fee (05: 5e0f4e0). Batch 59: manuscripts 01–06, printed as sources 08–13, pinned to 6996fee (09: b2b626d).

Abstract

A finite polytope with surreal coordinates has an ordinary real realizable face lattice: real-closed-field transfer rules out new finite combinatorial types. What the surreal numbers add is the organization of that finite geometry across infinitesimal and transfinite scales, and this report collects twelve independent answers to the question of how much of it can be classified. Part I fixes a real configuration of n points of affine dimension d and varies only the heights: arbitrary surreal heights compress to at most $n - d - 1$ real layers that keep the leading term of every real affine-dependence evaluation and every common initial normal-form history; the complete sign type is an oriented partial flag whose minimum polynomial germ degree is exactly $m - 1$ (or $m - r_0$ with a prescribed standard part), while finite lifted data need degree at most one. Part II shows that truncating the *coordinates* instead is uncontrolled: four bounded planar points realize every prescribed triangle/quadrilateral history, and $\omega^{-\omega}$ selects the final hull independently. Part III, over any set-sized real closed field $K \supseteq \mathbb{R}$ with its natural valuation, introduces affine residues and scale lattices and proves that simultaneous inner segments on a finite rational grid preserve every pure mixed-volume valuation. Part IV counts lattice points: a quadrilateral with the vertex shadows of the unit square can have counting function $(n + 1)^2 - \lceil \alpha n \rceil$ for any irrational $0 < \alpha < 1$, whose generating function has the unit circle as a natural boundary. Part V constructs polygons whose coordinates agree to all orders of an infinitesimal with a polygon of logarithmic extension complexity, while their own complexity grows like a power of n ; such gaps exist exactly when the infinitesimal is not an order unit of the value group. Parts VI–X add mixed-volume tomography (no fixed finite probe family recognizes planar scale lattices), realization of mixed-volume valuation profiles (a strictly Lorentzian Fano cubic is unrealizable), rational vertex arcs of degree at least $\lceil 2^k / (16k + 48) \rceil$ for $8k + 16$ vertices, omnific integer hulls (finitely generated for all boxed right-hand sides exactly when every row is rational in direction), and set-sized presentations of convex classes.

Status and scope. Twelve AI-assisted, unrefereed research manuscripts (six of batch 58, six of batch 59), merged without selection: every result, proof, example, question and limitation of each is printed, a result proved by two sources once with both named. The underlying tools (Conway normal forms, real-closed-field transfer, secondary polytopes, curve selection, mixed volumes, Ehrhart theory, extension complexity) are classical; each Part states what it claims. [write] No statement of this report is formalized in Lean or Rocq, and the report makes no claim to literature-wide priority. Labels carry the prefix `poly::`; conventions, the relation between the manuscripts and the provenance are in [section 1](#) and [Appendix A](#); text added in the merge is marked [write].

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1 Guide to the report

[write] This section, the formalization chapter’s framing, the provenance appendix and every paragraph marked [write] were written when the six manuscripts were merged. Everything else is the manuscripts’ text. Six further manuscripts became Parts VI–X in the batch-59 write phase (section 1.6).

1.1 Six answers to one question

Manuscripts 02–07 of batch 58 were written independently, on 30 September 2026, in answer to one prompt about finite polytopes over the surreal numbers. They share a setting: a set-sized real closed field $\mathbb{R} \subseteq K \subseteq \mathbf{No}$ (or, in manuscript 05, an ordered subfield), its ring of finite elements, and the standard part st . They share three basic facts, each proved by several of them and printed here once: $\text{st conv} = \text{conv st}$ (proposition 13.1), real-closed-field transfer of a finite configuration (theorem 18.1), and an explicit real threshold for finitely many sign conditions (lemma 10.3). Beyond that they answer three different questions, the three spines of this report.

Part	Subject	Sources (base first)	Labels
I	Liftings of a fixed real configuration	05, 02	<code>poly:</code> , <code>poly:ts:</code>
II	Coordinate-truncation histories	02	<code>poly:ts:</code>
III	Affine residues and mixed-volume scales	06	<code>poly:mv:</code>
IV	Real traces and lattice-point counting	03	<code>poly:ehr:</code>
V	Extension complexity beyond all orders	07, 04	<code>poly:ic:</code> , <code>poly:bao:</code>
VI–X	Six further manuscripts, batch 59 (section 1.6)	08–13	six sub-prefixes

- *The lifting spine* (Parts I and II). Manuscripts 05 and 02 prove the same theorem: surreal heights on a fixed real base compress to at most $q = n - d - 1$ real layers. Manuscript 05 is the base (weaker hypotheses, stronger conclusions, complete classification of the invariant); 02’s compatibility with every common initial cut, its history classification, planar sharpness, polar and band identities, Puiseux model and query-model bound are printed beside it. 02’s coordinate-truncation theory is Part II.
- *The residue and volume spine* (Parts III and IV). Manuscripts 06 and 03 are complementary: they share no theorem beyond the three basic facts. Manuscript 06 moves the coordinates and keeps volume information; manuscript 03 counts lattice points.
- *The complexity spine* (Part V). Manuscripts 07 and 04 prove the same phenomenon: explicit polygons that agree to all orders of an infinitesimal with a polygon of small extension complexity but have much larger complexity themselves. Manuscript 07 is the base (every $n \geq 4$, linear and semidefinite lifts, descent for every convex valuation ring); 04’s field- and scale-general criterion, its same-circle regular and cyclic family with a two-sided $\sqrt{n}$ order, and its rank-one and factorization results are printed beside it.

1.2 How the manuscripts fit together

No manuscript contradicts another. Three places look like conflicts and are not.

Remark 1.1 (Three minimum degrees). [write] Part I prints three “minimum degree” or “number of layers” statements that can look incompatible: exactly $m - 1$ (theorem 9.1), at most $r \leq p$ layers (theorem 12.2), and one real vector or degree at most one (theorem 10.4 and the paragraph after figure 1). They are three different invariants of one lifting:

- (a) the *complete real dependence sign type* χ_h (definition 8.1): its rank is m (05’s letter; 02’s p), and its minimum polynomial germ degree is $m - 1$, or $m - r_0$ when the standard part is prescribed (theorems 9.1 and 9.2);
- (b) the *strict geometric history* of lower subdivisions along the common initial cuts: it has $r \leq m$ strict steps and needs exactly r layers (theorems 11.2 and 12.2);
- (c) the *final finite data* (the lifted face lattice, the marked subdivision, finitely many signs): one real height vector reproduces it, and with a prescribed standard part an affine-linear path of degree at most one does (theorem 10.4; the paragraph after figure 1).

The tempting false reading is that 05’s “degree $m - 1$ ” and 02’s “ r layers” or “one real vector” measure the same thing with different answers. They do not: (a) asks for *every* real dependence sign, including the uncountably many that no finite face test sees; (b) asks only for the finitely many subdivisions met along the cuts; (c) asks for one of them. Theorem 10.6 shows that (a) and (c) can be arbitrarily far apart. In 02’s hexagon, (a) and (b) both need three layers, while (c) needs one real vector.

Remark 1.2 (Valuation rank three and rank two). [write] Manuscript 07 builds its polygons in one field of valuation rank three, while manuscript 04 uses fields K_n of rank two, and 07 also proves that rank two is sharp for gaps invisible to all-order jets. These are consistent. An invisible gap needs a value group of rank at least two: the jet scale, and elements beyond all its powers. Manuscript 07’s third archimedean layer comes only from keeping $\varepsilon = \omega^{-1}$ as a separate layer above its jet scale $\omega^{-\omega}$, for its extra assertion that each polygon is flat-close to a fixed triangle at the larger scale ε ; 07’s own rank-two variant (corollary 72.4) drops that assertion, and 04’s construction never makes it. 04’s fields K_n all lie in one rank-two field, the real closure of $\mathbb{R}(\omega^{-1}, \omega^{-\omega\sqrt{q}} : q \text{ prime})$ (remarks 70.8 and 72.5). The criterion common to both (Part V) says that invisible gaps exist exactly when the valuation of the jet scale is not an order unit of the value group, which already happens in rank two.

Remark 1.3 (Directions of the transfer statements). [write] Several Parts pass between K and $\mathbb{R}$, in different directions. Transfer (theorem 18.1) keeps a finite combinatorial type exactly but forgets the scales. Standard part and residue maps keep the scale data of Part I (proposition 13.3) but can only coarsen, and in Part V can only lower extension complexity; manuscript 04’s real-algebraic transfer keeps the exact extension complexity. Manuscript 03’s descent theorem (Part IV) lands in the real closure of $\mathbb{R}(t)$, a field of valuation rank one where the all-order jet kernel of Part V vanishes; 03 makes no valuation-theoretic claim, so it does not conflict with Part V.

Part III’s theory holds for every set-sized real closed field $K \supseteq \mathbb{R}$ with its natural convex valuation; its surreal specifics are examples, and the word “surreal” in manuscript 06’s title is not a hypothesis.

1.3 Conventions and notation

Each Part keeps its source’s definitions and hypotheses and opens with a paragraph “Notation in this Part”. The following conventions hold throughout; the table lists the letters and words that mean different things in different Parts, with the reading to avoid.

- **No** is the class of surreal numbers, ω^γ Conway's monomials. Every construction takes place in a set-sized subfield K with $\mathbb{R} \subseteq K \subseteq \mathbf{No}$ (real closed unless a Part says otherwise).
- $\varepsilon = \omega^{-1}$ and $\tau = \omega^{-\omega}$ wherever a fixed surreal is meant; sources also use ε for an arbitrary positive infinitesimal in statements valid for all of them, and say so. The one exception is [Part III](#), which keeps manuscript 06's η for $\omega^{-\omega}$ because there τ names mixed-volume profiles. The renamed source letters are listed in the Parts' notation paragraphs and in [Appendix A](#).
- $\mathcal{O}$ is the standard part on the ring of finite elements ($\mathcal{O}$; manuscript 02 writes $\mathcal{O}_K$, 03 $\mathcal{O}_K$). As in the collection's notation file, it is the scalar map $\text{st} : \mathcal{O} \rightarrow \mathbb{R}$, applied coordinatewise.

Symbol or word	Meanings, by Part	Reading to avoid
ε	ω^{-1} wherever a fixed surreal is meant (I–V); an arbitrary positive infinitesimal in statements valid for all of them (I; in IV, 03's ε, δ are always arbitrary); [write] in VI, IX and X always an arbitrary positive infinitesimal (IX: ω^{-2} in one proposition); VIII: ω^{-1}	that 03's ε is ω^{-1}
τ	$\omega^{-\omega}$ in I, II and V (in V it replaces 07's ρ of its main construction; in III, 06's η is kept); in III, $\tau_I, \tau(\alpha)$ are 06's mixed-volume profiles; [write] VI: the value $v \det B$ of a reference volume; VIII: the variable of a Puiseux field, $v(\tau) = 1$	that a profile τ_I is a power of $\omega^{-\omega}$
η	I: explicit thresholds $\eta(f)$ (05), η (02), and the band width of proposition 14.2 ; III: $\omega^{-\omega}$ (06's letter, kept); IV: a real cutoff; V: 07's perturbation monomials η_ℓ , 04's flat bound or rational tolerance	that III's η is a threshold
t	I: the real parameter of polynomial models, $t \rightarrow 0^+$ (05); II and 02's introduction: the power-series variable, evaluated at $t = \varepsilon$; III: formal Hahn monomials t^γ ; IV: the real parameter or infinitesimal of $\mathbb{R}(t)$; [write] VI, VII: a positive infinitesimal (the variable of $\mathbb{Q}(t)$ in the checks); VIII: a real parameter in $(0, 1/6)$ or a positive infinitesimal, with s the parameter of a competing arc	$t = \omega^{-1}$ in Part I
m	I: the comparison rank $\dim_{\mathbb{R}} L_h(V)$ (05) = p (02), but in 02's blocks the stage index of H_m ; III: the number of bodies; V: an ordinary integer exponent (07) and $n = 2^m$ (04); [write] VI: the number of listed displacements m_i of one polytope; VII: the number of bodies; VIII: $m = 4k + 8$ configuration points; IX: the number of rows of A ; X: 13's $m_\varepsilon, m_d(t)$	that 02's m is the rank; that VIII's m is Part I's
p	I, II: 02's comparison rank (= 05's m)	
q	I: $n - d - 1 = \dim V$; III: $q(B) = v \det W_B$; V: 07's easy vertices q_i , 04's primes q_i	
r	I: 02's strict history length $r \leq p$; 05's $r_0 \in \{0, 1\}$; 02's gauge residuals $r_i(h)$; III: affine dimensions r, r_i	that $r_i(h)$ are history steps

Symbol or word	Meanings, by Part	Reading to avoid
L	I: 05's map L_h , 02's space L_A ; III: the scale lattice $L(P)$, an $\mathcal{O}$ -module; IV: the lattice-point count $L_P(n)$; [write] VI: $L(P)$ is Part III's scale lattice (source 08's displacement module $\mathcal{L}(P)$, renamed); VIII: $L(B)$ is a Lawrence polytope and $\mathcal{L}, \mathcal{L}_k$ are labeled face lattices	$L_A = L_h$; $L(P) = L_P$; that VIII's $L(B)$ is a lattice
V	I: the dependence space $\ker \tilde{A}$ (05), a configuration (02's transfer and Puiseux theorems); III: mixed volume V ; V: 07's coarse valuation ring V_τ ; [write] VI, VII: mixed volume V ; X: the small hull $V(S)$ (sans-serif)	
S, T	I: a sign cell, the tail sum $T(f)$; III: segments S_i, T_i ; IV: the real trace set $S = \text{Tr}_{\mathbb{R}}(P)$, an implanted polytope T ; V: the slack matrix S , 07's triangle T , a nonnegative factor T in $S = TU$	
ρ	II: a sign $\rho \in \{-1, 1\}$; III: the threshold ρ_g ; IV: separated real scales $\rho_1 \gg \dots \gg \rho_r$; V: a jet scale that 07 re-chooses (corollary 72.4, theorem 74.2)	that III's ρ_g is a jet scale
residue	I, II: 02's word for the standard part; III: the affine residue $\text{Res}(P)$, renormalised and dimension-preserving; V: residue maps of convex valuation rings and their residue fields k (07); [write] VI: $\text{res} = \text{st}$ and the residue flag $E_\gamma(P)$; VII: the residue configuration of a minimum-valuation basis	$\text{Res}(P) = \text{st}(P)$ (III's thin triangle has $\text{st } P$ a segment and $\text{Res } P$ a triangle)
shadow	I: the standard part of heights or of a configuration (05); IV, V: $\text{st}(P)$; V: 04's "algebraic finite-accuracy shadows" are real-algebraic approximants and are printed as approximants	
trace	IV: the real trace $\text{Tr}_{\mathbb{R}}(P) = P \cap \mathbb{R}^d$; V: the matrix trace tr	that $\text{Tr}_{\mathbb{R}}(P)$ is $\text{st}(P)$ or tr
valuation	I: $\text{le}(x)$, the leading exponent (larger means larger); III: v with larger value meaning smaller element, value group Γ not necessarily inside $\mathbb{R}$; IV: none used; V: ν , with larger value meaning smaller element (07; 04's v renamed); [write] VI, VII: ν (sources 08–10), the same valuation as III's v ; VIII: v and the order ord_0 at 0, positive meaning infinitesimal	a common sign convention across Parts
$\mathcal{O}$	the ring of finite elements, one ring in every Part ($\mathcal{O}_K$ in 02, $\mathcal{O}_K$ in 03, $\mathcal{O}$ in 04, 06, 07), with maximal ideal $\mathfrak{m}$	
flat ideal	V: $\mathfrak{j}_x = \bigcap_{m \geq 1} x^m \mathcal{O}$ for a positive infinitesimal x (07's $\mathfrak{j}_s$; 04's $\mathcal{I}_\varepsilon$ is printed $\mathfrak{j}_\varepsilon$)	
minimum degree	I: three different invariants (remark 1.1)	that $m - 1, r$ and "one real vector" conflict
degree	[write] I: least degree of a polynomial height vector; VIII: δ_{rat} , the homogeneous degree of a rational vertex arc; X: the degree of non-exposedness (remark 1.4)	that VIII's exponential bound contradicts Part I's
α	[write] a multi-index in III and VII; VI: the scale spectrum $\alpha(P) = (\alpha_1, \dots, \alpha_d)$ of one polytope	that VI's $\alpha(P)$ indexes a coefficient

Symbol or word	Meanings, by Part	Reading to avoid
profile	[write] III: $\tau(\alpha)$, τ_I ; VI: the values $\mu(P_1, \dots, P_d)$, $\mu_k(P, Q)$ and $s_k(P)$; VII: 09's $h_P(\alpha) = v(c_\alpha)$ with volume polynomial F_P , 10's $q_P(\alpha) = v(V_\alpha(P))$ with f_P (the same objects)	
compression	[write] I: 02's cut-compatible greedy compression (theorem 7.1); III: one-parameter specialization of finite-support Hahn certificates (theorem 41.2); VII: parallelotope compression of polytopes	that these three theorems share a statement
Z, T, B	[write] III: $Z = \sum[0, u]$, probe segments $T^{(j)}$, a basis matrix B ; VI: $Z(P; v_0) = \sum[-w, w]$, probe values $T_i(Q)$; VII: 09's $\sum[-b, b]$ and 10's $\sum[0, a]$, mixing matrices T_i , 10's parallelotopes B_i	
G_A, A, h	[write] I: A the base configuration, h the heights, G_A a Graver basis of $\ker_{\mathbb{Z}} A$; IX: $A \in \mathbb{Z}^{m \times d}$ a constraint matrix, h a purely infinite part, G_A the Graver set of $[A \ -I]$ projected to $\mathbb{Z}^d$; X: G_A a face of the polar moment hull	that IX's G_A is Part I's
$\mathbf{Oz}, \Pi$	[write] IV, IX: the omnific integers and (IX) their purely infinite part, with the omnific floor $\lfloor \cdot \rfloor_{\mathbf{Oz}}$	
small	[write] X: indexed by a set in a fixed NBG universe with global choice; a small hull $V(S)$ is generally a proper class	that a small hull is a set
rank	I: the comparison rank $m = p$; V: the valuation rank (the number of archimedean layers of the value group), matrix rank, nonnegative rank rank_+ and psd rank rank_{psd}	valuation rank three versus two (remark 1.2)

1.4 What is claimed, and what is not

Every Part states its own claims and non-claims, and the merge keeps all of them. In summary: every theorem has a written conventional proof in its source; the verification programs check finite examples only; no manuscript claims literature-wide priority or the solution of a named published problem; the classical ingredients are attributed in each Part. [write] None of the report's results is formalized. The only ProveIt declaration that any of Parts I–V uses as an input is `Surreal.Foundations.SignSequence.omnific_bounded_iff` (Algebra/SurrealNumbers/Surreal/Foundations/OmnificUnits.lean; bounded omnific integers are ordinary integers), in [Part IV](#). In Parts VI–X no theorem is formalized either. Source 10 (Part VII) names the valuation and standard-part declarations of `Surreal/Foundations/SignSequenceValuation.lean` and `SignSequenceStandardPart.lean` as the repository's versions of its basic objects; source 12 (Part IX) states its integer-part interface as proposed contracts, and the repository's omnific Lean modules already prove that interface for the actual omnific ring (the note after its formalization table names the declarations); source 11 (Part VIII) inspected `Geometry.lean` as background. Manuscripts 05 and 02 inspected the coordinate-geometry file `Algebra/SurrealNumbers/Surreal/Algebra/Geometry.lean` as background; no result depends on it. Placement of this report beside the collection's Lean development confers no formal status on it.

Manuscripts 05 and 07 correct a privately supplied draft, *Finite Convexity and Exact Lexicographic Optimization over the Surreals* (22 September 2026), which is not part of ProveIt and could not be checked ([section 20](#) and [Part V](#)).

1.5 Labels, numbering and marks

Every label carries the prefix `poly::`; material from manuscripts 02, 03, 04, 06 and 07 carries the sub-prefixes `poly:ts:`, `poly:ehr:`, `poly:bao:`, `poly:mv:` and `poly:ic:`, followed by the delivered label unchanged. Sections are numbered consecutively through the report, and theorems, equations and questions within sections, so a source’s numbers are not those printed here. A result printed once for two sources carries both labels. Citation keys carry the prefixes `fs:` (05), `ts:` (02), `ehr:` (03), `bao:` (04), `mv:` (06) and `ic:` (07); the references are listed at the end, grouped by Part. [write] Parts VI–X use the sub-prefixes `poly:tom:` (source 08), `poly:trop:` (10), `poly:fmv:` (09), `poly:cd:` (11), `poly:oih:` (12) and `poly:bf:` (13), and the citation-key prefixes `tom:`, `trop:`, `fmv:`, `cd:`, `oih:` and `bf:`.

1.6 Six further manuscripts: Parts VI–X

[write] Six more manuscripts, all dated 30 September 2026, arrived in batch 59 and were merged in its write phase as Parts VI–X. Each continues a theorem or question of Parts I–IV; none saw this report, and only one cites a batch-58 source (manuscript 05, by title). Batch-59 manuscripts are numbered 01–06 within their batch, which would collide with the batch-58 numbers used above, so the report calls each by the prefix of its shipped files: *sources 08–13*.

Part	Subject	Sources (base first; batch-59 no.)	Labels
VI	Mixed-volume tomography	08 (01)	<code>poly:tom:</code>
VII	Realization and obstruction of mixed-volume profiles	10 (03), 09 (02)	<code>poly:trop:</code> , <code>poly:fmv:</code>
VIII	Moving vertices and exponential contact depth	11 (04)	<code>poly:cd:</code>
IX	Omnific integer hulls and the rational-normal dichotomy	12 (05)	<code>poly:oih:</code>
X	Beyond finite presentations	13 (06)	<code>poly:bf:</code>

The new Parts extend the spines of [section 1](#):

- *The residue and volume spine* continues from [Part III](#) into [Parts VI](#) and [VII](#). Source 08’s displacement module is Part III’s scale lattice; its colorful determinant formula, adaptive test and probe classification re-derive Part III’s theorems independently, and are printed once with both sources named. Sources 10 and 09 answer one prompt twice; source 10 is the base (sharper constants and the general matroid criterion), and 09’s additional theorems are printed beside it. Parts VI and VII partly answer five of Part III’s questions.
- *The lifting spine* continues from [Part I](#) into [Part VIII](#): once every vertex may move, the degree-one realization of [theorem 10.4](#) fails exponentially.
- *The lattice-point spine* continues from [Part IV](#) into [Part IX](#): integer hulls instead of lattice-point counts, over any integer part and with infinite coordinates and right-hand sides.
- *Foundations and infinite presentations*: [Part X](#) continues Part I’s scalar-extension correction and Part III’s question on convex bodies beyond finite vertex sets.

Remark 1.4 (Three degrees). [write] “Degree” means three different things. Part I’s degree ([theorem 9.1](#)) is the least degree of a polynomial height vector over a fixed real base. Part VIII’s δ_{rat} ([definition 120.2](#)) is the homogeneous degree of a rational arc along which every vertex

moves; it can be exponential in the number of vertices ([theorem 120.3](#)), while Part I's is at most $n - d - 1$. Part X's degree ([theorem 159.2](#)) is the degree of non-exposedness of a face, Martínez-Legaz's invariant. None bounds another.

Placement. Parts VI–X follow the roadmap chapter of Parts I–V ([section 83](#)) rather than precede it, so that no section, theorem or equation number of Parts I–V, of that chapter or of the provenance appendix changed when they were added. Their own formalization proposals stay inside the Parts. Paragraphs written in the batch-59 write phase are marked [write] like those of batch 58; their provenance is in [section A](#).

Part I

Liftings of a fixed real configuration

Source and scope. [write] This Part prints manuscript 05 of batch 58, *Finite-Scale Geometry of Surreal Liftings: Sharp dependence compression, exact polynomial degree bounds, and linear realization of finite combinatorics* (research manuscript prepared for Vladimir Reshetnikov, an AI-assisted contribution to the ProveIt research program, 30 September 2026; a 25-page A4 PDF; pinned to 5e0f4e0), whole and in its order. It is the base of this Part and of the report. Merged into it is the lifting material of manuscript 02, *Surreal Polytopes at Finite and Transfinite Scales: Sharp lifting compression and universal coordinate-truncation histories* (research article prepared for Vladimir Reshetnikov in connection with the ProveIt research repository, 30 September 2026; a 28-page Letter PDF; pinned to 6996fee): its introduction, its working-field conventions, its cut-compatible compression theorem, its fixed-base lifting and history sections, its standard-part and Puiseux sections, its certificate workflow and its query-model bound. Manuscript 02's Section 6, on coordinate truncation, is [Part II](#).

The two manuscripts prove one theorem from two sides: surreal heights on a fixed real base compress to at most $q = n - d - 1$ real layers that preserve every real affine-dependence sign. Manuscript 05 is the base because it assumes no more (a set-sized ordered subfield $\mathbb{R} \subseteq K \subseteq \mathbf{No}$, with real closure only where a statement needs it, against 02's real closed K) and concludes more (exact leading terms rather than signs, the complete flag classification, and exact germ degrees). A result proved by both is printed once, with both sources named; a genuinely different proof from 02 is kept as a second proof; the statements of 02 that go beyond 05 (compatibility with every common initial cut, the classification of histories, planar sharpness, the polar and band identities, the Puiseux model and the query-model bound) are printed whole. Each block from 02 opens with a [write] note that names it and relates it to 05.

Notation in this Part. [write] (The report-wide table is [section 1.3](#).) ε is ω^{-1} wherever a fixed surreal is meant; both sources also write ε for an arbitrary positive infinitesimal in statements that hold for every one, and say so. Manuscript 02 writes $t = \omega^{-1}$; in the 02 material of this Part that letter is replaced by ε , because here t is manuscript 05's *real* parameter of polynomial models ($H(t)$, $t \rightarrow 0^+$). Only 02's introduction ([section 3](#)), which previews [Part II](#), keeps t : there, as in [Part II](#), t is the variable of power series evaluated at $t = \varepsilon$. $\tau = \omega^{-\omega}$ (02's letter; 05's ε_2 in [section 16.5](#), renamed). The comparison rank is written m by 05 and p by 02; it is one number, $m = \dim_{\mathbb{R}} L_h(V) = \text{rank}_{\mathbb{R}}(c \mapsto c \cdot h) = p$ ([corollary 6.3](#) and [theorem 11.2\(d\)](#)). The letter p is kept in 02's material because 02 uses m for the stage index $0 \leq m \leq p$ of its polynomials H_m : in 02's blocks m is *not* the comparison rank. 05's dependence space $V = \ker_{\mathbb{R}} \tilde{A}$ is 02's L_A (02's L_A is a space, not 05's map L_h); in 02's transfer and Puiseux theorems V is a configuration of points. 05's slack $s_{B,i}(h)$ is 02's $\delta_{B,j}(h)$, 05's equality set $I_B(h)$ is 02's cell $D_B(h)$, and 05's

gauge residuals g_i are 02's $r_i(h)$ (not the history length r). 02's ring $\mathcal{O}_K$ of finite elements is the ring written $\mathcal{O}$ elsewhere in the report. In 02, “residue” means the standard part st ; it is not the affine residue Res of [Part III](#). Three different minimum degrees occur in this Part; they are compared in [remark 1.1](#).

Abstract of manuscript 05. A finite surreal polytope has an ordinary realizable face lattice, but its standard part can erase faces and their incidences. We isolate a setting in which the scale information can be classified exactly: surreal heights on a fixed real configuration of n labeled points of affine dimension d . Let V be the real affine-dependence space and let $L_h(c) = \sum_i c_i h_i$. If $m = \dim_{\mathbb{R}} L_h(V)$, then $m \leq n - d - 1$. After an affine height is removed, at most m monomial coefficient slices preserve the leading term of *every* real dependence evaluation, including its exact vanishing. This remains true for arbitrary set-sized Conway supports, not just finite series. The complete dependence sign type is classified by an oriented partial flag of length m .

A real polynomial height vector represents that entire sign type as a germ at 0^+ with minimum degree $m - 1$ when $m > 0$. For bounded heights, requiring the correct standard part changes the exact minimum to $m - r_0$, where $r_0 \in \{0, 1\}$ is the rank of $\text{st} \circ L_h$. By contrast, preserving only the finite lifted combinatorics and the standard part always requires degree at most one. For integral bases, the same linear deformation can also preserve the entire toric initial ideal. We prove both bounds sharply and explain why they do not conflict. We give determinantal and barycentric certificates, an explicit rational specialization threshold, counterexamples to naive truncation and shadow arguments, and corrections to two statements in an earlier convexity draft. Exact verification code and a formalization plan accompany the text. The underlying normal-form, ordered-field, and symbolic-perturbation tools are classical; the proposed contribution is this sharp, unified lifting package, not a claim to have invented higher-rank secondary geometry.

Status (manuscript 05). The statements below have mathematical proofs in this manuscript. The program checks finite examples only. No new Lean proof or independent refereeing is claimed. Literature-wide priority of the package has not been established.

Abstract of manuscript 02. (It covers this Part and [Part II](#).) Finite polytopes over the surreal numbers have no new face-lattice types: real-closed-field transfer already gives real algebraic realizations. The interesting question is therefore not which single finite combinatorial types exist, but how much information an infinitesimal or transfinite hierarchy can carry.

We establish a precise contrast. For a fixed real configuration of n points spanning dimension d , an arbitrary surreal lifting vector has a canonical support-adapted compression with at most $n - d - 1$ real coefficient vectors. This compression preserves the signs of *all* real affine-dependence functionals and the complete history of regular subdivisions obtained by common initial normal-form truncation. The number of effective comparison layers is an intrinsic real rank; the number of strict subdivision changes is at most this rank. Every marked coherent refinement chain occurs, and parabolic polygons attain the bound already in dimension two. Explicit finite sign certificates turn the compressed history into real polynomial liftings. These positive statements are constructive consequences and refinements of classical lexicographic-order and secondary-polytope ideas, not new versions of those foundational theories.

By contrast, four bounded planar points suffice to encode an arbitrary prescribed triangle/quadrilateral history under common coordinate truncation, from degree two onward. The construction uses reciprocal convergent power series with rational coefficients. A monomial

beyond all ordinary powers makes the full configuration uniform and independently selects its final hull type without changing any finite truncation. An elementary rational example already alternates indefinitely. We also prove standard-part identities, explain the failure of a naive map on faces, and give a semialgebraic Puiseux model for any finite collection of incidence data. Full conventional proofs, an exact-arithmetic verification program, a formalization roadmap, and further research questions are supplied.

Status (manuscript 02). AI-assisted, unrefereed research manuscript. The new arguments have not been independently certified or checked in Lean. Finite computations are regression tests, not substitutes for the proofs. No claim of exhaustive literature priority or resolution of a named published major conjecture is made.

2 The problem, its scope, and its relation to earlier work

2.1 What can genuinely change over the surreals?

A finite list of points with surreal coordinates may exhibit arbitrarily separated scales. Nevertheless, its face lattice is controlled by finitely many polynomial signs. Real-closed-field transfer therefore prevents new *finite realizable combinatorial types* from appearing solely because the ambient field is **No**. This observation does not make the geometry empty. It redirects attention to specialization, leading terms, and which comparisons must be retained simultaneously.

Three different demands must not be conflated:

- (i) retain the finite face structure, or the finite regular subdivision;
- (ii) retain the signs of all real affine-dependence evaluations;
- (iii) retain their actual leading monomials and leading coefficients.

The second demand involves an uncountable family of tests, even when the point configuration is finite. The third is stronger still. This manuscript shows that these distinctions lead to different, exactly computable notions of scale complexity.

The real-base hypothesis is essential. Fix

$$A = (a_1, \dots, a_n), \quad a_i \in \mathbb{R}^d, \quad \text{aff}\{a_1, \dots, a_n\} = \mathbb{R}^d.$$

We may take the points distinct; every label is retained, including labels that are not vertices of a resulting polytope. Only the heights $h_i \in \mathbf{No}$ vary. The lifted configuration consists of $p_i = (a_i, h_i)$. Determinants involving its height row are *linear* in h . This simple fact is the source of the sharp results. General moving surreal coordinates lead to polynomial, not merely linear, sign conditions; Section 18 treats the corresponding weaker conclusion.

2.2 Connection with the repository and the previous convexity draft

The inspected ProveIt source `Algebra/SurrealNumbers/Surreal/Algebra/Geometry.lean` develops coordinate geometry over an ordered base field, including dot and cross products, Gram identities, and Ptolemy-related identities [8]. The surrounding project documents a constructed surreal field, normal forms, and its real closedness; its guide explicitly separates unrefereed research manuscripts from statements mapped to Lean declarations [9]. The present paper adds a higher-dimensional *affine-dependence and lifting* layer. It does not depend

on the more speculative results elsewhere in that collection, and it does not infer machine verification from a repository README. [write] The inspected source is unchanged at the report's placement: the blob of `Geometry.lean` is `450fd4b` at `5e0f4e0`, at `6996fee` and at `b30441a`. Its declarations in the namespace `Surreal.Complexify` include `dot`, `cross`, `cross_def`, `cross_swap`, `dot_sq_add_cross_sq` (the Gram identity), `ptolemy_identity`, `ptolemy`, `triangleArea`, `heron_factorization` and `heron`. No result of this report uses them, and none of the report's results is formalized (section 1).

An earlier companion draft, *Finite Convexity and Exact Lexicographic Optimization over the Surreals*, dated 22 September 2026, already treats finite objective hierarchies and a sufficiently-small-parameter compiler [10]. We do not present that compiler as new. Our questions instead concern compression of arbitrary height normal forms, preservation of *all* real dependence tests, and optimal polynomial degree. Section 20 also corrects its global validity-interval statement and clarifies the field of convex combinations in its shadow statement. [write] That draft is not part of ProveIt; see the note at the head of section 20.

2.3 Literature boundary

Conway normal forms and the ordered-field foundations are classical [1]. Regular liftings, subdivisions, and secondary polytopes are standard subjects [2]. Joswig and Smith develop higher-rank tropical geometry and explicitly ask how regular-subdivision duality, multiplicities, and combinatorial structure should extend; their rank-two decomposition already exhibits successive minimizer data [3].

The literature check must include a very recent overlap: Chirivì, Costa Cesari, Fang, and Littellmann's *Higher rank Gelfand–Kapranov–Zelevinsky fans*, arXiv:2604.20250v2, revised 13 September 2026, develops higher-rank GKZ cones and fans and associated quasi-valuations and degenerations [4]. Thus this article does *not* claim the first higher-rank secondary theory or a complete solution of the broader questions in [3]. The explicit scale and degree classification below has a narrower scope. Its linear-algebraic mechanism is elementary once stated correctly; its priority as a package requires further literature and referee scrutiny.

2.4 Main answers at a glance

Put $q = n - d - 1$. The quantities m and r_0 will be defined precisely below. For $m > 0$, the conclusions are:

Information to preserve	Exact bound or conclusion
All real dependence leading terms	$m \leq q$ selected monomial slices
All real dependence signs, as a polynomial germ	Minimum degree $m - 1$
Same signs and the labeled real shadow	Minimum degree $m - r_0 \leq q$
Finite lifted sign data and the shadow	Minimum degree 0 or 1
A fixed positive real specialization for every real dependence	Impossible when $m \geq 2$

The first count is taken *after removing an affine height*. The polynomial models preserve signs, not numerical surreal exponents. The finite-data degree is zero exactly when the shadow already satisfies all the required signs. These qualifications are part of the theorems, not exceptions left implicit.

3 Questions, answers, and provenance

[write] This section is the introduction of manuscript 02, printed whole. It previews both the fixed-base lifting results, printed in this Part beside manuscript 05's, and the coordinate-truncation results of [Part II](#); in its second subsection $t = \omega^{-1}$ is the variable of the power series of [Part II](#). Its three operations (i)–(iii) are the settings of [theorem 18.1](#) (for (i)), of the lifting results of this Part (for (ii)), and of [Part II](#) (for (iii)).

Replacing real coordinates by surreal coordinates introduces exact infinite and infinitesimal scales. It does not automatically introduce new finite combinatorial objects. A polygon with one infinitesimal edge is still a polygon; its limiting picture may forget an edge, but the edge is present before taking a residue. To obtain a meaningful research problem one must specify which information is retained and which operation changes the scale.

This article separates three operations:

- (i) replacing one finite configuration by another with the same signed incidences;
- (ii) keeping a real base configuration fixed and truncating its lifting heights at common initial normal-form cuts;
- (iii) truncating the coordinates of the base configuration itself.

The first operation has a classical real-algebraic transfer theorem. The second admits a sharp finite-dimensional compression. The third can have arbitrary infinite combinatorial histories even with four planar generators. Conflating these operations would erase the main phenomenon.

3.1 The main positive statement

Let $A = (a_1, \dots, a_n) \in (\mathbb{R}^d)^n$ affinely span $\mathbb{R}^d$, and let $h \in \mathbf{No}^n$ be a vector of heights. Its lower lifted hull defines a marked regular subdivision of A . Set

$$L_A = \left\{ c \in \mathbb{R}^n : \sum_i c_i = 0, \quad \sum_i c_i a_i = 0 \right\}, \quad q = \dim L_A = n - d - 1.$$

Theorems [7.1](#) and [11.2](#) construct vectors $w^1, \dots, w^p \in \mathbb{R}^n$, with $p \leq q$, such that

$$\text{sgn}(c \cdot h) = \text{lexsgn}(c \cdot w^1, \dots, c \cdot w^p) \quad (c \in L_A). \quad (3.1)$$

The same list, read through its initial segments, preserves *every* initial normal-form lifting subdivision, including those reached only after transfinitely many redundant coefficient positions. The least possible number of vectors preserving the complete profile (3.1) is

$$p = \text{rank}_{\mathbb{R}}(L_A \longrightarrow \mathbf{No}, c \mapsto c \cdot h).$$

There are at most p strict subdivision changes, and q can occur in the plane. Theorem [12.2](#) characterizes the subdivision histories, with repetitions suppressed, as the finite marked regular refinement chains starting at the trivial subdivision.

3.2 The main negative statement

For a positive infinitesimal $t = \omega^{-1}$, consider four points

$$A_0 = (0, 0), \quad B = (1, f(t)), \quad C = (g(t), 1), \quad D = (0, 2).$$

Theorem 26.3 shows that, given any binary sequence from degree two onward, one can choose reciprocal convergent power series f, g with rational coefficients so that the hull of the degree- N coordinate truncations is a triangle or a quadrilateral according to the prescribed N th bit. Theorem 26.6 changes the full endpoint to either uniform configuration type by inserting $\omega^{-\omega}$, without changing any finite truncation.

This disproves a natural extension of the positive theorem: there is no bound depending only on n and d for the number of combinatorial changes under arbitrary common coordinate truncation. It also shows that requiring the full surreal configuration to be uniform is insufficient. By contrast, uniform ordinary power-series configurations do stabilize; their relevant nonzero determinants have finite orders (Proposition 26.7).

3.3 What is classical, and what is claimed here

The surreal normal form and its leading-coefficient order are classical; we use Gonshor’s account [1]. Real-closed-field transfer is Tarski’s theory [11]. Linear programming and finite convex-hull arguments over ordered fields, including finite Puiseux specialization, are established tools; see Joswig–Loho–Lorenz–Schröter [12]. Finite-dimensional total orders on real vector spaces have lexicographic models; Wortel’s Proposition 4.4 is a convenient precise reference [13]. The secondary polytope has dimension $n - d - 1$ and encodes regular subdivisions by refinement; Lee–Santos, Theorem 16.4.1, summarizes this classical structure [14].

Consequently, neither the existence of lexicographic representations nor the dimension bound on coherent refinement chains is claimed as a new foundational discovery. The contribution developed here is a single explicit framework that starts with arbitrary surreal normal-form coefficients, preserves all support-initial histories, gives a sharp intrinsic comparison rank and finite certificates, and contrasts this with a universal four-point coordinate-truncation construction. The arbitrary prescribed history, its convergent reciprocal realization, and the independent uniform transfinite endpoint are the principal specific constructions advanced by this manuscript. They come with proofs; their historical priority has not been established by an exhaustive search.

The questions answered here are formulated in this article. We do not attribute them to a published conjecture unless explicitly stated, and we do not reclassify classical transfer as the solution of a new open problem.

3.4 Connection with the repository

The inspected repository snapshot is

6996fee43cc97b6c16351def7507d59a95bf62f0.

Its commit timestamp is 30 September 2026, 16:07:39 UTC. The Algebra/SurrealNumbers subproject connects surreal normal forms, ordered-field algebra, finite geometry, and research reports. In particular, we inspected the actual source

Surreal/Algebra/Geometry.lean

within that subproject. It defines dot and cross products over a general coefficient ring and supplies Gram, Ptolemy, and Heron results under explicitly stated ordered-field and square-root hypotheses [16]. Its `cross_def` and `cross_swap` identities are exactly the kind of finite determinant algebra used in our planar examples.

The present article extends this direction toward finite configurations, lifting circuits, normal-form comparison ranks, and scale-sensitive convex hulls. It does not rely on unreviewed repository research claims as axioms. The repository’s claim ledger and individual source declarations must be consulted separately from its narrative reports. We did not run its Lean build, and no theorem of this article is presented as already formalized. The focused source inspection is not a proof that no related idea appears elsewhere in the large repository. Detailed source identities and the verification boundary are recorded in the accompanying notes. [write] Those notes are shipped with this report as `02-transfinite-scales-sources.md` and `02-transfinite-scales-proof_audit.md` (delivered as `notes/sources.md` and `notes/proof_audit.md`).

4 Foundations, normal forms, and the affine quotient

4.1 Finite mathematics inside a class field

Although **No** is a proper class, each normal form has set-sized support, and this paper uses only finitely many numbers at a time. Statements about convex hulls can be interpreted in a set-sized ordered subfield K containing $\mathbb{R}$ and the finitely many coordinates in question. When needed, enlarge K to its real closure inside **No**. Selected monomials can also be added to this field. We do not apply first-order model theory directly to a proper-class structure, or appeal to compactness of a surreal interval.

For points $v_1, \dots, v_N \in K^D$, our convention is

$$\text{conv}_K(v_1, \dots, v_N) = \left\{ \sum_{i=1}^N \lambda_i v_i : \lambda_i \in K_{\geq 0}, \sum_i \lambda_i = 1 \right\}.$$

A nonempty exposed face is the set maximizing a K -linear functional. For a finite polytope these give the usual faces. One elementary justification is finite linear elimination: feasible nonnegative combinations and supporting inequalities admit finite certificates, and redundant inequalities can be removed until facet inequalities remain. Alternatively all fixed-size assertions can be transferred from real closed fields. The later lifting certificates will be proved directly.

4.2 Normal forms and standard part

Every nonzero surreal has a unique Conway normal form

$$x = \sum_{\gamma \in S} x_\gamma \omega^\gamma, \quad x_\gamma \in \mathbb{R} \setminus \{0\}, \quad (4.1)$$

where S is reverse well-ordered: every nonempty subset has a greatest member [1]. We write

$$\text{le}(x) = \max S, \quad \text{LT}(x) = x_{\text{le}(x)} \omega^{\text{le}(x)}, \quad \text{LT}(0) = 0.$$

Throughout, ω^γ denotes the Conway omega-map monomial in normal form; no identity with $\exp(\gamma \log \omega)$ is assumed for general surreal γ . The sign of x is the sign of its leading coefficient. A finite union of reverse well-ordered supports is reverse well-ordered, so real linear combinations of a finite list are computed coefficientwise. These are the only infinite-series operations needed in the central proof.

A surreal x is *finite* when $|x| < M$ for some real $M > 0$. Its normal form has no positive exponent, and its standard part is its coefficient at exponent zero. The map st on finite surreals

is an order-preserving ring homomorphism to $\mathbb{R}$. In particular $x \geq 0$ implies $\text{st}(x) \geq 0$, not necessarily $\text{st}(x) > 0$. The loss of strictness is responsible for geometric collapse.

4.3 Removing affine heights

Let

$$\tilde{A} = \begin{pmatrix} 1 & \cdots & 1 \\ a_1 & \cdots & a_n \end{pmatrix}, \quad V = \ker_{\mathbb{R}} \tilde{A}, \quad \dim_{\mathbb{R}} V = q.$$

An element $c = (c_i) \in V$ is a real affine dependence: $\sum_i c_i = 0$ and $\sum_i c_i a_i = 0$. Define

$$L_h : V \longrightarrow \mathbf{No}, \quad L_h(c) = \sum_i c_i h_i, \quad m = \dim_{\mathbb{R}} \text{im } L_h. \quad (4.2)$$

Thus $0 \leq m \leq q$. Adding the values of any surreal-valued affine function on $\mathbb{R}^d$ to the heights leaves L_h unchanged.

Choose an affine basis $B \subseteq \{1, \dots, n\}$ with $|B| = d + 1$. Write the real barycentric coordinates of a_i relative to B as

$$a_i = \sum_{b \in B} \lambda_{B,i,b} a_b, \quad \sum_{b \in B} \lambda_{B,i,b} = 1.$$

They are uniquely defined, without a positivity requirement. Set

$$\ell(a_i) = \sum_{b \in B} \lambda_{B,i,b} h_b, \quad g_i = h_i - \ell(a_i). \quad (4.3)$$

The function ℓ is affine and $g_b = 0$ for $b \in B$.

Lemma 4.1 (The affine gauge). *Projection $c \mapsto (c_i)_{i \notin B}$ is a real-linear isomorphism $V \simeq \mathbb{R}^q$. Under it,*

$$L_h(c) = \sum_{i \notin B} c_i g_i, \quad m = \dim_{\mathbb{R}} \text{span}_{\mathbb{R}} \{g_i : i \notin B\}.$$

The shear $(a, z) \mapsto (a, z - \ell(a))$ preserves the labeled lifted combinatorics and the projected lower subdivision.

Proof. The equations $\tilde{A}c = 0$ determine c_B uniquely from c_{B^c} because the columns indexed by B form an invertible square matrix. Affine functions annihilate affine dependencies, yielding the displayed formula. The shear is an invertible affine transformation with positive determinant one. An inequality $z \geq u \cdot a + v$ becomes another inequality with coefficient $+1$ on z , so lower supporting hyperplanes remain lower supporting hyperplanes. General faces are preserved by affine invertibility. $\square$

5 Ordered fields, finite configurations, and normal forms

[write] Manuscript 02's conventions. Its working field is a set-sized *real closed* K , whereas manuscript 05 (section 4) needs only a set-sized ordered subfield and passes to the real closure where a statement requires it; every 02 result printed in this Part is stated over 02's K . The common initial normal-form cut (5.2) is the operation on which 02's cut-compatibility clauses (theorem 7.1(c),(d) and theorem 11.2(b)) and all of Part II rest.

5.1 A set-sized working field

Although **No** is a proper class, the data used by any one construction here form a set. Choose a set-sized real-closed subfield

$$\mathbb{R} \subseteq K \subseteq \mathbf{No}$$

containing those data. When all initial normal-form truncations of finitely many coordinates are used, include that set of truncation values as well. One obtains such a field by taking the real closure inside **No** of the field generated by $\mathbb{R}$ and the specified set. Whenever needed, include $\varepsilon = \omega^{-1}$ and $\tau = \omega^{-\omega}$.

For $v_1, \dots, v_n \in K^d$, write

$$P_K = \text{conv}_K(v_1, \dots, v_n) = \left\{ \sum_{i=1}^n \lambda_i v_i : \lambda_i \geq 0, \sum_i \lambda_i = 1 \right\}.$$

All linear maps displayed with target **No** below have image in such a working field; their ranks are ordinary set-sized real vector-space ranks. All faces in this article are exposed polytope faces, including the empty and full faces. A proper nonempty face is the maximizer set of a linear functional. The usual equivalence with the algebraic notion of a face holds for finite polytopes over an ordered field. Supporting-hyperplane existence, lower-hull descriptions, and linear feasibility can be proved by finite elimination or the ordered-field Farkas lemma [12]. Nothing here uses compactness in the order topology of K .

The finite face labels are independent of enlarging the real-closed working field: whether a specified finite system of linear equalities and inequalities has a solution is unchanged by a real-closed extension. This justifies speaking informally of a finite “surreal polytope” while carrying out every set-theoretic construction in K .

5.2 Common initial normal-form truncation

Every surreal number has a Conway normal form with real coefficients and reverse-well-ordered monomial support [1]. For a finite vector $h \in \mathbf{No}^n$, take the union of the coordinate supports and write

$$h = \sum_{\alpha < \lambda} v_\alpha \omega^{\gamma_\alpha}, \quad v_\alpha \in \mathbb{R}^n, \quad \gamma_\alpha > \gamma_{\alpha'} \quad (\alpha < \alpha'). \quad (5.1)$$

Here λ is a set ordinal, and zero entries in a coefficient vector are allowed. The finite union of the supports is again reverse well ordered. For $0 \leq \beta \leq \lambda$, define

$$h^{<\beta} = \sum_{\alpha < \beta} v_\alpha \omega^{\gamma_\alpha}. \quad (5.2)$$

The cut is common to all coordinates and retains the large monomials first. It is not a truncation of Conway sign sequences. Nor is it an operation that deletes leading terms and retains a tail.

A real linear functional commutes with this normal-form sum. After any cancellations, the sign of its value is the sign of its first nonzero coefficient. This elementary fact, rather than real-closedness, supplies the positive compression theorem. In a power-series example we write $\varepsilon = \omega^{-1}$ [write] (in Part II: t , evaluated at $t = \varepsilon$) and use the increasing exponents $0, 1, 2, \dots$

5.3 Lexicographic sign

For a finite real tuple $(b_1, \dots, b_p)$, let $\text{lexsgn}(b_1, \dots, b_p)$ be the sign of its first nonzero entry, or zero when all entries are zero. The empty tuple has sign zero. If ε is a positive infinitesimal, then

$$\text{sgn}\left(\sum_{j=1}^p b_j \varepsilon^{j-1}\right) = \text{lexsgn}(b_1, \dots, b_p).$$

This identity holds for every real tuple. Replacing ε by a positive *real* number preserves it for a prescribed finite collection of nonzero tuples when that number is sufficiently small, but not uniformly for all real tuples. This quantifier distinction will matter.

6 A finite coefficient-pivot theorem

Theorem 6.1 (Simultaneous leading-term compression). *Let $g = (g_1, \dots, g_q) \in \mathbf{No}^q$. There exist distinct exponents $\gamma_1 > \dots > \gamma_m$ and linearly independent real vectors $b_1, \dots, b_m \in \mathbb{R}^q$ such that*

$$g^\# = \sum_{j=1}^m b_j \omega^{\gamma_j} \tag{6.1}$$

satisfies

$$\text{LT}(\lambda \cdot g^\#) = \text{LT}(\lambda \cdot g) \quad \text{for every } \lambda \in \mathbb{R}^q. \tag{6.2}$$

Here $m = \dim_{\mathbb{R}} \text{span}_{\mathbb{R}}\{g_1, \dots, g_q\}$. The exponents and vectors may be chosen among the actual coefficient slices of g . Moreover, the nonzero real combinations of the coordinates of g have exactly m different leading exponents.

Proof. Take the finite union S of the supports of the coordinates and write

$$g = \sum_{\gamma \in S} b_\gamma \omega^\gamma, \quad b_\gamma \in \mathbb{R}^q \setminus \{0\}.$$

If S is empty, the conclusion holds with $m = 0$. Otherwise choose $\gamma_1 = \max S$. Having chosen $\gamma_1, \dots, \gamma_j$, put $U_j = \text{span}_{\mathbb{R}}\{b_{\gamma_1}, \dots, b_{\gamma_j}\}$. If every coefficient vector belongs to U_j , stop. If not, the set

$$\{\gamma \in S : b_\gamma \notin U_j\}$$

is nonempty and has a greatest element; choose that as γ_{j+1} . Each choice strictly increases the real dimension of U_j , so at most q choices are made. They have strictly decreasing exponents. Write $b_j = b_{\gamma_j}$.

Fix $\lambda \in \mathbb{R}^q$. If $\lambda \cdot b_j = 0$ for every selected vector, then λ annihilates every coefficient vector, by the stopping condition. Hence $\lambda \cdot g = \lambda \cdot g^\# = 0$. Otherwise let j be the first selected index with $\lambda \cdot b_j \neq 0$. Every coefficient vector at an exponent greater than γ_j belongs to U_{j-1} : this is precisely the maximality in the choice of γ_j . Those coefficients vanish after pairing with λ . The coefficient at γ_j is $\lambda \cdot b_j \neq 0$. Consequently both combinations in (6.2) have leading term $(\lambda \cdot b_j) \omega^{\gamma_j}$.

The maps $\lambda \mapsto \lambda \cdot g$ and $\lambda \mapsto (\lambda \cdot b_1, \dots, \lambda \cdot b_m)$ have the same kernel. The latter has rank m since the selected vectors are independent. Rank-nullity identifies m with the dimension of the span of the coordinates of g . Finally, for every j there exists a real λ annihilating U_{j-1} but not b_j . Thus γ_j actually occurs as a leading exponent. No other leading exponent can occur by the preceding argument. $\square$

Remark 6.2 (Finite rank is not finite support scanning). The proof makes finitely many *rank-increasing choices*. It need not inspect finitely many consecutive terms of each input normal form. There may be an infinite block of coefficient slices in U_j before the next slice outside U_j . Without an effective representation and a certificate that no further independent slice exists, the theorem is not an algorithm for arbitrary surreal names. Section 21 gives the exact finite-support algorithm that is implemented here.

Corollary 6.3 (Sharp affine-height compression). *For a lifting h of A , remove an affine height as in (4.3). There is a height vector*

$$h_i^\# = \ell(a_i) + \sum_{j=1}^m b_{j,i} \omega^{\gamma_j}, \quad b_{j,b} = 0 \quad (b \in B),$$

with $m \leq n - d - 1$, such that

$$\text{LT}(L_{h^\#}(c)) = \text{LT}(L_h(c)) \quad (c \in V).$$

Exactly m distinct leading exponents occur among the nonzero $L_h(c)$. In particular m is the minimum number of distinct monomial scales in a real-coefficient representation preserving all these leading terms modulo affine heights.

Proof. Apply Theorem 6.1 to the q residual coordinates, and use Lemma 4.1. A representation with s monomial scales can have at most s different leading exponents under real linear combinations; the last assertion of the theorem proves the lower bound $s \geq m$. $\square$

Example 6.4 (Sharpness in every codimension). For any integer $q \geq 1$, take $d = 1$, $n = q + 2$, and $a_i = i - 1$ for $1 \leq i \leq q + 2$. Use $B = \{1, 2\}$ and set

$$h_1 = h_2 = 0, \quad h_{j+2} = \omega^{-j} \quad (1 \leq j \leq q).$$

The residual heights are real-linearly independent, since their leading exponents are distinct. Hence $m = q$. The fundamental dependence for the j th nonbasis point evaluates to ω^{-j} . This is a sharpness example for a full dependence invariant; it does *not* assert that a face lattice requires q independent infinitesimals.

7 Support-adapted finite-rank compression

[write] This section is manuscript 02's form of [theorem 6.1](#). Both select, among the coefficient vectors of a normal form and in decreasing order of exponents, each one that increases the real span. Clauses (a) and (b) of [theorem 7.1](#) and (7.1) restate [theorem 6.1](#) for signs (manuscript 05 proves the stronger equality of leading terms and the exact count of leading exponents), and the minimality assertion at its end is the lower bound of [corollary 6.3](#). Clauses (c) and (d), agreement with every common initial cut (5.2) and exactly $p + 1$ distinct profiles along the cuts, are new against 05 and are proved by the same argument as (b); so the theorem is printed whole. Here E is any finite-dimensional real vector space; [theorem 11.2](#) applies the theorem to the quotient Q_A .

7.1 A linear theorem independent of convexity

Let E be a real vector space of finite dimension q . A normal-form E -vector can be interpreted by choosing a basis of E and using a finite vector of surreal coordinates. This interpretation is independent of the basis.

Theorem 7.1 (Greedy support compression). *Suppose*

$$x = \sum_{\alpha < \lambda} v_\alpha \omega^{\gamma_\alpha}, \quad v_\alpha \in E,$$

with the exponents ordered as in (5.1). Select the least nonzero coefficient, and thereafter the least coefficient outside the real span of the previously selected coefficients. Let the selected indices be $\alpha_1 < \dots < \alpha_p$, and put $w^j = v_{\alpha_j}$. Then:

(a) $p \leq q$ and $\text{span}_{\mathbb{R}}\{w^1, \dots, w^p\} = \text{span}_{\mathbb{R}}\{v_\alpha : \alpha < \lambda\}$;

(b) for every $\ell \in E^*$,

$$\text{sgn } \ell(x) = \text{lexsgn}(\ell(w^1), \dots, \ell(w^p));$$

(c) for every initial cut β , the sign profile of $x^{<\beta}$ is that of the initial selected list consisting of the w^j with $\alpha_j < \beta$;

(d) the complete real-linear sign profiles along these cuts have exactly $p + 1$ distinct values, including the zero profile.

If $T_x : E^* \rightarrow \mathbf{No}$ sends ℓ to $\ell(x)$, then

$$p = \text{rank}_{\mathbb{R}} T_x. \quad (7.1)$$

Moreover, any list of real vectors producing the same complete sign profile as x has length at least p .

Proof. At every selection the span strictly increases. Thus at most q coefficients are selected. If an unspanned coefficient remains, the well-order on coefficient positions provides a least such coefficient, so the procedure cannot stop before the selected vectors span all coefficients. This proves (a), including the possibility $p = 0$.

Fix ℓ . If it kills every selected vector, it kills their span and hence every coefficient, so $\ell(x) = 0$. Otherwise let j be the first selected index for which $\ell(w^j) \neq 0$. By the definition of the selection, every coefficient before α_j belongs to the span of the earlier selected vectors. It is therefore killed by ℓ . The first nonzero coefficient of $\ell(x)$ occurs at α_j and equals $\ell(w^j)$, proving (b). Exactly the same argument with the coefficient positions restricted to $\alpha < \beta$ proves (c).

Whenever w^j is added, there exists a real functional that vanishes on $w^1, \dots, w^{j-1}$ but not on w^j . Its sign changes from zero to nonzero. No other profile change is possible by (c). This proves (d).

Finally, $\ker T_x$ is precisely the annihilator of the coefficient span. Its codimension in E^* is p , proving (7.1). If a list of r vectors gives the same sign profile, its common annihilator is $\ker T_x$, since zero is one of the signs that must be preserved. The map from E^* to $\mathbb{R}^r$ obtained by evaluation on that list has rank at most r and kernel of codimension p . Hence $r \geq p$. $\square$

Remark 7.2 (Scope of the word canonical). The selected *positions and coefficient vectors* are uniquely specified once the normal form and quotient space are fixed. Representatives of a quotient vector, coordinate gauges, and other equivalent lexicographic lists need not be canonical. The rank p is intrinsic. We do not assert that this construction preserves valuations, monomial magnitudes, or multiplicative relations.

7.2 Transfinite delays are not extra rank

Example 7.3. Let w^1, w^2 be independent real vectors. With $\varepsilon = \omega^{-1}$ and $\tau = \omega^{-\omega}$, put

$$x = \frac{w^1}{1 - \varepsilon} + \tau w^2 = w^1 + \varepsilon w^1 + \varepsilon^2 w^1 + \dots + \tau w^2.$$

There are two selected coefficient positions. The second occurs only after all the ordinary power-series positions, but the compression is simply $w^1 + \varepsilon w^2$. Thus the theorem does not say “keep the first q terms.” It says “keep the first q possible rank increases,” and a rank increase can be delayed through a transfinite interval of redundant positions.

The underlying existence of a finite lexicographic description is consistent with classical ordered-vector-space theory [13]. What the present proof adds to that abstract description is explicit compatibility with the *given* normal-form initial cuts, together with an elementary optimality certificate through the kernel of T_x .

8 Oriented flags classify the complete dependence sign type

Definition 8.1. The *complete real dependence sign type* of a lifting is the function

$$\chi_h : V \longrightarrow \{-1, 0, 1\}, \quad \chi_h(c) = \operatorname{sgn} L_h(c).$$

It is not just the list of signs of the finitely many circuits of A .

By the compression theorem, there are independent functionals $f_1, \dots, f_m \in V^*$ such that

$$\chi_h(c) = \operatorname{sgn}_{\text{lex}}(f_1(c), \dots, f_m(c)), \quad (8.1)$$

where the sign on the right is that of the first nonzero entry, or zero if all entries vanish. The selected coefficient rows give $f_j(c) = \sum_i c_i b_{j,i}$. Define

$$W_0 = V, \quad W_j = \bigcap_{k=1}^j \ker f_k \quad (1 \leq j \leq m).$$

Each inclusion has codimension one. On the one-dimensional quotient W_{j-1}/W_j , the restriction of f_j specifies a positive ray.

Theorem 8.2 (Flag classification). *The function χ_h determines, and is determined by, the oriented partial flag*

$$V = W_0 \supsetneq W_1 \supsetneq \dots \supsetneq W_m = \ker L_h.$$

Every such flag on V occurs for a surreal lifting of A . Two independent lists $f_1, \dots, f_m$ and $f'_1, \dots, f'_{m'}$ produce the same complete sign type if and only if $m = m'$ and

$$f'_j = \alpha_j f_j + \sum_{k < j} \beta_{jk} f_k, \quad \alpha_j > 0. \quad (8.2)$$

Proof. Given the oriented flag, find the first j for which $c \notin W_j$; the orientation of $c + W_j$ in W_{j-1}/W_j determines its sign. Vectors in W_m have sign zero, so the flag determines (8.1).

Conversely, suppose the sign type is not identically zero and let $P = \{c : \chi_h(c) = 1\}$. In the ordinary finite-dimensional real topology,

$$\overline{P} = \{c : f_1(c) \geq 0\}.$$

Indeed positive points lie in this closed halfspace. Every point with $f_1 > 0$ is positive, and each point with $f_1 = 0$ is a limit of points with $f_1 > 0$. Thus χ_h determines $\ker f_1$ and the positive orientation of its quotient. Restrict the sign function to this kernel and repeat. After exactly m steps it is identically zero. This reconstructs the oriented flag without choosing a coordinate system.

For two lists defining the same flag, the restrictions of f'_j and f_j to W_{j-1} have the same kernel and the same positive ray. They differ there by a positive factor α_j . Their difference annihilates W_{j-1} and is therefore in the span of $f_1, \dots, f_{j-1}$, proving (8.2). These transformations plainly preserve the first nonzero sign, proving the converse.

Finally extend functionals realizing a prescribed flag to coefficient rows in $\mathbb{R}^n$, or use the affine-gauge identification $V \simeq \mathbb{R}^q$. Choose heights with residual vector $\sum_{j=1}^m b_j \omega^{-j}$. Their dependence type is the required flag. $\square$

Remark 8.3 (What is canonical and what is not). The flag and its orientations are canonical for χ_h ; the normalized functionals are not. The actual leading exponents are additional data not determined by the sign type. Replacing ω^{y_j} by any other strictly decreasing list of surreal monomials leaves (8.1) unchanged. Formula (8.2) is also the familiar triangular invariance of lexicographic orders, not a new group action discovered here.

9 Polynomial realization and exact minimum degree

For a real polynomial height vector $H(t) \in \mathbb{R}[t]^n$, define $\chi_{H,0^+}(c)$ to be the eventual sign of $L_{H(t)}(c)$ as $t \rightarrow 0^+$. A polynomial is eventually zero precisely when it is identically zero. We assign degree zero to constant paths, including the zero vector. The quantifiers are

$$\forall c \in V \exists \eta_c > 0 \forall t \in (0, \eta_c) : \text{sgn } L_{H(t)}(c) = \chi_h(c).$$

They do not assert a uniform η_c for all real c .

Theorem 9.1 (Exact unrestricted polynomial degree). *If $m > 0$, the minimum degree of a real polynomial height vector H with $\chi_{H,0^+} = \chi_h$ is exactly $m - 1$. Such a vector is*

$$H(t) = b_1 + b_2 t + \dots + b_m t^{m-1}, \quad (9.1)$$

where the coefficient rows vanish on the chosen affine basis. If $m = 0$, a constant affine height vector suffices.

Proof. For every c , the first nonzero coefficient of $L_{H(t)}(c)$ is the first nonzero $f_j(c)$. This gives the upper bound.

For the lower bound write an arbitrary degree- D candidate as $H(t) = u_0 + u_1 t + \dots + u_D t^D$. Its identically vanishing dependence space is

$$\bigcap_{j=0}^D \ker(c \mapsto c \cdot u_j).$$

Equality of complete sign types makes this space exactly $\ker L_h$, which has codimension m . An intersection of $D + 1$ linear kernels has codimension at most $D + 1$. Thus $m \leq D + 1$. $\square$

Theorem 9.2 (Exact degree with prescribed standard part). *Suppose all h_i are finite and put $h^0 = \text{st}(h)$ componentwise. Define*

$$r_0 = \text{rank}_{\mathbb{R}}(\text{st} \circ L_h : V \rightarrow \mathbb{R}) = \text{rank}_{\mathbb{R}} L_{h^0} \in \{0, 1\}.$$

The minimum degree of a real polynomial height vector satisfying

$$H(0) = h^0, \quad \chi_{H,0^+} = \chi_h$$

is exactly $m - r_0$. This includes $m = 0$, where the minimum is zero.

Proof. Use the affine gauge. All residual heights are finite, so the selected exponents satisfy $\gamma_j \leq 0$.

If $r_0 = 1$, the coefficient slice at exponent zero in the residual vector is nonzero. It is the first pivot, so $\gamma_1 = 0$ and $b_1 = \text{st}(g)$. Then

$$H_i(t) = \text{st}(\ell(a_i)) + b_{1,i} + \sum_{j=2}^m b_{j,i} t^{j-1}$$

has constant term h_i^0 and degree $m - 1$.

If $r_0 = 0$, then $\text{st}(g) = 0$. All pivots, if any, have strictly negative exponents. Set

$$H_i(t) = \text{st}(\ell(a_i)) + \sum_{j=1}^m b_{j,i} t^j.$$

Its constant term is h_i^0 and its degree is m . In both cases the first nonzero dependence coefficient has the required sign.

For a degree- D candidate, the coefficient at t^0 induces the fixed map L_{h^0} of rank r_0 . Each of the remaining D coefficient rows increases the rank of the joint coefficient map by at most one. Since that map must have the same kernel as L_h , its rank is m , yielding $m \leq r_0 + D$. This proves optimality. $\square$

Corollary 9.3 (Sharp uniform bounds). *For real-base liftings, the complete dependence type has a polynomial model of degree at most $q - 1$ when it is nonzero. For bounded liftings with the correct shadow, degree at most q always suffices. Both bounds are attained by Example 6.4; in its shadow-preserving version $r_0 = 0$.*

Proposition 9.4 (A fixed real parameter cannot do the same job). *If $m \geq 2$, there is no real height vector $w \in \mathbb{R}^n$ such that $\text{sgn } L_w(c) = \chi_h(c)$ for every $c \in V$. In particular no single positive real specialization of a polynomial model can preserve all real dependence signs at once.*

Proof. The real-valued linear map L_w has rank at most one. Equality of the sign types would force $\ker L_w = \ker L_h$, which has codimension $m \geq 2$. This contradicts rank-nullity. $\square$

For a concrete picture, compare $x + \varepsilon y$ at a positive infinitesimal ε with $x + t_0 y$ at a fixed real $t_0 > 0$. The real pair $(x, y) = (-t_0, 1)$ makes the second expression zero and the first strictly negative. The exceptional test depends on t_0 . This is exactly why pointwise stabilization for every fixed test is compatible with failure of a uniform real specialization.

10 Finite combinatorics: determinants, slacks, and a linear path

10.1 Determinants of a lifted configuration

The augmented matrix of the lifted points is

$$M(h) = \begin{pmatrix} 1 & \cdots & 1 \\ a_1 & \cdots & a_n \\ h_1 & \cdots & h_n \end{pmatrix}.$$

For an ordered $(d + 2)$ -subset J , expansion along the last row gives

$$\det M(h)_J = \sum_{i \in J} c_{J,i} h_i = L_h(c_J), \quad c_J \in V. \quad (10.1)$$

The coefficients are real signed cofactors. The relation $\tilde{A}c_J = 0$ follows by expanding a determinant with two equal rows. Some such vectors may be zero, and those minors vanish for every height vector.

Lemma 10.1 (Finite determinant signs determine the labeled faces). *Suppose $m > 0$. The signs of the finitely many determinants in (10.1) determine the labeled face lattice of the lifted polytope. When $m = 0$, the lifted polytope is an affine copy of the base polytope and its labeled face structure is already fixed by A .*

Proof. When $m > 0$, the height row is not in the span of the rows of $\tilde{A}$, so $M(h)$ has rank $d + 2$ and the lifted affine dimension is $d + 1$. An ordered list I of $d + 1$ lifted points is affinely independent exactly when it extends to a nonzero augmented maximal minor. For such I , the signs of $\det M(h)_{I \cup \{j\}}$, with a consistently fixed column order, tell on which side of its affine hyperplane each other label lies. The hyperplane supports a facet precisely when all nonzero signs agree. The zero labels are exactly the labels on that facet. Every facet contains an affine basis of this size, so this enumerates all facets with their labels. Every proper nonempty face is an intersection of facets, completing the reconstruction. If $m = 0$, Lemma 4.1 identifies the whole configuration with the graph of an affine function on A . $\square$

In particular equality of complete dependence types implies equality of the finite lifted sign data. The converse fails dramatically: finitely many hyperplanes cannot detect all the levels of an oriented flag.

10.2 Barycentric certificates for lower cells

For each affine basis B of A and each label i , define

$$s_{B,i}(h) = h_i - \sum_{b \in B} \lambda_{B,i,b} h_b. \quad (10.2)$$

This is another real dependence evaluation. Its coefficient vector is a fundamental circuit vector when zero entries are removed; the zero vector occurs for $i \in B$.

Proposition 10.2 (Exact lower-cell certificate). *A basis B supports a lower cell of full base dimension precisely when $s_{B,i}(h) \geq 0$ for all labels i . Its equality-label set is*

$$I_B(h) = \{i : s_{B,i}(h) = 0\}.$$

The maximal marked cells of the induced regular subdivision are exactly the distinct sets $I_B(h)$ arising this way, interpreted geometrically as $\text{conv}_{\mathbb{R}}\{a_i : i \in I_B(h)\}$ together with their labels. Their faces supply the remaining cells. This convention includes the full cell when all heights are affine.

Proof. The graph of the affine function interpolating h on B is a lower supporting hyperplane exactly when every lifted point lies on or above it. These are the inequalities in (10.2). The equalities identify its contact labels, and the projection has dimension d because it contains B . Conversely every full-dimensional projected lower face contains such a base affine basis, so every maximal cell is obtained. The resulting finite sign system has a real specialization by the polynomial construction and the finite threshold below; thus the asserted subdivision and its face incidences are also those of an ordinary real regular lifting. This avoids any appeal to topological compactness over $\mathbf{No}$. $\square$

We use *marked* subdivisions deliberately. A point lying in the interior of a real cell may be on its lower supporting hyperplane, or may lie above that hyperplane. The two possibilities need not alter the underlying geometric region but are distinguished by I_B .

10.3 An explicit simultaneous real threshold

Lemma 10.3 (Finite sign stabilization). *For a nonzero real polynomial $f(t) = \sum_{j=0}^D a_j t^j$, let $k = \min\{j : a_j \neq 0\}$ and $T(f) = \sum_{j>k} |a_j|$. Set*

$$\eta(f) = \begin{cases} 1, & T(f) = 0, \\ \min\{1, |a_k|/(2T(f))\}, & T(f) > 0. \end{cases}$$

Then $\operatorname{sgn} f(t) = \operatorname{sgn} a_k$ whenever $0 < t < \eta(f)$. For a finite list of polynomials, the minimum of these bounds for its nonzero members preserves every sign, with identically zero members remaining zero.

Proof. For $0 < t < 1$,

$$\left| \sum_{j>k} a_j t^{j-k} \right| \leq t \sum_{j>k} |a_j| = tT(f).$$

When the tail is nonzero this is less than $|a_k|/2$ under the stated bound. The sign of a_k is therefore unchanged; multiplication by $t^k > 0$ finishes the proof. A finite minimum of positive real bounds is positive. $\square$

Apply the lemma to the determinant polynomials and all the slack polynomials of any model from Section 9. The resulting η_0 gives the same labeled lifted faces and the same marked regular subdivision for every $0 < t < \eta_0$. Rational coefficients give a rational η_0 and a rational specialization $t_0 = \eta_0/2$.

The threshold is sufficient, not asserted optimal. Section 20 explains why even an optimal *initial* interval must not be confused with the entire set of valid parameters.

10.4 The finite-data degree collapses to one

Theorem 10.4 (Linear realization with the exact shadow). *Let $h \in \mathbf{No}^n$ be finite and $h^0 = \operatorname{st}(h)$. Let $\mathcal{F} \subseteq V$ be any finite collection of real dependence vectors. There exists $w \in \mathbb{R}^n$ such that the affine-linear path*

$$K(s) = (1-s)h^0 + sw \tag{10.3}$$

satisfies

$$\operatorname{sgn} L_{K(s)}(c) = \operatorname{sgn} L_h(c) \quad (c \in \mathcal{F}, 0 < s \leq 1).$$

In particular one path preserves the entire labeled lifted face lattice, all its determinant signs, and the marked regular subdivision while starting at the exact labeled standard part.

Among polynomial paths starting at h^0 and realizing the chosen finite sign data for all sufficiently small positive parameters, the minimum degree is zero when h^0 already has those signs, and one otherwise.

Proof. Choose a shadow-preserving polynomial model H from Theorem 9.2. The finite sign lemma provides a real $t_0 > 0$ such that $w = H(t_0)$ has the required signs on $\mathcal{F}$. For any $c \in \mathcal{F}$ with $L_h(c) > 0$ we have

$$L_{h^0}(c) = \operatorname{st}(L_h(c)) \geq 0, \quad L_w(c) > 0.$$

Their convex combination in (10.3) is strictly positive for $0 < s \leq 1$. The negative case follows by replacing c with $-c$. If $L_h(c) = 0$, both real endpoint values are zero. This proves the simultaneous claim. Use the finite list of determinants and slacks for the geometric consequence.

A constant path starting at h^0 can realize only the signs of h^0 . When those signs suffice, take that constant path. Otherwise degree zero is impossible and the exhibited path has degree one. $\square$

Remark 10.5 (Why the conclusion is stronger than a limit argument). The path has the correct signs on the *whole* punctured segment $0 < s \leq 1$, not merely on an unspecified small interval. After the real endpoint has been chosen, this is a convexity argument in a finite linear sign chamber. It does not claim to preserve χ_h on all of V . Proposition 9.4 rules that out when $m \geq 2$.

Theorem 10.6 (Arbitrarily large separation of the two degrees). *For every real configuration with $q = n - d - 1 \geq 1$, there is a bounded lifting with affine shadow for which the minimum degree for the complete dependence type with that shadow is q , whereas the minimum degree for all finite lifted determinant and slack signs with that shadow is one.*

Proof. There are finitely many nonzero determinant and slack vectors in V . Choose $f_1 \in V^*$ nonzero on all of them: a finite union of proper hyperplanes does not cover V^* . Extend f_1 to a basis $f_1, \dots, f_q$ of V^* , represented by real height rows b_j in the affine gauge. Put

$$h = \varepsilon b_1 + \varepsilon^2 b_2 + \dots + \varepsilon^q b_q, \quad \varepsilon = \omega^{-1}.$$

Then $m = q$ and $h^0 = 0$, so $r_0 = 0$. Theorem 9.2 gives minimum complete-type degree q . Every nonzero vector in the finite test list already has its sign decided by f_1 , so $K(s) = s b_1$ realizes that list for all $s > 0$. At least one lifted determinant is nonzero because $m > 0$; at the shadow every such determinant is zero. Hence a constant path cannot realize the finite list, and its exact minimum degree is one. $\square$

This separation is one reason not to use the phrase “the number of surreal scales of a polytope” without specifying the invariant. Even a single nontrivial combinatorial degeneration can carry an arbitrarily long complete dependence flag.

11 Fixed real bases: circuits, liftings, and certificates

[write] Manuscript 02’s fixed-base sections. The setting is 05’s (a fixed real configuration A of n distinct points spanning $\mathbb{R}^d$, heights $h \in \mathbf{No}^n$), over 02’s real closed working field; 02’s comparison rank p is 05’s m , and its space L_A is 05’s V . Its slack certificate is proposition 10.2, proved by both manuscripts (02’s formulation is remark 11.1 below); its threshold theorem 11.4 is lemma 10.3 with 02’s constants; the remark after that theorem is proposition 9.4. New against 05 are clause (b) of theorem 11.2, the positive-sum lemma 12.1, the classification theorem 12.2 of support-initial histories, and the sharp planar family theorem 12.3 with its hexagon.

Throughout this section, $A = (a_1, \dots, a_n)$ is a fixed configuration of *distinct real points* spanning $\mathbb{R}^d$. We use labeled cells even when some points are interior to the convex hull. The statements extend to repeated labels with suitable conventions, but no such extension is needed here.

11.1 Marked lower cells

For heights $h \in K^n$, lift a_i to $(a_i, h_i) \in K^{d+1}$. A maximal lower cell is the label set

$$C = \{i : h_i = \ell(a_i)\}, \quad h_i \geq \ell(a_i) \text{ for every } i,$$

for an affine function $\ell : K^d \rightarrow K$, with $\dim \text{Aff}\{a_i : i \in C\} = d$. The resulting collection of maximal marked cells is denoted $\mathcal{S}_A(h)$, or $\mathcal{S}(h)$ when A is fixed. Their convex hulls cover $\text{conv}_K(A)$ and meet along faces. This is the marked regular subdivision induced by h .

A point of A may be omitted from all lower cells. Even if it lies geometrically in another cell, its omission is part of the marked data. The trivial subdivision has the single marked cell $\{1, \dots, n\}$. We say that $\mathcal{T}$ *refines* $\mathcal{S}$ when each maximal marked cell of $\mathcal{T}$ is contained as a label set in a maximal marked cell of $\mathcal{S}$. Strict refinement means that the collections differ. For configurations in convex position this is the usual geometric refinement relation. Keeping the marking avoids an ambiguity when interior points are inactive.

11.2 A finite test using affine bases

Let B be any affine basis of A , that is, a $(d+1)$ -element index set with affinely independent points. For each $j \notin B$, let $\lambda_{B,j,b} \in \mathbb{R}$ be the unique barycentric coefficients satisfying

$$a_j = \sum_{b \in B} \lambda_{B,j,b} a_b, \quad \sum_{b \in B} \lambda_{B,j,b} = 1.$$

Define a slack functional

$$\delta_{B,j}(h) = h_j - \sum_{b \in B} \lambda_{B,j,b} h_b = c_{B,j} \cdot h. \quad (11.1)$$

Its coefficient vector $c_{B,j}$ lies in L_A . Its support is the fundamental affine circuit associated with B and j , with zero coefficients omitted. It is harmless to retain all these vectors even when several represent the same circuit.

Remark 11.1 (Manuscript 02's slack certificate). [write] Manuscript 02 states [proposition 10.2](#) (source 05) as its Lemma “Complete slack certificate”, whose label `poly:ts:lem:slack` is attached to [proposition 10.2](#), in the notation (11.1): *An affine basis B lies on a lower facet exactly when $\delta_{B,j}(h) \geq 0$ for every $j \notin B$. In that case the associated maximal marked cell is*

$$D_B(h) = B \cup \{j \notin B : \delta_{B,j}(h) = 0\}. \quad (11.2)$$

Enumerating all lower bases and removing duplicate sets (11.2) recovers $\mathcal{S}_A(h)$. Consequently, the signs of the finite family of functionals (11.1) determine the whole subdivision. Thus $D_B(h) = I_B(h)$ and $\delta_{B,j} = s_{B,j}$. Manuscript 02's proof, which is the proof of [proposition 10.2](#), reads: There is a unique affine function ℓ_B taking the prescribed values h_b on B . At a_j its value is $\sum_b \lambda_{B,j,b} h_b$. The condition that its graph support the lifted points from below is exactly the nonnegativity of all the slacks. The equality labels are exactly (11.2). They span dimension d since they contain B , so they form a maximal lower cell. Conversely, every maximal lower cell contains an affine basis, and its supporting function agrees with the interpolating function of that basis.

This proof works over any ordered extension of the real coefficient field. It is the direct finite algebra behind the usual lifting and circuit tests; no topological limiting argument is hidden in it.

11.3 The sharp comparison-rank theorem

The vectors of affine height values form the subspace

$$\text{Aff}(A) = \{(b + u \cdot a_i)_{i=1}^n : b \in \mathbb{R}, u \in \mathbb{R}^d\} \subseteq \mathbb{R}^n.$$

Set $Q_A = \mathbb{R}^n / \text{Aff}(A)$. Its dimension is $q = n - d - 1$, and its dual is naturally L_A . Adding affine heights does not change the lower subdivision or any functional in L_A .

[write] **Theorem 11.2** is [corollary 6.3](#) (source 05) in manuscript 02's form. Its clauses (a) and (d) are the sign and minimality parts of [corollary 6.3](#), with $p = m$; 05 also keeps exact leading terms and exact leading exponents. Clause (b), agreement with every common initial cut, is new against 05. Clause (c) is 05's canonical refinement chain [proposition 13.4](#) read along the cuts: 05's chain keeps the first k pivot rows, 02's the first m selected positions, and both have at most $m = p$ strict steps.

Theorem 11.2 (All-scale lifting compression). *Let $h \in \mathbf{No}^n$ have common normal form (5.1). Apply Theorem 7.1 to the coefficient classes in Q_A , and choose representatives $w^1, \dots, w^p \in \mathbb{R}^n$ of the selected classes. Define*

$$H_m(\varepsilon) = \sum_{j=1}^m \varepsilon^{j-1} w^j \quad (0 \leq m \leq p), \quad H_0 = 0.$$

Then:

(a) $p \leq n - d - 1$, and for every $c \in L_A$,

$$\text{sgn}(c \cdot h) = \text{sgn}(c \cdot H_p(\varepsilon)) = \text{lexsgn}(c \cdot w^1, \dots, c \cdot w^p);$$

(b) every common initial normal-form cut $h^{<\beta}$ has the same complete L_A sign profile, and hence the same marked subdivision, as $H_m(\varepsilon)$, where m is the number of selected positions before β ;

(c) the sequence

$$\mathcal{S}_A(H_0), \mathcal{S}_A(H_1(\varepsilon)), \dots, \mathcal{S}_A(H_p(\varepsilon))$$

is a refinement chain, possibly with repetitions;

(d) p is the least number of real coefficient vectors that preserve all L_A signs. It equals the real rank of $c \mapsto c \cdot h$.

In particular, the whole initial-cut lifting history has at most $n - d - 1$ strict changes.

Proof. Assertions (a), (b), and (d), except for the subdivision conclusion, are Theorem 7.1 applied to Q_A . The subdivision conclusion follows from [proposition 10.2](#), because each slack belongs to L_A .

To prove (c), take an affine basis B that is lower for $H_{m+1}(\varepsilon)$. If any slack for $H_m(\varepsilon)$ were negative, its first nonzero coefficient would stay negative when one smaller coefficient is appended. This would contradict the choice of B . Thus B was also lower for $H_m(\varepsilon)$. Furthermore a nonzero slack at stage m cannot become zero at stage $m + 1$: its first nonzero coefficient is unchanged. Therefore

$$D_B(H_{m+1}(\varepsilon)) \subseteq D_B(H_m(\varepsilon)).$$

Each maximal cell at the new stage has a lower basis, so this is exactly the marked refinement assertion. There are p transitions, proving the last bound. $\square$

Remark 11.3 (Zero rank and invisible comparison layers). If $q = 0$, all heights are affine and the only subdivision is trivial. More generally $p = 0$ means that h is affine over its working field. For $p > 0$, the complete comparison profile changes at every selected position, but the subdivision need not change there. A comparison not visible in the lower hull can acquire a new nonzero sign. Thus the full profile rank p and the strict geometric history length r are different invariants; only $r \leq p$ is asserted.

11.4 A concrete quotient gauge

Choose one affine basis B_0 . Subtract the affine interpolant through its height values. For every $i \notin B_0$, the residual is

$$r_i(h) = h_i - \sum_{b \in B_0} \lambda_{B_0,i,b} h_b. \quad (11.3)$$

The q residual coordinates identify Q_A with $\mathbb{R}^q$ at the coefficient level. The comparison rank is also

$$p = \dim_{\mathbb{R}} \text{span}_{\mathbb{R}} \{r_i(h) : i \notin B_0\} \subseteq \mathbf{No}. \quad (11.4)$$

Indeed the vectors $c_{B_0,i}$ form a basis of L_A , so the image of $c \mapsto c \cdot h$ is the span in (11.4). For finite rational coefficient arrays this yields an exact Gaussian elimination algorithm: scan the gauged rows in their supplied order, retaining precisely the rank-increasing rows.

11.5 An explicit real specialization bound

The polynomial with infinitesimal parameter preserves every real affine comparison. For the finite geometry, an actual real parameter suffices, and can be certified explicitly.

[write] [Theorem 11.4](#) and [lemma 10.3](#) (source 05) give the same kind of explicit threshold with different normalizations: 02 caps it at $\frac{1}{2}$ and allows $0 < s \leq \eta$, while 05 caps it at 1 and requires $0 < t < \eta(f)$. Both are valid, and they agree on the report's numerical examples. Recomputed exactly in the write phase over all basis slacks and all augmented determinants, both give $\eta = 1/34$ for 02's hexagon (12.4) and $\eta_0 = 1/10$ for 05's three-scale example in [section 16](#). 02's form is kept because it holds simultaneously for all stages H_m of a history.

Theorem 11.4 (Simultaneous finite sign specialization). *Let $C \subseteq L_A$ be a finite family, for example all the slack vectors $c_{B,j}$. Put $a_{c,j} = c \cdot w^j$. For each c with some nonzero entry, let k_c be its first such index and put*

$$B_c = \sum_{j > k_c} |a_{c,j}|.$$

With an empty inner minimum omitted, define

$$\eta = \min \left(\frac{1}{2}, \min_{c: B_c > 0} \frac{|a_{c,k_c}|}{2B_c} \right) > 0. \quad (11.5)$$

For every real s with $0 < s \leq \eta$, every $c \in C$, and every $0 \leq m \leq p$,

$$\text{sgn}(c \cdot H_m(s)) = \text{lexsgn}(a_{c,1}, \dots, a_{c,m}).$$

Taking all slack vectors therefore realizes the complete compressed subdivision history by the real polynomial liftings $H_m(s)$.

Proof. If all entries through m vanish, the assertion is exact equality to zero. Otherwise $m \geq k_c$, and

$$s^{1-k_c} c \cdot H_m(s) = a_{c,k_c} + \sum_{j=k_c+1}^m a_{c,j} s^{j-k_c}.$$

Since $s \leq 1/2 < 1$, the absolute value of the tail is at most sB_c . If $B_c = 0$ there is no tail. If $B_c > 0$, the choice of η makes the tail at most $|a_{c,k_c}|/2$, so it cannot change the leading sign. The argument is simultaneous in m and c . $\square$

When A and the selected height rows are rational, the bound is a positive rational number obtained by exact arithmetic. The list of basis slacks has at most

$$\binom{n}{d+1}(n-d-1)$$

entries, with fewer when some candidate bases are dependent. Enumeration is a finite certificate construction, not a claimed dimension-uniform polynomial-time algorithm.

Remark 11.5 (A real parameter cannot preserve all comparisons). [write] This is [proposition 9.4](#) (source 05) in manuscript 02's notation. The finite restriction on C is essential. If $p > 1$, no single real height vector can reproduce all L_A signs of h . Its evaluation map $L_A \rightarrow \mathbb{R}$ has rank at most one, whereas $c \mapsto c \cdot h$ has rank p . Equality of all signs would imply equality of the kernels, which would contradict their codimensions. Thus Theorem 11.4 preserves finite geometry, not an entire non-Archimedean ordered vector space inside $\mathbb{R}$.

12 Exactly which lifting histories occur?

12.1 Positive sums of compatible heights

Lemma 12.1 (Compatible positive sums). *Suppose real height vectors u, v induce marked subdivisions $\mathcal{S}, \mathcal{T}$, respectively, and $\mathcal{T}$ refines $\mathcal{S}$. For positive scalars a, b in any ordered extension of $\mathbb{R}$,*

$$\mathcal{S}_A(au + bv) = \mathcal{T}.$$

Proof. Let C be a maximal cell of $\mathcal{T}$, contained in a maximal cell D of $\mathcal{S}$, and choose an affine basis $B \subseteq C$. Every slack $\delta_{B,j}(u)$ and $\delta_{B,j}(v)$ is nonnegative. For $j \in C$ both vanish, since $C \subseteq D$. For $j \notin C$, the second is strictly positive, because the zero set for v is exactly C . Consequently the lower cell of $au + bv$ containing B is exactly C .

All maximal cells of $\mathcal{T}$ therefore occur as maximal lower cells of the sum. Their convex hulls cover the base polytope. There can be no additional different maximal lower cell: on a full-dimensional overlap, the two supporting affine functions would agree and hence define the same equality labels. Equivalently one may use the usual finite lower-envelope description. The finite covering by real cells remains a covering over an ordered real-closed extension, by linear elimination. The same elementary description proves the assertion over any ordered extension for these real cell data. $\square$

Theorem 12.2 (Classification of support-initial lifting histories). *Fix A . After repetitions are suppressed and the initial trivial subdivision is included, the histories of common initial normal-form surreal liftings are exactly the finite strict chains*

$$\mathcal{S}_0 > \mathcal{S}_1 > \cdots > \mathcal{S}_r, \quad \mathcal{S}_0 = \{\{1, \dots, n\}\}, \quad (12.1)$$

of marked regular subdivisions, ordered from coarse to fine. Every such chain has $r \leq n - d - 1$. It has a realization

$$h = u_1 + \varepsilon u_2 + \cdots + \varepsilon^{r-1} u_r \quad (12.2)$$

with real u_j inducing $\mathcal{S}_j$, and the u_j are independent modulo $\text{Aff}(A)$. If A is rational, the u_j can be chosen rational. The least number of nontrivial coefficient layers required to realize the strict chain is r .

Proof. Theorem 11.2 shows that every support-initial history is a finite refinement chain. Its subdivisions are realizable over $\mathbb{R}$ by Theorem 11.4, so they are regular in the ordinary sense.

Conversely choose real u_j realizing S_j . The refinement relation and Lemma 12.1 show inductively that every positive linear combination of $u_1, \dots, u_j$ with all coefficients positive induces exactly S_j . Apply this to the coefficients $1, \varepsilon, \dots, \varepsilon^{j-1}$. Each initial segment of (12.2) therefore realizes the prescribed stage. Notice that no smallness bound is needed for this compatibility argument: positivity is enough.

For independence, consider the strict transition from S_{j-1} to S_j . There is a maximal marked cell D of S_{j-1} on which the entries of u_j are not the values of one affine function. Otherwise, take a full-dimensional new cell inside each old cell. Its supporting affine function would agree with u_j on every label of the old cell, forcing that entire old cell to persist; the refinements would not be strict. Linear algebra now provides a relation $c \in L_A$, supported in D , with $c \cdot u_j \neq 0$. Every earlier u_i is affine on D , since D lies inside a cell of S_i . Hence $c \cdot u_i = 0$ for $i < j$. Thus u_j cannot lie in the span of its predecessors modulo $\text{Aff}(A)$. Induction gives r independent classes and therefore $r \leq q$.

For rational A , a height vector realizing a specified marked regular subdivision can be described by a finite rational system of linear conditions. One way to see this without a general secondary-fan theorem is to choose a real witness and retain the signs of all its basis slacks. The zero conditions define a rational linear subspace, and the other conditions are strict rational inequalities. Rational points are dense in that subspace, so a sufficiently close rational point is a witness with the same subdivision. Choose such a point for each u_j .

Finally, each appended coefficient layer can cause at most one new stage. A strict chain of length r therefore needs at least r layers, and (12.2) achieves that number. $\square$

The dimension assertion is also visible from the secondary polytope: its face poset is the marked coherent refinement poset and its dimension is $n - d - 1$ [14, Theorem 16.4.1]. The proof above isolates the precise coefficient mechanism and handles arbitrary surreal normal-form starting data without treating that classical dimension fact as new.

12.2 Sharpness in dimension two

Theorem 12.3 (A sharp parabolic polygon family). *For every $n \geq 4$, let*

$$a_i = (i, i^2), \quad 0 \leq i \leq n - 1.$$

For $2 \leq k \leq n - 2$, define

$$w_i^{(k)} = \max\{0, i(i - k)\}. \quad (12.3)$$

These $n - 3$ height vectors are independent modulo affine heights. The successive prefixes of

$$h = w^{(2)} + \varepsilon w^{(3)} + \dots + \varepsilon^{n-4} w^{(n-2)}$$

have exactly $n - 3$ strict subdivision changes. At stage j , for $0 \leq j \leq n - 3$, the maximal cells are the triangles

$$\{0, \ell, \ell + 1\} \quad (1 \leq \ell \leq j)$$

together with the remaining polygon

$$\{0, j + 1, j + 2, \dots, n - 1\}.$$

Thus both the comparison-rank bound and the strict-history bound are sharp already for planar configurations in convex position.

Proof. All the a_i are vertices of their convex hull: the graph of x^2 is strictly convex, so each point has a supporting tangent line meeting no other listed point. The convex hinge

$$\psi_k(x, y) = \max\{0, y - kx\}$$

takes the values (12.3) on the vertices. Its crease cuts the polygon along the chord from a_0 to a_k . On each side it is affine, and its convexity puts its graph below every convex combination of lifted vertices. Since the two vertex polygons cover the base polygon, it is exactly the lower piecewise-affine roof for these heights.

The chords from a_0 to $a_2, \dots, a_{n-2}$ do not cross in their relative interiors. A positive sum of the corresponding hinges has the common refinement of their domains of affinity. No crease disappears: across the interior of one chord only its own hinge changes gradient, and its coefficient is positive. The first j chords give precisely the listed fan triangles and remaining polygon. This proves the subdivision statement and its strictness.

For independence, suppose a real linear combination of the height vectors is affine on A . Each height vector vanishes at $i = 0, 1, 2$. An affine function of (i, i^2) is a polynomial in i of degree at most two; those three zeros force it to vanish identically. Evaluate the purported zero combination at $i = 3$: only $w^{(2)}$ can be nonzero and its value is 3, so its coefficient vanishes. Evaluation at $i = 4$ then isolates $w^{(3)}$, with value 4. Continue up to $i = n - 1$. Every coefficient vanishes, giving independence modulo affine heights. $\square$

12.3 A fully explicit hexagon

For $n = 6$, the three rows are

$$\begin{aligned} w^1 &= (0, 0, 0, 3, 8, 15), \\ w^2 &= (0, 0, 0, 0, 4, 10), \\ w^3 &= (0, 0, 0, 0, 0, 5). \end{aligned} \tag{12.4}$$

The cells are shown in Table 2. The all-basis-slack certificate (11.5) gives $\eta = 1/34$. Thus the ordinary rational height vectors w^1 , $w^1 + w^2/34$, and $w^1 + w^2/34 + w^3/34^2$ provide a single explicitly certified real specialization of the three stages. This bound is not optimal. In this compatible hinge family every choice of positive layer weights works, as the geometric proof shows.

Table 2: The maximal marked cells of the compressed hexagon history. Braces contain vertex labels.

Stage	Maximal marked cells
0	$\{0, 1, 2, 3, 4, 5\}$
1	$\{0, 1, 2\}, \{0, 2, 3, 4, 5\}$
2	$\{0, 1, 2\}, \{0, 2, 3\}, \{0, 3, 4, 5\}$
3	$\{0, 1, 2\}, \{0, 2, 3\}, \{0, 3, 4\}, \{0, 4, 5\}$

The example also distinguishes two tasks. To preserve only the final triangulation, a single ordinary real height vector is enough. To preserve all three strict transitions requires three layers. To preserve every real affine comparison of the original non-Archimedean height vector requires its full comparison rank. Those numbers agree in this example, but need not agree in general. [write] These are three of the invariants compared in [remark 1.1](#): the final triangulation (one real vector here; with a prescribed shadow, degree at most one by [theorem 10.4](#)), the strict geometric history (r layers, [theorem 12.2](#)), and the complete sign type

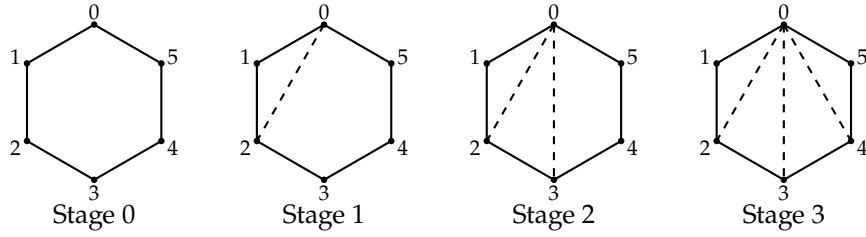


Figure 1: Successively introducing three fan diagonals. This regular-hexagon drawing is a combinatorial schematic; the actual rational coordinates are (i, i^2) , and the heights are (12.4).

(rank $p = m$, minimum germ degree $m - 1$ by [theorem 9.1](#)). [Theorem 10.6](#) shows how far apart they can be.

13 Shadows and canonical refinement hierarchies

Proposition 13.1 (Standard part of a finite convex hull). *For finite points $v_1, \dots, v_N \in K^D$,*

$$\text{st}(\text{conv}_K(v_1, \dots, v_N)) = \text{conv}_{\mathbb{R}}(\text{st}(v_1), \dots, \text{st}(v_N)). \quad (13.1)$$

Proof. Every convex coefficient lies between zero and one and is therefore finite. Standard part takes a convex combination to the convex combination with real coefficients $\text{st}(\lambda_i)$. Conversely every real convex combination of the shadows is the standard part of the combination of the original points with those same real coefficients. $\square$

[write] Manuscript 02 proves the same identity (its Proposition “Standard part of a finite hull”, for points of $\mathcal{O}_K^d$) by the same argument; its labels `poly:ts:prop:stconv` and `poly:ts:eq:stconv` are attached here. [Parts III to V](#) use the identity and cite it here.

The proposition is an equality of convex sets after applying `st`. It does not say that standard part sends each face to a face.

Example 13.2 (A face whose shadow is not a face). Lift the four corners of a unit square with heights $(0, 0, 0, \varepsilon)$, where the ordering is $(0, 0), (1, 0), (0, 1), (1, 1)$ and $\varepsilon > 0$ is infinitesimal. The first three lifted points form a lower triangular facet. The shadow of the entire polytope is a square in the plane $z = 0$, whereas the shadow of that facet is the filled triangle with vertices $(0, 0), (1, 0), (0, 1)$. That triangle is not a face of the square. Thus there is no unrestricted face-lattice map $F \mapsto \text{st}(F)$.

Proposition 13.3 (The shadow subdivision is a coarsening). *For a bounded lifting of a fixed real base, its marked regular subdivision refines the one induced by $h^0 = \text{st}(h)$: every maximal source cell is contained in a maximal shadow cell, including inclusion of its contact labels.*

Proof. Choose a supporting basis B of a maximal source cell. All $s_{B,i}(h) \geq 0$, so all $s_{B,i}(h^0) = \text{st}(s_{B,i}(h)) \geq 0$. Thus the same basis supports a shadow cell. Every source equality remains an equality after standard part, giving $I_B(h) \subseteq I_B(h^0)$. $\square$

More generally the complete flag yields a finite hierarchy, even when the heights are not bounded. Let $b_1, \dots, b_m$ be its coefficient rows, and for $0 \leq k \leq m$ use the lexicographic height with the first k rows. Equivalently take the polynomial $H_k(t) = \sum_{j=1}^k b_j t^{j-1}$ at a sufficiently small positive real t , with $H_0 = 0$. Denote its marked subdivision by $\mathcal{S}_k$.

Proposition 13.4 (Canonical finite refinement chain). *The subdivisions $\mathcal{S}_k$ do not depend on the choice of coefficient functionals realizing the same oriented flag, and*

$$\mathcal{S}_0 \geq \mathcal{S}_1 \geq \cdots \geq \mathcal{S}_m = \mathcal{S}(h),$$

where $\geq$ means “is coarser than.” Every subdivision in the chain is an ordinary real regular subdivision of A . There are at most m strict refinement steps. The oriented flag need not be recoverable from this chain.

Proof. Positive triangular transformations (8.2) preserve the first-nonzero signs of every truncated list, so they preserve all truncated slack signs. Hence the chain depends only on the flag. If B supports a lower cell for the first $k+1$ rows, every slack has nonnegative lexicographic sign through that level. Its first k entries are either all zero or have the same positive first nonzero entry. Thus B also supports a lower cell at level k , and its equality labels can only increase when the last row is removed. This proves refinement. Finite sign stabilization realizes every level over $\mathbb{R}$; a common real parameter can be chosen for the finitely many levels and slacks.

For the last assertion, choose f_1 nonzero on every nonzero slack vector as in Theorem 10.6. All later rows are invisible to the subdivision, so $\mathcal{S}_1 = \cdots = \mathcal{S}_m$ while the remaining flag on $\ker f_1$ can vary whenever its dimension permits. $\square$

This is a flag-refined version of the usual secondary data, not a claim that the secondary fan is the entire space of complete dependence types. Already in a generic real secondary chamber, further infinitesimal coefficients may encode nontrivial orders on its invisible real dependence directions.

14 Standard parts: exact identities and lost faces

[write] Manuscript 02’s standard-part section. Its first proposition, $\text{st conv} = \text{conv st}$, is [proposition 13.1](#) (both sources) and is not repeated. The finite polar identity, the lost face and the optimality band are new against 05. Manuscript 05’s [example 13.2](#) and 02’s (14.2) are two instances of one failure, a face whose standard part is not a face; [Part III](#) adds a pentagon ([section 40](#)), and [proposition 14.2](#) bears on the questions of [Part III](#) about face specialization.

Let $K \supseteq \mathbb{R}$ be an ordered real-closed working field containing a positive infinitesimal. Write

$$\mathcal{O}_K = \{x \in K : |x| < m \text{ for some } m \in \mathbb{N}\}$$

for its ring of finite elements. Completeness of $\mathbb{R}$ gives a unique standard part $\text{st}(x) \in \mathbb{R}$ for each finite x : the difference $x - \text{st}(x)$ is infinitesimal. Standard part is a weakly order-preserving ring homomorphism on $\mathcal{O}_K$. Apply it coordinatewise.

14.1 Convex hull and polar commute with residue

[write] The hull identity, 02’s first proposition here, is [proposition 13.1](#), printed once above for both sources.

Proposition 14.1 (Finite polar residue). *Let $P = \text{conv}_K(v_1, \dots, v_n)$ with all v_i finite, and define $P^\circ = \{y \in K^d : y \cdot x \leq 1 \text{ for all } x \in P\}$. No assumption that 0 is interior is required. Then*

$$\text{st}(P^\circ \cap \mathcal{O}_K^d) = (\text{st } P)^\circ. \tag{14.1}$$

Proof. If y is a finite polar point, standard part of each inequality $y \cdot v_i \leq 1$ gives the inclusion from left to right. For the reverse inclusion, let $y \in (\text{st } P)^\circ \subseteq \mathbb{R}^d$ and put

$$\delta = \max(0, \max_i (y \cdot v_i - 1)).$$

All terms are finite and their standard parts are nonpositive, so δ is a nonnegative infinitesimal. The point $y' = y/(1 + \delta)$ satisfies $y' \cdot v_i \leq 1$ for all i , is finite, and has standard part y . Thus $y' \in P^\circ \cap O_K^d$. $\square$

The finite-part restriction in (14.1) matters because a polar can be unbounded and standard part is not defined on infinite coordinates. The proof does not need a Hausdorff convergence theorem or a metric on all of K^d .

14.2 A face need not have a face as its standard part

Consider

$$P = \text{conv}_K\{(-1, 0), (0, -\varepsilon), (1, 0), (1, 1), (-1, 1)\}. \quad (14.2)$$

Its unique point maximizing $(x, y) \mapsto -y$ is the vertex $(0, -\varepsilon)$. The standard-part hull is the rectangle $Q = [-1, 1] \times [0, 1]$. The standard part of that vertex is $(0, 0)$, which lies in the relative interior of the bottom edge of Q and is not a face by itself.

Thus the assignment $F \mapsto \text{st}(F)$ is not a map from the face lattice of P to the face lattice of Q . This is stronger than the observation that two vertices can merge. Even an exposed face can map to a proper nonface subset of a limiting face.

14.3 Near-optimal points recover the lost limiting face

Proposition 14.2 (An exact infinitesimal optimality band). *Let $P = \text{conv}_K(v_1, \dots, v_n)$ be finite, $Q = \text{st}(P)$, and $c \in \mathbb{R}^d$. Put $M = \max_i c \cdot v_i$ and $m = \text{st}(M)$. There is a positive infinitesimal η such that*

$$\text{st}\{x \in P : c \cdot x \geq M - \eta\} = \arg\max_{y \in Q} c \cdot y. \quad (14.3)$$

Proof. Standard part commutes with a finite maximum, so $m = \max_{y \in Q} c \cdot y$. Let

$$J = \{i : c \cdot \text{st}(v_i) = m\}, \quad D = \max_{i \in J} (M - c \cdot v_i).$$

The set J is nonempty, and D is a nonnegative infinitesimal. Choose any positive infinitesimal $e \in K$ and set $\eta = D + e$.

A point in the band on the left of (14.3) has standard-part objective value at least m , hence exactly m because its standard part lies in Q . Conversely, every v_i with $i \in J$ belongs to the band, and so does their K -convex hull. Its standard part is the real convex hull of the maximizing standard-part vertices. That is the entire maximizer face of Q , proving the other inclusion. $\square$

For (14.2), exact optimization remembers just the infinitesimally lower vertex, but an appropriate infinitesimal optimality band recovers the whole bottom edge. This suggests that residue geometry should retain near-optimal fibers or scale-decorated cells rather than attempt to force a false homomorphism between ordinary face lattices.

15 Higher-rank tropical minimizers: an elementary bridge

This section records a finite-rank interpretation of the hierarchy, not a new theory of higher-rank tropical multiplicities. It is useful for comparing the lifting package with [3, 4] without identifying different notions of topology or convexity.

Take real height rows $b_1, \dots, b_r \in \mathbb{R}^n$. For $x = (x_1, \dots, x_r) \in (\mathbb{R}^d)^r$, compare the vectors

$$\Phi_i(x) = (b_{1,i} + a_i \cdot x_1, \dots, b_{r,i} + a_i \cdot x_r) \in \mathbb{R}^r$$

in lexicographic order. Define $I_0 = \{1, \dots, n\}$ and successively

$$I_j(x) = \arg \min_{i \in I_{j-1}(x)} (b_{j,i} + a_i \cdot x_j). \quad (15.1)$$

The final minimizers of $\Phi_i(x)$ are exactly $I_r(x)$.

For nonempty $T \subseteq S \subseteq \{1, \dots, n\}$ set

$$\mathcal{D}_b^S(T) = \{y \in \mathbb{R}^d : \arg \min_{i \in S} (b_i + a_i \cdot y) = T\}.$$

This is a relatively open polyhedron, possibly empty. Indeed choose $i_0 \in T$, impose equalities

$$(a_i - a_{i_0}) \cdot y = b_{i_0} - b_i \quad (i \in T),$$

and strict inequalities saying that every $i \in S \setminus T$ has a larger value. If nonempty, it has dimension $d - \dim \text{aff}(A_T)$.

Proposition 15.1 (Product strata for a fixed minimizer history). *For a specified chain of nonempty sets $I_0 \supseteq I_1 \supseteq \dots \supseteq I_r$, the points x with this exact successive minimizer history form*

$$\mathcal{D}_{b_1}^{I_0}(I_1) \times \mathcal{D}_{b_2}^{I_1}(I_2) \times \dots \times \mathcal{D}_{b_r}^{I_{r-1}}(I_r). \quad (15.2)$$

A nonempty such stratum has dimension

$$rd - \sum_{j=1}^r \dim \text{aff}(A_{I_j}).$$

There are finitely many histories, and the lexicographic tie locus is the disjoint union of those strata with $|I_r| \geq 2$.

Proof. Lexicographic comparison first discards all labels not minimizing the first coordinate, then discards nonminimizers of the second among the survivors, and continues. For a prescribed chain, the j th condition involves only x_j , which proves the product formula. The equalities defining a nonempty factor have rank $\dim \text{aff}(A_{I_j})$. Its strict inequalities cut out an open subset of that affine solution space and do not decrease its dimension. Dimensions add under products. The finiteness and disjointness follow because every x has a unique chain of finite sets. $\square$

This derivation makes no claim that the Euclidean closures of these strata constitute a pure polyhedral complex. Higher-rank examples can fail the naive closure and purity expectations; such issues are treated explicitly in [3]. Its rank-two product decomposition is a close precedent for (15.2). The newer framework [4] uses higher-rank GKZ cones and its own face and coface structure. A scalar-field Farkas lemma over K and a separation statement for lexicographically

ordered matrix halfspaces are different assertions; one cannot import one as the other without checking the ambient structure.

The present finite compression concerns the *height dependence space*. It does not preserve the full ordered multiplicative group of all surreal monomials. Nor does it supply tropical weights, balancing, or the algebraic degeneration machinery of [4]. Those are additional structures beyond the sign and leading-term invariants proved here.

16 Examples that locate the exact boundary

16.1 Identical leading heights, opposite square diagonals

For the square ordered as above, the affine-dependence space is spanned by $c = (1, -1, -1, 1)$. Consider

$$h^+ = (1, 1, 1, 1 + \varepsilon), \quad h^- = (1, 1, 1, 1 - \varepsilon), \quad 0 < \varepsilon \ll 1.$$

All four individual heights have leading term 1 in both liftings and both standard parts are $(1, 1, 1, 1)$. Yet

$$L_{h^+}(c) = \varepsilon, \quad L_{h^-}(c) = -\varepsilon.$$

The positive case uses diagonal a_2a_3 , with maximal cells $\{1, 2, 3\}$ and $\{2, 3, 4\}$. The negative case uses diagonal a_1a_4 , with maximal cells $\{1, 2, 4\}$ and $\{1, 3, 4\}$. These exact contact-label sets are also returned by the supplied checker.

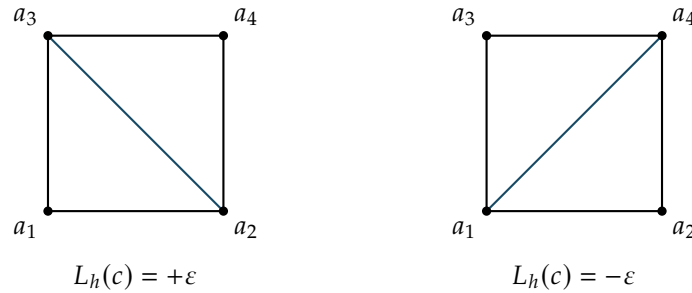


Figure 2: The coordinatewise leading heights and the real shadows agree; the first nonzero affine-dependence coefficient determines different regular triangulations. The figure shows the projected lower cells.

16.2 A nontrivial three-scale compression

Take $A = (0, 1, 2, 3, 4)$ in one dimension, and gauge the first two heights to zero. On the last three coordinates put

$$g = (1, 1, 0)\varepsilon + (2, 2, 0)\varepsilon^2 + (0, 1, 1)\varepsilon^5 \\ + (1, -2, -3)\varepsilon^6 + (1, 0, 1)\varepsilon^9, \quad \varepsilon = \omega^{-1}.$$

The coefficient at ε^2 is twice the first; the coefficient at ε^6 is $(1, 1, 0) - 3(0, 1, 1)$. The coefficient-pivot procedure therefore selects powers 1, 5, 9, and

$$g^\# = (1, 1, 0)\varepsilon + (0, 1, 1)\varepsilon^5 + (1, 0, 1)\varepsilon^9.$$

The three selected rows are independent. Every real linear combination of the three original coordinates has exactly the same leading term after compression, not merely the same sign on the finitely many circuits. A shadow-preserving polynomial model is

$$H(t) = (0, 0, t + t^3, t + t^2, t^2 + t^3).$$

Its minimum degree for the complete sign type with zero shadow is three. The exact checker gives the sufficient bound $\eta_0 = 1/10$, the sample $t_0 = 1/20$, and lower maximal contact sets $\{1, 2\}$ and $\{2, 5\}$. Thus a degree-one path from zero to $H(1/20)$ already retains the finite lifted geometry, while losing some of the full real dependence type.

The gaps between the powers could be replaced by much larger surreal exponent gaps. The proof depends on their order and coefficient spans, not on the existence of a smallest step in the exponent group.

16.3 Unsigned determinant valuations also lose combinatorics

Let $\delta > 0$ be infinitesimal and $u = (1, 1)$. For a real configuration A put $p_i = u + \delta a_i$. Every coordinate of every p_i has leading term 1. Every nonzero augmented 3×3 determinant is δ^2 times the corresponding real determinant. Compare

$$A = \{(0, 0), (2, 0), (0, 2), (2, 2)\}$$

with

$$A' = \{(0, 0), (2, 0), (0, 2), (1/2, 1/2)\}.$$

In both configurations every triple is affinely independent, so all these determinants have the same leading exponent. Nevertheless one hull has four vertices and the other has three, with a labeled interior point. Neither coordinate leading terms nor the unsigned determinant valuations recover the labeled face lattice. The signs of leading coefficients matter.

16.4 Finite truncation can destroy an exact equality

Let $y = \varepsilon$ and $x = \varepsilon + \varepsilon^2 + \dots = \varepsilon/(1 - \varepsilon)$. The exact algebraic relation $(1 - y)x - y = 0$ holds. Replacing x by the first N terms gives

$$(1 - \varepsilon)(\varepsilon + \dots + \varepsilon^N) - \varepsilon = -\varepsilon^{N+1} \neq 0.$$

The pivot theorem preserves real *linear* combinations. It does not preserve arbitrary polynomial relations between the coordinates, and it must not be advertised as a finite truncation algorithm for a surreal field.

16.5 One parameter does not preserve the entire scale field

The positive infinitesimals $\varepsilon = \omega^{-1}$ and $\tau = \omega^{-\omega}$ satisfy $\tau < \varepsilon^N$ for every positive ordinary integer N . Replacing them by t^a, t^b with fixed positive real a, b in a rank-one monomial group cannot preserve all these inequalities: choose $N > b/a$. Our finite-height models preserve a specified real-linear dependence type, not every multiplicative comparison in the field generated by the original scales. This is a boundary on what the theorem says, not a contradiction to its polynomial realization conclusion.

17 A toric application: preserving the whole initial ideal

Suppose now that the base points are integral, $a_i \in \mathbb{Z}^d$, and let k be any field. The homogeneous toric ideal is

$$I_A = \ker(k[x_1, \dots, x_n] \longrightarrow k[z_0, z_1^{\pm 1}, \dots, z_d^{\pm 1}], \quad x_i \longmapsto z_0 z^{a_i}).$$

For a polynomial, define its h -initial form to be the sum of the terms whose weights $\sum_i u_i h_i$ are *maximal*. These weights lie in the ordered additive group of $\mathbf{No}$; no infinite sum or coefficient valuation is involved. The ideal generated by the initial forms of all members of I_A is denoted $\text{in}_h I_A$.

Put $\Lambda = \ker_{\mathbb{Z}} \tilde{A}$. For $c \in \Lambda$ write $c = c^+ - c^-$ with disjoint nonnegative integer supports, and set $B_c = x^{c^+} - x^{c^-}$. Its initial form depends only on $\chi_h(c)$. A vector $g \in \Lambda \setminus \{0\}$ is *conformally indecomposable* when it is not a sum of two nonzero lattice vectors lying in its closed orthant, with their absolute values adding coordinatewise. The set G_A of these vectors, with either both signs or one representative of each pair, is the Graver basis. Its relationship with universal Gröbner bases is classical; see, for example, [7]. We include the finite argument needed for arbitrary ordered weights.

Lemma 17.1 (A finite certificate for arbitrary ordered weights). *The set G_A is finite. For heights in any totally ordered abelian group,*

$$\text{in}_h I_A = \langle \text{in}_h B_g : g \in G_A \rangle.$$

Proof. In each orthant, conformal indecomposables are the minimal nonzero elements of the lattice under coordinatewise comparison of absolute values. Dickson's lemma makes this set finite. One proof of that lemma is induction on the number of coordinates: any infinite sequence in $\mathbb{N}^n$ contains an infinite subsequence whose first coordinate is nondecreasing; apply the induction hypothesis to the remaining coordinates to find a comparable pair. An infinite set of distinct minimal elements would give an infinite sequence with no such pair.

Repeatedly splitting a decomposable lattice vector strictly decreases the positive integer ℓ^1 -size of each resulting summand. Thus every $c \in \Lambda \setminus \{0\}$ has a conformal decomposition $c = g_1 + \dots + g_s$ into Graver vectors. Let $\alpha_j = h \cdot g_j$. If $h \cdot c > 0$, then some $\alpha_j > 0$. The monomial $\text{in}_h B_{g_j} = x^{g_j^+}$ divides $x^{c^+} = \text{in}_h B_c$. The negative case is symmetric. If $h \cdot c = 0$ and not all α_j vanish, there are both a positive and a negative α_j . The two terms of B_c are then separately divisible by initial monomials of Graver binomials. If all $\alpha_j = 0$, use the identity, valid for conformal sums,

$$B_{g+g'} = x^{g'^+} B_g + x^{g^-} B_{g'}.$$

It expresses B_c in the ideal generated by the tied Graver binomials. Hence every initial form of a lattice binomial belongs to the stated ideal. A common monomial factor handles binomials whose exponents overlap.

It remains to justify that binomials suffice. Decompose any $f \in I_A$ into its $\tilde{A}$ -homogeneous parts. Within each part, the coefficients sum to zero, because all monomials have the same image under the toric map. If the terms of one part have different weights, every maximal-weight monomial is the initial monomial of its difference with a lower-weight monomial in that same part. If all its terms have equal weight, the whole part is a linear combination of tied differences with one fixed monomial. Thus its initial form belongs to the ideal of initial binomials. The initial form of the original f is a sum of the initial forms of those homogeneous parts whose maximal weights attain the global maximum, so it belongs as well. The reverse inclusion follows because every B_g belongs to I_A . $\square$

Theorem 17.2 (Toric compatibility of compression and real shadows). *For an integral base configuration, the compressed lifting satisfies*

$$\text{in}_{h^\#} I_A = \text{in}_h I_A.$$

A polynomial model of its complete dependence type has the same initial ideal for every sufficiently small positive real parameter. If h is bounded, a real affine-linear path starting at $\text{st}(h)$ can preserve this initial ideal, all the lifted determinant signs, and the marked lower subdivision simultaneously on its whole punctured segment.

Proof. Compression preserves the sign of $L_h(g)$ for every $g \in G_A \subseteq V$, so Lemma 17.1 gives the first equality. The finite sign lemma applied to these Graver comparisons proves stabilization along a polynomial model. For the last assertion, add G_A to the finite determinant and slack test list in Theorem 10.4. $\square$

This preserves the initial ideal itself, not only its radical or the underlying subdivision. The initial algebra and its scheme-theoretic invariants therefore agree. The statement is an application of classical finite Graver control together with our lifting construction; it is not a new construction of the higher-rank degenerations of [4]. For the square, $I_A = (x_1x_4 - x_2x_3)$: the two infinitesimal choices above give initial ideals (x_1x_4) and (x_2x_3) , respectively.

18 General surreal polytopes: what transfer gives and what it does not

Theorem 18.1 (Classical finite realization and shadow arcs). *Every finite surreal point configuration has the same labeled oriented matroid, and hence the same labeled polytope face structure, as a real configuration. Without prescribed real coordinates or other real parameters, a realization can be chosen with real algebraic coordinates. If the original coordinates are finite, there is a real semialgebraic arc of configurations with the same finite determinant signs that converges to their labeled standard parts. Additional finitely many polynomial sign conditions can be preserved along the same arc. [write] In manuscript 02's form (its Theorem "Classical real-algebraic transfer"): for any finite labeled configuration $V = (v_1, \dots, v_n)$ in a real-closed field, there is a labeled configuration of real algebraic points with the same signed affine-hyperplane patterns and the same affine dimension. In particular, the two convex hulls have isomorphic labeled face lattices.*

Proof. List the signs of all minors of the homogeneous coordinate matrix, keeping each zero as an equality. This is a finite system of polynomial equalities and strict inequalities with integer coefficients. It has a solution in a set-sized real closed subfield of $\mathbf{No}$ containing the coordinates. Completeness of the theory of real closed fields transfers the existential sentence to $\mathbb{R}$, and also to the real algebraic numbers when there are no prescribed real parameters. The minor signs determine the rank and oriented matroid; the facet reconstruction in Lemma 10.1, in its appropriate affine rank, gives the face structure. These are standard real-algebraic facts [5].

For the bounded assertion, let v^0 be the vector of all standard-part coordinates and let S be the corresponding real semialgebraic sign cell, including the extra predicates when present. For every positive real r , the surreal tuple lies in the ball of radius r about v^0 and satisfies the sign conditions. Transfer over the named real parameters shows that S contains a real tuple within that ball. Thus $v^0 \in \bar{S}$. If a constant configuration already suffices, take a constant arc. Otherwise semialgebraic curve selection yields an arc entering S and tending to v^0 [5, 6]. Its one-variable semialgebraic coordinates are algebraic-function branches; one may use a finite

power reparameterization to obtain analytic branches at the endpoint. The determinant signs on the punctured arc are the desired ones. $\square$

Second proof of the realization assertion, in manuscript 02's covector form (source 02). For each $\sigma \in \{-1, 0, 1\}^n$, let $\Psi_\sigma(V)$ be the formula

$$\exists u_1, \dots, u_d, b \quad \bigwedge_{i=1}^n \operatorname{sgn}(u \cdot v_i + b) = \sigma_i.$$

Replace each sign assertion by the corresponding polynomial equality or strict inequality. There are only 3^n patterns. Conjoin Ψ_σ for all patterns realized by V and $\neg\Psi_\sigma$ for all other patterns. Also conjoin the finite minor conditions expressing its affine dimension. Finally existentially quantify the coordinates of V .

The resulting sentence is a first-order sentence of ordered rings with integer coefficients. It is true in the given real-closed field. By the completeness of the theory of real-closed fields, equivalently by Tarski's elimination theorem [11], it is true in the real algebraic numbers, which themselves form a real-closed field. Choose a witnessing configuration there.

A supporting affine functional has all its values nonpositive after subtracting its maximum. Its zero labels are exactly the labels on its maximizing face. Thus the all-nonpositive sign patterns recover all nonempty face label sets; adding the empty face gives the entire labeled face lattice. The construction preserves these patterns and their inclusions. $\square$

[write] The next two paragraphs follow the transfer theorem in manuscript 02; the family announced at their end is the lifting compression of [section 7](#).

This theorem is intentionally stronger than a statement about the numbers of faces: it preserves the entire affine covector system. It is also intentionally weaker than an exact coordinate representation. There is no order-preserving field map from **No** into $\mathbb{R}$ fixing $\mathbb{R}$, since a positive infinitesimal would have to become a positive real number smaller than every $1/m$. Nor does the theorem guarantee rational coordinates.

Consequently, such familiar finite combinatorial questions as which f -vectors occur do not acquire additional examples merely by allowing surreal coordinates. Any genuinely additional information must reside in scales, residue maps, simultaneous structures, or infinite families of observations. We turn to one such family.

The realization statement is classical transfer, not a new surreal realizability theorem. Likewise the general arc existence is an application of curve selection; [6] supplies a quantitative framework. The special contribution of the fixed-base setting is the elementary polynomial model with sharp degree and the linear finite-data path.

Example 18.2 (Polynomial arcs need not preserve nonlinear constraints). Take the finite surreal point

$$(x, y) = (\sqrt{1 - \varepsilon^2}, \varepsilon), \quad 0 < \varepsilon \ll 1.$$

It lies on the unit circle, has $y > 0$, and has standard part $(1, 0)$. There is no real polynomial arc $(X(t), Y(t))$ with $(X(0), Y(0)) = (1, 0)$, $X(t)^2 + Y(t)^2 = 1$, and $Y(t) > 0$ for small positive t . Indeed, if either polynomial is nonconstant, the coefficient of the largest even degree in $X^2 + Y^2$ is a positive sum of squares and cannot vanish. Both must therefore be constant, contradicting the endpoint and $Y(t) > 0$. An algebraic analytic arc exists, but a polynomial one does not. This example concerns the permitted extra polynomial constraints; it is not claimed as an obstruction arising from determinant signs alone.

19 A finite Puiseux model with the same residue

[write] [Theorem 19.1](#) is the bounded half of [theorem 18.1](#) (source 05) in manuscript 02's form: it keeps the full signed affine-hyperplane patterns and any finite list of real polynomial signs along a semialgebraic arc starting at the standard part, with convergent algebraic Puiseux germs. Both proofs are the same transfer-and-curve-selection argument (05 cites [5, 6], 02 cites [15, 6]). The circle example after [theorem 18.1](#) shows that such arcs cannot in general be taken polynomial. The finite-support specialization [theorem 41.2](#) of [Part III](#) is an effective counterpart for finite-support Hahn certificates.

The universal truncation theorem does not prevent a *different* well-chosen real one-parameter model from preserving any finite collection of incidence data. This is a useful distinction between an arbitrary coordinate truncation and a controlled semialgebraic specialization.

Theorem 19.1 (Finite incidence-and-residue realization). *Let $V \in O_K^{dn}$ be a finite bounded configuration and let $b = \text{st}(V) \in \mathbb{R}^{dn}$. Fix any finite collection of polynomials with real coefficients in the coordinates of V . There is a continuous semialgebraic arc $\gamma : [0, a) \rightarrow \mathbb{R}^{dn}$ such that*

- (a) $\gamma(0) = b$;
- (b) for every sufficiently small $s > 0$, all the prescribed polynomial signs at $\gamma(s)$ equal those at V ;
- (c) for those s , the configuration $\gamma(s)$ has the same signed affine-hyperplane patterns and affine dimension as V .

The coordinate germs of γ can be taken to be convergent algebraic Puiseux germs over $\mathbb{R}(s)$. In particular, they give a finite geometric model with the full face lattice of V and its prescribed standard-part limit.

Proof. Encode the prescribed polynomial signs and the complete finite list of affine sign patterns exactly as in [Theorem 18.1](#), now keeping real parameters. Include the rank conditions. Quantifier elimination makes the resulting condition a semialgebraic set $S \subseteq \mathbb{R}^{dn}$, and the original configuration satisfies the same defining condition in K .

For every real $\delta > 0$, the field K satisfies

$$\exists z \in S(K) \quad \max_i |z_i - b_i| < \delta,$$

since $z = V$ is a witness. The inclusion $\mathbb{R} \subseteq K$ is elementary for ordered-ring formulas, by real-closed-field quantifier elimination. Hence the displayed assertion has a real witness. It follows that $b \in \bar{S}$ in the ordinary Euclidean topology.

If $b \in S$, the constant arc is sufficient. Otherwise apply the semialgebraic curve selection lemma to S at b . It gives a continuous semialgebraic arc with initial point b and positive-time values in S . This proves (a)–(c). A one-variable semialgebraic function has, after restricting to a sufficiently short interval, an algebraic Puiseux germ; its Puiseux expansion is convergent. Equivalently, after a common finite ramification $s = u^m$, all the coordinate germs are analytic at the endpoint. These are the standard curve-selection and algebraic-Puiseux facts from real algebraic geometry [15, 6]. $\square$

Here “algebraic” means algebraic over $\mathbb{R}(s)$, not that every coefficient is a real algebraic number. The limiting configuration may itself contain transcendental real coordinates. [Theorem 19.1](#) is a synthesis of standard transfer and curve-selection tools, not a new curve selection lemma. It supplies neither explicit complexity bounds nor a coordinate embedding of the original surreal field.

Most importantly, its quantifier is *finite*. The chosen arc can preserve the final four-point hull and its residue, or any finite list of additional polynomial observations. It need not preserve the arbitrary infinite sequence of coordinate-truncation hulls constructed in Section 26. This is exactly how finite realizability and infinite scale complexity coexist.

20 Corrections to the earlier convexity draft

[[write] The draft corrected here, *Finite Convexity and Exact Lexicographic Optimization over the Surreals* (22 September 2026), was supplied privately to the authors of manuscripts 05 and 07; it is not part of ProveIt and could not be checked. The statements below are printed as standalone results.]

The earlier draft [10] contains a valid sufficiently-small comparison argument. Its Proposition 7.1, however, identifies the *entire* positive validity set with an interval. Its Theorem 9.1 and following explanation also require a distinction between a real domain and its scalar extension. Neither issue affects the proofs above.

20.1 The global validity set need not be an interval

On $P = [0, 1]$, take the objective hierarchy $c_0x = x$, $c_1x = -3x$, and $c_2x = 2x$. Lexicographic maximization selects $x = 1$. The compiled objective is

$$(1 - 3t + 2t^2)x = (1 - t)(1 - 2t)x.$$

It selects exactly the same maximizer for

$$t \in (0, 1/2) \cup (1, \infty), \quad (20.1)$$

not for a single interval. At $t = 1/2$ and $t = 1$ it is constant on the whole segment; between those points it selects $x = 0$.

Proposition 20.1 (Correct initial-radius statement). *For finitely many nonzero comparison polynomials with their required signs at 0^+ , the component of the simultaneous validity set adjacent to zero is $(0, \rho)$, where ρ is the least positive zero of one of the normalized nonzero comparison polynomials, or $+\infty$ when none has a positive zero. Over a real closed coefficient field, a finite ρ belongs to that field. The entire validity set may have additional components beyond ρ .*

Proof. Divide each polynomial by its lowest power of t ; its constant term is nonzero. Before its first positive zero its sign is constant, by the intermediate value property. At a positive zero the required strict sign fails, even if the polynomial merely touches zero and later has the same sign again. Taking the least positive zero over the finite list gives exactly the component adjacent to zero. Signs on subsequent root intervals can agree again, as (20.1) shows. $\square$

For a general ordered field the first zero may lie only in a real closure, so the endpoint may define a cut rather than an element of the original field. A sufficient algebraic bound such as Lemma 10.3 avoids this issue. We never use a global-interval assertion.

20.2 A real face and its scalar extension are different sets

Let $P_{\mathbb{R}} = \text{conv}_{\mathbb{R}}(v_1, \dots, v_N)$ with real vertices, and let $P_K = \text{conv}_K(v_1, \dots, v_N)$. Suppose a compiled objective makes exactly the labels in J optimal. Then

$$\arg \max_{x \in P_K} c \cdot x = \text{conv}_K\{v_j : j \in J\}, \quad (20.2)$$

not generally $\text{conv}_{\mathbb{R}}\{v_j : j \in J\}$. To prove (20.2), express x as a convex combination and subtract its objective value from the optimal vertex value. The result is a sum of nonnegative terms. Equality forces every positive coefficient to occur at an optimal vertex, exactly as over $\mathbb{R}$.

For example, maximizing the first coordinate on $[0, 1]_K^2$ gives the whole edge $\{1\} \times [0, 1]_K$, including $(1, \varepsilon)$ for a positive infinitesimal $\varepsilon \in K$. This point is not in the ordinary real edge. The correct shadow identity is

$$\text{st}\left(\arg \max_{P_K} c \cdot x\right) = \text{conv}_{\mathbb{R}}\{v_j : j \in J\},$$

by Proposition 13.1. The same optimal *vertex labels* and the same base-changed face are selected. If the old statement is read literally with optimization only over the subset $P_{\mathbb{R}} \subset \mathbf{No}^d$, then equality with the real face is correct; it is not the assertion about K -convex polytopes used elsewhere. This is a scalar-extension ambiguity, not a reason to discard the finite compiler.

21 Computation, certificates, and a formalization route

21.1 What the executable verifier actually does

The accompanying `code/05-finite-scale-liftings-verify.py` [write] (delivered as `code/verify.py`) uses only Python's standard library and exact rational arithmetic. It accepts the finite examples encoded there as Laurent polynomials in a positive infinitesimal $\varepsilon = \omega^{-1}$. An input coefficient list is complete, not an oracle that may reveal further terms later. The procedure is:

1. find affine bases by exact determinants and compute barycentric coefficients by exact elimination;
2. subtract the affine gauge coefficientwise and retain exactly the rank-increasing coefficient rows;
3. build the polynomial model, every affine-basis slack, and every augmented maximal minor;
4. compare leading terms and signs before and after compression, compute the rational threshold, and verify the real specialization and several points on the linear shadow path.

The theorem proves validity on the whole segment; the sampled points are only additional implementation checks. Exact leading-term equality is checked on the finite determinant/slack list and on 60 additional rational dependence vectors per example. No finite test can replace the universal proof in Theorem 6.1.

The executed suite contains seven named cases, six members of the sharpness family, and forty randomized rational configurations using seed 20260930. All fifty-three cases passed, including the three-scale pivot list $[1, 5, 9]$ and the validity-set counterexample. This gives 3,180 extra sampled dependence checks, in addition to the complete finite geometry test lists for each case. Machine-readable data are in `data/05-finite-scale-liftings-certificates.json`; the execution summary is in `data/05-finite-scale-liftings-verification_report.txt`. The program does not implement arbitrary surreal arithmetic, semialgebraic curve selection, or a Graver basis algorithm. The corresponding general results rest on the written proofs, not on a claimed implementation.

If the finite support has s coefficient rows and the affine-dependence dimension is q , an incremental elimination implementation can perform the pivot selection using $O(sq^2)$ field operations, after gauging. The supplied reference verifier recomputes some small ranks for simplicity; it is not presented as achieving that optimized complexity. Enumerating all affine bases involves at most $\binom{n}{d+1}$ candidates, and rational bit growth must be accounted for in any bit-complexity claim.

21.2 Why coefficient access alone does not make the bound computable

Proposition 21.1 (An explicit representation obstruction). *There is no algorithm that always computes m for rational-coefficient height series given by uniformly computable coefficient functions, even under the promise that each series actually has finite support and that the real base configuration is fixed.*

Proof. Use $A = (0, 1, 2, 3)$ and gauge the first two heights to zero. For a Turing machine e , let $x_e = \varepsilon^{T+1}$ if it halts for the first time at step T , and let $x_e = 0$ if it never halts. The coefficient of ε^N in x_e is computable uniformly from e, N by simulating at most N steps. Every resulting series has at most one nonzero coefficient. Take heights $(0, 0, 1, x_e)$. Their dependence-image dimension is one if the machine never halts and two if it halts, since a nonzero positive infinitesimal is real-linearly independent of 1. An algorithm computing m from these coefficient functions would decide the halting problem. $\square$

Thus a complete finite support list and a computable coefficient function with a finite-support promise are very different input representations. The latter supplies no computable bound on the last possible nonzero coefficient. The rank theorem does not erase that distinction.

21.3 A finite certificate workflow

[write] This subsection and the next are from manuscript 02’s section “Algorithms, verification, and proof boundaries” (section 27 prints the rest). Its query-model bound [proposition 21.2](#) and 05’s [proposition 21.1](#) are different obstructions and both are kept: 02’s is an adversary argument in the query model (no algorithm that halts after finitely many coefficient queries computes the rank of every finite-support input), 05’s a reduction of the halting problem for uniformly computable coefficient functions. For a real base with a supplied finite rational array of height coefficients, the positive results suggest the following reproducible workflow. Compute affine bases and their barycentric slack vectors. Fix one basis to gauge away affine coefficient rows by (11.3). Scan the ordered row array, retaining each rank-increasing residual row. For every retained prefix, enumerate lower bases by lexicographic slack signs and deduplicate their zero label sets. Finally compute (11.5) and check the same signs after real rational specialization.

Every operation in this workflow is finite linear algebra and rational arithmetic. The output consists of selected row positions, cell label sets, and a positive rational sign threshold. Each claim can be rechecked without trusting a floating-point convex-hull package. The supplied program implements this workflow for test configurations. It does not purport to parse arbitrary surreal sign sequences or decide arbitrary normal-form supports.

21.4 A rank bound is not a coefficient-query bound

Proposition 21.2 (No finite-query extraction for unrestricted streams). *There is no algorithm which, given only exact coefficient queries to an arbitrary power-series vector, always halts and computes*

its complete comparison rank, even if the coefficient vectors belong to a fixed rational two-dimensional space and the input is promised to have finite support without a supplied degree bound.

Proof. Let w^1, w^2 be independent rational vectors. Run a purported algorithm on the coefficient stream of $x = w^1$. If it always halts, it makes only finitely many coefficient queries. Choose a positive integer N larger than every queried index. The finite-support input $x' = w^1 + \varepsilon^N w^2$ gives exactly the same answers to all queries, including adaptive queries, so the algorithm follows the same computation and returns the same result. But the comparison ranks are 1 and 2, respectively, by Theorem 7.1. This is a contradiction. $\square$

Thus “at most q selected rows” is not “at most q coefficient inspections.” Effective treatment of infinite support needs an additional representation promise, a degree bound, or a method for deciding whether a new coefficient direction will ever occur. The obstruction is already present for ordinary finite-support series with an unknown cutoff; it is not solely a proper-class issue.

[write] Manuscript 05’s theorem-by-theorem Lean plan, a subsection here in the source, is printed in [section 83](#).

22 Further research questions and concrete routes

[write] Manuscript 02’s eight research directions are printed in [section 28](#). Four of them are close to questions here: 05’s question 1 and 02’s “Effective controlled specialization” both ask for effective bounds on arcs from a standard part; question 2 and 02’s “Beyond linear comparisons: realizability constraints” both ask when compressed minors come from actual points; question 4 and 02’s “Minimal geometric certificates versus full comparison rank” both ask for certificates smaller than the all-basis list; question 5 and 02’s “Formalization and representation-aware computation” both ask for representations in which the next rank increase can be certified. Each is printed where its source puts it, with this cross-reference.

The questions here are proposed projects, not assertions that every one is a previously published open problem. Their scopes exclude the finite-data and complete-type degree questions already answered above.

1. Moving-base determinant degenerations. For n bounded surreal points in dimension d , seek sharp complexity bounds for a real algebraic or Puiseux arc preserving all determinant signs and the labeled shadow. General quantitative curve selection already gives effective bounds; the task is to exploit the determinant structure rather than reprove mere existence. A useful first case is a planar configuration with one prescribed collinearity collapse and all other minors nonzero. Determine when polynomial coordinates suffice without adding extra nonlinear constraints.

[[write] *Re-scoped.* For rational arcs the answer is negative: [Part VIII](#) constructs labeled polytopes with $8k + 16$ vertices in dimension $4k + 10$ whose rational vertex arcs preserving the labeled face lattice and the shadow need homogeneous degree at least $\lceil 2^k / (16k + 48) \rceil$ (at most 2^k suffices; [theorem 120.3](#)). Algebraic and Puiseux arcs are covered only through the normalized ramification bound ([theorem 127.2](#)).]

2. More than one nonreal lifting coordinate. Replace one height by $r \geq 2$ surreal coordinates over a fixed real base. Maximal minors now involve products of heights. Compressing their coefficient vectors in an exterior-power representation is finite-dimensional, but the compressed

data may fail the decomposability relations needed to come from actual points. Determine the sharp realizable compression bound already for $r = 2$, and identify the Plücker-type obstruction explicitly.

3. Singular fibers of the flag-to-secondary map. A complete dependence flag maps to a regular subdivision, and for integral A to a toric initial ideal. Generic first rows leave later flags invisible; thus the fibers need not be points or contractible. Describe the topology and incidence structure of fibers when the first row lies on a secondary or Gröbner wall. The coefficient-kernel description gives explicit semialgebraic conditions from which to start.

4. Certificates smaller than all bases or a Graver basis. The proofs give finite certificates, but the evident lists can be large. Can a certificate containing only a controlled set of supporting bases, redundancy witnesses, and essential lattice relations verify simultaneously the face lattice, the lower subdivision, and the initial ideal? Separate certificate size from the complexity of finding that certificate. A practical target is an output-sensitive checker rather than an efficient universal enumeration algorithm.

5. Effective infinite-support subclasses. Proposition 21.1 rules out uniform extraction from arbitrary coefficient programs. Identify normal-form languages with enough support structure to certify that no later independent coefficient exists. Examples to investigate include rational generating functions and prescribed finite-dimensional coefficient recurrences. The desired result is an algorithm with a proved stopping criterion, not merely symbolic access to individual coefficients.

6. Relating flags to higher-rank algebraic degenerations. Compare the canonical real-dependence flag here with the higher-rank quasi-valuations in [4]. Determine precisely which information is lost when a matrix of weights is replaced by the flag modulo positive triangular and affine transformations. Equality of toric initial ideals is proved above; equality of valuation structures or multi-parameter families is a stronger requirement and should be tested separately.

7. Arithmetic constraints on a real replacement. With rational A and rational coefficient slices, the present construction gives rational specializations, but denominators may grow substantially. Optimize coefficient height and the size of a rational endpoint while retaining specified signs and a prescribed rational shadow. For an integral endpoint one must account for the shadow and allowed affine rescalings, rather than silently multiplying the entire path by a scalar. This is a concrete quantitative refinement of the existence theorem.

8. A small trusted formalization. Implement the coefficient-pivot and finite-slack certificate theorems first, with no surreal class machinery in the checker. Then connect them to the existing ordered-field geometry and normal forms. An especially useful milestone is a machine-checked version of the separation theorem: a family with linear finite combinatorial realizations but arbitrarily large minimum degree for the complete dependence type.

23 Conclusion

The real-base lifting problem admits an exact three-level answer. Arbitrary surreal support collapses to $m \leq n - d - 1$ selected coefficient slices when all real dependence leading terms are the invariant. The resulting complete sign type is an oriented partial flag, and its minimum polynomial germ degree is determined by m and, when a shadow is fixed, r_0 . Finite combinatorics is less demanding: a single real segment suffices, and in the integral case that segment may also preserve the entire toric initial ideal.

These results do not produce a new finite face lattice unavailable over the reals. They describe exactly what can be hidden behind a familiar face lattice, what standard part destroys, and which additional data restore it. The strongest claim of this manuscript is the proved sharp classification in this stated setting. Its possible novelty lies in that assembled theorem package and its implementation boundary; independent proof review and a broader priority audit remain appropriate.

24 Dependency and verification ledger

[write] Appendix A of manuscript 05. It records manuscript 05's results only; manuscript 02's proof-dependency audit is [table 5](#).

Result	Evidence and dependencies
Coefficient-pivot theorem	Full proof; Conway coefficient uniqueness, leading-sign rule, and finite-dimensional linear algebra.
Affine compression and flags	Full proofs from the pivot theorem and ordinary real linear algebra.
Exact polynomial degrees	Full constructions and rank lower bounds; no transfer or numerical approximation.
Finite face and slack certificates	Full determinant and barycentric proofs; finite rational examples checked.
Linear shadow realization	Full proof from standard part and finite sign stabilization; arbitrary finite test list.
Arbitrary degree separation	Explicit full-rank construction and exact lower bounds; finite family members checked.
Toric initial ideal	Full conformal-decomposition proof, with classical Graver-background attribution; no Graver implementation claimed.
General semialgebraic arcs	Application of cited real-closed-field transfer and curve selection, not a new general existence theorem.
Validity-set correction	Explicit quadratic counterexample and corrected initial-component proof; exact signs checked.
Coefficient-oracle obstruction	Direct reduction from the halting problem.
New Lean declarations	None supplied or claimed.

25 Source and reproduction notes

The fixed geometry source inspected was

`Algebra/SurrealNumbers/Surreal/Algebra/Geometry.lean`

at repository commit `5e0f4e05046bacda3c6ba2643f19b91aa632436b`, with blob `450fd4bbfc017e0da24c28e538af1cc02037ad5a`. The surrounding README was read on 30 September 2026; its fetched blob was `cf63fc5ff9ca9d8877e90f8e69ec34e6ded19d23`. These identify inspected sources, not a claim that the repository has stopped changing. The comparison covers these files, the reports guide, and targeted searches; it is not a complete audit of every manuscript in the repository.

The earlier companion draft was recovered from the research library under `article(20260923-000126).pdf`; the relevant statements are on its printed pages 5–6. It was consulted for comparison and correction, not imported as a theorem dependency. The current literature check included the September 2026 revision of [4]. No exhaustive priority search or independent referee report is represented by this record.

To reproduce the executable checks, run

```
python3 code/verify.py
```

from the package directory. To rebuild this document, run

```
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

A third run may be needed after a substantial change to the contents or cross-references. The bibliography is included below; no external `.bib` file is required. Standard TeX Live packages provide the fonts; no font files are distributed with this package.

[write] In this report: the verifier is shipped as `code/05-finite-scale-liftings-verify.py` and its outputs as `data/05-finite-scale-liftings-certificates.json` and `data/05-finite-scale-liftings-verification_report.txt`. The program writes its outputs under the delivered names, into `data/` beside its parent directory, so it must be rerun on a copy laid out as delivered; the report’s README gives the recipe. The manuscript’s own README and PDF are not shipped (this report and its README.md replace them); its `SOURCES.md` and `PROOF_REVIEW.md` are shipped as `05-finite-scale-liftings-SOURCES.md` and `05-finite-scale-liftings-PROOF_REVIEW.md`, and its checksum file `SHA256SUMS.txt` is not shipped. The research-library copy of the earlier draft named above is not in ProveIt.

Part II

Coordinate-truncation histories

Source and scope. [write] This Part prints Section 6 of manuscript 02 (named in [Part I](#), which also prints its abstract and status), followed by the parts of its Sections 9–11 and its two appendices that are not printed in [Part I](#): the recorded tests and the proof-dependency audit, its research directions, its conclusion, the exact recursion for the universal construction, and its reproducibility notes. Its operation, truncating the *coordinates* of a configuration at a common degree, is operation (iii) of 02’s introduction ([section 3](#)). It is not the common initial cut of lifting heights of [Part I](#), and the finiteness theorems of [Part I](#) do not apply to it. No other manuscript of the batch proves a result of this kind. Manuscript 05’s example “Finite truncation can destroy an exact equality” ([section 16.4](#)) and the examples of [Part III](#) showing that no truncation depth depends only on dimension and vertex count ([section 40](#)) are weaker instances of the phenomenon of [theorems 26.2](#) and [26.3](#).

Notation in this Part. [write] t is the variable of the power series u, f, g, H, J ; every sign is evaluated at $t = \varepsilon = \omega^{-1}$ (manuscript 02's convention), and the same series are also real-analytic functions of a real t near 0 (theorem 26.3). $\tau = \omega^{-\omega}$. $\Delta = 1 - fg$ is an orientation determinant; $\rho \in \{-1, 1\}$ in theorem 26.6 is a sign, not a threshold or a jet scale; H is a sign-generating series, not a height vector; M in proposition 26.7 is a maximal order.

26 Four moving points can encode arbitrary histories

The preceding finiteness theorem is fundamentally linear: the base A is real and fixed, and every relevant test is a real linear functional of its heights. If the point coordinates move, orientation tests become polynomials in those coordinates. Truncation does not preserve exact multiplicative relations. The following examples exploit that difference.

For a power series $u(t) = \sum_{k \geq 0} u_k t^k$, write $u_{\leq N} = \sum_{k=0}^N u_k t^k$. A degree- N coordinate truncation means that this operation is applied to *every* coordinate at the same N . All signs are evaluated at the positive infinitesimal $t = \omega^{-1}$, or equivalently by the first nonzero coefficient.

26.1 A four-point orientation test

Lemma 26.1 (Triangle versus quadrilateral). *Let f, g be elements of an ordered field, with*

$$g > 0, \quad 2g - 2 + \Delta > 0, \quad \Delta = 1 - fg.$$

For the four points

$$A_0 = (0, 0), \quad B = (1, f), \quad C = (g, 1), \quad D = (0, 2), \quad (26.1)$$

the following hold. If $\Delta > 0$, they are the four vertices of a strictly convex quadrilateral in the order displayed. If $\Delta < 0$, B is strictly inside the triangle A_0CD . If $\Delta = 0$ and $g > 1$, B lies in the relative interior of the edge A_0C .

Proof. The four consecutive oriented triangle determinants are

$$\begin{aligned} [A_0, B, C] &= \Delta, \\ [B, C, D] &= 2g - 2 + \Delta, \\ [C, D, A_0] &= 2g, \\ [D, A_0, B] &= 2. \end{aligned} \quad (26.2)$$

Here $[X, Y, Z] = \det(Y - X, Z - X)$. For $\Delta > 0$ these signs give the strict convex ordering. This can also be checked without a polygon-order criterion: the line through each displayed consecutive pair has both other points strictly on its left, using the same four determinants.

For $\Delta < 0$, the edge tests for membership in the positively oriented triangle A_0CD are $-\Delta$, $2g - 2 + \Delta$, and 2 , all positive. Thus B is strictly inside that triangle. For $\Delta = 0$, the equality $fg = 1$ gives $B = C/g$; if $g > 1$, the scalar $1/g$ lies strictly between zero and one. $\square$

26.2 The simplest infinite alternation

Theorem 26.2 (Rational alternating truncations). *Let $g = 1 + t$ and $f = (1 + t)^{-1}$ in $\mathbb{R}((t)) \subseteq \mathbf{No}$, and let P_N be the convex hull of the common degree- N coordinate truncations of (26.1). For every $N \geq 1$,*

P_N is a quadrilateral when N is odd, and a triangle when N is even.

The full configuration has triangular hull, with B in the relative interior of A_0C . All points are finite, all these hulls have dimension two, and there are only four labeled generators.

Proof. The geometric-series identity gives

$$f_{\leq N} = \sum_{k=0}^N (-t)^k, \quad (1+t)f_{\leq N} = 1 + (-1)^N t^{N+1}.$$

Consequently

$$\Delta_N = 1 - (1+t)f_{\leq N} = (-1)^{N+1} t^{N+1}. \quad (26.3)$$

The other nonconstant side test is

$$2g - 2 + \Delta_N = 2t + (-1)^{N+1} t^{N+1} > 0.$$

Lemma 26.1 proves the parity statement. For the full coordinates $fg = 1$ and $g > 1$, so the final assertion about B follows from the same lemma. $\square$

The theorem already rules out a bound on the number of coordinate-truncation changes in terms of n and d . The alternating points are rational functions of t . They do not require arbitrary surreal exponents or an infinite-dimensional base configuration. The infinitude is in the list of truncation levels.

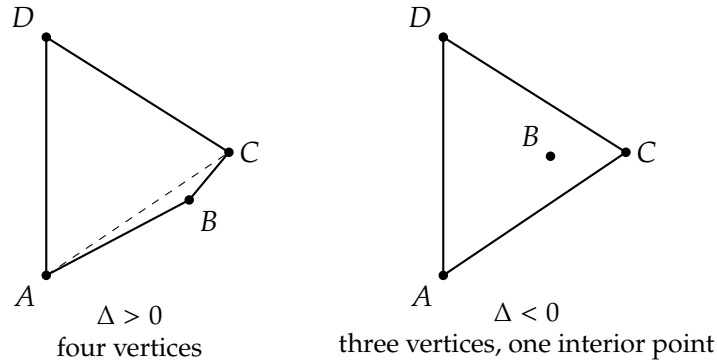


Figure 3: The two hull types detected by $1 - fg$. Here the schematic label A denotes A_0 . The drawing exaggerates the displacement of B from the diagonal; it is not a real-scale plot of infinitesimal coordinates.

26.3 A universal reciprocal-series construction

The alternating example can be generalized from a periodic pattern to an arbitrary prescribed pattern.

Theorem 26.3 (Universal binary coordinate-truncation histories). *Let $(\sigma_N)_{N \geq 1}$ be any sequence in $\{-1, 1\}$ with $\sigma_1 = 1$. There are power series $f, g \in \mathbb{Q}[[t]]$, convergent on a common real neighborhood of zero, such that*

$$fg = 1, \quad f = 1 - t + O(t^2), \quad g = 1 + t + O(t^2), \quad (26.4)$$

and, for every $N \geq 1$,

$$1 - f_{\leq N} g_{\leq N} = \sigma_N t^{N+1} + O(t^{N+2}). \quad (26.5)$$

The hull of the common degree- N truncations of (26.1) is therefore a quadrilateral for $\sigma_N = 1$ and a triangle for $\sigma_N = -1$. Every prescribed binary history from degree two onward occurs in this way.

Proof. Define

$$H(t) = \sum_{N \geq 1} \sigma_N t^{N-1}, \quad J(t) = H(t) + \frac{t^2}{4} H(t)^2. \quad (26.6)$$

Since $H(0) = J(0) = 1$, there is a unique formal square root of J with constant term 1. Put

$$\begin{aligned} g(t) &= 1 + \frac{t^2}{2} H(t) + t \sqrt{J(t)}, \\ f(t) &= 1 + \frac{t^2}{2} H(t) - t \sqrt{J(t)}. \end{aligned} \quad (26.7)$$

The coefficients are rational: the formal square-root recursion divides by 2, and all the coefficients of J are rational. Direct multiplication gives

$$fg = \left(1 + \frac{t^2}{2} H\right)^2 - t^2 \left(H + \frac{t^2}{4} H^2\right) = 1.$$

The constant and linear terms give (26.4), while addition gives the particularly useful identity

$$f + g = 2 + t^2 H. \quad (26.8)$$

Write $f = \sum_{k \geq 0} b_k t^k$ and $g = \sum_{k \geq 0} a_k t^k$, so $a_0 = b_0 = 1$. Through degree N , the product of the truncations agrees with the full product 1. At degree $N+1$, the only terms of the full product omitted by multiplying the two degree- N truncations are $a_{N+1} b_0 + a_0 b_{N+1} = a_{N+1} + b_{N+1}$. By (26.8), this sum equals σ_N . The coefficient of degree $N+1$ in the truncated product is therefore $-\sigma_N$, proving (26.5).

It remains to verify convergence and the hull interpretation. The series H converges absolutely for $|t| < 1$. One can use a common quantitative neighborhood, independent of the sign sequence. For complex $|t| \leq 1/8$,

$$|H(t) - 1| \leq \frac{1}{7}, \quad |H(t)| \leq \frac{8}{7}, \quad |J(t) - 1| \leq \frac{1}{7} + \frac{1}{196} = \frac{29}{196} < 1.$$

The binomial series for $\sqrt{1 + (J - 1)}$ consequently converges near this closed disk and selects the root with constant term 1. The two series in (26.7) are convergent near zero. The construction thus works both analytically and formally, and its formal values at $t = \omega^{-1}$ are legitimate surreal normal forms.

For every $N \geq 1$, $g_{\leq N} = 1 + t + O(t^2)$ and $f_{\leq N} = 1 - t + O(t^2)$. The remaining side test in Lemma 26.1 is

$$2g_{\leq N} - 2 + \Delta_N = 2t + O(t^2) > 0.$$

Also $g_{\leq N} > 0$. The first nonzero term of Δ_N is exactly (26.5), so the lemma proves the prescribed hull type. $\square$

Remark 26.4 (Why the first bit is fixed). The normalization $f = 1 - t + O(t^2)$ and $g = 1 + t + O(t^2)$ forces $1 - f_{\leq 1} g_{\leq 1} = t^2$. Thus $\sigma_1 = 1$ is not an overlooked restriction: the first truncation is automatically a quadrilateral in this normalized family. All later bits are free. This is enough to realize every infinite binary pattern after one fixed initial stage.

Remark 26.5 (No contradiction with analytic nonoscillation). For each fixed analytic family, the full identity $fg = 1$ holds at every sufficiently small real t . The histories above are not the signs of one analytic function evaluated at a sequence of real arguments. Instead the polynomials $1 - f_{\leq N} g_{\leq N}$ themselves change with N . They approach the zero function while retaining prescribed leading signs. There is no accumulation of isolated zeros of a nonzero analytic function at an interior point.

26.4 The final hull can be selected independently

The full reciprocal configuration is degenerate along A_0C . A scale beyond all ordinary powers removes that degeneracy without constraining any finite truncation.

Theorem 26.6 (An independent uniform transfinite endpoint). *For the series in Theorem 26.3, set*

$$t = \omega^{-1}, \quad \tau = \omega^{-\omega}, \quad f^{(\rho)} = f - \rho\tau, \quad \rho \in \{-1, 1\}.$$

Use $f^{(\rho)}$ instead of f in (26.1). Every finite degree- N common coordinate truncation is unchanged. The full hull is a quadrilateral if $\rho = 1$ and a triangle with B strictly interior if $\rho = -1$. In both cases every three of the four full points are affinely independent. Thus the full configuration is uniform, while its finite coordinate-truncation history remains arbitrary.

Proof. The monomial τ is positive and smaller than t^m for every ordinary integer $m \geq 0$. Its exponent position is beyond every finite degree cut, so those cuts ignore it. The exact full determinant becomes

$$1 - gf^{(\rho)} = 1 - gf + \rho g\tau = \rho g\tau.$$

It has sign ρ . The remaining side test is $2g - 2 + \rho g\tau$, whose leading term is $2t > 0$. The other two oriented determinants are $2g > 0$ and $2 > 0$. Apply Lemma 26.1. The four determinants in (26.2) are all nonzero and enumerate all triples up to orientation, proving uniformity. $\square$

The theorem uses transfinite scale separation only to choose the endpoint. The arbitrary finite-degree history itself already uses ordinary convergent power series. For the simple alternating history, one may retain the rational $f = (1 + t)^{-1}$, $g = 1 + t$ of Theorem 26.2. Then the full uniform example lies in $\mathbb{Q}(t, \tau)^2$ coordinatewise. It is not necessary to use an arbitrary analytic series to obtain infinitely many changes with a uniform endpoint.

26.5 Where ordinary power-series stability returns

Proposition 26.7 (Finite determinacy for uniform ordinary series). *Let $V(t)$ be a configuration of $n \geq d + 1$ points with coordinates in $\mathbb{R}[[t]]$. Suppose every $(d + 1)$ -point affine determinant is nonzero. Let M be the maximum of the finite orders of these determinants. For every $N \geq M$, the common degree- N coordinate truncation has the same signs of all these determinants, and hence the same oriented matroid and face lattice, as the full configuration.*

Proof. All coordinate entries have nonnegative order. Truncating them at degree N changes each entry by a series of order at least $N + 1$. Expand a homogeneous affine determinant by multilinearity in its columns, including the constant homogeneous row. In every term of the difference between the full determinant and the truncated determinant, at least one entry is a truncation error and all remaining entries have nonnegative order. The determinant error therefore has order at least $N + 1$.

Each original nonzero determinant has order at most $M \leq N$, so its first nonzero coefficient survives unchanged. The sign pattern of the maximal homogeneous minors determines the oriented matroid of a uniform configuration and its supporting facets. In this uniform case one can also recover facets directly: a d -point candidate facet is supporting exactly when all determinants obtained by appending another point have one weak orientation, which here is strict. Hence the face lattice is unchanged. $\square$

The two hypotheses explain the boundary sharply. In the rational alternating example, a full affine determinant is identically zero, so it has no finite leading order to protect. In

the transfinite uniform example, the previously zero determinant has order ω relative to the monomials $\omega^{-\gamma}$; no ordinary N reaches that order. Thus ordinary uniformity plus ordinary power-series support is a genuine stability regime, while full surreal uniformity alone is not.

26.6 A second truncation convention that must be excluded

Even with a fixed real base, deleting the *leading* part and retaining a tail is not covered by the lifting theorem. For example, take a square and place a height $u(t) = \sum_{k \geq 0} (-t)^k$ at one vertex, with the other three heights zero. The unique circuit evaluates to u up to a fixed nonzero real factor. The tail starting at degree N has leading sign $(-1)^N$, so the induced diagonal alternates. This is a tail history, not an initial-prefix history. The distinction is mathematical, not terminological.

27 Algorithms, verification, and proof boundaries

[write] Manuscript 02’s finite certificate workflow and its query-model bound [proposition 21.2](#) are printed in [section 21](#), and its proposed Lean development in [section 83](#). The program is shipped as `code/02-transfinite-scales-verify.py`; its outputs are shipped as `data/02-transfinite-scales-verification.json`, `data/02-transfinite-scales-hexagon_example.json` and `data/02-transfinite-scales-universal_history_example.json`, and the captured run as `data/02-transfinite-scales-verification_run.txt`.

27.1 Recorded exact-arithmetic tests

The accompanying code `code/02-transfinite-scales-verify.py` uses Python’s standard-library `fractions.Fraction`. It requires no external Python package or network access. The recorded run passed **163,914 exact assertions**. These assertions cover the following finite cases.

Table 4: Scope of the supplied finite verification.

Family	What was checked
Sharp polygons	All nine sizes $4 \leq n \leq 12$: selected quotient rank, all prefix cell sets, refinement, and all basis-slack signs after the explicit rational specialization.
Redundant-row examples	Thirty seeded arrays of thirteen rational rows, including affine components and redundant coefficient directions: original and compressed prefix signs, cells, and real specialization.
Universal histories	All $2^7 = 128$ sign histories of length eight with first bit positive; twelve seeded histories of length thirty-two; one additional length-twelve example. Reciprocal jets, leading defect coefficients, and side-orientation signs were checked exactly.
Rational alternation	The full polynomial identity (26.3) for each $1 \leq N \leq 64$.

The count measures individual assertions, not independent theorems or statistical confidence. In particular, none of these tests proves the quantification over all real functionals, arbitrary infinite sign sequences, transfinite supports, or all dimensions. Those claims rest on the written proofs. The deterministic seeds are 20260930 for the compression tests and 31415926 for the longer history samples. The machine-readable output is

data/02-transfinite-scales-verification.json; the hexagon and a prescribed-history coefficient example have separate JSON files.

The program uses a coefficient recursion equivalent to the closed construction (26.7); Appendix 30 derives it. This supplies a second algebraic view of the main negative example. There is no numerical approximation to ω in the program.

27.2 A proof dependency audit

The delicate assumptions are summarized in Table 5. These are boundaries of the results, not optional technicalities.

Table 5: Hypotheses whose omission would change or invalidate the claim.

Result	Essential boundary
Lifting compression	The base coordinates and comparison coefficients are real and fixed. Moving coordinates lead to nonlinear determinant expressions and are not covered.
History preservation	Cuts retain common initial normal-form segments. Independent coordinate cuts, sign-sequence cuts, and tail deletion are different operations.
Rank minimality	All real affine-dependence signs, including zero, are preserved. A finite circuit list or one final subdivision may require less information.
Real specialization	The explicit bound protects a finite family of functionals. It does not embed the entire higher-rank ordered comparison space into $\mathbb{R}$.
Refinement chains	Marked cell-label inclusion is used. Geometric containment alone can conceal whether an interior point is on the lower hull.
Universal histories	The degree-one bit is forced positive by the normalization. The freely prescribed history starts at degree two. There are four generators, not necessarily four extreme vertices.
Uniform endpoint	An added monomial beyond all ordinary powers chooses the endpoint. This is not an ordinary analytic correction with a finite Taylor order.
Ordinary stability	All relevant maximal determinants are nonzero and have finite orders, and coordinates have nonnegative orders.
Standard-part geometry	Coordinates are finite. The polar identity uses only finite polar points; a face image need not be a face.
Puiseux realization	Only finitely many polynomial observations and incidences are preserved. Curve selection and convergent algebraic Puiseux expansions are classical imported theorems.
Computations	Exact finite arithmetic tests selected examples. No Lean build, general surreal algorithm, or independent proof certification was performed.

28 Further research questions

The following are proposed research directions arising from this article. They are not assertions that each topic is absent from the literature, nor are they labeled as longstanding published open problems. The first three would most directly deepen the results proved here.

28.1 Which histories admit rational-function coordinates?

The universal construction permits arbitrary convergent reciprocal power series, while rational functions already give a periodic alternating history. Characterize the binary histories realizable by four points of the form (26.1) with $f, g \in \mathbb{Q}(t)$ and $fg = 1$. In the normalized reciprocal setting, the degree- $N + 1$ coefficient of the degree- N truncation defect equals that of $f + g$. When nonzero it determines the history bit; a zero requires looking at higher degrees. For rational functions, the coefficients of $f + g$ satisfy a linear recurrence after finitely many indices. The problem therefore links geometric truncation histories to sign patterns of rational generating functions, with the additional constraint that the generating function is a reciprocal sum. A useful first target is a classification of eventually periodic histories and a precise description of the extra reciprocal constraint.

28.2 Stability in the presence of exact incidences

Proposition 26.7 treats configurations whose maximal determinants are all nonzero. The universal examples exploit a zero full determinant whose truncations have arbitrary leading signs. Find a necessary and sufficient condition, in a specified algebraic or analytic representation class, for the *zero* determinant relations to have eventually stable truncation signs. A condition merely on the final oriented matroid cannot suffice. Possible additional data include how an incidence ideal interacts with the coordinate truncation maps, and bounds on cancellation in products of truncated coordinates.

28.3 Effective controlled specialization

Theorem 19.1 is qualitative. Given rational polynomial incidence data, a compatible algebraic standard-part configuration, and a realizable sign pattern, obtain explicit bounds on a Puiseux arc's ramification, algebraic degree, and coefficient height. An especially useful version would preserve selected vanishing orders or a finite sequence of near-face optimality bands in addition to the final face lattice. Such a theorem must distinguish geometric sign complexity from the bit complexity of an exact coefficient representation. [write] Manuscript 05's question 1 (section 22) asks the same for determinant signs; theorem 41.2 of Part III gives an effective specialization for finite-support Hahn certificates. [[write] Part VIII gives a lower-bound family for rational arcs (theorem 120.3); no upper bound of the kind asked is proved.]

28.4 Minimal geometric certificates versus full comparison rank

The rank p classifies how many coefficient layers are needed to preserve all real affine comparisons, whereas the strict history length r can be smaller. Determine an efficient canonical quotient of the comparison data that preserves just the marked subdivision history. How small can a slack or circuit certificate be compared with the all-basis list? What is the complexity of extracting it for a fixed dimension, and which families admit polynomial-size certificates without enumerating every candidate basis? A single final subdivision cannot determine p ; any proposed invariant must respect that information loss. [write]

Manuscript 05’s question 4 ([section 22](#)) asks the same for face lattices, lower subdivisions and toric initial ideals together.

28.5 A replacement for the false residue map on faces

The image of a face under standard part need not be a face. Develop a polyhedral object whose cells retain the fibers, infinitesimal widths, and near-optimality data discarded by ordinary residue. The identity in Proposition [14.2](#) suggests indexing such data by objective functionals and admissible infinitesimal error levels. The challenge is to obtain an intrinsic finite description when one exists, and to state precisely when transfinite scale labels are unavoidable. [write] Related: the questions of [Part III](#) on face specialization and oriented scale data and on joint residue data ask for structures of the same kind; [proposition 14.2](#) is the single-objective case.

28.6 Beyond linear comparisons: realizability constraints

One can apply linear compression to a finite list of determinant values regarded as unrelated surreal scalars. That does *not* show that the compressed values are minors of an actual point configuration: Plücker relations and other polynomial dependencies may fail. Determine which rank-compression operations preserve realizability of valued or oriented matroid data. A useful theorem would identify a nontrivial class in which support compression can be lifted back to coordinate matrices without changing prescribed signs and selected valuation comparisons. [write] Manuscript 05’s question 2 ([section 22](#)) asks this for two nonreal lifting coordinates over a fixed real base.

28.7 Infinite locally finite configurations

The proof of Theorem [7.1](#) depends on finite dimension of the coefficient quotient, not on a bound for support length. For infinite configurations the affine relation space is generally infinite dimensional. Which local finiteness, support, or uniform-rank assumptions imply local stabilization of coherent subdivisions? A satisfactory formulation should specify whether stabilization is pointwise on finite subconfigurations, uniform on bounded regions, or global. The four-point examples show that coordinatewise convergence alone is too weak even before infinitude is introduced.

28.8 Formalization and representation-aware computation

Carry out the layered Lean program in [section 83.2](#) and attach a checked theorem ledger. In parallel, classify effective representations for which the next rank-increasing coefficient can be found or its absence certified. Finite support with a known bound, rational functions, algebraic series, and general coefficient oracles have materially different capabilities. Proposition [21.2](#) shows that an implementation cannot hide those differences behind a single unrestricted “surreal input” interface. [write] Manuscript 02’s layered Lean program is printed in [section 83](#); manuscript 05’s question 5 ([section 22](#)) asks for such representations from the side of [proposition 21.1](#).

29 Conclusion

Finite surreal geometry has two distinct levels. At the level of a single finite incidence pattern, real-closed-field transfer gives an ordinary real algebraic model. At the level of a fixed real base with surreal lifting heights, finite-dimensional linear algebra still controls all normal-form prefix histories: at most $n - d - 1$ effective comparison layers are needed, and that bound is sharp.

Neither fact controls arbitrary coordinate truncation. Four bounded planar generators can encode any binary triangle/quadrilateral history from degree two onward, and a transfinite monomial can independently choose a uniform final endpoint. Thus the relevant additional structure of surreal polytopes is not a previously impossible finite face lattice. It is the organization of incidence data across scales, together with the precise operations used to expose or forget those scales.

30 An exact recursion for the universal construction

[write] Appendix A of manuscript 02.

The closed formula (26.7) is convenient for proving convergence. For exact finite calculations, the following recursion avoids formal square-root software. Write

$$g = \sum_{k \geq 0} a_k t^k, \quad f = \sum_{k \geq 0} b_k t^k, \quad a_0 = b_0 = 1, \quad a_1 = 1, \quad b_1 = -1.$$

For $n \geq 2$, assume the previous coefficients have been chosen and set

$$\begin{aligned} c_n &= - \sum_{k=1}^{n-1} a_k b_{n-k}, \\ M_n &= \sum_{k=2}^{n-1} a_k b_{n+1-k}, \\ a_n &= \frac{\sigma_n + c_n + M_n}{2}, \quad b_n = c_n - a_n. \end{aligned} \tag{30.1}$$

The equality $a_n + b_n = c_n$ forces the degree- n coefficient of fg to vanish. The coefficient of degree $n + 1$ in the product of the two degree- n truncations is

$$a_1 b_n + a_n b_1 + M_n = b_n - a_n + M_n = c_n - 2a_n + M_n = -\sigma_n.$$

Thus the leading truncation defect is σ_n . At the next step the condition of being reciprocal forces $a_{n+1} + b_{n+1} = \sigma_n$. Together with $a_2 + b_2 = 1 = \sigma_1$, this reproduces $f + g = 2 + t^2 H$.

Conversely, the identities $fg = 1$, $f + g = 2 + t^2 H$, and the linear terms $g = 1 + t + O(t^2)$, $f = 1 - t + O(t^2)$ select exactly the branch in (26.7). Hence the recursion computes that same pair. Every requested sign prefix determines the corresponding finite jets; an arbitrary continuation of the signs completes them to the convergent series of Theorem 26.3.

For example, $\sigma_2 = 1$ gives $a_2 = 1$, $b_2 = 0$, whereas $\sigma_2 = -1$ gives $a_2 = 0$, $b_2 = 1$. These yield, respectively, positive and negative leading degree-three defects. Later sign choices are independent and are obtained by the same rational recursion.

31 Reproducibility and source inventory

[write] Appendix B of manuscript 02; in this report its files are shipped under prefixed names. `code/verify.py` is `code/02-transfinite-scales-verify.py`; the three generated JSON files are `data/02-transfinite-scales-verification.json`, `data/02-transfinite-scales-hexagon_example.json` and `data/02-transfinite-scales-universal_history_example.json`; the Makefile is `code/02-transfinite-scales-Makefile`; the source and proof-audit notes are `02-transfinite-scales-sources.md` and `02-transfinite-scales-proof_audit.md`; the build record notes/`validation.json` is `data/02-transfinite-scales-validation.json` and describes the delivered PDF. The manuscript’s `article.tex`, PDF and README are not shipped: this report replaces them. The commands below name the delivered layout; run them on a copy (the report’s README gives the recipe).

The archive contains `article.tex`, its compiled PDF, `code/verify.py`, the three generated JSON data files, a Makefile, a README, and separate source and proof-audit notes. From the archive directory, run

```
python3 code/verify.py
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The verification script resolves its output paths relative to itself. The PDF build uses ordinary TeX packages and no shell escape, external bibliography processor, or network call. The Makefile performs an extra TeX pass for stable cross-references. No font files are distributed.

The repository geometry file inspected at the pinned commit has blob identifier

```
450fd4bbfc017e0da24c28e538af1cc02037ad5a.
```

The code-search index observed during the source check pointed to parent commit

```
5e0f4e05046bacda3c6ba2643f19b91aa632436b.
```

The actual geometry source was then read at the full pinned commit. This distinction is recorded to avoid presenting search-index freshness as a repository-wide proof audit. No original repository files were modified.

Part III

Affine residues and mixed-volume scales

Source and scope. [write] This Part prints, whole, manuscript 06 of batch 58: *Surreal Polytopes: Affine Residues, Mixed-Volume Scales, and Simultaneous Segment Models*, headed “ProveIt research program”, author line “Research article prepared for Vladimir Reshetnikov”, dated 30 September 2026 (archive `Surreal_Polytopes_Research_Package.zip`, main file `Surreal_Polytopes/article.tex`; delivered PDF of 27 pages, not shipped). It was written against ProveIt commit 6996fee43. Its program, recorded run, build file and provenance note are shipped as `code/06-affine-residues-verify.py`, `data/06-affine-residues-verification.json`, `code/06-affine-residues-Makefile` and `06-affine-residues-PROVENANCE.md`. Its README and checksum list SHA256SUMS are not shipped; they survive in the delivery archive of arrival commit b2b626d85. Every section,

result, proof, example, remark, figure, table and research question of the manuscript is printed here, in the manuscript's order, with three structural changes: its appendix is printed as the ordinary [section 45](#); its subsection "A modular Lean development" is printed in the formalization chapter ([section 83.3](#)); and its bibliography is printed as the list "References of Part III". This Part is its own spine (the residue/volume spine of [section 1](#)); it shares no theorem with the other Parts beyond two foundations, both marked where they occur: the standard part of a finite hull ([lemma 34.2](#), proved in Part I as [proposition 13.1](#)) and finite real-closed-field transfer ([proposition 33.3](#), Part I's [theorem 18.1](#)). The Part keeps its own definitions and hypotheses, fixed in [sections 32](#) and [33](#). In this Part "the article" and "the package" mean manuscript 06 and its delivery. Where manuscript 06 says "the report", it means itself or, in [section 42](#), its recorded run. It never means this merged report.

Generality. Although manuscript 06 and this report have "surreal" in their titles, every theorem of this Part is proved for an *arbitrary* set-sized real closed ordered field $K \supseteq \mathbb{R}$ with its natural convex valuation (the standing hypothesis stated at the end of the first subsection below). The surreal field enters only as the source of examples: finitely many surreal coordinates lie in such a K ([section 33](#)), and ω^{-1} , $\omega^{-\omega}$ occur only in examples and in the audit. The word "surreal" in the titles is therefore not a hypothesis of any result here.

Notation in this Part. [write] The following symbols mean something different elsewhere in the report ([section 1.3](#)); the Part-local meaning is the one in force here.

- $\mathcal{O}$ is the ring of elements of K bounded by real constants and $\mathfrak{m}$ its maximal ideal: the same ring as $\mathcal{O}_K$ (Parts I–II) and $\mathcal{O}_K$ (Part IV). $\text{st} : \mathcal{O} \rightarrow \mathbb{R}$ is the standard part, as everywhere in the report.
- $v : K \rightarrow \Gamma \cup \{\infty\}$, $\Gamma = K^\times / \mathcal{O}^\times$, is ordered so that *larger values mean smaller elements*: $v(x) \geq v(y)$ iff $x/y \in \mathcal{O}$, and $v(x) > 0$ iff $x \in \mathfrak{m} \setminus \{0\}$. The value group need not embed in $\mathbb{R}$. (Part I's leading exponent measures growth, in the opposite direction; Part V writes v and v_* .)
- The *affine residue* $\text{Res}(P)$ ([definition 35.2](#)) is a renormalised, dimension-preserving real polytope, the image of P in $(p_0 + L(P))/\mathfrak{m}L(P)$. *It is not the standard part* $\text{st}P$: the thin triangle of [section 40](#) has $\text{st}P$ a segment but $\text{Res}(P)$ a triangle. Where this Part says "residue field", "reduction" or "volume residue" ([proposition 34.3](#)), it means the standard part. The report reserves "affine residue" for Res .
- $L(P) = \text{span}_{\mathcal{O}}(P - P)$ is the *scale lattice*, a free $\mathcal{O}$ -module, not a discrete subgroup; it is unrelated to Part I's L_h and L_A and to Part IV's lattice-point count $L_P(n)$.
- $V(P_1, \dots, P_d)$ (printed V) is mixed volume, not Part I's dependence space V or Part V's coarse ring. Vol_d is determinant (triangulation) volume; Part IV writes vol_d for Euclidean volume of real polytopes.
- m is the number of bodies (not Part I's comparison rank, not Part V's m with $n = 2^m$); r, r_i are affine dimensions (not Part II's history length or Part I's shadow rank r_0); $q(B) = v \det W_B$ in [lemma 38.2](#) (not $q = n - d - 1$); H is the grid bound $\binom{m-1}{d-1}$ in [theorem 37.2](#) and a finitely generated group in [lemma 41.1](#); S_i are inner segments and T_i probe segments (not Part IV's trace S and implanted polytope T , not Part V's slack matrix and triangle); ρ_g is the sign threshold of a specialized Laurent polynomial in [section 41](#) (Part I's threshold is η); s is the real specialization parameter and $t^\vee$ a formal Hahn monomial there (Part II's t is a power-series variable evaluated at $t = \omega^{-1}$).
- $\tau_I, \tau(\alpha)$ and "the profile τ " are the pure and full mixed-volume valuation profiles; they

are *not* powers of the surreal $\tau = \omega^{-\omega}$ of the other Parts. Manuscript 06 writes η for $\omega^{-\omega}$ (in its examples, in [section 41](#) and in its audit), and this Part keeps that letter so that τ means only a profile here; $\varepsilon = \omega^{-1}$ as in the source. Part I's η is a different object (a sign threshold).

- Macro change only: manuscript 06's `\Span` (plain span over the ring or field in context) is typeset with the report's `\spanplain`; the printed output is unchanged.

Abstract of manuscript 06. Finite polytopes over the surreal numbers have no new finite face-lattice possibilities merely because the coordinates are surreal. Their genuinely non-Archimedean information lies instead in relative scales, cancellation, and degeneration. We develop an algebraic treatment of that information inside set-sized real closed subfields of the surreal field.

The principal result is a simultaneous inner-segment theorem: for every finite family of polytopes, a segment can be chosen inside each member so that all mixed-volume valuations involving distinct members are preserved at once. Consequently the pure mixed-volume valuation configurations are exactly the determinant-valuation configurations realizable over the given ordered field. A finite rational grid suffices for the segment choices. Applying the theorem to labelled copies of the bodies produces a representable valuated-matroid model of the entire mixed-volume profile, and proves its exchange-convexity for arbitrary ordered value groups.

The underlying invariant is the translation-invariant module $L(P) = \text{span}_{\mathcal{O}}(P - P)$, where $\mathcal{O}$ is the ring of finite elements. A maximum vertex simplex gives an explicit basis of this module and a full-dimensional real affine residue, even when ordinary standard part collapses the original polytope. Mixed-volume valuations depend only on these modules and admit certificates using at most d^d determinants per coefficient after basis selection. Conversely, a full-dimensional module is determined by mixed-volume probes, with a $d + 1$ -measurement comparison test relative to a reference polytope.

Detailed proofs, higher-rank examples, finite-support specialization, exact computational audits, and a research agenda are provided. The classical ingredients and the unverified priority of the combined results are identified explicitly; no solution of the full real mixed-volume configuration problem or new Lean formalization is claimed.

Scope (title page of manuscript 06). Finite vertex sets; fixed finite dimension; the natural convex valuation; exact algebraic volume. No proper-class integration, Archimedean valuation hypothesis, or sequential compactness argument is used.

32 Research target, principal results, and status

32.1 Why the finite combinatorics is not the whole question

Replacing real coordinates by surreal coordinates does not, by itself, create new finite combinatorial types of polytopes. The incidence data of a fixed finite vertex configuration can be encoded in the first-order theory of real closed fields. That observation is useful, but it does not explain a triangle of height ω^{-1} , a determinant whose first nonzero term occurs at $\omega^{-\omega}$, or the mixed volume of two bodies thin in different directions.

The appropriate questions are more precise. Can one normalize a finite surreal polytope without destroying its affine dimension? Which finite algebraic data determine the scale of a mixed volume? Can all the scales of a family be realized simultaneously by much simpler

objects? Can the resulting scale information be recovered from measurements? The article answers these questions under explicit hypotheses.

Throughout the main results, K is a set-sized real closed ordered field containing $\mathbb{R}$, $\mathcal{O}$ is its ring of elements bounded by real constants, and $v : K \rightarrow \Gamma \cup \{\infty\}$ is the corresponding valuation. The surreal application takes K inside **No**. Our convention is that larger valuations mean smaller quantities. The value group Γ need not embed in $\mathbb{R}$.

32.2 The principal theorem in a compact form

For nonempty finite polytopes $P_1, \dots, P_m \subset K^d$, put

$$\tau_I = v(V(P_{i_1}, \dots, P_{i_d})), \quad I = \{i_1 < \dots < i_d\} \subseteq [m].$$

We call this the *pure* profile: every body occurs at most once in an entry. The mixed-volume convention is $V(P, \dots, P) = \text{Vol}_d(P)$.

Theorem 32.1 (Simultaneous inner-segment model, preview). *For $m \geq d \geq 1$, there are segments $S_i \subseteq P_i$, possibly reduced to points, such that*

$$v(V(S_{i_1}, \dots, S_{i_d})) = \tau_I \quad \text{for every } I \in \binom{[m]}{d}.$$

The segments may be chosen with endpoints that are rational convex combinations of vertices of their respective polytopes.

The proof in Section 37 does more than choose a generic point informally. After selecting at most d vertex directions in each body, it gives a finite rational grid containing a successful simultaneous choice. If $S_i = [p_i, p_i + w_i]$, the segment formula becomes

$$\tau_I = v \det(w_{i_1}, \dots, w_{i_d}),$$

with $v(0) = \infty$. Thus every pure profile has a single matrix realization, not merely a different determinant witness for each entry.

Repetitions matter. Replacing a full-dimensional body by one segment cannot preserve its ordinary volume in dimension greater than one. For the *full* profile, which includes repetitions, we use either parallelotopes with at most d generators or d separately labelled segment copies of each body. The clone construction retains all repeated-body valuations and implies a higher-rank discrete exchange inequality (Theorem 38.4).

32.3 What the other theorems add

For a polytope P , define its *scale lattice*

$$L(P) = \text{span}_{\mathcal{O}}(P - P).$$

The word lattice here means a finitely generated free $\mathcal{O}$ -module in a finite-dimensional K -space; it does not mean a discrete subgroup in the Euclidean sense. The following results describe the geometry behind the segment theorem.

Result	Content and location
Affine normalization	A maximum vertex simplex generates $L(P)$; normalization lies between a standard simplex and a cube. The residue preserves affine dimension (Section 35).
Mixed determinant formula	Every mixed-volume valuation is the minimum valuation of one-direction-per-body determinants; selected bases need at most d^d such determinants (Section 36).
Simultaneous simplification	Inner segments preserve all pure entries; parallelotopes preserve the full profile (Sections 36–37).
Discrete convexity	A clone realization gives the exchange axiom over any ordered value group, without a real-valued grading (Section 38).
Reconstruction	All segment probes characterize $L(P)$; a reference basis gives a finite $d + 1$ -measurement test (Section 39).
Finite certification	Finite-support higher-rank input admits a one-parameter model preserving specified finite certificates, but not the whole higher-rank order (Section 41).

32.4 Repository provenance and dependence

The inspected ProveIt snapshot is commit 6996fee43cc97b6c16351def7507d59a95bf62f0, retrieved on 30 September 2026. The relevant development is Algebra/SurrealNumbers; its README records the constructed surreal ordered field, real-closedness, and a statement-level formalization ledger [17]. The report also draws a methodological connection to the finite support certificates in *Reversion Beyond Archimedean Valuations* under Analysis/Transseries [18].

The inspection was targeted, not a complete audit of the repository. Search results were used for navigation; the principal source files were read at the immutable commit above. In particular, this article does not assume the correctness of every research claim in the repository. Its mathematical prerequisites are the classical real-closedness of the surreal field, elementary valuation algebra, and finite real polytope theory. The proofs below are independent of the repository’s unrefereed advanced analytic claims.

[[write] Re-verified at the write phase. The Transseries report cited here is at Analysis/Transseries/docs/series-and-transseries/Reversion_Beyond_Archimedean_Valuations/, with title and subtitle exactly as in the bibliography entry [18]; its finite positive-grading result, credited below as the antecedent of lemma 41.1, is the theorem “Finite supports admit a positive rational grading” there. The SurrealNumbers README still describes the constructed ordered field, its real-closedness and the formalization ledger. Nothing in this Part is formalized in Lean (see section 83.3).]

32.5 Literature boundary and novelty

Ordered-field polyhedral geometry is established mathematics; a modern formal treatment of face lattices over ordered fields is given by Allamigeon, Katz, and Strub [21]. Classical mixed-volume theory, including its determinant special case, is treated by Schneider [22]. Averkov, von Dichter, Richard, and Soprunov connect mixed volumes of zonoids to absolute Grassmannian coordinates and explicitly propose investigating a connection with tropical Grassmannians in their November 2024 version [23]. Our simultaneous segment theorem gives a concrete valued-field connection for *all finite polytopes*, with a precise realization field and no closure convention hidden in the claim.

The broader link between volume polynomials, Lorentzian polynomials, and M -convex tropicalizations is already part of the theory of Brändén and Huh [24]. Lattices, norms, buildings, and realizable valuated matroids also have an extensive theory; see, for example, Battistella, Kühn, Kuhrs, Ulirsch, and Vargas [25]. We do not claim to have invented these connections. The contribution developed here is the explicit package of simultaneous inner models, dimension-only determinant certificates, affine residues, reconstruction, and arbitrary-value-group proofs adapted to finite surreal geometry.

Remark 32.2 (Status of the claims). Every numbered theorem presented as a result is proved below, either from identified classical facts or by an explicit algebraic argument. Their correctness and the extent of their novelty have not been independently refereed, and an exhaustive priority search has not been completed. The computational supplement checks finite instances; it is not a Lean proof. No solution of the complete real mixed-volume configuration problem is claimed. These qualifications concern verification and priority, not a missing step being concealed in the stated theorems.

33 Set-sized foundations and finite geometry

33.1 Working inside the surreal field without a class-sized domain

The surreal numbers form a real closed ordered class field; standard references discuss this field and its universality properties [19]. For finitely many coordinate values, start with the set-sized field generated by those values and $\mathbb{R}$, then take its real closure inside $\mathbf{No}$. This yields a suitable set-sized K containing all input. The elementary algebraic constructions used here remain in such a field.

Consequently a polytope in the article is the ordinary set

$$P = \text{conv}_K\{p_1, \dots, p_n\} = \left\{ \sum_{i=1}^n \lambda_i p_i : \lambda_i \in K_{\geq 0}, \sum_i \lambda_i = 1 \right\} \subset K^d.$$

This is not an assertion that the collection of *all* surreal points in the corresponding class convex hull is a set. Finite face data and volumes are compatible with enlarging K : their definitions use finite linear algebra and first-order formulas over the coefficient field.

33.2 Finite elements, residue, and order of magnitude

Define

$$\begin{aligned} \mathcal{O} &= \{x \in K : |x| \leq C \text{ for some } C \in \mathbb{R}_{>0}\}, \\ \mathfrak{m} &= \{x \in K : |x| < c \text{ for every } c \in \mathbb{R}_{>0}\}. \end{aligned}$$

The ring $\mathcal{O}$ is convex in K . For $x \neq 0$, either x is finite or $1/x$ is finite, so it is a valuation ring. Its nonunits are precisely $\mathfrak{m}$: a finite nonzero x is a unit exactly when $|x|$ is bounded below by a positive real constant.

Lemma 33.1 (Standard part). *For each $x \in \mathcal{O}$ there is a unique $r \in \mathbb{R}$ such that $x - r \in \mathfrak{m}$. The resulting map $\text{st} : \mathcal{O} \rightarrow \mathbb{R}$ is an order-preserving surjective ring homomorphism with kernel $\mathfrak{m}$.*

Proof. The real set $\{a \in \mathbb{R} : a \leq x\}$ is nonempty and bounded above. Let r be its supremum. For every positive real c , the definition of supremum gives $r - c < x < r + c$, allowing harmless

weak inequalities first and then replacing c by $c/2$. Thus $x - r$ is infinitesimal. A nonzero real number cannot be infinitesimal, proving uniqueness.

Finite sums of infinitesimals are infinitesimal, and the product of a finite element with an infinitesimal is infinitesimal. Expanding $(r + \epsilon)(s + \delta)$ proves preservation of multiplication; addition is immediate. Constants give surjectivity, the kernel description follows from uniqueness, and positivity follows from convexity. $\square$

Write $\Gamma = K^\times / \mathcal{O}^\times$ additively and $v(x)$ for the class of $x \neq 0$. Order it so that

$$v(x) \geq v(y) \iff x/y \in \mathcal{O}.$$

Set $v(0) = \infty$, with $\gamma + \infty = \infty$. Positive real constants have value zero. A nonzero element belongs to $\mathfrak{m}$ exactly when its value is positive.

Lemma 33.2 (Finite positive sums and bounded comparison). *If $a_1, \dots, a_N \in K_{\geq 0}$, then*

$$v\left(\sum_{i=1}^N a_i\right) = \min_i v(a_i).$$

If $0 < c \leq C$ are real and $0 \leq a, b \in K$ satisfy $cb \leq a \leq Cb$, then $v(a) = v(b)$, including the case $a = b = 0$.

Proof. When some a_i is nonzero, choose a largest member M . The inequalities $M \leq \sum_i a_i \leq NM$ show that the quotient of the sum by M is a unit of $\mathcal{O}$. The same unit argument proves the second assertion. The all-zero case is immediate. $\square$

No assertion about infinite sums is involved. The absence of cancellation for a finite positive sum will replace all appeals to a real-valued valuation or an Archimedean growth parameter.

33.3 The precise finite transfer principle

Real closed fields satisfy quantifier elimination and have a complete first-order theory in the ordered-field language; semialgebraic geometry provides the usual framework [20]. In particular, $\mathbb{R} \subset K$ is an elementary extension. Only *finite-dimensional, finitely presented* instances are transferred here.

Proposition 33.3 (No new finite face types). *Every finite polytope over K has the face lattice of a polytope whose coordinates are real algebraic numbers. Any prescribed finite collection of rational-polynomial equalities, inequalities, and sign tests on the vertex coordinates can be preserved simultaneously when it is included in the realization formula.*

Proof. Fix the number n of labelled input points and their affine dimension. For a nonempty subset $S \subseteq [n]$, the assertion that exactly the labels in S lie on a supporting face is expressed by

$$\exists a \in K^d \exists c \in K : \quad a \cdot p_i = c \ (i \in S), \quad a \cdot p_i < c \ (i \notin S).$$

The full polytope corresponds to $S = [n]$; the empty face is added separately. Form the conjunction of this assertion or its negation for each of the finitely many subsets S . Include a rank formula for the affine dimension and the requested additional tests. Existentially quantify the vertex coordinates. The resulting sentence has rational coefficients and holds in K , hence in the real algebraic field, which is also real closed. Its realization has exactly the same labelled face sets and therefore the same inclusion lattice. $\square$

This does not preserve the natural valuation, infinitesimal relations to *every* real constant, or a whole infinite type. Nor does it assert a rational realization. The face-lattice result is the classical boundary condition for the rest of the article, rather than its proposed novel contribution.

[[write] This is the finite real-closed-field transfer that Part I states and proves for finite configurations as [theorem 18.1](#) (manuscripts 05 and 02). Manuscript 06's statement and proof are kept as printed because they are phrased for face lattices together with prescribed sign tests. The same principle appears again as Part IV's [proposition 48.1](#) and, strengthened to oriented matroids and extension complexity, as Part V's [theorem 74.3](#). The limitations in the preceding paragraph apply to every one of these statements.]

34 Algebraic volume and reduction to the residue field

34.1 Determinant volume, not surreal measure theory

For an affine d -simplex set

$$\text{Vol}_d \text{conv}(p_0, \dots, p_d) = \frac{1}{d!} |\det(p_1 - p_0, \dots, p_d - p_0)|.$$

A full-dimensional finite polytope admits a finite triangulation; summing these simplex volumes defines $\text{Vol}_d(P)$. Set the ambient d -volume to zero for a polytope of smaller affine dimension. This construction is independent of triangulation and has the ordinary finite additivity, monotonicity, translation invariance, and determinant covariance.

Here is a careful reason classical facts apply. For fixed vertex count and dimension there are finitely many candidate simplicial complexes on the input vertices. The assertion that a given complex triangulates the convex hull is a first-order statement: containment, covering, and intersection conditions are expressed by barycentric coordinates, with finitely many cells. The candidate volume is a finite sum of absolute determinants. Existence of a triangulation and equality of the sums from any two candidates are true over $\mathbb{R}$, so they transfer to K . Equivalently, the graph of the fixed-vertex-count volume function is semialgebraic.

This argument is specific to *polytopes*. It does not claim that volumes of arbitrary semialgebraic sets, or arbitrary analytic integrals, are ordered-field-definable.

34.2 Mixed volume and its finite identities

Define mixed volume by polarization:

$$d! \, V(P_1, \dots, P_d) = \sum_{I \subseteq [d]} (-1)^{d-|I|} \text{Vol}_d \left(\sum_{i \in I} P_i \right), \quad (34.1)$$

where the empty Minkowski sum is $\{0\}$. For $d \geq 1$ its volume is zero. The classical real theory [22] and the preceding finite transfer argument give symmetry, Minkowski multilinearity, homogeneity for nonnegative scalars, independent translation invariance, nonnegativity, and monotonicity in every argument. They also give

$$d! \, V([0, u_1], \dots, [0, u_d]) = |\det(u_1, \dots, u_d)|. \quad (34.2)$$

For a matrix $A \in \text{GL}_d(K)$,

$$V(AP_1, \dots, AP_d) = |\det A| \, V(P_1, \dots, P_d). \quad (34.3)$$

For m bodies and a multi-index $\alpha \in \mathbb{N}^m$ of total degree d , use $P_i[\alpha_i]$ to denote α_i repeated arguments. The volume polynomial identity is

$$\text{Vol}_d \left(\sum_{i=1}^m t_i P_i \right) = \sum_{|\alpha|=d} \frac{d!}{\alpha_1! \cdots \alpha_m!} V(P_1[\alpha_1], \dots, P_m[\alpha_m]) t^\alpha \quad (t_i \in K_{\geq 0}). \quad (34.4)$$

The formula is a polynomial identity after the mixed coefficients have been defined. All transfers have bounded, finite vertex counts: a Minkowski sum is presented by finitely many sums of input vertices.

Remark 34.1 (A computational warning). Formula (34.1) is an exact definition but can suffer severe leading-term cancellation. It is not a safe way to infer a valuation by taking the minimum of the valuations of its signed terms. The positive determinant comparison in Section 36 provides a cancellation-safe alternative.

34.3 Standard part commutes with finite polytope volume

For a polytope whose vertices lie in $\mathcal{O}^d$, write $\text{st } P$ for its coordinatewise standard-part image.

Lemma 34.2 (Reduction of convex hulls and sums). *If $p_1, \dots, p_n \in \mathcal{O}^d$, then*

$$\text{st conv}_K(p_1, \dots, p_n) = \text{conv}_{\mathbb{R}}(\text{st } p_1, \dots, \text{st } p_n).$$

For bounded finite polytopes, $\text{st}(P + Q) = \text{st } P + \text{st } Q$.

Proof. Every convex coefficient belongs to $[0, 1] \subset \mathcal{O}$. Reducing a convex combination gives a real convex combination. Conversely a real convex combination of the reduced vertices is the reduction of the combination with the same real coefficients upstairs. The assertion about sums follows from additivity of standard part and independent choices of summands. $\square$

[[write] The first identity, $\text{st conv} = \text{conv st}$ for a finite bounded hull, is Part I's [proposition 13.1](#), where it is proved (manuscripts 05 and 02); Part IV's [lemma 47.2](#) states it once more. The Minkowski-sum clause, the proof above, and the volume and mixed-volume clauses built on it in [proposition 34.3](#) are manuscript 06's own and are printed only here.]

Proposition 34.3 (Volume and mixed-volume residue). *For finite polytopes with vertices in $\mathcal{O}^d$,*

$$\text{st Vol}_d(P) = \text{Vol}_d(\text{st } P), \quad (34.5)$$

$$\text{st } V(P_1, \dots, P_d) = V(\text{st } P_1, \dots, \text{st } P_d). \quad (34.6)$$

Dimension loss in the residue is allowed.

Proof. For fixed n, d , let f denote the semialgebraic function taking an n -tuple of points to the volume of its convex hull. Over $\mathbb{R}$, this function is continuous, including at lower-dimensional configurations: vertex convergence gives Hausdorff convergence, and volume is continuous on real compact convex sets [22].

Let $a = (\text{st } p_1, \dots, \text{st } p_n)$. For every positive real e , real continuity supplies a positive real δ for which

$$\|x - a\|_\infty < \delta \implies |f(x) - f(a)| < e.$$

This is a first-order assertion, because the graph of f is semialgebraic. It remains true in K . The tuple p satisfies its hypothesis for every such real δ , so $f(p) - f(a)$ is infinitesimal. Both quantities are finite, since all vertices fit in a real bounded box. This proves (34.5). Apply it to every sum in (34.1) and use Lemma 34.2 to obtain (34.6). $\square$

[[write] [Part VII](#) gives two further proofs, by triangulation (source 10) and by thickening (source 09); see [remark 104.1](#).]

35 The scale lattice and canonical affine residue

35.1 A maximum vertex simplex is a module basis

Let P have affine dimension $r \geq 1$. Choose affine coordinates on its affine hull. Among the finitely many r -simplices formed by its input vertices, select one with largest absolute determinant, and label it $p_0, p_1, \dots, p_r$. Set

$$u_j = p_j - p_0, \quad B = (u_1, \dots, u_r).$$

A change of affine coordinates multiplies every determinant by the same nonzero factor, so maximality does not depend on that preliminary choice. The maximum exists by finite comparison; no compactness theorem is used.

Theorem 35.1 (Maximum-simplex normalization). *For the above choice,*

$$\Delta_r \subseteq \tilde{P} := B^{-1}(P - p_0) \subseteq [-1, 1]^r, \quad \Delta_r = \text{conv}(0, e_1, \dots, e_r). \quad (35.1)$$

Moreover

$$L(P) := \text{span}_O(P - P) = \sum_{i=1}^n O(p_i - p_0) = BO^r. \quad (35.2)$$

Thus $L(P)$ is free of rank r .

Proof. The first containment holds because the selected simplex is in P . For an input vertex p , Cramer's rule expresses the j th coordinate of $B^{-1}(p - p_0)$ as

$$\frac{\det(u_1, \dots, u_{j-1}, p - p_0, u_{j+1}, \dots, u_r)}{\det B}.$$

The numerator is the determinant of another vertex simplex, possibly degenerate. Its absolute value is at most $|\det B|$. Therefore every coordinate lies in $[-1, 1]$; convexity gives the outer containment for all of P .

The vertex differences consequently lie in BO^r . Every difference of two convex combinations is an O -linear combination of these vertex differences, so $L(P) \subseteq BO^r$. Conversely each column of B is in $P - P$, and hence $BO^r \subseteq L(P)$. The columns are K -independent, so they are also an O -basis. $\square$

The equality with $\text{span}_O(P - P)$ proves independence from the chosen vertex presentation, including redundant points. Notice that freeness has been proved directly. No Noetherian hypothesis on O , and no general structure theorem for finitely generated modules over valuation rings, is needed.

35.2 An intrinsic affine residue

The coset $p_0 + L(P)$ is independent of the chosen point $p_0 \in P$. Reduce this affine space modulo $\mathfrak{m}L(P)$. Its translation space $L(P)/\mathfrak{m}L(P)$ is an r -dimensional real vector space.

Definition 35.2 (Affine residue polytope). The *affine residue* $\text{Res}(P)$ is the image of P in the affine quotient

$$(p_0 + L(P))/\mathfrak{m}L(P).$$

The quotient is an affine $\mathbb{R}$ -space, with no preferred origin.

Theorem 35.3 (Full-dimensional and coordinate-independent residue). *The affine residue of an r -dimensional finite polytope is a real r -dimensional polytope. In maximum-simplex coordinates it is represented by $\text{st } \tilde{P}$ and contains Δ_r . Different maximum-simplex choices give real affinely isomorphic representatives. Any invertible K -affine change of coordinates induces an affine isomorphism of the residue polytopes.*

Proof. Equation (35.2) identifies the affine quotient with $\mathbb{R}^r$, and Lemma 34.2 identifies the image of P with the convex hull of its reduced normalized vertices. Equation (35.1) shows that this image contains a real r -simplex.

For another basis B' and origin p'_0 , the change of normalized coordinates has linear part $A = B^{-1}B'$ and translation $b = B^{-1}(p'_0 - p_0)$. Since $BO^r = B'O^r$, both A and A^{-1} have entries in O ; also $b \in O^r$. Reducing gives an invertible real affine map, because $\text{st } A \text{ st } A^{-1} = I$. Finally a K -linear isomorphism T maps $L(P)$ onto $L(TP)$ and $\text{m}L(P)$ onto $\text{m}L(TP)$. Translations behave compatibly with the affine cosets. $\square$

For a point, $L(P) = 0$ and the affine residue is a point; all assertions extend with $r = 0$. The construction preserves dimension, not the full face lattice. Several vertices may coalesce in the residue, and a small face can disappear.

35.3 A leading-volume formula

For a full-dimensional $P \subset K^d$, let $D = |\det B|$ for its selected maximum simplex.

Corollary 35.4 (Volume scale and residue coefficient). *One has*

$$\frac{D}{d!} \leq \text{Vol}_d(P) \leq 2^d D, \quad (35.3)$$

$$v \text{Vol}_d(P) = vD, \quad (35.4)$$

$$\text{st} \left(\frac{\text{Vol}_d(P)}{D} \right) = \text{Vol}_d(\text{st } \tilde{P}) > 0. \quad (35.5)$$

Proof. Apply monotonicity of volume to (35.1) and then use determinant covariance. Lemma 33.2 gives the valuation identity. For the last statement, $\text{Vol}_d(P)/D = \text{Vol}_d(\tilde{P})$, and Proposition 34.3 applies. $\square$

The positive coefficient in (35.5) is expressed relative to the chosen determinant scale D . A valuation alone has no canonical leading coefficient without such a normalization or a chosen cross-section of the value group. The residue polytope supplies geometric information beyond the valuation.

36 Parallelotope models and finite determinant certificates

36.1 Dimension-only inner and outer models

For a positive-dimensional polytope P_i of affine dimension r_i , choose maximum-simplex directions $u_{i1}, \dots, u_{ir_i}$ as in Theorem 35.1, and put

$$Z_i = \sum_{j=1}^{r_i} [0, u_{ij}].$$

This is an r_i -dimensional parallelotope in its linear span. We use the term also for such a lower-dimensional object embedded in K^d .

Lemma 36.1 (Parallelotope sandwich). *After translating the selected origin p_{i0} to zero,*

$$\frac{1}{r_i}Z_i \subseteq P_i - p_{i0} \subseteq 2Z_i - \sum_{j=1}^{r_i} u_{ij}. \quad (36.1)$$

Proof. In normalized coordinates, $[0, 1/r_i]^{r_i} \subseteq \Delta_{r_i}$, since its coordinates are nonnegative and sum to at most one. The other inclusion is $\widetilde{P}_i \subseteq [-1, 1]^{r_i} = 2[0, 1]^{r_i} - (1, \dots, 1)$. Apply the basis matrix. $\square$

Theorem 36.2 (Simultaneous full-profile parallelotope model). *For every multi-index $\alpha \in \mathbb{N}^m$ with $|\alpha| = d$, if every body occurring in that entry has positive affine dimension, then*

$$\left(\prod_{i:\alpha_i > 0} r_i^{-\alpha_i} \right) V(Z_1[\alpha_1], \dots, Z_m[\alpha_m]) \leq V(P_1[\alpha_1], \dots, P_m[\alpha_m]) \leq 2^d V(Z_1[\alpha_1], \dots, Z_m[\alpha_m]). \quad (36.2)$$

Consequently all entries have the same valuations for the two families. A point may be replaced by $Z_i = \{0\}$; any entry containing it is zero on both sides. Thus one choice of parallelotopes, each with at most d generators, preserves the entire mixed-volume valuation profile.

Proof. Apply monotonicity separately in all d argument slots of the mixed volume, using (36.1). Remove the independent translations and collect the positive scalar factors. The comparison constants are positive reals, so Lemma 33.2 gives equality of valuations, also when the compared mixed volumes vanish. An argument that is a point can be translated to zero, and multilinearity or polarization makes its mixed volume zero. $\square$

The same models work for every entry, including entries in which a body is repeated. There is no claim that their leading coefficients or face lattices agree with those of the input polytopes.

[write] Sources 10 and 09 prove this theorem again, independently, in Part VII: source 10 with the same constants (remark 101.1), source 09 with a symmetric parallelotope and the weaker lower constant $\prod_i (r_i(r_i + 1))^{-\alpha_i}$ (remark 101.2); source 09 adds that the total of dm generators is optimal (corollary 101.3). Source 08's colorful determinant formula (theorem 87.2) is a form of theorem 36.3 below for arbitrary generating lists.]

36.2 The determinant formula and explicit constants

For d bodies of positive affine dimensions $r_1, \dots, r_d$, define

$$M = \max_{1 \leq j_i \leq r_i} |\det(u_{1j_1}, \dots, u_{dj_d})|, \quad R_0 = \prod_{i=1}^d r_i. \quad (36.3)$$

When an argument is a point, use $M = 0$.

Theorem 36.3 (Finite mixed-determinant certificate). *For nonempty finite polytopes in K^d ,*

$$\frac{M}{d!} \leq V(P_1, \dots, P_d) \leq \frac{2^d}{d!} \sum_{j_1, \dots, j_d} |\det(u_{1j_1}, \dots, u_{dj_d})| \leq \frac{2^d R_0}{d!} M, \quad (36.4)$$

with the zero convention for a point argument. In particular,

$$v V(P_1, \dots, P_d) = \min_{j_1, \dots, j_d} v \det(u_{1j_1}, \dots, u_{dj_d}). \quad (36.5)$$

At most d^d determinants are required after the maximum-simplex bases have been selected.

Proof. Each segment $[p_{i0}, p_{i0} + u_{ij}]$ lies in P_i . Monotonicity and the segment formula give the lower bound for every determinant in (36.3), hence for their maximum.

By the upper half of Theorem 36.2, $V(P_1, \dots, P_d) \leq 2^d V(Z_1, \dots, Z_d)$. Expanding each zonotope by multilinearity and applying (34.2) gives the sum in (36.4). It contains R_0 terms, each at most M . If $M = 0$, the inequalities force the mixed volume to vanish. Otherwise the extreme comparison constants are positive real units and give (36.5). Finally $r_i \leq d$. $\square$

The lower constant is attained by independent segment arguments. No optimality is asserted for the upper constant. The bound counts the *final mixed determinants*, not the work needed to find a maximum simplex in a large vertex list. Brute-force basis selection can examine $\binom{n_i}{r_i+1}$ candidate simplices for body i .

36.3 An intrinsic determinant ideal

For finitely generated $\mathcal{O}$ -modules $L_1, \dots, L_d \subset K^d$, define

$$\mathfrak{D}(L_1, \dots, L_d) = \langle \det(w_1, \dots, w_d) : w_i \in L_i \rangle_{\mathcal{O}} \subset K.$$

This is an $\mathcal{O}$ -submodule of K , or fractional ideal in this usage; it need not be contained in $\mathcal{O}$.

Proposition 36.4 (Basis independence and the intrinsic formula). *For $L_i = L(P_i)$, the module $\mathfrak{D}(L_1, \dots, L_d)$ is generated by the finitely many determinants of the selected bases. It is either zero or a principal fractional ideal $\delta\mathcal{O}$, and*

$$v V(P_1, \dots, P_d) = v \delta \quad \text{when it is nonzero.}$$

Thus the mixed-volume valuation depends only on the scale lattices of its arguments. The minimum formula remains true if one uses all vertex differences instead of the selected bases.

Proof. Expand each w_i in its $\mathcal{O}$ -basis. Multilinearity expresses its determinant as an $\mathcal{O}$ -linear combination of the displayed finite basis determinants. Conversely each basis determinant is a generator of the intrinsic module. Among the finitely many nonzero determinants select one δ of least valuation. Every other determinant divided by δ lies in $\mathcal{O}$, so the generated module is $\delta\mathcal{O}$. Theorem 36.3 identifies its value with the mixed-volume value. Every vertex difference belongs to its corresponding module, while selected basis directions are vertex differences, proving the last assertion. $\square$

This is the decisive cancellation distinction. Individual determinants may have substantial cancellation and must be computed correctly. Once they are known, the mixed-volume scale is obtained by a finite minimum; there is no additional cancellation between those determinant scales.

37 Simultaneous inner segments and the valued Grassmannian

37.1 Avoiding finitely many leading-term cancellations

We first record the elementary finite-grid fact used to make the word “generic” effective.

Lemma 37.1 (A finite grid detects a nonzero polynomial). *Let $F \in \mathbb{R}[X_1, \dots, X_N]$ be nonzero and have degree at most H in each individual variable. If each $A_j \subset \mathbb{Q}$ contains $H + 1$ distinct numbers, some point of $A_1 \times \dots \times A_N$ is not a zero of F .*

Proof. For one variable this is the bound on the number of roots of a nonzero polynomial. Induct on N . If F vanished on the whole product grid, fixing the first $N - 1$ coordinates would give a polynomial in X_N with $H + 1$ roots, so all its coefficients would vanish on the smaller grid. The induction hypothesis would make every coefficient the zero polynomial, a contradiction. $\square$

Theorem 37.2 (Constructive simultaneous inner-segment theorem). *Let $m \geq d \geq 1$ and let $P_1, \dots, P_m \subset \mathbb{K}^d$ be nonempty finite polytopes. Choose maximum-simplex bases in their positive-dimensional affine hulls and let $r_i = \dim P_i$. Put*

$$H = \binom{m-1}{d-1}.$$

There are rational coefficients

$$c_{ij} \in \left\{ \frac{1}{r_i(H+2)}, \frac{2}{r_i(H+2)}, \dots, \frac{H+1}{r_i(H+2)} \right\} \quad (r_i > 0) \quad (37.1)$$

such that the segments

$$S_i = [p_{i0}, p_{i0} + w_i], \quad w_i = \sum_{j=1}^{r_i} c_{ij} u_{ij},$$

with $w_i = 0$ for a point, satisfy

$$S_i \subseteq P_i, \quad v \mathbf{V}(S_i : i \in I) = v \mathbf{V}(P_i : i \in I) \quad \left(I \in \binom{[m]}{d} \right). \quad (37.2)$$

A successful choice exists in a grid of at most $(H + 1)^{md}$ points.

Proof. Introduce separate variables X_{ij} for each positive-dimensional body and set $w_i(X) = \sum_j X_{ij} u_{ij}$. For a d -subset $I = \{i_1 < \dots < i_d\}$, expand

$$D_I(X) = \det(w_{i_1}(X), \dots, w_{i_d}(X)) = \sum_{j_1, \dots, j_d} \det(u_{i_1 j_1}, \dots, u_{i_d j_d}) \prod_{k=1}^d X_{i_k j_k}.$$

Distinct choices $(j_1, \dots, j_d)$ produce distinct monomials, because the bodies in I are distinct labelled arguments.

Let $\tau_I = v \mathbf{V}(P_i : i \in I)$. If $\tau_I = \infty$, Theorem 36.3 says that every coefficient of D_I is zero; then no condition on the choice is necessary. Otherwise select a coefficient δ_I of valuation τ_I . All coefficients of D_I/δ_I lie in $\mathcal{O}$, and at least one is a unit. Reducing its coefficients gives a nonzero real polynomial

$$F_I(X) = \text{st}(D_I(X)/\delta_I) \in \mathbb{R}[X_{ij}].$$

Here standard part is applied coefficientwise, before any evaluation. Nonzeroness follows from distinctness of the monomials and from the chosen unit coefficient.

Take the product $F = \prod_{I:\tau_I < \infty} F_I$, using $F = 1$ if there are no such subsets. It is a nonzero real polynomial. Each variable X_{ij} occurs with degree at most one in F_I , and only when $i \in I$. There are at most H such subsets, so its degree in F is at most H . Lemma 37.1 applied to the grids in (37.1) gives a rational point c with $F(c) \neq 0$.

For each finite entry, evaluation at real c commutes with standard part, and $F_I(c) \neq 0$ says that $D_I(c)/\delta_I$ is a unit of $\mathcal{O}$. Hence $vD_I(c) = \tau_I$. Infinite entries remain zero identically. The segment formula gives all equalities in (37.2). Finally every selected coefficient is positive and

$$\sum_{j=1}^{r_i} c_{ij} \leq \frac{H+1}{H+2} < 1.$$

Thus $p_{i0} + w_i$ is a convex combination of the selected simplex vertices, and the whole segment lies in P_i . $\square$

The theorem proves the preview in Section 32. It applies even when no nonzero integer multiple of one positive valuation is cofinal in the value group. The only genericity argument occurs over the ordinary real residue field, and there it is replaced by a finite grid.

37.2 An exact equality of configuration images

Let $\mathcal{T}_{m,d}^{\text{pure}}(K)$ be the set of all vectors

$$(v \mathbf{V}(P_i : i \in I))_{I \in \binom{[m]}{d}}$$

obtained from nonempty finite polytopes in K^d . Define the determinant image

$$\mathcal{G}_{m,d}(K) = \left\{ (v \det W_I)_{I \in \binom{[m]}{d}} : W \in K^{d \times m} \right\},$$

where W_I is the submatrix with columns indexed by I in increasing order. Rank-deficient matrices contribute the all-infinite vector.

Corollary 37.3 (Pure profile realization). *One has the exact equality*

$$\boxed{\mathcal{T}_{m,d}^{\text{pure}}(K) = \mathcal{G}_{m,d}(K).} \quad (37.3)$$

After omitting the all-infinite vector and quotienting by common addition of a value, the right side is the valuation image of the K -realizable Grassmannian in Plücker coordinates.

Proof. Theorem 37.2 produces the matrix $W = (w_1, \dots, w_m)$ for every polytope profile. Conversely take $P_i = [0, w_i]$ and use (34.2); the factor $d!$ has valuation zero. $\square$

Several restrictions in this statement are essential. It is an equality of *valuation images over the specified ordered field K* , not a claim that every abstract valuated matroid is realizable. It does not replace K by an algebraically closed field, forget ordered-field realizability, or silently take a topological closure. The use of absolute determinants makes no difference to the value, since $v|a| = v(a)$. Thus (37.3) gives a concrete, field-specific link from polytope mixed-volume scales to the absolute and valued Grassmannian viewpoints discussed in [23].

37.3 Tropical Plücker constraints

Corollary 37.4 (A universal pure-profile relation). *Let $A \subseteq [m]$ have size $d - 2$, and let a, b, c, e be four distinct indices outside A . For any finite polytope family, the minimum of*

$$\tau_{Aab} + \tau_{Ace}, \quad \tau_{Aac} + \tau_{Abe}, \quad \tau_{Aae} + \tau_{Abc} \quad (37.4)$$

is attained at least twice, with the usual convention when all terms are infinite.

Proof. Represent the entire pure profile by one matrix, using Corollary 37.3. Its minors satisfy the three-term Plücker relation, with signs determined by the chosen ordering. In a valued field a sum of nonzero terms cannot vanish when exactly one term has strictly least valuation: dividing by that term gives a unit plus an infinitesimal. Apply this observation to the three products of minors. $\square$

For four planar bodies, the conclusion says that the minimum of $\tau_{12} + \tau_{34}$, $\tau_{13} + \tau_{24}$, and $\tau_{14} + \tau_{23}$ is attained at least twice. It is stronger than merely testing one mixed coefficient at a time. The same matrix realization supplies all determinantal relations simultaneously.

38 Full profiles, cloned segments, and higher-rank exchange

38.1 A colored determinant formula

Concatenate the maximum-simplex bases for $P_1, \dots, P_m$ into a matrix U , and give a column from body i the color i . For a d -element column subset B , let $\text{col}(B) \in \mathbb{N}^m$ record the counts of its colors. Put

$$\tau(\alpha) = v \, V(P_1[\alpha_1], \dots, P_m[\alpha_m]), \quad |\alpha| = d.$$

The zonotope expansion gives the precise multiplicity identity

$$V(Z_1[\alpha_1], \dots, Z_m[\alpha_m]) = \frac{\prod_i \alpha_i!}{d!} \sum_{B: \text{col}(B)=\alpha} |\det U_B|. \quad (38.1)$$

Indeed, choosing one generator in each ordered argument slot gives zero when the same column is chosen twice. Every surviving unordered column subset occurs $\prod_i \alpha_i!$ times. Consequently

$$\tau(\alpha) = \min_{B: \text{col}(B)=\alpha} v \det U_B, \quad (38.2)$$

where the minimum over no nonzero determinant is infinite.

This formula describes the profile as a colorwise minimum of determinant values. To prove its full exchange structure without invoking a real-valued tropicalization theorem, it is convenient to strengthen the realization.

Theorem 38.1 (Color-symmetric clone realization). *Make d labelled copies of each P_i . There exist vectors $w_{i,k} \in P_i - p_{i0}$, $1 \leq k \leq d$, such that for every d -element set B of distinct clone labels,*

$$v \det(w_{i,k} : (i, k) \in B) = \tau(\text{col}(B)). \quad (38.3)$$

Thus the determinant valuation depends only on the color counts, not on which clone labels are chosen. The vectors can be obtained from the finite rational grids of Theorem 37.2, applied to md labelled bodies.

Proof. Apply the simultaneous inner-segment theorem to the family in which each of the m bodies occurs as d separately labelled members. A pure mixed volume of this larger family is exactly the corresponding repeated-body mixed volume of the original family. Taking the segment direction vectors proves (38.3). $\square$

The number d of clones per body allows all multiplicities up to d . It is a statement about representing a finite profile, not about representing a polytope itself by the union of those segments.

38.2 A determinant exchange lemma

Lemma 38.2 (Valued basis exchange). *Let columns of a matrix over K be indexed by a finite set. For a d -element subset B , write $q(B) = v \det W_B$. If $q(I)$ and $q(J)$ are finite and $e \in I \setminus J$, there is $f \in J \setminus I$ for which*

$$q(I) + q(J) \geq q(I - e + f) + q(J - f + e). \quad (38.4)$$

Both terms on the right are finite for the selected f .

Proof. The determinant exchange identity expresses

$$\det W_I \det W_J = \sum_{f \in J \setminus I} \pm \det W_{I-e+f} \det W_{J-f+e}.$$

For completeness, expand the column w_e in the basis indexed by J using Cramer's rule, then use linearity in its slot in $\det W_I$ and multiply by $\det W_J$. Terms indexed by $f \in I \cap J$ have a repeated column in the first determinant and vanish. Reordering the columns accounts for the signs.

The left side is nonzero. The valuation of a finite sum is at least the minimum of its summands' valuations. Hence some nonzero summand has valuation at most $q(I) + q(J)$, which is exactly the assertion. $\square$

38.3 Exchange-convexity over an arbitrary value group

Definition 38.3. A function $f : \{\alpha \in \mathbb{N}^m : |\alpha| = d\} \rightarrow \Gamma \cup \{\infty\}$ will be called *exchange-convex* if, whenever $f(x), f(y) < \infty$ and $x_i > y_i$, there exists j with $x_j < y_j$ such that

$$f(x) + f(y) \geq f(x - e_i + e_j) + f(y + e_i - e_j).$$

This is the usual M -convex exchange axiom, stated with values in the ordered abelian group Γ rather than necessarily in $\mathbb{R}$.

Theorem 38.4 (Higher-rank mixed-volume exchange). *The full mixed-volume valuation profile τ is exchange-convex.*

Proof. Use the clone matrix from Theorem 38.1. For two multi-indices x, y with finite values, choose the clone subsets

$$I = \{(k, s) : 1 \leq s \leq x_k\}, \quad J = \{(k, s) : 1 \leq s \leq y_k\}.$$

They have size d . The initial-segment choice ensures that their common clones in each color are exactly the first $\min(x_k, y_k)$ labels.

If $x_i > y_i$, choose a clone e of color i in $I \setminus J$. Lemma 38.2 supplies $f \in J \setminus I$. Its color j satisfies $x_j < y_j$, precisely because the clone sets were nested within each color. The exchanged subsets have color counts $x - e_i + e_j$ and $y + e_i - e_j$. Substitute (38.3) into (38.4) to obtain the desired inequality. $\square$

This proof uses only finite determinants, reduction to an infinite residue field, and ordered-group comparisons. It needs no embedding $\Gamma \hookrightarrow \mathbb{R}$, no limiting process, and no assumption that a positive value has cofinal integer multiples. For real-valued settings, its conclusion lies within the established Lorentzian and M -convex framework [24]; the point here is an explicit matrix proof that continues unchanged at higher rank.

For example, taking $x = \alpha + e_i - e_j$ and $y = \alpha - e_i + e_j$ when both are admissible gives

$$2\tau(\alpha) \leq \tau(\alpha + e_i - e_j) + \tau(\alpha - e_i + e_j).$$

Thus the familiar midpoint inequality is only one consequence of the stronger exchange axiom. Infinite entries encode genuine vanishing, not a finite but unusually small mixed volume.

38.4 Exactly which entries are nonzero

Let $U_i = \text{span}_K(P_i - P_i)$. The following linear-algebra criterion makes the support of the full profile explicit.

Lemma 38.5 (Independent representatives of subspaces). *For subspaces $E_1, \dots, E_s$ of a finite-dimensional vector space over a field, there exist independent vectors $e_i \in E_i$ if and only if*

$$\dim \left(\sum_{i \in I} E_i \right) \geq |I| \quad (I \subseteq [s]).$$

Proof. Necessity is immediate. For sufficiency induct on s . If there is a nonempty proper subset I with equality, choose independent representatives for its subspaces by induction. They form a basis of $W = \sum_{i \in I} E_i$. In the quotient by W , the remaining subspaces satisfy the dimension condition, since for $J \subseteq [s] \setminus I$,

$$\dim \left((W + \sum_{j \in J} E_j) / W \right) \geq |I| + |J| - |I| = |J|.$$

Choose independent representatives in the quotient and lift them.

If no such tight proper subset exists and $s > 1$, choose any nonzero $e_s \in E_s$. Every nonempty subset of the other indices spans dimension at least its size plus one. Quotienting by Ke_s decreases that dimension by at most one, so induction applies to the remaining $s - 1$ subspaces. The case $s = 1$ is immediate. $\square$

[[write] Sources 10 and 09 state and use the same lemma in [Part VII \(remark 105.2\)](#).]

Corollary 38.6 (Support of the full profile). *For $|\alpha| = d$,*

$$\tau(\alpha) < \infty \iff \dim_K \left(\sum_{i \in J} U_i \right) \geq \sum_{i \in J} \alpha_i \text{ for all } J \subseteq [m]. \quad (38.5)$$

Proof. By the determinant formula, positivity of the mixed volume is equivalent to choosing independent representatives from the d slots, with U_i repeated α_i times. Lemma 38.5 applies. Its condition for all slot subsets is equivalent to (38.5): necessity follows by taking all slots of the colors in J ; sufficiency follows because an arbitrary slot subset uses no more than all α_i slots of each color it meets. $\square$

This is a finite subspace-rank condition. It separates the zero pattern, which uses only K -linear spans, from the nonzero scale values, which require the finer O -lattices.

39 Recovering a polytope's scale class from probes

39.1 All mixed responses classify the scale lattice

Fix full-dimensional polytopes $P, Q \subset K^d$. A *segment probe* is a tuple of $d - 1$ segments in K^d ; for $d = 1$ it is the empty tuple.

Theorem 39.1 (Scale-lattice classification). *The following are equivalent:*

(i) $L(P) = L(Q)$.

(ii) For every segment probe $T_1, \dots, T_{d-1}$,

$$v \, V(P, T_1, \dots, T_{d-1}) = v \, V(Q, T_1, \dots, T_{d-1}).$$

(iii) Replacing P by Q in any one slot of any mixed volume of finite polytopes leaves the valuation unchanged.

Proof. Proposition 36.4 gives (i) $\Rightarrow$ (iii), and segment arguments give (iii) $\Rightarrow$ (ii).

Assume (ii). Choose a maximum-simplex basis $B = (u_1, \dots, u_d)$ for P and write $D = \det B$. For each j , use as probe the segments in all directions u_k with $k \neq j$. For P the determinant minimum is vD : all basis choices except u_j give a repeated column, and u_j gives $\pm D$.

Let $C = (z_1, \dots, z_d)$ be a basis for $L(Q)$, and set $A = B^{-1}C$. By the determinant formula, the corresponding probe for Q has value

$$vD + \min_k v(A_{jk}).$$

Equality with vD forces every entry of row j of A to lie in $\mathcal{O}$. For all rows together, this gives $L(Q) \subseteq L(P)$. Repeating the argument with a basis of Q gives the reverse inclusion. $\square$

The probes range over directions in K^d , not merely real directions. Restricting to a smaller collection is a different question: hidden non-Archimedean shear data must not be discarded without proof.

[[write] Printed once for sources 06 and 08: source 08 (Part VI) proves the equivalence of (i) with equality of all first-slot values against arbitrary finite polytopes, by the same probe argument (remark 94.3). Part VI answers the question of the previous sentence negatively in the plane (theorem 91.2): no fixed finite family of mixed-area probes recognizes every scale lattice.]

39.2 A finite comparison test

Theorem 39.2 (A reference-basis $d + 1$ -measurement test). *Choose a maximum-simplex basis $B = (u_1, \dots, u_d)$ of P , and let $T^{(j)}$ be the $(d - 1)$ -tuple of segments $[0, u_k]$, $k \neq j$. For a full-dimensional Q , the equality $L(Q) = L(P)$ holds if and only if*

$$v \, \text{Vol}_d(Q) = v \, \text{Vol}_d(P), \tag{39.1}$$

$$v \, V(Q, T^{(j)}) \geq v \, \text{Vol}_d(P) \quad (1 \leq j \leq d). \tag{39.2}$$

When these conditions hold, every inequality in (39.2) is an equality. Thus, relative to the known reference basis and scale of P , one volume value and d probe values suffice to test its entire scale class.

Proof. For a maximum-simplex basis C of Q , write $A = B^{-1}C$ as above. The threshold conditions say that all entries of A lie in $\mathcal{O}$. The volume condition and Corollary 35.4 say that $v \det A = 0$. Hence $\det A$ is a unit and

$$A^{-1} = \frac{\text{adj}(A)}{\det A}$$

has entries in $\mathcal{O}$. Therefore $A \in \text{GL}_d(\mathcal{O})$ and the lattices are equal. Conversely lattice equality makes A and its inverse integral, so its determinant is a unit and every row has some unit entry. The volume equality and all probe equalities follow. $\square$

No minimality of the number $d + 1$ is asserted. It is a uniform explicit upper bound for a comparison test, not a reconstruction algorithm from unlabelled real measurements.

[[write] Printed once for sources 06 and 08: source 08 (Part VI) proves the same test independently, for an arbitrary $\mathcal{O}$ -basis of $L(P)$ in place of a maximum-simplex basis, by the same adjugate argument; its formulation is remark 90.3.]

39.3 A geometric interpretation of equal lattices

Proposition 39.3 (Bounded real distortion). *For full-dimensional $P, Q \subset K^d$, $L(P) = L(Q)$ if and only if there are $a \in P$, $b \in Q$ and positive real constants c, C such that*

$$c(P - a) \subseteq Q - b \subseteq C(P - a). \quad (39.3)$$

Proof. For a maximum-simplex normalization of a d -polytope, center at the centroid $h = (1/(d+1), \dots, 1/(d+1))$ of Δ_d . With $\delta = 1/(d(d+1))$, the simplex contains $h + [-\delta, \delta]^d$, while the normalized polytope minus h is contained in $[-2, 2]^d$. Thus its centered shape lies between two real scalar multiples of the cube in its $\mathcal{O}$ -basis.

If $L(P) = L(Q)$, the basis change belongs to $\text{GL}_d(\mathcal{O})$. All entries of it and its inverse are bounded by a common positive real number. Matrix multiplication therefore sends a cube into a real scalar multiple of a cube in both directions. Combining the two cube comparisons gives (39.3).

Conversely take differences of points in the two inclusions. The resulting inclusions of difference sets imply $cL(P) \subseteq L(Q) \subseteq CL(P)$. Positive real constants are units of $\mathcal{O}$, so both outer modules equal $L(P)$. $\square$

The equivalence class detected by all mixed-volume valuations is exactly this bounded-distortion class. The affine residue then records a real shape inside the class, and finer infinitesimal data may distinguish polytopes even when their residue shapes agree.

40 Examples, counterexamples, and what the invariants forget

40.1 A triangle whose ordinary shadow is only a segment

Let $\varepsilon = \omega^{-1} > 0$ and

$$P = \text{conv}\{(0, 0), (1, 0), (0, \varepsilon)\}.$$

Its ordinary standard part is the horizontal unit segment, but

$$L(P) = \mathcal{O}e_1 \oplus \mathcal{O}(\varepsilon e_2), \quad B = \begin{pmatrix} 1 & 0 \\ 0 & \varepsilon \end{pmatrix}, \quad B^{-1}P = \Delta_2.$$

Hence $\text{Res}(P)$ is a real triangle, not a segment. Its area is $\text{Vol}_2(P) = \varepsilon/2$, its volume value is $v\varepsilon$, and its normalized residue area is $1/2$. The rescaling has not removed the infinitesimal information: that information is retained in the embedded scale lattice and its determinant.

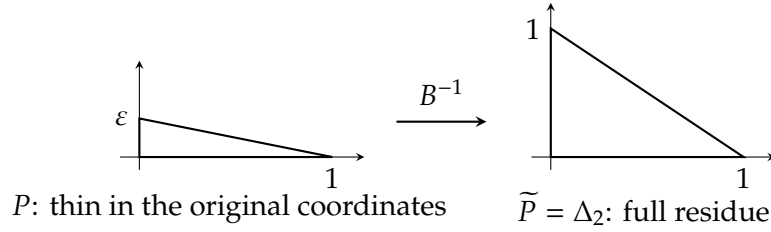


Figure 4: Affine normalization of an infinitesimally thin triangle. The left vertical scale is exaggerated; the picture is schematic, not a real-scale rendering of an infinitesimal height.

40.2 Cancellation can lie beyond every fixed power scale

Take $\eta = \omega^{-\omega}$ and $u = (1, 1)$, $w = (1, 1 + \eta)$. Then

$$\det(u, w) = \eta.$$

Keeping only the standard part of each coordinate makes both columns $(1, 1)$ and destroys the determinant. Moreover $0 < \eta < \varepsilon^N$ for every positive integer N , so no fixed finite order in powers of ε reveals it. The correct finite certificate is the determinant itself, computed with enough exact coefficient information to resolve its cancellation.

Even in rank one there is no truncation depth depending only on dimension and vertex count. For each integer $N > 0$, the three-vertex triangle with height ε^N has nonzero area $\varepsilon^N/2$. A rule discarding all terms above a fixed power cutoff fails on some such triangle.

[[write] These two observations are weaker instances of Part II's results on coordinate truncation. There, [theorem 26.2](#) gives four labelled points with coordinates in $\mathbb{R}((\varepsilon))$ whose degree- N truncated hulls alternate between a quadrilateral and a triangle for every N . [theorem 26.3](#) realizes *every* prescribed binary history in the same way. So no truncation depth works, and moreover the truncated combinatorics need not stabilize at all. Part I's [section 16.4](#) records the corresponding failure for an exact algebraic equality. The examples here are kept because they concern a determinant (a scale) rather than a face lattice.]

40.3 Identical individual scales do not determine a mixed scale

Use the rank-two convention

$$v\varepsilon = (0, 1), \quad v\eta = (1, 0) \quad \text{in lexicographically ordered } \mathbb{Z}^2.$$

Define three parallelograms

$$\begin{aligned} P &= [0, e_1] + [0, \varepsilon e_2], \\ Q &= [0, e_1 + \varepsilon e_2] + [0, \eta e_2], \\ Q' &= [0, \eta e_1] + [0, e_2]. \end{aligned}$$

Their individual areas satisfy

$$\text{Vol}_2(P) = \varepsilon, \quad \text{Vol}_2(Q) = \text{Vol}_2(Q') = \eta.$$

Each is affinely a square, and each has an affine residue affinely isomorphic to a square. Nevertheless expansion into segments gives

$$\begin{aligned} V(P, Q) &= \frac{1}{2}(\varepsilon + \eta + \varepsilon + 0) = \varepsilon + \frac{1}{2}\eta, \\ V(P, Q') &= \frac{1}{2}(0 + 1 + \varepsilon\eta + 0) = \frac{1}{2}(1 + \varepsilon\eta). \end{aligned}$$

The mixed values are respectively $(0, 1)$ and $(0, 0)$. Thus individual volume values and separate affine residue isomorphism classes do not determine a family's mixed profile. The *relative positions* of its scale lattices in the common ambient vector space matter.

40.4 A four-body Plücker example

Let

$$\begin{aligned} P_1 &= [0, e_1] + [0, \eta e_2], & P_2 &= [0, e_1 + \varepsilon e_2] + [0, \eta e_2], \\ P_3 &= [0, e_2] + [0, \eta e_1], & P_4 &= [0, e_2 + \varepsilon e_1] + [0, \eta e_1]. \end{aligned}$$

All four bodies have positive area. Their pure mixed values are

$$\tau_{12} = \tau_{34} = v\varepsilon, \quad \tau_{13} = \tau_{14} = \tau_{23} = \tau_{24} = 0.$$

[write] Here $\tau_{ab} = v V(P_a, P_b)$ is an entry of the pure profile of [corollary 37.4](#); the η in the generators is the surreal $\omega^{-\omega}$, written τ elsewhere in the report.) In this example the first displayed generator of each body already gives a simultaneous inner-segment model. The three Plücker sums are $2v\varepsilon, 0, 0$, so the minimum is attained exactly twice. The example makes visible the difference between two nearly parallel pairs and the transverse cross-pairs.

40.5 The affine residue need not retain all faces

Let

$$P_\varepsilon = \text{conv}\{(0, 0), (1, 0), (1, 1), (0, 1), (-\varepsilon, 1/2)\}.$$

It is a pentagon over K , of area $1 + \varepsilon/2$. Its scale lattice is $\mathcal{O}^2$, and its affine residue is the square $[0, 1]^2$. Thus a full-dimensional residue can lose a vertex even though it loses no dimension. Maximum-simplex normalization does not repair this loss of fine face information; it is deliberately a first residue invariant.

[write] This is the same phenomenon as Part I's lost-face example ([14.2](#)) (manuscript 02). There a pentagon has a vertex infinitesimally below the bottom edge of the rectangle $[-1, 1] \times [0, 1]$, and that vertex's standard part is not a face. Here the vertex $(-\varepsilon, 1/2)$ lies infinitesimally outside an edge of the square. The loss occurs for the standard part and for the affine residue alike: the scale lattice here is already $\mathcal{O}^2$, so $\text{Res}(P_\varepsilon)$ is affinely isomorphic to $\text{st } P_\varepsilon = [0, 1]^2$. Part I's [proposition 14.2](#) shows how an infinitesimal optimality band recovers the whole limiting face; see the re-scoped [question 43.6](#).]

The real triangle Δ_2 and the real unit square also have the same scale lattice $\mathcal{O}^2$, but areas $1/2$ and 1 . The common lattice fixes the value, not the leading coefficient. This is why a determinant minimum is an exact scale formula rather than an exact mixed-volume formula.

40.6 Why the pure/full distinction cannot be dropped

For $d > 1$, a full-dimensional P has $\text{Vol}_d(P) > 0$, while every segment S has $\text{Vol}_d(S) = 0$. Therefore no single-segment replacement can preserve the repeated entry $V(P[d])$. The simultaneous theorem is formulated for *distinct labels* for this reason. The clone theorem solves the repeated-entry issue by providing independent segment choices for different copies of the same body.

40.7 An exact characterization of full profiles

The clone theorem also has a converse that is useful when formulating further realization problems.

Corollary 40.1 (Full profiles as color-symmetric determinant data). *An array $t(\alpha) \in \Gamma \cup \{\infty\}$, $|\alpha| = d$, is the full mixed-volume valuation profile of finite polytopes in K^d if and only if there is a $d \times md$ matrix over K , with d columns of each of m colors, such that every d -element column subset B satisfies*

$$v \det W_B = t(\text{col}(B)).$$

Proof. Necessity is Theorem 38.1. For sufficiency, make Z_i the zonotope generated by the d columns of color i . For any α , formula (38.1) expresses the corresponding mixed volume as a positive real factor times a finite sum of absolute determinants. All eligible determinants have value $t(\alpha)$, or all are zero when $t(\alpha) = \infty$. There is at least one eligible column subset for every $|\alpha| = d$, since every color has d labels. The positive-sum lemma gives the required value. $\square$

This is a realizability characterization with matrix witnesses, not a claim that the exchange axiom alone is sufficient. Removing the matrix witnesses in favor of a usable intrinsic list of conditions is one of the research questions below.

[[write] Part VII re-derives this corollary independently, as the equivalence (i) $\Leftrightarrow$ (iii) of theorem 103.3, by generic real block mixing (source 10) and by integer mixing with the explicit bound $d \binom{dm}{d}$ (source 09, remark 103.4). Its Fano cubic (theorem 108.4) shows that exchange-convexity (theorem 38.4, re-proved there by source 09 with the same nested-clone argument) is not sufficient.]

41 Finite higher-rank certificates and real specialization

41.1 The finite-support input model

The preceding results apply to arbitrary finite tuples of elements of K ; they are not an algorithm for comparing arbitrary surreal expressions. For an effective specialization theorem, restrict the coordinates to finite sums

$$f = \sum_{\gamma \in A} a_\gamma t^\gamma, \quad a_\gamma \in \mathbb{R},$$

where A is a finite subset of a totally ordered abelian group. Rational functions in such sums are also allowed when their denominators are certified nonzero. The order is the Hahn leading-term order: the least exponent determines the sign and value of a nonzero sum.

A finite certificate may contain coordinate expressions, determinants, nonzero denominators, polynomial differences encoding comparisons, and products needed for specified additive comparisons of valuations. Expand these expressions exactly first; cancellation within an expression must already have been resolved. For comparisons of absolute determinants, one may use the polynomial difference of their squares.

41.2 Finite order data admit an integer-valued model

Lemma 41.1 (Finite positive grading). *Let H be a finitely generated subgroup of a totally ordered abelian group, and let $D \subset H_{>0}$ be finite. There exists a group homomorphism $\lambda : H \rightarrow \mathbb{Z}$ with $\lambda(\delta) > 0$ for every $\delta \in D$.*

Proof. The group H is finitely generated and torsion-free, so choose an isomorphism $H \cong \mathbb{Z}^s$. No nontrivial nonnegative integer combination of members of D can be zero, because all its summands are positive in H .

The origin is not in the real convex hull of the corresponding integer vectors. Otherwise the finite linear system $\sum a_\delta \delta = 0$, $\sum a_\delta = 1$, $a_\delta \geq 0$ would have a real solution and hence a rational solution: a nonempty polytope defined over $\mathbb{Q}$ has a rational vertex, or a rational point in its minimal supporting face, by Gaussian elimination. Clearing denominators would give a forbidden nonnegative integer relation.

Strictly separate this finite convex hull from zero in $\mathbb{R}^s$. The separating linear functional is positive on all members of D . Approximate its coefficients rationally closely enough to preserve these finitely many strict inequalities, and clear denominators. The resulting integer linear functional defines λ . $\square$

The finite positive-grading principle has a close antecedent in the repository's support-certificate discussion [18]; the application here is to determinant and polytope sign certificates. It is not asserted for an infinite support, and it is not asserted to preserve the full order of H .

Theorem 41.2 (One-parameter preservation of a finite certificate). *Let $\mathcal{F}$ be a finite collection of finite-support Hahn polynomials. There is an integer-valued homomorphism on the group generated by their exponents such that the substitution*

$$t^\gamma \mapsto s^{\lambda(\gamma)} \quad (41.1)$$

preserves, for every member of $\mathcal{F}$, zero versus nonzero, sign, leading coefficient, and the ordered list of its exponents. It also preserves the order of the valuations of the nonzero members of $\mathcal{F}$. The resulting expressions are ordinary one-variable Laurent polynomials. Finite rational expressions are covered by including their nonzero numerators and denominators in the certificate.

Proof. Let A be the union of all supports occurring in $\mathcal{F}$. Apply Lemma 41.1 to the positive differences $\beta - \alpha$ of all distinct pairs $\alpha, \beta \in A$ with $\alpha < \beta$. The resulting homomorphism is strictly order-preserving on the finite set A . Thus no two terms within a certificate polynomial acquire the same exponent, and the term ordering, leading coefficient, sign, and nonzeroness are unchanged. Zero expressions remain zero because the substitution is a group-algebra homomorphism. Comparing leading exponents across different members gives the assertion about their valuations. Including denominators ensures that specialization of each rational expression is defined in $\mathbb{R}(s)$. $\square$

An equality of exponent sums is automatically preserved by a group homomorphism. To preserve a specified *inequality* between such sums, include the corresponding products or monomials among the finitely many certificate expressions before applying the theorem. This prevents an unwarranted claim that every order comparison in the infinite group has been retained.

[[write] Theorem 41.2 is the effective, finite-support counterpart of Part I's theorem 19.1 (manuscript 02). That theorem takes an arbitrary finite bounded configuration over K and a finite list of polynomial sign conditions. By real-closed-field transfer and semialgebraic curve selection it gives a convergent algebraic Puiseux arc over $\mathbb{R}(s)$ that also keeps the prescribed standard-part limit, but it is not effective. The theorem here assumes finite-support Hahn input. In exchange it gives an explicit integer grading, ordinary Laurent polynomials with the same leading coefficients and exponent order, and the explicit threshold ρ_g below. Part IV's theorem 50.1 is a third one-parameter statement, preserving a whole real trace by algebraic germs. None of the three preserves the entire higher-rank order (see the example below).]

41.3 A certified small positive real parameter

Suppose a nonzero specialized expression is

$$g(s) = c_0 s^{m_0} + c_1 s^{m_1} + \cdots + c_k s^{m_k}, \quad m_0 < m_1 < \cdots < m_k \in \mathbb{Z}.$$

For $k > 0$ set

$$\rho_g = \min \left(1, \frac{|c_0|}{2 \sum_{j=1}^k |c_j|} \right),$$

and use $\rho_g = 1$ for a single monomial. If $0 < s < \rho_g$, then

$$|s^{-m_0} g(s) - c_0| \leq s \sum_{j=1}^k |c_j| < |c_0|/2,$$

so $g(s)$ has the sign of its leading coefficient. A common parameter smaller than all the finitely many thresholds preserves all certificate signs simultaneously. For rational input coefficients, a rational choice of s gives a rational finite configuration satisfying those tests.

This result does not contradict non-Archimedean behavior. It reproduces a *specified finite collection* of inequalities and equalities. It does not reproduce the assertion $0 < \eta < \varepsilon^N$ for every integer N .

Example 41.3 (Why an entire rank-two order cannot be compressed). For lexicographic $\mathbb{Z}^2$ with $v\eta = (1, 0)$ and $v\varepsilon = (0, 1)$, an order-preserving homomorphism to $\mathbb{Z}$ would have $a = \lambda(1, 0) > N\lambda(0, 1) = Nb$ for every $N > 0$, with $b > 0$. No two integers a, b satisfy this. In contrast, finitely many comparisons are preserved by $\lambda(a, b) = Ma + b$ for a sufficiently large integer M . The required M depends on the certificate, not only on the geometric dimension.

[[write] This is the same phenomenon as Part I's [section 16.5](#) (manuscript 05): the scales ω^{-1} and $\omega^{-\omega}$ satisfy $\omega^{-\omega} < \omega^{-N}$ for every N , so replacing them by t^a, t^b with fixed positive real exponents in a rank-one monomial group cannot preserve all these comparisons. There, as here, the finite theorems preserve only a specified finite list of comparisons.]

41.4 An explicit finite workflow

A proof-producing implementation for finite-support coordinates can use the following stages. First establish affine ranks and select maximum vertex simplices by exact determinant comparisons. Second compute the mixed determinant lists and their least valuations. Third, when a pure matrix model is requested, enumerate the rational grids in Theorem 37.2 and retain a choice passing every determinant value test. Fourth collect all the resulting finite polynomial tests, construct a positive grading, and produce a real-parameter sign bound.

The certificate consists of the selected simplex indices, exact comparison expressions, basis determinants, chosen rational coefficients, and the finite grading inequalities. Its verification is finite algebra. The theorem gives a terminating exhaustive-grid method when exact coefficient and order operations are available. It does not give a polynomial-time algorithm, a universal truncation depth, or a decision procedure for unrestricted surreal-number notation.

42 Exact verification and a formalization route

42.1 What the supplied program actually checks

The accompanying code/`06-affine-residues-verify.py` uses only the Python standard library. Its coefficient ring is

$$\mathbb{Q}[\eta, \eta^{-1}, \varepsilon, \varepsilon^{-1}],$$

with rational coefficients and lexicographically ordered exponent pairs. Finite sums in this ring embed into the surreal example $\eta = \omega^{-\omega}$, $\varepsilon = \omega^{-1}$. All sign tests use exact leading terms; no floating-point arithmetic is used. ([write] The program and its recorded output name $\eta = \omega^{-\omega}$ eta, for example "exponent_order": "Z^2 lexicographic, eta=(1,0), epsilon=(0,1)" and the key lambda_eta.)

For planar polytopes, the program computes convex hulls by exact orientation tests, areas by the shoelace formula, and mixed areas by polarizing the area of the pairwise-sum convex hull. It then compares these independently computed mixed areas with the maximum-simplex mixed-determinant certificate. Thus the audit does not define the mixed area by the very minimum formula it is checking.

The included report (`data/06-affine-residues-verification.json`) records the following run, using seed 20260930:

Audit category	Count
Random full-dimensional polygon pairs	160
Normalized residue checks from those pairs	320
Scale-lattice/probe equivalence comparisons	320
Exact common-affine-map covariance checks	24
Additional lower-dimensional or zero cases	4
Explicit rank-two mixed-area examples	2
Four-body simultaneous inner-segment families	20
Four-body tropical Plücker checks	20
Three-color families with six labelled clones	12
Individual M -convex exchange tests	396

The program also checks the thin triangle, the cancellation determinant, the pentagon with square residue, and a finite compression certificate. Its successful inner-segment choices are searched randomly within the proved rational grids, with a fixed finite audit budget; that convenience is distinguished in the source from the exhaustive-grid existence proof. The run used at most four trials for a four-body family and at most three for a six-clone family.

To reproduce the default report, run

```
python3 code/06-affine-residues-verify.py \
  --output /path/to/scratch/verification.json
```

([write] Manuscript 06 printed `python3 verify.py`. Without `-output` the program writes `verification.json` next to itself, that is, into `code/`; pass an output path outside the report directory, as above, so that the shipped record is never overwritten. The delivered `code/06-affine-residues-Makefile` still names the delivery paths `verify.py` and `article.tex`.) The output, shipped as `data/06-affine-residues-verification.json`, contains the exact example expressions, counts, and specialization data. For more instances, the command-line options `--pairs`, `--seed`, and `--output` are available. These are finite-instance tests, not a formal proof of the universal statements and not an exhaustive comparison with the literature.

42.2 A modular Lean development

[write] The formalization roadmap of manuscript 06 is printed in [section 83](#), together with its statement that the package contains no Lean source and reports no new Lean build.

43 Further research questions and topics

The following questions are proposed directions arising from the proved results. Their formulation here is not a claim that every item is a previously published open problem or that no partial answer exists in related literature. Each question isolates something the current theorems do not settle.

Question 43.1 (Intrinsic full-profile realization). Give a practical intrinsic characterization of the arrays that admit the color-symmetric matrix realization of [Corollary 40.1](#). Which additional conditions, beyond support rank constraints and M -convex exchange, express realizability over an ordered field? A useful first target is a complete analysis for three or four colors in dimension three, keeping the realization field explicit.

[[write] *Re-scoped: partly answered in [Part VII](#)*. Exchange-convexity with full support, even with strictly Lorentzian volume polynomials, is not sufficient ([theorem 108.4](#), $d = 3$, seven bodies); every exposed initial support is a real-representable polymatroid base set ([theorem 105.4](#)); every finite-valued M -convex profile with $d = 2$ is realizable ([theorem 110.2](#)); and a matroid-type profile is realizable exactly when the matroid is representable over $\mathbb{R}$ ([theorem 106.1](#)). Dimension three with three or four colours remains open.]

Question 43.2 (Leading coefficients above a fixed scale profile). Fix a realizable valuation profile τ . Describe the possible positive leading coefficients of the mixed volumes after choosing a compatible system of determinant scales. The simultaneous segment theorem determines the scales but not these coefficients. Can the coefficient problem be organized by real residue configurations attached to the lowest-valued minors, rather than by the entire original coordinate series?

[[write] *Re-scoped: partly answered in [Part VII](#)*. Each weighted initial form is the real volume polynomial of the residue configuration in a minimum-valuation basis ([theorem 104.2](#); unique up to one element of $GL_d(\mathbb{R})$, [proposition 104.5](#)), and a realizable profile need not have liftable leading coefficients ([theorem 109.2](#)). Compatibility across weights remains open. Source 08's question "Leading coefficients beyond valuations" ([section 95](#)) is the same question.]

Question 43.3 (Joint residue data for several lattices). A single affine residue is full-dimensional, but separate residue shapes do not determine mixed scales. Construct a canonical finite joint reduction for a family of embedded scale lattices that retains their relative position and supports leading-coefficient calculations. For higher-rank valuations, determine when a finite flag of successive reductions is sufficient and how its length depends on the family.

[[write] Related material elsewhere in the report, which leaves the question open. Part II's research direction [section 28.5](#) (manuscript 02) asks for a polyhedral object that keeps the fibres, infinitesimal widths and near-optimality data discarded by the standard part; a joint reduction of several scale lattices would be one such object. Finite flags of successive reductions are proved to suffice in two narrower settings. For the rows of one halfspace system, Part IV's [theorem 49.2](#) needs at most $d + 1$ separated real levels per row. For the heights of a lifting of a fixed real configuration, Part I's [theorem 6.1](#) needs at most $n - d - 1$ real layers. Neither is a joint reduction of several scale lattices.]

[[write] [Proposition 104.5 \(Part VII\)](#) gives a common real reduction of a scaled family, canonical up to one element of $\mathrm{GL}_d(\mathbb{R})$; the higher-rank flag part remains open.]

Question 43.4 (Efficient deterministic inner-segment selection). The rational grid gives an explicit finite existence bound, but exhaustive search is generally large. Find a deterministic selection algorithm with substantially better complexity for the structured product of normalized determinant polynomials in the proof. Separate arithmetic-operation complexity from bit complexity of exponents and coefficients, and compare with the randomized exact search used in the supplement.

Question 43.5 (Minimal probe systems). [Theorem 39.2](#) gives $d + 1$ measurements relative to a reference basis. Determine optimal bounds under various access models: known reference lattice, unknown basis, restricted directions, and uniform tests independent of the reference polytope. The distinction between arbitrary K -directions and real directions should be explicit.

[[write] *Re-scoped*. For uniform tests independent of the reference polytope the answer is negative in the plane: no fixed finite family of mixed-area probes recognizes every scale lattice, even with the area, the spectrum, the shadow and the full residue flag supplied ([theorem 91.2](#)). Optimality of $d + 1$ for the reference-dependent test remains open.]

Question 43.6 (Face specialization and oriented scale data). Characterize when reduction maps faces of P onto faces of $\mathrm{Res}(P)$, and when vertices or entire face intervals disappear. Enrich the lattice and residue data by signed leading minors, and compare the resulting structure with valuated oriented-matroid and regular-subdivision data. A first useful result would be a finite certificate guaranteeing that all faces survive a chosen normalization.

[[write] *Re-scoped: partly answered in this report*. Part I's [proposition 14.2](#) (manuscript 02) shows, for every real functional c , that the standard part of a suitable infinitesimal optimality band $\{x \in P : c \cdot x \geq M - \eta\}$ is exactly the maximizing face of $\mathrm{st} P$. So every exposed face of the reduction is recovered from near-optimal points of P , even when the exact maximizing face of P reduces to a nonface ([\(14.2\)](#), and the pentagon in [section 40](#)). Applied to the bounded normalized polytope $\tilde{P} = B^{-1}(P - p_0)$, whose standard part represents $\mathrm{Res}(P)$ ([theorem 35.3](#)), the same identity describes the faces of the affine residue. Still open: when the faces of P themselves map onto faces of $\mathrm{Res}(P)$, the signed-minor enrichment, the comparison with valuated oriented matroids, and the finite survival certificate. Part II's research direction [section 28.5](#) (manuscript 02) asks for the same kind of replacement for the false face map, and cites the same band identity as its starting point.]

Question 43.7 (Adaptive cancellation certificates). For coordinates supplied as truncatable infinite Hahn series, find hypotheses under which determinant-leading terms can be certified by an adaptive terminating procedure. The height- ε^N examples exclude a bound depending only on vertex count and dimension. The appropriate parameters may involve support geometry, coefficient dependencies, or an input-specific separation certificate rather than a valuation cutoff.

[[write] Part II's [theorem 26.3](#) (manuscript 02) sharpens the obstruction. It gives convergent rational power series coordinates for which the sign of the truncated defect $1 - f_{\leq N} g_{\leq N}$, which decides the truncated hull type, follows any prescribed pattern in N . So a procedure that stops once successive truncations agree for a while can be fooled. The question itself is not answered there.]

Question 43.8 (Dual geometric invariants). Develop mixed-volume and residue invariants involving polar bodies, widths, and successive directional scales. Simple cube comparisons

already control coarse polar lattices in suitably centered situations; the harder target is joint leading-coefficient and incidence information under duality, especially for nonsymmetric bodies and families with different scale lattices.

[[write] *Re-scoped*. For centrally symmetric bodies the coarse polar lattice is exactly the dual module, with reversed spectrum and $v \operatorname{Vol}(P) + v \operatorname{Vol}(P^\circ) = 0$ ([theorem 93.3](#)); nonsymmetric bodies and coefficient-level duality remain open.]

Question 43.9 (Beyond finite vertex sets). Extend the theory to selected definable convex bodies or controlled limits of polytopes. A satisfactory formulation must first specify the volume object and its coefficient field: the finite determinant definition does not automatically extend to arbitrary semialgebraic volume. Determine which additional hypotheses retain a finite inner-model theorem and which introduce genuinely infinite residue or support phenomena.

[[write] *Re-scoped: partly answered*. [Part X](#) treats set-sized presentations: a class that is both a set-generated hull and a set-indexed halfspace intersection has finite subpresentations of both ([theorem 157.2](#)), so definable small hulls are polytopes; the volume half of this question is untouched.]

Question 43.10 (Formal certificates in the repository). Implement and independently check the maximum-simplex module theorem, the mixed determinant squeeze, and the simultaneous grid theorem as separate generic Lean modules. Connect finite-support computational certificates to those statements without assuming unrestricted comparison of surreal expressions. The concrete goal is an auditable end-to-end example in which a higher-rank mixed-volume profile is verified from vertex data and then instantiated in the constructed surreal field.

44 Conclusion

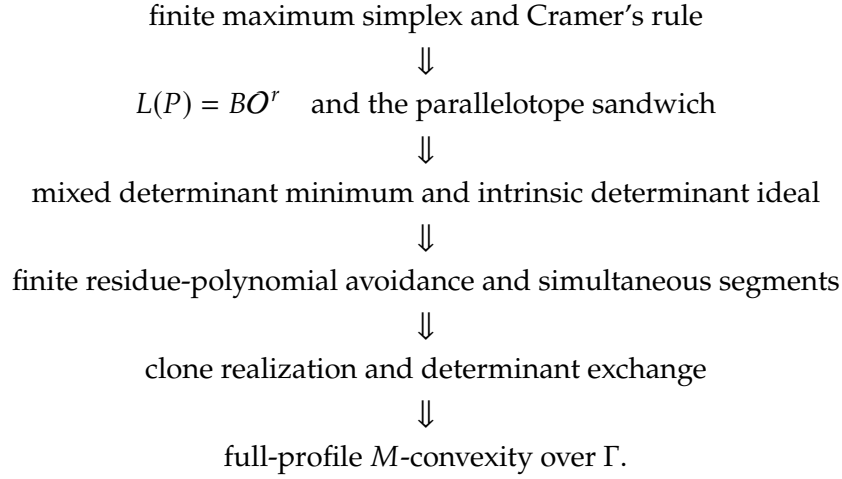
The finite face lattice alone does not distinguish real from surreal polytope geometry. The first meaningful extra layer is the embedded scale lattice; a maximum vertex simplex makes it explicit and supplies a full-dimensional affine residue. The scale of every mixed volume is then a finite determinant minimum.

The simultaneous inner-segment theorem adds a global compatibility fact: all pure mixed-volume scales of a finite family come from one matrix. The clone construction extends this compatibility to repeated entries and produces the full exchange axiom over arbitrary ordered value groups. These results give exact simplification and reconstruction statements, while the examples identify what they deliberately forget: residue coefficients, fine faces, and unrestricted higher-order information.

The resulting research program is finite and algebraic at its core. It can use the surreal field to accommodate very large hierarchies of scales without relying on class-sized measure theory or on a nonexistent universal real-valued parameter. Its next substantive challenge is to recover the coefficient and incidence geometry above those scale data, not merely to reprove the ordinary finite combinatorics over a larger field.

45 Proof dependencies and audit boundaries (Appendix A of manuscript 06)

The main scale results have the following dependency chain:



The affine residue and its leading-volume identity use the separate standard-part volume lemma. The probe classification uses the determinant ideal and the adjugate formula, not the genericity argument. Finite one-parameter specialization is an additional effective result for finite-support coordinates; it is not needed for any existence theorem about arbitrary finite surreal coordinates.

Potential failure mode	How the statements handle it
Proper-class domain	All inputs and constructions are placed in a set-sized real closed subfield of No .
Lower-dimensional body	Its module has its affine rank; a point argument makes the mixed volume zero. Support is decided by Corollary 38.6.
Infinite or infinitesimal size	Maximum-simplex coordinates put the body in a real bounded cube without changing its affine rank.
Cancellation	Determinants are computed exactly. Generic segment selection avoids the finitely many leading cancellations simultaneously.
Higher-rank value group	Proofs use finite comparisons and real residue polynomials, never cofinal multiples of a positive value.
Repeated arguments	One segment per original body is insufficient; parallelotope models or separately labelled clones are used.
Unjustified transfer	Only fixed finite polytope formulas and their semialgebraic determinant volumes are transferred.
Overstated effectiveness	Algorithms assume exact finite-support arithmetic. Arbitrary surreal comparison and polynomial-time bounds are not claimed.
Overstated novelty	Classical ingredients are credited. The combined results are research proofs supplied for review, not certified historical priority claims.

Part IV

Real traces and lattice-point counting

Source and scope. [write] This Part prints, whole, manuscript 03 of batch 58: *Infinitesimal Boundary Geometry of Surreal Polytopes: Lexicographic traces, one-parameter descent, and Ehrhart universality*, headed “Surreal geometry / lattice enumeration”, author line “Research manuscript prepared for Vladimir Reshetnikov” (the PDF metadata of the delivered file reads “Research manuscript prepared with ChatGPT”), dated September 30, 2026 (archive `surreal_polytope_research.zip`, main file `surreal_polytopes/surreal_polytopes.tex`; delivered PDF of 28 pages on Letter paper, not shipped). It was written against ProveIt commit 6996fee43. Every section, result, proof, example, remark, figure, table and research question of the manuscript is printed here, in the manuscript’s order, with three structural changes: its two appendices are printed as the ordinary [sections 60](#) and [61](#); its subsection “A realistic formalization sequence” is printed in the formalization chapter ([section 83.4](#)); and its bibliography is printed as the list “References of Part IV”. This Part is its own spine ([section 1](#)): it shares no theorem with the other Parts beyond two foundations, both marked where they occur. These are the shadow of a finite hull ([lemma 47.2](#), proved in Part I as [proposition 13.1](#)) and finite real-closed-field transfer ([proposition 48.1](#); Part I’s [theorem 18.1](#) and Part V’s [theorem 74.3](#)). The Part keeps its own definitions and hypotheses, fixed in [section 47](#). In this Part “the article” and “this manuscript” mean manuscript 03. As in Part III, every theorem is stated for an arbitrary set-sized real closed field $K \supseteq \mathbb{R}$ with no rank or valuation hypothesis; the surreal field supplies the examples and, through bounded omnific integers, the reason for counting ordinary lattice points.

Notation in this Part. [write] The following symbols mean something different elsewhere in the report ([section 1.3](#)); the Part-local meaning is the one in force here. No symbol of manuscript 03 has been renamed.

- O_K and $\mathfrak{m}_K$ are the finite elements and the infinitesimals: the same ring and ideal as O_K (Parts I–II) and $O, \mathfrak{m}$ (Parts III and V). st is the standard part, and the *shadow* $Q = \text{st}(P)$ is the standard part of the polytope, as in Part I.
- The *real trace* $S = \text{Tr}_{\mathbb{R}}(P) = P \cap \mathbb{R}^d$ (printed $\text{Tr}_{\mathbb{R}}$) is the set of real points of P . It is *not* the shadow: it can be empty or non-closed while the shadow is a full polytope ([example 47.4](#)); in general $\text{int } Q \subseteq S \subseteq Q$ only for a full-dimensional shadow. Nor is it Part V’s matrix trace tr , or Part III’s affine residue Res .
- $L_P(n) = \#(nP \cap \mathbb{Z}^d)$ counts lattice points; it is unrelated to Part III’s scale lattice $L(P)$ and to Part I’s L_h . $\mathbf{Oz}$ is the ring of omnific integers.
- ε and δ denote an *arbitrary* positive infinitesimal of K , not necessarily ω^{-1} ; t is a real parameter tending to 0^+ , or the positive infinitesimal of the ordered field $\mathbb{R}(t)$ (not Part II’s $t = \omega^{-1}$); $\eta > 0$ in [theorem 50.1](#) is a *real* cutoff for that parameter (not Part I’s sign threshold, not $\tau = \omega^{-\omega}$).
- $\rho_1 \gg \cdots \gg \rho_r$ is a separated basis of positive elements of a real subspace of K ([lemma 49.1](#)); it is not Part V’s monomial ρ or Part III’s threshold ρ_g .
- m is the number of rows in [theorem 49.2](#), a period in the proof of [theorem 52.1](#), and a nonsquare integer in [section 57](#) (not Part III’s number of bodies); S, T, G, Q, B, E, H are the trace, an implanted real polytope, a face, the shadow, a bounding box, a realization space or generating function, and a halfspace intersection or generating function (not

Part V's slack matrix S or triangle T); $A(Q)$ is the lattice-normalized boundary measure and μ_F the normalized facet measure; V_i are coefficient spaces (not Part I's dependence space); $\beta(P)$ is the surface coefficient.

- vol_d is Euclidean volume of real sets; on real polytopes it agrees with Part III's determinant volume Vol_d .
- No valuation is used in this Part (see the remark after [theorem 50.1](#)).

Abstract of manuscript 03. A finite surreal polytope has an ordinary realizable face lattice, but its infinitesimal boundary placement can carry arithmetic information invisible in that lattice and in its standard-part shadow. We give a finite lexicographic description of its real trace, with at most $d + 1$ levels per inequality in dimension d . Combining this description with semialgebraic curve selection gives a one-parameter algebraic realization preserving the entire real trace, the shadow, the labeled face lattice, and every ordinary lattice-counting value. We then construct a face-implantation operation that imports arbitrary lower-dimensional real-polytope counting functions into a fixed rational shadow. For a lattice shadow, we prove an explicit second asymptotic coefficient and determine its full attainable interval. If the shadow is simple, every value in this interval is attained without changing the labeled face lattice or any vertex standard part. Finally, a quadrilateral with the four integral vertex shadows of the unit square has counting function $(n + 1)^2 - \lceil \alpha n \rceil$ for any irrational $0 < \alpha < 1$; its generating function has the unit circle as a natural boundary and is not D-finite.

Status and scope (title page of manuscript 03). This manuscript supplies mathematical proofs and exact finite checks. It has not been independently refereed or formalized in Lean. Its classical inputs are explicitly attributed; priority for the assembled results is not certified. No named published open conjecture is claimed solved. All counting statements concern ordinary integer dilations of polytopes bounded by a real box.

46 The research questions and their answers

The relevant starting point in ProveIt is its construction of the surreal field and its omnific-integer subring, together with its distinction between research manuscripts and mapped formal theorems [26, 27]. The particularly useful bridge is that an omnific integer bounded in absolute value by a real number is an ordinary integer. The repository contains an explicit Lean declaration of this statement [28]. Consequently, finite polytopes in bounded surreal space lead to a precise, ordinary lattice-enumeration problem. Their coordinates may be non-Archimedean; the lattice points being counted are not.

The organizing question is not whether arbitrary finite combinatorics becomes possible over the surreals. Real-closed-field transfer already rules that out for finite face lattices. Rather, we ask what information survives when one retains the real numbers and their integer lattice while moving supporting hyperplanes through infinitesimal distances.

46.1 Four questions posed and resolved here

Does the shadow determine lattice enumeration? No. Even a surreal quadrilateral with exactly the four integral vertex shadows of the unit square can have non-quasipolynomial counts. Its generating function can have infinitely many analytic obstructions in the strongest possible form: every point of the unit circle is a singular boundary point. The construction and proof appear in [Section 52](#).

Can arbitrarily deep surreal scales be necessary for the real trace? Not for a finite system in fixed dimension. An individual affine inequality in d coordinates needs at most $d + 1$ lexicographic levels on real inputs. A stronger global theorem replaces the whole polytope by one algebraic parameter while preserving its trace and finite combinatorics simultaneously; see Sections 49–50. This is a statement about a specified finite geometric object, not about collapsing the surreal field or its full value group.

How much counting behavior can be hidden on a face? Every real polytope lying in that face can be implanted. If G is a proper face of a rational polytope Q and $T \subseteq G$ is any real polytope, we construct P with shadow Q and

$$L_P(n) = L_Q(n) - L_G(n) + L_T(n) \quad (n \geq 1).$$

The last summand need not be an Ehrhart quasipolynomial. This isolates a universality mechanism rather than merely an isolated counterexample.

What remains of the first two Ehrhart coefficients? For a full-dimensional lattice shadow Q , the leading coefficient is always $\text{vol}_d(Q)$. The next coefficient is the sum of the normalized measures of the portions of facets retained in the real trace, minus half the normalized boundary measure of Q . Its complete attainable interval is

$$\left[-\frac{1}{2} \sum_F \mu_F(F), \frac{1}{2} \sum_F \mu_F(F) \right].$$

For a *simple* lattice shadow every value is attainable by perturbing its existing facets, with no change in the labeled face lattice and no change in any vertex standard part. Sections 53 and 54 prove these statements.

46.2 What is classical, and what is being assembled

The real-closed-field transfer principle and semialgebraic curve selection are classical [30, 31, 32]. Lexicographic inequalities are also established objects in optimization [33]. Classical Ehrhart theory and reciprocity apply to rational polytopes [34]. Irrationality does not by itself preclude polynomial enumeration: period collapse already occurs for irrational triangles [35]. Finally, the natural-boundary step below is an application of the classical Pólya–Carlson theorem, not a new complex-analytic theorem [36].

The contributions developed here are the finite trace normal form and its trace-preserving descent application, the face-implantation construction, the surface-coefficient formula and its sharp realization theorem, and the explicit connection of these statements to bounded surreal and omnific geometry. Their proofs are given in full, except for the clearly named classical inputs. These results may overlap consequences of existing non-Archimedean convexity or real-algebraic geometry; the literature checks recorded here are not an exhaustive priority search.

47 Fields, finite polytopes, shadows, and traces

47.1 A set-sized setting

Throughout, K is a set-sized real closed ordered field containing $\mathbb{R}$ as an ordered subfield. When an infinitesimal is needed we suppose that $0 < \varepsilon < r$ for every $r \in \mathbb{R}_{>0}$. Define

$$O_K = \{x \in K : |x| \leq r \text{ for some } r \in \mathbb{R}_{>0}\}, \quad \mathfrak{m}_K = \{x \in K : |x| < r \text{ for every } r \in \mathbb{R}_{>0}\}.$$

Completeness of $\mathbb{R}$ gives a unique standard part $\text{st}(x) \in \mathbb{R}$ for $x \in O_K$, characterized by $x - \text{st}(x) \in \mathfrak{m}_K$. To see existence, take the supremum of the real numbers below x ; the defining property of the supremum excludes a positive real distance in either direction. Uniqueness follows because no nonzero real number is infinitesimal. Directly from this characterization,

$$\text{st}(x + y) = \text{st}(x) + \text{st}(y), \quad \text{st}(xy) = \text{st}(x) \text{st}(y),$$

and $x \geq 0$ implies $\text{st}(x) \geq 0$. Thus O_K is a convex subring, $\mathfrak{m}_K$ is its maximal ideal, and its residue field is $\mathbb{R}$.

A finite tuple of surreal coordinates can be placed in such a field: take inside **No** the real closure of the field generated by $\mathbb{R}$ and that tuple. This uses the classical real closedness of the surreal field [29]. All constructions below take place in this set-sized field. No application of first-order model theory treats the proper class **No** as a set-sized structure, and no saturation hypothesis is used.

Definition 47.1. A *finite K -polytope* is the convex hull of finitely many points of K^d , with convex combinations using nonnegative coefficients in K . It is *real-bounded* if it is contained in $[-M, M]^d$ for some $M \in \mathbb{R}_{>0}$. For such a nonempty polytope P , its *shadow* and *real trace* are

$$Q = \text{st}(P) = \{\text{st}(x) : x \in P\} \subseteq \mathbb{R}^d, \quad S = \text{Tr}_{\mathbb{R}}(P) = P \cap \mathbb{R}^d.$$

Standard part is applied coordinatewise.

Finite-dimensional polyhedral linear algebra works over real closed fields. In particular a finite convex hull has a finite halfspace representation, allowing pairs of opposite inequalities to encode its affine hull, and a nonempty bounded intersection of finitely many closed halfspaces is a finite convex hull. These statements can be obtained from finite elimination, or from their fixed-size first-order real versions by transfer [30]. We never invoke topological compactness of a bounded subset of K^d .

Lemma 47.2 (The shadow of a finite convex hull). *If $P = \text{conv}_K(v_1, \dots, v_N)$ is real-bounded, then*

$$\text{st}(P) = \text{conv}_{\mathbb{R}}(\text{st}(v_1), \dots, \text{st}(v_N)).$$

In particular, its shadow is an ordinary closed real polytope.

Proof. If $x = \sum_j \lambda_j v_j$ with $\lambda_j \geq 0$ and $\sum_j \lambda_j = 1$, then every λ_j is finite and its standard part is nonnegative. Taking standard parts produces a real convex combination. Conversely, a real convex combination of the shadow vertices is the standard part of the same real combination of their lifts. $\square$

[[write] This is Part I's [proposition 13.1](#), where it is proved (manuscripts 05 and 02), and the first clause of Part III's [lemma 34.2](#). Manuscript 03's statement and proof are kept because the rest of this Part cites them.]

Lemma 47.3 (Interior points survive). *If $Q = \text{st}(P)$ has dimension d , then*

$$\text{int } Q \subseteq S \subseteq Q, \quad \text{cl}_{\mathbb{R}} S = Q.$$

Proof. The second inclusion is immediate because $\text{st}(x) = x$ for real x . For $x \in \text{int } Q$, choose a real d -simplex in Q containing x in its interior. Its vertices $q_0, \dots, q_d$ have lifts $p_0, \dots, p_d \in P$ by Lemma 47.2. The matrix whose columns are $(1, p_j)$ has determinant with nonzero standard part. It is therefore invertible over K . The solution of

$$\sum_{j=0}^d \lambda_j p_j = x, \quad \sum_{j=0}^d \lambda_j = 1$$

has standard part equal to the strictly positive barycentric coordinates of x in the real simplex. Hence every $\lambda_j > 0$, so $x \in P$. Taking closures gives the last assertion. $\square$

Example 47.4 (A shadow is not a trace). The segment $\{(x, \varepsilon) : 0 \leq x \leq 1\}$ has shadow $[0, 1] \times \{0\}$ and empty real trace. Thus the dimension assumption in Lemma 47.3 cannot be dropped. Even for full-dimensional shadows the trace need not be closed: $[\varepsilon, 1 - \varepsilon]$ has trace $(0, 1)$ and shadow $[0, 1]$.

47.2 Ordinary and omnific lattice points

Conway's omnific integers are the surreal normal forms with no negative exponents and with an ordinary integer constant coefficient [29]. A nonconstant such form has a nonzero leading term of positive exponent, so it is not bounded in absolute value by a real number. It follows that

$$\mathbf{Oz} \cap \mathbf{O}_{\mathbf{No}} = \mathbb{Z}. \quad (47.1)$$

This is also the content of the repository theorem `omnific_bounded_iff`; Section 56 records the exact path and the verification boundary [28].

[[write] `omnific_bounded_iff` is the one Lean declaration that any Part of this report uses as an input; the other Parts cite Lean declarations at most as background. It states that an omnific integer whose surreal value is bounded in absolute value by a real number is the cast of an ordinary integer. Re-verified at the write phase: it is at line 65 of `Algebra/SurrealNumbers/Surreal/Foundations/OmnificUnits.lean`, in namespace `Surreal.Foundations.SignSequence`, next to `omnific_isFinite_iff` (line 55) and `omnific_eq_intConstant_of_finite` (line 42), and the file is unchanged since the pin. Manuscript 03 also proves (47.1) directly from normal forms, as above. *None of this Part's own results is formalized*: the Lean statement covers only the bridge (47.1), not the trace, descent, implantation, counting or surface-coefficient theorems.]

For any real-bounded K -polytope and an *ordinary* positive integer n , define $L_P(n) = \#(nP \cap \mathbb{Z}^d)$. In the surreal case,

$$L_P(n) = \#(nP \cap \mathbf{Oz}^d) = \#(nP \cap \mathbb{Z}^d) = \#(nS \cap \mathbb{Z}^d). \quad (47.2)$$

The omnific and integer sets agree coordinatewise by (47.1); the last equality holds over any K because $x/n \in \mathbb{R}^d$ for real x . The counts are finite, and we set $L_P(0) = 1$ for nonempty P . For a real polytope or bounded real set T , write $L_T(n) = \#(nT \cap \mathbb{Z}^d)$ when the ambient space is understood.

Equation (47.2) is a reduction to the *real trace*, not to the shadow. It also explains why the integer-part structure does not add new bounded lattice points. Infinite omnific dilations, infinite-coordinate polytopes, and counts indexed by a proper class are outside our scope.

48 The classical limit: no new finite face lattices

Proposition 48.1 (Finite combinatorial transfer). *Every finite polytope over a real closed ordered field has a realization over the real algebraic numbers with the same labeled vertex–facet incidences, and consequently the same face lattice. Real boundedness is not required.*

Proof. Choose a finite vertex list and a finite inequality description, including affine-hull equations when necessary. With their sizes fixed, the following requirements form a finite first-order formula in the language of ordered fields: the convex hull equals the halfspace intersection; the listed points are pairwise distinct vertices; and each specified vertex–row slack is zero or strictly positive according to the original incidence data. Convex-hull membership is expressed with finitely many existential barycentric coordinates. Extremality is expressed by exclusion from the convex hull of the other listed vertices. H/V equality is expressed by quantifying a point of the fixed ambient dimension. None of these formulas names a nonrational parameter: all coordinates and coefficients are variables.

The formula has a solution in the given real closed field. Completeness of the theory of real closed fields gives a solution in the real algebraic numbers [30]. Every face of a polytope is an intersection of its supporting facets, so the preserved incidence data determine the labeled face lattice. Equivalently one can transfer the finite determinant sign pattern of a homogeneous vertex configuration. $\square$

This observation prevents a misleading research claim: finite surreal polytopes do not create previously unrealizable finite face lattices. It does not say that all face lattices have rational realizations, that given coordinates descend, or that standard parts and valuations are preserved. Most importantly, quantification over $\mathbb{Z}$ is not part of the pure ordered-field language. Proposition 48.1 gives no transfer theorem for lattice-counting functions.

[[write] Proposition 48.1 is the finite real-closed-field transfer that Part I proves as theorem 18.1 (manuscripts 05 and 02); Part III’s proposition 33.3 is another instance. Part V’s theorem 74.3 (manuscript 07) strengthens it for polytopes over a real closed field: a real-algebraic realization can keep the oriented matroid and the linear and semidefinite extension complexities as well. The limitations just stated apply to all of these: none transfers standard parts, valuations or lattice counts.]

The rest of this article retains exactly the data discarded by that proposition: the distinguished real subfield, its standard-part map, and the boundary trace on the ordinary integer lattice.

49 A finite lexicographic normal form for real traces

For positive $a, b \in K$, write $a \gg b$ if b/a is infinitesimal. For a finite tuple of real numbers, lexicographic nonnegativity means either that every entry is zero or that its first nonzero entry is positive.

Lemma 49.1 (Separated real basis). *Let $V \subseteq K$ be a nonzero real vector space of finite dimension r . There is a basis of positive elements*

$$\rho_1 \gg \rho_2 \gg \cdots \gg \rho_r > 0$$

such that, for all $c_1, \dots, c_r \in \mathbb{R}$,

$$\operatorname{sgn}\left(\sum_{j=1}^r c_j \rho_j\right) = \operatorname{sgn}(c_k)$$

when k is the first index with $c_k \neq 0$.

Proof. Choose a real basis $v_1, \dots, v_r$ and let ρ_1 be the largest of $|v_1|, \dots, |v_r|$. It belongs to V , is positive, and every v/ρ_1 with $v \in V$ is finite: the coefficients of a fixed finite real linear combination give a real bound. Therefore

$$c_1 : V \longrightarrow \mathbb{R}, \quad c_1(v) = \text{st}(v/\rho_1)$$

is real linear and takes ρ_1 to 1. Its kernel has dimension $r - 1$, and every element of the kernel is infinitesimal relative to ρ_1 . Moreover,

$$V = \mathbb{R}\rho_1 \oplus \ker c_1.$$

Apply the same argument inside the kernel. Induction gives an exact basis expansion after r steps. Each later basis element divided by an earlier one is infinitesimal. After dividing a nonzero combination by its first active ρ_k , its leading coefficient is a nonzero real number and the remaining finite sum is infinitesimal. This proves the sign rule. $\square$

This argument uses exact finite-dimensional linear algebra. It does not truncate a Hahn series, assume that its support is finite, or require that the value group have finite rank.

Theorem 49.2 (Lexicographic trace theorem). *Let*

$$P = \{x \in K^d : b_i - a_i \cdot x \geq 0, \ 1 \leq i \leq m\}$$

be any finite halfspace system. For every nonzero row there exist real affine forms $\ell_{i1}, \dots, \ell_{ir_i}$, where $r_i \leq d + 1$, such that

$$P \cap \mathbb{R}^d = \{x \in \mathbb{R}^d : (\ell_{i1}(x), \dots, \ell_{ir_i}(x)) \geq_{\text{lex}} 0 \text{ for all } i\}. \quad (49.1)$$

More precisely, there are positive separated scales ρ_{ik} with

$$b_i - a_i \cdot x = \sum_{k=1}^{r_i} \rho_{ik} \ell_{ik}(x) \quad (x \in K^d). \quad (49.2)$$

Zero rows may be deleted.

Proof. Apply Lemma 49.1 to

$$V_i = \text{span}_{\mathbb{R}}\{b_i, a_{i1}, \dots, a_{id}\}.$$

Its dimension is at most $d + 1$. Expand each coefficient in the separated basis and collect the real affine forms multiplying the basis elements. This gives the exact identity (49.2). When x is real, all $\ell_{ik}(x)$ are real, so the sign rule is precisely (49.1). $\square$

Corollary 49.3 (Finite semilinear structure). *The real trace of a real-bounded finite K -polytope is a convex semilinear set. It is a finite disjoint union of relatively open real polytopes.*

Proof. A lexicographic condition is a finite union of conditions of the form

$$\ell_1 = \dots = \ell_{k-1} = 0, \quad \ell_k > 0,$$

together with the all-zero condition. Thus it is semilinear. Subdivide a real bounding box by all the affine hyperplanes in the description and by its boundary hyperplanes. Each relatively open cell has a constant truth value for the trace formula. The selected cells give the stated finite disjoint union; boundedness makes their closures polytopes. Convexity follows from convexity of P and the real subfield. $\square$

Example 49.4 (Why leading standard parts are insufficient). The inequality

$$1 - y + \varepsilon(x - \alpha) \geq 0$$

on real (x, y) first tests $1 - y$. Only on the line $y = 1$ does it test $x - \alpha$. Taking the standard part of its coefficients loses the second test and incorrectly keeps the entire line segment on that boundary.

49.1 Polynomial compression, without a combinatorial guarantee

Corollary 49.5 (A degree- d one-infinitesimal trace description). *Suppose P is real-bounded and its shadow is full-dimensional. Fix a real bounding box B containing P . For any positive infinitesimal δ , set*

$$P_{\text{flat}} = \left\{ x \in B_K : \sum_{k=1}^{r_i} \delta^{k-1} \ell_{ik}(x) \geq 0 \text{ for all } i \right\}.$$

Then P_{flat} is a nonempty real-bounded polytope with the same real trace and shadow as P . Its row coefficients are polynomials in δ of degree at most d . In the surreal setting all its ordinary counting values agree with those of P .

Proof. On real inputs powers of δ implement exactly the same lexicographic sign rule. Thus the traces agree. The common trace contains $\text{int } Q$ by Lemma 47.3, so P_{flat} is nonempty and its shadow is full-dimensional. Applying that lemma to both polytopes shows that both shadows are the closure of the common trace. Equation (47.2) proves the counting claim. $\square$

This compression need not preserve the original face lattice. Different rows may interact through their relative infinitesimal scales at nonreal points. The next theorem preserves those interactions by selecting an algebraic curve in an appropriate realization space rather than replacing each row independently.

50 One-parameter algebraic descent preserving the entire trace

An *algebraic semialgebraic germ* below is a germ at $t = 0^+$ of a real semialgebraic function algebraic over $\mathbb{R}(t)$. It can be chosen Nash on a sufficiently small positive interval. The coefficients of its defining polynomial may be arbitrary real numbers; “algebraic” here does not mean “defined over $\mathbb{Q}$.”

Theorem 50.1 (Trace-preserving algebraic descent). *Let $P \subseteq K^d$ be a nonempty real-bounded finite polytope, with its distinct vertices labeled $v_1, \dots, v_N$. There exist $\eta > 0$ and a family of ordinary real polytopes P_t , $0 < t < \eta$, whose labeled vertices and inequality coefficients are algebraic semialgebraic germs, such that:*

- (i) *every P_t has the same labeled vertex–inequality incidences and face lattice as P ;*
- (ii) *its labeled vertices $v_j(t)$ tend to $\text{st}(v_j)$;*
- (iii) *for every fixed real point x , membership $x \in P_t$ is eventually constant and equals membership $x \in \text{Tr}_{\mathbb{R}}(P)$.*

Evaluating these germs at a positive infinitesimal gives a polytope P^ in a real closure of $\mathbb{R}(t)$, embedded in a real closed extension, with exactly the same real trace, shadow, and labeled face lattice as P . All ordinary lattice counts agree. For every fixed ordinary $n \geq 1$,*

$$L_{P_t}(n) = L_P(n)$$

for all sufficiently small positive real t , where the cutoff may depend on n . No full-dimensional-shadow hypothesis is needed.

Proof. We give the finite encoding because preserving all real points at once is stronger than preserving a finite determinant sign list.

Step 1: encode every row with adjacent ratios. Use the exact expansion (49.2) and divide row i by its positive ρ_{i1} . Introduce one variable u_{ik} for each adjacent ratio, and write

$$f_i(u, x) = \ell_{i1}(x) + u_{i1}\ell_{i2}(x) + u_{i1}u_{i2}\ell_{i3}(x) + \cdots + (u_{i1} \cdots u_{i,r_i-1})\ell_{ir_i}(x). \quad (50.1)$$

The original ratio tuple $u_{ik}^0 = \rho_{i,k+1}/\rho_{ik}$ consists of positive infinitesimals. Substituting it recovers the original normalized rows exactly. Choose a closed real box B containing P strictly inside its boundary.

Step 2: construct a semialgebraic realization space. Let $z = (z_1, \dots, z_N)$ be variable points. Let E be the set of real pairs (u, z) satisfying the following finite first-order conditions: $0 < u_{ik} < 1$; all $z_j \in B$; the polyhedron

$$H(u) = \{x \in B : f_i(u, x) \geq 0 \text{ for every } i\}$$

equals $\text{conv}(z_1, \dots, z_N)$; the z_j are distinct and each lies outside the convex hull of the others; and each slack $f_i(u, z_j)$ is zero or strictly positive according to the original row-vertex table. The box inequalities can be included in this table as well.

H/V equality and exclusion from a finite convex hull have first-order formulas with finitely many barycentric variables. Quantifier elimination therefore makes E a semialgebraic set over $\mathbb{R}$ [30]. The same formula interpreted over K contains the original point (u^0, v) .

Step 3: approach the standard-part endpoint over the reals. Put $a = (0, \text{st}(v_1), \dots, \text{st}(v_N))$. For every real $r > 0$, the point (u^0, v) lies in the radius- r ball about a . Hence the existential sentence asserting a point of E in that ball is true over K and, by elementary transfer over the named real parameters, over $\mathbb{R}$. Therefore $a \in \text{cl } E$.

When there are ratio variables, $a \notin E$ because they must be positive. Semialgebraic Nash curve selection gives a curve

$$(u(t), z(t)) \in E, \quad (u(t), z(t)) \longrightarrow a \quad (t \downarrow 0),$$

with algebraic semialgebraic coordinates [31, 32]. If there are no ratio variables, all rows are positive multiples of real rows, the polytope and its vertices are already real, and the constant family suffices. Define $P_t = H(u(t))$.

Step 4: preserve geometry and the whole trace. By the definition of E , the labeled vertices and all their row incidences are unchanged for every sufficiently small positive t . These data preserve the face lattice, including when the shadow loses dimension. The limit of the vertex list gives (ii).

Fix any real x . For a given row, suppose k is the first index with $\ell_{ik}(x) \neq 0$. Divide (50.1) by the positive product $u_{i1}(t) \cdots u_{i,k-1}(t)$. Its value tends to the nonzero real number $\ell_{ik}(x)$, since every later term contains a product of ratios tending to zero. If all the forms vanish, the row vanishes identically. Thus each row eventually has exactly its original lexicographic truth value at x . There are only finitely many rows, which proves (iii). Points outside the fixed box are excluded throughout.

Step 5: evaluate at an infinitesimal. The ordered field $\mathbb{R}(t)$ with t positive infinitesimal embeds by $t \mapsto \delta$ into any real closed extension containing such a δ . Here δ is transcendental over the real closed field $\mathbb{R}$. The embedding extends to a real closure; the selected real algebraic germs are elements of that ordered real closure. Semialgebraic eventual identities and inequalities are preserved by this evaluation. In particular the defining formula for E remains true, and the row argument now gives exact membership agreement for every real x . Vertex limits become standard parts, proving equality of shadows by Lemma 47.2. Equation (47.2) gives the counting claim.

Finally $P_t \subseteq B$ for all small t . For fixed n there are only finitely many candidates in $nB \cap \mathbb{Z}^d$. Apply (iii) to their quotients by n and take the minimum of finitely many positive cutoffs. This proves the last assertion without asserting a cutoff uniform in n . $\square$

Remark 50.2 (What descent does not preserve). The theorem does not preserve arbitrary valuations of all rational expressions, the original field embedding, or the infinite hierarchy of scales outside the finite description. It supplies algebraic germs, not necessarily polynomial coordinate functions or an effective degree bound. The finite degree bound of Corollary 49.5 concerns a weaker, trace-only construction. Confusing these two conclusions would erase a substantive distinction.

Remark 50.3 ([write] Descent and Part V). The polytope P^* of theorem 50.1 lies in a real closure of $\mathbb{R}(t)$, whose natural value group is $\mathbb{Q}$, of rank one. There every positive infinitesimal has an order-unit value, so Part V's flat ideal of beyond-all-orders elements (its jet kernel) is zero (theorem 72.2, manuscript 07; theorem 72.7, manuscript 04). Finite coordinate tuples of P^* are determined by their power jets, and the invisible-complexity phenomena of Part V cannot occur for P^* . This does not conflict with Part V. As the preceding remark says, descent keeps the real trace, the shadow, the labeled face lattice and the lattice counts, and it expressly does not keep valuations, the field embedding or the hierarchy of scales. Manuscript 03 uses no valuation at all, and claims nothing about the jets or the extension complexity of P .

Corollary 50.4. *Every ordinary lattice-counting sequence of a real-bounded finite surreal polytope is already the sequence of a polytope over a real closure of a one-infinitesimal extension $\mathbb{R}(t)$, with the same labeled finite combinatorics and the same shadow.*

This is a reduction of the field needed for a finite observation. It does not turn the sequence into an ordinary Ehrhart quasipolynomial: the next sections show why one infinitesimal is already sufficient for substantial arithmetic complexity.

51 Face implantation and boundary universality

Let $Q \subseteq \mathbb{R}^d$ be a full-dimensional rational polytope and G a nonempty proper face. Choose a real affine function s_G nonnegative on Q whose zero set there is exactly G . Such a function can be obtained by adding the nonnegative facet slacks of the facets whose intersection is G . For a real polytope $T \subseteq G$, choose finitely many real affine functions $h_1, \dots, h_r$ on $\mathbb{R}^d$ such that

$$T = \{x \in G : h_j(x) \geq 0 \text{ for all } j\}.$$

An inequality description on $\text{aff } G$ extends to the ambient space by linear algebra. Lower-dimensional T is allowed.

Theorem 51.1 (Face implantation). *For a positive infinitesimal ε , set*

$$P_\varepsilon = \{x \in Q_K : s_G(x) + \varepsilon h_j(x) \geq 0 \text{ for } 1 \leq j \leq r\}. \quad (51.1)$$

Then P_ε is a nonempty real-bounded polytope with

$$\text{st}(P_\varepsilon) = Q, \quad \text{Tr}_{\mathbb{R}}(P_\varepsilon) = (Q \setminus G) \cup T. \quad (51.2)$$

For every ordinary $n \geq 1$,

$$L_{P_\varepsilon}(n) = L_Q(n) - L_G(n) + L_T(n). \quad (51.3)$$

The empty choice $T = \emptyset$ is also possible by taking the single function $h_1 = -1$.

Proof. For a real $x \in Q \setminus G$, the positive real number $s_G(x)$ dominates each infinitesimal $\varepsilon h_j(x)$, so all the added inequalities hold. For real $x \in G$, they hold exactly when all $h_j(x) \geq 0$. This proves the trace identity. In particular the trace contains $\text{int } Q$, so P_ε is nonempty. It is a bounded halfspace intersection, hence a polytope.

Its shadow is contained in Q because $P_\varepsilon \subseteq Q_K$. It contains $\text{int } Q$ and is closed by Lemma 47.2; hence it contains Q . The shadow identity follows. For $n > 0$, dilation is injective and the trace identity is a disjoint replacement of G by T . Counting lattice points gives (51.3). The argument is unchanged when T is empty. $\square$

Corollary 51.2 (A fixed-cube universality formula). *For every real polytope $T \subseteq [0, 1]^{d-1}$, there is a real-bounded K -polytope with unit-cube shadow and*

$$L_P(n) = n(n+1)^{d-1} + L_T(n) \quad (n \geq 1). \quad (51.4)$$

Proof. Implant $T \times \{1\}$ on the top facet of $[0, 1]^d$. Then $L_Q(n) = (n+1)^d$ and $L_G(n) = (n+1)^{d-1}$. $\square$

A boundary perturbation can therefore contain the entire counting sequence of an arbitrary real polytope of one lower dimension, modulo the explicit polynomial in (51.4). Rationality of the shadow alone cannot control quasipolynomiality, generating-function rationality, or recurrence properties of the resulting sequence. This is a structural explanation of the counterexamples, not a claim that irregular enumeration of irrational polytopes is a newly discovered phenomenon.

Proposition 51.3 (Localization by face dimension). *If the implanted face has dimension k , then*

$$L_{P_\varepsilon}(n) - L_Q(n) = O(n^k).$$

Thus implantation on a face of codimension c cannot change terms of order strictly higher than n^{d-c} .

Proof. Both nG and nT lie in a fixed k -dimensional affine space dilated by n , and in a ball of radius $O(n)$ within it. Their distinct integer points are at least unit distance apart. Packing disjoint balls of fixed radius in that affine space gives $O(n^k)$ points. Apply (51.3). For $k = 0$ the assertion simply says $O(1)$. $\square$

52 An integral-shadow quadrilateral with a natural boundary

The general implantation formula already gives a pentagonal example by cutting a thin wedge from a square. A more economical construction perturbs one existing facet and uses only four vertices.

Theorem 52.1 (The irrational boundary quadrilateral). *Fix an irrational real number $0 < \alpha < 1$ and a positive infinitesimal ε . Define*

$$P_{\alpha,\varepsilon} = \{(x, y) \in K^2 : 0 \leq x \leq 1, \quad 0 \leq y \leq 1 + \varepsilon(x - \alpha)\}. \quad (52.1)$$

It is a quadrilateral with cyclically ordered vertices

$$(0, 0), \quad (1, 0), \quad (1, 1 + \varepsilon(1 - \alpha)), \quad (0, 1 - \varepsilon\alpha). \quad (52.2)$$

Its shadow is $[0, 1]^2$, every vertex shadow is integral, and its labeled face lattice is that of the square. Nevertheless,

$$\boxed{L_{P_{\alpha,\varepsilon}}(n) = (n + 1)^2 - \lceil \alpha n \rceil} \quad (n \geq 0). \quad (52.3)$$

For $n \geq 1$ this is

$$L_{P_{\alpha,\varepsilon}}(n) = n^2 + (2 - \alpha)n + \{\alpha n\}. \quad (52.4)$$

The sequence is not eventually quasipolynomial. Its ordinary generating function has the unit circle as a natural boundary; it is not D-finite, and the sequence is not P-recursive.

Proof. The upper bounds on the two vertical sides in (52.1) are positive. The four intersections in (52.2) are the four vertices, with the stated incidences. Their standard parts prove the shadow assertion. On real points the upper inequality tests $1 - y$ first and $x - \alpha$ only when $y = 1$. Thus

$$\mathrm{Tr}_{\mathbb{R}}(P_{\alpha,\varepsilon}) = ([0, 1] \times [0, 1)) \cup ([\alpha, 1] \times \{1\}). \quad (52.5)$$

For a positive ordinary n , the admissible integer columns are $x = 0, 1, \dots, n$. The rows $y = 0, 1, \dots, n - 1$ occur in every column. At $y = n$, the last inequality is $\varepsilon(x - \alpha n) \geq 0$, so the top row retains exactly $n + 1 - \lceil \alpha n \rceil$ points. There are no higher integer rows. Hence

$$L(n) = n(n + 1) + n + 1 - \lceil \alpha n \rceil.$$

The convention at $n = 0$ agrees with (52.3). Since αn is nonintegral for $n > 0$, expanding the ceiling gives (52.4).

Suppose the sequence were eventually quasipolynomial with period m . On each progression $n \equiv r \pmod{m}$ it would eventually equal a polynomial $p_r(n)$. By (52.4), the polynomial $p_r(n) - n^2 - (2 - \alpha)n$ is bounded on that progression and so is constant. Consequently $\{\alpha(n + m)\} = \{\alpha n\}$ for sufficiently large n in the progression. This would imply $\alpha m \in \mathbb{Z}$, a contradiction. The argument even excludes constituent polynomials with arbitrary real coefficients.

Write $a_n = \lceil \alpha n \rceil$ for $n \geq 0$, and $w_n = a_{n+1} - a_n$. Then $w_n \in \{0, 1\}$ and

$$\frac{1}{N} \sum_{n=0}^{N-1} w_n = \frac{a_N}{N} \longrightarrow \alpha. \quad (52.6)$$

If $E(z) = \sum_{n \geq 0} L(n)z^n$ were rational, so would be $H(z) = \sum_{n \geq 0} a_n z^n$, because

$$\sum_{n \geq 0} (n + 1)^2 z^n = \frac{1 + z}{(1 - z)^3}.$$

Since $a_0 = 0$, the series of the increments is

$$W(z) = \sum_{n \geq 0} w_n z^n = \frac{1 - z}{z} H(z),$$

which would also be rational. The coefficients of a rational series satisfy an eventual fixed-length linear recurrence with constant coefficients. For a finite-valued sequence, the finitely many possible recurrence state vectors evolve deterministically. A state must repeat, after which the sequence is periodic. An eventually periodic 0–1 sequence has a rational limiting mean, contradicting (52.6). Thus E is not rational.

Its coefficients are integers and grow quadratically, so its radius of convergence is exactly 1. The Pólya–Carlson theorem now implies that the unit circle is a natural boundary [36, Theorem 1.1]. An analytic D-finite germ satisfies a linear differential equation with polynomial coefficients. Away from the finitely many zeros of its leading coefficient, solutions analytically continue locally along paths. Such a germ cannot have an entire circle as a natural boundary. Hence E is not D-finite. The coefficient correspondence between D-finite series and P-recursive sequences gives the final assertion [38]. $\square$

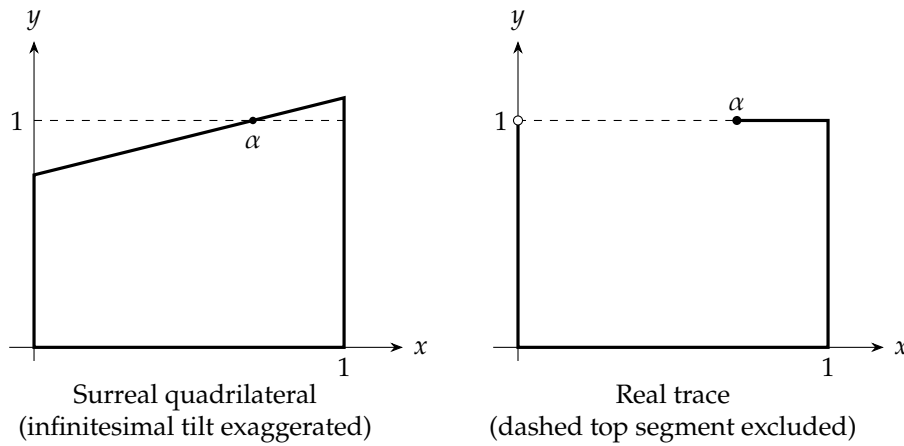


Figure 5: A four-vertex perturbation with a square shadow. The trace removes $[0, \alpha) \times \{1\}$, including the upper-left corner, while retaining the rest of the square. The left drawing is schematic, not a numerical representation of an infinitesimal.

Corollary 52.2 (Algebraic input already suffices). *Taking $\alpha = 1/\sqrt{2}$ gives the phenomenon over a real closure of $\mathbb{Q}(\sqrt{2})(\varepsilon)$, with coefficients linear in one infinitesimal. Thus neither real-algebraic flag coefficients, integral vertex shadows, nor simplicity of the shadow guarantees a quasipolynomial counting function.*

Corollary 52.3 (Every dimension at least two). *For every $d \geq 2$ there is a d -polytope with the labeled combinatorics and integral vertex shadows of the unit d -cube whose counting series has the unit circle as a natural boundary and whose sequence is not P-recursive.*

Proof. Take $P_{\alpha,\varepsilon} \times [0, 1]^{d-2}$. Its counts are

$$(n+1)^{d-2}((n+1)^2 - \lceil \alpha n \rceil).$$

If these counts satisfied a polynomial-coefficient recurrence, substitution would give one for $L_{P_{\alpha,\varepsilon}}(n)$: replace each coefficient of the shifted product by that coefficient times $(n+j+1)^{d-2}$. Nonzero polynomial coefficients remain nonzero. This contradicts Theorem 52.1. The series is therefore nonrational; its integer coefficients have polynomial growth and radius 1, so Pólya–Carlson applies again. Non-quasipolynomiality follows as well, since every quasipolynomial sequence is P-recursive. $\square$

52.1 Why a sufficiently small real number is not a uniform substitute

Replace ε in (52.1) by a fixed small positive real t . Ordinary Euclidean area is

$$\text{vol}_2(P_{\alpha,t}) = \int_0^1 (1 + t(x - \alpha)) dx = 1 + t \left(\frac{1}{2} - \alpha \right). \quad (52.7)$$

The elementary lattice-cell estimate proved in Lemma 53.1 below gives

$$\lim_{n \rightarrow \infty} \frac{L_{P_{\alpha,t}}(n)}{n^2} = 1 + t \left(\frac{1}{2} - \alpha \right) \neq 1.$$

In contrast the surreal count has leading coefficient 1. Hence no fixed positive real t reproduces *all* counting values. For each fixed n , a sufficiently small t does reproduce the value, as guaranteed by Theorem 50.1.

For $\alpha = 1/\sqrt{2} > 1/2$ the discrepancy is especially explicit:

$$\lim_{n \rightarrow \infty} \frac{L_{P_{\alpha,t}}(n) - n^2}{n} = -\infty \quad (t > 0 \text{ fixed and small}),$$

whereas taking the eventual $t \downarrow 0$ value first yields $2 - 1/\sqrt{2}$ after $n \rightarrow \infty$. The issue is the order of quantifiers, not a numerical accuracy requirement.

53 The universal surface coefficient

We now describe what always survives from the first two Ehrhart coefficients. The shadow in this section is a full-dimensional *lattice* polytope $Q \subseteq \mathbb{R}^d$, with $d \geq 2$. A facet F has direction lattice

$$\Lambda_F = \text{lin}(F - F) \cap \mathbb{Z}^d.$$

Let μ_F be $(d-1)$ -dimensional measure on $\text{aff } F$, normalized so that a fundamental parallelepiped of Λ_F has measure one. Define

$$A(Q) = \sum_{F \text{ facet of } Q} \mu_F(F). \quad (53.1)$$

For the real trace S of P , put

$$R_F = \text{cl}_{\text{aff } F}(S \cap \text{relint } F). \quad (53.2)$$

An empty or lower-dimensional R_F has μ_F -measure zero.

53.1 The elementary counting estimates

Lemma 53.1 (Volume estimate for real polytopes). *For a full-dimensional bounded real polytope $T \subseteq \mathbb{R}^k$ and a full-rank lattice Λ , with lattice-normalized measure μ ,*

$$\#(nT \cap \Lambda) = \mu(T)n^k + O(n^{k-1}).$$

The estimate also holds for a bounded convex semilinear set whose closure is full-dimensional. A bounded subset of a fixed j -dimensional affine space has at most $O(n^j)$ points of any fixed ambient integer lattice in its n -dilate.

Proof. Choose a half-open fundamental parallelepiped of Λ . Its translates tile the space and each contains exactly one lattice point under a fixed consistent boundary convention. Associate such a tile with every lattice point of nT . The symmetric difference between their union and nT lies in a neighborhood of the boundary of fixed width depending only on the tile. A polytope has finitely many facets. The volume of that neighborhood is $O(n^{k-1})$, by bounding a fixed-width neighborhood of each dilated facet. Comparing the two volumes proves the first estimate. No rationality of the vertices is used.

A bounded convex semilinear set has a closure that is a polytope: its finite cell decomposition has finitely many closure vertices, and the convex closure is their convex hull. The cells missing from the set but lying in its closure have lower dimension, since a convex set contains the relative interior of its closure. They contribute $O(n^{k-1})$ lattice points. That bound, and the last assertion of the lemma, follow by placing disjoint balls of fixed radius around the lattice points within the affine space containing them. Distinct integer points have distance at least one, and the dilated set lies in a ball of radius $O(n)$ in that space. $\square$

Lemma 53.2 (Classical interior coefficient). *For a full-dimensional lattice polytope Q and $d \geq 2$,*

$$\#(n \operatorname{int} Q \cap \mathbb{Z}^d) = \operatorname{vol}_d(Q)n^d - \frac{A(Q)}{2}n^{d-1} + O(n^{d-2}). \quad (53.3)$$

Proof. We use the classical Ehrhart polynomial and reciprocity [34]. Write $L_Q(n) = c_d n^d + c_{d-1} n^{d-1} + O(n^{d-2})$. Lemma 53.1 gives $c_d = \operatorname{vol}_d(Q)$, while reciprocity gives $L_{\operatorname{int} Q}(n) = (-1)^d L_Q(-n)$. Their difference, which counts the boundary, has leading term $2c_{d-1}n^{d-1}$.

On the other hand the boundary is the union of its facets. Since every facet contains a lattice vertex, its affine lattice on dilation is a translate of Λ_F . Lemma 53.1 gives its leading count $\mu_F(F)n^{d-1}$. Intersections of facets have dimension at most $d-2$ and contribute $O(n^{d-2})$. Thus the boundary count has leading term $A(Q)n^{d-1}$, proving $c_{d-1} = A(Q)/2$ and then (53.3). $\square$

53.2 The coefficient formula

Theorem 53.3 (Boundary-trace surface coefficient). *Let $P \subseteq K^d$ be a real-bounded polytope whose shadow Q is a full-dimensional lattice polytope, with $d \geq 2$. Then*

$$L_P(n) = \operatorname{vol}_d(Q)n^d + \beta(P)n^{d-1} + O(n^{d-2}), \quad (53.4)$$

where

$$\beta(P) = \sum_F \left(\mu_F(R_F) - \frac{1}{2} \mu_F(F) \right). \quad (53.5)$$

In particular

$$-\frac{A(Q)}{2} \leq \beta(P) \leq \frac{A(Q)}{2}. \quad (53.6)$$

Proof. By Lemma 47.3, every real interior point of Q belongs to the trace and no trace point lies outside Q . Decompose S into its interior part, its intersections with the relative interiors of the facets, and its intersection with the union of the faces of codimension at least two. The last part has $O(n^{d-2})$ lattice points after dilation, by Lemma 53.1.

By Corollary 49.3, each $S \cap \operatorname{relint} F$ is bounded, convex, and semilinear. Its closure R_F is a real polytope, possibly empty or lower-dimensional. Choose a lattice vertex $z_F \in F$. The lattice in $n \operatorname{aff} F$ is

$$\mathbb{Z}^d \cap n \operatorname{aff} F = n z_F + \Lambda_F.$$

Translating by nz_F and using the normalized measure, the preceding lemma gives

$$\#(n(S \cap \text{relint } F) \cap \mathbb{Z}^d) = \mu_F(R_F)n^{d-1} + O(n^{d-2}).$$

If R_F has smaller dimension, the same statement follows from the packing bound with zero leading measure. Adding these finitely many facet contributions to (53.3) proves (53.4)–(53.5). Since $R_F \subseteq F$, each of its measures lies between zero and $\mu_F(F)$, proving the bounds. $\square$

Example 53.4 (The quadrilateral revisited). For the unit square, $A(Q) = 4$. Three facets are retained up to sets of one-dimensional measure zero; the retained portion of the top facet has length $1 - \alpha$. Therefore

$$\beta(P_{\alpha,\varepsilon}) = (4 - \alpha) - 2 = 2 - \alpha,$$

in agreement with (52.4). The remaining term is the bounded nonperiodic sequence $\{\alpha n\}$. Even in dimension two the $O(1)$ remainder need not approach a limit.

Remark 53.5 (Why the shadow is assumed lattice). If Q is only rational, a dilated facet can meet the integer lattice on some residue classes of n and miss it on others. A single constant surface coefficient need not exist even for $P = Q$. Theorem 53.3 is therefore not stated for arbitrary rational shadows. One can formulate residue-class versions after tracking affine-lattice congruences, but that additional bookkeeping is separate from the theorem proved here. For any full-dimensional real shadow the simpler inclusions $\text{int } Q \subseteq S \subseteq Q$ still give $L_P(n) = \text{vol}_d(Q)n^d + O(n^{d-1})$ by Lemma 53.1.

54 The full attainable interval, with fixed combinatorics

The bounds in (53.6) are sharp not merely at their endpoints: there are no gaps anywhere between them. More strongly, simplicity of the shadow permits this freedom while preserving its entire labeled face lattice.

Theorem 54.1 (Complete surface-coefficient range). *Fix a full-dimensional lattice polytope $Q \subseteq \mathbb{R}^d$ with $d \geq 2$. Among real-bounded K -polytopes with shadow Q , the values of $\beta(P)$ are exactly*

$$[-A(Q)/2, A(Q)/2].$$

Every value can be attained using one positive infinitesimal and inequality coefficients affine in that infinitesimal.

Proof. The upper and lower bounds were proved in Theorem 53.3. Fix $0 \leq \theta \leq 1$. For each facet F , choose a point in its relative interior and contract F toward that point by the real factor $\theta^{1/(d-1)}$. This gives a real polytope $T_F \subseteq F$ with

$$\mu_F(T_F) = \theta \mu_F(F).$$

At $\theta = 0$ it is a point. Choose affine functions h_{Fj} cutting out T_F inside F , and let s_F be the original nonnegative facet slack. Define

$$P = \{x \in Q_K : s_F(x) + \varepsilon h_{Fj}(x) \geq 0 \text{ for all } F, j\}.$$

Every real interior point of Q belongs to P . Its shadow is Q by the same closed-shadow argument as in Theorem 51.1. For a real point in $\text{relint } F$, all other facet slacks are strictly positive, so only the inequalities associated with F matter. Its trace there is exactly $T_F \cap \text{relint } F$, whose closure has measure $\theta \mu_F(F)$. Consequently

$$\beta(P) = (\theta - 1/2)A(Q).$$

As θ ranges through $[0, 1]$, this gives the full claimed interval. All rows are affine in the single parameter ε . $\square$

54.1 Stability of a simple polytope under infinitesimal facet motion

A d -polytope is simple if exactly d facets meet at each vertex. Write an irredundant facet representation of a fixed full-dimensional simple real polytope as

$$Q = \{x \in \mathbb{R}^d : s_F(x) = b_F - a_F \cdot x \geq 0 \text{ for every facet } F\}.$$

For arbitrary real affine forms $h_F = c_F + g_F \cdot x$, perturb only these existing inequalities:

$$P_\varepsilon = \{x \in K^d : s_F(x) + \varepsilon h_F(x) \geq 0 \text{ for all } F\}. \quad (54.1)$$

Unlike (51.1), this construction is *not* intersected with Q_K .

Lemma 54.2 (Simple labeled stability). *The polytope in (54.1) is nonempty, full-dimensional, and real-bounded. It has the same labeled face lattice as Q . Every vertex has the standard part of its correspondingly labeled vertex in Q , and its shadow is Q .*

Proof. First we establish boundedness; this must precede taking standard parts of hypothetical vertices. Since Q is bounded, its recession cone $\{u : a_F \cdot u \leq 0 \text{ for all } F\}$ is zero. Hence for every real unit vector u ,

$$f(u) = \max_F a_F \cdot u > 0.$$

By ordinary real compactness of the unit sphere, its minimum is a real $c > 0$. The first-order polynomial statement

$$\forall u \left(\sum u_j^2 = 1 \implies \bigvee_F a_F \cdot u \geq c \right)$$

transfers to K . Choose real bounds for all the finitely many coefficients of g_F . For unit $u \in K^d$, the error $|\varepsilon g_F \cdot u|$ is less than $c/2$, so some perturbed normal $A_F = a_F - \varepsilon g_F$ satisfies $A_F \cdot u \geq c/2$. Choose a real $B > 0$ exceeding every $|b_F + \varepsilon c_F|$. For nonzero feasible x , apply this to $u = x/\|x\|$; a positive square root exists because K is real closed. Feasibility then gives

$$\frac{c}{2} \|x\| \leq A_F \cdot x \leq b_F + \varepsilon c_F < B.$$

Thus $\|x\| < 2B/c$. The polyhedron is real-bounded. An ordinary interior point of Q has strictly positive real original slacks and therefore strictly positive perturbed slacks, proving nonemptiness and full dimension.

Let v be a vertex of Q , and $I(v)$ its d incident facets. The corresponding original normal matrix is nonsingular. The perturbed matrix has the same nonzero determinant standard part, so its system has a unique solution v_ε with $\text{st}(v_\varepsilon) = v$. At every other facet the original slack at v is strictly positive; hence its perturbed slack at v_ε is positive. The point v_ε is feasible and is a vertex, being the intersection of d independent active hyperplanes.

Conversely, a vertex w of the perturbed full-dimensional polytope has d independent active rows. It is finite by the preceding bound. Taking standard parts shows that $\text{st}(w) \in Q$ lies on all of those d distinct original facets. In a simple polytope a point on d distinct facets must be a vertex: the facets through any positive-dimensional face are fewer than d . One way to see this is to choose a vertex of that face; its d incident facet normals are independent, and the intersection of any d of them is that vertex alone. Thus these are exactly the incident facets of an original vertex, and the corresponding original matrix is nonsingular. Its unique perturbed solution is the already constructed v_ε . This proves that there are no other vertices.

At the constructed vertices the zero slacks are exactly the original zero slacks. The vertex-facet table, and hence the labeled face lattice, is unchanged. The vertex standard parts and Lemma 47.2 give the shadow assertion. $\square$

Theorem 54.3 (Full range with fixed labeled simple combinatorics). *Let $Q \subseteq \mathbb{R}^d$, $d \geq 2$, be a full-dimensional simple lattice polytope. Every*

$$b \in [-A(Q)/2, A(Q)/2]$$

is realized as $\beta(P)$ by a real-bounded K -polytope P satisfying all of the following simultaneously: its shadow is Q ; its labeled face lattice equals that of Q ; its labeled vertex standard parts are exactly the original integral vertices of Q ; and it uses one affine-in- ε perturbation of each original facet inequality, with no added facets.

Proof. Put $\theta = 1/2 + b/A(Q) \in [0, 1]$. For every facet F , choose a real affine function h_F for which

$$\mu_F(F \cap \{h_F \geq 0\}) = \theta \mu_F(F). \quad (54.2)$$

Such a choice exists: a nonconstant linear functional on the $(d-1)$ -dimensional facet has sublevel sections whose volume varies continuously from zero to full measure as the level traverses its range. Continuity follows since the moving level hyperplane has measure zero and the facet is bounded. For the endpoints use $h_F = -1$ or $h_F = 1$.

Form P_ε as in (54.1). Lemma 54.2 proves all the geometric assertions. At a real point x on relint F , exactly the original F -slack is zero; the others are strictly positive. Consequently trace membership there is precisely $h_F(x) \geq 0$. The closure of that retained part has the measure in (54.2); deleting the boundary of F does not change it. Theorem 53.3 therefore gives

$$\beta(P_\varepsilon) = \sum_F (\theta - 1/2) \mu_F(F) = (\theta - 1/2) A(Q) = b.$$

□

Theorem 54.3 says more than that one can choose open or closed facets. Intermediate real portions of a facet, selected by an infinitesimal tilt, continuously interpolate its contribution. Neither the finite face lattice nor the full labeled list of vertex standard parts records that selection. Simplicity is used only in the combinatorial stability lemma; Theorem 54.1 establishes the interval without it.

54.2 Dimension one behaves differently

A real-bounded interval with a positive-length lattice shadow $[a, b]$, with $a, b \in \mathbb{Z}$, retains all points of (a, b) and independently retains or excludes its two endpoints. Hence for $n \geq 1$ its count is

$$(b - a)n - 1 + e_a + e_b, \quad e_a, e_b \in \{0, 1\}.$$

Its constant term is one of $-1, 0, 1$, not an arbitrary value in an interval. There is no positive-dimensional facet along which a threshold can slide. This explains the $d \geq 2$ condition in the continuous-range theorems.

55 A rationality criterion that restores Ehrhart behavior

The preceding examples identify a failure of a natural extension of Ehrhart theory. There is nevertheless a useful sufficient condition that replaces rational vertex coordinates by rational data in the trace flags.

Definition 55.1. A real-bounded polytope has a *rational flag presentation* if its real trace is described by finitely many lexicographic nonnegativity conditions on real affine forms all of whose coefficients are rational, together with a rational bounding box. This is a property of a possible presentation, not a claim about the canonicity of the basis in Lemma 49.1.

Theorem 55.2 (Rational flags imply positive-dilation quasipolynomiality). *If P has a rational flag presentation, then $L_P(n)$ is quasipolynomial for positive integers n . Its ordinary generating function, with the convention $L_P(0) = 1$, is rational.*

Proof. Use the hyperplane arrangement of the rational affine forms, including the rational box boundary. The selected cells form a finite disjoint union of relatively open rational polytopes $C_1, \dots, C_s$. Each is the relative interior of its closure, a rational polytope. For $n > 0$ the dilation map is injective and the dilated cells remain disjoint, so

$$L_P(n) = \sum_{j=1}^s \#(nC_j \cap \mathbb{Z}^d).$$

Classical rational Ehrhart theory and reciprocity imply that the count of the relative interior of a rational polytope is quasipolynomial on positive integers [34]. This statement includes lower-dimensional rational affine hulls and their residue-class restrictions. A finite sum of quasipolynomial is quasipolynomial, with period dividing a common multiple of their periods. Splitting into residue classes shows that its positive-index generating function is rational. Changing or supplying the single coefficient at $n = 0$ preserves rationality. $\square$

Example 55.3 (Why $n = 0$ needs a convention). Let $P = [\varepsilon, 1 - \varepsilon]$. Its trace is $(0, 1)$, described by the rational strict inequalities $x > 0$ and $1 - x > 0$. Then

$$L_P(n) = n - 1 \quad (n \geq 1), \quad L_P(0) = 1.$$

The positive-dilation polynomial does not take the conventional value at zero. Any theorem using relatively open trace cells must either treat $n = 0$ separately or adopt a different valuation convention there.

Remark 55.4 (Not a necessary condition). The theorem is a sufficient criterion, not an equivalence. Irrational real polytopes can have polynomial counting functions; the examples of Cristofaro-Gardiner, Li, and Stanley show that arithmetic cancellations and period collapse must be taken seriously [35]. Conversely, the choice $\alpha = 1/\sqrt{2}$ in Theorem 52.1 shows that replacing “rational” by “real algebraic” in the hypothesis is insufficient. An exact classification would have to detect cancellation among boundary pieces rather than inspect each coefficient separately.

55.1 Three notions of rationality that must not be conflated

A polytope may have rational or integral *vertex shadows*; its trace may admit a description by *rational flags*; and its counting *generating function* may be rational. These are different properties. The quadrilateral has integral vertex shadows but irrational trace data and a nonrational generating function. Rational flags imply a rational generating function by Theorem 55.2. The manuscript does not assert any converse from generating-function rationality to a specific kind of geometric presentation.

The stronger descent theorem does not conflict with this distinction. Its one-parameter algebraic coefficients can contain arbitrary real constants. Eliminating extra infinitesimal scales is not the same as eliminating irrational real thresholds.

55.2 A classical analytic dichotomy for every bounded trace count

The analytic consequence of Pólya–Carlson is more general than the quadrilateral example. It supplies a useful complete dichotomy once the geometric rationality question has been answered.

Corollary 55.5 (Analytic classification of bounded counts). *Let P be any nonempty real-bounded finite K -polytope, without a dimension assumption on its shadow, and let $E_P(z) = \sum_{n \geq 0} L_P(n)z^n$. The following are equivalent:*

- (i) E_P is rational;
- (ii) E_P is D-finite;
- (iii) $L_P(n)$ is P-recursive;
- (iv) $L_P(n)$ is eventually quasipolynomial.

If these conditions fail, the unit circle is a natural boundary of E_P . This is a direct application of classical results, not a new analytic classification theorem.

Proof. A real bounding box gives $0 \leq L_P(n) = O(n^d)$, and the values are integers. If only finitely many are nonzero then E_P is a polynomial. Otherwise integrality and polynomial growth imply radius of convergence exactly one. Pólya–Carlson says that a nonrational such series has the unit circle as a natural boundary [36, 37]. The finite-singularity argument in Theorem 52.1 then excludes D-finiteness. Conversely a rational series is D-finite. The standard coefficient correspondence identifies D-finiteness with P-recursiveness [38].

In the rational case, the integer-coefficient form of Pólya–Carlson also says that every pole is a root of unity [36, Theorem 1.1]. Partial fractions therefore express the coefficients, after a finite initial segment, as finite sums of terms $n^j \zeta^n$ with ζ a root of unity. They are polynomial on residue classes modulo a common order. Conversely, splitting an eventually quasipolynomial sequence into its residue classes gives a rational series. $\square$

[[write] The root-of-unity clause is classically the Fatou–Kronecker argument, not part of the natural-boundary dichotomy proper. By Fatou’s lemma a rational power series with integer coefficients is $A(z)/C(z)$ with $A, C \in \mathbb{Z}[z]$ and $C(0) = 1$. Radius of convergence 1 puts every root of the reversed polynomial, which is monic with integer coefficients, in the closed unit disc. Kronecker’s theorem then makes each nonzero root a root of unity. Manuscript 03 cites this through the integer-coefficient statement it attributes to [36, Theorem 1.1] and does not prove it; the equivalence of (i) and (iv) rests on it.]

Thus for the present unweighted, one-variable counting problem there is no intermediate nonrational D-finite case. The genuine remaining task is to determine from the boundary geometry which side of the dichotomy occurs.

56 Repository integration and the exact proof boundary

56.1 Snapshot and source use

The repository tree inspected for this work had commit

6996fee43cc97b6c16351def7507d59a95bf62f0.

The root and surreal-project readmes were consulted for orientation. The surreal report guide and the omnific source below were then fetched at this explicit commit. Code-search results may refer to older indexed commits; they were used as locators, not as proof of the contents of the pinned snapshot.

The key source is

```
Algebra/SurrealNumbers/Surreal/Foundations/OmnificUnits.lean
```

Its namespace is `Surreal.Foundations.SignSequence`. The three relevant declarations are

```
omnific_eq_intConstant_of_finite
omnific_isFinite_iff
omnific_bounded_iff
```

The last states that an omnific integer has a real bound on its absolute value exactly when it is an ordinary integer cast [28]. This is the bridge in (47.1).

The source was read; this work did *not* rebuild the repository, check its complete transitive axiom dependencies, or independently audit its surreal-field implementation. The mathematical article can be read independently using the standard normal-form argument given in Section 47. None of its new trace, descent, implantation, or surface-coefficient theorems is being represented as an existing Lean theorem.

56.2 A realistic formalization sequence

[write] The formalization roadmap of manuscript 03 is printed in [section 83](#).

56.3 A theorem dependency map

Result	Principal input	Status in this package
Bounded omnific reduction	Normal form; repository boundedness theorem	Source inspected; direct mathematical argument
Lexicographic trace	Standard part; finite real linear algebra	Full proof, no Lean certificate
Algebraic descent	Trace expansion; real-closed transfer; curve selection	Full application proof with classical inputs
Face implantation	Dominance of a real slack over an infinitesimal	Full constructive proof
Surface coefficient	Trace cells; lattice-cell estimate; Ehrhart reciprocity	Full application proof
Fixed-combinatorics range	Simple facet stability; continuous section volume	Full constructive proof
Quadrilateral count	Exact lexicographic membership; finite row count	Full proof and exact finite checks
Natural boundary	Nonrationality; Pólya–Carlson	Classical analytic input explicitly used

57 Exact computational checks and reproducibility

The accompanying code/03-boundary-geometry-verify.py uses only the Python standard library. All assertions use integers, rational numbers, exact arithmetic in $\mathbb{Q}(\sqrt{2})$, or polynomials in a formal positive infinitesimal ordered by their first nonzero coefficient. No floating-point arithmetic is used to resolve a boundary comparison.

57.1 What was actually checked

For $\alpha = 1/\sqrt{2}$, the program verifies the 16 vertex–facet slacks of the quadrilateral, its four positive cyclic orientation determinants, and the exact shoelace identity

$$2 \operatorname{vol}_2(P_{\alpha,\varepsilon}) = 2 + \varepsilon(1 - \sqrt{2}).$$

It independently enumerates integer candidates in an enlarged bounding box for every $1 \leq n \leq 150$: 1,205,575 candidate membership tests in total. Every result agrees with (52.3). The ceiling formula and its 0–1 increments are checked for $1 \leq n \leq 10,000$.

For rational thresholds $\alpha = p/q$ with $2 \leq q \leq 12$ and $1 \leq p < q$, it checks 3,960 threshold/dilation cases, including vanishing third forward differences on each tested progression of step q . For the three-dimensional face implantation of

$$T = [1/\sqrt{2}, 1] \times [1/\sqrt{3}, 1]$$

on the top of the unit cube, it checks all $1 \leq n \leq 35$ by 443,555 candidate tests against the exact formula

$$n(n+1)^2 + (n+1 - \lceil n/\sqrt{2} \rceil)(n+1 - \lceil n/\sqrt{3} \rceil).$$

Finally it verifies real specialization at $t_n = 1/(4(n+1))$ for $1 \leq n \leq 100$, using 353,700 exact $\mathbb{Q}(\sqrt{2})$ sign comparisons. The specialization parameter deliberately depends on n .

For nonsquare positive integer m and $n > 0$, the program evaluates

$$\lceil n/\sqrt{m} \rceil = \operatorname{isqrt}(\lfloor n^2/m \rfloor) + 1$$

and verifies the defining inequalities $m(c-1)^2 < n^2 < mc^2$. Thus the tests near the irrational cut do not depend on a decimal approximation.

n	$\lceil n/\sqrt{2} \rceil$	$L_{P_{1/\sqrt{2},\varepsilon}}(n)$	$(n+1)^2$
0	0	1	1
1	1	3	4
2	2	7	9
3	3	13	16
4	3	22	25
5	4	32	36
6	5	44	49
7	5	59	64
8	6	75	81
9	7	93	100
10	8	113	121

The successful test report is in `data/03-boundary-geometry-verification.json`; `data/03-boundary-geometry-counts.csv` records values for $0 \leq n \leq 100$. To reproduce the checks and compile the article, run

```
python3 code/03-boundary-geometry-verify.py --out-dir /path/to/scratch
latexmk -pdf -interaction=nonstopmode article.tex
```

A double invocation of `pdflatex`, with an additional pass when cross-references change, can replace `latexmk`. All figures are contained in the TeX source; no external image assets are needed.

[[write] Delivery names replaced. Manuscript 03 printed `verify.py`, `verification.json`, `counts.csv` and `python3 verify.py`; the shipped names are those above. Without `-out-dir` the program writes `verification.json` and `counts.csv` next to itself, that is, into `code/`. The report's recorded outputs are under `data/` and should never be overwritten, so pass an output directory outside the report. The program writes the CSV with Python's `csv` module, so its lines end in CRLF on every platform. The shipped `data/03-boundary-geometry-counts.csv` keeps those bytes: 102 lines (a header and $n = 0, \dots, 100$), each ending in CRLF, and marked `-text` for Git. Manuscript 03's own TeX file `surreal_polytopes.tex` is not shipped: its text is this Part of the report's `article.tex`, which the second command builds.]

57.2 What finite checks do not establish

Agreement through 10,000 does not prove nonperiodicity or exclude a large period. Those conclusions use the irrationality argument in Theorem 52.1. The checks do not establish the general trace, descent, or surface-coefficient theorems. They do not prove Pólya–Carlson, certify a natural boundary, or substitute for a proof assistant. Their role is to catch arithmetic, incidence, dilation, and endpoint mistakes in the explicit constructions.

58 Further research questions

The following are proposed research problems arising from this article. Unless an external source is explicitly attached to a question, it is not being represented as an established named open problem in the literature.

Question 58.1 (An intrinsic classification of traces). Which bounded convex semilinear sets $S \subseteq \mathbb{R}^d$ are the real traces of finite polytopes over a real closed extension? Can one characterize them without choosing a halfspace presentation over the extension?

Theorem 49.2 provides a necessary finite lexicographic description. Every finite conjunction of such descriptions has an infinitesimal halfspace realization, but an arbitrary convex semilinear set is initially given by a Boolean combination, not necessarily such a conjunction. The useful target is a geometric characterization, including compatibility of the selected portions on intersecting faces and a criterion for prescribing the shadow when it loses dimension.

Question 58.2 (Minimal trace depth and description size). What is the least maximum lexicographic depth among all descriptions of a fixed trace, and how does it trade off against the number of rows?

The $d + 1$ bound proved here is a rowwise coefficient-space bound. It does not prove that $d + 1$ is an optimal invariant of traces after changing their presentations. A normal form with canonical depth would be useful for certificate design and for comparing independent constructions.

Question 58.3 (Effective algebraic descent). Given exact algebraic coefficients and a finite certificate for a lexicographic presentation and incidence data, what degree, coefficient height, and complexity bounds suffice for the descent curve in Theorem 50.1?

Quantitative curve selection supplies relevant general machinery [31]. The open task here is to exploit the special multi-affine ratio structure in (50.1), account for the realization-space quantifiers, and produce bounds suitable for an executable certificate. The degree- d trace-only compression is not already such a bound for incidence-preserving descent.

[[write] Related, not an answer: for coordinates given as finite-support Hahn polynomials, Part III's [theorem 41.2](#) (manuscript 06) specializes to ordinary Laurent polynomials in one parameter, with an explicit integer grading and explicit sign thresholds. It keeps any specified *finite* certificate. Whether such a certificate can be chosen to keep the whole real trace, and so give an effective descent with degree bounds, is not proved in this report. Part I's [theorem 19.1](#) is the non-effective real-algebraic analogue for finite sign data.]

[[write] [Part VIII](#) gives a lower-bound family for rational arcs ([theorem 120.3](#)); no upper bound of the kind asked is proved.]

Question 58.4 (Surface freedom for nonsimple fixed combinatorics). For a nonsimple lattice polytope Q , which values of β can be realized while preserving its labeled face lattice and all vertex shadows?

[Theorem 54.1](#) gives the full interval when combinatorics is allowed to change, and [Theorem 54.3](#) gives it with fixed combinatorics when Q is simple. At a nonsimple vertex, preserving several dependent facet intersections imposes compatibility equations on the perturbations. Determining whether these constraints shrink the interval, and in which examples, is a sharply delimited next problem.

Question 58.5 (The next asymptotic layer). Which conditions ensure a coefficient of n^{d-2} , and can it be expressed using traces on ridges and second-level facet data?

The surface term always exists for lattice shadows, but the bounded fractional-part remainder in dimension two already prevents a universal constant next term. A suitable theorem may require rational flags, averaging in n , residue-class restrictions, or a discrepancy term rather than an ordinary coefficient. Face implantation provides test cases with defects localized at each codimension.

Question 58.6 (Rationality and cancellation). Within traces defined by real-algebraic flags, can one decide when the counting series is rational, or characterize when irrational boundary contributions cancel to give a quasipolynomial?

Rational flags give a sufficient condition; irrational period collapse prevents a naive converse [35]. The face-implantation identity reduces many examples to lower-dimensional real-polytope enumeration, offering both obstructions and constructive test families.

Question 58.7 (Weighted and multivariate counting). What replaces [Corollary 55.5](#) for algebraic or real weights, several dilation parameters, or generating functions recording individual lattice coordinates?

The unweighted one-variable dichotomy is already resolved by that corollary, including lower-dimensional shadows. Arbitrary real weights remove the integer-coefficient hypothesis, while several variables require an appropriate notion of boundary and multivariate D-finiteness. Existing multivariate Pólya–Carlson results provide relevant analytic input [37]; the task is to identify geometric hypotheses that let these results be applied to the proposed weighted or multivariate enumeration.

Question 58.8 (Reciprocity with boundary data). Is there a useful reciprocity formula for trace counts stated in terms of constructible weights on faces rather than just open versus closed polytopes?

A trace may retain a proper region of a facet and may make additional choices on its boundary. Ordinary closed-polytope reciprocity cannot simply be reused without recording those choices. The rational cell decomposition in [Theorem 55.2](#) provides a finite starting point, but an intrinsic formula should be independent of the chosen subdivision.

Question 58.9 (Symmetry-constrained perturbations). What parts of the surface-coefficient interval remain attainable under central symmetry or under a prescribed finite affine symmetry group?

The independent facet choices in Theorem 54.3 need not respect a group action. Averaging perturbations preserves some algebraic symmetries but may change the selected boundary regions. Orbitwise constraints and self-paired facets should be analyzed explicitly rather than inferred from the unconstrained theorem.

Question 58.10 (Finite enriched combinatorial invariants). What finite augmentation of the face lattice and vertex shadows suffices to reconstruct the real trace and its lattice counts?

The lexicographic affine flags are one sufficient description, but not an intrinsic minimal invariant. Ordinary oriented-matroid signs alone forget the real threshold data, while unrestricted valuation information is more than the counting sequence needs. An invariant adapted to boundary flags may bridge these extremes.

Question 58.11 (Other integer parts and unbounded geometry). How do the conclusions change for a different integer part of an ordered field, or when coordinates and dilation parameters are allowed to be infinite?

The present reduction depends on the exact bounded-element property (47.1). Infinite dilations do not automatically yield finite sets of lattice points, so a new counting invariant must be specified before an analogue of Ehrhart theory can be meaningfully formulated. Ordinary cardinality, polynomial evaluation in an infinite argument, and surreal-valued numerosity are not interchangeable definitions.

[[write] *Re-scoped: partly answered for hulls.* For integer hulls rather than counts, Part IX treats arbitrary integer parts (corollary 141.4) and infinite coordinates and right-hand sides (theorems 141.2 and 142.4); the counting half remains open.]

Question 58.12 (A checked geometry pipeline). Can the exact row-expansion, incidence, and finite-count certificates be assembled into a Lean-checked pipeline with an explicitly small trust boundary?

The immediate milestone is a checked proof of the quadrilateral count and the bounded omnific reduction, followed by trace equivalence. A general symbolic discovery procedure for arbitrary surreal inputs is a much stronger demand and is not implied by the finite certificate theorems.

59 Conclusion

Finite surreal convex geometry has a classical combinatorial boundary: its face lattices are already realizable over the real algebraic numbers. Its arithmetic boundary is different. The real trace, not the standard-part shadow, determines ordinary and bounded omnific lattice counts. That trace is finitely lexicographic, and a single algebraic parameter can preserve it together with the entire finite incidence structure.

Inside this finite framework, infinitesimal boundary motion is expressive. It implants arbitrary lower-dimensional real-polytope counting functions, changes the surface coefficient through its whole allowable interval, and can do so without changing a simple lattice shadow or any of its labeled vertex standard parts. A four-vertex, one-infinitesimal example already escapes quasipolynomial and P-recursive enumeration. These statements give a concrete research program and an exact formalization target without requiring genuinely class-sized convexity or a new notion of infinite lattice cardinality.

60 Additional scope checks and potential failure modes (Appendix A of manuscript 03)

Finite generators versus topological compactness. “Polytope” means a finite convex hull, equivalently a bounded finite halfspace system in the setting used here. Proofs over K use linear algebra or first-order transfer. The only compactness argument in the simple-stability proof takes place on the *real* unit sphere and is then converted to a polynomial statement before transfer.

The interior-lifting hypothesis. A full-dimensional shadow ensures that its interior points belong to the trace. It is not enough that P itself have positive dimension, and even full dimension of P need not imply full dimension of its shadow. The one-parameter descent theorem deliberately avoids relying on this lemma and therefore allows rank loss in the shadow.

A representation-dependent flag is not an invariant. The separated basis exists for every finite coefficient space, but it is not canonically chosen. The trace theorem asserts existence of a bounded-depth presentation. It does not assert uniqueness of the scales or optimality among all presentations.

Zero standard part does not mean zero slack. When a leading slack vanishes at a real point, its infinitesimal terms can still decide membership. Dropping these terms is exactly the error exposed by (52.5). Conversely, a strictly positive real slack cannot be overturned by a finite sum of infinitesimal corrections.

One parameter is not a uniform real approximation. An infinitesimal evaluation and a fixed positive real evaluation have different quantifier meanings. Descent gives eventual agreement for each fixed real point and each fixed finite lattice query. Equation (52.7) shows why a single real evaluation generally cannot answer all dilations.

Real-algebraic is not rational. All defining constants of the explicit quadrilateral can lie in $\mathbb{Q}(\sqrt{2})$, yet its count is not quasipolynomial. The rational flag theorem requires rational coefficients for the real affine tests, not merely algebraic coefficients or integral shadow vertices.

No general reciprocity theorem is smuggled in. Classical Ehrhart reciprocity is used for the lattice shadow in Lemma 53.2 and for rational cells in Theorem 55.2. It is not assumed for an arbitrary irrational trace or for an arbitrary surreal polytope.

The analytic step has an external dependency. The arithmetic proof establishes nonrationality directly. The assertion of a natural boundary uses Pólya–Carlson, and the non-P-recursive conclusion uses its analytic consequence together with the standard D-finite correspondence. Finite experimental evidence is not used as a substitute for any of these implications.

61 Reproducibility and source ledger (Appendix B of manuscript 03)

The archive contains the TeX source, compiled PDF, standard-library verification program, its JSON report, a CSV of example counts, and a README. The bibliography gives stable DOIs

or arXiv identifiers for the classical inputs and commit-pinned paths for the two repository documents used substantively. The source review was performed on September 30, 2026. No repository files were modified or committed as part of this work.

[[write] In this report the delivered verification program is shipped as `code/03-boundary-geometry-verify.py`, its JSON report as `data/03-boundary-geometry-verification.json`, and the CSV as `data/03-boundary-geometry-counts.csv` (CRLF line endings, as delivered). The TeX source is printed as this Part, not shipped as a file. The compiled PDF and the member README are not shipped; they survive only in the delivery archive `surreal_polytope_research.zip` of arrival commit `b2b626d85`.]

The source `OmniFicUnits.lean` was returned with blob identifier

`c9ce3771e76d4b21ff76814682f37f63cc4664f8`.

This identifies the inspected file; it is not a claim that the entire repository was locally compiled. ([write] The blob of this file at the report's placement is still `c9ce3771`.) The article imports no unreviewed research-report theorem as a hidden prerequisite: its surreal-specific bounded-integer fact is also justified directly, and its remaining external inputs are the classical mathematical results named in the text.

Part V

Extension complexity beyond all orders

Source and scope. [write] This Part is built from two batch-58 manuscripts. Its base, printed whole and in its own order, is manuscript 07, *Invisible Complexity in Surreal Polytopes*, subtitle *Explicit polygons with identical all-order jets and identical initial slack matrices*, author line “Research manuscript prepared for Vladimir Reshetnikov”, dated 30 September 2026 (archive `surreal_polytopes.zip`, main file `surreal_polytopes.tex`; repository pin `6996fee43cc97b6c16351def7507d59a95bf62f0`; its delivered PDF, which is not shipped, has 26 pages according to its README). Into it are merged the results of manuscript 04, *Beyond-All-Orders Extension Complexity in Surreal Polygon Geometry*, subtitle *Flat deformations, classical face lattices, and a sharp-order complexity gap*, author line “Research manuscript prepared for Vladimir Reshetnikov / Developed with ChatGPT; independent review and formalization pending”, dated September 30, 2026 (archive `surreal_polytopes_beyond_all_orders.zip`, main file `article.tex`; the same pin `6996fee43`; delivered PDF of 26 pages, not shipped). The two manuscripts answer the same question: explicit polygons whose coordinates agree to all orders of a chosen infinitesimal with a polygon of small extension complexity, but whose own extension complexity is much larger. Manuscript 07 is the base because it covers every $n \geq 4$ (04: $n = 2^m$), linear and semidefinite lifts, residue descent for every convex valuation ring (04: standard part and nonnegative rank only), every fixed dimension, and a sharper parameter count. Manuscript 04 contributes a second construction on one circle with a two-sided square-root order, a field- and scale-general classification of when all-order jets lose information, a finite-order rank-one analogue, failure of lifting coherent finite-order factorizations, algebraic finite-accuracy approximants, a rational-chart generalization and an in-principle decision procedure. Every item of 04 is printed where it belongs and marked “source 04”; items proved by both manuscripts are printed once and name both. Labels of 07 carry the prefix `poly:ic:`, labels of 04 the prefix `poly:bao:`. This Part keeps its own definitions and hypotheses: the finite-element ring $\mathcal{O}$, standard part `st` and convex hulls are those of [Part I](#), restated here in the words of 07 and 04. The formalization roadmap of 07 is printed in [section 83](#); nothing in this Part is formalized.

Notation in this Part. [write] The valuation is written v throughout this Part, with the convention of 04: $v(x) > v(y)$ precisely when x/y is infinitesimal, so a *larger* value means a *smaller* element and $v(\omega^{-\gamma}) = \gamma$. (Manuscript 07 writes v without defining it; manuscript 04 writes v ; we renamed 04's v to v . Manuscript 04 also uses v_i for monomials in the proof of [theorem 74.1](#); those are renamed μ'_i .) The natural value group is G (07's letter; 04's Γ in its Section 3 is renamed G); Γ , Γ_2 and Γ_n denote the specific groups ([72.1](#)), [corollary 72.4](#) and ([70.11](#)). The flat ideal of a positive infinitesimal scale x is written with one symbol, $\mathfrak{i}_x = \bigcap_{m \geq 1} x^m \mathcal{O}$ (07's $\mathfrak{j}$); manuscript 04's $\mathcal{I}_\varepsilon$ is printed as $\mathfrak{j}_\varepsilon$. The jet scale is always the subscript of $\mathfrak{j}$: a general positive infinitesimal is s in statements from 07 and ε in statements from 04 ([section 63.4](#) and [theorems 72.7](#) and [76.1](#)); in 04's explicit family it is specialized to $\varepsilon = \omega^{-1}$, the report's ε . Manuscript 07's main construction uses the fixed scale $\rho = \omega^{-\omega}$; following the report-wide convention it is printed as $\tau = \omega^{-\omega}$ wherever it denotes that surreal. Where 07 re-chooses the scale, the letter ρ is kept as a general parameter: in [corollary 72.4](#) ($\rho = \omega^{-1}$) and in [theorem 74.2](#) (a chosen infinitesimal); the auxiliary bound called τ in the proof of [theorem 74.2](#) is renamed σ , and the tolerance set $\mathcal{T} \ni \tau$ of 04's [theorem 74.2](#) is renamed $\mathcal{D} \ni \delta$ (07's letters). The two quoted abstracts keep their own letters: ρ in 07's abstract is this Part's τ , and P_n in 04's abstract is this Part's P_n° . Tempting false readings, with the true one beside them:

- P_n, Q_n (07) are polygons in the triangle T ; manuscript 04's cyclic polygon on the unit circle, also called P_n there, is printed as P_n° with vertices p_i° , and its regular reference polygon is R_n with vertices r_i . R_N in ([66.1](#)) is 07's reflection map, not a polygon.
- st is the standard part. $\mathcal{O}/\mathfrak{j}_s$ is a different quotient and is not $\mathbb{R}$; the coarse residue field of V_τ ([proposition 72.10](#)) is a third one. Manuscript 04's "algebraic finite-accuracy shadows" are real-algebraic approximants, not standard parts; we call them *approximants* ([corollary 74.6](#)).
- tr is the matrix trace (07), not Part IV's real trace $\text{Tr}_{\mathbb{R}}$.
- q_i are 07's easy vertices; in 04's text q_i are distinct ordinary primes, entering only through $\sqrt{q_i}$ and $\mu_i = \omega^{-\omega\sqrt{q_i}}$. a_n (07) is the real root of $X^{2n+1} - 2$; a_i (04) is $\tan(\theta_i/2)$.
- η_ℓ (07) are perturbation monomials and $\eta = \eta_1$; in 04, η is a flat bound ([proposition 71.1](#)) or a positive rational tolerance ([corollary 74.6](#)).
- D : 07's orientation determinant $D(X, Y, Z)$ of points and $D = h(h+1)/2$ in [theorem 67.4](#); 04's $D(a, b, c)$ is the orientation determinant in circle parameters ([70.9](#)). H_N (07) is a parabola hull; $H(V)$ (04) is a $3 \times n$ point matrix.
- m : an ordinary integer exponent in 07 ($\mathfrak{j}_s, <_{\text{pow}}$) and in 04 $n = 2^m$; M is 04's exponent letter. $\mathfrak{m}$ is the maximal ideal of $\mathcal{O}$. T is 07's triangle, but also a nonnegative factor matrix in 04 ($S = TU$) and in the proof of [theorem 73.5](#); $U(n)$ is 07's easy bound, U also a factor matrix. S is always a slack matrix.
- K is a set-sized real closed field; K_0, K_n and F_n are specific fields of 07 and 04; k is a residue field in [section 73](#) and $k = n - 2$ in 07's construction.

Across the report: ρ here is not Part III's threshold ρ_g nor Part IV's separated scales $\rho_1 \gg \dots \gg \rho_r$; V_τ is not Part I's dependence space nor Part III's mixed volume.

Abstract of manuscript 07. Can the real shadow, the complete expansion at a chosen infinitesimal scale, or the leading terms of a slack matrix determine the exact complexity of a surreal polytope? We give explicit negative answers for both linear and positive-semidefinite

lifts. For every ordinary finite $n \geq 4$ we construct two strictly convex, labeled surreal n -gons Q_n, P_n inside the same real triangle. They have the same real shadow and oriented matroid; their vertex coordinates and canonical slack matrices agree modulo every power of a specified infinitesimal ρ ; and corresponding nonzero slack entries have identical valuations and leading coefficients. Nevertheless,

$$\text{xc}(Q_n) \leq 2\lceil \log_2(n-2) \rceil + 2, \quad \text{xc}(P_n)(\text{xc}(P_n) - 1) \geq 2n,$$

and $\text{sxc}(P_n) \geq (2n)^{1/4}$, while $\text{sxc}(Q_n) \leq \text{xc}(Q_n)$. Every coordinate is a sum of at most three Conway monomials, and all examples fit one set-sized real closed subfield of valuation rank three.

The proof combines a binary reflection lift with explicit algebraically independent, beyond-all-powers perturbations. We supply a non-Archimedean proof of the generic semidefinite lower bound using finite definability, rather than rational exposing objectives, which can fail over the surreals. We also prove residue descent for nonnegative and semidefinite ranks, identify the exact kernel of power-jet observations, establish a sharp rank-one/rank-two boundary for all-order ambiguity, and extend the construction to arbitrary set-sized precision requirements.

These are proved statements within the manuscript, not a claim to have settled a previously named classical conjecture. The generic lower bounds, reflection method, cone-factorization correspondence and real-closed-field transfer are classical. The proposed contribution is their explicit multiscale separation package. Priority and correctness have not been independently certified; no Lean formalization is claimed.

Keywords (manuscript 07): surreal polytope; extension complexity; nonnegative rank; semidefinite rank; convex valuation ring; infinitesimal perturbation; real closed field; formal jets.

Abstract of manuscript 04. Infinitesimal coordinates do not create new finite polytopal face lattices: real-closed-field transfer already prevents this. They can, however, conceal large changes in extension complexity beneath every power of a prescribed infinitesimal. We construct explicit cyclic surreal n -gons P_n , for $n = 2^m$, and regular n -gons R_n on the same unit circle such that their labeled vertex coordinates agree modulo $\varepsilon^M \mathcal{O}$ for every ordinary integer $M \geq 1$, where $\varepsilon = \omega^{-1}$. Nevertheless,

$$m \leq \text{xc}(R_n) \leq 2m, \quad \lceil \sqrt{n} \rceil \leq \text{xc}(P_n) \leq \min\{n, \lceil 24\sqrt{n} \rceil\}.$$

The polygons also have identical standard parts, leading affine Plücker data, and leading slack matrices; every finite linear objective has an optimal-value difference smaller than every power of ε . The construction uses algebraically independent Conway monomials and a rational parametrization of the circle. A general theorem identifies the exact boundary: over any real closed $K \supseteq \mathbb{R}$, unbounded such gaps in all-order jet fibers occur precisely when $\bigcap_{M \geq 1} \varepsilon^M \mathcal{O} \neq 0$, equivalently when $v(\varepsilon)$ is not an order unit of the natural value group. We give full proofs of these statements, a finite-order rank-one analogue, a rational-chart generalization, and an explicit account of why finite real-algebraic transfer is compatible with the obstruction. The classical extension-complexity bounds are credited separately from the proposed flat-geometric synthesis.

Status (manuscript 04). This is a research manuscript, not a claim of a refereed breakthrough or a Lean-checked result. The flat-fiber theorems and explicit all-order examples are offered as proposed contributions. Their priority has not been independently certified. No named published conjecture is claimed solved. The auxiliary exact checks verify identities and finite examples, not the asymptotic lower bounds.

[write] Manuscript 07 has no separate status paragraph; its status is the last paragraph of its abstract above and its table in [section 62](#). In 04’s abstract the valuation symbol is printed as v (04: v).

62 The questions, the answer, and the research boundary

A polytope can be geometrically almost indistinguishable from another and still require a much larger exact optimization model. Surreal coefficients make it possible to ask a stronger version: can the difference remain invisible not merely at a small numerical tolerance, but at *every* power of a prescribed infinitesimal? The answer is yes, even with the same number of vertices, the same full-dimensional real shadow and the same leading slack data.

Here a linear extended formulation is an exact description of a polytope as an affine image of a polytope in a possibly larger space. Its size is the number of facet inequalities; equations do not count. A semidefinite lift uses an affine slice of the cone of positive-semidefinite symmetric matrices. Its size is the matrix order, not the number of scalar entries. Both notions are exact and allow arbitrary coefficients in the ambient real closed field.

The concrete questions answered below are the following.

- (1) Does equality of all power jets of labeled vertices force comparable linear or semidefinite extension complexity? No: the ratios are unbounded.
- (2) Does the zero pattern together with every leading valuation and leading coefficient of a slack matrix determine these complexities? No, even for rank-three slack matrices of polygons.
- (3) Does taking a real or more general ordered residue lose complexity in a controlled direction? Yes: both complexities can only decrease.
- (4) Is non-Archimedean order alone sufficient for nontrivial all-order ambiguity? No: the chosen scale must fail to be an order unit in the value group. Valuation rank two is necessary and sufficient for examples with one layer of such ambiguity.

These are questions formulated and answered here. They are not presented as a list of previously published open conjectures.

62.1 The question of manuscript 04

[write] The following is the opening of manuscript 04 (its Section 1, “The question and the main result”); its explicit family and main theorem are printed in [section 70](#).

A polytope can be complicated in two different ways. Its vertices and faces can have an intricate incidence pattern, or its particular coordinates can make it difficult to express as the projection of a simpler polytope. The first notion is combinatorial. The second is measured by *extension complexity*: the minimum number of facets of a higher-dimensional polytope that projects onto the given one.

For a polygon, every face lattice is a cycle. This leaves considerable room for coordinate-dependent behavior. Regular polygons have logarithmic extension complexity, whereas suitable generic polygons require at least square-root complexity. These are classical facts, not discoveries of this manuscript [43, 44]. Kwan, Sauermann, and Zhao proved that every *cyclic* n -gon—one with all vertices on a circle—has extension complexity at most $24\sqrt{n}$ [53]. Here “cyclic polygon” does not refer to the higher-dimensional cyclic polytopes on the moment curve.

The question investigated here is more refined:

Can a polygon be indistinguishable from a regular polygon at every order of a fixed infinitesimal scale, while retaining square-root rather than logarithmic extension complexity?

The answer is affirmative in the surreal field, and its exact valuation theoretic boundary can be identified. The interesting phenomenon is not a new abstract face lattice. It is loss of information about an exact existential problem under an all-order asymptotic observation map.

[write] Question (1) of 07 and the question of 04 are the same question for two reference polygons: a logarithmic-size polygon in a triangle (07) and a regular polygon on the unit circle (04). Question (4) of 07 and 04’s “exact valuation theoretic boundary” are the same order-unit theorem ([theorem 72.2](#), with its two-sided consequences [corollary 72.4](#) and [theorem 72.7](#)).

62.2 Relationship to ProveIt and the previous convexity report

The selected repository documentation describes a surreal field development, real closedness, surcomplex algebraic closedness, finite geometry, and a large collection of research reports. It explicitly separates research claims from statements mapped to Lean declarations [39, 40]. The repository snapshot recorded during this investigation is 6996fee43cc97b6c16351def7507d59a95bf62f0, dated 30 September 2026. The audit consulted the root README, the surreal README, the report catalogue, and selected directory metadata; it was not an exhaustive theorem-by-theorem audit or an independent rebuild.

[[write] The draft corrected here, *Finite Convexity and Exact Lexicographic Optimization over the Surreals* (22 September 2026), was supplied privately to the authors of manuscripts 05 and 07; it is not part of ProveIt and could not be checked. The statements below are printed as standalone results.]

The earlier draft *Finite Convexity and Exact Lexicographic Optimization over the Surreals* already treats finite Farkas theory, linear-programming duality, finite convexity theorems and exact lexicographic compilation [41]. That draft was located in the supplied research library; its presence in the maintained repository tree was not established in this audit. The present paper does not reclassify that material as new. Instead, it studies *the size of exact lifts and the information lost by scale reduction*.

62.3 Relation to ProveIt (source 04)

The inspected snapshot of VladimirReshetnikov/ProveIt contains a surreal-number library and a collection of research reports that explicitly distinguish manuscript arguments, finite computational tests, and mapped Lean declarations [54, 55]. Its computer-algebra report also separates exact denotation from approximation and from decidable equality [56]. The present work develops a geometric consequence of that separation: even complete information at every power of one selected scale need not determine a finite exact optimization invariant.

We use the established ordered-field and normal-form foundations. We do not assume the correctness of the repository’s unreviewed new research claims. The source snapshot and the limited inspection scope are recorded in [section 80](#). The current paper supplies no Lean development and makes no assertion about having rebuilt the repository.

[write] Both descriptions were re-checked against the current tree. The surreal README (Algebra/SurrealNumbers/README.md, lines 1–27 and 147) describes the report collection,

the real closed surreal field, the algebraically closed surcomplex numbers, finite geometry among the formalized topics, and the ledger that maps statements to Lean declarations; the computer-algebra report’s README (Algebra/SurrealNumbers/docs/foundations-and-computation/computer-algebra/README.md, lines 24–28) separates “exact denotation from effective coefficient access from decidable equality/order from certified approximation”. The Lean module `Surreal.Surcomplex.RegularPolygon` (Algebra/SurrealNumbers/Surreal/Surcomplex/RegularPolygon.lean: `vertex`, `vertex_radius`, `side_length`, `area_eq`) formalizes regular polygons at surreal radius, which is related to 04’s regular R_n , but neither manuscript uses it; nothing in this Part is formalized.

62.4 What is classical and what is proposed here

Fiorini, Rothvoß and Tiwary established generic polygon lower bounds and small extended formulations for special polygons [43]. Reflection relations are an established tool for compact formulations [44]. Gouveia, Robinson and Thomas proved the generic semidefinite lower bound [47]. The slack-factorization interpretation originates with Yannakakis and its cone generalization [45, 46]; normalization and lower-semicontinuity methods for positive ranks are also established [48].

Component	Status in this manuscript
Generic rank bounds	Classical mechanism, reproved over arbitrary real closed fields; the semidefinite proof avoids an Archimedean step.
Binary reflection lift	Explicit specialization of the classical reflection-relation method.
Residue descent	Valuation-ring formulation with algebraic proofs; not claimed as a new closure theorem.
Explicit P_n, Q_n	Proposed separation package: finite monomial support, all-order agreement, identical initial slacks, and diverging lift sizes.
Rank and precision boundary	Consequences proved here, with the exact scale and set-size hypotheses stated.
Formal verification	Finite exact computational checks only; no machine-checked proof of the manuscript.

Searches of the selected documentation and primary literature did not locate this exact explicit separation package. This is a limited novelty check, not evidence that every related formulation is absent from the literature. All uses of “new” should therefore be understood as *candidate new formulations and consequences*, subject to expert review.

62.5 What is new here, and what is imported (source 04)

The proposed contribution is the combination of exact circular geometry, explicit nonreal algebraic independence, and agreement to *all* orders, together with a necessary-and-sufficient condition for the same phenomenon in arbitrary real closed extensions of $\mathbb{R}$. The resulting failure of reconstruction persists even after one supplies all leading Plücker coordinates, all leading slack entries, and every finite-direction linear optimization value to all orders.

The following ingredients are established mathematics: real closedness of the surreal field and uniqueness of Conway normal forms [52]; quantifier elimination for real closed fields [50, 51]; Yannakakis’ slack-factorization theorem [45]; reflection constructions for regular

polygons [44]; and the polygon bounds just cited. We prove the needed elementary extensions and calculations rather than treating the phrase “by transfer” as a substitute for specifying the finite formula.

63 Fields, polytopes, lifts, and power-jet information

63.1 Set-sized fields first

Unless otherwise stated, K is a set-sized real closed ordered field. For standard parts we assume $K \supseteq \mathbb{R}$. A finite surreal configuration, and any finite collection of putative lift certificates, lie in a set-sized real closed subfield of **No**: generate an ordered field from their coefficients and take its relative real closure in **No**. Thus every first-order argument below is made in a genuine set-sized field.

We use the classical real closedness and normal-form arithmetic of **No** [42]. In particular,

$$\omega^a \omega^b = \omega^{a+b}, \quad a < b \implies \omega^a \ll \omega^b, \quad (63.1)$$

where $x \ll y$ means $|x| < y/m$ for every positive ordinary integer m . A nonzero finite sum of distinct Conway monomials cannot vanish. All arithmetic inside an exponent is surreal *field* arithmetic; no unstated ordinal-arithmetic convention is used.

Every n, m, r, k in this paper is an ordinary finite integer. No result concerns a polytope with an infinite number of vertices. A statement about lifts “over **No**” means a statement about finite coefficient certificates; it does not apply first-order model theory directly to a proper class.

63.2 Finite convexity and the two lift sizes

For $v_1, \dots, v_N \in K^d$ define

$$\text{conv}_K(v_1, \dots, v_N) = \left\{ \sum_j \lambda_j v_j : \lambda_j \geq 0, \sum_j \lambda_j = 1 \right\}.$$

Dimension and facets are relative to the affine hull. A vertex is an extreme point. The finite vertex–facet correspondence, linear-programming duality and the finite Farkas alternative hold over real closed fields. They follow from their finite algebraic proofs or from real-closed-field transfer. No order-topological compactness is assumed.

We allow size zero for a point (a constant affine image of the zero-dimensional cone). For a positive-dimensional polytope P , let $\text{xc}_K(P)$ be the smallest number of facets of a polytope E admitting an affine surjection onto P . The usual definition allowing polyhedral extensions gives the same number. Indeed, a nonnegative factorization of a slack matrix gives a bounded extension after zero factor columns are removed: each remaining nonnegative auxiliary coordinate is bounded by a bounded slack divided by a fixed positive entry. This also explains why unbounded lifts cannot circumvent the lower bound proved later.

Let $\mathcal{S}^k(K)$ denote the symmetric $k \times k$ matrices. Write $\mathcal{S}_+^k(K)$ for the positive-semidefinite cone; equivalently, every principal minor is nonnegative. Define $\text{sxc}_K(P)$ as the least k for which an affine slice of $\mathcal{S}_+^k(K)$ has an affine image equal to P . The slice itself need not be bounded.

Definition 63.1 (Two factorization ranks). For a nonnegative matrix $M \in K^{f \times N}$, $\text{rank}_+^K(M)$ is the least r such that $M = UV$ with $U \in K_{\geq 0}^{f \times r}$ and $V \in K_{\geq 0}^{r \times N}$. Its semidefinite rank $\text{rank}_{\text{psd}}^K(M)$ is the least k admitting $A_i, B_j \in \mathcal{S}_+^k(K)$ with $M_{ij} = \text{tr}(A_i B_j)$.

A facet–vertex slack matrix of $P = \{x : Ax \leq b\} = \text{conv}(v_j)$ has entries $S_{ij} = b_i - A_i v_j$. The classical factorization theorems imply

$$\text{xc}_K(P) = \text{rank}_+^K(S), \quad \text{sx}_K(P) = \text{rank}_{\text{psd}}^K(S), \quad \text{sx}_K(P) \leq \text{xc}_K(P). \quad (63.2)$$

We import the real factorization theorem [45, 46]. Its extension to K needs no analytic limiting argument: for each fixed finite dimension, vertex count, facet count and proposed rank, existence of a factorization and existence of a lift are first-order statements in ordered-field arithmetic. Positive semidefiniteness is a finite list of principal-minor inequalities. The universal equivalence over $\mathbb{R}$ therefore holds over every real closed field. Quantifier elimination for these fields is the standard algebraic input; see also its formalized constructive treatment in [49].

[write] Manuscript 04 proves the linear half of (63.2) directly over K , checking the field and boundedness issues instead of transferring; it is printed as [proposition 63.6](#) in [section 63.6](#) below.

63.3 Finite elements, standard part, and all power jets

Suppose now that $K \supseteq \mathbb{R}$. Its finite-element ring and infinitesimal ideal are

$$\mathcal{O} = \{x \in K : |x| \leq c \text{ for some } c \in \mathbb{R}_{>0}\}, \quad \mathfrak{m} = \{x : |x| < 1/m \text{ for every } m \geq 1\}.$$

Completeness of the *ordinary real coefficient field* identifies $\mathcal{O}/\mathfrak{m}$ with $\mathbb{R}$; write the residue map as st . This does not assert completeness of K .

For a positive infinitesimal s , define

$$\mathfrak{j}_s = \bigcap_{m \geq 1} s^m \mathcal{O}. \quad (63.3)$$

Equality of all s -power jets means equality in every quotient $\mathcal{O}/s^m \mathcal{O}$, or equivalently that the difference lies in $\mathfrak{j}_s$. For a vector, $\mathfrak{j}_s^M$ denotes the Cartesian power of the ideal. These are quotients of the *valuation ring*, not of the field K . In a field the ideal generated by a nonzero s is the whole field.

[write] Manuscript 04 (its Section 3) fixes a positive infinitesimal $\varepsilon \in K$ and sets

$$\mathfrak{j}_\varepsilon = \bigcap_{M \geq 1} \varepsilon^M \mathcal{O}, \quad (63.4)$$

the same ideal for $s = \varepsilon$ (04 writes $\mathcal{I}_\varepsilon$). Here and throughout 04's text M ranges over ordinary finite integers.

For positive infinitesimals a, b we write

$$a <_{\text{pow}} b \iff 0 < a < b^m \text{ for every } m \geq 1.$$

This notation means “smaller than every power,” not merely $a < b$.

Lemma 63.2 (The power-flat ideal; sources 07 and 04). *For every positive infinitesimal s ,*

$$\mathfrak{j}_s = \{x \in K : |x| < s^m \text{ for every } m \geq 1\}.$$

It is a prime ideal of $\mathcal{O}$ and the kernel of

$$\mathcal{O} \longrightarrow \varprojlim_m \mathcal{O}/s^m \mathcal{O}.$$

Moreover $s^{-h} \mathfrak{j}_s = \mathfrak{j}_s$ for each integer $h \geq 0$. [source 04] For $x \in \mathcal{O}$, membership $x \in \mathfrak{j}_s$ is also equivalent to $v(x) > Mv(s)$ for every $M \geq 1$, with $v(0) = +\infty$; and $\mathfrak{j}_s$ is a convex prime ideal of $\mathcal{O}$.

Proof. If $x \in s^{m+1}\mathcal{O}$, write $x = s^{m+1}u$ with u finite. Then $|su| < 1$, so $|x| < s^m$. Conversely, $|x| < s^m$ makes x/s^m finite. The ideal and kernel assertions follow from the intersection definition. If $x, y \in \mathcal{O} \setminus \mathfrak{j}_s$, there are m, n with $|x| \geq s^m$ and $|y| \geq s^n$; hence $xy \notin \mathfrak{j}_s$. Finally, shifting every exponent m by a fixed integer does not change the defining all-orders condition. $\square$

Second proof (source 04), with the valuation and convexity clauses. [write] (Written for $s = \varepsilon$, as in 04.) If $x = \varepsilon^{M+1}u$ with u finite, then $\varepsilon|u| < 1$, and hence $|x| < \varepsilon^M$. Conversely, $|x| < \varepsilon^M$ implies $x/\varepsilon^M \in \mathcal{O}$. Membership in $\varepsilon^M\mathcal{O}$ means $v(x) \geq Mv(\varepsilon)$, and using $M + 1$ instead of M turns the intersection of these weak inequalities into the asserted family of strict ones.

The intersection in (63.4) is an ideal. It is convex because $|y| \leq |x|$ and $x \in \mathfrak{j}_\varepsilon$ imply $|y| < \varepsilon^M$ for every M . If $a, b \notin \mathfrak{j}_\varepsilon$, choose $r, s \geq 1$ with $|a| \geq \varepsilon^r$ and $|b| \geq \varepsilon^s$. Then $|ab| \geq \varepsilon^{r+s}$, so $ab \notin \mathfrak{j}_\varepsilon$. This proves primeness. $\square$

In particular, a nonzero error can disappear in the *entire* s -adic completion. This is failure of separatedness, not slow convergence. Exact equality tests in K still distinguish such an error from zero.

63.4 The all-order observation map (source 04)

[write] This subsection prints the rest of the first part of 04's Section 3 ("The all-order observation kernel"); its second part, on the valuation boundary, is printed with the same theorem of 07 in [section 72](#). In this subsection ε is an arbitrary positive infinitesimal of K .

Consequently $\mathcal{O}/\mathfrak{j}_\varepsilon$ is an ordered integral domain. It is generally not $\mathbb{R}$ and need not be a field. In fact $\varepsilon \notin \mathfrak{j}_\varepsilon$, so the class of ε is a nonzero element in the maximal ideal $\mathfrak{m}/\mathfrak{j}_\varepsilon$. It is important not to conflate this quotient with standard part.

Definition 63.3 (All-order jet data). The all-order observation map is the ring homomorphism

$$J_\varepsilon : \mathcal{O} \longrightarrow \varprojlim_{M \geq 1} \mathcal{O}/\varepsilon^M\mathcal{O}, \quad x \longmapsto (x \bmod \varepsilon^M\mathcal{O})_M. \quad (63.5)$$

Two finite coordinate arrays have the same ε -jet data if their images under this map agree coordinatewise.

The kernel of J_ε is exactly $\mathfrak{j}_\varepsilon$. This language does not assume that an arbitrary surreal has a convergent or formal expansion in integer powers of ε . The individual rings $\mathcal{O}/\varepsilon^M\mathcal{O}$ may have nilpotents; they are not being treated as ordered fields.

[write] The map J_ε (capital italic J) is 04's jet map; its kernel is the flat ideal $\mathfrak{j}_\varepsilon$ (fraktur j). The map of [lemma 63.2](#) is the same map for the scale s .

Lemma 63.4 (Regular rational expressions preserve flat agreement; source 04). *Let $b \in \mathbb{R}^s$, $z - b \in \mathfrak{j}_\varepsilon^s$, and $f \in \mathbb{R}(X_1, \dots, X_s)$ admit a presentation $f = g/h$ with $h(b) \neq 0$. Then $f(z)$ is finite and*

$$f(z) - f(b) \in \mathfrak{j}_\varepsilon.$$

Proof. The ideal property gives $g(z) - g(b), h(z) - h(b) \in \mathfrak{j}_\varepsilon$. Since $\mathfrak{j}_\varepsilon \subseteq \mathfrak{m}$, the standard part of $h(z)$ is $h(b) \neq 0$. Consequently $h(z)$ and $h(b)$ are units of $\mathcal{O}$. The difference

$$\frac{g(z)}{h(z)} - \frac{g(b)}{h(b)} = \frac{(g(z) - g(b))h(b) - g(b)(h(z) - h(b))}{h(z)h(b)}$$

belongs to $\mathfrak{j}_\varepsilon$. $\square$

The condition on the denominator is essential. Division by a sufficiently small flat quantity can reveal information that J_ε discards.

[write] In lemma 63.4 the letter s is a number of variables (04's usage), not a scale. The lemma is the real-base-point case, at an arbitrary scale, of the rational-function clause of 07's lemma 65.1, which is stated at the scale τ with base points in $\mathbb{R}(\tau)$.

63.5 Conventions of manuscript 04

[write] Manuscript 04's Section 2 ("Fields, polytopes, and size conventions") restates the setting above in its own words; it is printed here in full.

A set-sized field is enough (source 04). Unless specifically stated otherwise, K is a *set-sized* real closed ordered field containing the usual $\mathbb{R}$ as an ordered subfield. A finite list of surreal coordinates is contained in such a field: take the field generated by $\mathbb{R}$ and those coordinates and then its real algebraic closure inside the real closed surreal field. This is a set. Accordingly, every use of first-order logic below takes place in a set-sized structure.

The notation $\mathbf{x}_{\mathbb{N}_0}$ means that a finite extension is allowed arbitrary surreal coefficients. It will follow from scalar-extension invariance in corollary 74.7 that the answer is already attained in any real closed subfield containing the input data. No first-order model is made out of a proper class.

For finite $V = (v_1, \dots, v_n) \subset K^d$, define

$$\text{conv}_K(V) = \left\{ \sum_{i=1}^n \lambda_i v_i : \lambda_i \in K_{\geq 0}, \sum_i \lambda_i = 1 \right\}.$$

A polytope is such a convex hull. Throughout, polytopes are nonempty. Faces are the subsets on which a linear functional achieves its maximum, together with the empty face. A linear functional on a polytope attains its maximum at a listed vertex because

$$c \cdot \sum_i \lambda_i v_i \leq \max_i c \cdot v_i.$$

This is finite algebra. We never invoke topological compactness of a closed bounded subset of a non-Archimedean field.

The usual finite halfspace description, vertex criterion, and linear Farkas lemma hold over real closed fields. One may prove them by finite linear elimination. Alternatively, each fixed-size statement is a first-order assertion in the language of ordered fields and transfers from $\mathbb{R}$ by Tarski's theorem [50]. We use these finite facts in the slack-factorization argument below.

Finite elements and standard part (source 04). Define

$$\mathcal{O} = \{x \in K : |x| \leq N \text{ for some } N \in \mathbb{N}\}, \quad \mathfrak{m} = \{x \in K : |x| < 1/N \text{ for all } N \geq 1\}.$$

The finite elements form a convex valuation ring; $\mathfrak{m}$ is its maximal ideal. Its units are exactly the finite elements with finite reciprocal. Every $x \in \mathcal{O}$ is infinitely close to a unique real number: take the supremum in $\mathbb{R}$ of the reals not exceeding x . This defines the ring homomorphism $\text{st} : \mathcal{O} \rightarrow \mathbb{R}$ with kernel $\mathfrak{m}$. Nonnegative elements have nonnegative standard part, and st fixes $\mathbb{R}$.

[write] This is the same ring $\mathcal{O}$ as 07's (a bound by a positive real is a bound by an integer).

Proposition 63.5 (Standard part of a finite convex hull; source 04). *If all $v_i \in \mathcal{O}^d$, then*

$$\text{st}(\text{conv}_K(v_1, \dots, v_n)) = \text{conv}_{\mathbb{R}}(\text{st}(v_1), \dots, \text{st}(v_n)).$$

Proof. A convex coefficient belongs to $[0, 1] \subset \mathcal{O}$. Applying st to a convex combination gives a real convex combination. Conversely, use any real convex coefficients as elements of K . These two observations give the two inclusions. $\square$

The proposition does not assert preservation of the face lattice or of extension complexity. Our examples retain the former but not the latter.

[write] This is Part I's [proposition 13.1](#) and 07's (65.6); all three proofs are the same argument.

63.6 Slack factorization over a real closed field (source 04)

[write] From 04's Section 4 ("Extension complexity over ordered fields"). Its scalar-extension proposition is printed with 07's [corollary 74.7](#); its parameter count after 07's [theorem 67.1](#); its reflection lemma and cyclic upper bound in [section 70](#).

For a full-dimensional polytope $P \subset K^d$ with vertices v_j and irredundant facet inequalities $a_i \cdot x \leq b_i$, its slack matrix is

$$S_{ij} = b_i - a_i \cdot v_j \geq 0.$$

The nonnegative rank $\text{rank}_+^K(S)$ is the least r for which $S = TU$, with $T \in K_{\geq 0}^{f \times r}$ and $U \in K_{\geq 0}^{r \times n}$. Positive rescaling of a row or column does not change it. [write] (rank_+^K is 07's rank_+^K of [definition 63.1](#).)

Proposition 63.6 (Yannakakis over a real closed field; source 04, the linear half of 07's (63.2)). *If P has positive dimension, then*

$$\text{xc}_K(P) = \text{rank}_+^K(S).$$

Here extension complexity counts facets of a bounded polytope in its affine hull, and allows an affine projection.

Proof. This is the classical factorization theorem [45]; we check the field and boundedness issues. Suppose $S = TU$ and discard any zero columns of T . Consider

$$Q = \{(x, y) : Ax + Ty = b, y \geq 0\}.$$

Every point of Q projects into P . Each vertex v_j lifts using the j th column of U , and convex combinations of these lifts show that all of P is the projection. The set Q is bounded: x ranges over P , and for each column ℓ choose i with $T_{i\ell} > 0$. Then

$$0 \leq y_\ell \leq \frac{b_i - a_i \cdot x}{T_{i\ell}},$$

whose right-hand side is bounded on the finite convex hull P . Thus Q is a polytope with at most r facets in its affine hull.

Conversely, let Q have r facet slacks $g_\ell \geq 0$ and project onto P . Pull back the i th facet slack of P to a nonnegative affine function on Q . The affine Farkas lemma writes it as $c_i + \sum_\ell T_{i\ell} g_\ell$, with all coefficients nonnegative. There is a point of Q above that facet of P , where the function is zero; hence $c_i = 0$. Choose one lift w_j of each vertex v_j and set $U_{\ell j} = g_\ell(w_j)$. Evaluation gives $S = TU$. $\square$

For a matrix with fixed dimensions and an ordinary integer r , introduce the finite formula

$$\text{NF}_r(S) : \exists(T_{i\ell}, U_{\ell j}) \left[T_{i\ell}, U_{\ell j} \geq 0 \wedge \bigwedge_{i,j} S_{ij} = \sum_{\ell=1}^r T_{i\ell} U_{\ell j} \right]. \quad (63.6)$$

All conjunctions and sums here are finite. This is the precise first-order content of a nonnegative-rank upper bound.

64 The explicit separation theorem

[write] In this section and in [sections 65 to 69](#) and [72](#) manuscript 07's scale $\rho = \omega^{-\omega}$ is printed as τ (see "Notation in this Part"). A second construction, on one circle, from manuscript 04 is printed in [section 70](#).

Set

$$\varepsilon = \omega^{-1}, \quad \tau = \omega^{-\omega}. \quad (64.1)$$

For $n \geq 4$, put $k = n - 2$ and $u_j = j/(k + 1)$, $1 \leq j \leq k$. Define

$$A = (\tau, 1 - 2\tau), \quad B = (1 - 2\tau, \tau), \quad (64.2)$$

$$C_j = (\tau u_j, \tau(1 - u_j)^2). \quad (64.3)$$

The easy configuration, in cyclic order, is

$$q_1 = A, \quad q_{j+1} = C_j \ (1 \leq j \leq k), \quad q_n = B. \quad (64.4)$$

Let a_n be the positive real root of $X^{2n+1} - 2$, and set

$$\eta_\ell = \omega^{-\omega^2 a_n^{\ell-1}} \quad (1 \leq \ell \leq 2n), \quad p_i = q_i + (\eta_{2i-1}, \eta_{2i}). \quad (64.5)$$

Finally,

$$Q_n = \text{conv}(q_1, \dots, q_n), \quad P_n = \text{conv}(p_1, \dots, p_n), \quad T = \text{conv}((0, 0), (1, 0), (0, 1)).$$

A subscript on conv or T specifies the coefficient field when needed.

Theorem 64.1 (Explicit all-order separation). *The configurations (64.4)–(64.5) have the following properties for every ordinary finite $n \geq 4$.*

- (i) P_n, Q_n are strictly convex n -gons contained in the strict interior of T . Every increasing labeled triple has positive orientation in both. Their real shadows are the same full-dimensional triangle: $\text{st}(P_n) = \text{st}(Q_n) = T_{\mathbb{R}}$.

- (ii) The configurations have identical τ -power jets: $p_i - q_i \in \mathfrak{i}_\tau^2$. In fact, writing $\eta = \eta_1$,

$$P_n \subseteq Q_n + [-\eta, \eta]^2, \quad Q_n \subseteq P_n + [-\eta, \eta]^2, \quad \eta \prec_{\text{pow}} \tau.$$

Also $T \subseteq P_n + [-3\tau, 3\tau]^2$ and $T \subseteq Q_n + [-2\tau, 2\tau]^2$, while $3\tau < \varepsilon^m$ for every $m \geq 1$.

- (iii) For the canonical edge slacks

$$S(v)_{ij} = \det(v_{i+1} - v_i, v_j - v_i), \quad v_{n+1} = v_1,$$

$S(p) - S(q)$ is entrywise in $\mathfrak{i}_\tau$. The two slack matrices have the same zero pattern, the same valuation of each entry, and the same leading coefficient of each nonzero entry. Both have ordinary rank three.

(iv) With $U(n) = 2\lceil \log_2(n-2) \rceil + 2$,

$$\text{xc}(Q_n) \leq U(n), \quad \text{sxc}(Q_n) \leq U(n), \quad (64.6)$$

$$r = \text{xc}(P_n) \implies r(r-1) \geq 2n, \quad (64.7)$$

$$h = \text{sxc}(P_n) \implies \left(\frac{h(h+1)}{2} \right)^2 \geq 2n. \quad (64.8)$$

In particular $\text{xc}(P_n) \geq \sqrt{2n}$ and $\text{sxc}(P_n) \geq (2n)^{1/4}$.

(v) The bounds remain valid over every real closed ordered extension containing these coordinates, including the finite-certificate interpretation over all of **No**. Every coordinate has at most three Conway monomials. All the examples, for all finite n , lie in one set-sized real closed field of valuation rank three.

Consequently neither the labeled order type, the full system of power jets, nor the initial slack matrix controls exact lift complexity up to any uniform multiplicative constant. The ratios $\text{xc}(P_n)/\text{xc}(Q_n)$ and $\text{sxc}(P_n)/\text{sxc}(Q_n)$ both tend to infinity along these examples. The first ratio is at least of order $\sqrt{n}/\log n$; the second is at least of order $n^{1/4}/\log n$.

[write] Manuscript 07 proves no upper bound for its hard polygon beyond the trivial $\text{xc}(P_n) \leq n$, so its ratio is bounded below only. The same-circle construction of manuscript 04 (theorem 70.1) has a two-sided order $\Theta(\sqrt{n}/\log n)$, because its hard polygon is cyclic and the Kwan–Sauermann–Zhao bound applies; it covers only $n = 2^m$ and linear lifts.

64.1 Proof architecture

The easy polygon is a tiny affine image of equally spaced points on a parabola, with two additional vertices near the other corners of T . Binary reflection gives a logarithmic-size extension. The hard polygon adds one algebraically independent monomial to each coordinate. These monomials are smaller than every power of τ , so they change no τ -power jet or slack leading term, but they force large algebraic parameter counts in any exact lift. The following four sections give the individual proofs; Section 72 proves the common-field assertion.

65 Geometry of the cap and stability beyond all powers

65.1 Four exact orientation identities

For three planar points write $D(X, Y, Z) = \det(Y - X, Z - X)$. Temporarily let $C(u) = (\tau u, \tau(1-u)^2)$. Direct expansion gives

$$D(C(u), C(v), C(w)) = \tau^2(v-u)(w-u)(w-v), \quad (65.1)$$

$$D(A, C(u), C(v)) = \tau(v-u)[1 + \tau(uv - u - v - 1)], \quad (65.2)$$

$$D(C(u), C(v), B) = \tau(v-u)[2 - u - v + \tau(uv + 2u + 2v - 4)], \quad (65.3)$$

$$D(A, C(u), B) = (1 - 3\tau)[1 - \tau(u^2 - u + 2)]. \quad (65.4)$$

For $0 < u < v < w < 1$ every displayed expression has positive leading coefficient. These four cases exhaust the increasing triples of (64.4). Hence every cyclic edge has all other vertices strictly on its left, and every point is an exposed vertex. This proves strict convexity and the stated cyclic order without an appeal to a picture.

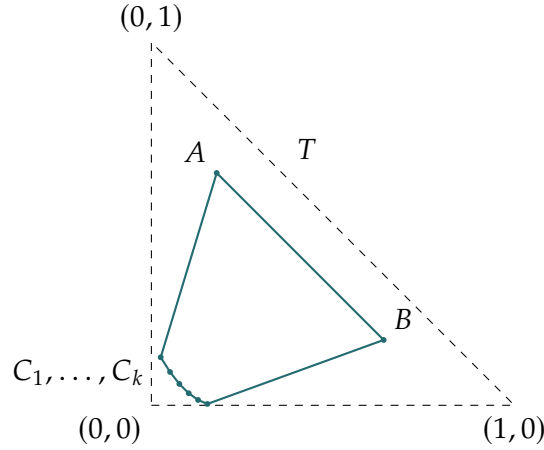


Figure 6: A real specialization of the easy construction, drawn with $\tau = 0.18$ and $k = 6$ to make the cap visible. This schematic does not model the surreal scale inequalities; those are proved algebraically. The hard perturbations would occur at a separate, still smaller scale. [write] (Source 07; its ρ is printed as τ .)

All coordinates are positive. At A, B the coordinate sum is $1 - \tau$. At $C(u)$ it is $\tau(1 - u + u^2) < 1$. Therefore Q_n lies in the strict interior of T .

65.2 A stability lemma with an explicit scope

Lemma 65.1 (Flat perturbations of finite algebraic observations). *Let $q \in \mathbb{R}(\tau)^M$ have finite coordinates, and let $e \in \mathfrak{j}_\tau^M$. If $f \in \mathbb{R}(\tau)[X_1, \dots, X_M]$, then*

$$f(q + e) - f(q) \in \mathfrak{j}_\tau.$$

If $f(q) \neq 0$, the two values have the same sign and the same Conway leading monomial and coefficient. If F is a rational function over $\mathbb{R}(\tau)$ whose denominator is nonzero at q , its denominator is also nonzero at $q + e$, and $F(q + e) - F(q) \in \mathfrak{j}_\tau$.

Proof. Every nonzero $c \in \mathbb{R}(\tau)$ has the form $\tau^h u$ for some $h \in \mathbb{Z}$, where u is finite with nonzero real standard part. This follows by factoring the lowest nonzero power of τ from the numerator and denominator. Thus multiplication by c preserves $\mathfrak{j}_\tau$.

Expanding $f(q + e) - f(q)$ produces a finite sum of terms, each containing an e -factor. All other coordinate factors are finite; the coefficient factors preserve $\mathfrak{j}_\tau$. The sum is therefore flat. If $f(q) = \tau^h u \neq 0$, the error divided by τ^h is still flat and in particular infinitesimal. It cannot change $\text{st}(u)$ or its sign.

For rational functions, apply the polynomial assertion to numerator and denominator. The two denominators have the same nonzero leading term, and their inverses grow at most like a fixed negative power of τ . Cross-multiplication then shows that the difference of the quotients is flat. $\square$

The lemma controls a specified family of observations. It does not say that exact algebra cannot detect the perturbation. For example,

$$2q_{1,1} + q_{1,2} - 1 = 0, \quad 2p_{1,1} + p_{1,2} - 1 = 2\eta_1 + \eta_2 > 0. \quad (65.5)$$

The right side is invisible in every τ -power quotient, but an exact zero test distinguishes it immediately. Thus this is an information-loss theorem for specified reductions, not a lower bound for an unrestricted symbolic equality oracle.

65.3 Perturbed convexity and the full-dimensional shadow

For the scales in (64.5), $a_n > 1$ and

$$0 < \eta_{2n} < \cdots < \eta_1 = \omega^{-\omega^2} < \tau^m \quad (m \geq 1).$$

Apply Lemma 65.1 to the orientation determinants and to the strict inequalities $x > 0$, $y > 0$, $1 - x - y > 0$. All their reference values are nonzero elements of $\mathbb{R}(\tau)$, with the required signs. It follows that P_n is also a strictly convex n -gon in the strict interior of T , with the same labeled order type.

For completeness, standard part commutes with a finite convex hull of finite points:

$$\text{st}(\text{conv}_K(v_1, \dots, v_N)) = \text{conv}_{\mathbb{R}}(\text{st}(v_1), \dots, \text{st}(v_N)). \quad (65.6)$$

Indeed, convex coefficients lie in $[0, 1] \subseteq O$, so one inclusion follows by taking their standard parts. Conversely, real convex coefficients themselves are admissible lifts. [write] (This is proposition 63.5 of source 04 and Part I's proposition 13.1.) Here

$$\text{st}(A) = (0, 1), \quad \text{st}(B) = (1, 0), \quad \text{st}(C_j) = (0, 0),$$

and the perturbations have zero standard part. This proves the common triangular shadow.

Matching the same convex coefficients on p_i and q_i proves the two inclusions with error box $[-\eta_1, \eta_1]^2$ in Theorem 64.1. Each vertex of T is within 2τ in the maximum norm of a reference vertex: use A, B for the nonzero corners and any C_j for the origin. Interpolating these three matches with real or K -valued convex weights proves $T \subseteq Q_n + [-2\tau, 2\tau]^2$. The perturbed inclusion follows because $2\tau + \eta_1 < 3\tau$. Finally, $3\tau < \varepsilon^m$ for each finite m , as $\omega - m$ is positive infinite. This proves parts (i) and (ii) of Theorem 64.1.

66 An explicit logarithmic-size reflection lift

The following elementary recurrence is the source of the small upper bound. Its mechanism is an affine instance of the reflection-relation method [44]. We give the whole construction because it is a short, exact certificate over any ordered field.

Lemma 66.1 (A two-inequality reflection step). *Let*

$$H_N = \text{conv}\{(j, j^2) : j = 0, \dots, N\}, \quad M = \lfloor N/2 \rfloor.$$

If H_M is the affine image of a polytope with r inequalities, then H_N has an extension with at most $r + 2$ inequalities.

Proof. The affine involution

$$R_N(x, y) = (N - x, y + N^2 - 2Nx) = (x, y) + (N - 2x)(1, N) \quad (66.1)$$

sends (j, j^2) to $(N - j, (N - j)^2)$. The two index sets $\{0, \dots, M\}$ and $\{N - M, \dots, N\}$ cover $\{0, \dots, N\}$, so $H_N = \text{conv}(H_M \cup R_N(H_M))$.

Suppose the given extension uses a variable z with affine output $(x(z), y(z))$. Introduce one scalar λ and impose exactly

$$0 \leq \lambda \leq N - 2x(z). \quad (66.2)$$

The upper bound is nonnegative because $0 \leq x(z) \leq M$. Project the new extension by

$$(z, \lambda) \mapsto (x(z) + \lambda, y(z) + N\lambda).$$

For fixed z , this traces the segment joining $(x(z), y(z))$ and its image under R_N . Thus every output lies in the required convex hull. Conversely, the new feasible set is convex, and its image contains both endpoint sets, so its image contains their convex hull. The extension remains bounded: z was bounded and $0 \leq \lambda \leq N$. $\square$

[write] Manuscript 04's reflection lemma (lemma 70.2, a Euclidean reflection in a hyperplane through the origin) is the same two-inequality step for a different involution; it is printed with 04's regular polygons in section 70.

Corollary 66.2 (Parabola formulation). *For every $N \geq 1$,*

$$\text{xc}(H_N) \leq 2\lceil \log_2(N+1) \rceil.$$

More explicitly, follow the chain $N, \lfloor N/2 \rfloor, \dots, 0$ in reverse, beginning at $(0, 0)$, and apply (66.2) at each nonzero stage. The number of stages is $\lceil \log_2(N+1) \rceil$.

Proof. The starting point requires no inequalities. Each nonzero stage adds two. The binary length identity gives the stated count. $\square$

Lemma 66.3 (Adding one vertex). *If a positive-dimensional polytope P has an extension with r facets, then $\text{conv}(P \cup \{a\})$ has one with at most $r + 1$ facets.*

Proof. Let E be an extension and $\pi : E \rightarrow P$ its affine output. Form a pyramid over E in one extra dimension. The base and one pyramid facet over each facet of E give $r + 1$ facets. Extend π affinely so that the apex is sent to a . The image is precisely $\text{conv}(P \cup \{a\})$. $\square$

The cap $\text{conv}(C_1, \dots, C_k)$ is an invertible affine image of H_{k-1} : first replace j by $(j+1)/(k+1)$, then replace (u, u^2) by $(\tau u, \tau(1 - 2u + u^2))$. For $k \geq 2$ this cap has positive dimension. Adding A and B and applying Corollary 66.2 and lemma 66.3 yields

$$\text{xc}(Q_n) \leq 2\lceil \log_2 k \rceil + 2 = U(n).$$

A nonnegative orthant is a diagonal positive-semidefinite cone, so the same upper bound holds for $\text{sx}(Q_n)$. No approximation of the parabola or of the output polygon has been used.

67 Generic lower bounds over real closed fields

This section isolates the only complexity-lower-bound input. It is a field-general version of the classical algebraic parameter-counting method [43, 47]. Its hypotheses concern algebraic independence of coordinates, not independence of the inequalities of an unknown lift. All auxiliary coefficients of that lift remain unrestricted.

67.1 A linear parameter count

Theorem 67.1 (Generic linear lower bound). *Let $F \subseteq K$ be an ordered subfield, and let $P \subset K^d$ be a full-dimensional polytope with N vertices. Suppose its dN labeled vertex coordinates are algebraically independent over F . Then every linear extension of P with r facets satisfies*

$$dN \leq r(r-1). \tag{67.1}$$

Proof. Choose a bounded extension E and put $q = \dim E$. A bounded q -dimensional polytope has at least $q + 1$ facets, so $q \leq r - 1$. One elementary reason is that at most q inequality normals either have a common nonzero kernel direction or, if they form a basis, admit a direction making every scalar product negative; either case gives an unbounded ray from a feasible point.

The affine output $E \rightarrow P$ has rank d . Complete its d coordinate functions to an affine coordinate system on $\text{aff}(E)$. In these coordinates the output is projection onto the first d entries, so no separate output parameters need to be counted. Write the r facet inequalities using $q + 1$ coefficients each. Divide each row by the absolute value of one of its nonzero coefficients. One coefficient in every row is now 1 or -1 , both in F , leaving at most rq scalar parameters.

Let L be the field generated over F by these parameters. At each vertex of E , q linearly independent active equations determine the coordinates. Gaussian elimination puts all those coordinates in L . Every vertex v of P is the image of a vertex of E : expose v by a K -linear objective and take a vertex of the resulting nonempty face of E . Here there is no requirement that the objective have coefficients in F or in $\mathbb{R}$.

Therefore all dN coordinates of P lie in L , and

$$dN \leq \text{trdeg}_F L \leq rq \leq r(r - 1).$$

The equality between bounded-extension size and the usual formulation size was explained in [Section 63](#), so allowing unbounded polyhedral extensions does not evade the result. $\square$

[write] Manuscript 04 proves the same parameter count in a form that does not assume full independence: it bounds the complexity by the transcendence degree s of the coordinate field. That form is needed for 04's circle construction, whose $2n$ coordinates have transcendence degree only n ([remark 70.6](#)); it is printed next. [Theorem 67.1](#) supersedes it in the count: the proof above never uses independence before its last line, so it shows $\text{trdeg}_F F(\text{vertex coordinates}) \leq r(r - 1)$ for every full-dimensional P with an r -facet extension, which is sharper than 04's $s \leq r^2$.

Proposition 67.2 (Parameter count; source 04). *Let $F \subseteq K$ be any subfield, and let $P \subset K^d$ be a full-dimensional polytope. If the field generated over F by its vertex coordinates has transcendence degree s , then*

$$\text{xc}_K(P) \geq \lceil \sqrt{s} \rceil. \quad (67.2)$$

Proof. This is the parameter-count method in [\[43\]](#), with its base field made explicit. If $d = 0$, then $s = 0$ and the conclusion is immediate. Otherwise, let an e -dimensional polytope Q with r facets project onto P . Choose affine coordinates on the affine hull of Q in which the projection is onto the first d coordinates. Write $Q = \{z : Bz \leq c\}$. There are $r(e + 1)$ entries in (B, c) . Every vertex of Q is the unique solution of e independent tight facet equations, so its coordinates belong to $F(B, c)$ by Cramer's rule. Every vertex of P is the image of a vertex of Q : maximize an exposing functional and take a vertex of the resulting nonempty face. Hence

$$s \leq \text{trdeg}_F F(B, c) \leq r(e + 1).$$

A bounded full-dimensional e -polytope has $r \geq e + 1$, giving $s \leq r^2$. $\square$

The coordinate change does not require adding a separate parameter budget for the projection. All its effects are already encoded in the rewritten facet coefficients (B, c) . The proposition does not require $F = \mathbb{Q}$. We will use $F = \mathbb{R}$ for the explicit surreal family and $F = \mathbb{R}_{\text{alg}}$ for the one-parameter existence theorem.

[write] In [proposition 67.2](#) the letters s (a transcendence degree), Q , B and c are local to 04's proof. Its Q is not 07's easy polygon.

67.2 Finite definability replaces rational exposing objectives

Lemma 67.3 (Finite definable sets are algebraic). *If a finite subset $X \subset K^d$ is definable in the language of ordered fields with parameters from a subfield F , then every coordinate of every point of X is algebraic over F .*

Proof. A coordinate projection of X is finite and F -definable. By quantifier elimination it is described by a Boolean combination of finitely many polynomial sign conditions with coefficients in F . Discard identically zero polynomials. If a point a is transcendental over F , none of these polynomials vanishes at a . Their finitely many real roots partition K into intervals on which all signs are constant. The point a lies in one such open interval, so the defining formula would contain that whole interval, contradicting finiteness. Apply this argument to each coordinate projection. $\square$

For a polytope, its finite vertex set is definable using only its membership predicate. In particular,

$$x \in P \wedge \forall y, z \in P (2x = y + z \implies y = z) \quad (67.3)$$

defines the vertices. The midpoint characterization of extremality holds over any ordered field: a nontrivial finite convex representation permits a small positive transfer of weight in both directions.

Theorem 67.4 (Generic semidefinite lower bound). *Under the hypotheses of [Theorem 67.1](#), any semidefinite lift of P of matrix order h satisfies, with $D = h(h+1)/2$,*

$$dN \leq D^2 \leq h^4. \quad (67.4)$$

Proof. Let the lift be an affine slice of $\mathcal{S}_+^h(K)$ with affine output onto P . The affine space defining the slice may be parametrized injectively by ℓ variables, with $\ell \leq D$. Since the output has full-dimensional image, its linear part on that affine space has rank d . Make an affine change of these ℓ variables so that the first d variables are exactly the output coordinates. The lift is then

$$P = \left\{ x \in K^d : \exists z \in K^{\ell-d}, \quad G_0 + \sum_{i=1}^d x_i G_i + \sum_{j=1}^{\ell-d} z_j G_{d+j} \succeq 0 \right\}. \quad (67.5)$$

The matrices need not have bounded entries.

In fact $\ell \leq D - 1$. If $\ell = D$, the defining affine space is all of $\mathcal{S}^h(K)$. An affine image of the entire positive-semidefinite cone is a translated cone. A bounded translated cone is a point, contradicting $\dim P = d \geq 1$. Thus the pencil has at most D coefficient matrices, each with D independent entries. Let L be generated over F by those at most D^2 entries.

The set (67.5) is L -definable because positive semidefiniteness is given by principal minors. Formula (67.3) makes its vertex set finite and L -definable. By [Lemma 67.3](#), all dN vertex coordinates are algebraic over L . Algebraic independence over F now gives

$$dN \leq \text{trdeg}_F L \leq D^2 \leq h^4.$$

This argument does not require Slater duality, an attained optimum in the lift, or a rational exposing objective. $\square$

67.3 Why the Archimedean proof needs care

The real proof in [47, Theorem 2.1] uses a rational objective uniquely exposing a chosen vertex. Such objectives exist for real polytopes but can fail over $K \supsetneq \mathbb{R}$. The following exact example isolates the issue without challenging the real theorem.

Example 67.5 (A vertex invisible to all real objectives). Let $0 < s < 1/m$ for every $m \geq 1$ and set

$$P = \text{conv}((0, 0), (-1, s), (1, -2s), (0, -1)).$$

The origin is a vertex. It uniquely maximizes $c_x x + c_y y$ exactly when

$$c_y > 0, \quad s < \frac{c_x}{c_y} < 2s.$$

The vector $(3s/2, 1)$ satisfies this condition, but no nonzero vector in $\mathbb{R}^2$ does. Its real shadow is the full-dimensional triangle with vertices $(-1, 0), (1, 0), (0, -1)$; the standard part of the origin lies in the relative interior of its upper edge.

Proof. Strictly beating the other three points is equivalent to $-c_x + sc_y < 0$, $c_x - 2sc_y < 0$, and $-c_y < 0$. These are exactly the conditions stated. A positive real number cannot lie between s and $2s$, while a nonpositive one does not satisfy the left inequality. $\square$

Thus one must not transport the rational-objective step unchanged into a non-Archimedean field. [Lemma 67.3](#) supplies an algebraic replacement. It also shows why a single surreal coefficient carrying a long normal form cannot secretly encode arbitrarily many independent coordinates for a *semialgebraic* lift: finite definability only recovers elements algebraic over the field of its scalar parameters.

[write] The proofs of [theorem 67.1](#) and [proposition 67.2](#) use a K -linear exposing objective, not a real one, and are therefore unaffected by [example 67.5](#).

68 Explicit independence and the hard lower bounds

A genericity argument alone would only assert the existence of a suitable perturbation. Here the independence has a finite, exact normal-form proof.

Lemma 68.1 (A monomial independence certificate). *For each fixed n , the $(2n + 1)$ -tuple*

$$\tau, \eta_1, \dots, \eta_{2n}$$

is algebraically independent over $\mathbb{R}$. Consequently, the $2n$ coordinates of $p_1, \dots, p_n$ are algebraically independent over $\mathbb{R}(\tau)$, and in particular over $\mathbb{R}$.

Proof. Eisenstein's criterion at 2 makes $X^{2n+1} - 2$ irreducible over $\mathbb{Q}$. Hence $1, a_n, \dots, a_n^{2n-1}$ are linearly independent over $\mathbb{Q}$. Consider a polynomial in formal variables $R, E_1, \dots, E_{2n}$ with real coefficients. A monomial $R^b \prod_\ell E_\ell^{m_\ell}$ evaluates to

$$\omega^{-b\omega - \omega^2 \sum_\ell m_\ell a_n^{\ell-1}}.$$

Two distinct exponent tuples cannot give the same surreal exponent. Indeed, equality would give

$$(b - b')\omega + \omega^2 \sum_\ell (m_\ell - m'_\ell) a_n^{\ell-1} = 0.$$

Divide by ω^2 . The second summand is real and the first is an infinitesimal real multiple of ω^{-1} . Thus each must vanish. The independence of the powers of a_n then gives $m_\ell = m'_\ell$ for every ℓ , and also $b = b'$.

A nonzero polynomial therefore becomes a finite sum of distinct Conway monomials with nonzero real coefficients. It cannot vanish. This proves the first assertion. Clearing denominators shows that the η_ℓ are independent over $\mathbb{R}(\tau)$. Finally, coordinatewise translation by $q_i \in \mathbb{R}(\tau)^2$ is a polynomial automorphism over $\mathbb{R}(\tau)$, so it preserves algebraic independence. $\square$

[write] Manuscript 04's independence certificate (lemma 70.5, rationally independent exponents give algebraically independent Conway monomials) is the general form of the step used here; 04 applies it with the exponents $\omega\sqrt{q_i}$, q_i distinct primes, where 07 uses $\omega^2 a_n^{\ell-1}$ and Eisenstein's criterion.

Applying Theorems 67.1 and 67.4 with $d = 2$, $N = n$, and $F = \mathbb{R}$ proves (64.7) and (64.8). The argument allows the coefficients of a proposed lift to lie in any larger real closed field. Enlarging that field does not create an algebraic relation among the fixed independent vertex coordinates. Thus no choice of extra surreal scales in the lift can defeat the parameter count.

The word *explicit* here has a precise meaning. All coordinates are specified by rational arithmetic, the unique positive root of one given integer polynomial, and finitely many named Conway monomials. It does not assert an algorithm for arbitrary surreal comparison. Nor does it supply a small rational numerical realization with the same provable lower bounds: those are different effectivity questions.

68.1 Numerical sizes of the proved bounds

Table 6 lists bounds, not computed exact complexities. The semidefinite lower column uses the stronger inequality $D^2 \geq 2n$, not only $h^4 \geq 2n$. The easy column bounds *both* lift sizes of Q_n .

n	Easy upper bound	Hard LP lower bound	Hard PSD lower bound
258	18	24	7
4 098	26	92	13
65 538	34	363	27
1 048 578	42	1 449	54
16 777 218	50	5 794	108

Table 6: Consequences of the proved inequalities. The large examples are symbolically defined families, not explicitly enumerated in the finite verification script. A lower bound smaller than the easy upper bound does not establish a gap at that particular n . [write] (Source 07. For example, the row $n = 258$ is a linear gap, $24 > 18$, but not a semidefinite one, $7 < 18$.)

Remark 68.2 (Where the linear gap starts). [write] Recomputing from the stated formulas $U(n) = 2\lceil \log_2(n-2) \rceil + 2$ and " $r(r-1) \geq 2n$ " (least such r), the first $n \geq 4$ at which the proved hard linear lower bound exceeds the easy upper bound is $n = 121$: there $U(121) = 16$ and the least r with $r(r-1) \geq 242$ is 17. The gap is not yet monotone: for $131 \leq n \leq 153$ the bound $U(n) = 18$ is not exceeded, and from $n = 154$ on (checked to $n = 2 \cdot 10^5$, and for all larger n because $\sqrt{2n} > 2 \log_2 n + 4$ there) every n has a strict linear gap. The semidefinite columns of table 6 give a strict gap only from the row $n = 1\,048\,578$ on ($54 > 42$). All five rows reproduce exactly. These are bounds, as the caption says, not computed complexities. (Computation by the write phase, not by source 07.)

69 Identical initial slack matrices and identical jets

69.1 Canonical normalization matters

A facet slack row can be multiplied by any positive scalar without changing its factorization ranks. Its valuation and leading coefficient do change under such a multiplication. A comparison of initial slack matrices is therefore meaningful only after a normalization has been fixed. For our cyclically labeled polygons we use the same determinant convention on both sides:

$$S(v)_{ij} = D(v_i, v_{i+1}, v_j), \quad i, j \in \{1, \dots, n\}, \quad v_{n+1} = v_1. \quad (69.1)$$

The strict orientation statements already proved show that this is a facet–vertex slack matrix. The zeros occur exactly at $j = i, i + 1$, with cyclic indices.

Theorem 69.1 (Initial-slack and jet agreement). *For P_n, Q_n the matrices (69.1) satisfy*

$$S(p)_{ij} - S(q)_{ij} \in \mathfrak{i}_\tau \quad \text{for every } i, j.$$

For each nonzero corresponding pair,

$$v(S(p)_{ij}) = v(S(q)_{ij}), \quad \text{lc}(S(p)_{ij}) = \text{lc}(S(q)_{ij}), \quad (69.2)$$

and even $S(p)_{ij}/S(q)_{ij} - 1 \in \mathfrak{i}_\tau$. The only nonzero reference valuations are $0, v(\tau), 2v(\tau)$. Both matrices have ordinary rank three, but their nonnegative and semidefinite ranks have the separations of Theorem 64.1.

Proof. Each slack is a polynomial of degree two in the coordinates. Its perturbation is flat by Lemma 65.1. Reference entries are polynomials of degree at most two in τ , because every reference coordinate is affine in τ . The orientation formulas show that every nonzero one has a positive leading real coefficient. Thus its valuation is one of $0, v(\tau), 2v(\tau)$, and a flat error can change neither that valuation nor its leading coefficient. Dividing by the reference entry, which has growth of a fixed power of τ , preserves flatness. The incident zeros remain zero by the determinant identity itself, not by a perturbation estimate.

Every row is an affine function of the vertex coordinates, so the ordinary rank is at most three. The submatrix on the first three rows and columns has the pattern

$$\begin{pmatrix} 0 & 0 & a \\ b & 0 & 0 \\ c & d & 0 \end{pmatrix}, \quad a, b, d > 0.$$

Its determinant is $abd > 0$, including when $n = 4$. Thus the rank is exactly three. The final assertion follows from (63.2) and the bounds already proved. $\square$

If $s \neq 0$ has normal form beginning $c\omega^{-\gamma}$, write $\text{in}_v(s) = c\omega^{-\gamma}$ and put $\text{in}_v(0) = 0$. Then the theorem says that $\text{in}_v(S(p)) = \text{in}_v(S(q))$ entrywise. The result is stronger than coincidence of tropical valuations or of a Boolean support matrix: it also preserves the actual real coefficients of those leading monomials.

[write] The same-circle family of source 04 has the analogous properties with all nonzero slack valuations equal to 0 (section 71, (71.1)).

69.2 What the negative answer does and does not cover

The theorem rules out a function of the *initial slack matrix alone* that recovers either exact factorization rank for all these inputs. Together with (64.6)–(64.8), it rules out a uniform multiplicative estimate from that information: the same input data occur with an unbounded ratio of possible complexities.

The all-orders assertion is stronger in a different direction. The labeled vertex tuple, the labeled slack tuple, and every fixed polynomial observation over $\mathbb{R}(\tau)$ have the same images modulo all powers of τ whenever the observation has finite values; the rational-function version is given in Lemma 65.1. These statements do not preserve all exact polynomial *zero predicates*: (65.5) is a counterexample. There is no contradiction between a generic hard configuration and an algebraically special easy one.

Nor does this assert a failure of all tropical methods. A method that retains extra initial forms of polynomial relations, higher coarsenings, or additional exact certificates has strictly more information than the entrywise initial slack matrix used here. The precise insufficiency statement must not be broadened beyond its input model.

70 A second construction: regular and cyclic polygons on one circle (source 04)

[write] This section prints manuscript 04's explicit family and main theorem (its Subsection 1.1), the regular-polygon and cyclic bounds (from its Section 4) and its Section 5 proper ("The explicit cyclic construction"). It is the second construction of this Part, beside 07's theorem 64.1. Its hard polygon, called P_n in 04, is printed as P_n° with vertices p_i° to keep it apart from 07's P_n ; in 04's text q_i are distinct ordinary primes; the flat ideal $\mathcal{I}_\varepsilon$ of 04 is printed $\mathfrak{j}_\varepsilon$ and its valuation v as v .

70.1 An explicit family

Let $n = 2^m$ with $m \geq 2$, and put

$$\theta_i = \frac{(2i+1)\pi}{n}, \quad a_i = \tan \frac{\theta_i}{2}, \quad 0 \leq i < n. \quad (70.1)$$

These a_i are ordinary real algebraic numbers. No vertex has angle π , so they are all defined. The trigonometric notation specifies ordinary constants only; no trigonometric function on infinite surreals is used. Define the rational circle parametrization

$$\varphi(z) = \left(\frac{1-z^2}{1+z^2}, \frac{2z}{1+z^2} \right). \quad (70.2)$$

Let q_i be pairwise distinct ordinary primes. In **No**, set

$$\varepsilon = \omega^{-1}, \quad \mu_i = \omega^{-\omega\sqrt{q_i}}, \quad r_i = \varphi(a_i), \quad p_i^\circ = \varphi(a_i + \mu_i), \quad (70.3)$$

and let $R_n = \text{conv}(r_0, \dots, r_{n-1})$ and $P_n^\circ = \text{conv}(p_0^\circ, \dots, p_{n-1}^\circ)$. Every expression ω^α in this manuscript denotes a *Conway monomial*; it is not an unqualified use of an analytic exponential with base ω .

Theorem 70.1 (Explicit all-order cyclic gap; source 04). *The polygons in (70.3) have the following properties.*

- (i) R_n is regular, P_n° has exactly n vertices in the same cyclic labeling, and every vertex of both polygons lies on $x^2 + y^2 = 1$.
- (ii) For every ordinary $M \geq 1$ and every i ,

$$\|p_i^\circ - r_i\|_\infty < \varepsilon^M.$$

Consequently their coordinates have the same image in every $O/\varepsilon^M O$, and their standard parts agree.

- (iii) All 3×3 affine orientation determinants have the same nonzero leading real coefficient and valuation for the two configurations. Their determinant-normalized slack matrices have the same exact zero pattern and the same leading entries.
- (iv) Their extension complexities satisfy

$$m \leq \text{xc}(R_n) \leq 2m, \quad \lceil \sqrt{n} \rceil \leq \text{xc}(P_n^\circ) \leq \min\{n, \lceil 24\sqrt{n} \rceil\}. \quad (70.4)$$

In particular,

$$\frac{\text{xc}(P_n^\circ)}{\text{xc}(R_n)} = \Theta\left(\frac{\sqrt{n}}{\log n}\right).$$

The proof is given in [section 70.6](#). The square-root lower bound comes from transcendence degree, using the classical parameter-count method of Fiorini–Rothvoß–Tiwary [43]. The square-root upper bound is the theorem of Kwan–Sauermaann–Zhao transferred to a real closed field [53]. Thus the order of the gap is sharp within the cyclic family relative to a regular reference polygon. No improved universal bound for ordinary polygons is asserted.

[write] Comparison with 07’s [theorem 64.1](#): 07 separates both linear and semidefinite lift sizes for every $n \geq 4$ inside a fixed triangle, with a one-sided lower bound for the hard polygon; 04 separates linear lift sizes for $n = 2^m$ on a fixed circle, with the two-sided order above. The jet scale is $\tau = \omega^{-\omega}$ in 07 and $\varepsilon = \omega^{-1}$ in 04; 04’s standard parts are the regular polygon itself, not a triangle.

70.2 A direct logarithmic extension for a regular polygon

The following standard reflection construction [44] also supplies the constant in (70.4).

Lemma 70.2 (Adding a reflected copy; source 04). *Let $Q \subset K^d$ lie in $u \cdot x \geq 0$, with $u \neq 0$, and let $\sigma(x) = x - 2(u \cdot x)u/(u \cdot u)$. Then*

$$\text{xc}_K(\text{conv}(Q \cup \sigma Q)) \leq \text{xc}_K(Q) + 2.$$

Proof. Take an extension E of Q with projection π . Adjoin a variable λ and the two inequalities

$$0 \leq \lambda \leq \frac{2u \cdot \pi(z)}{u \cdot u}.$$

Project the resulting bounded polytope by $(z, \lambda) \mapsto \pi(z) - \lambda u$. Its image is convex and consists of segments joining x to $\sigma(x)$ for $x \in Q$. It contains both Q and σQ , and every such segment belongs to their convex hull. Thus the image is exactly the claimed polytope. $\square$

[write] In [lemma 70.2](#), Q is any polytope and σ a hyperplane reflection; 07’s two-inequality step [lemma 66.1](#) is the same device for the affine involution R_N of the parabola.

Corollary 70.3 (Source 04). *For $n = 2^m$, $m \leq \text{xc}_K(R_n) \leq 2m$.*

Proof. Starting with the vertex at angle π/n , suppose that the first t vertices at angles $(2j+1)\pi/n$, $0 \leq j < t$, have been obtained, where $t \leq n/2$. Reflect their convex hull in the line at angle $2t\pi/n$. The reflected vertex indices are $2t - 1 - j$, so this doubles the list to the first $2t$ vertices. The original t vertices lie strictly on one side of the reflecting line, because their angle differences from it are between $-\pi$ and 0 . Applying lemma 70.2 for $t = 1, 2, \dots, 2^{m-1}$, where t doubles at each step, uses $2m$ inequalities. A point has zero facets in its affine hull.

For the lower bound, a polytope with r facets has at most 2^r vertices: a vertex is determined by its set of tight facets. Each vertex of a projection is the image of some vertex of the original polytope. Thus $n \leq 2^r$ and $r \geq m$. $\square$

[write] The lower bound $m \leq xc$ is a general counting bound; 07 states no lower bound for its easy polygon Q_n .

70.3 The cyclic upper bound and its transfer

Proposition 70.4 (Transferred cyclic bound; source 04). *Every cyclic n -gon over a real closed ordered field satisfies*

$$xc_K(P) \leq \min\{n, \lceil 24\sqrt{n} \rceil\}.$$

Proof. Over $\mathbb{R}$, the $24\sqrt{n}$ bound is [53, Theorem 1.3]. We explain why it applies in K . Fix the ordinary integers n and $r = \lceil 24\sqrt{n} \rceil$. Translate and rescale a circle to the unit circle; a positive square root of its squared radius exists in K . For vertices in counterclockwise order set

$$S_{ij} = \det \begin{pmatrix} 1 & p_{i,1} & p_{i,2} \\ 1 & p_{i+1,1} & p_{i+1,2} \\ 1 & p_{j,1} & p_{j,2} \end{pmatrix}, \quad i+1 \pmod{n}. \quad (70.5)$$

The circle equations, distinctness of the vertices, and positivity of all nonincident S_{ij} are polynomial conditions. They specify the desired cyclic polygon and its facet slacks. The sentence saying that every such configuration satisfies $NF_r(S)$ is first-order, with rational coefficients. It holds in $\mathbb{R}$ by the cited theorem and proposition 63.6, and hence in every real closed K . Finally a polygon itself has n facets, giving the other upper bound. $\square$

The integer r is chosen externally for each fixed n . No quantification over a set of integers inside the ordered field, and no assertion about randomness in a non-Archimedean field, is involved.

[write] In (70.5) the p_i are the vertices of an arbitrary cyclic polygon; the matrix is the same edge-normalized slack as 07's (69.1), written as an affine 3×3 determinant.

70.4 Rational identities on the circle

For every z in an ordered field, $1 + z^2 > 0$ and

$$\varphi_1(z)^2 + \varphi_2(z)^2 = 1, \quad 1 + \varphi_1(z) = \frac{2}{1 + z^2}, \quad z = \frac{\varphi_2(z)}{1 + \varphi_1(z)}. \quad (70.6)$$

Thus the coordinate field of a list of points $\varphi(z_i)$ is exactly the field of its parameters z_i .

Two further identities make all scale assertions explicit:

$$\varphi_1(s) - \varphi_1(a) = -\frac{2(s-a)(s+a)}{(1+s^2)(1+a^2)}, \quad (70.7)$$

$$\varphi_2(s) - \varphi_2(a) = \frac{2(s-a)(1-as)}{(1+s^2)(1+a^2)}. \quad (70.8)$$

The affine orientation determinant is

$$D(a, b, c) := \det \begin{pmatrix} 1 & \varphi_1(a) & \varphi_2(a) \\ 1 & \varphi_1(b) & \varphi_2(b) \\ 1 & \varphi_1(c) & \varphi_2(c) \end{pmatrix} = \frac{4(b-a)(c-a)(c-b)}{(1+a^2)(1+b^2)(1+c^2)}. \quad (70.9)$$

For instance, multiplying the three rows by their denominators turns the numerator determinant into one with rows $(1+a^2, 1-a^2, 2a)$; elementary column operations reduce it to the Vandermonde determinant, with the factor and sign in (70.9). These are rational identities, not asymptotic expansions.

[write] In (70.7)–(70.8) the letter s is a free variable, not a scale.

70.5 Algebraic independence below every observed scale

Lemma 70.5 (Independent monomials; source 04). *If $\alpha_1, \dots, \alpha_s \in \mathbf{No}$ are linearly independent over $\mathbb{Q}$, then the Conway monomials $\omega^{-\alpha_1}, \dots, \omega^{-\alpha_s}$ are algebraically independent over $\mathbb{R}$.*

Proof. Substitution into a nonzero polynomial gives a finite sum

$$\sum_{e \in E} c_e \omega^{-\sum_i e_i \alpha_i}, \quad c_e \in \mathbb{R} \setminus \{0\}, \quad e \in \mathbb{N}^s,$$

where E is finite. Distinct exponent vectors e give distinct surreal exponents, by rational linear independence. After sorting these exponents, this is a nonempty finite Conway normal form. Uniqueness of normal forms [52] says that it is nonzero. $\square$

The numbers $\sqrt{q_1}, \dots, \sqrt{q_s}$ for distinct primes are linearly independent over $\mathbb{Q}$; in fact 1 together with these numbers is linearly independent. One elementary proof works in the multiquadratic field generated by the prime square roots. The independent prime square classes give automorphisms that change the sign of any one square root and fix the others. Subtracting a putative relation from its image under one such automorphism kills every term except the chosen coefficient. The independence of the prime square classes follows from unique factorization in $\mathbb{Z}$.

Since multiplication by the nonzero surreal ω preserves rational linear independence, lemma 70.5 applies to $\alpha_i = \omega\sqrt{q_i}$. Also $\omega\sqrt{q_i} > M$ for every ordinary M , and hence

$$0 < \mu_i = \omega^{-\omega\sqrt{q_i}} < \omega^{-M} = \varepsilon^M. \quad (70.10)$$

We have therefore exhibited finitely many algebraically independent nonzero elements of the flat ideal, not merely assumed their existence.

70.6 Proof of the main theorem

Proof of theorem 70.1. Choose a set-sized real closed field containing $\mathbb{R}$, ε , and the finitely many μ_i . All subsequent operations occur there. The ordinary half-angle identities show that r_i is the regular polygon vertex $(\cos \theta_i, \sin \theta_i)$. Equation (70.6) puts each p_i° on the same unit circle.

By (70.10) and lemma 63.4, or directly by (70.7)–(70.8), both coordinates of $p_i^\circ - r_i$ belong to j_ε . Their maximum absolute value is still in j_ε . This proves part (ii). The standard parts r_i are distinct, so the p_i° are distinct.

Every affine orientation determinant of three distinct r_i is nonzero: a line intersects a circle in at most two points, as is also clear from (70.9). The corresponding determinant for P_n° differs

from it by an element of $\mathbf{j}_\varepsilon$. It consequently has the same sign, valuation zero, and nonzero leading real coefficient. In particular the edge slacks (70.5) are positive at every nonincident vertex and zero exactly at the two incident vertices. Each p_i° is uniquely exposed by adding its two adjacent edge slack functionals and minimizing the sum. All p_i° are vertices, the edges occur in the same cyclic order, and part (iii) follows as well.

Put $z_i = a_i + \mu_i$. Translation by constants in $\mathbb{R}$ preserves algebraic independence, so the z_i are algebraically independent over $\mathbb{R}$. The inverse identity in (70.6) gives

$$\mathbb{R}(p_{i,1}^\circ, p_{i,2}^\circ : 0 \leq i < n) = \mathbb{R}(z_0, \dots, z_{n-1}).$$

Its transcendence degree over $\mathbb{R}$ is exactly n . Applying [proposition 67.2](#) gives the lower bound for P_n° ; applying [proposition 70.4](#) gives its upper bound. The two bounds for R_n come from [corollary 70.3](#). Scalar-extension invariance permits the notation $\text{xc}_{\mathbf{No}}$ throughout.

Finally the ratio lies between $\sqrt{n}/(2m)$ and $\lceil 24\sqrt{n} \rceil/m$. Since $m = \log_2 n$, these have the same asymptotic order. $\square$

Remark 70.6 (The circle consumes one parameter per vertex; source 04). The coordinate transcendence degree here is n , not $2n$; every vertex satisfies $x_i^2 + y_i^2 = 1$. Applying the unconstrained generic-polygon lower bound $\sqrt{2n}$ without accounting for these relations would be an error. The square-root order remains sharp because the cyclic upper bound applies.

Remark 70.7 (The circle count against 07's parameter count). [write] This is the count $s = n$ that 04's [proposition 67.2](#) needs and 07's [theorem 67.1](#) does not provide as stated (the $2n$ coordinates of P_n° are not independent). The proof of [theorem 67.1](#), which bounds the transcendence degree of the coordinate field by $r(r-1)$, gives with $s = n$ the slightly sharper $\text{xc}(P_n^\circ)(\text{xc}(P_n^\circ) - 1) \geq n$. This observation combines the two sources and is made by the write phase; neither manuscript states it. The order $\sqrt{n}$ is unchanged.

70.7 Only two valuation ranks are needed

The construction does not require the full hierarchy of surreal scales. Let

$$K_n = \text{the real closure inside } \mathbf{No} \text{ of } \mathbb{R}(\omega^{-1}, \mu_0, \dots, \mu_{n-1}).$$

Before real closure the values belong to the additive group generated by 1 and $\omega\sqrt{q_i}$. This follows directly for polynomials from normal-form uniqueness, and for rational functions by subtraction of values. After real closure the values belong to

$$\Gamma_n = \mathbb{Q} + \omega \text{span}_{\mathbb{Q}}\{\sqrt{q_0}, \dots, \sqrt{q_{n-1}}\}. \quad (70.11)$$

Indeed, if $x \neq 0$ is algebraic and $\sum_j b_j x^j = 0$, at least two nonzero summands have the same least value; otherwise the uniquely least one could not cancel. Thus $(j-k)v(x) = v(b_k) - v(b_j)$, so its value lies in the divisible hull of the base value group. Conversely, real closure supplies positive roots and hence all rational multiples of base values. This proves equality with (70.11).

The low component $\mathbb{Q}$ is a convex subgroup, and the quotient is an Archimedean subgroup of $\mathbb{R}$. Both components are nonzero. The natural valuation therefore has rank two. Its element $v(\varepsilon) = 1$ lies in the low component and is not an order unit. The all-order effect is already present in this small ordered-field setting.

Remark 70.8 (One rank-two field for every n). [write] Put $\mu_q = \omega^{-\omega\sqrt{q}}$ for each ordinary prime q (04 indexes the same monomials by vertex, $\mu_i = \mu_{q_i}$). The real closure L inside $\mathbf{No}$ of $\mathbb{R}(\omega^{-1}, \mu_q : q \text{ prime})$ has value group $\mathbb{Q} + \omega \text{span}_{\mathbb{Q}}\{\sqrt{q} : q \text{ prime}\}$ and so is one rank-two field for every n : it contains every K_n , whatever distinct primes are chosen, and when the q_i are the

first n primes the fields K_n form a chain whose union is L . Manuscript 04 states a field K_n for each n . The proof is 04's: the exponents 1 and $\omega\sqrt{q}$ (q prime) are linearly independent over $\mathbb{Q}$, because 1, $\sqrt{q}$ are and ω is infinite, so the least-value argument above gives the divisible hull; its convex subgroups are $0 \subsetneq \mathbb{Q} \subsetneq$ the whole group, with Archimedean quotients.

71 What the asymptotic data cannot recover (source 04)

[write] Manuscript 04's Section 7, for its same-circle family. Its bounded-factor normalization (04's Proposition 7.1) is printed with 07's residue theorem as [theorem 73.2](#), together with the paragraph that follows it in 04. The corresponding statements for 07's family are [theorem 69.1](#) and [lemma 65.1](#).

71.1 Leading Plücker data and slack entries

For a labeled planar configuration V , put

$$H(V) = \begin{pmatrix} 1 & \cdots & 1 \\ v_{0,1} & \cdots & v_{n-1,1} \\ v_{0,2} & \cdots & v_{n-1,2} \end{pmatrix}.$$

Its maximal minors are the affine orientation determinants. In our examples all minors indexed by three distinct columns are nonzero and have valuation zero. Their leading coefficients are exactly the minors of $H(R_n)$. Thus the signs, valuations, and leading coefficients of all these Plücker coordinates are identical for P_n° and R_n .

The edge-normalized slack matrix [\(70.5\)](#) obeys the stronger entrywise identity

$$S(P_n^\circ) - S(R_n) \in \mathfrak{j}_\varepsilon^{n \times n}. \quad (71.1)$$

All nonzero entries have valuation zero and the same positive leading coefficient; the zero pattern is exactly $j = i$ or $j = i + 1$ in both matrices. It follows from [proposition 63.6](#) that this identical leading slack data is compatible with nonnegative ranks of order $\log n$ and $\sqrt{n}$.

These statements concern *entries of the chosen slack matrices and maximal minors of the point matrices*. They do not say that every polynomial identity, or every minor of every derived matrix, is preserved. A polynomial that vanishes at the regular configuration may acquire a nonzero flat value. Such exact equalities are precisely information that an all-order observation can lose.

71.2 Standard part cannot increase nonnegative rank

There is also a useful positive statement, proved without taking limits of possibly unbounded factors. [write] It is printed as the second proof of 07's [theorem 73.2](#) (label `poly:bao:prop:normalization`), of which it is the case $V = \mathcal{O}$, $k = \mathbb{R}$.

71.3 All finite linear optimization values agree to all orders

For a finite vertex list V and $c \in K^d$, define the support function

$$h_V(c) = \max_i c \cdot v_i.$$

This is well-defined without any topological hypothesis.

Proposition 71.1 (Flat stability of bounded observations; source 04). *Suppose $V, W \in (\mathcal{O}^d)^n$ and $v_i - w_i \in \mathfrak{j}_\varepsilon^d$ for each i . Then for every $c \in \mathcal{O}^d$,*

$$h_V(c) - h_W(c) \in \mathfrak{j}_\varepsilon.$$

There is also $\eta \in \mathfrak{j}_\varepsilon \cap K_{\geq 0}$ such that every point of $\text{conv}_K(V)$ is within η in the maximum norm of a point of $\text{conv}_K(W)$, and conversely.

Proof. For finite lists of scalars, $|\max_i \alpha_i - \max_i \beta_i| \leq \max_i |\alpha_i - \beta_i|$. Apply this to $\alpha_i = c \cdot v_i$ and $\beta_i = c \cdot w_i$. Each difference is in $\mathfrak{j}_\varepsilon$, and so is the finite maximum of their absolute values. For the second statement set $\eta = \max_i \|v_i - w_i\|_\infty \in \mathfrak{j}_\varepsilon$. To $\sum_i \lambda_i v_i$ associate $\sum_i \lambda_i w_i$ with the same convex coefficients. The norm of the difference is at most η . Interchanging V, W proves the other direction. $\square$

In particular every finite-direction linear program on P_n° and R_n has the same all-order optimal-value data. This does not mean that the optimal faces always agree, nor does it cover objectives with coefficients as large as reciprocals of the discarded flat perturbations.

[write] The second clause is the analogue of 07's [theorem 64.1](#) (ii) (Hausdorff-type inclusions with error box $[-\eta, \eta]^2$); in [proposition 71.1](#) the scale ε is arbitrary.

The signed area is a polynomial in the vertex coordinates and therefore also agrees modulo $\mathfrak{j}_\varepsilon$. Squared side lengths do as well. For a side whose length has positive real standard part, the difference of lengths is the difference of squared lengths divided by the sum of the lengths; that denominator is a unit in $\mathcal{O}$. Thus the perimeters of our two polygons agree modulo $\mathfrak{j}_\varepsilon$, while the circumradius is exactly one for both. None of these metric observations determines their extension complexities.

71.4 Coherent low-rank factorizations at every finite order

The obstruction can be stated as an explicit failure of lifting a coherent family of approximate solutions to an exact one.

Theorem 71.2 (All-order factorization does not imply exact factorization; source 04). *For $n = 2^m$ with $m \geq 10$, the matrix $S(P_n^\circ)$ has nonnegative rank strictly greater than $2m$, but there are fixed nonnegative real matrices T_0, U_0 of inner dimension at most $2m$ such that*

$$S(P_n^\circ) - T_0 U_0 \in \varepsilon^M \mathcal{O}^{n \times n} \quad \text{for every } M \geq 1.$$

In particular their reductions give compatible algebraic factorizations of every finite jet of $S(P_n^\circ)$, induced by nonnegative lifts, without an exact nonnegative factorization of that inner dimension over K .

Proof. By [corollary 70.3](#) and [proposition 63.6](#), choose a nonnegative real factorization $S(R_n) = T_0 U_0$ of inner dimension at most $2m$. Then use (71.1). The lower bound $\text{rank}_+^K(S(P_n^\circ)) \geq \lceil 2^{m/2} \rceil > 2m$ for $m \geq 10$ excludes the exact factorization. $\square$

Remark 71.3 (The hypothesis $m \geq 10$ is conservative). [write] The inequality $\lceil 2^{m/2} \rceil > 2m$ already holds at $m = 9$ ($\lceil \sqrt{512} \rceil = 23 > 18$); it fails for $2 \leq m \leq 8$ (at $m = 8$, $16 = 16$). The theorem is therefore true for $m \geq 9$ by the same proof. This is a conservative hypothesis, not an error. (Checked by the write phase.)

No positivity structure is being imposed on the possibly nonreduced rings $\mathcal{O}/\varepsilon^M \mathcal{O}$. The factors there are simply the reductions of actual nonnegative real factors. Their compatibility is especially strong: the same pair T_0, U_0 works for every M . The remaining error is not zero;

it lies in the nonzero kernel j_ε . This is a nonseparatedness phenomenon, not an appeal to a failure of numerical precision.

[write] This theorem bears on 07's question on lifting a residue factorization ([question 82.4](#)): coherent finite-order solutions, even one fixed pair for all orders, are not a sufficient lifting criterion. Manuscript 07 makes the same point in words after that question.

Corollary 71.4 (An information-theoretic obstruction; source 04). *There is no function of the labeled all-order vertex jets alone that returns extension complexity for all cyclic surreal polygons. The same is true after adjoining their standard parts, leading Plücker data, leading slack entries, and all finite-direction support-function jets.*

Proof. For sufficiently large n , [theorem 70.1](#) and [proposition 71.1](#) give two inputs with identical such data but different answers. $\square$

This is not a computability claim about every representation of surreal numbers. Exact finite expressions for the μ_i , for example, contain information that the observation map intentionally discards.

[write] (The corollary is unlabelled in 04; the label `poly:bao:cor:information` is added here. 07's counterpart is the discussion after [theorem 69.1](#), which also covers semidefinite rank.)

72 A common field and a sharp valuation-rank boundary

72.1 All examples in one rank-three field

Let $\mathbb{R}_{\text{alg}}$ be the real algebraic numbers and define the divisible ordered subgroup of the additive surreal field

$$\Gamma = \{c\omega^2 + b\omega + d : c \in \mathbb{R}_{\text{alg}}, b, d \in \mathbb{Q}\}. \quad (72.1)$$

Its order is lexicographic by the coefficients c, b, d . Let

$$K_0 = \mathbb{R}(\omega^{-\gamma} : \gamma \in \Gamma) \subseteq \mathbf{No}, \quad K = \{x \in \mathbf{No} : x \text{ is algebraic over } K_0\}.$$

The latter is the relative real closure, with the inherited order.

Proposition 72.1 (Fixed field and fixed rank). *The field K is set-sized and real closed, with real residue field and natural value group exactly Γ . In particular its valuation rank is three. It contains every P_n, Q_n , together with ε and τ , for all ordinary finite $n \geq 4$.*

Proof. The set of monomial generators is a set; rational expressions in finitely many such generators and real coefficients form a set. Nonzero polynomials over K_0 have finitely many roots, so the collection of their surreal roots is also set-sized. It is a field. Positive square roots and roots of odd-degree polynomials over it exist in $\mathbf{No}$ and remain algebraic over K_0 , proving real closedness.

Finite sums and quotients of monomials from Γ have valuations in Γ , and every $\gamma \in \Gamma$ is realized by a generator. If a nonzero x is algebraic over K_0 , choose a polynomial relation $\sum_i a_i x^i = 0$. At least two nonzero summands have the same least valuation; otherwise the uniquely dominant summand could not cancel. Thus

$$v(a_i) + iv(x) = v(a_j) + jv(x) \quad \text{for some } i \neq j.$$

Since Γ is divisible, this puts $v(x)$ in Γ . Therefore algebraic closure inside the ordered field adds no values. The residue field is $\mathbb{R}$, as $K \subseteq \mathbf{No}$ contains all real constants. The convex subgroup chain is exactly

$$0 \subsetneq \mathbb{Q} \subsetneq \mathbb{Q}\omega + \mathbb{Q} \subsetneq \Gamma;$$

each successive quotient is Archimedean. This gives valuation rank three. Every $a_n^{\ell-1}$ is real algebraic, so every exponent in (64.1) and (64.5) belongs to Γ . $\square$

This avoids any infinite-summation issue. The proof does not need a full Hahn-series field, a support-completion theorem, or a changing universe as n increases. The exponent group has countably infinite rational rank but only three Archimedean layers. These are different invariants.

72.2 The exact all-orders kernel condition

For a nontrivial ordered abelian group, a positive element γ is an *order unit* if every positive group element is at most $m\gamma$ for some positive ordinary integer m .

The valuation boundary (source 04). The natural value group is denoted by $G = v(K^\times)$, ordered so that $v(x) > v(y)$ precisely when x/y is infinitesimal. A positive $\gamma \in G$ is an *order unit* if for every $\beta \in G$ there is an integer $M \geq 1$ with $\beta \leq M\gamma$. [write] (04 writes Γ and v ; renamed G and v . The two definitions of order unit agree.)

Theorem 72.2 (When all power jets lose information; sources 07 and 04). *Let $K \supseteq \mathbb{R}$ be real closed, with natural value group G , and let s be a positive infinitesimal. Then*

$$j_s \neq 0 \iff v(s) \text{ is not an order unit in } G. \quad (72.2)$$

Thus all s -power jets determine a finite coordinate tuple exactly if and only if $v(s)$ is an order unit.

Proof. By Lemma 63.2, a nonzero $x \in j_s$ has $v(x) > mv(s)$ for every m : use the bound with $m+1$ if a strict valuation comparison is needed. Conversely, any value $\beta > mv(s)$ for every m is represented by a nonzero field element x with $|x| < s^m$ for every m . This is exactly the negation of the order-unit condition. Equality of all coordinate jets is equivalent to each coordinate difference belonging to this kernel. $\square$

Second proof (source 04; its Proposition 3.3, “Exact nonseparatedness criterion”, stated for $s = \varepsilon$).

A nonzero flat element has value larger than every integer multiple of $v(\varepsilon)$, by lemma 63.2. Conversely, a value with this property is realized by a nonzero element of K by the definition of the value group; that element is finite and belongs to j_ε . $\square$

In a rank-one valued ordered field, the value group is Archimedean and every positive value is an order unit. Thus $j_\varepsilon = 0$. A field of higher rank can have $j_\varepsilon = 0$ for one choice of ε and $j_\varepsilon \neq 0$ for another. The distinction is relative to the observation scale, not just to the rank of the field. [write] (Source 04.)

Example 72.3 (The basic surreal flat element; source 04). For $\varepsilon = \omega^{-1}$, the nonzero element $\omega^{-\omega}$ is smaller than $\varepsilon^M = \omega^{-M}$ for every ordinary M . Thus J_ε is not injective. Equality of all its observations is not equality in $\mathbf{No}$.

Corollary 72.4 (Rank two is the sharp existence threshold). *No valued field of natural valuation rank one has distinct labeled polytopes with identical all-order power jets at a positive infinitesimal scale. There is, however, a single set-sized real closed field of valuation rank two containing pairs of polygons with all the jet, initial-slack, shadow and complexity-separation properties above.*

Proof. A rank-one value group is Archimedean, so every positive element is an order unit. Then [Theorem 72.2](#) makes the jet kernel zero.

For the converse, repeat the construction with

$$\rho = \omega^{-1}, \quad \eta_\ell = \omega^{-\omega a_n^{\ell-1}}, \quad \Gamma_2 = \mathbb{R}_{\text{alg}}\omega + \mathbb{Q}.$$

[write] (That is, with the scale τ of (64.1) replaced by $\rho = \omega^{-1}$; here ρ is 07's general scale letter.) The same orientation, reflection and independence proofs apply. The exponents of ρ and the η_ℓ are independent in exactly the way used in [Lemma 68.1](#). The common field is the relative real closure of $\mathbb{R}(\omega^{-\gamma} : \gamma \in \Gamma_2)$, and the proof of [Proposition 72.1](#) now gives rank two. The separate assertion that the entire polygon is flat-close to T at a *larger* scale ε is omitted: its distance to T is $O(\rho)$. All-order agreement *between the two polygons* remains unchanged. $\square$

The scale condition is sharper than a condition on the whole field. A rank-two field can contain a scale whose value is itself an order unit; power jets at that scale are separated. Our examples deliberately choose a lower layer, leaving room for a nonzero higher-layer error.

Remark 72.5 (Rank three in 07, rank two in 04). [write] Manuscript 07's "valuation rank three" ([theorem 64.1](#) (v), [proposition 72.1](#)) comes only from keeping $\varepsilon = \omega^{-1}$ as a third layer, below the layers of τ and of the η_ℓ , so that both polygons are also flat-close to the triangle T at the larger scale ε ; [corollary 72.4](#) drops exactly that assertion and works in rank two. Manuscript 04's construction needs only rank two from the start: its jet scale is $\varepsilon = \omega^{-1}$ itself, in the low layer $\mathbb{Q}$ of Γ_n ((70.11)), and the perturbations μ_i live in the upper layer. The two statements are consistent, and both fit a single rank-two field for all n ([corollary 72.4](#); [remark 70.8](#)). 04's [corollary 72.9](#) below is 07's paragraph just printed, made explicit: in one rank-two field one scale has a nonzero flat ideal and another has none.

72.3 An exact criterion for hidden complexity (source 04)

[write] Manuscript 04's Section 6. In this subsection ε is an arbitrary positive infinitesimal, as in 04, except in [corollary 72.9](#).

The explicit prime-monomial construction gives independence even over all of $\mathbb{R}$. Surprisingly, the existence of an arbitrary single nonzero flat element suffices for the qualitative and quantitative gap.

Lemma 72.6 (One infinitesimal parameter and generic real coefficients; source 04). *Let $\delta \in K \setminus \mathbb{R}$, and let $F \subseteq \mathbb{R}$ be a subfield. Assume δ is transcendental over $\mathbb{R}$, $b_i \in F$, and $c_1, \dots, c_s \in \mathbb{R}$ are algebraically independent over F . Then*

$$b_1 + \delta c_1, \dots, b_s + \delta c_s$$

are algebraically independent over F .

Proof. For a nonzero $f \in F[X_1, \dots, X_s]$ of total degree D , regard $f(b_1 + Tc_1, \dots, b_s + Tc_s)$ as a polynomial in T with real coefficients. Its leading coefficient is $f_D(c_1, \dots, c_s)$, where f_D is the highest homogeneous part of f . It is nonzero by algebraic independence of the c_i . Substitution $T = \delta$ therefore cannot give zero, since δ is transcendental over $\mathbb{R}$. $\square$

Every nonzero infinitesimal in an ordered extension of $\mathbb{R}$ is transcendental over $\mathbb{R}$: a real closed field has no proper ordered algebraic extension. This observation supplies the hypothesis on δ .

Theorem 72.7 (Flat-ideal classification; source 04). *Let $K \supseteq \mathbb{R}$ be real closed and let $\varepsilon > 0$ be infinitesimal. The following conditions are equivalent.*

- (a) $j_\varepsilon \neq 0$.
- (b) $v(\varepsilon)$ is not an order unit of the natural value group.
- (c) For every $n = 2^m$, $m \geq 2$, there is a cyclic n -gon $P_n^\circ \subset K^2$ on the unit circle, with the same labeled ε -jet data as the regular polygon R_n , and

$$\lceil \sqrt{n} \rceil \leq \text{xc}_K(P_n^\circ) \leq \min\{n, \lceil 24\sqrt{n} \rceil\}.$$

- (d) The ratios $\text{xc}_K(P)/\text{xc}_K(R_n)$ are unbounded among cyclic n -gons P with the same labeled ε -jet data as R_n , with $n = 2^m$ allowed to vary.

Proof. The equivalence of (a) and (b) is [theorem 72.2](#). Assume (a) and choose $0 < \delta \in j_\varepsilon$. Let $F = \mathbb{R}_{\text{alg}}$, the field of real algebraic numbers. For each n , choose real numbers $c_0, \dots, c_{n-1}$ algebraically independent over F , and set

$$z_i = a_i + \delta c_i, \quad p_i^\circ = \varphi(z_i).$$

Such finite independent lists exist because $\mathbb{R}$ has infinite transcendence degree over $\mathbb{R}_{\text{alg}}$. The a_i from (70.1) belong to F , so [lemma 72.6](#) makes the z_i algebraically independent over F . The rational inverse (70.6) makes the vertex-coordinate field have transcendence degree n over F . [Proposition 67.2](#) and [proposition 70.4](#) give the desired bounds. Because $\delta c_i \in j_\varepsilon$, the rational-flat lemma and the strict orientation signs of the base polygon show that these are cyclic n -gons with the prescribed jet data. This proves (c).

Condition (c), together with $\text{xc}_K(R_n) \leq 2 \log_2 n$, implies (d). If (a) fails, equality of jet data is literal coordinate equality by (63.5); hence every polygon occurring in (d) equals its labeled regular reference and every ratio is one. Thus (d) implies (a). $\square$

The theorem is not a criterion for extension complexity of an individual polygon. It is a complete criterion for whether *unbounded complexity gaps can occur within the specified all-order observation fibers*. In the sufficiency proof the lower bound uses the real algebraic base field, not all of $\mathbb{R}$; this distinction is essential because the construction uses generic real coefficients.

Remark 72.8 (One order-unit theorem in both sources). [write] The equivalence (a) $\Leftrightarrow$ (b) is [theorem 72.2](#), printed once above for both sources. 04's (a) $\Leftrightarrow$ (c) $\Leftrightarrow$ (d) (gaps occur iff $j_\varepsilon \neq 0$ iff $v(\varepsilon)$ is not an order unit) and 07's [corollary 72.4](#) (rank one never, rank two already suffices) are the same order-unit theorem read in two directions: 04 fixes an arbitrary field and scale and characterizes the scales, 07 asks which valuation ranks admit some scale. Since a rank-one value group has only order units, 04's criterion implies the first half of 07's corollary; 04's condition (c), applied in one fixed rank-two field with $j_\varepsilon \neq 0$ (for instance the field L of [remark 70.8](#)), gives its second half for linear lifts of cyclic polygons. 07's own proof of the second half also covers semidefinite lifts and the slack and shadow properties.

Corollary 72.9 (Two scales in one rank-two field; source 04). *In the field K_n of (70.11), ω^{-1} has a nonzero flat ideal, while μ_0 has zero flat ideal.*

Proof. The value of ω^{-1} lies in the lower convex component and is bounded above by $v(\mu_0)$ after any integer multiplication. The value $v(\mu_0) = \omega\sqrt{q_0}$, on the other hand, is an order unit of Γ_n : every leading real coefficient in the upper component is bounded by an integer multiple of $\sqrt{q_0}$, and a further increment also dominates any lower component. Apply [theorem 72.2](#). $\square$

Thus changing the observation scale can restore injectivity without changing the field. The obstruction should not be described simply as “higher rank makes all asymptotic information incomplete.”

72.4 Coarsening explains the lost information

Define the convex subring

$$V_\tau = \bigcup_{m \geq 0} \tau^{-m} \mathcal{O}. \quad (72.3)$$

It consists of elements whose growth is bounded by some finite negative power of τ , up to a real finite factor.

Proposition 72.10 (The coarse residue behind all-order agreement). *The ring V_τ is a convex valuation ring of K , its maximal ideal is $\mathfrak{j}_\tau$, and its residue field is real closed. The field $\mathbb{R}(\tau)$ embeds into that residue field, and p_i and q_i have identical residues for every i .*

Proof. The union in (72.3) is a convex ring. If x does not belong to it, then x^{-1} is smaller than every power of τ , so $x^{-1} \in \mathfrak{j}_\tau \subseteq V_\tau$. Hence it is a valuation ring. An element $x \in V_\tau$ is a nonunit precisely when it is smaller than every power of τ ; this identifies its maximal ideal with $\mathfrak{j}_\tau$. The general real-closed residue assertion is proved in [Lemma 73.1](#) below. Every nonzero element of $\mathbb{R}(\tau)$ has valuation a finite integer multiple of $v(\tau)$ and is therefore a unit of V_τ , proving injectivity on that field. Finally, $p_i - q_i \in \mathfrak{j}_\tau^2$. $\square$

Thus the hard polygon becomes the easy configuration after a suitable coarse residue, while ordinary standard part takes both to T . The next section proves that the direction of the resulting complexity inequality is forced in every such residue reduction.

[write] For 04’s family the analogous coarse ring is $\bigcup_{m \geq 0} \varepsilon^{-m} \mathcal{O}$, with maximal ideal $\mathfrak{j}_\varepsilon$; manuscript 04 uses instead the ordered quotient $\mathcal{O} \rightarrow \mathcal{O}/\mathfrak{j}_\varepsilon$ (see the paragraph after [theorem 73.2](#)). The coarse residue is not the standard part, and neither is $\mathcal{O}/\mathfrak{j}_\varepsilon$.

73 Residue descent for positive ranks and polytope lifts

The main example shows that residue reduction can erase a large amount of complexity. It cannot create complexity. We prove this for an arbitrary convex valuation ring, not just the finite-element ring associated with $\mathbb{R}$. These are algebraic counterparts of familiar factor-normalization and rank-closure arguments [48]; novelty is not claimed for the underlying lower-semicontinuity principle.

Let V be a convex valuation ring in a real closed ordered field K , with maximal ideal $\mathfrak{n}$ and ordered residue field $k = V/\mathfrak{n}$. Write $\bar{x}$ for the residue of $x \in V$. A bar over a vector or matrix is entrywise reduction.

Lemma 73.1 (Real closedness of the residue). *The ordered residue field k is real closed.*

Proof. A positive $\bar{a} \in k$ has a positive lift $a \in V$. Its positive square root in K is bounded by $1 + a \in V$, hence belongs to V and reduces to a square root of $\bar{a}$.

Lift a monic odd-degree polynomial over k coefficientwise to one over V . It has a root $x \in K$. Every such root satisfies

$$|x| \leq 1 + \sum_{i < d} |a_i|,$$

where the a_i are its nonleading coefficients: outside this bound the leading power dominates the sum of all others. Convexity puts x in V , and its residue is a root of the original polynomial. The square-root and odd-degree-root characterization now proves real closedness. $\square$

73.1 Nonnegative factors can be made integral

Theorem 73.2 (Nonnegative rank under residue; sources 07 and 04). *If M is a finite nonnegative matrix over V , then*

$$\text{rank}_+^k(\bar{M}) \leq \text{rank}_+^K(M).$$

More strongly, every length- r nonnegative factorization of M can be rescaled to one of length at most r whose factors lie in V .

Proof. Write $M = UW$ with nonnegative factors, and remove zero columns of U together with the matching rows of W . For each remaining factor index α , set

$$s_\alpha = \sum_i U_{i\alpha} > 0, \quad U'_{i\alpha} = U_{i\alpha}/s_\alpha, \quad W'_{\alpha j} = s_\alpha W_{\alpha j}.$$

Then $M = U'W'$, $0 \leq U'_{i\alpha} \leq 1$, and

$$0 \leq W'_{\alpha j} \leq \sum_\beta W'_{\beta j} = \sum_i M_{ij}.$$

Convexity of the valuation ring implies that all entries of U' , W' belong to it. Reducing the finite matrix identity gives a nonnegative factorization of $\bar{M}$ of the same or smaller length. $\square$

Remark 73.3 (Manuscript 04's form: bounded factor normalization). [write] Manuscript 04 proves the case $V = \mathcal{O}$, $k = \mathbb{R}$ (standard part) as its Proposition 7.1 ("Bounded factor normalization"), with a different normalization (column maxima instead of column sums) and one extra conclusion. Its statement: *Let $S \in \mathcal{O}_{\geq 0}^{f \times n}$. Then $\text{rank}_+^{\mathbb{R}}(\text{st } S) \leq \text{rank}_+^K(S)$. Moreover, any nonnegative factorization $S = TU$ can be replaced, after omitting zero columns, by one in which every entry of both factors is finite and every column of the first factor has maximum one.* Its proof follows.

Second proof (source 04), for $V = \mathcal{O}$. For a nonzero column ℓ of T , let $a_\ell = \max_i T_{i\ell} > 0$ and replace

$$T_{i\ell} \mapsto T'_{i\ell} = T_{i\ell}/a_\ell, \quad U_{\ell j} \mapsto U'_{\ell j} = a_\ell U_{\ell j}.$$

The product is unchanged and $0 \leq T'_{i\ell} \leq 1$. Choose i_ℓ with $T'_{i_\ell \ell} = 1$. Nonnegativity of all other summands gives

$$0 \leq U'_{\ell j} \leq S_{i_\ell j}.$$

Thus both factors are finite. Applying st entrywise produces a nonnegative real factorization of the same inner dimension. $\square$

The same normalization permits reduction through the ordered quotient $\mathcal{O} \rightarrow \mathcal{O}/\mathfrak{j}_\epsilon$. For the matrices in (71.1), the reduction is literally the real matrix $S(R_n)$ embedded in this quotient. Its nonnegative rank there equals its real nonnegative rank: a factorization over the quotient would also be one over the real closure of its fraction field, where [corollary 74.7](#) applies. Hence passage to the quotient can strictly lower the rank by the stated asymptotic factor, but cannot raise it for a finite matrix with nonnegative factors. [write] (Source 04.)

The normalization is necessary. A factorization such as $1 = \omega \cdot \omega^{-1}$ cannot simply be reduced term by term in the finite-element ring: one factor is not in the ring. Nonnegativity allows normalization without cancellations, which is why the proof works.

73.2 Semidefinite normalization

Theorem 73.4 (Semidefinite rank under residue). *For every finite nonnegative matrix M over V ,*

$$\text{rank}_{\text{psd}}^k(\bar{M}) \leq \text{rank}_{\text{psd}}^K(M).$$

Any size- h factorization can be replaced, after a possible compression, by positive-semidefinite factors of size at most h with entries in V .

Proof. Take $M_{ij} = \text{tr}(A_i B_j)$ with $A_i, B_j \geq 0$, and put $A = \sum_i A_i$. If $A = 0$, then every $A_i = 0$ and $M = 0$, which is immediate. Otherwise compress to the range of A using an orthogonal basis over K . For positive-semidefinite matrices,

$$\ker A = \bigcap_i \ker A_i :$$

if the quadratic form of their sum vanishes, each nonnegative summand vanishes, and a zero quadratic value for a positive-semidefinite matrix puts the vector in its kernel. Thus the A_i have no components on the removed subspace. Compressing each B_j preserves the traces and positive semidefiniteness.

On the resulting space A is positive definite. Real closedness supplies its positive square root and inverse. Define

$$C_i = A^{-1/2} A_i A^{-1/2}, \quad D_j = A^{1/2} B_j A^{1/2}.$$

Then $C_i, D_j \geq 0$, $\sum_i C_i = I$, and

$$\text{tr}(C_i D_j) = M_{ij}, \quad \text{tr}(D_j) = \sum_i M_{ij}.$$

Every diagonal entry of C_i lies between 0 and 1. Every diagonal entry of D_j lies between 0 and $\text{tr}(D_j) \in V$. If $H \geq 0$, its quadratic forms on $e_a \pm e_b$ give

$$2|H_{ab}| \leq H_{aa} + H_{bb}.$$

Hence all off-diagonal entries of the C_i, D_j also lie in V . Reducing preserves every nonnegative principal minor, so the reduced matrices are positive semidefinite over k . Reduction also preserves the trace identities, yielding the desired factorization. $\square$

All ingredients in this proof are finite linear algebra over a real closed field. The normalization matrices themselves need not lie in V ; only the resulting factors must. No extraction of convergent subsequences or compactness in a surreal topology is involved.

73.3 Polytope shadows cannot have larger lift size

For $P = \text{conv}_K(v_1, \dots, v_N)$ with $v_j \in V^d$, define $\bar{P} = \{\bar{x} : x \in P\}$. Every convex coefficient belongs to V . Conversely, nonnegative residue coefficients summing to one can be lifted to nonnegative elements of V and normalized by their sum, which is a unit with residue one. Therefore

$$\bar{P} = \text{conv}_k(\bar{v}_1, \dots, \bar{v}_N). \tag{73.1}$$

Theorem 73.5 (Residue descent of extension complexity). *Let $P \subset K^d$ be full-dimensional with a finite vertex set in V^d . If $\bar{P}$ is positive-dimensional, then*

$$\text{xc}_k(\bar{P}) \leq \text{xc}_K(P), \quad \text{sxc}_k(\bar{P}) \leq \text{sxc}_K(P). \quad (73.2)$$

The dimension of $\bar{P}$ may be smaller than d . For a point shadow the same conclusions hold with the convention that the lift size of a point is zero.

Proof. Take a relative facet inequality $\bar{a}_i \cdot x \leq \bar{b}_i$ of $\bar{P}$, extending its linear functional from $\text{aff}(\bar{P})$ to k^d . Lift the coefficients of $\bar{a}_i$ to a vector $a_i \in V^d$, and set

$$\beta_i = \max_j a_i \cdot v_j.$$

It belongs to V , and $\bar{\beta}_i = \bar{b}_i$ because the residue inequality is tight. The row

$$M_{ij} = \beta_i - a_i \cdot v_j$$

is a finite nonnegative row over V . Its reduction is the slack of the residue points in that facet inequality.

Let S be a facet–vertex slack matrix of P . Finite linear-programming duality, applied to the tight valid inequality $a_i \cdot x \leq \beta_i$, expresses its slack row as a nonnegative combination of the rows of S . The constant term matches exactly because the inequality is tight. Thus $M = TS$ for some $T \geq 0$ over K . Left multiplication by a nonnegative matrix cannot increase either factorization rank: for the semidefinite rank, replace row factors by the corresponding nonnegative sums. Consequently,

$$\text{rank}_+^K(M) \leq \text{xc}_K(P), \quad \text{rank}_{\text{psd}}^K(M) \leq \text{sxc}_K(P).$$

Apply [Theorems 73.2](#) and [73.4](#) to M .

The columns of $\bar{M}$ include every vertex of $\bar{P}$, possibly with repetitions and extra nonvertex points. Removing those extra columns cannot increase a factorization rank. Adding them back also cannot increase it, because each is a convex combination of vertex slack columns. Hence $\bar{M}$ has exactly the positive ranks of a usual slack matrix of $\bar{P}$. Equation (63.2) proves (73.2). The point case is immediate from the stated convention. $\square$

Corollary 73.6 (Real standard part). *If $K \supseteq \mathbb{R}$ and a full-dimensional polytope has finite coordinates, then neither linear nor semidefinite extension complexity increases under standard part, even if its dimension drops.*

[write] For 04’s family the standard part of P_n° is R_n itself, so [corollary 73.6](#) gives $\text{xc}(R_n) \leq \text{xc}(P_n^\circ)$, consistent with (70.4); for 07’s family the standard part is the triangle $T_{\mathbb{R}}$.

Remark 73.7 (Why facetwise reduction is not a proof). Reducing a chosen list of facet inequalities of P may erase an entire row or produce redundant inequalities. Likewise, a vertex can reduce to a nonvertex, as in [Example 67.5](#). The proof above starts instead with facets of the *residue polytope* and lifts each to a tight inequality of P . This avoids any unsupported claim that standard part induces a face-lattice homomorphism.

74 Prescribed precision and finite conservativity

74.1 Every fixed precision can fail in rank one (source 04)

[write] Manuscript 04’s Section 8 (“Finite precision and arbitrary observation scales”); its second theorem is printed with 07’s [theorem 74.2](#) below.

The all-order ambiguity is absent in rank one, but no *fixed finite* order is sufficient without additional restrictions.

Theorem 74.1 (Explicit finite-order gaps in rank one; source 04). *For every integer $M \geq 1$ and $n = 2^m$, $m \geq 2$, there is a rank-one real closed ordered subfield $K \subset \mathbf{No}$ containing $\mathbb{R}$ and $\varepsilon = \omega^{-1}$, and a cyclic n -gon $P_{n,M}^\circ \subset K^2$, such that*

$$\|p_i^\circ - r_i\|_\infty < \varepsilon^M, \quad \lceil \sqrt{n} \rceil \leq \text{xc}_K(P_{n,M}^\circ) \leq \min\{n, \lceil 24\sqrt{n} \rceil\}.$$

The all-order map J_ε on this K is nevertheless injective.

Proof. Use

$$\mu'_i = \omega^{-(M+1+\sqrt{q_i})}, \quad p_i^\circ = \varphi(a_i + \mu'_i),$$

and take the real closure of $\mathbb{R}(\varepsilon, \mu'_0, \dots, \mu'_{n-1})$ inside $\mathbf{No}$. The real exponents $M+1+\sqrt{q_i}$ are rationally independent: a relation among them would be a relation among 1 and the distinct prime square roots. Thus the μ'_i are algebraically independent over $\mathbb{R}$, giving the lower bound exactly as before. They are smaller than ε^{M+1} ; the finite coefficients in (70.7)–(70.8) then give the strict coordinate bound by ε^M . Cyclicity gives the upper bound.

The value group of the rational-function field lies in the additive subgroup of $\mathbb{R}$ generated by 1 and these real exponents. The least-value cancellation argument following (70.11) shows that the real closure still has value group in their rational span. It is nontrivial and Archimedean, hence of rank one. Now use [theorem 72.2](#) to conclude that $j_\varepsilon = 0$ and J_ε is injective. $\square$

[write] In this proof 04's monomials v_i are printed μ'_i , since v is the valuation in this Part.

This theorem distinguishes two assertions that should not be merged. A particular finite truncation can omit important information even in a separated setting. Equality of *all* truncations forces equality in a separated setting; its failure in [theorem 70.1](#) needs a nonzero flat ideal.

74.2 Beyond an arbitrary set of tolerances

The two named scales in [Theorem 64.1](#) are not essential. The surreal set-cut property permits a version that escapes every member of any prescribed *set* of positive tolerances.

[write] There is also a proper-class version of the choice of scale, while the constructed polygons and every proof still use only set-sized data (source 04's introduction to its version, part (b) below). In part (b) 04's tolerance set $\mathcal{T} \ni \tau$ is printed $\mathcal{D} \ni \delta$.

Theorem 74.2 (Set-sized precision escape; sources 07 and 04). *Two versions are proved.*

(a) [source 07] *Let $F \subseteq \mathbf{No}$ be a set-sized subfield containing $\mathbb{R}$, let $\mathcal{D} \subseteq \mathbf{No}_{>0}$ be a set, and fix $n \geq 4$. There exist surreal n -gons P, Q and a positive infinitesimal ρ such that:*

- (i) P, Q lie in the strict interior of T , have shadow $T_\mathbb{R}$, and have the same labeled order type;
- (ii) for every $\delta \in \mathcal{D}$, $T \subseteq P + [-\delta, \delta]^2$ and $T \subseteq Q + [-\delta, \delta]^2$;
- (iii) their labeled coordinates and canonical slacks agree modulo every power of ρ , and their corresponding nonzero slacks have the same leading terms;
- (iv) $\text{xc}(Q), \text{sx}(Q) \leq U(n)$, while the coordinates of P are algebraically independent over F and therefore satisfy the lower bounds (64.7)–(64.8).

The statement concerns existence over all of $\mathbf{No}$; each resulting pair and each finite certificate still live in a set-sized real closed field.

- (b) [source 04: set-indexed tolerance strengthening] Let $\mathcal{D}$ be any set of positive surreal numbers. There are cyclic polygons P_n° , $n = 2^m$, on the unit circle with the bounds (70.4), such that

$$\|p_i^\circ - r_i\|_\infty < \delta \quad \text{for every } \delta \in \mathcal{D} \text{ and every vertex } i.$$

The exponent scale used in the construction may be chosen once for all n .

Proof of (a) (source 07). Choose $\rho > 0$ below every $1/m$ and every $\delta/4$ with $\delta \in \mathcal{D}$. The right-hand constraints form a set, so the surreal cut property supplies such a ρ . Construct Q from (64.2)–(64.4) [write] (with ρ in place of τ). Choose $\sigma > 0$ below every ρ^m .

Every interval $(0, \sigma)$ in $\mathbf{No}$ is a proper class: the injection $x \mapsto \sigma/(1+x)$ maps the positive surreal class into it. On the other hand, the elements algebraic over a set-sized field form a set. Choose successively $2n$ elements $e_1, \dots, e_{2n}$ in $(0, \sigma)$, each transcendental over $F(\rho)$ and all earlier choices. This is a finite sequence of choices; no class-length choice principle is needed. The resulting tuple is algebraically independent over $F(\rho)$.

Put $p_i = q_i + (e_{2i-1}, e_{2i})$. Every perturbation lies in i_ρ , so the stability and slack proofs apply. Translation over $F(\rho)$ preserves algebraic independence. The reflection lift of Q and the generic lower bounds of P give (iv). Finally, $T \subseteq P + [-3\rho, 3\rho]^2$ and $3\rho < \delta$ for every prescribed δ , proving the tolerance assertion. $\square$

Proof of (b) (source 04). For each $\delta \in \mathcal{D}$, let $c_\delta \omega^{b_\delta}$ be its leading Conway term, with $c_\delta > 0$. Choose a positive surreal Λ larger than every element of the set

$$\{N : N \geq 1\} \cup \{|b_\delta| + N : \delta \in \mathcal{D}, N \geq 1\}.$$

Such a surreal exists by the set-sized cut construction. Put $\mu_i = \omega^{-\Lambda\sqrt{q_i}}$ and $p_i^\circ = \varphi(a_i + \mu_i)$. For each δ , the leading exponent of μ_i/δ is $-\Lambda\sqrt{q_i} - b_\delta$, which is less than $-N$ for every ordinary N . In particular μ_i/δ is infinitesimal. The difference identities (70.7)–(70.8), whose remaining factors are finite, give the stated strict tolerance bound.

Multiplication by nonzero Λ preserves rational independence of the $\sqrt{q_i}$, so the monomials are algebraically independent over $\mathbb{R}$. The proof of theorem 70.1 gives the same cyclic complexity bounds. The choice of Λ was independent of n . $\square$

The set qualification cannot be removed. No nonzero positive surreal is smaller than *every* positive surreal, since half of it is another positive surreal. The theorem is not a construction inside an arbitrary preassigned small subfield, which might not have enough transcendence or coinital scales.

The requirement that $\mathcal{D}$ be a set is indispensable. Taking all positive surreals as tolerances would force every coordinate difference to be zero. The theorem does not say that a fixed set-sized subfield contains polygons below all of its own positive tolerances; the newly chosen scale can lie outside that subfield. [write] (Source 04, on part (b).)

[write] The two parts differ in what they control: (a) fixes n and makes two polygons flat-close to each other at a chosen ρ and close to the triangle below every tolerance; (b) makes one cyclic polygon for every $n = 2^m$ close to the regular polygon below every tolerance, with one scale Λ for all n , which (a) does not assert.

74.3 Finite algebraic polytope data remain conservative

Non-Archimedean geometry in this paper does not create new finite oriented-matroid realizability types. The phenomenon belongs to infinite scale requirements and the information discarded by residue, not to a failure of real-closed-field transfer.

[write] This subsection also prints manuscript 04's Section 9 ("Finite algebraic transfer and its precise limits"), which opens: "The negative results concern an infinitary observation scheme. Every fixed finite semialgebraic question still behaves classically."

Theorem 74.3 (Simultaneous finite conservativity; sources 07 and 04). *For any positive-dimensional finite polytope over a real closed field, there is a polytope with real algebraic coordinates having the same labeled oriented matroid and exactly the same linear and semidefinite extension complexities. Its real extension complexities have those same values. Numerical coordinates and infinite scale requirements need not be preserved.*

Proof. Fix the finite dimension, vertex count and facet count of the given polytope, and choose vertex and facet descriptions. They can be asserted to describe the same full-dimensional polytope by an ordered-field formula. The signs of the finitely many homogeneous determinants encode the labeled oriented matroid. Facet incidence and nonredundancy can also be expressed by finite sign and rank conditions.

Write r and h for the two exact complexities. By (63.2), exact nonnegative rank r is the conjunction of existence of a length- r nonnegative factorization and nonexistence of one of length $r - 1$. Exact semidefinite rank h is expressed in the same way with positive-semidefinite factors. These are first-order conditions; the nonexistence clauses are allowed to be universal.

Quantify the unknown vertex and facet coefficients, keeping only rational constants in the formula. The resulting sentence is true in the original real closed field, hence also in the real algebraic numbers by completeness of real closed fields. This gives the required algebraic realization. Finally, an inclusion of real closed fields is elementary, so its ranks do not change on passing from the real algebraic numbers to $\mathbb{R}$. $\square$

Remark 74.4 (Manuscript 04's form, including the point case). [write] Manuscript 04 states the linear half as its Theorem 9.1 ("Real-algebraic realization with exact extension complexity"): *Let P be a finite-dimensional polytope over a real closed ordered field, with finitely many labeled vertices. There is a polytope P^{alg} with coordinates in $\mathbb{R}_{\text{alg}}$ having the same labeled face lattice and the same exact extension complexity. The result applies in particular to finite surreal polytopes.* The labeled face lattice is determined by the labeled oriented matroid, so for positive dimension this is a special case of theorem 74.3; 04 also covers the zero-dimensional case, a point, where 07's statement assumes positive dimension. 04's proof, which writes the face-lattice conditions as explicit formulas (74.1), follows.

Second proof (source 04), including the point case. A zero-dimensional polytope is a point; its face lattice and extension complexity zero are realized by the origin. Otherwise work in the affine hull so that the dimension is $d \geq 1$. Let there be n vertices, f facets, and let $k = \text{xc}(P)$. These are fixed ordinary integers. Write variables $V = (v_1, \dots, v_n)$, A , and b for a vertex list and an f -row halfspace description.

For each nonempty proper subset $I \subset \{1, \dots, n\}$, the assertion that I is exactly the set of vertices of an exposed face is the finite formula

$$E_I(V) : \quad \exists(c, \beta) \left[\bigwedge_{i \in I} c \cdot v_i = \beta \wedge \bigwedge_{j \notin I} c \cdot v_j < \beta \right]. \quad (74.1)$$

There are only finitely many subsets I . Impose E_I or $\neg E_I$ according to the face lattice of P , and impose a nonzero affine d -minor for full dimension. Singletons in this pattern ensure that all listed points are vertices.

The halfspace and convex-hull descriptions agree if

$$\forall x \left[Ax \leq b \iff \exists \lambda_1, \dots, \lambda_n \geq 0 \left(\sum_j \lambda_j = 1 \wedge x = \sum_j \lambda_j v_j \right) \right].$$

Require each row to expose its prescribed facet, and define $S_{ij} = b_i - a_i \cdot v_j$. Finally impose

$$\text{NF}_k(S) \wedge \neg \text{NF}_{k-1}(S).$$

These conditions, preceded by existential quantifiers for V, A, b , form one finite first-order sentence with rational coefficients. It is true in the original real closed field. Completeness of the theory of real closed fields makes it true in $\mathbb{R}_{\text{alg}}$ [50]. The resulting polytope has the prescribed faces by (74.1), and exact extension complexity k by [proposition 63.6](#). $\square$

Thus surreal coordinates introduce neither new finite polytopal face lattices nor new attainable pairs consisting of a labeled face lattice and an exact extension-complexity value. The new feature is how different realizable values can coexist inside an asymptotic fiber, not an extension-complexity number impossible over the reals. [write] (Source 04.)

Remark 74.5 (Descent and transfer go in different directions). [write] 07's residue descent ([theorem 73.5](#), [corollary 73.6](#)) changes the polytope, replacing it by its residue, and complexity can only drop. 04's real-algebraic transfer ([theorem 74.3](#), [corollary 74.6](#)) also changes the polytope, replacing it by a real-algebraic realization, and the exact complexity is kept. The two maps are different (a residue map versus a choice of algebraic witness by transfer) and the statements do not conflict: the approximant of [corollary 74.6](#) is not the standard part of P_n° (which is R_n , of logarithmic complexity), and a real-algebraic realization with exact complexity $\text{xc}(P_n^\circ)$ is not a residue of P_n° .

Corollary 74.6 (Algebraic finite-accuracy approximants; source 04). *Fix $n = 2^m$ and a positive rational η . There is a cyclic polygon with real algebraic coordinates within η of the corresponding regular vertices and with extension complexity exactly $\text{xc}(P_n^\circ)$ from [theorem 70.1](#). In particular it has extension complexity at least $\lceil \sqrt{n} \rceil$.*

Proof. In the formula of [theorem 74.3](#), additionally impose the unit-circle equations, the cyclic slack signs, and $|p_{i,j} - r_{i,j}| < \eta$. The $r_{i,j}$ lie in $\mathbb{R}_{\text{alg}}$, so these are allowed parameters. The explicit surreal configuration is a witness, and $\mathbb{R}_{\text{alg}}$ is an elementary substructure of the real closed field containing it. Transfer gives the requested algebraic witness. $\square$

[write] Manuscript 04 titles this corollary “Algebraic finite-accuracy shadows”. The polygon it produces is a real-algebraic approximant of the regular polygon, chosen by transfer; it is not a standard part (the standard part of P_n° is R_n) and not a residue in the sense of [section 73](#). We therefore say “approximant”.

The transcendence-degree certificate itself is not preserved by this specialization: all coordinates of the resulting polygon are algebraic. What is preserved is the finite statement $\neg \text{NF}_{k-1}(S)$, whose truth was established using the transcendental witness. This is a useful way to separate a proof device from the property it certifies.

Why there is no contradiction (source 04). The assertion $0 < \mu < \varepsilon^M$ for *every ordinary* M is an infinite family of formulas, not one formula obtained by quantifying over ordinary integers in the language of ordered fields. Nor can $\mathbb{R}$ contain a positive real ε smaller than every positive rational. Consequently [theorem 74.3](#) neither preserves the entire hierarchy of infinitesimals nor embeds the coefficient field into $\mathbb{R}$.

The theorem is a finite realization statement. It is not a statement that one may substitute a single small real number for all surreal monomials and preserve every possible equality, inequality, and complexity bound simultaneously.

Corollary 74.7 (Scalar extension does not simplify fixed real data; sources 07 and 04). *If a polytope is defined over a real closed subfield $F \subseteq K$, its linear and semidefinite extension complexities are identical over F and over K . In particular, a fixed real polytope does not acquire a smaller exact lift merely by allowing surreal coefficients.*

Proof. For each proposed factorization rank, existence is a first-order existential statement with parameters in F . Elementarity transfers its truth in either direction. Take the least successful finite rank. $\square$

Remark 74.8 (Manuscript 04's form: scalar-extension invariance). [write] Manuscript 04 states the linear half as its Proposition 4.2 ("Scalar-extension invariance"): *If $K \subseteq L$ are real closed ordered fields and S has entries in $K_{\geq 0}$, then $\text{rank}_+^K(S) = \text{rank}_+^L(S)$. The extension complexity of a polytope defined over K is likewise unchanged by extension to L . For a finite surreal input, it is unchanged by allowing the entire surreal field as the coefficient domain.* It adds the matrix form and the clause on the whole class **No**, which is how 04 justifies writing xc_{No} . Its proof follows.

Second proof (source 04). An inclusion of real closed ordered fields is elementary by quantifier elimination [50, 51]. Apply this to (63.6) for each r . The polytope assertion follows from [proposition 63.6](#). A putative factorization with surreal entries uses only finitely many coefficients; these and K lie in a set-sized real closed extension, where the same argument applies. $\square$

The corollary and the main example concern different situations. Scalar extension keeps a polytope's exact coordinates unchanged. Our perturbation changes them by nonzero elements of a jet kernel. Neither operation can be substituted for the other.

For a fixed n , one may preserve any *finite* subcollection of the scale inequalities along with the two exact complexities in a real algebraic realization of the pair. One cannot preserve the requirement $0 < \eta < \rho^m$ for every m with positive ordinary real $\rho < 1$ and η [write] (here ρ and η are real stand-ins for the scale τ and the perturbation η_1). That requirement is an infinite scheme, not one of the finite ordered-field sentences used in the proof. The conclusion also does not supply rational-coordinate realizations: $\mathbb{Q}$ is not real closed.

75 The same separation in every fixed dimension

The construction is not confined to a two-dimensional shadow. Iterated pyramids preserve the invisible perturbations while replacing the triangle by a simplex of any fixed dimension.

Lemma 75.1 (Faces do not increase lift size). *If F is a nonempty face of a polytope P , then $\text{xc}(F) \leq \text{xc}(P)$ and $\text{sxc}(F) \leq \text{sxc}(P)$.*

Proof. Pull a supporting equality for F back to an extension of P . For a polyhedral extension this gives a face of the extension, which has at most as many relative facets. For a semidefinite

lift it simply adds an affine equality to the defining slice without changing the matrix order. The resulting images are exactly F . $\square$

Corollary 75.2 (Fixed-dimensional simplex shadows). *For every fixed $d \geq 2$ and every $n \geq 4$ there are d -dimensional polytopes $\widehat{P}_{n,d}, \widehat{Q}_{n,d}$ with $n + d - 2$ vertices, the same labeled face structure, and standard shadow the standard d -simplex, such that their coordinates agree to every order in τ . They admit slack matrices with identical entrywise leading terms and identical all-order jets, while*

$$\text{xc}(\widehat{Q}_{n,d}), \text{sxc}(\widehat{Q}_{n,d}) \leq U(n) + d - 2,$$

and both hard lower bounds from [Theorem 64.1](#) hold with $\widehat{P}_{n,d}$ in place of P_n . All these examples remain in the same rank-three field.

Proof. Apply $d - 2$ ordinary pyramids to each polygon, using the same successive apices at the new standard coordinate vectors. Each pyramid adds one vertex and at most one inequality to an extension. The original polygon is a face, so [Lemma 75.1](#) preserves its lower bounds. The standard part of an iterated pyramid over the standard triangle is the standard d -simplex.

For a pyramid, put the base at last coordinate zero and the apex at the last unit vector. From base inequalities $b_i - a_i \cdot x \geq 0$, the side inequalities are $b_i(1 - z) - a_i \cdot x \geq 0$ and the base inequality is $z \geq 0$. Ordering old vertices before the apex and side facets before the base gives a slack matrix

$$\begin{pmatrix} S & 0 \\ 0 & 1 \end{pmatrix}.$$

Thus common zero patterns, initial entries and flat differences persist at every step. The new vertex coordinates are ordinary constants, so no larger coefficient field is needed. $\square$

Strict interior containment in the simplex is not asserted in this higher-dimensional corollary: the pyramid apices are simplex vertices. The shadow is nevertheless full-dimensional and fixed, and the gaps remain unbounded as the ordinary finite vertex count grows.

76 A rational-chart generalization (source 04)

[write] Manuscript 04's Section 10. It generalizes the lower-bound half of 04's circle construction to rationally parametrized families in any dimension, beside 07's pyramid extension [corollary 75.2](#). Here ε is an arbitrary positive infinitesimal and s a number of parameters.

The circle is a particularly clean realization space because it supplies one exact free parameter per vertex and a sharp-order upper bound. The lower-bound construction extends to rationally parametrized families.

Theorem 76.1 (Generic configurations in a flat rational-chart fiber; source 04). *Let $K \supseteq \mathbb{R}$ be real closed with $\mathfrak{j}_\varepsilon \neq 0$. Let $F \subseteq \mathbb{R}$ have at least s algebraically independent real elements over it. Suppose*

$$\Phi \in F(Z_1, \dots, Z_s)^{dn}$$

is regular at $b \in F^s$, its value $\Phi(b)$ is the labeled vertex list of a full-dimensional real d -polytope P_0 with exactly n vertices, and Φ has a rational left inverse defined over F and regular at $\Phi(b)$. Then there exists a K -polytope P with the same labeled vertex jets as P_0 , exactly n vertices, and

$$\text{xc}_K(P) \geq \lceil \sqrt{s} \rceil.$$

Every rational identity satisfied identically by the parametrization is still exact at P . Every strict affine determinant sign of the base configuration is preserved. In particular, if the base configuration is in affine general position, its labeled oriented matroid and face lattice are preserved.

Proof. Choose $0 < \delta \in \mathfrak{j}_\varepsilon$ and $c_1, \dots, c_s \in \mathbb{R}$ algebraically independent over F . Put $z = b + \delta c$ and take $\Phi(z)$ as the vertex list. Its coordinates agree with $\Phi(b)$ modulo $\mathfrak{j}_\varepsilon$ by [lemma 63.4](#). Each real vertex of P_0 has an exposing real functional with a strictly positive gap from all other vertices. Infinitesimal perturbation preserves these finitely many gaps, so all n new points remain vertices. A nonzero affine d -minor likewise preserves full dimension.

By [lemma 72.6](#), the parameters $z_1, \dots, z_s$ are algebraically independent over F . The rational map and its rational left inverse identify their generated field with the vertex-coordinate field, so this field has transcendence degree s over F . Apply [proposition 67.2](#). Rational identities remain identities under substitution; regularity ensures the denominators used here do not vanish. Every originally nonzero determinant retains its sign because its change is flat. In affine general position this covers all maximal minors and hence the full oriented-matroid data and its face lattice. $\square$

A base configuration with zero maximal minors needs separate care. Flat perturbation can turn such a zero into a nonzero flat determinant, so preserving all originally *strict* signs does not generally preserve a nongeneric face lattice. The cyclic planar case avoids this problem because every triple of distinct circle points is noncollinear.

The theorem suggests a systematic strategy: choose a rational realization chart with many recoverable parameters, choose a low-complexity base point, and move generically in an invisible flat direction. The dimension of the chart gives a lower bound; a separate geometric theorem is needed for any matching upper bound. One must not infer a universal upper bound from the chart argument alone.

[write] As in [remark 70.7](#), the proof of 07's [theorem 67.1](#) would give $\text{xc}_K(P)(\text{xc}_K(P) - 1) \geq s$ here. 07's semidefinite count [theorem 67.4](#) is likewise stated only for fully independent coordinates; 04 states no semidefinite version.

77 Exact computation versus discarded information (source 04)

[write] Manuscript 04's Section 11. Its F_n is a field of 04, not 07's subfield letter F .

77.1 A decidable coefficient core for the explicit examples

The explicit family need not be described by an oracle for arbitrary surreals. Consider the ordered subfield

$$F_n = \mathbb{R}_{\text{alg}}(\varepsilon, \mu_0, \dots, \mu_{n-1}) \subset \mathbf{No}$$

with the monomials from (70.3). The generators are algebraically independent over $\mathbb{R}_{\text{alg}}$: the values 1 and $\omega\sqrt{q_i}$ are rationally independent, and the proof of [lemma 70.5](#) applies.

A polynomial is a finite sum of terms

$$c_{k,e} \varepsilon^k \mu_0^{e_0} \cdots \mu_{n-1}^{e_{n-1}}, \quad c_{k,e} \in \mathbb{R}_{\text{alg}}.$$

After combining equal exponent vectors, the leading term is the one with least value

$$k + \omega \sum_i e_i \sqrt{q_i}.$$

Comparison of these values is lexicographic comparison of

$$\left(\sum_i e_i \sqrt{q_i}, k \right), \tag{77.1}$$

and comparison in the first coordinate is exact comparison of ordinary real algebraic numbers. Distinct exponent vectors cannot have the same pair. The sign of the whole polynomial is therefore the sign of its unique leading coefficient. A rational function is compared by the signs of numerator and denominator. This gives exact field operations, zero testing, and ordering for this finite rational-monomial core.

Proposition 77.1 (In-principle decidability for the explicit input family; source 04). *For a specified $n = 2^m$, the exact value $\text{xc}_{\mathbf{No}}(P_n^\circ)$ is computable from the finite expression (70.3). No effective presentation of the entire surreal field is required.*

Proof. The constants a_i are effectively real algebraic: for $n = 2^m$ they can be computed by the usual real half-angle radical formulas, with signs fixed by their specified angles. The coordinates and edge slacks of P_n° therefore lie effectively in F_n . Apply an effective real-closed-field quantifier-elimination procedure [50, 51] to $\text{NF}_r(S(P_n^\circ))$, for successive integers $3 \leq r \leq \min\{n, \lceil 24\sqrt{n} \rceil\}$. Treat the finitely many coefficients as parameters during elimination and evaluate the resulting polynomial sign conditions by the exact ordered-field procedure above. The first successful r is the nonnegative rank over a real closure of F_n . By [corollary 74.7](#) it is also the nonnegative rank over $\mathbf{No}$, and [proposition 63.6](#) identifies it with extension complexity. $\square$

This proposition asserts termination, not practicality. It does not supply a good running-time bound, a compact minimal extension, or an implemented quantifier-elimination backend. It explains why the information-theoretic obstruction is compatible with an effective exact representation: the expression retains the independent flat generators, whereas all-order ε -jet data erase them.

77.2 What the accompanying program actually checks

The archive includes `code/04-beyond-all-orders-verify.py` and its recorded output `data/04-beyond-all-orders-verification.json`. The program uses exact SymPy arithmetic for the circle and inverse identities, both difference formulas, the orientation determinant, reflection interpolation, rational cyclic polygon slack signs and rank, bounded factor normalization, and the finite index-doubling scheme for regular polygons. [write] (Delivered as `code/verify.py` and `data/verification.json`; shipped in this report under the prefixed names printed here. Run in place, the script would write an unprefixed `data/verification.json` one level above its own directory; rerun it on a copy with the delivered layout, as the report README describes.)

The recorded run used Python 3.13.5 and SymPy 1.14.0 and passed **114 of 114** exact finite checks. The rational test polygons are only test cases for the algebra and sign conventions. They are not claimed to instantiate the transcendence-degree lower bound. No optimization solver was used to certify a minimum nonnegative rank; no computer search is offered as evidence of algebraic independence or of any statement quantified over all orders. The proof of the main lower bound is instead the exact normal-form and parameter-count argument in [section 70](#).

For orientation, [table 7](#) lists theorem bounds, not computed extension complexities. Large values of n make the distinction between low-order observations and exact formulation size conspicuous even though the vertex perturbations are flat for every n .

m	$n = 2^m$	$\text{xc}(R_n) \leq$	$\text{xc}(P_n^\circ) \geq$	$\text{xc}(P_n^\circ)/\text{xc}(R_n) \geq$
8	256	16	16	1
10	1 024	20	32	8/5
12	4 096	24	64	8/3
16	65 536	32	256	8
20	1 048 576	40	1 024	128/5
24	16 777 216	48	4 096	256/3

Table 7: Bounds supplied by the proofs. No exact minimum extension complexity is numerically computed here. The row $m = 8$ alone does not certify a strict gap. [write] (Source 04; its P_n is printed P_n° . All six rows reproduce from (70.4). The last column divides the lower bound for P_n° by the upper bound $2m$ for R_n ; compare 07’s table 6.)

78 Reproducibility and final conclusions

[write] The formalization roadmap of manuscript 07 is printed in section 83 (section 83.5); in 07 it stood between its research questions and this section. Manuscript 04 has no roadmap section. Nothing in this Part is formalized.

78.1 What the executable checks establish

The companion code/07-invisible-complexity-verify.py uses rational arithmetic and SymPy, without floating-point optimization. The recorded run checks:

1. the four symbolic orientation identities and three reflection identities;
2. 132 088 positive triple orientations and 32 384 slack entries in reference and rationally perturbed polygons for $4 \leq n \leq 36$;
3. all 4 686 candidate active bases of the binary reflection lifts for $1 \leq N \leq 31$, with the feasible vertices projecting exactly onto $\{(j, j^2) : 0 \leq j \leq N\}$;
4. 636 symbolic canonical slack entries for $4 \leq n \leq 12$, including their zero pattern and positive lowest coefficients;
5. sample factor-normalization identities and the integer rounding used in Table 6.

The reflection index identity is additionally checked for $1 \leq N \leq 256$. The rational perturbations in these tests are deliberately *not* claimed to be algebraically independent. The all-orders theorem, the surreal independence proof and the universal lower bounds are not validated by substituting very small real numbers.

The run used Python 3.13.5 and SymPy 1.14.0. The JSON output records the counts and exact scope. Running the script again regenerates it. The LaTeX source compiles directly with `latexmk -pdf`; it does not need external image assets or downloaded fonts.

[write] File names: the script was delivered as `verification/verify.py`, its JSON output as `verification/results.json` and its notes as `verification/README.md`; they are shipped here as `code/07-invisible-complexity-verify.py`, `data/07-invisible-complexity-results.json` and `07-invisible-complexity-verification-README.md` (whose text still says “Run `python verification/verify.py`”). The script writes `results.json` next to itself, so running it in place would add an unprefixed `code/results.json`; rerun it on a copy. The delivered LaTeX source `surreal_polytopes.tex` and its 26-page PDF are not shipped; this Part is their merged print, built with the report’s `article.tex`.

78.2 One fully expanded small lift

For H_7 , the binary chain is $0, 1, 3, 7$. The complete formulation is

$$\begin{aligned} 0 &\leq \lambda_1 \leq 1, \\ 0 &\leq \lambda_2 \leq 3 - 2\lambda_1, \\ 0 &\leq \lambda_3 \leq 7 - 2\lambda_1 - 2\lambda_2, \end{aligned}$$

with output

$$x = \lambda_1 + \lambda_2 + \lambda_3, \quad y = \lambda_1 + 3\lambda_2 + 7\lambda_3.$$

This uses six inequalities to produce the eight parabola vertices. For the cap with $k = 8$, apply the affine map

$$\Phi_k(x, y) = \left(\frac{\tau(x+1)}{k+1}, \frac{\tau(y-2kx+k^2)}{(k+1)^2} \right).$$

Two pyramid steps then add A and B . This example illustrates that the logarithmic upper bound is an actual formulation compiler, not only a counting argument.

78.3 Conclusion

The central distinction is between an exact finite polytope and its scale-reduced information. A surreal polygon can retain a large exact linear or semidefinite lift complexity after its labeled coordinates have become indistinguishable, to every power of a chosen scale, from a polygon with a logarithmic-size lift. Even the complete entrywise initial slack matrix fails to recover what was lost. This happens with uniformly finite normal-form support and inside one fixed small real closed field.

The positive counterpart is equally important: every convex residue reduction can only lower these lift sizes. The obstruction is not a failure of real-closed-field algebra, but the nonzero kernel of the chosen power-jet map. Valuation rank two already permits that kernel; a third layer allows both polygons themselves to be flat-close to a common full-dimensional real simplex at a larger scale.

The manuscript supplies explicit constructions, complete arguments for its stated results relative to identified classical inputs, and finite verification artifacts. Independent review and formalization remain the next checks on the proposed contribution, rather than conclusions assumed from the act of generating the article.

78.4 Conclusion of manuscript 04

Finite surreal polytopes obey the classical finite realizability theory, but their exact formulation sizes need not be visible at a chosen scale. A regular polygon and an explicit cyclic perturbation can agree in every power jet, every leading affine Plücker coordinate, every leading slack entry, and every finite-direction linear optimal value, while their extension complexities have different asymptotic orders.

The exact obstruction is the nonzero kernel $\bigcap_M \varepsilon^M \mathcal{O}$. Its presence is equivalent to unbounded cyclic complexity gaps within all-order jet fibers; its absence makes the full jet map injective. This supplies both a negative theorem about reconstruction and a positive boundary theorem identifying when that particular loss of information cannot occur. The main lesson is not that surreal geometry is disconnected from classical geometry. It is that exact algebraic relations and asymptotic observations are different kinds of data, even for finite polygons on a fixed circle.

79 Source and dependency audit (Appendix A of manuscript 07)

The repository was accessed on 30 September 2026. The branch metadata reported the following commit and timestamp:

```
6996fee43cc97b6c16351def7507d59a95bf62f0
2026-09-30T16:07:39Z
```

The selected documentation paths were `README.md`, `Algebra/SurrealNumbers/README.md`, and `Algebra/SurrealNumbers/docs/README.md`. A large tree response was truncated, so it is not treated as an exhaustive inventory. No claim of novelty relies on an assertion that every repository file was searched.

The earlier finite-convexity draft was retrieved by its title and abstract from the research library. Its role here is to mark previously covered topics. None of its unreviewed theorems is a necessary new assumption: the finite linear-programming facts used above are standard ordered-field facts, and the manuscript specifies where the classical factorization and real-closed-field results enter. [write] (The draft is private and not part of ProvelIt; see the note in [section 62.](#))

The external audit used the primary papers [43, 44, 47, 46, 48], the classical surreal reference [42], and the real-algebraic formalization reference [49]. Searches also combined “surreal polytopes,” “surreal extension complexity,” “nonnegative rank Hahn,” and related valuation terms. The search did not establish priority. In particular, the paper does not advertise the generic lower bounds, reflection relations, slack factorization, or rank lower-semicontinuity as original discoveries.

[write] The three documentation paths exist in the current tree, and the pin and its timestamp agree with the repository history. The delivered `SOURCE_AUDIT.md` of 07 is shipped as `07-invisible-complexity-SOURCE_AUDIT.md`; its file list names the delivered layout.

80 Repository, literature, and verification audit (Appendix A of manuscript 04)

80.1 Pinned repository scope

The repository was inspected through its GitHub connection at the snapshot

```
6996fee43cc97b6c16351def7507d59a95bf62f0.
```

The directly read sources relevant to this article were:

1. `Algebra/SurrealNumbers/README.md`, for the stated foundational library and the distinction between formalization and research drafts.
2. `Algebra/SurrealNumbers/docs/README.md`, for the report guide and its verification boundaries.
3. Under `Algebra/SurrealNumbers/docs/`, the report `foundations-and-computation/computer-algebra/README.md`, for the distinction between exact representations, coefficient access, and approximation.

The tree was inspected and a scoped indexed search for the phrase “extension complexity” returned no results. This is not an assertion that every repository file was read or that no related observation exists anywhere in the repository. No change was made to the repository.

[write] Re-checked in the current tree: `rg -il "extension complexity"` over the whole repository, `.lake` excluded, finds no file outside this report's own directory (and `rg -il "extended formulation"` likewise none). This report, and in particular this Part, is now the repository's treatment of extension complexity. The three paths above exist; the computer-algebra README is described accurately (see [section 62](#)).

80.2 Literature dependence and novelty boundary

The crucial external inputs are recorded in [table 8](#). The exact all-order flat-fiber formulation and prime-monomial cyclic construction were not found in the sources inspected here. This limited search is not a certification of originality. The constituent methods are intentionally exposed so that an expert can assess whether the synthesis is new, implicit in earlier work, or best viewed as a useful corollary.

Table 8: Dependence and claim status (source 04).

Ingredient or conclusion	Source or argument	Status
Real closed fields; finite transfer	Tarski; Collins [50, 51]	Classical input
Conway monomials and normal forms	Standard surreal theory, as in [52]	Classical input
Slack-factorization equality	Yannakakis [45]; field check in proposition 63.6	Classical theorem, adapted proof
Regular polygon upper bound	Reflection method [44]; direct proof in corollary 70.3	Classical bound
Transcendence-degree lower bound	Parameter-count method [43]; arbitrary base in proposition 67.2	Classical method
Cyclic square-root upper bound	Kwan–Sauermann–Zhao [53]; transfer in proposition 70.4	Imported theorem
Explicit all-order cyclic gap	Theorem 70.1	Proposed synthesis with proof
Flat-ideal classification	Theorem 72.7	Proposed structural contribution
Invisible exact factorization obstruction	Theorem 71.2	Consequence proved here
Finite-accuracy and set-tolerance versions	Theorems 74.1 and 74.2	Consequences proved here
Exact finite algebraic shadow [write] (approximant)	Theorem 74.3	Standard transfer mechanism, explicit formulation
Exact finite checks	<code>code/04-beyond-all-orders-verify.py</code> ; 114/114 passed	Symbolic and finite, not formal proof
Lean verification of this article	No files or build supplied	Not claimed

80.3 Reproduction

The LaTeX source is standalone and uses an internal bibliography. From the archive root one may run

```
python code/04-beyond-all-orders-verify.py
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

`pdflatex -interaction=nonstopmode -halt-on-error article.tex`

The delivered PDF was compiled from this source. The program exits with an error if any check fails and rewrites its JSON report only on success. A Python environment with SymPy is needed; no internet access or proprietary solver is required for the finite checks. The exact quantifier-elimination algorithm described in [proposition 77.1](#) is not implemented by this script.

[write] Delivered as python code/verify.py, writing data/verification.json; the shipped script is code/04-beyond-all-orders-verify.py and its recorded output data/04-beyond-all-orders-verification.json (with data/04-beyond-all-orders-requirements.txt and code/04-beyond-all-orders-build.sh, which still names article.tex and build/). Run in place, the script writes an unprefixd data/verification.json; use a copy with the delivered layout. The delivered article.tex and its 26-page PDF are not shipped: the report's article.tex is the merged report, which the pdflatex lines above now build, and this Part is 04's merged print.

81 Notation and possible misreadings (Appendix B of manuscript 04)

[write] Printed with this Part's symbols (j_ε for 04's I_ε , P_n° for 04's P_n); see also "Notation in this Part" at the opening of [Part V](#).

Notation	Meaning and restriction
K	A set-sized real closed ordered field; it contains $\mathbb{R}$ in the flat-ideal theorems, but need not do so in the general scalar-extension or effective-core arguments.
No	The surreal class field, used only through finite input lists and set-sized subfields when applying first-order logic.
$\mathbb{R}_{\text{alg}}$	The real algebraic numbers, not all reals.
ω^α	Conway monomial, with normal-form multiplication and ordering.
$\mathcal{O}, \mathfrak{m}$	Finite elements and ordinary infinitesimals relative to the embedded real field.
st	Ordinary real standard part; not the quotient by j_ε .
j_ε	$\bigcap_{M \geq 1} \varepsilon^M \mathcal{O}$, the all-order flat ideal.
J_ε	The compatible list of reductions modulo $\varepsilon^M \mathcal{O}$; no integer-power expansion is assumed to exist.
xc_K	Minimum facet count of a bounded extension polytope over K , allowing affine projection and measuring facets in the affine hull.
rank_+^K	Nonnegative factorization rank over the specified field.
Cyclic polygon	A planar polygon with vertices on one circle; not a higher-dimensional moment-curve polytope.
Leading slack data	Entrywise values and leading coefficients for the specific determinant-normalized slack matrix, not all its minors.
All-order agreement	Equality under J_ε , not equality in K unless $j_\varepsilon = 0$.

82 Further research questions and concrete next steps

[write] The ten questions of manuscript 07 (its Section 13) and the nine of manuscript 04 (its Section 12, "Further research questions"), with duplicates merged and both sources named. Three pairs are merged: the exact orders of the hard families, lifting a reduced factorization,

and the formalization path. 04's remaining questions follow 07's in their own subsections, as 04 prints them. Where a result of the other source bears on a question, a [write] note re-scopes it.

The following questions are proposed by the present results. They are a research program, not a claim that each has already appeared as a named open problem. In particular, the existence and valuation-rank threshold questions settled above are not listed again as unresolved.

The questions below are proposed directions growing directly out of the proved results. They are not presented as an exhaustive catalogue of published open problems, and no priority claim is attached to their formulation. [write] (Source 04.)

82.1 Questions of manuscript 07, with the matching questions of manuscript 04

Question 82.1 (Exact orders for the explicit hard families; sources 07 and 04). [source 07] What are the asymptotic orders of $\text{xc}(P_n)$ and $\text{sx}(P_n)$ for the particular monomial construction (64.5)? [source 04, "Exact complexity inside a fixed jet fiber"] For the explicit prime-monomial polygon P_n° , determine the exact extension complexity, or improve the constants between $\sqrt{n}$ and $24\sqrt{n}$. More generally, which nonnegative ranks occur among cyclic polygons in the all-order jet fiber of a fixed regular polygon?

[source 07] The bounds proved here leave a substantial range below the trivial n -facet and size- n diagonal lifts. The first task is to apply and make explicit the available polygon upper-bound constructions over the chosen real closed field. A lower-bound attack could exploit the structured hierarchy of the exponents rather than only their algebraic independence. No assertion that the elementary upper bounds used here are the best bounds in the literature is intended.

[source 04] The lower bound only uses how many independent parameters are present. It does not exploit their ordered exponents or the exact circle slack identities. A stronger lower-bound certificate might distinguish this family from an arbitrary generic cyclic configuration. Conversely, an explicit small factorization could exploit its highly organized leading regular geometry.

[write] For 04's cyclic family the order is already known to be $\Theta(\sqrt{n})$ ([theorem 70.1](#)); only constants remain. For 07's family, which is not cyclic, the Kwan–Sauermann–Zhao bound does not apply and no upper bound beyond n is proved, so the order of $\text{xc}(P_n)$ is open between $\sqrt{2n}$ and n .

Question 82.2 (Short formulas at finite rational value-group rank). Can comparably strong, uniform separations be obtained from equally short formulas with algebraic real coefficients in a field whose value group has fixed *finite rational rank*?

Our field has valuation rank three but unbounded available rational rank, and the explicit genericity proof uses $2n$ independent exponent directions. An alternative should not merely replace these directions by unspecified algebraically independent real constants. It would need a different lower-bound mechanism, for example a controlled finite combinatorial or elimination certificate. The issue is explicitness and uniform certification, not the bare possibility of scaling a hard finite configuration.

[write] Manuscript 04's [theorem 72.7](#) does give uniform linear separations (order $\sqrt{n}$ against $\log n$) in one field of rational value-group rank two, for instance the real closure of $\mathbb{R}(\omega^{-1}, \omega^{-\omega})$ with $\varepsilon = \omega^{-1}$, but only by the route this question excludes: a single flat element times unspecified real constants algebraically independent over $\mathbb{R}_{\text{alg}}$. The question as posed, with algebraic real coefficients, stays open.

Question 82.3 (Effective precision under restricted input models). For coordinates given by bounded-degree algebraic expressions in a fixed finite list of monomials, can one bound the truncation information needed to decide a specified lift-size inequality?

A useful theorem must state bounds on the expression degrees, coefficient encoding and permitted exponents, and specify whether exact zero tests are available. Quantifier elimination gives finite polynomial tests for a fixed rank question, but a practical theorem would relate their required scale resolution to the input representation. The unrestricted all-orders negative result does not answer this bounded-input problem.

[write] Manuscript 04's [proposition 77.1](#) shows that the exact extension complexity of its explicit family is computable from the finite expression (termination only, with exact zero tests), and its [theorem 74.1](#) shows that no fixed finite truncation order suffices without restrictions, even in rank one. Neither bounds the truncation information; 04's question "Quantitative stability with controlled certificates" below is a companion.

Question 82.4 (Lifting a residue factorization without increasing rank; sources 07 and 04). [source 07] Given S over a convex valuation ring and a factorization of $\bar{S}$, when does that factorization lift to S with the same nonnegative or semidefinite rank? [source 04, "A lifting criterion for positive factorizations"] Find verifiable hypotheses on a factorization of a reduced slack matrix that guarantee lifting to an exact factorization of the same inner dimension, despite a nonzero flat ideal.

[source 07] The descent theorems give only one direction. A first attack is to fix the zero pattern of the factors and study the linearization of the bilinear equations $UV = S$ or $\text{tr}(A_i B_j) = S_{ij}$, together with tangent cones to the positive constraints. The main example warns that agreement in a nonseparated completion cannot itself be a sufficient lifting criterion.

[source 04] [Theorem 71.2](#) rules out lifting based solely on coherent finite-order solutions. A plausible positive theory would need nondegeneracy conditions on the factorization map $(T, U) \mapsto TU$, control of active zero entries, and a genuine Henselian or implicit-function argument in an appropriate local ring. Positivity lies on the boundary for polygon slack matrices, so an ordinary interior implicit-function argument cannot simply be quoted without analysis.

[write] Re-scoped: [theorem 71.2](#) gives, in explicit form, the negative instance that 07's last sentence anticipates. Take for V the coarse valuation ring $\bigcup_{m \geq 0} \varepsilon^{-m} \mathcal{O}$, whose maximal ideal is $\mathfrak{j}_\varepsilon$ (as for V_τ in [proposition 72.10](#)). Then $\overline{S(P_n^\circ)} = \overline{S(R_n)}$, and the reduction of a real factorization $S(R_n) = T_0 U_0$ of inner dimension at most $2m$ is a factorization of the residue matrix that does not lift to $S(P_n^\circ)$ with the same inner dimension (for $m \geq 10$ as stated, and by the same proof for $m \geq 9$, [remark 71.3](#)). So a residue factorization need not lift, even when it is compatible with every finite jet. What remains open is the positive direction: sufficient, checkable hypotheses for lifting.

Question 82.5 (Explicit polynomial obstructions to small lifts). Can the transcendence lower bound for P_n be replaced by a practically checkable polynomial or semialgebraic obstruction for a useful range of n and proposed ranks?

The factorization equations supply a concrete elimination problem. The challenge is to extract a modest certificate rather than a very large quantifier-elimination output. Such a certificate would permit strong lower bounds for chosen algebraic specializations, where full coordinate independence is unavailable.

[write] 04's [corollary 74.6](#) shows that algebraic specializations with the same exact complexity exist near the regular polygon, but by transfer, without such a certificate; see 04's question "Effective real-algebraic approximants" below.

Question 82.6 (Geometry of low-lift loci in a realization space). Inside the realization space of a fixed labeled oriented matroid, what are the valuation strata along which linear or semidefinite lift size can jump?

For each fixed proposed size, the lift-existence locus is semialgebraic by its factorization description. The additional problem is to describe its interaction with valuations and its degenerations at the special reflection locus. One should retain initial information about the *defining relations of the locus*, not only the entries of a slack matrix.

Question 82.7 (Iterated residue chains and complexity profiles). Which finite nonincreasing sequences of lift sizes can occur along a chain of convex residue reductions of one finite polytope?

The current construction gives a large hard stage, a logarithmic-size reference stage, and a fixed simplex shadow. A systematic theory should separate restrictions caused by the common vertex count from restrictions caused by the ordered value group. Constructing prescribed profiles is more informative than producing just one drop.

Question 82.8 (Normal directions and hidden faces). Can one classify the flags of normal cones whose directions disappear under standard part, and relate them to the complexity lost by residue?

[Example 67.5](#) shows that even a vertex of a polytope with a full-dimensional shadow can have no real exposing direction. A refined object should record residue faces together with the next-scale normal cones over them. It must not assume a naive face-poset map, which the same example disproves.

Question 82.9 (Beyond orthants and positive-semidefinite cones). For which ordered-field cones can every factorization of an integral matrix be normalized to integral factors without increasing its size?

The proofs in [Section 73](#) use a bounded normalization cross-section: column sums for the orthant and trace after congruence for the positive-semidefinite cone. Other cones may or may not admit a compatible normalization. A precise criterion would extend the residue-descent theorem to other exact conic lifts without importing unsupported analytic compactness assumptions.

[write] 04's question "Semidefinite and other lift sizes" below asks for the flat-fiber phenomenon for other cones; its normalization clause is this question.

Question 82.10 (A complete certificate path to formal verification; sources 07 and 04). [source 07] What is the smallest reusable real-closed-field library that formally checks the reflection construction, the explicit genericity witness, and the two complexity lower bounds? [source 04, "Formalization milestones"] Can the entire dependency chain be formalized over an abstract real closed ordered field, leaving only the explicit monomial construction to the surreal library?

[source 07] The proof should separate basic ordered-field algebra, finite linear algebra, real-closed-field definability, and surreal monomial semantics. A certificate format for monomial independence is especially promising: the only algebraic-number input here is Eisenstein irreducibility of $X^{2^{n+1}} - 2$, followed by a finite exponent-separation argument.

[source 04] A useful decomposition is: finite convex combinations and exposing functionals; slack matrices and Farkas factorization; the parameter-count lemma; flat ideals and rational congruence; the rational circle identities; and finally the normal-form independence lemma.

The transferred cyclic upper bound is a substantial imported theorem and should remain visibly separate until its proof or a certified transfer route is formalized. Proving the lower-bound and flat-congruence package first would already give a meaningful verified result without pretending that all published polyhedral machinery had been mechanized.

[write] 07's staged design is printed in [section 83.5](#). The repository's `Surreal.Surcomplex.RegularPolygon` (regular polygons at surreal radius) would be a natural source for 04's R_n ; no part of this Part is formalized.

82.2 Separated completion versus flat kernels (source 04)

Question 82.11. In a separated valued field, can a compatible system of positive factorizations in all finite quotient rings fail to lift because the factors do not belong to the field, even though there is no flat kernel? Which completeness or definability assumptions exclude such failure?

Our examples isolate nonseparatedness by using one fixed pair of approximate factors. In a separated but incomplete field, a distinct obstruction could be nonexistence of limits of the compatible factor entries. These mechanisms should be distinguished rather than combined under a general assertion that “formal solutions do not lift.”

82.3 Quantitative stability with controlled certificates (source 04)

Question 82.12. Can finite precision determine extension complexity within a restricted family whose factorization certificates satisfy explicit separation, height, or conditioning bounds?

The counterexamples impose no such uniform bounds. A positive result would have to quantify distance from the relevant semialgebraic rank strata, not merely distance between vertex lists. This connects geometric approximation to exact certificate complexity rather than to coordinate error alone.

[write] Companion to 07's question “Effective precision under restricted input models” above.

82.4 Effective real-algebraic approximants (source 04)

[write] 04 titles this subsection “Effective real-algebraic shadows”; the objects are the approximants of [corollary 74.6](#), not standard parts.

Question 82.13. Replace [corollary 74.6](#) by explicit real algebraic or rational coordinates, with useful degree, height, and running-time bounds, while retaining a provable square-root lower bound near a regular polygon.

The transfer argument certifies existence but hides its complexity. Rational coordinates require additional care: real-algebraic realizability does not automatically imply rational realizability with the same exact extension complexity. A constructive elimination or height argument would be a substantial strengthening.

82.5 Beyond the circle (source 04)

Question 82.14. Identify rational realization charts in higher-dimensional spherical, inscribed, or other constrained polytope families where the chart lower bound of [theorem 76.1](#) can be paired with an equally sharp upper bound.

Preserving a nonsimplicial face lattice may force the perturbation to remain on a proper realization stratum. The dimension relevant to the lower bound is then the dimension of a recoverable chart on that stratum, not the total number of ambient vertex coordinates.

[write] 07's [corollary 75.2](#) gives fixed-dimensional examples (pyramids over its polygons, with simplex shadow), but with a one-sided lower bound only, so it does not answer this question.

82.6 Semidefinite and other lift sizes (source 04)

Question 82.15. What is the corresponding flat-fiber phenomenon for positive semidefinite rank, second-order-cone lifts, or other finite-dimensional conic extension measures?

Finite semialgebraic formulations still support transfer in settings where the cone is first-order definable. However, the parameter budget, the normalization needed before reduction, and the best bounds for regular and cyclic polygons are different. None of the linear extension-complexity estimates in this paper can be relabeled as a semidefinite estimate without a new proof.

[write] Partly answered by manuscript 07: for positive semidefinite rank the phenomenon exists, with unbounded ratio $\text{sxc}(P_n)/\text{sxc}(Q_n) \geq (2n)^{1/4}/U(n)$ inside one all-order jet fiber ([theorem 64.1](#), parameter budget [theorem 67.4](#)), and the normalization needed before reduction is [theorem 73.4](#). Open: the semidefinite sizes of 04's regular and cyclic polygons on the circle, second-order-cone and other cones (see 07's question "Beyond orthants and positive-semidefinite cones" above).

82.7 Which additional exact data are sufficient? (source 04)

Question 82.16. What compact enrichment of leading Plücker and slack data can certify extension-complexity bounds when all ordinary power jets are identical?

Recording a second scale detects the present examples, but a fixed set of scale tolerances cannot uniformly cover the entire surreal class, by [theorem 74.2](#). A representation-sensitive answer may instead record a finite independent monomial basis and exact algebraic relations. Such data are closer to the effective core of [section 77](#) than to numerical truncations.

[write] Compare 07's remark that a method retaining extra initial forms of polynomial relations or higher coarsenings has strictly more information than the initial slack matrix (end of [section 69](#)), and 07's question "Geometry of low-lift loci in a realization space" above.

Formalization roadmaps of Parts I–V

83 Formalization roadmaps (proposals only)

[write] Five of the six manuscripts end with a plan for a Lean development: 05 and 02 (Parts I and II), 06 (Part III), 03 (Part IV) and 07 (Part V); manuscript 04 has no roadmap section, and its formalization proposal is a research question of [Part V](#). The plans are collected here in the order of the Parts, each with a note on where it came from. They are proposals. None of them has been carried out: no declaration of the ProveIt Lean development states a result of this report, and no manuscript ran a Lean build. The collection does contain ingredients that some plans name: the ordered-field coordinate geometry of `Algebra/SurrealNumbers/Surreal/Algebra/Geometry.lean` (`Surreal.Complexify.dot`, `cross`, `cross_def`, `ptolemy`, `heron` and their relatives) and `Surreal.Foundations.SignSequence.omnific_bounded_iff`, the one declaration that any Part uses as an input ([Part IV](#)). Whether the other prerequisites a plan names exist in Mathlib was not checked in the merge. [write] Parts VI–X follow this chapter ([section 1.6](#)) and keep their sources’ formalization proposals inside the Parts; none of them has been carried out either.

83.1 Roadmap of manuscript 05 (finite-scale geometry of surreal liftings)

[write] Printed from [section 21](#) of [Part I](#), where it is the source’s subsection “A theorem-by-theorem Lean plan”. It is a proposal: no statement of Parts I and II is formalized.

The most reusable starting point is not the full class of surreals. Work first with a set-sized Hahn-style normal-form structure over $\mathbb{R}$, equipped with coefficient uniqueness, leading signs, and the fact that every nonempty support subset has a largest exponent in the growth convention. The desired modules are proposed components, not existing declarations:

Coefficient compression. Formalize the rank-increasing selection as finite-dimensional linear algebra on coefficient vectors. Prove the key invariant: before the next selected exponent, every coefficient row belongs to the already selected span. Derive kernel equality, leading-term equality, and the exact count of attained leading exponents.

Affine quotient. Use the kernel of the augmented real matrix as the dependence space. Formalize the affine-basis projection isomorphism and coefficientwise gauge. This part is independent of normal forms and should work over any suitable coefficient field.

Certificates. Prove cofactor vectors and barycentric slack vectors belong to the kernel. Prove that their signs imply the listed supporting faces, rather than trusting an external convex-hull routine. The external Python program may produce rational witnesses, but a Lean checker should verify those witnesses by rational equalities and inequalities.

Optimality and shadows. The degree lower bound is a rank-nullity theorem about polynomial coefficient maps. The linear shadow path needs only an order-preserving real-linear standard-part map and a finite real specialization theorem. Neither proof should require a formal theory of tropical fans.

Toric extension. Formalize conformal decomposition and the finite Graver certificate separately. Its ordered-weight proof does not need real closedness or a surreal-specific polynomial ring.

Only after these modules are proved should they be instantiated with the repository’s constructed surreal field. The normal-form, real-embedding, and standard-part interfaces must be checked against that implementation. No claim is made that the whole plan is already supported by existing Mathlib declarations, and no Lean build was run for this manuscript.

83.2 Roadmap of manuscript 02 (surreal polytopes at finite and transfinite scales)

[write] Manuscript 02’s subsection “A proposed Lean development, not an implemented one”, from its section on algorithms and verification (section 27). It is a proposal as well.

The most useful first formalization target is not all of surreal convexity. It is the finite coefficient compression lemma over a real vector space, followed by an abstract leading-coefficient sign interface. One can then separate the theory into three modules.

Finite linear algebra. Define the affine relation space L_A , the affine-height quotient, barycentric slacks, and zero-label cells. Prove the quotient dimension, annihilator description, rank minimality, and finite real sign bound. These statements make sense independently of surreal arithmetic.

An ordered-series interface. Provide a type of well-ordered coefficient families, prefix restriction, and a theorem that the sign of a nonzero real linear evaluation is its first nonzero coefficient sign. The greedy construction needs a least unspanned position and finite-dimensional rank induction. Normal-form integration should occur only after this interface is proved for the repository’s actual surreal construction.

Geometric and counterexample interfaces. Formalize the lower-basis certificate and its marked refinement relation. For the negative examples, the finite identities $1 - (1 + t) \sum_{k=0}^N (-t)^k = (-1)^{N+1} t^{N+1}$ and (26.2) should come first. The inspected `Surreal/Algebra/Geometry.lean` already provides the relevant cross-product algebra, but not the polytope statements of this paper. An ordinary power-series realization of the reciprocal construction can precede its embedding into surreal normal forms. The final transfinite endpoint needs the explicit order comparison $0 < \omega^{-\omega} < \omega^{-m}$ for ordinary m .

No uncompiled Lean skeleton is included as supposed evidence. A future implementation should map each theorem to a checked declaration and record all imported axioms, rather than infer verification from a shared notation or neighboring file.

83.3 Roadmap of manuscript 06 (affine residues and mixed-volume scales)

[write] Printed from manuscript 06, Section “Exact verification and a formalization route” (section 42), subsection “A modular Lean development”; “the article” below is manuscript 06, printed as Part III. This is a proposal only: at the write phase no result of Part III is formalized, and no Lean file in the ProveIt tree mentions mixed volumes. The research question “Formal certificates in the repository” stays with Part III’s questions in section 43.

A modular Lean development

The existing repository’s constructed real closed field provides a potential target for instantiation, but most of the article should be formalized generically first. The mathematical dependencies suggest four layers.

Ordered-field finite geometry. Represent an input polytope by a finite vertex family. Prove the maximum-simplex Cramer bound, the simplex/cube inclusions, and the parallelotope sandwich over a linearly ordered field. These steps require only finite maxima, matrix determinants, convex combinations, and linear independence. They do not require surreal arithmetic.

Convex valuation algebra. Introduce the finite-element subring, its maximal ideal, and residue map, or use an abstract convex valuation ring with an ordered residue field. Prove the positive-sum value rule and the module-basis theorem. Package the mixed determinant ideal independently of polytopes. The probe reconstruction theorem then reduces to matrix integrality and the adjugate formula.

Finite mixed volume. Either develop determinant triangulation volume and its finite geometric properties, or provide a separately verified abstract mixed-volume interface containing the classical properties explicitly used here. The latter is an intermediate theorem package, not a substitute for constructing mixed volume. The determinant squeeze requires only monotonicity, translation invariance, positive homogeneity, multilinearity, and the segment determinant formula.

Genericity and discrete exchange. Formalize the multivariate finite-grid lemma, coefficientwise residue of finite polynomials, the inner-segment construction, and the determinant exchange identity. The clone proof of M -convexity is then finite combinatorics on labelled sets. This stage does not require the full machinery of Lorentzian polynomials or buildings.

The present package contains no Lean source and reports no new successful Lean build. The repository’s own formalization statements remain subject to its ledger and pinned sources [17]. The outline above identifies concrete formalization tasks rather than treating a prospective interface as an already proved theorem.

83.4 Roadmap of manuscript 03 (real traces and lattice-point counting)

[write] Printed from manuscript 03, Section “Repository integration and the exact proof boundary” (section 56), subsection “A realistic formalization sequence”; “the article” below is manuscript 03, printed as Part IV. This is a proposal only. At the write phase the one existing Lean input is `omnific_bounded_iff` (Algebra/SurrealNumbers/Surreal/Foundations/OmnificUnits.lean, line 65), none of Part IV’s own results is formalized, and no Lean file in the ProveIt tree mentions Ehrhart theory, Pólya–Carlson or curve selection. The research question “A checked geometry pipeline” stays with Part IV’s questions in section 58.

A realistic formalization sequence

The most useful integration order follows the finite proof boundary, not the most sophisticated theorem first.

First: the counting bridge and quadrilateral. Represent the integer candidates in a real bounding box by a finite set. Use `omnific_bounded_iff` to identify all bounded omnific lattice points with integer tuples. Prove the two-level sign lemma

$$a + \varepsilon b \geq 0 \iff (a > 0) \vee (a = 0 \wedge b \geq 0) \quad (a, b \in \mathbb{R}),$$

where ε is positive infinitesimal. This makes the quadrilateral count an elementary finite sum and a ceiling identity. It is the smallest self-contained target with a genuinely nontrivial consequence.

Second: finite flags and cell semantics. Formalize the standard-part functional on a finite-dimensional real subspace of the ordered field, its codimension-one kernel, and induction producing a separated basis. Then prove the row expansion and trace formula. A checker need not discover a separated basis automatically: it can accept a supplied finite expansion and certify its algebraic identities, positivity, and separation conditions.

Third: geometry and asymptotics. Prove the shadow convex-hull lemma, interior lifting by barycentric coordinates, and simple-polytope stability. The surface coefficient also requires ordinary lattice-volume estimates and Ehrhart reciprocity. These should be separate reusable results rather than bundled assumptions inside a surreal-specific theorem.

Fourth: imported global inputs. The one-parameter descent proof uses quantifier elimination and Nash curve selection; the analytic conclusion uses Pólya–Carlson. These are distinct large formalization tasks. A conditional formal theorem may expose them as named assumptions, but that is not an unconditional kernel verification of the whole article. The elementary finite counterexample and its exact count do not depend on either curve selection or Pólya–Carlson.

83.5 Roadmap of manuscript 07 (invisible complexity)

[write] Section 14 of manuscript 07, “Formalization design and the trusted boundary”, printed from Part V (Part V). It is a proposal: no statement of Part V is formalized. The scale written ρ in 07 is this report’s $\tau = \omega^{-\omega}$. Manuscript 04, the other source of Part V, has no roadmap section; its formalization proposal is its research question “Formalization milestones”, merged with 07’s last question in section 82.

The repository’s documented real closed surreal field is a natural target, but this manuscript has not been compiled into Lean and does not extend any existing formalization ledger. A staged implementation should avoid beginning with a proper-class polytope type.

83.5.1 A field-generic core

The first layer should work over an abstract ordered field and formalize finite vertex configurations, affine images, orientation determinants, and the binary reflection relation. The four identities in (65.1)–(65.4) are polynomial identities; after denominators are cleared, ring normalization is enough to check them. Their positivity consequences require only the specified inequalities on u, v, w, τ .

An extension certificate can store a finite matrix of inequalities, affine output maps, and a proof that the image is the required hull. For the reflection step, the two crucial facts are the affine endpoint identity and convexity of the relation (66.2). The procedure should expose these facts rather than hide them inside a numerical optimizer.

83.5.2 The valuation and linear-algebra layers

A convex-subring interface should provide the ideal j_s , its intersection characterization, and closure under bounded multiplication. The power-jet kernel, the coarsening ring and the scalar perturbation lemma can then be proved before any surreal-specific instantiation. The nonnegative normalization theorem is finite sums and inequalities.

The semidefinite layer needs positive-semidefinite matrices over a real closed field, kernel/range decomposition, positive square roots and trace invariance. The normalization proof has a clear endpoint: every factor entry is bounded by an integral scalar. This is a better formal interface than a topological sequential-compactness theorem.

83.5.3 The genuine logical dependency

The semidefinite lower bound depends on the statement that a finite ordered-field definable set has coordinates algebraic over its parameter field. One may prove this from quantifier elimination, or isolate it as a separately verified theorem of real closed fields. Replacing it by an unproved claim that every vertex has a rational normal would be incorrect. [Example 67.5](#) should be included as a regression example.

The linear lower bound is more elementary: it requires dimension bounds, Cramer’s rule or Gaussian elimination, and transcendence-degree counting. The PSD factorization–lift correspondence is an additional imported mathematical theorem unless the formal library proves it. A proof of the rank lower bound is not, by itself, a formal proof of the geometric lift lower bound without this bridge.

83.5.4 Surreal instantiation

Only the final layer needs the actual surreal monomial operation. Its interface consists of multiplication of monomials, domination by larger exponents, and nonvanishing of finite sums with distinct supports. The algebraic-number lemma supplies linear independence of the exponent coefficients. All individual configurations can be instantiated inside a small real closed subfield; the set-sized precision theorem can remain a separate class-level result.

Artifact or dependency	Evidence actually supplied here
Orientation and reflection identities	Symbolic exact checks and written proofs.
Finite reference and perturbed polygons	Exact rational checks; not transcendental witnesses.
All-orders stability and independence	Written mathematical proofs.
Generic linear and PSD lower bounds	Written field-generic proofs, with stated classical imports.
Repository foundations	Documentation consulted; no independent rebuild.
Lean implementation	Proposed architecture only; no Lean source represented as checked.

[write] The one related Lean module, `Surreal.Surcomplex.RegularPolygon` (regular polygons at surreal radius: `vertex`, `vertex_radius`, `side_length`, `area_eq`), is used by neither manuscript.

Part VI

Mixed-volume tomography

Source and scope. [write] This Part prints, whole, batch 59 manuscript 01, *Mixed-Volume Tomography of Surreal Polytopes: Determinant ideals, complete scale spectra, and an adaptive–nonadaptive dichotomy* (archive `surreal_mixed_volume_tomography.zip`, main file `surreal_mixed_volume_tomography/article.tex`; “Research manuscript prepared for Vladimir Reshetnikov”, 30 September 2026; the delivered PDF has 24 pages, US Letter). In this report it is called *source 08*, after the prefix of its shipped files. It inspected ProveIt at the pin 6996fee and could not see [Part III](#): the two manuscripts arrived in consecutive batches and neither cites the other.

The Part continues [Part III](#). Source 08’s *displacement module* $\mathcal{L}(P) = \text{span}_O(P - P)$ is Part III’s *scale lattice* $L(P)$ ([theorem 35.1](#)); it is printed here with Part III’s letter L , and source 08’s determinant ideal $\mathfrak{d}$ with Part III’s $\mathfrak{D}$. Five items re-derive Part III results independently:

- the colorful determinant formula and module factorization ([theorem 87.2](#) and [corollary 87.3](#)) are [theorem 36.3](#) and [proposition 36.4](#) of Part III. Source 08’s statement allows arbitrary finite generating lists and needs no full-dimensionality, with the generator-count comparison constant $\prod_i m_i(m_i + 1)$; Part III’s is the dimension-only form, with at most d^d determinants after the maximum-simplex bases are chosen. Both are printed, with both proofs;
- the adaptive $d + 1$ -value test is [theorem 39.2](#), with the same test and the same adjugate proof; it is printed once, in Part III, and [remark 90.3](#) below gives source 08’s formulation;
- the separation of modules by all first-slot scales is [theorem 39.1](#), printed once in Part III ([remark 94.3](#) below);
- the bounded-distortion description ([corollary 86.3](#)) is [proposition 39.3](#) in difference-body form, and is kept;
- the pentagon of [example 94.1](#) is the pentagon of [section 40](#), itself the lost face of Part II ((14.2)).

The positive-sum lemma and the transfer appendix repeat foundations printed in Parts I and III ([lemma 33.2](#) and [theorem 18.1](#)); they are kept, with pointers. New here: the classification of one-body scale spectra, singular lengths, the two stability theorems, the nonadaptive obstruction ([theorem 91.2](#)), the exact two-body interval ([theorem 92.1](#)), the complementary-minor formula and polarity ([theorem 93.3](#)). The nonadaptive theorem answers, in the plane, the question that Part III leaves open after [theorem 39.1](#) and the reference-free case of its question “Minimal probe systems”; polarity answers the symmetric coarse case of its question “Dual geometric invariants”. Both questions carry re-scoping notes in [section 43](#).

Notation in this Part. [write] Source 08’s letters are kept except the three renamed above ($\mathcal{L} \rightarrow L$, $\mathfrak{d} \rightarrow \mathfrak{D}$, and its macro for the valuation, printed v as in the source).

- v is the natural valuation, with larger value meaning smaller element, $v(r\omega^a) = -a$; it is Part III’s v under another letter.
- $\alpha(P) = (\alpha_1, \dots, \alpha_d)$ is the *scale spectrum* of one polytope, not a multi-index as in Parts III and VII.
- $\mu(P_1, \dots, P_d) = vV(P_1, \dots, P_d)$ and $\mu_k(P, Q) = \mu(P[d - k], Q[k])$ are mixed-volume

values; Part III's profile τ and Part VII's h_P, q_P are the same quantities indexed by multi-indices. Here $\tau = \nu \det B$ is the value of the *reference volume* in [section 90](#); it is neither $\omega^{-\omega}$ nor a profile.

- m, m_i count the listed displacements of one polytope (Part III: number of bodies); $D_P = P - P$ is the difference body and $Z(P; v_0)$ the symmetric zonotope $\sum_j [-w_j, w_j]$ (Part III's Z uses segments $[0, u]$); $T_i(Q)$ are probe *values* (Part III's $T^{(i)}$ are the probe *segments*); h is a shear (Part III: a centroid); δ is a strip thickness and $\delta_k(W)$ a least minor value.
- $\text{res} = \text{st}$ is the residue map $\mathcal{O} \rightarrow \mathbb{R}$, and $E_\gamma(P)$ the residue flag; neither is Part III's affine residue $\text{Res}(P)$.
- ε and t are arbitrary positive infinitesimals, not ω^{-1} (in the checks, t is the variable of $\mathbb{Q}(t)$).

Abstract of source 08. Finite polytopes over the surreal numbers have metric information that their real standard parts do not retain. We identify exactly the part of this information measured by valuations of mixed volumes. If each polytope is given by a finite list of vertex displacements, the valuation of its mixed volume with other polytopes is the minimum valuation of a determinant choosing one displacement from each list. The proof uses an explicit, ordinary-factor comparison with zonotopes and involves no completeness or rank-one hypothesis.

This formula connects ordinary order-convex polytopes to finitely generated modules over the natural valuation ring. It yields a complete classification of one-polytope scale spectra, an algebraic singular-value interpretation, sharp perturbation bounds, and polarity. More substantially, a full-dimensional reference polytope admits a finite, reference-dependent test for equality of displacement modules: d mixed-volume probes and one volume valuation suffice. In contrast, in dimension two no fixed finite list of mixed-area valuation probes works for all polytopes, even when the complete one-body scale spectrum and all residue flags are supplied. We also characterize exactly the possible mixed-area valuations of two planar polytopes with prescribed individual spectra, and reduce the higher-dimensional interaction problem to valuations of complementary minors of an integral invertible matrix.

The results answer explicit structural questions formulated here; they are not presented as solutions of a named, previously published major conjecture. They use standard real-closed-field and convex-geometric inputs, not unreviewed conjectural results from the repository.

Status. This is an AI-assisted, unrefereed research manuscript. The proofs below are mathematical arguments, not Lean proofs. The accompanying exact computations test finite instances, not the universal statements. Historical priority of the synthesis and tomography results is not certified.

Keywords: surreal numbers; non-Archimedean geometry; mixed volume; valuation ring; determinant ideals; tomography; infinitesimal degeneration.

Mathematical setting: finite dimension, finitely many vertices, arbitrary-rank natural value group.

[write] The delivered README's section "Mathematical and verification status" adds: the manuscript has complete written proofs of its 27 numbered results; it is not Lean-verified, and independent correctness review and historical priority are not certified; standard ordered-field polyhedral theory, mixed-volume facts, valuation-ring elimination and the prior singular-value connection are credited rather than presented as new; no named major open conjecture is claimed solved; the 3,196 assertions include low-level checks and are not a count of independent

theorems, nor a proof of the arbitrary-rank statements; the nonadaptive obstruction requires a non-Archimedean field; and no proper-class volume measure or spherical-completeness assumption is used. The README itself is not shipped (see the report’s README).

84 The questions and the contribution

84.1 Why metric scale, rather than a new face lattice?

An infinitesimally thin parallelogram can have the same real shadow as a segment and nevertheless have positive area. Two such parallelograms can have identical thicknesses and identical shadows, yet their mixed area can live at a much larger scale than either area. The missing datum is their relative position at infinitesimal precision.

This paper asks four precise questions. What finite algebraic data determine the *valuation* of a mixed volume? Which lists of mixed-volume scales can occur for a single polytope? How many geometric measurements identify the resulting scale object? Finally, how strongly can two objects with prescribed individual scale data interact? Theorems 87.2, 88.4, 39.2, 91.2, and 92.1 answer these questions in the forms stated below.

A face lattice records incidences, not infinitesimal widths. We do not assert that a finite surreal polytope has a combinatorial type unavailable over the reals. Nor do we identify a polytope from the measurements studied here. Our object is its displacement module, a precise quotient of its metric geometry by ordinary bounded multiplicative distortion.

84.2 Relation to ProveIt and the literature

The inspected repository snapshot is

6996fee43cc97b6c16351def7507d59a95bf62f0.

Its surreal-number project describes a construction of the surreal ordered field and real closedness, and explicitly separates research reports from declarations mapped to formal proofs [57, 58]. Its Euclidean-three-space report concerns coordinate geometry, rotations, and compact groups [59]. We use this as a research setting, not as independent certification of every repository claim. No repository build was run for this manuscript.

[[write] Re-checked at the write phase: the four READMEs cited here (root, surreal project, report guide, Euclidean three-space) are unchanged between the pin and the current tree, and their descriptions above are accurate. This report is one of the research reports that the surreal project separates from its formally mapped declarations; nothing in this Part is formalized.]

Polyhedral geometry over ordered fields is established mathematics. Joswig, Loho, Lorenz, and Schröter give a computational treatment over Puiseux fractions, including the ordered-field foundations behind linear optimization and hull computation [60]. The ordinary mixed-volume facts we use are classical [61]. The transfer of these facts for finitely presented polytopes is explained in section 97 [write] (Appendix A of source 08); it is not itself claimed as a new result.

There is a second, different notion of convexity over a valued field: affine combinations with coefficients in its valuation ring, without the nonnegativity requirement. Chernikov and Mennen study this geometry through valuation-ring modules [62]. Our polytopes use *order convexity*; the module associated with a polytope is a different, generally much larger set. The bridge between their module geometry and mixed-volume scale measurements is the

subject here. All our modules are finitely generated, so no spherical-completeness hypothesis is invoked.

Invariant factors and limiting singular values are connected in the Laurent-series setting by Kaveh and Makhnatch [63]. We give a direct finite Cauchy–Binet argument for the valuation identity needed here. Brändén and Huh place volume polynomials and their tropicalizations in the wider theory of Lorentzian polynomials [64]. The elementary determinant approach below does not replace that theory, and we make no claim to a new general tropical Alexandrov–Fenchel theorem.

84.3 Results at a glance

Let K be a set-sized real-closed ordered field with $\mathbb{R} \subseteq K$, let $\mathcal{O}$ be its natural valuation ring, and write ν for its additive valuation, with smaller valuation meaning larger magnitude. For a polytope P , put

$$L(P) = \text{span}_{\mathcal{O}}(P - P).$$

[[write] Source 08 writes $\mathcal{L}(P)$ and calls it the displacement module; it is Part III’s scale lattice $L(P)$ (theorem 35.1), printed with Part III’s letter.] The central chain of results is

$$\text{finite order-convex polytope} \mapsto L(P) \mapsto \text{determinant ideals} \mapsto \text{mixed-volume valuations}.$$

The first arrow forgets bounded real shape information. The middle object, however, retains infinitesimal shears which can be invisible to every layer of the associated residue flag.

The constructive upper bound and the obstruction have different quantifiers. For *each* reference P there exists a finite test identifying $L(P)$ among candidate modules. There does not exist *one fixed finite test* identifying every planar module. Neither statement asserts optimality of the measurement count, and the negative statement concerns the specified mixed-area valuations, not all imaginable nonlinear observations.

85 Scalars, positivity, and finite polyhedral volume

85.1 A set-sized workspace

Throughout, $d \geq 1$ and all list lengths are ordinary finite integers; $[n] = \{1, \dots, n\}$. A finite list of surreal coordinates belongs to a set-sized real-closed subfield $K \subseteq \mathbf{No}$ containing $\mathbb{R}$: take the field generated by the coordinates and $\mathbb{R}$, then its real closure inside $\mathbf{No}$. Thus no proper-class integration, proper-class vector space, or class-sized compactness argument is needed. The statements are proved for K and apply to every finite surreal instance.

Define

$$\begin{aligned} \mathcal{O} &= \{x \in K : |x| \leq n \text{ for some ordinary } n \in \mathbb{N}\}, \\ \mathfrak{m} &= \{x \in K : |x| < 1/n \text{ for every ordinary } n \geq 1\}. \end{aligned}$$

The ring $\mathcal{O}$ is a convex valuation ring with maximal ideal $\mathfrak{m}$. Indeed, if $x \notin \mathcal{O}$, then $|x| > n$ for every ordinary positive integer, hence $x^{-1} \in \mathfrak{m}$. Its residue field is $\mathbb{R}$: a finite element determines a Dedekind cut in $\mathbb{R}$ with a unique real point differing from it by an infinitesimal. We write $\text{st} = \text{res} : \mathcal{O} \rightarrow \mathbb{R}$.

Let $\Gamma = K^\times / \mathcal{O}^\times$, written additively and ordered so that

$$\nu(x) \geq 0 \iff x \in \mathcal{O}, \quad \nu(x) > 0 \iff x \in \mathfrak{m} \setminus \{0\}.$$

Set $v(0) = +\infty$. In surreal normal-form notation, if the leading term of $x \neq 0$ is $r\omega^a$ with $r \in \mathbb{R}^\times$, then $v(x) = -a$. The value group need not embed in $\mathbb{R}$. Whenever $\gamma \in \Gamma$ is used, a representative of that value exists in K by definition. No global choice of a multiplicative section is required.

Lemma 85.1 (Positive sums and ordinary factors). *For a finite nonempty family $x_i > 0$ in K ,*

$$v\left(\sum_i x_i\right) = \min_i v(x_i).$$

If $0 < x \leq y \leq nx$ for an ordinary positive integer n , then $v(x) = v(y)$. More generally, $v(x) = v(y)$ for positive x, y exactly when x/y and y/x are bounded by ordinary positive integers.

Proof. Writing $M = \max_i x_i$ and N for the number of summands gives $M \leq \sum_i x_i \leq NM$. The quotient by M is a positive unit of $\mathcal{O}$. The remaining assertions are the definitions of $\mathcal{O}$ and its units. $\square$

[[write] The first two assertions are Part III's [lemma 33.2](#) (source 06); the characterization in the last sentence is source 08's addition. The lemma is repeated here because the whole Part cites it.]

85.2 Polytopes and normalization

A polytope is a nonempty convex hull

$$P = \text{conv}_K(v_0, \dots, v_m) = \left\{ \sum_{j=0}^m \lambda_j v_j : \lambda_j \geq 0, \sum_j \lambda_j = 1 \right\} \subset K^d.$$

A listed point need not be an extreme vertex, and repetitions are harmless. A convex body in this manuscript always means such a finite polytope; no assertion is silently extended to arbitrary closed bounded subsets of K^d .

For a d -simplex use the algebraic volume

$$\text{Vol}_d \text{conv}(v_0, \dots, v_d) = \frac{1}{d!} |\det(v_1 - v_0, \dots, v_d - v_0)|.$$

Triangulation gives $\text{Vol}_d(P)$, with value zero in lower dimension. The mixed volume is normalized by

$$\text{Vol}_d \left(\sum_{i=1}^r \lambda_i P_i \right) = \sum_{k_1 + \dots + k_r = d} \frac{d!}{k_1! \dots k_r!} V(P_1[k_1], \dots, P_r[k_r]) \lambda_1^{k_1} \dots \lambda_r^{k_r}. \quad (85.1)$$

Here $P[k]$ means k repeated arguments, and $V(P[d]) = \text{Vol}_d(P)$. The coefficient of $\lambda_1 \dots \lambda_d$ is $d!V(P_1, \dots, P_d)$, not V itself.

We use symmetry, nonnegativity, translation invariance, Minkowski multilinearity, monotonicity in each argument, and the segment identity

$$V([0, u_1], \dots, [0, u_d]) = \frac{1}{d!} |\det(u_1, \dots, u_d)|. \quad (85.2)$$

These are the classical finite-polytope facts [61]. They remain true here by finite semialgebraic transfer, detailed in [section 97](#). In particular, no measure on all surreal subsets has been postulated.

Definition 85.2 (Mixed-volume scale). Write

$$\mu(P_1, \dots, P_d) = \nu V(P_1, \dots, P_d) \in \Gamma \cup \{+\infty\}.$$

The value $+\infty$ means the mixed volume vanishes. A positive infinitesimal mixed volume has a finite, positive valuation, not $+\infty$.

86 The displacement module and bounded-distortion geometry

Definition 86.1. For $P = \text{conv}(v_0, \dots, v_m)$ define its displacement module by

$$L(P) = \sum_{j=1}^m \mathcal{O}(v_j - v_0) \subset K^d. \quad (86.1)$$

Let $D_P = P - P$ be the difference body.

Proposition 86.2 (Intrinsic description). *The module in (86.1) is independent of the displayed generating points and basepoint. More precisely,*

$$L(P) = \text{span}_{\mathcal{O}}(P - P) = \bigcup_{n \geq 1} n(P - P), \quad (86.2)$$

where the union is over ordinary positive integers.

Proof. Every point $x \in P$ satisfies $x - v_0 = \sum_{j=1}^m \lambda_j (v_j - v_0)$ with $0 \leq \lambda_j \leq 1$, so $P - P$ lies in the right-hand module of (86.1). Conversely each listed displacement belongs to $P - P$. This proves the first equality.

The set D_P is symmetric and convex and contains zero. If $x = \sum_{j=1}^s a_j z_j$, where $z_j \in D_P$ and $a_j \in \mathcal{O}$, choose an ordinary integer n larger than $\sum_j |a_j|$. Then

$$\frac{x}{n} = \sum_j \frac{|a_j|}{n} \text{sgn}(a_j) z_j + \left(1 - \sum_j \frac{|a_j|}{n}\right) 0 \in D_P.$$

Thus $x \in nD_P$. The reverse inclusion is immediate because $n \in \mathcal{O}$. □

Corollary 86.3 (Exact geometric meaning). *For finite polytopes P, Q ,*

$$L(P) = L(Q) \iff D_P \subseteq nD_Q \text{ and } D_Q \subseteq mD_P \text{ for some ordinary } n, m \geq 1.$$

Proof. If the modules agree, each of the finitely many points $v_i - v_j$ whose convex hull is D_P lies in an ordinary integer dilate of D_Q by (86.2). Take the maximum of these finitely many integers and use convexity. Reverse the roles for the other inclusion. The converse follows by taking $\mathcal{O}$ -spans. □

[[write] This is Part III's [proposition 39.3](#) (source 06) in difference-body form: there the comparison is between translates $P - a$ and $Q - b$ with positive real constants, here between the difference bodies with ordinary integer factors. The two descriptions are equivalent by [lemma 87.1](#); both proofs are kept.]

Proposition 86.4 (Operations). For a K -linear map T and $c \in K$,

$$\begin{aligned} L(P + Q) &= L(P) + L(Q), & L(cP) &= cL(P), \\ L(TP) &= TL(P), & L(P + a) &= L(P). \end{aligned}$$

If $p_0 \in P$ and $q_0 \in Q$, then

$$L(\text{conv}(P \cup Q)) = L(P) + L(Q) + \mathcal{O}(q_0 - p_0).$$

Proof. The differences of points of $P + Q$ are sums of differences from P and Q . Conversely a difference from P is a difference in $P + Q$ after adding the same point of Q to both terms. The other formulas follow by choosing the indicated basepoints and writing the generators explicitly. $\square$

Example 86.5 (Intersection does not commute with the scale quotient). Let $0 < \varepsilon \in \mathfrak{m}$, $P = [0, 1]$, and $Q = [1 - \varepsilon, 2 - \varepsilon]$. Both displacement modules equal $\mathcal{O}$, whereas

$$L(P \cap Q) = \varepsilon\mathcal{O} \neq \mathcal{O} = L(P) \cap L(Q).$$

A module records displacement size, not the relative translations locating an intersection.

87 The colorful determinant theorem

87.1 An explicit ordinary-factor zonotope comparison

For $P = \text{conv}(v_0, \dots, v_m)$ with $m \geq 1$, put $w_j = v_j - v_0$ and

$$Z(P; v_0) = \sum_{j=1}^m [-w_j, w_j], \quad c_P = \frac{1}{m+1} \sum_{j=0}^m v_j.$$

Lemma 87.1 (Finite sandwich). With $N = m + 1$,

$$P - c_P \subseteq D_P \subseteq Z(P; v_0) \subseteq mD_P \subseteq mN(P - c_P). \quad (87.1)$$

All factors are ordinary integers, independent of the sizes of the coordinates.

Proof. The first inclusion follows because $c_P \in P$. If $x, y \in P$, write them using convex coefficients. Then $x - y = \sum_{j=1}^m (\lambda_j - \eta_j)w_j$, where each coefficient lies in $[-1, 1]$. This proves $D_P \subseteq Z$. Every segment $[-w_j, w_j]$ is contained in D_P by symmetry and convexity. Hence their sum is contained in mD_P .

Finally, for each listed v_j ,

$$-(v_j - c_P) = \sum_{\ell \neq j} (v_\ell - c_P) \in m(P - c_P).$$

By convexity, $-(P - c_P) \subseteq m(P - c_P)$, so $D_P \subseteq (P - c_P) + m(P - c_P) = N(P - c_P)$. $\square$

Theorem 87.2 (Colorful determinant formula). Let $P_i = \text{conv}(v_{i0}, \dots, v_{i,m_i}) \subseteq K^d$ and put $w_{ij} = v_{ij} - v_{i0}$. Then

$$\mu(P_1, \dots, P_d) = \min_{1 \leq j_i \leq m_i} \nu \det(w_{1j_1}, \dots, w_{dj_d}). \quad (87.2)$$

An empty minimum, or a minimum consisting only of zero determinants, is $+\infty$. No full-dimensionality assumption is needed.

Proof. If one list has $m_i = 0$, the associated polytope is a point and its mixed volume in a positive-degree argument is zero by translation invariance and multilinearity. Otherwise let $Z_i = Z(P_i; v_{i0})$. Monotonicity, translation invariance, homogeneity, and Lemma 87.1 give

$$V(P_1, \dots, P_d) \leq V(Z_1, \dots, Z_d) \leq \left(\prod_{i=1}^d m_i(m_i + 1) \right) V(P_1, \dots, P_d). \quad (87.3)$$

The two volumes vanish simultaneously, and otherwise have the same valuation by Lemma 85.1.

Expand the middle volume by Minkowski multilinearity. Since $[-w, w]$ is a translate of $[0, 2w]$, (85.2) gives the exact identity

$$V(Z_1, \dots, Z_d) = \frac{2^d}{d!} \sum_{j_1, \dots, j_d} |\det(w_{1j_1}, \dots, w_{dj_d})|. \quad (87.4)$$

All summands are nonnegative. The factor $2^d/d!$ is an ordinary nonzero rational and has valuation zero. The positive-sum lemma now gives (87.2), including the all-zero case. $\square$

The adjective “colorful” simply means that exactly one displacement is chosen from each polytope. The theorem concerns the *valuation* of V , not an equality between V and the determinant sum for an arbitrary polytope. Equation (87.4) is exact for the comparison zonotopes; the ordinary-factor bounds are essential for passing to P_i .

[write] Theorem 87.2 re-derives Part III’s theorem 36.3 (source 06) independently; the two manuscripts arrived in consecutive batches and neither cites the other. The minimum formulas agree. They differ in their comparison constants and in what they enumerate: here any finite generating lists, with the generator-count factor $\prod_i m_i(m_i + 1)$ of (87.3); in Part III maximum-simplex bases, with the dimension-only squeeze $M/d! \leq V \leq 2^d R_0 M/d!$ and at most d^d determinants after the bases are chosen. Part VII’s parallelotope compression (Part VII) gives a third form of the same comparison.]

87.2 Determinant ideals and the exact quotient being measured

For finitely generated modules $L_i \subseteq K^d$ define

$$\mathfrak{D}(L_1, \dots, L_d) = \sum_{x_i \in L_i} \mathcal{O} \det(x_1, \dots, x_d) \subseteq K. \quad (87.5)$$

This is a finitely generated fractional ideal: choose finite generating lists for L_i and use determinant multilinearity. A finitely generated nonzero ideal in a valuation ring is principal, generated by an element of least valuation among a finite generating list. We write $v(\mathfrak{D})$ for this least value and put $v(0) = +\infty$.

Corollary 87.3 (Module factorization). *For finite polytopes,*

$$\mu(P_1, \dots, P_d) = v(\mathfrak{D}(L(P_1), \dots, L(P_d))).$$

Thus replacing any argument by a polytope with the same displacement module leaves every mixed-volume valuation unchanged.

Proof. Expand each module vector as an $\mathcal{O}$ -linear combination of the listed displacements. Its determinant is an $\mathcal{O}$ -linear combination of the colorful determinants, and each colorful determinant is itself among the generators in (87.5). The two ideals are equal. Apply Theorem 87.2. $\square$

[[write] This is Part III's [proposition 36.4](#), proved there from the selected bases; the proofs are the same expansion.]

Corollary 87.4 (Tropical multilinearity). *With all other arguments fixed,*

$$\begin{aligned}\mu(P + Q, P_2, \dots, P_d) &= \min\{\mu(P, P_2, \dots, P_d), \mu(Q, P_2, \dots, P_d)\}, \\ \mu(cP, P_2, \dots, P_d) &= \nu(c) + \mu(P, P_2, \dots, P_d) \quad (c \neq 0).\end{aligned}$$

Moreover, applying an invertible $T \in \mathrm{GL}_d(K)$ to every argument adds $\nu \det T$.

Proof. The first assertion follows either from positivity and classical mixed-volume additivity or from the sum of determinant ideals. Scaling and simultaneous change of coordinates follow directly from the determinant formula; $\nu|c| = \nu(c)$. $\square$

Remark 87.5 (A finite positivity certificate). The same formula shows that the mixed volume is positive exactly when one can select one displacement from each list to obtain a basis of K^d . Thus every positive instance has a certificate consisting of d selected displacements and one nonzero determinant. This is an instance of the familiar colorful independence criterion, not a new general positivity theorem for mixed volumes.

88 Complete classification of one-polytope scale spectra

88.1 Mixed volumes are determinantal minima

Let $P \subseteq K^d$ be full-dimensional, and let W be the $d \times m$ matrix of listed displacements from a basepoint. For $0 \leq k \leq d$ define

$$\delta_0(W) = 0, \quad \delta_k(W) = \min_{\substack{I \subseteq [d], J \subseteq [m] \\ |I|=|J|=k}} \nu \det W_{I,J}. \quad (88.1)$$

All these values are finite. Let $C = [0, 1]^d$ and put

$$s_k(P) = \mu(P[k], C[d - k]), \quad 0 \leq k \leq d.$$

Proposition 88.1 (Complete mixed-width profile). *For every k ,*

$$s_k(P) = \delta_k(W). \quad (88.2)$$

In particular these numbers do not depend on the chosen generating list or basepoint. The cube can be replaced by any polytope B with $L(B) = O^d$.

Proof. The module of C is generated by the coordinate vectors. In the determinant ideal choose k columns of W and $d - k$ coordinate vectors. Repeated chosen columns or coordinate vectors give zero; a nonzero determinant is, up to sign, a $k \times k$ minor of W . Conversely each such minor is obtained by filling the complementary coordinate directions. [Corollary 87.3](#) proves the assertion and the independence of B . $\square$

88.2 Finite valuation-ring elimination

The following elementary form of Smith reduction is included to make clear that no discreteness or Noetherian property of O is being used.

Lemma 88.2 (Valuation diagonalization). *If $W \in K^{d \times m}$ has row rank d , there are $U \in \mathrm{GL}_d(\mathcal{O})$ and $T \in \mathrm{GL}_m(\mathcal{O})$ such that*

$$UWT = [\mathrm{diag}(a_1, \dots, a_d) \ 0], \quad v(a_1) \leq \dots \leq v(a_d). \quad (88.3)$$

For every k ,

$$\delta_k(W) = \sum_{i=1}^k v(a_i). \quad (88.4)$$

Proof. Choose a nonzero entry of least valuation among all entries, and move it to position $(1, 1)$ by row and column permutations. It divides every entry in the valuation-ring sense: all quotients by the pivot lie in $\mathcal{O}$. Clear its column by integral elementary row operations, then clear its row by integral elementary column operations. The remaining entries are differences of quantities whose valuations are at least that of the pivot; hence they still have at least that valuation. Repeat in the remaining block. Full row rank supplies exactly d nonzero pivots. All operations and their inverses have entries in $\mathcal{O}$.

Left or right multiplication by an integral matrix expresses each k -minor as an integral linear combination of the old k -minors by Cauchy–Binet. Applying this also to the inverse matrices proves equality of the corresponding determinantal ideals. For the diagonal matrix in (88.3), the smallest k -minor valuation is the sum of the k smallest pivot valuations. This proves (88.4) and uniqueness of their ordered list. $\square$

Definition 88.3 (Scale spectrum). The scale spectrum of a full-dimensional polytope is the ordered tuple

$$\alpha(P) = (\alpha_1, \dots, \alpha_d), \quad \alpha_i = s_i(P) - s_{i-1}(P). \quad (88.5)$$

Its entries lie in Γ and satisfy $\alpha_1 \leq \dots \leq \alpha_d$.

Theorem 88.4 (Classification and realization). *The scale spectrum has the following complete interpretation.*

- (i) *Two full-dimensional polytopes P, Q have the same spectrum if and only if some $T \in \mathrm{GL}_d(\mathcal{O})$ satisfies $L(TP) = L(Q)$.*
- (ii) *Equivalently, after such a T , their difference bodies contain each other up to ordinary positive integer factors.*
- (iii) *A tuple $(s_0, \dots, s_d) \in \Gamma^{d+1}$ occurs as the mixed-width profile precisely when $s_0 = 0$ and the differences $s_i - s_{i-1}$ are nondecreasing. Every admissible tuple is realized by a rectangular box.*

Proof. By Lemma 88.2, $L(P) = U_P^{-1} D_P \mathcal{O}^d$ for a diagonal D_P with diagonal valuations $\alpha_i(P)$. If the two spectra agree, corresponding diagonal entries differ by units of $\mathcal{O}$, so their diagonal modules agree. Consequently $T = U_Q^{-1} U_P \in \mathrm{GL}_d(\mathcal{O})$ maps one module to the other. Conversely such a map preserves all determinantal ideals and hence all s_k .

Part (ii) is Corollary 86.3. For (iii), necessity follows from the ordered pivot valuations. Given nondecreasing $\alpha_i = s_i - s_{i-1}$, choose $a_i > 0$ in K with $v(a_i) = \alpha_i$, and take $P = \prod_i [0, a_i]$. Its displacement module is $\mathrm{diag}(a_i) \mathcal{O}^d$, so its profile is the prescribed one. $\square$

For surreal inputs the realizing box can be written $\prod_i [0, \omega^{-\alpha_i}]$ in an enlarged set-sized workspace. This applies even if the finite list of α_i occupies several Archimedean ranks. The theorem classifies modules up to bounded invertible changes of coordinates, not up to exact Euclidean congruence, and not up to equality in fixed coordinates.

Corollary 88.5 (Orthogonal invariance). *Every K -orthogonal transformation preserves the scale spectrum.*

Proof. Each entry of an orthogonal matrix has absolute value at most one, because the sum of the squares in its column is one. The inverse is its transpose and has the same property. Thus the matrix lies in $\mathrm{GL}_d(\mathcal{O})$. $\square$

88.3 Standard parts and the residue flag

Assume $P - v_0 \subseteq \mathcal{O}^d$. Its standard-part polytope means $\mathrm{st}(P - v_0) \subseteq \mathbb{R}^d$. Taking standard parts of the convex coefficients, and conversely using real coefficients, gives

$$\mathrm{st}(P - v_0) = \mathrm{conv}_{\mathbb{R}}(\mathrm{st}(v_0 - v_0), \dots, \mathrm{st}(v_m - v_0)).$$

Proposition 88.6 (Visible dimension). *Under the boundedness assumption, all $\alpha_i(P) \geq 0$ and*

$$\dim_{\mathbb{R}} \mathrm{st}(P - v_0) = \#\{i : \alpha_i(P) = 0\}.$$

Proof. The displacement matrix is integral, so its pivot valuations are nonnegative. Reducing the diagonalization modulo $\mathfrak{m}$ preserves ranks because the reductions of U, T are invertible. Exactly the pivots of valuation zero survive. This rank is the affine dimension of the real hull of the reduced displacement vectors. $\square$

More generally, for $\gamma \in \Gamma$ choose $r > 0$ of value γ and define

$$E_{\gamma}(P) = \mathrm{res}((r^{-1}L(P)) \cap \mathcal{O}^d) \subseteq \mathbb{R}^d. \quad (88.6)$$

Proposition 88.7 (Finite residue flag). *The subspace $E_{\gamma}(P)$ is independent of the chosen r , increases with γ , and satisfies*

$$\dim E_{\gamma}(P) = \#\{i : \alpha_i(P) \leq \gamma\}.$$

There are at most d jumps, even if Γ has arbitrary rank.

Proof. Replacing r by another representative multiplies by a unit, which multiplies the residue subspace by a nonzero real scalar and leaves it unchanged. Write $L(P) = U^{-1} \mathrm{diag}(a_i) \mathcal{O}^d$. Since U and its inverse preserve $\mathcal{O}^d$, it suffices to work diagonally. In coordinate i , the intersection of $r^{-1}a_i\mathcal{O}$ with $\mathcal{O}$ has nonzero residue precisely when $\alpha_i \leq \gamma$; then its residue is all of $\mathbb{R}$. Transform back by $\mathrm{res}(U^{-1})$. $\square$

The residue flag is not a complete invariant of the module in fixed coordinates. Section 91 exhibits modules with the same entire flag, not merely the same flag dimensions, but different mixed-volume interactions.

89 Euclidean meaning and sharp perturbation thresholds

89.1 A finite algebraic singular-value identity

For a full-row-rank displacement matrix W , the symmetric matrix $WW^{\top}$ is positive definite. The real-closed-field spectral theorem gives eigenvalues $\lambda_1 \geq \dots \geq \lambda_d > 0$. Put $\sigma_i = \sqrt{\lambda_i}$, so the singular lengths are ordered from largest to smallest.

Theorem 89.1 (Singular lengths equal scale exponents). *For $1 \leq i \leq d$,*

$$v(\sigma_i) = \alpha_i(P).$$

Thus the valuations of the singular lengths are independent of the displayed list of generators, although the singular lengths themselves need not be.

Proof. Let e_k be the k th elementary symmetric polynomial. Summing Cauchy–Binet over the principal k -minors gives

$$e_k(WW^T) = \sum_{\substack{I \subseteq [d], J \subseteq [m] \\ |I|=|J|=k}} (\det W_{I,J})^2. \quad (89.1)$$

There is no cancellation in this positive sum, so its valuation is $2\delta_k(W)$. On the other hand,

$$e_k(WW^T) = \sum_{|I|=k} \prod_{i \in I} \lambda_i.$$

Since the eigenvalues are positive and decreasing, their valuations are nondecreasing. The valuation of this sum is $\sum_{i=1}^k v(\lambda_i)$. Taking consecutive differences yields $v(\lambda_i) = 2\alpha_i(P)$. Since $\lambda_i = \sigma_i^2$, the result follows. $\square$

The spectral theorem is finite-dimensional and algebraic; it is valid over a real-closed ordered field by transfer of the corresponding fixed-size polynomial matrix statement. This proof does not take a limit $t \rightarrow 0$. The analytic Laurent-series connection in [63] is relevant prior work rather than a novelty claim of this manuscript.

89.2 Partial precision and full-module precision

For a matrix E write $v_{\min}(E) = \min_{i,j} v(E_{ij})$; for the zero matrix take this value to be $+\infty$.

Theorem 89.2 (Partial-profile stability). *Let W have full row rank and spectrum $\alpha_1 \leq \dots \leq \alpha_d$. If E has the same size and*

$$\rho = \min_{i,j} v(E_{ij}) > \alpha_k,$$

then

$$\delta_j(W + E) = \delta_j(W) \quad (1 \leq j \leq k).$$

When $W + E$ remains full rank, its first k scale exponents are unchanged. The minor equalities also apply if its rank drops but remains at least k . No condition on the finer exponents $\alpha_{k+1}, \dots, \alpha_d$ is needed.

Proof. The zero-error case is immediate. Fix $j \leq k$ and a j -minor. Expanding its determinant in the columns of $W + E$, the difference from the old minor is a sum of terms using $\ell \geq 1$ error columns. Laplace expansion along those columns expresses each term as a signed sum of products of an ℓ -minor of E and a $(j - \ell)$ -minor of W . Every product has valuation at least

$$\ell\rho + \delta_{j-\ell}(W) > \alpha_{j-\ell+1} + \dots + \alpha_j + \delta_{j-\ell}(W) = \delta_j(W).$$

The strict inequality uses $\rho > \alpha_k \geq \alpha_j$. Every minor therefore changes by a term of valuation strictly greater than $\delta_j(W)$. No new minor has a smaller value, and a minor attaining the old minimum retains its value. This proves equality. $\square$

Theorem 89.3 (Module stability). *Let L be generated by the columns of a full-row-rank W . If every column of E belongs to $\mathfrak{m}L$, then the columns of $W + E$ generate exactly L . A sufficient coordinatewise condition is*

$$\min_{i,j} v(E_{ij}) > \alpha_d(W).$$

Consequently this condition preserves every mixed-volume valuation in which the perturbed polytope occurs, against arbitrary fixed other arguments.

Proof. Choose a basis matrix B of L over $\mathcal{O}$. Then $A = B^{-1}W$ is integral and its columns generate $\mathcal{O}^d$. Choose an integral right inverse C with $AC = I_d$: for each coordinate vector choose a preimage under the surjective map given by A . The hypothesis gives $H = B^{-1}E \in \mathfrak{m}^{d \times m}$. Therefore

$$(A + H)C = I_d + HC.$$

Its determinant is 1 modulo $\mathfrak{m}$, so the matrix and its inverse are integral. Thus $A + H$ has an integral right inverse and generates $\mathcal{O}^d$, proving the module equality.

For the coordinatewise sufficient condition choose $B = U^{-1} \text{diag}(a_i)$ from Lemma 88.2. Every entry of $B^{-1} = \text{diag}(a_i^{-1})U$ has valuation at least $-\alpha_d$. Hence every entry of $B^{-1}E$ has positive valuation. The mixed-volume assertion is Corollary 87.3. $\square$

Example 89.4 (The strict inequality cannot be relaxed). In an ordered field containing a positive infinitesimal t , take

$$W = \begin{pmatrix} 1 & 0 \\ 0 & t^3 \end{pmatrix}, \quad W + E = \begin{pmatrix} 1 & 0 \\ 0 & t^7 \end{pmatrix}.$$

The least entry valuation of E is $3v(t)$, exactly the finer original exponent. Both matrices remain full rank, but their spectra are $(0, 3v(t))$ and $(0, 7v(t))$. This refutes the version with $\rho \geq \alpha_d$. The corresponding simplices give the same counterexample geometrically.

These thresholds are mathematical statements about exact valuations. They do not imply that arbitrary surreal coordinates have a uniform computable approximation scheme. A representation and a valuation oracle, or an effective restricted field, must be supplied before the finite algorithms become executable.

90 Adaptive finite tomography

90.1 Selecting a module basis from the input

Diagonalization is not required to construct a reference test. A least-valuation full determinant suffices.

Lemma 90.1 (Minimum-determinant basis). *Let W have full row rank, and choose d of its columns whose determinant has the minimum valuation $\delta_d(W)$. If B is the resulting invertible matrix, then*

$$L(P) = BO^d.$$

Proof. For any column w of W , Cramer's rule expresses each coordinate of $B^{-1}w$ as the determinant obtained by replacing one column of B , divided by $\det B$. The numerator is zero or a full minor of W , up to sign. Its valuation is at least $v \det B$, so every coordinate is integral. Thus all generating columns lie in BO^d . The reverse inclusion holds because the columns of B were selected from W . $\square$

[[write] Part III's [theorem 35.1](#) proves the same with a simplex of largest absolute determinant, which in particular has the least determinant valuation; this lemma needs only the least valuation.]

90.2 The reconstruction criterion

Fix a full-dimensional reference polytope P and an $\mathcal{O}$ -basis $b_1, \dots, b_d$ of $L(P)$. Set

$$B = (b_1 \ \cdots \ b_d), \quad \tau = \nu \det B = \nu \operatorname{Vol}_d(P).$$

For any polytope Q define the d reference-dependent probes

$$T_i(Q) = \mu(Q, [0, b_1], \dots, \widehat{[0, b_i]}, \dots, [0, b_d]). \quad (90.1)$$

The hat means omission. Each expression has exactly d arguments. If $d = 1$, the list of segment arguments is empty.

Lemma 90.2 (What the probes measure). *If w_j are a generating displacement list of Q , then*

$$T_i(Q) = \tau + \min_j \nu((B^{-1}w_j)_i). \quad (90.2)$$

In particular,

$$L(Q) \subseteq L(P) \iff T_i(Q) \geq \tau \text{ for every } i. \quad (90.3)$$

Also $T_i(P) = \tau$ for every i .

Proof. The determinant formed from w_j and every b_ℓ except b_i is, up to sign, $\det(B)(B^{-1}w_j)_i$. The colorful determinant formula proves (90.2). All coordinates of every displacement are integral precisely when the module is contained in BO^d . Finally the coordinates of the displacements of P generate O^d , so their projection to every coordinate generates $\mathcal{O}$ and has minimum valuation zero. $\square$

Remark 90.3 (Adaptive tomography in $d + 1$ values; source 08's form of [theorem 39.2](#)). [write] Source 08's Theorem "Adaptive tomography in $d + 1$ values" is Part III's [theorem 39.2](#): the same test, proved by the same adjugate argument. It is printed once, in Part III, which carries source 08's label; source 08 states it for an arbitrary $\mathcal{O}$ -basis $b_1, \dots, b_d$ of $L(P)$, Part III for a maximum-simplex basis. In the notation of this section it reads as follows. For a full-dimensional candidate polytope Q ,

$$L(Q) = L(P) \iff \begin{cases} T_i(Q) \geq \tau & (1 \leq i \leq d), \\ \nu \operatorname{Vol}_d(Q) = \tau. \end{cases} \quad (90.4)$$

Thus d mixed-volume valuations and one volume valuation, with probes chosen from the reference module, identify that module exactly. Equivalently they identify the reference difference body up to ordinary mutual dilation in fixed coordinates.

Proof (source 08; Part III's argument). Equality of modules implies equality of all measurements by Corollary 87.3. Conversely the d inequalities give $L(Q) \subseteq L(P)$. Let C be an $\mathcal{O}$ -basis matrix of $L(Q)$. Then $A = B^{-1}C$ has entries in $\mathcal{O}$. The volume equality gives

$$\nu \det A = \nu \det C - \nu \det B = 0.$$

Hence $\det A$ is a unit. The adjugate formula gives $A^{-1} \in O^{d \times d}$, so $AO^d = O^d$. Therefore $CO^d = BO^d$. The final interpretation is Corollary 86.3. $\square$

Remark 90.4 (Meaning of “adaptive”). The reference P is known before the probes are chosen. No iterative feedback protocol is asserted, and the $d + 1$ measurements are not claimed to reconstruct an unknown module without a reference. The theorem supplies a finite membership test for the equivalence class of a specified reference. Section 91 proves that reference dependence cannot be removed from this kind of finite universal test in the plane.

90.3 Why both parts of the test matter

The volume condition alone is insufficient. The box with edge lengths $\varepsilon, \varepsilon^{-1}$ has the same area valuation as the unit square but a different displacement module.

The d probe conditions alone are also insufficient. In dimension two compare O^2 with

$$L = O(1, 1) + O(0, \varepsilon).$$

Both coordinate projections of L are all of O , so both coordinate probes have exactly the reference valuation. Nevertheless $L \neq O^2$, and its determinant has positive valuation $v\varepsilon$. The volume measurement detects this deficit. These examples justify the two ingredients, but do not prove that every conceivable adaptive scheme needs $d + 1$ values.

91 The nonadaptive obstruction: invisible shears

91.1 A fully explicit family

Let $c \in \mathbb{R}$, $0 < \delta \in \mathfrak{m}$, and $h \in \mathfrak{m}$. Define

$$P_h = [0, (1, c + h)] + [0, (0, \delta)] \subseteq K^2. \quad (91.1)$$

This is a full-dimensional parallelogram. Its area is exactly δ , its scale spectrum is $(0, v\delta)$, and its real shadow is the segment $[0, (1, c)]$. Its residue flag is

$$E_\gamma(P_h) = \begin{cases} 0, & \gamma < 0, \\ \mathbb{R}(1, c), & 0 \leq \gamma < v\delta, \\ \mathbb{R}^2, & \gamma \geq v\delta. \end{cases} \quad (91.2)$$

Indeed its basis matrix is a lower-triangular integral unit matrix times $\text{diag}(1, \delta)$, and the reduction of that unit matrix sends e_1 to $(1, c)$. Thus all these assertions follow from the determinant, spectrum, and residue-flag formulas, independently of the particular infinitesimal h .

To see the module more accurately, compare its basis with that of P_0 :

$$\begin{pmatrix} 1 & 0 \\ c & \delta \end{pmatrix}^{-1} \begin{pmatrix} 1 & 0 \\ c + h & \delta \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ h/\delta & 1 \end{pmatrix}. \quad (91.3)$$

Consequently

$$L(P_h) = L(P_0) \iff vh \geq v\delta, \quad (91.4)$$

with $h = 0$ included by convention. An infinitesimal shear larger than the thickness changes the module while leaving all the individual data in (91.2) unchanged.

Proposition 91.1 (Exact mixed area of two strips). *For the family (91.1),*

$$V(P_0, P_h) = \delta + \frac{|h|}{2}, \quad \mu(P_0, P_h) = \min\{v\delta, vh\}. \quad (91.5)$$

Proof. Use the two segment generators in each argument and expand the mixed area. The four determinants are $h, \delta, -\delta$, and zero, up to sign according to the ordering. In dimension two the segment normalization is $1/2$. Their absolute values therefore sum to $|h| + 2\delta$, giving the displayed exact formula. Apply the positive-sum lemma for the valuation. $\square$

For example, with $\delta = t^8$ and $h = t^3$,

$$V(P_0, P_h) = t^8 + \frac{1}{2}t^3.$$

Both individual spectra are $(0, 8vt)$ but the mixed-area value is $3vt$, rather than the self-interaction value $8vt$.

91.2 No fixed finite family works

Theorem 91.2 (Nonadaptive finite tomography is impossible in the plane). *Assume K is non-Archimedean. For every finite list of fixed polytopes $T_1, \dots, T_N \subseteq K^2$, there exist full-dimensional bounded parallelograms P, Q such that*

- (i) $\text{Vol}_2(P) = \text{Vol}_2(Q) > 0$ and $\alpha(P) = \alpha(Q)$;
- (ii) $\text{st}(P) = \text{st}(Q)$ and $E_\gamma(P) = E_\gamma(Q)$ for every $\gamma \in \Gamma$;
- (iii) $\mu(P, T_j) = \mu(Q, T_j)$ for $1 \leq j \leq N$;
- (iv) $L(P) \neq L(Q)$.

Thus a fixed finite family of mixed-area valuations cannot universally determine the displacement module, even with exact area, the complete individual spectrum, the real shadow, and the entire residue flag added to the data.

Proof. Choose finite displacement lists for the T_j and collect all their nonzero vectors in a finite set S . For $z = (z_1, z_2) \in S$, put

$$f_z(x_1, x_2) = \det((x_1, x_2), z) = z_2x_1 - z_1x_2, \quad \kappa_z = \min\{vz_1, vz_2\}.$$

Choose $s_z \neq 0$ of value κ_z . The two coefficients of $s_z^{-1}f_z$ lie in $\mathcal{O}$ and at least one is a unit. Its residue is a nonzero real linear form. There is at most one $c \in \mathbb{R}$ for which

$$\text{res}(s_z^{-1}f_z)(1, c) = 0;$$

if the coefficient of c vanishes, there are none. Avoiding finitely many forbidden values, choose a real c for which all these residues are nonzero.

Take any positive infinitesimal δ and choose $h > 0$ with $0 < vh < v\delta$, for example $h = \sqrt{\delta}$. Let $P = P_0$ and $Q = P_h$ as in (91.1). For every $z \in S$,

$$\begin{aligned} v f_z(1, c) &= \kappa_z, \\ v f_z(1, c + h) &= \kappa_z, \\ v f_z(0, \delta) &\geq \kappa_z + v\delta > \kappa_z. \end{aligned}$$

For the second equality, the perturbation divided by s_z is infinitesimal while the original value has nonzero residue. The third inequality includes the zero case.

The displacement module of each strip is generated by its two displayed segment directions. Corollary 87.3 therefore gives, for every nonpoint test T_j ,

$$\mu(P, T_j) = \min_{z \text{ in its nonzero list}} \kappa_z = \mu(Q, T_j).$$

A point test gives $+\infty$ for both. The area, spectrum, shadow, and residue-flag equalities were established in (91.1)–(91.2). Finally $h/\delta \notin \mathcal{O}$, so (91.3) shows that the modules differ. $\square$

The proof uses finite avoidance over the real residue field, not compactness or a limit. A fixed probe is blind because its leading determinant does not cancel on the chosen long direction. An adaptive probe can be chosen *to force this cancellation* and thereby expose the smaller-scale shear.

Example 91.3 (How the adaptive test sees the invisible shear). For reference P_0 , choose $b_1 = (1, c)$ and $b_2 = (0, \delta)$. The two probes of P_h satisfy

$$T_1(P_h) = v\delta, \quad T_2(P_h) = \min\{vh, v\delta\}.$$

The first follows from $V(P_h, [0, b_2]) = \delta/2$. The second follows from $V(P_h, [0, b_1]) = (|h| + \delta)/2$. If the shear is larger than the thickness, $T_2(P_h) < \tau$ and the module-containment test fails immediately.

Remark 91.4 (What the obstruction does not say). The theorem is not an impossibility result for exact mixed-area values, infinite probe families, arbitrary nonlinear measurements, or recovery of a finite-dimensional family with a priori bounded precision. It does not contradict the finite adaptive theorem: the reference-dependent exceptional direction is selected after the reference is known. Its non-Archimedean hypothesis is necessary for the construction.

92 A complete rank-two interaction theorem

For two planar polytopes the one-body spectra do not determine their mixed-area valuation. Nevertheless they determine its *exact attainable interval*.

Theorem 92.1 (All mixed-area scales for prescribed spectra). *Fix $a_1 \leq a_2$ and $b_1 \leq b_2$ in Γ . As P, Q range over full-dimensional planar polytopes with spectra (a_1, a_2) and (b_1, b_2) , the attainable values of $\mu(P, Q)$ are exactly*

$$\boxed{[a_1 + b_1, \min\{a_1 + b_2, a_2 + b_1\}] \cap \Gamma.} \quad (92.1)$$

Every value in the interval is realized by a pair of parallelograms.

Proof. A simultaneous change of coordinates in $\text{GL}_2(\mathcal{O})$ does not change either spectrum or the mixed-area valuation. By Lemma 88.2 we may therefore write

$$L(P) = D_a \mathcal{O}^2, \quad L(Q) = U D_b \mathcal{O}^2,$$

where $D_a = \text{diag}(A_1, A_2)$, $D_b = \text{diag}(B_1, B_2)$, the entries are positive representatives of the indicated values, and $U = (u_{ij}) \in \text{GL}_2(\mathcal{O})$. The colorful formula gives

$$\begin{aligned} \mu(P, Q) = \min\{ & a_1 + b_1 + vu_{21}, \quad a_1 + b_2 + vu_{22}, \\ & a_2 + b_1 + vu_{11}, \quad a_2 + b_2 + vu_{12} \}. \end{aligned} \quad (92.2)$$

Every matrix entry is integral, so all four terms are at least $a_1 + b_1$.

If u_{21} is a unit, the first term is exactly $a_1 + b_1$. Otherwise $u_{21} \in \mathfrak{m}$. Since the determinant of U is a unit, reduction modulo $\mathfrak{m}$ forces u_{11} and u_{22} to be units. The middle two terms then give $\mu(P, Q) \leq \min\{a_1 + b_2, a_2 + b_1\}$. This proves necessity.

Conversely let $\mu = a_1 + b_1 + \beta$ lie in the indicated interval. Then

$$0 \leq \beta \leq \min\{a_2 - a_1, b_2 - b_1\}.$$

Choose $h > 0$ of value β and take

$$U = \begin{pmatrix} 1 & 0 \\ h & 1 \end{pmatrix}.$$

Use the parallelograms generated respectively by the columns of D_a and UD_b . Their spectra are the prescribed ones because $U \in \text{GL}_2(\mathcal{O})$. Formula (92.2) becomes

$$\min\{a_1 + b_1 + \beta, a_1 + b_2, a_2 + b_1\} = \mu.$$

Thus every value is attained, including both endpoints and intervals in arbitrary-rank value groups. $\square$

Example 92.2 (An interval rather than a single answer). Suppose the spectra are $(0, 8\gamma)$ and $(2\gamma, 7\gamma)$ for $\gamma > 0$. Their mixed-area value can be any

$$\mu \in [2\gamma, 7\gamma] \cap \Gamma.$$

The endpoint 2γ corresponds to a unit-sized change in long direction. The upper endpoint occurs when the directions are aligned closely enough that one thickness scale becomes dominant. Intermediate values are controlled by a shear.

This is stronger than the tropicalized planar Minkowski inequality alone, which only gives $2\mu(P, Q) \leq (a_1 + a_2) + (b_1 + b_2)$. The exact upper endpoint in (92.1) can be strictly smaller. The proof also supplies realization, which a necessary inequality does not.

[[write] For two bodies the union of these intervals over all spectra with fixed sums $a_1 + a_2 = A$ and $b_1 + b_2 = B$ is $\{\mu \leq (A + B)/2\} \cap \Gamma$ (the value group of a real closed field is divisible; take $a_1 - b_1 = (A - B)/2$ and a_1, b_1 small). That is the $m = 2$ case of Part VII's quadratic sufficiency theorem (theorem 110.2): the profile (A, μ, B) of two planar bodies is realizable exactly when $2\mu \leq A + B$. Theorem 92.1 refines it by fixing the individual spectra.]

93 Higher-dimensional interactions and polarity

93.1 Complementary minors give the complete finite reduction

For full-dimensional $P, Q \subseteq K^d$, put

$$\mu_k(P, Q) = \mu(P[d - k], Q[k]), \quad 0 \leq k \leq d.$$

Use a simultaneous integral invertible change of coordinates to arrange

$$L(P) = D_a \mathcal{O}^d, \quad L(Q) = UD_b \mathcal{O}^d, \quad U \in \text{GL}_d(\mathcal{O}),$$

where $v(D_{a,ii}) = a_i$ and $v(D_{b,jj}) = b_j$ are the ordered individual spectra.

Theorem 93.1 (Complementary-minor interaction formula). *In this notation,*

$$\mu_k(P, Q) = \min_{\substack{I, J \subseteq [d] \\ |I| + |J| = k}} \left(\sum_{i \notin I} a_i + \sum_{j \in J} b_j + v \det U_{I, J} \right). \quad (93.1)$$

For $k = 0$ the empty determinant is one. In particular $\mu_0 = \sum_i a_i$ and $\mu_d = \sum_j b_j$.

Proof. Choose the module bases as generating lists in the determinant ideal. A nonzero colorful determinant selects $d - k$ different columns of D_a and k different columns of UD_b . Let I be the coordinate indices not used by the first selection and J those used in the second. Expanding along the coordinate columns gives, up to sign,

$$\left(\prod_{i \notin I} D_{a,ii} \right) \left(\prod_{j \in J} D_{b,jj} \right) \det U_{I,J}.$$

Every pair (I, J) occurs. Taking minimum valuations proves the formula. At $k = d$, $\nu \det U = 0$. $\square$

This is an exact reduction to finite linear algebra, not a complete higher-dimensional realization theorem. Minor valuations of a single matrix are constrained jointly. Treating them as independent nonnegative parameters would give an unjustified enlargement of the attainable set. Likewise the valuation of a minor is not generally the minimum of its permutation-product valuations: cancellation inside that determinant can increase its value. Only the later sum of *absolute* determinants is cancellation-free.

Corollary 93.2 (Classical tropical Alexandrov–Fenchel restriction). *For $1 \leq k \leq d - 1$,*

$$2\mu_k(P, Q) \leq \mu_{k-1}(P, Q) + \mu_{k+1}(P, Q).$$

Proof. The classical Alexandrov–Fenchel inequality for polytopes, transferred as in [section 97](#), gives

$$V(P[d - k], Q[k])^2 \geq V(P[d - k + 1], Q[k - 1])V(P[d - k - 1], Q[k + 1]).$$

All terms are positive for full-dimensional inputs. Applying the order-compatible valuation reverses the weak inequality on magnitudes, yielding the result. $\square$

This corollary is attributed to the classical inequality [61]; the broader Lorentzian framework is developed in [64]. We do not assert that this single family of inequalities characterizes all arrays in (93.1).

93.2 Polarity is exact module duality

Let $P = -P$ be a full-dimensional centrally symmetric polytope. Its polar is

$$P^\circ = \{y \in K^d : \langle y, x \rangle \leq 1 \text{ for every } x \in P\}.$$

It is again a full-dimensional polytope. For a full-rank finitely generated module L , define

$$L^\vee = \{y \in K^d : \langle y, L \rangle \subseteq \mathcal{O}\}.$$

Theorem 93.3 (Polar duality of scale modules). *For such P ,*

$$L(P^\circ) = L(P)^\vee. \tag{93.2}$$

Consequently

$$\alpha_i(P^\circ) = -\alpha_{d+1-i}(P), \quad \nu \operatorname{Vol}_d(P) + \nu \operatorname{Vol}_d(P^\circ) = 0. \tag{93.3}$$

Proof. Since $0 \in P$ and $P = -P$, $L(P) = \text{span}_O P$, and the same holds for P° . If $y \in P^\circ$, symmetry gives $|\langle y, x \rangle| \leq 1$ for every $x \in P$. The pairing with every O -linear combination is therefore in O . Taking O -spans proves one inclusion in (93.2).

Conversely take $y \in L(P)^\vee$. Its pairings with the finitely many listed vertices of P are finite, so a single ordinary positive integer n bounds all their absolute values. By convexity this bounds all pairings with P , and $y/n \in P^\circ$. Since $n \in O$, $y \in L(P^\circ)$.

If $L(P) = BO^d$, the definition of the dual gives $L(P)^\vee = B^{-\top}O^d$. Write $B = U^{-1} \text{diag}(a_i)$ with $U \in \text{GL}_d(O)$. Then $B^{-\top} = U^\top \text{diag}(a_i^{-1})$. Its ordered exponent list is the reversed negative of that for B . Summing the exponents and using $\nu \text{Vol}_d(P) = \delta_d(P)$ proves the volume identity. $\square$

The last identity says that the volume product has valuation zero. It does not determine its real leading coefficient and does not solve Mahler's volume-product conjecture or sharpen a classical real volume-product bound. This is a boundary between scale geometry and coefficient-level convex geometry.

[write] [Theorem 93.3](#) answers the symmetric coarse case of Part III's question "Dual geometric invariants" ([section 43](#)); nonsymmetric bodies and coefficient-level duality remain open there. Part II's [proposition 14.1](#) is a different statement, about the standard part of a finite polar.]

94 Finite algorithms, certificates, and examples

94.1 What an implementation needs

The field-generic algorithms assume exact arithmetic in the input field, equality testing, and comparisons of valuations of nonzero elements. The proofs do not provide these operations uniformly for arbitrary surreal normal forms. In restricted fields such as rational functions ordered at a positive infinitesimal, the required operations are effective. The supplied program works in $\mathbb{Q}(t)$, not in the whole surreal field and not in an implementation of arbitrary-rank Hahn series.

Three different computational tasks should be separated.

A single mixed-volume scale. Given finite displacement lists $W_1, \dots, W_d$, enumerate one column from each list, compute the corresponding determinant exactly, and take the minimum valuation. There are at most $\prod_i m_i$ such determinants. With a cubic determinant algorithm, this is at most $O(d^3 \prod_i m_i)$ field operations, aside from valuation costs. This simple bound is not claimed to be optimal. It avoids convex-hull enumeration entirely.

A complete one-body spectrum. Apply the minimum-valuation pivot procedure of [Lemma 88.2](#). At each step find the least-valuation remaining entry, swap it into the next pivot position, and clear the pivot row and column with integral quotients. Stop if the remaining block is zero. For a dense $d \times m$ matrix with $m \geq d$, the elimination and pivot searches use $O(d^2 m)$ field operations and valuation comparisons, with ordinary matrix bookkeeping. Track the left row-operation matrix U , but not the full right $m \times m$ matrix. The output pivots give the spectrum and $U^{-1} \text{diag}(a_i)$ gives a basis of the displacement module. This is an arithmetic-operation count, not a bit-complexity result.

A reference comparison. Once the basis B of the reference is fixed, compute $B^{-1}W_Q$. The minimum valuation in each row is $T_i(Q) - \tau$. These d minima test containment. Compute

one determinant-ideal value $\delta_d(W_Q)$, by the same elimination or by a minimum-determinant basis, to test equality. Theorems 39.2 and 89.3 then give exact logical certificates.

94.2 A certificate boundary suitable for formalization

A certificate can consist of a nonsingular $d \times d$ basis matrix B , integral matrices expressing the displayed generators in that basis and expressing B in the displayed generators, and exact identities verifying the expressions. Checking these identities proves equality of generated modules without trusting the search that found them.

For containment $L(Q) \subseteq L(P)$, a matrix A with entries in $\mathcal{O}$ satisfying $W_Q = B_P A$ suffices. For equality between full-rank basis modules, one can give A and A^{-1} over $\mathcal{O}$. Alternatively give integral A and a unit certificate for its determinant. These are finite matrix identities and valuation inequalities. To turn them into proofs about actual mixed-volume valuations, the checker must also be connected to Theorem 87.2; exact matrix tests by themselves do not formalize the geometric bridge.

94.3 Worked scale data

Let $\gamma = v(t) > 0$. Table 10 records simple examples. Each row uses a generating matrix for the module; taking the sum of the associated segments realizes the displayed module by a parallelepiped.

Generating matrix	Spectrum	Interpretation
$\text{diag}(1, t^3, t^7)$	$(0, 3\gamma, 7\gamma)$	One visible direction; two hidden widths.
$\text{diag}(t^{-2}, 1, t^5)$	$(-2\gamma, 0, 5\gamma)$	One infinite direction and one thin direction.
$\begin{pmatrix} 1 & 0 \\ t^3 & t^8 \end{pmatrix}$	$(0, 8\gamma)$	Same spectrum as the unsheared strip, different fixed-coordinate module.
$\text{diag}(t^{-7}, t^{-3}, 1)$	$(-7\gamma, -3\gamma, 0)$	Spectrum dual to the first row under symmetric polarity.

Table 10: Scale spectra retain magnitudes, but not all relative infinitesimal positions. The polar row asserts equality of spectra; the polar of a box need not itself be a box.

For the first row, the entire mixed-width profile is

$$(s_0, s_1, s_2, s_3) = (0, 0, 3\gamma, 10\gamma).$$

Thus reading only the volume scale 10γ would lose the distinction between widths $3\gamma, 7\gamma$ and widths $4\gamma, 6\gamma$. Reading the full profile recovers the ordered width exponents exactly. Even that profile does not recover the hidden shear in the third row.

Example 94.1 (The face lattice can change while the scale module does not). Start with the unit square in K^2 and include $(1/2, 0)$ as a redundant generating point. Move only this point to $(1/2, -t)$ for a positive infinitesimal t . The convex hull becomes a pentagon, but its displacement module remains $\mathcal{O}^2$. This follows directly from its generators, or from Theorem 89.3 because the original finer exponent is zero and the displacement error has positive valuation. Consequently all mixed-volume valuations against any fixed other polytopes are unchanged, despite the change from four to five vertices.

[[write] The same phenomenon as Part III’s pentagon P_ε (section 40, a vertex infinitesimally outside the left edge of the unit square, with scale lattice $\mathcal{O}^2$) and Part II’s lost face (14.2); here the moved point lies below the bottom edge.]

Example 94.2 (Arbitrary-rank data beyond a single power-series scale). Within the surreals, take

$$\delta = \omega^{-\omega^2}, \quad h = \omega^{-\omega}.$$

For the strips in (91.1), both exponents exceed every ordinary integer, and no finite truncation in ordinary powers of ω^{-1} sees either term. Nevertheless their individual finer exponent is ω^2 while their mixed-area value is ω . The proof is still the finite identity $V(P_0, P_h) = \delta + |h|/2$; no transseries summation is needed. This example illustrates the rank-independent scope, not a new general theorem about beyond-all-orders asymptotics.

94.4 The full observation quotient and scalar extension

Remark 94.3 (All first-slot mixed-volume scales separate modules; source 08’s form of theorem 39.1). [write] Source 08’s Corollary “All first-slot mixed-volume scales separate modules” lies between conditions (ii) and (iii) of Part III’s theorem 39.1, and is printed once there, with source 08’s label. Its statement reads: for full-dimensional $P, Q \subseteq K^d$, the following are equivalent:

$$L(P) = L(Q) \iff \mu(P, R_2, \dots, R_d) = \mu(Q, R_2, \dots, R_d) \text{ for all finite polytopes } R_2, \dots, R_d.$$

For $d = 1$, the empty probe tuple has the expected meaning.

Proof (source 08; Part III’s argument). The forward direction is Corollary 87.3. For the reverse direction choose the segment probes from a basis of $L(P)$. Their equality gives $L(Q) \subseteq L(P)$ by Lemma 90.2. Repeat with a basis of $L(Q)$ to obtain the reverse inclusion. In dimension one the single volume valuation already gives equality of the principal displacement modules. $\square$

Thus the module is exactly the quotient of full-dimensional finite polytopes seen by *all* these first-slot observations. It is neither too coarse nor unnecessarily fine for this measurement problem. The finite adaptive theorem replaces the unbounded probe family by a reference-dependent containment test and one determinant-volume check.

If the scalar workspace is enlarged from K to a real-closed ordered extension K' containing it, its natural valuation restricts to the original one, with the original value group embedded in the new group. Every determinant minimum and every basis-coordinate integrality condition in this paper is unchanged. Therefore the spectra, mixed-volume scales, and containment or equality of the finitely generated modules are compatible with scalar extension. The literal sets $L(P)$ grow by extending scalars; it is their presentations and comparison statements that are preserved. This is the precise sense in which the set-sized results give statements about finite surreal inputs independently of the chosen workspace.

95 Further research questions and a concrete agenda

The questions below are proposed research directions, not claims that a literature-wide search has established each as a new published open problem. They distinguish issues settled here from those still requiring work.

Question 95.1 (The first unsolved interaction dimension). For $d = 3$, characterize the pairs (μ_1, μ_2) attainable for prescribed spectra a, b . More generally, characterize all arrays in (93.1) as U ranges over $\mathrm{GL}_d(\mathcal{O})$.

The rank-two theorem is complete because only four entries and a unit determinant are involved. Already in dimension three the two-minor valuations interact with the entries through algebraic identities. A concrete starting point is to fix rational exponents in a Puiseux field, enumerate low-complexity unit matrices, and derive candidate inequalities, then prove realization rather than stopping at necessary bounds. The target is not merely convexity of (μ_k) , which Corollary 93.2 already supplies.

Question 95.2 (Optimal reference-dependent measurement counts). What is the least number of measurements needed to recognize $L(P)$ among all full-rank candidates when the reference P is known? How does the answer change if every probe must be first-slot mixed-linear in the candidate, if candidate volume is allowed, or if arbitrary mixed self-interactions are allowed?

Theorem 39.2 gives an upper bound of $d + 1$ values in a specified model. It is not a lower bound. The distinction between a first-slot valuation and the candidate's volume must be retained: the latter repeats the candidate d times and carries a determinant-rank deficit which finitely many coordinate projections can miss.

[[write] The same question as Part III's "Minimal probe systems" (section 43), restricted to a known reference; its reference-free case is answered negatively in the plane by theorem 91.2. The optimality of $d + 1$ remains open.]

Question 95.3 (Recovering the hidden shear data). Which finite or finitely presented enhancement of the residue flag, together with the scale spectrum, determines a given finitely generated module in fixed coordinates? Can the missing data be organized by successive residue-valued extension parameters in an affine-building or lattice-moduli description?

The strips show that ordinary associated-graded subspaces forget the ratio h/δ relevant to module equality. A satisfactory enhancement must record relative lifts across scales, not just the dimensions or locations of leading residue subspaces. Any finite representation claim must specify the scalar representation, since a basis entry can itself have arbitrarily complicated surreal support.

[[write] Related: Part III's question question 43.3 asks for a canonical joint reduction of several scale lattices; Part VII's proposition 104.5 gives a common real reduction of a scaled family, canonical up to one element of $\mathrm{GL}_d(\mathbb{R})$. Neither recovers the shear data asked for here, which stays open.]

Question 95.4 (Leading coefficients beyond valuations). After the determinant minima are known, which additional data determine the leading coefficient of a mixed volume? When can a common subdivision or a bounded-distortion normalization reduce this question to real mixed volumes of residue polytopes?

The zonotope sandwich proves equality of valuations but generally not equality of leading coefficients. Recovering coefficients requires keeping more than the displacement module: even real polytopes of different shapes can have the same module $\mathcal{O}^d$ and different volumes. This is the natural next boundary, rather than an unsupported claim that the present quotient solves coefficient-level convex geometry.

[[write] *Re-scoped: partly answered in Part VII.* Each weighted initial form of a volume polynomial is the real volume polynomial of the residue configuration in a minimum-valuation basis (theorem 104.2, unique up to one element of $\mathrm{GL}_d(\mathbb{R})$ by proposition 104.5), which is the

reduction to real mixed volumes asked for in the second sentence; and a realizable profile need not have liftable leading coefficients ([theorem 109.2](#)). A description of the leading coefficients by additional data remains open. Part III's question "Leading coefficients above a fixed scale profile" is the same question.]

Question 95.5 (Efficient colorful determinant optimization). Can the minimum in ([87.2](#)) be found without enumerating all colorful choices, with a provable arithmetic or bit-complexity bound for useful represented fields? Which valuated-matroid or linear-matroid intersection algorithms can be connected to an exact certificate checker?

For one body the pivot algorithm is already efficient in the arithmetic model. For several independent color classes, a faster method must respect exact cancellations inside determinants. A proof-producing implementation should output both the selected basis and a certificate that no lower valuation is possible; a good heuristic selection alone proves only an upper bound on the minimum.

Question 95.6 (Formalizing the bridge rather than only its examples). Can the finite sandwich theorem and the determinant-ideal formula be formalized over a generic real-closed ordered field, and only then instantiated in the repository's surreal field?

A useful first milestone is the module and matrix part: integral generation, Cramer's-rule bases, the adaptive criterion, and the hidden-shear witness. The next milestone is a finite-polytope mixed-volume interface with the needed axioms proved, not admitted. This order isolates a small logical boundary. A Lean file containing tested determinants would not by itself formalize [Theorem 87.2](#).

[[write] Part III asks the same in "Formal certificates in the repository" and prints its own plan in [section 83.3](#). Nothing of this Part is formalized.]

Question 95.7 (Beyond finite generation). Which parts of the theory survive for suitable infinite-generated order-convex sets or for limits of polytopes? What replacements for minima, principal determinant ideals, and finite adaptive tests are required?

Finite generation is used essentially in the ordinary-factor sandwich, in the existence of determinant minima, and in basis selection. The distinction between finite-generated modules and more general valuation-ring modules, including the role of spherical completeness in the latter setting, is already important in [\[62\]](#). Extending the present conclusions by simply deleting the word "finite" would be unjustified.

[[write] Related: [Part X](#) studies set-sized presentations of surreal convex classes (finite combinations of a set of generators, or a set of halfspaces) and shows that a class with both kinds of presentation has finite subpresentations of both ([theorem 157.2](#)). It does not treat displacement modules or mixed volumes, so this question stays open.]

Question 95.8 (Combinatorics coupled to scale geometry). Can one classify which face-lattice degenerations occur inside a fixed displacement-module class, and which additional determinant sign data control them? Is there a useful stratification combining a fixed module with an oriented-matroid realization space?

[Example 94.1](#) shows that the module class alone is too coarse. Conversely the strips show that a fixed combinatorial type can have different scale modules. A coupled theory should keep these independent layers separate rather than treating either as a substitute for the other.

[[write] Related: Part III's question [question 43.6](#) (face specialization and oriented scale data), with its re-scoping note.]

Question 95.9 (Coarsened valuations and equality cases). How does the full family of mixed-volume scales behave under coarsening Γ by convex subgroups? Which equality cases of tropical Alexandrov–Fenchel lift to coefficient-level equalities, and which reflect merely that several different real factors have valuation zero?

The determinant formula is compatible with an order-preserving quotient of the value group, so a finite instance has a natural hierarchy of coarsened observations. The nontrivial issue is the reverse passage: recovering lost layers and interpreting equality. Equality of valuations is much weaker than equality of positive quantities.

A focused continuation would begin with Question 95.1 and a checker for the module certificates of Section 94. These have finite, explicit inputs and sharply stated success criteria. The general infinite-generated and coefficient-reconstruction problems require additional hypotheses before an effective research program can be specified.

96 Verification, provenance, and boundaries

96.1 What was checked computationally

The accompanying code/verify.py [write] (shipped as code/08-mixed-volume-tomography-verify.py) uses exact rational functions in $\mathbb{Q}(t)$, ordered at positive infinitesimal t . It completed **3,196 assertions** with fixed seed 2026093017; the machine-readable report is data/verification.json [write] (shipped as data/08-mixed-volume-tomography-verification.json; the build metadata data/build_report.json as data/08-mixed-volume-tomography-build_report.json).

The most independent check computes mixed areas from exact planar convex hulls and the polarization identity

$$V(P, Q) = \frac{1}{2}(\text{Vol}_2(P + Q) - \text{Vol}_2(P) - \text{Vol}_2(Q)),$$

and compares their valuations with the colorful determinant minimum. One hundred random finite pairs passed. Hull orientation and rational-function signs use leading coefficients at $t = 0^+$, not floating-point approximations.

Ninety matrices in dimensions two through four compare all-minor profiles with minimum-valuation pivot elimination. Other checks cover integral basis certificates, redundant generators, dual spectra, fifty adaptive comparisons, partial and full stability examples, 415 rank-two interaction realizations, the sharp threshold counterexample, and the hidden-shear mixed-area identity. An eighteen-body fixed test family illustrates the nonadaptive construction, with generic slope $c = 4$, thickness t^8 , and shear t^3 . Twenty further checks exercise the cube/projection identity. The assertion count includes low-level checks such as nonnegativity and integrality; it is not a count of independent theorems.

These computations can detect many normalization, sign, and example errors. They do not prove universal statements over K , validate arbitrary-rank valuations, or supply Lean kernel verification. The universal evidence offered here is the written proof of each stated result.

96.2 Claim and dependency ledger

Statement family	Mathematical dependency	Status in this manuscript
Finite polyhedral volume and mixed-volume properties	Classical real polytope theory; finite real-closed transfer	Imported, with transfer boundary explained; not claimed new.
Valuation-ring diagonalization	Finite integral row and column operations; Cauchy–Binet	Complete proof included; standard algebraic mechanism.
Colorful scale formula	Explicit zonotope sandwich; positive-sum valuation	Complete proof included; principal bridge used by later results.
Spectrum classification and realization	Determinantal minima and diagonalization	Complete proofs included; synthesis with mixed-volume measurements.
Adaptive tomography	Cramer’s rule, determinant-unit criterion, colorful formula	Complete proof included; finite exact tests supplied.
Nonadaptive obstruction	Residue-field avoidance and infinitesimal shear	Complete proof included; requires a non-Archimedean field.
Rank-two interval	Four determinant values and a unit-matrix construction	Necessary and sufficient conditions proved, with realization.
Higher-dimensional interaction	Complementary-minor expansion	Exact reduction proved; general realization problem not solved here.
Singular lengths and polarity	Positive Cauchy–Binet, spectral theorem, module duality	Complete connecting proofs included; relevant prior theory credited.
Formal verification and historical novelty	Lean build, independent refereeing, and exhaustive priority review	Not performed or claimed.

96.3 What the repository inspection establishes

The repository references in the bibliography identify the pinned root and project READMEs and the Euclidean-three-space report’s README. These document the reported scope and verification boundary. They are not an exhaustive inspection of the repository’s thousands of statements, nor a proof that no similar argument appears elsewhere in it. The proofs in this manuscript depend only on the explicitly stated finite-field, polyhedral, and valuation facts; no unrefereed claimed breakthrough elsewhere in the repository is imported as a premise.

The external literature review establishes relevant prior frameworks, not a priority certificate for every consequence formulated here. Accordingly “proved here” means a proof is supplied in this manuscript. It does not mean that the statement is asserted to be absent from all previous literature. In particular, neither diagonalization over a valuation ring, the singular-value connection, nor classical Alexandrov–Fenchel theory should be credited as a new invention of this work.

96.4 The precise outcome

The resulting theory identifies a complete scale object for finite order-convex polytopes: their displacement module. It proves a finite geometric formula for all of its mixed-volume observations, a classification of its one-body spectrum, a reference-dependent finite recognition theorem, and a genuinely different nonadaptive obstruction. The planar *two-body* relative-position problem *with prescribed individual spectra* is completely solved at the valuation level ([theorem 92.1](#)) [[write] the delivered sentence says “The planar relative-position problem”, which overstates it: the many-body planar profile problem is Part VII’s [theorem 110.2](#), for finite-valued profiles]; its higher-dimensional counterpart is reduced to a concrete finite matrix problem. These are the substantive mathematical outcomes. A claim to have settled a named major conjecture, established historical priority, or produced a fully formally verified article would go beyond the evidence supplied.

97 Why the mixed-volume inputs transfer (Appendix A of source 08)

[[write] Another copy of the finite transfer argument printed in Part III ([section 33](#)) and, for configurations, in Part I ([theorem 18.1](#)). It is kept because source 08’s volume-graph formulation differs in detail.]

This appendix makes explicit the finite logical boundary behind [Section 85](#). It is not necessary to transfer a theory of measures or an infinite collection of convex bodies in one sentence.

97.1 The graph of finite-polytope volume

Fix the dimension d and an ordinary upper bound N on the number of displayed points. Membership in their convex hull has the first-order ordered-field description

$$x \in \text{conv}(v_1, \dots, v_N) \iff \exists \lambda_1, \dots, \lambda_N \geq 0 : \sum_i \lambda_i = 1, \quad x = \sum_i \lambda_i v_i.$$

A finite collection of simplices with vertices drawn from this list can be declared to triangulate the hull by first-order conditions: each simplex is contained in the hull; their union is the hull; and full-dimensional interiors of distinct simplices are disjoint. Degenerate simplices have zero determinant. There are only finitely many candidate simplex collections for fixed d, N . For each collection, the sum of absolute determinant volumes is definable by an ordered-field formula. Taking the finite disjunction gives a formula for the graph of the volume function.

Over $\mathbb{R}$, triangulations exist and give a unique sum. Existence and uniqueness of this finite formula are sentences of the theory of real-closed fields. Hence they hold in K . Lower-dimensional hulls can be handled in the same definition by a separate rank condition and volume zero. This specifies a unique algebraic K -valued finite-polytope volume.

97.2 Mixed volumes are finite combinations of such volumes

For exactly d argument lists, define

$$V(P_1, \dots, P_d) = \frac{1}{d!} \sum_{S \subseteq [d]} (-1)^{d-|S|} \text{Vol}_d \left(\sum_{i \in S} P_i \right), \quad (97.1)$$

where the empty sum is the point $\{0\}$. Over $\mathbb{R}$ this is the usual mixed volume: finite differences of the degree- d volume polynomial retain precisely the monomial containing all d variables. Each nonempty Minkowski sum has a vertex-generating list formed by sums of one point from each argument list. Its size is bounded by the product of the displayed list sizes. Consequently (97.1) is a finite ordered-field definition for each fixed input format.

Classical symmetry, translation invariance, nonnegativity, monotonicity, the segment identity, and multilinearity become first-order sentences after replacing each volume term by its definable graph. A sum $P + Q$ can increase the bound on list size, but only by an ordinary finite amount. Universal quantification over the finitely many vertex coordinates and any nonnegative scalar parameters is allowed. The real statements therefore transfer to K . The same reasoning applies to any one fixed-dimensional Alexandrov–Fenchel instance for finitely many finite polytopes.

No assertion is made that an arbitrary topologically compact set in K^d has a volume, that ordinary compactness theorems hold in the order topology, or that a proper-class measure exists on $\mathbf{No}^d$. Those issues are not part of the finite construction. The ordered-field polyhedral viewpoint is also the one used in the computational work of [60].

98 Reproducing the checks and the document (Appendix B of source 08)

[[write] Shipped names: `code/verify.py` is `code/08-mixed-volume-tomography-verify.py`, `requirements.txt` is `data/08-mixed-volume-tomography-requirements.txt`, `data/verification.json` is `data/08-mixed-volume-tomography-verification.json`, and the build helpers are `code/08-mixed-volume-tomography-build.sh` and `code/08-mixed-volume-tomography-build.ps1`. The delivered `article.tex`, PDF, README and checksum list are not shipped; this Part is the article. The program writes `data/verification.json` in the parent of its own directory, which for the shipped file is the report's `data/` under an unprefix name; it must therefore be run on a copy with the delivered layout, as the report's README shows.]

The archive is self-contained except for standard software dependencies. It contains the standalone LaTeX source, its compiled PDF, the exact verification program, the verification report, a short source audit, and build instructions. There is no separate bibliography file: the bibliography below is included in the LaTeX source.

To rerun the mathematics checks, use Python 3.10 or later with SymPy. The delivered run used SymPy 1.14.0. From the archive's root directory:

```
python -m pip install -r requirements.txt
python code/verify.py
```

The program writes its report under `data/`. Its fixed random seed makes the generated examples reproducible. All comparisons are exact in the rational-function field. The tests do not depend on a numerical value chosen for t .

To compile the article with a standard TeX Live installation:

```
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

A third pass can be used if cross-reference warnings remain after a source edit. The supplied build script performs the document build. Recompiling may update PDF metadata without changing the mathematical content.

Part VII

Realization and obstruction of mixed-volume profiles

Source and scope. [write] This Part prints two manuscripts of batch 59 that are independent answers to one prompt. The base is batch 59, manuscript 03, printed as *source 10: Tropical Mixed Volumes of Surreal Polytopes: Parallelotope compression, Grassmannian realization, and full-support matroid obstructions*, author line “Research manuscript prepared for the ProveIt research program”, dated 30 September 2026 (archive `surreal_mixed_volumes_research.zip`, main file `surreal_mixed_volumes/article.tex` of 1,884 lines; delivered PDF of 25 A4 pages, not shipped), written against ProveIt commit 6996fee43. The member is batch 59, manuscript 02, printed as *source 09: Finite Models and Real Degenerations of Surreal Mixed Volumes: Parallelotope compression, representable initial supports, and explicit Lorentzian non-liftability*, headed “Research manuscript”, author line “Prepared for Vladimir Reshetnikov”, dated 30 September 2026 (archive `surreal_mixed_volumes.zip`, main file `surreal_mixed_volumes/article.tex` of 1,948 lines; delivered PDF of 28 A4 pages, an unnumbered title page and 27 numbered pages, not shipped), written against ProveIt commit b2b626d85, the child of 6996fee43 that added the batch-58 archives. Both arrived in commit ed80eba06 and were placed in f2cb2034e. Source 10’s program, recorded outputs, build script and source audit are shipped as `code/10-tropical-mixed-volumes-verify.py`, `data/10-tropical-mixed-volumes-verification.json`, `data/10-tropical-mixed-volumes-verification.txt`, `data/10-tropical-mixed-volumes-fano_coefficients.csv`, `data/10-tropical-mixed-volumes-rank_two_example.txt`, `data/10-tropical-mixed-volumes-requirements.txt`, `code/10-tropical-mixed-volumes-build.sh` and `10-tropical-mixed-volumes-source_audit.md`; source 09’s as `code/09-finite-mixed-volume-models-verify.py`, `data/09-finite-mixed-volume-models-verification_results.json`, `data/09-finite-mixed-volume-models-requirements.txt`, `code/09-finite-mixed-volume-models-Makefile` and `09-finite-mixed-volume-models-SOURCE_AUDIT.md`. The two manuscripts’ PDFs and README files are not shipped; they survive in the delivery archives of ed80eba06, and the README status paragraphs are printed below.

The two sources have identical hypotheses: a set-sized real closed ordered field $K \supseteq \mathbb{R}$ with its natural valuation, of arbitrary rank. They prove largely the same theorems, often in the same order. Source 10 is the base because its compression constants are the sharper ones, its rank-two lemma is an equivalence, and its initial-support theorem is followed by a general matroid criterion ([theorem 106.1](#)) of which source 09 has only the Fano instance. Source 09’s own material is absorbed where it adds something: the sharpness of the dm generator bound as a theorem ([corollary 101.3](#)), the explicit integer mixing bound $d\binom{dm}{d}$ ([remark 103.4](#)), an explicit form of the initial-polynomial theorem with its coordinate independence ([theorem 104.4](#) and [proposition 104.5](#)), rank recovery and its converse ([theorem 105.7](#) and [corollary 105.8](#)), the affine-shift clause of the Fano theorem, the “even up to a positive scalar” form of the coefficient obstruction ([theorem 109.2](#)), the boxes example, its literature discussion, the degree-boundary subsection, and its own questions. A result proved by both is printed once, names both sources and carries both labels; a genuinely different proof is printed as “Second proof (source 09)”; where one source’s statement is the weaker, its extra content (constant, clause, count) is kept in a marked remark. The merge had to choose in these places, and each is marked where it occurs.

Relation to Part III. Neither source cites manuscript 06 of batch 58 (Part III); the two batches arrived one after the other, and neither manuscript cites the other. Both re-derive several

Part III results independently. Where the statement is the same, the result is printed once, in Part III, and this Part keeps a marked pointer that carries the sources' labels, together with their different proofs: the parallelotope compression (lemma 36.1 and theorem 36.2, with exactly source 10's constants), the positive-sum lemma (lemma 33.2), the standard part of volume and mixed volume (proposition 34.3; the two sources give a second and a third proof), linear independent representatives (lemma 38.5), and exchange-convexity of realizable profiles (theorem 38.4; source 09's proof is Part III's nested-clone proof). Where the statement differs it is printed here in full with a marked sentence relating it to Part III; the principal case is the exact realization criterion theorem 103.3, whose equivalence (i) $\Leftrightarrow$ (iii) is Part III's corollary 40.1. No Part III claim is contradicted: the Fano cubic of section 108 shows that Part III's theorem 38.4 has no converse, which Part III never claimed. The results here partly answer three questions of Part III (intrinsic full-profile realization, leading coefficients above a fixed scale profile, and joint residue data), as recorded after those questions. Part VI (source 08, the same batch) is a sibling: its two-body interval theorem, theorem 92.1, refines the case $m = 2$ of theorem 110.2.

Generality. As in Part III, every theorem of this Part holds for an arbitrary set-sized real closed ordered field $K \supseteq \mathbb{R}$ with its natural valuation; surreal numbers enter through the localization of finite data and through examples such as $t = \omega^{-1}$.

Arrangement. Source 10's sections are printed in its order, with source 09's sections absorbed into the matching ones and marked [source 09]; source 09's section on examples and algorithms and its section on the repository and reproducible verification are printed as sections of their own. The four appendices are printed as ordinary sections at the end of the Part (sections 116 to 119); the formalization subsections stay in this Part. Both bibliographies are printed as one list, "Part VII (sources 10 and 09)"; a work cited by both is listed once under source 10's key, with source 09's differing details added to the entry. In this Part "this paper", "this manuscript", "the article" and "the package" mean the source in whose text they occur, never this report.

Notation in this Part. [write] The following symbols mean something different elsewhere in the report (section 1.3); the Part-local meaning is the one in force here.

- $\nu : K \rightarrow \Gamma \cup \{\infty\}$ is the natural valuation, written as in both sources (and in Part VI). It is Part III's v , with the same direction: larger values mean smaller elements, $\nu(\omega^{-a}) = a$, and $\nu(x) > 0$ exactly for nonzero infinitesimals. The report's macro for v is not used here, so that the sources' ν is printed.
- The *mixed-volume valuation profile* has three names in the report. Source 10 writes f_P for the volume polynomial, $V_\alpha(P)$ for the mixed volume $V(P_1[\alpha_1], \dots, P_m[\alpha_m])$ and $q_P(\alpha) = \nu(V_\alpha(P))$; source 09 writes F_P , $c_\alpha(P) = [x^\alpha]F_P = \frac{d!}{\alpha!}V_\alpha(P)$ and $h_P(\alpha) = \nu(c_\alpha(P))$; Part III writes $\tau(\alpha)$. Since multinomial factors are real units, $q_P = h_P = \tau$. The sources' letters are kept.
- $\Delta_m^d = \Delta_m^d = \{\alpha \in \mathbb{N}^m : |\alpha| = d\}$ is the degree- d integer simplex, not Part III's simplex Δ_r .
- *Compression* here is *parallelotope compression*: every body is replaced by a parallelotope with at most d generators without changing any profile value. It is neither Part II's greedy support compression nor Part III's one-parameter specialization (theorem 41.2).
- *Residue* and *reduction* mean the standard part st ("residue configuration", "residue map"), not Part III's affine residue $\text{Res}(P)$.
- $\text{Pol}(q)$ (source 10) and p_h (source 09) are the same *canonical polarization*, a function on

d -subsets of $E = [m] \times [d]$; “polarization formula” (100.4) is the unrelated inclusion–exclusion formula for mixed volume.

- B : source 10’s parallelotopes B_i and real-representable base set $\mathcal{B}(r)$; source 09’s base set $B(r)$ (the same set), its basis matrix B in [theorem 104.4](#), and its Fano basis polynomial B_F . Z_i is $\sum_j [0, a_{ij}]$ in source 10 and the symmetric $\sum_j [-b_{ij}, b_{ij}]$ in source 09 (Part III: $\sum [0, u]$). T_i are mixing matrices, not Part III’s probe segments. D is a minimizing basis matrix in source 10 and a positive scalar $|\det B|$ in source 09. E is the colored index set, and in source 09 also the polynomial e_2 of [section 109](#).
- H is source 10’s Fano polynomial; source 09 writes B_F for it and H_A for a colored minimum (Part III: a grid bound). S is $x_1 + \cdots + x_m$ in source 10 and an index set elsewhere; source 09 writes s for the sum and S_w for an initial support. $L = I + tJ$ in [proposition 108.2](#); source 09’s L_i are subspaces (Part III’s $L(P)$ is the scale lattice). Source 10’s Hessian matrices M_k, N_k are source 09’s H_j, M_j .
- t is a positive infinitesimal with $\gamma = v(t) > 0$ (Part III: volume variables t_i and a Hahn variable); $\delta_r = 1/(r(r+1))$ is source 09’s sandwich constant.
- The two sources label the Fano plane differently; the dictionary is in [section 108](#).
- Macro changes only: source 09’s `\vol` is typeset as `\Vo1` and both sources’ plain `\Span` as `\spanplain`; source 09’s “Research question” environment is the report’s question environment, numbered with the theorems. The printed output is otherwise unchanged.

Abstract of source 10. We study the orders of magnitude of all mixed volumes of a finite family of polytopes over the surreal numbers. The arguments take place in set-sized real closed subfields with their natural valuation and real residue field, and therefore do not require class-sized convexity, analytic integration, or a rank-one value group.

A maximal-simplex argument replaces each polytope by a parallelotope without changing *any* mixed-volume valuation. Consequently, for m bodies in dimension d , at most md segment directions suffice. We prove an exact realization criterion: a function on the degree- d integer simplex is a mixed-volume valuation profile if and only if its canonical polarization is the valuation vector of the maximal minors of a $d \times md$ matrix over the scalar field. This identifies a realization locus, not merely a set satisfying tropical exchange inequalities.

A complementary normalization theorem shows that every exposed, coefficientwise initial polynomial is a volume polynomial of real polytopes. Its support is the base set of a real-representable polymatroid; squarefree initial supports must be real-representable matroids. We use this to prove that, for a loopless rank- d matroid M , the full-support profile $q_M(\alpha) = d - r_M(\text{supp } \alpha)$ is realizable precisely when M is representable over $\mathbb{R}$.

An explicit cubic in seven variables is strictly Lorentzian, has all 84 coefficients positive, and satisfies every tropical exchange condition, but its valuation profile cannot arise from any finite surreal polytopes. A finite rank-two construction establishes the planar sufficiency boundary. A second, quadratic example separates realizability of valuations from realizability of leading coefficients. Complete proofs, reproducible exact checks, a source audit, and a program of further research are included.

Status (title page of source 10). This is an unrefereed research manuscript. The structural statements are proved below; priority for their precise formulation has not been established. Classical matroid, Lorentzian, and mixed-volume results are identified as such. No new Lean formalization or independent proof-assistant verification is claimed.

Status paragraph of source 10's README ("Read first"). [write] Source 10's delivery README is not shipped; its "Read first" section reads:

`article.pdf` is the 25-page article. `article.tex` is its standalone LaTeX source, with an internal bibliography. The scalar hypotheses are a set-sized real closed ordered field containing the real numbers, equipped with its natural valuation. Finite surreal configurations are handled by localization to such fields.

The manuscript proves simultaneous parallelotope compression, an exact canonical-polarization determinant criterion, preservation of volume polynomials under normalized standard-part degeneration, and a full-support profile test for real matroid representability. An explicit Fano cubic is strictly Lorentzian but has an unrealizable mixed-volume valuation profile. A finite rank-two construction establishes planar sufficiency for full-support M-convex profiles. A four-variable quadratic separately shows that realizability of a profile does not imply realizability of prescribed leading coefficients.

These are conventional mathematical proofs in an unrefereed manuscript. Priority for the exact structural formulations has not been established. The classical Fano/support obstruction, Lorentzian background, and rank-two tropical Grassmannian facts are credited in the article. No new Lean formalization or Lean build is claimed.

([write] Neither `article.pdf` nor `article.tex` of the delivery is shipped; the manuscript is printed in this Part.)

Abstract of source 09. For finitely many polytopes over a set-sized real closed subfield of the surreal numbers, we study the valuations of all coefficients of their Minkowski volume polynomial. The natural valuation can have arbitrary rank. We prove that every such valuation profile is realized by parallelotopes with at most d segment generators per polytope in dimension d . This bound is sharp. Equivalently, the entire profile is a colored minimum of maximal-minor valuations of a matrix with at most dm columns. Generic integer mixing realizes its full polarized minor table. In degree two, every finite-valued M-convex profile is realized by parallelograms, without extending the field.

A different normalization preserves leading coefficients rather than only their valuations. After any positive rescaling of the summands, a single maximum-determinant change of coordinates makes all vertices finite and gives a nonzero real volume polynomial by standard part. Consequently every weighted initial support is a real-linearly-representable polymatroid base set.

We exhibit a strictly Lorentzian cubic with all 84 coefficients positive whose valuation profile cannot arise from any seven surreal three-dimensional polytopes. Its coefficients occupy just three valuation levels, and its initial support is the Fano matroid. An exact Hessian decomposition proves strict Lorentzianity. Thus degree three is the first degree allowing a full-support M-convex non-liftability obstruction. A second, quadratic example has a realizable valuation profile but impossible initial coefficients, separating the two obstructions.

The real mixed-volume, matroid, and Lorentzian ingredients are classical and are credited explicitly. The contribution developed here is their constructive, arbitrary-rank synthesis, with sharp finite witnesses and explicit infinitesimal families. No priority claim, solution of the distinct projective volume-polynomial problem, or Lean verification is asserted. Full conventional proofs, exact computational certificates, and a further research program are included.

Verification status (title page of source 09). AI-assisted research manuscript, not independently refereed. The accompanying program checks the finite coefficient identities, seven Hessian congruences, 12,047 exchange obligations, 23 rational sandwich examples, and 15

quadratic reconstructions. It does not certify the general theorems. Repository context is pinned to the commit recorded in [section 113](#); no repository-wide proof review or build is claimed.

Status paragraph of source 09’s README (“Scope and status”). [write] Source 09’s delivery README is not shipped; its “Scope and status” section reads:

This is an AI-assisted research manuscript, not independently refereed and not Lean-checked. It gives full conventional proofs of its stated results. Historical priority has not been established. Classical mixed-volume identities, maximum-determinant normalization, Lorentzian theory, real matroid obstructions, and the rank-two tropical Grassmannian are explicitly credited rather than claimed as new. The research questions answered here are formulated in the manuscript; no previously named open conjecture is claimed solved.

The arbitrary-rank finite constructions and their consequences are developed as the manuscript’s contribution. They concern ordinary convex polytopes over an ordered field, not max-plus tropical polytopes and not the distinct projective volume-polynomial problem discussed by Huh.

The repository context is pinned to ProveIt commit `b2b626d8537f7dea8f20393ee4714c7201a7001e`. Only the sources and ranges described in `SOURCE_AUDIT.md` were used as repository context. No full repository audit, review of all incoming archives, or Lean build was performed.

([write] `SOURCE_AUDIT.md` is shipped as `09-finite-mixed-volume-models-SOURCE_AUDIT.md`.)

Reading routes (source 09). **Reading routes.** For the finite realization theorem, read Sections [101–102](#). For actual leading coefficients, read Sections [104–105](#). The sharp degree distinction is established in Sections [110–108](#); Section [109](#) separates valuation data from coefficient data. The final sections give algorithms, source and verification boundaries, and ten further research questions.

99 The problem and the contribution

A surreal polytope can have ordinary finite combinatorics while carrying volumes on many incomparable Archimedean scales. A face lattice records which incidences occur; it does not record whether a mixed volume is appreciable, infinitesimal of order ω^{-1} , or smaller than every integer power of another infinitesimal. This manuscript studies that missing information.

Fix nonempty finite polytopes $P_1, \dots, P_m \subset K^d$, where K is a real closed ordered field containing $\mathbb{R}$. Its natural valuation is written $v : K \rightarrow \Gamma \cup \{\infty\}$. The mixed-volume polynomial is

$$f_P(x) = \text{Vol}_d(x_1 P_1 + \dots + x_m P_m) = \sum_{\alpha \in \Delta_m^d} \frac{d!}{\alpha!} V_\alpha(P) x^\alpha, \quad \Delta_m^d = \{\alpha \in \mathbb{N}^m : |\alpha| = d\}. \quad (99.1)$$

Its *valuation profile* is

$$q_P(\alpha) = v(V_\alpha(P)) = v([x^\alpha] f_P). \quad (99.2)$$

The equality holds because nonzero rational numbers have valuation zero. The central question is which functions q occur in this way.

99.1 What is proved

The results form a progression from convex geometry to finite algebra.

1. **Uniform compression.** Every family has the same profile as a family of parallelotopes, with at most d independent directions per body (Theorem 36.2). The comparison constants are explicit and independent of the vertices' magnitudes.
2. **An exact determinant criterion.** Polarize the profile by replacing each color i by d labeled copies. The resulting function on d -subsets is realizable by maximal-minor valuations if and only if the original profile comes from polytopes (Theorem 103.3).
3. **Initial geometry and obstructions.** Every exposed initial polynomial is an actual real-polytope volume polynomial (Theorem 104.2). Every exposed initial support is a real-representable polymatroid base set (Theorem 105.4).
4. **A matroid test and explicit separation.** The profile $d - r_M(\text{supp } \alpha)$ recognizes real representability of M (Theorem 106.1). The Fano specialization gives a strictly Lorentzian full-support cubic with an impossible mixed-volume profile (Theorem 108.4).

The determinant criterion is useful because its matrix size depends only on (m, d) , not on the original numbers of vertices. It is not an assertion that checking realizability is easy: it reduces one genuine realization problem to another, and makes otherwise hidden matroid obstructions visible.

99.2 Source 09's setting and principal results

[write] Source 09 opens with the same problem in its own notation (F_P, c_α, h_P) ; see "Notation in this Part"). Its introduction follows.

A polytope with surreal coordinates can have sides of incomparable orders of magnitude. An infinitesimal perturbation can also leave every coordinate valuation unchanged while altering a determinant by an arbitrarily large valuation. Thus the interesting extra information is not captured by taking valuations of coordinates separately.

We focus on a global invariant that couples different polytopes. Fix positive integers d, m . Let $P = (P_1, \dots, P_m)$ be a finite collection of polytopes in K^d , where $K \supseteq \mathbb{R}$ is a set-sized real closed ordered field. Its volume polynomial is

$$F_P(x) = \text{Vol}_d(x_1 P_1 + \dots + x_m P_m) = \sum_{\alpha \in \Delta_m^d} c_\alpha(P) x^\alpha, \quad \Delta_m^d = \{\alpha \in \mathbb{N}^m : |\alpha| = d\}. \quad (99.3)$$

Here $+$ is ordinary Minkowski addition over the ordered field, not tropical addition. The coefficient c_α is a positive real multinomial factor times a mixed volume. We use the natural additive valuation v of K , with small positive elements having positive value, and set

$$h_P(\alpha) = v(c_\alpha(P)), \quad v(0) = +\infty.$$

We call h_P the *mixed-volume valuation profile*. It is not a volume assigned to a max-plus tropical polytope, and the word "valuation" here means a field valuation, not the inclusion-exclusion valuation on convex bodies.

99.3 Questions answered in this manuscript (source 09)

The following questions arise naturally from combining the repository's ordered-field geometry with mixed-volume theory. They are formulated here; they are not being represented as

previously published open conjectures.

Can arbitrarily complicated finite polytopes carry arbitrarily complicated valuation witnesses? Not in the number of segment generators. [Remarks 101.2](#) and [102.3](#) replace each r_i -dimensional polytope by an r_i -generator parallelotope without changing any mixed-volume valuation. Thus a $d \times N$ matrix with $N \leq dm$ suffices, irrespective of the original vertex counts. The entries may still have complicated surreal values.

Is a leading form only an abstract Lorentzian polynomial? No. [Theorem 104.4](#) constructs an actual collection of real polytopes whose volume polynomial is the normalized leading form. This holds after every positive rescaling of the summands, even when the value group has no rank-one or countable-cofinality hypothesis.

Do Lorentzian inequalities suffice to recognize valuation profiles of surreal polytopes? No. [Theorem 108.4](#) gives a full-support, strictly Lorentzian cubic for which no polytope collection has the same valuation profile. The obstruction is not a zero in the ordinary support: all monomials of degree three occur. It is the nonrepresentability of the set of monomials of least valuation. By contrast, [theorem 110.2](#) proves that every finite-valued quadratic M-convex profile has a parallelogram realization. Together these results identify the first degree at which such a valuation obstruction can occur. [[write] Here, and in both sources' abstracts, the obstruction and the degree/dimension boundary concern finite-valued (equivalently full-support) profiles; profiles with zero coefficients are not covered, as source 09 says itself ([section 110.1](#)).]

Does realizability of a valuation profile settle the leading coefficient problem? Again no. [Theorem 109.2](#) constructs a strictly Lorentzian quadratic whose profile is realized by four thin parallelograms, while its own initial coefficients violate the real Plücker relation.

99.4 A compact form of the main results (source 09)

Assume that $P_1 + \dots + P_m$ is full-dimensional. There exist independent vectors $b_{i1}, \dots, b_{ir_i}$ for each i , where $r_i = \dim P_i \leq d$, such that, on putting $A = (b_{ij})$ and denoting its color classes by E_i ,

$$h_P(\alpha) = \min_{\substack{|I|=d \\ |I \cap E_i| = \alpha_i \ (1 \leq i \leq m)}} v(\det A_I). \quad (99.4)$$

The minimum is $+\infty$ if there is no nonzero eligible minor. Conversely, every such matrix produces a polytope profile by taking $Z_i = \sum_j [-b_{ij}, b_{ij}]$. This is an equivalence, though it is not yet an intrinsic elimination of the matrix variables.

For weights $w_i \in \Gamma = v(K^\times)$, choose $\lambda_i > 0$ with $v(\lambda_i) = w_i$. There is a $D > 0$ with

$$v(D) = \min_{\alpha} \{h_P(\alpha) + \alpha \cdot w\}$$

and real polytopes $\bar{Q}_i$ such that

$$\text{st}(D^{-1}F_P(\lambda_1 x_1, \dots, \lambda_m x_m)) = F_{\bar{Q}}(x) \neq 0. \quad (99.5)$$

Standard part is applied coefficientwise. In particular, the minimum set in [equation \(99.5\)](#) has the form

$$\left\{ \alpha \in \Delta_m^d : \sum_{i \in J} \alpha_i \leq \dim_{\mathbb{R}} \sum_{i \in J} L_i \quad \text{for every } J \subseteq [m] \right\} \quad (99.6)$$

for real subspaces $L_i \subseteq \mathbb{R}^d$.

99.5 What is classical, and what is not claimed

The starting distinctions are classical. Brändén–Huh prove that volume polynomials and matroid basis polynomials are Lorentzian; they also identify real-representability obstructions for squarefree volume supports, explicitly including the Fano matroid [66, Theorem 4.1 and Remark 4.3]. Their substitution and tropicalization results supply additional background [66, Theorems 2.10 and 3.20]. The present Fano argument is therefore *not* the first separation of Lorentzian polynomials from volume polynomials. Menges studies that comparison in substantially greater generality [69].

The proposed contribution here is the simultaneous compression, canonical determinant realization criterion, coefficientwise normalization, and the explicit full-support valuation encoding. The proofs are supplied independently of the repository’s informal manuscripts. The literature search did not establish priority for these exact formulations, so they should be read as research results submitted for scrutiny, not as certified first discoveries.

Averkov, von Dichter, Richard, and Soprunov relate mixed volumes of zonoids to absolute Grassmannian coordinates and explicitly raise the question of a connection with tropical Grassmannians [65, Section 1.2]. Our finite-valued formulation gives one precise connection: taking valuations of positive determinant sums turns their aggregation into minima, while compression extends the result from zonotopes to arbitrary finite polytopes. We do not solve their full problem of determining all numerical mixed-volume inequalities.

Throughout this paper, a *volume polynomial* means the polynomial of a finite family of convex polytopes, unless stated otherwise. Intersection-number polynomials of projective varieties and their closures are different objects. In particular, this manuscript does not claim to answer the algebraic-geometric Fano question discussed in Huh’s 2026 survey [68, Example 3.4].

99.6 Relation to established work and limits of the novelty claim (source 09)

Mixed-volume polynomiality, monotonicity, and the independent-segments criterion are classical; Schneider is our reference [71]. Brändén and Huh prove that volume polynomials are Lorentzian and connect Lorentzian tropicalization to discrete convexity [66]. Their Remark 4.3 already distinguishes support representability from exact coefficient realizability and includes the four-variable quadratic obstruction used below. In particular, a Fano support obstruction is not a new discovery of this manuscript. Menges classifies the pairs of degree and number of variables for which the real convex-body volume-polynomial and Lorentzian classes coincide [69].

The finite normalization below is also related in spirit to the classical compactness of collections of convex bodies modulo common affine changes of coordinates, discussed by Shephard and in [66, Remark 4.3]. We do not claim that normalization by a large simplex is new. What we prove in the present formulation is an explicit arbitrary-rank field statement, a sharp dm -column realization criterion for *all* mixed-volume valuations at once, and the stated positive infinitesimal examples. The degree-two construction uses the classical rank-

two tropical Grassmannian and tree correspondence of Speyer and Sturmfels [72]; its finite arbitrary-value-group version is proved here. A targeted literature check does not establish historical priority for this synthesis.

One terminological boundary is essential. “Volume polynomial over a field” in algebraic geometry can mean an intersection polynomial of divisors, or a limit of such polynomials. This is a larger class than the convex-polytope class considered here. Huh’s 2026 survey asks a separate Fano question for projective volume polynomials in characteristic different from two [68, Example 3.4]. Our negative results do not answer that question.

99.7 The connection to the repository

The inspected ProveIt sources already provide a natural valuation, positive monomials of prescribed value, and an ordered standard-part homomorphism with real residue field [73, 74]. These are precisely the scalar operations used below. We do not need a complete Hahn-series implementation, analytic integration, or a surreal analogue of Hausdorff compactness.

The audited source snapshot is

6996fee43cc97b6c16351def7507d59a95bf62f0.

The subsequently observed head, b2b626d8537f7dea8f20393ee4714c7201a7001e, is its child and adds incoming archives. The mathematical source comparisons in this article are pinned to the former commit. A source-level audit is not a successful build; no build was performed. [write] Re-verified against the current ProveIt tree: the natural valuation valuation (an AddValuation), the positive monomials tMonomial and their value valuation_tMonomial are declared in Algebra/SurrealNumbers/Surreal/Foundations/SignSequenceValuation.lean, and the ordered standard-part homomorphism standardPartHom from finite elements to $\mathbb{R}$ in SignSequenceStandardPart.lean of the same directory; both files are unchanged since 6996fee43 (the blob identifiers recorded in section 116 are those of the current tree), and b2b626d85 is indeed a child of 6996fee43. These declarations are inputs that the source cites. No statement of this Part is formalized.

100 Finite scalar workspaces and algebraic volume

100.1 The natural valuation

The standing scalar hypotheses are:

$$\mathbb{R} \subseteq K, \quad K \text{ is a set-sized real closed ordered field,} \quad \nu \text{ is its natural valuation.} \quad (100.1)$$

Thus

$$\mathcal{O} = \{a \in K : |a| \leq n \text{ for some } n \in \mathbb{N}\}, \quad \mathfrak{m} = \{a \in K : |a| < 1/n \text{ for every } n \geq 1\}.$$

The residue map is the ordered ring homomorphism $\text{st} : \mathcal{O} \rightarrow \mathbb{R}$ with kernel $\mathfrak{m}$. We use the min-plus convention:

$$\nu(ab) = \nu(a) + \nu(b), \quad \nu(a + b) \geq \min\{\nu(a), \nu(b)\}, \quad \nu(0) = \infty.$$

No embedding of Γ into $\mathbb{R}$ is assumed. In particular, Γ may have arbitrarily high ordered rank.

Remark 100.1 ([write] The positive-sum lemma is Part III's). Both sources state and prove, as their Lemma "Positive sums and comparison" (source 10) and "No cancellation among positive terms" (source 09), exactly [lemma 33.2](#) of Part III: for finitely many $a_j \geq 0$ in K , $v(\sum_j a_j) = \min_j v(a_j)$, and $cb \leq a \leq Cb$ with positive real c, C forces $v(a) = v(b)$, including the case $a = b = 0$. It is printed once, in Part III; the three manuscripts arrived in consecutive batches and none cites another. Source 09's proof (take a largest term; the sum lies between it and n times it) is Part III's. Source 10's proof takes a term of least valuation instead:

Second proof (source 10) of lemma 33.2. The all-zero case is immediate. Otherwise discard zero terms and choose a term of least valuation, say a_1 . Every ratio a_j/a_1 belongs to $\mathcal{O}$, so the sum of the ratios belongs to $\mathcal{O}$. The sum is at least 1, hence is a unit of $\mathcal{O}$. This proves the first assertion. For the second, when $a > 0$ the ratio b/a is bounded above and bounded away from zero by positive real constants, so it is a unit. The zero case follows directly from the inequalities. $\square$

This positivity is the essential reason a valuation of a zonotope coefficient is a minimum rather than an expression involving unknown cancellation. Determinants themselves may cancel internally; we never replace their valuations by minima of their terms without justification.

100.2 The natural valuation and standard part (source 09)

Define

$$\mathcal{O} = \{a \in K : |a| \leq R \text{ for some } R \in \mathbb{R}_{>0}\}, \quad \mathfrak{m} = \{a \in K : |a| < r \text{ for every } r \in \mathbb{R}_{>0}\}.$$

The first is a convex valuation ring and the second its maximal ideal. Every $a \in \mathcal{O}$ has a unique standard part $\text{st}(a) \in \mathbb{R}$ with $a - \text{st}(a) \in \mathfrak{m}$. For existence, take the real supremum of $\{r \in \mathbb{R} : r \leq a\}$; real completeness is used only in $\mathbb{R}$, not in K . The usual estimates show that st is an order-preserving ring homomorphism $\mathcal{O} \rightarrow \mathbb{R}$ with kernel $\mathfrak{m}$.

Write $v : K^\times \rightarrow \Gamma$ for the associated additive valuation. We fix its sign by

$$a \in \mathcal{O} \iff v(a) \geq 0, \quad a \in \mathfrak{m} \setminus \{0\} \iff v(a) > 0.$$

Nonzero real numbers have valuation zero. For positive a, b ,

$$v(a) = v(b) \iff \exists c, C \in \mathbb{R}_{>0} \quad cb \leq a \leq Cb. \quad (100.2)$$

Here and below $+\infty$ is adjoined with the usual conventions. No embedding $\Gamma \hookrightarrow \mathbb{R}$ is assumed.

Finite sums are important. Neither the lemma nor any subsequent argument allows one to discard summability conditions on an infinite Hahn series.

100.3 Why finite surreal data fit the hypotheses

The surreal numbers form a real closed ordered field in the usual proper-class sense; see the standard theory in [67]. For a finite collection of surreal coordinates, start with the set-sized field they generate over $\mathbb{R}$ and take its real closure inside **No**. This produces a field satisfying (100.1). Further finitely many coordinates, normalizing scalars, or matrix entries can be added before taking the real closure. Every argument in this paper uses only such finite data.

On surreal monomials our convention is

$$v(\omega^{-a}) = a. \quad (100.3)$$

The repository realizes this convention with its `tMonomial` construction and `valuation_tMonomial` theorem [73]. The full surreal field is never treated as a set over which to form a class-sized convex hull or to invoke compactness.

100.4 A set-sized interpretation of surreal geometry (source 09)

The surreal numbers form a real closed ordered class field [67]. Given finitely many surreal coordinates, first take the set-sized field generated by them and $\mathbb{R}$, and then take its real closure inside **No**. All the arguments below take place in that field K . Enlarging K preserves the finite determinant identities and ordered-field statements used here.

Thus a K -polytope is the set

$$\text{conv}_K\{p_1, \dots, p_n\} = \left\{ \sum_{j=1}^n a_j p_j : a_j \in K_{\geq 0}, \sum_j a_j = 1 \right\}.$$

We do not integrate over the proper class **No**^{*d*}. A scalar volume is defined by a finite triangulation and a finite determinant sum. Its value belongs to K and agrees with the same construction after extending scalars. The notation “surreal polytope” refers to this finite-coordinate interpretation. A full class-sized theory of measures or arbitrary convex subsets is outside our scope.

100.5 Volume without analytic integration

A polytope is the convex hull of a nonempty finite set of points. For an affinely independent $(d + 1)$ -tuple, define

$$\text{Vol}_d \text{conv}(v_0, \dots, v_d) = \frac{1}{d!} |\det(v_1 - v_0, \dots, v_d - v_0)|.$$

Finite triangulations extend this definition to full-dimensional polytopes; lower-dimensional polytopes have d -volume zero. Triangulations here are algebraic, face-to-face subdivisions, not analytic decompositions requiring a measure on K^d .

We recall why the usual finite-polytope identities remain valid. Fix the dimension and the finite numbers of labeled vertices. A proposed triangulation, its covering property, its face intersections, and the absence of overlapping interiors can be expressed using finitely many polynomial equalities and inequalities, allowing first-order quantifiers for point membership. The volume attached to a fixed triangulation is a finite expression in absolute determinants. Equality of the expressions from two triangulations is therefore a first-order statement over real closed fields. Since the statement holds over $\mathbb{R}$, it holds over K . A pulling triangulation supplies a finite triangulation; its existence can likewise be expressed as a finite disjunction over possible simplices. This also deals with changes of oriented-matroid type and degenerate vertex configurations.

The same fixed-data argument applies to Minkowski polynomiality and the standard mixed-volume identities [71, Chapters 5 and 7]. For additional concreteness, mixed volume can be defined by the finite polarization formula

$$V(Q_1, \dots, Q_d) = \frac{1}{d!} \sum_{I \subseteq [d]} (-1)^{d-|I|} \text{Vol}_d \left(\sum_{i \in I} Q_i \right), \quad (100.4)$$

where the empty sum is the singleton $\{0\}$. Once the numbers of vertices have been fixed, each term is a semialgebraically specified finite determinant expression. Nonnegativity,

monotonicity in every argument, independent translation invariance, Minkowski multilinearity, homogeneity, and

$$V(Q, \dots, Q) = \text{Vol}_d(Q), \quad V(TQ_1, \dots, TQ_d) = |\det T|V(Q_1, \dots, Q_d)$$

are consequently valid over K . This transfer uses neither completeness nor an order-topological compactness theorem for K .

100.6 What transfers, and what does not (source 09)

A simplex with vertices $p_0, \dots, p_d$ has volume

$$\text{Vol}_d \text{conv}_K(p_0, \dots, p_d) = \frac{|\det(p_1 - p_0, \dots, p_d - p_0)|}{d!}.$$

Finite triangulations define the volume of a full-dimensional polytope; lower-dimensional polytopes have ambient d -volume zero. For fixed finite input sizes, the choice and validity of triangulations can be expressed by a finite first-order description over an ordered field. The real-closed-field transfer principle therefore transports the relevant finite identities from $\mathbb{R}$ to K [75]. A more explicit explanation is given in [section 118](#).

We use no Heine–Borel theorem for the order topology of K , no sequential compactness of a surreal interval, and no extension of Lebesgue integration to arbitrary surreal sets. The maximum determinants in our constructions are maxima of *finite lists*.

[write] Source 09 then states and proves an unlabelled proposition, “Finite combinatorial types do not require new scalars”: for fixed finite dimension and number of vertices, every face lattice of a K -polytope is realized by a real polytope. It is Part III’s [proposition 33.3](#) (and Part I’s [theorem 18.1](#)), with the same first-order transfer proof, and is printed there. Source 09 continues:

This classical transfer observation motivates studying the valuation profile: it retains scale comparisons that a face lattice alone does not record.

Remark 100.2 (Scope of the classical input). The general proofs below use only finite triangulations, the listed mixed-volume properties, finite-dimensional linear algebra, and the natural valuation. The Lorentzian discussion additionally invokes the classical results explicitly cited in [Section 107](#). The Fano cubic’s strict signature calculation is given directly.

100.7 Volume polynomials and the finite toolkit (source 09)

For nonempty polytopes $P_i \subseteq K^d$, Minkowski’s volume polynomial has nonnegative coefficients. We use the convention

$$c_\alpha(P) = \frac{d!}{\alpha_1! \cdots \alpha_m!} V(P_1[\alpha_1], \dots, P_m[\alpha_m]), \quad (100.5)$$

where V is symmetric and multilinear with respect to Minkowski addition and nonnegative scalar multiplication, and $V(P[d]) = \text{Vol}_d(P)$. The multinomial factors in [equation \(100.5\)](#) are real units; they matter for exact coefficient statements but not for h_P .

We need the following classical properties [71], in their finite-polytope ordered-field versions.

Lemma 100.3 (Mixed-volume operations; source 09). *Mixed volume is nonnegative, invariant under separate translations of its arguments, and monotone in each argument. For $T \in \text{GL}_d(K)$,*

$$F_{TP}(x) = |\det T|F_P(x).$$

Replacing P_i by $\lambda_i P_i$ replaces c_α by $\lambda^\alpha c_\alpha$. If an argument is a point, a mixed volume involving it is zero.

The monotonicity statement is a coefficient theorem, not an inference from pointwise inequalities between two polynomials. In general a polynomial inequality on a positive orthant does not imply a coefficientwise inequality. We use mixed-volume monotonicity itself whenever comparing coefficients.

Lemma 100.4 (Zonotope determinant formula; source 09). For $v_1, \dots, v_N \in K^d$ and $s_j \geq 0$,

$$\text{Vol}_d \left(\sum_{j=1}^N s_j [0, v_j] \right) = \sum_{\substack{I \subseteq [N] \\ |I|=d}} |\det(v_j : j \in I)| \prod_{j \in I} s_j, \quad (100.6)$$

$$\text{Vol}_d \left(\sum_{j=1}^N s_j [-v_j, v_j] \right) = 2^d \sum_{\substack{I \subseteq [N] \\ |I|=d}} |\det(v_j : j \in I)| \prod_{j \in I} s_j. \quad (100.7)$$

The order used for the determinant columns is immaterial after absolute value.

Proof. The volume polynomial is homogeneous of degree d . It is affine in each individual segment length s_j : the segment-extrusion identity says that adding $[0, s_j v_j]$ increases volume linearly in s_j . For polytopes that identity follows from a finite prism decomposition and transfers to K . Thus only squarefree monomials of degree d can occur. The coefficient of $\prod_{j \in I} s_j$ is found by setting all other lengths to zero; the remaining zonotope is a parallelotope of volume $|\det(v_j : j \in I)|$. This proves equation (100.6). A symmetric segment is a translate of $[0, 2v_j]$, which gives equation (100.7). $\square$

Our proofs will use positivity to turn this determinant *sum* into an exact valuation *minimum*. They do not replace a signed determinant by the minimum of the valuations of its summands.

101 Simultaneous parallelotope compression

Remark 101.1 ([write] Parallelotope compression is Part III's theorem). Source 10's Lemma "A finite maximal simplex" and Theorem "Simultaneous valuation compression" are Part III's lemma 36.1 and theorem 36.2, with the same maximal-simplex proof and the same constants; they are printed once, in Part III (the manuscripts arrived in consecutive batches and neither cites the other). In source 10's notation they read as follows. If $P \subset K^d$ has affine dimension $r > 0$, there are vertices $v_0, \dots, v_r$ of P maximizing the absolute $r \times r$ determinant in an affine chart, independent vectors $w_j = v_j - v_0$, the parallelotope $B = \sum_{j=1}^r [0, w_j]$ and $s = \sum_j w_j$ with

$$\frac{1}{r} B \subseteq P - v_0 \subseteq -s + 2B. \quad (101.1)$$

For nonempty finite polytopes $P_1, \dots, P_m \subset K^d$ with $r_i = \dim P_i$, the parallelotopes B_i so obtained (with $B_i = \{0\}$ when $r_i = 0$) satisfy, for every $\alpha \in \Delta_m^d$,

$$\left(\prod_i \max(1, r_i)^{-\alpha_i} \right) V_\alpha(B) \leq V_\alpha(P) \leq 2^d V_\alpha(B), \quad (101.2)$$

and in particular

$$q_P = q_B. \quad (101.3)$$

The same choice of $B_1, \dots, B_m$ works for all mixed coefficients. Source 09's version, with a symmetric parallelotope and a weaker lower constant, is [remark 101.2](#) below.

The coarser lower bound $d^{-d}V_\alpha(B) \leq V_\alpha(P)$ is valid for all α . Nothing depends on a real bound for the original coordinates: a simplex can be infinitely long or infinitesimally thin. Only a finite maximum of determinants was used.

Remark 101.2 ([write] Source 09's symmetric compression). Source 09 proves the compression with a parallelotope centred at the centroid of a maximal simplex. Its lower constant $\delta_r = 1/(r(r+1))$ is weaker than $1/r$ in (101.2), and it measures against the symmetric $Z = \sum_k [-b_k, b_k]$ instead of $\sum_k [0, b_k]$ (against $\sum_k [0, 2b_k]$, a translate of Z , its constants become $2^d \prod_i \delta_{r_i}^{\alpha_i}$ below and 4^d above). The profile conclusion is the same. Its lemma, theorem and proofs follow in its words.

A dimension-only sandwich.

Lemma (Maximum-minor sandwich). Let $P \subseteq K^d$ be a polytope of affine dimension $r \geq 1$. There are a point $c \in P$ and linearly independent vectors $b_1, \dots, b_r$ in the direction space of P such that, for

$$Z = \sum_{k=1}^r [-b_k, b_k], \quad \delta_r = \frac{1}{r(r+1)},$$

one has

$$\delta_r Z \subseteq P - c \subseteq 2Z. \quad (101.4)$$

The vectors can be chosen among differences of vertices from one anchor vertex. The constants do not depend on the field, its valuation rank, or the number of vertices.

Proof. Write $P = \text{conv}_K(p_0, p_1, \dots, p_n)$ and $v_j = p_j - p_0$. Choose r ambient coordinate rows whose projection π is injective on $L = \text{span}_K\{v_j\}$. Among the finitely many r -subsets of the columns $\pi(v_j)$, choose a basis with maximum absolute determinant. Let its lifts be $b_1, \dots, b_r$.

For each j , write $v_j = \sum_k a_{kj} b_k$. Cramer's rule expresses a_{kj} as a replacement determinant divided by the selected determinant. Maximality gives $|a_{kj}| \leq 1$. Hence, in the coordinates of this basis,

$$P - p_0 \subseteq [-1, 1]^r, \quad S := \text{conv}_K(0, e_1, \dots, e_r) \subseteq P - p_0.$$

Let $q = (1/(r+1), \dots, 1/(r+1))$, the centroid of S , and let $c = p_0 + \sum_k b_k/(r+1)$. Every coordinate of every point in $[-1, 1]^r - q$ has absolute value at most 2, giving the outer inclusion.

For the inner inclusion, take $z \in [-1, 1]^r$. Each coordinate of $q + \delta_r z$ is nonnegative, and

$$\sum_{k=1}^r (q_k + \delta_r z_k) \leq \frac{r}{r+1} + r\delta_r = 1.$$

Thus $q + \delta_r [-1, 1]^r \subseteq S$. Translating and mapping back by the basis gives the inner inclusion of [equation \(101.4\)](#). $\square$

For $r = 0$, take $c = P$ and $Z = \{0\}$. In formulas that do not involve a positive-dimensional argument, put $\delta_0 = 1$.

Simultaneous preservation of every mixed-volume valuation.

Theorem (Parallelotope compression). Let $P_1, \dots, P_m \subseteq K^d$ be nonempty polytopes, and put $r_i = \dim P_i$. There are parallelotopes

$$Z_i = \sum_{j=1}^{r_i} [-b_{ij}, b_{ij}]$$

with independent generators within each color such that, for every $\alpha \in \Delta_m^d$,

$$\left(\prod_{i=1}^m \delta_{r_i}^{\alpha_i} \right) c_\alpha(Z) \leq c_\alpha(P) \leq 2^d c_\alpha(Z). \quad (101.5)$$

Consequently

$$h_P(\alpha) = h_Z(\alpha) \quad (\alpha \in \Delta_m^d). \quad (101.6)$$

The assertion includes equality of zero patterns.

Proof. Apply [remark 101.2](#) separately to the polytopes. Translate P_i by $-c_i$, which leaves every mixed volume unchanged. Repeated use of monotonicity and homogeneity in [lemma 100.3](#) proves [equation \(101.5\)](#). An argument of dimension zero makes the relevant coefficient zero on both sides. Otherwise the comparison constants are strictly positive real numbers. [Lemma 33.2](#) gives [equation \(101.6\)](#); the two inequalities also imply that a coefficient vanishes on one side exactly when it vanishes on the other. $\square$

This replacement preserves valuations, not exact volumes, face lattices, angles, or normal fans. For example, in a single full-dimensional real polytope every nonzero volume has valuation zero; the profile alone cannot distinguish a simplex from a cube. The theorem becomes informative when several polytopes and several non-Archimedean scales interact.

Corollary 101.3 (Sources 10 and 09; source 09: “Optimal number of generators”). *The realization of a nonzero mixed-volume profile of m polytopes in K^d never requires more than md segment directions. When every entry of the profile is finite, every body may be chosen to be a full-dimensional parallelotope. [source 09] Every profile in dimension d on m polytopes has a segment-sum witness using at most dm generators in total. The bound is optimal within witnesses that assign a finite sum of segments to each of the same m colors.*

Proof. There are at most d generators in each compressed body. A finite profile has $q(de_i) < \infty$, hence $\text{Vol}_d(P_i) > 0$ and $r_i = d$. $\square$

Second proof (source 09), with the optimality. The upper bound is $\sum_i r_i \leq dm$. For the lower bound, consider any collection with $c_{de_i} > 0$ for every i , for example m full-dimensional cubes. Any other segment-sum collection with the same profile must have positive individual d -volume in each color. Fewer than d segment vectors span a space of dimension less than d , so each color needs at least d generators. $\square$

The sharpness statement concerns witness size, not optimal running time or bit complexity. A single surreal entry may have noncomputable or infinitely supported normal form data.

Remark 101.4 (What compression forgets). Compression preserves zero coefficients and valuation orders. It need not preserve a numerical coefficient, its leading real coefficient, the face lattice, or a preferred embedding. This distinction is essential in [Section 109](#).

102 The determinant model

Let

$$E = [m] \times [d], \quad A = (a_{ij})_{(i,j) \in E} \in K^{d \times md}, \quad Z_i = \sum_{j=1}^d [0, a_{ij}].$$

For $S \subseteq E$, write $c(S)_i = |S \cap (\{i\} \times [d])|$ for its color-count vector. Columns are always ordered using one fixed total order on E .

Proposition 102.1 (Positive determinantal coefficients). *For every $\alpha \in \Delta_m^d$,*

$$[x^\alpha]f_Z(x) = \sum_{\substack{S \subseteq E, |S|=d \\ c(S)=\alpha}} |\det A_S|. \quad (102.1)$$

Consequently,

$$q_Z(\alpha) = \min_{\substack{S \subseteq E, |S|=d \\ c(S)=\alpha}} v(\det A_S). \quad (102.2)$$

The minimum of an all-zero determinant family is ∞ .

Proof. For segments $L_j = [0, u_j]$,

$$V(L_1, \dots, L_d) = \frac{1}{d!} |\det(u_1, \dots, u_d)|.$$

Indeed, their Minkowski sum with separate weights is a parallelotope when the vectors are independent, and has zero volume otherwise. Expand $\text{Vol}_d(\sum_{i,j} x_i [0, a_{ij}])$ by Minkowski multilinearity. An ordered list of d segment labels contributes its determinant divided by $d!$. Repeated labels give determinant zero; every d -element subset occurs $d!$ times. Group by color counts to obtain (102.1). Lemma 33.2 gives (102.2). $\square$

The formula is for *ordinary polynomial coefficients*. For mixed volumes themselves one multiplies by $\alpha!/d!$. Omitting this normalization changes numerical coefficients but does not change a natural-valuation profile.

Example 102.2 (Two-scale rectangles). Suppose positive scalars a_i, b_i have values $A_i, B_i \in \Gamma$, and

$$P_i = [0, a_i e_1] + [0, b_i e_2].$$

Then

$$f_P(x) = \left(\sum_i a_i x_i \right) \left(\sum_i b_i x_i \right),$$

so

$$q_P(2e_i) = A_i + B_i, \quad q_P(e_i + e_j) = \min\{A_i + B_j, A_j + B_i\} \quad (i \neq j).$$

For $a_1 = b_2 = 1$ and $b_1 = a_2 = t^2$ with $v(t) = \gamma > 0$, the three values for two bodies are $(2\gamma, 0, 2\gamma)$. The mixed area is appreciable even though both areas are infinitesimal. The same formulas work in higher-rank value groups.

102.1 The colored determinant realization criterion (source 09)

Let $E = E_1 \sqcup \dots \sqcup E_m$ be a finite colored set, and let $A = (a_e)_{e \in E}$ be a $d \times |E|$ matrix of rank d over K . For $\alpha \in \Delta_m^d$, define

$$H_A(\alpha) = \min \{v(\det A_I) : |I| = d, |I \cap E_i| = \alpha_i \text{ for all } i\}. \quad (102.3)$$

Zero minors have valuation $+\infty$.

Remark 102.3 ([write] Source 09's "Exact characterization by at most dm columns"). Source 09 states the following theorem. It is the equivalence (i) $\Leftrightarrow$ (ii) of [theorem 103.3](#) below, read through [proposition 102.1](#) (in Part III, the colorwise minimum (38.2) together with compression), and is printed as a remark with source 09's proof: *A non-identically-infinite function $h : \Delta_m^d \rightarrow \Gamma \cup \{+\infty\}$ is the mixed-volume valuation profile of m finite K -polytopes if and only if $h = H_A$ for a rank- d matrix with color classes of size at most d . The columns in each color class may be required to be linearly independent.*

Proof. For the forward direction, use the generators of [remark 101.2](#) as the columns of A . The total Minkowski sum is full-dimensional because the profile is not identically infinite, so the columns have rank d . The symmetric zonotope formula gives

$$c_\alpha(Z) = 2^d \sum_{\substack{|I|=d \\ |I \cap E_i| = \alpha_i \forall i}} |\det A_I|. \quad (102.4)$$

Take valuations and apply [lemma 33.2](#). The factor 2^d is a real unit. This proves $h_P = H_A$.

Conversely, given A , define $Z_i = \sum_{e \in E_i} [-a_e, a_e]$. Equation (102.4) proves $h_Z = H_A$. With independent columns in a color, Z_i is a parallelotope. If a witness begins with dependent columns or more than d columns per color, [remark 101.2](#) replaces it by one satisfying the stated bounds. $\square$

102.2 What the criterion says in matroid language (source 09)

The nonzero $d \times d$ minors of A index bases of a realizable matroid. Their valuations satisfy the valued basis-exchange inequalities. One can see the latter directly from the determinant exchange identity: for two bases I, J and $e \in I \setminus J$, the product $\det A_I \det A_J$ is, up to signs, a sum of products

$$\det A_{I-e+f} \det A_{J-f+e}, \quad f \in J \setminus I.$$

At least one product on the right has valuation at most that on the left. Thus there is a simultaneous exchange whose total valuation does not increase. Here cancellation may occur in the identity; the ordinary valuation inequality still gives the conclusion.

The function H_A forgets the individual column labels except for colors and minimizes over bases with prescribed color multiplicities. In this sense the theorem describes profiles as colored projections of *realizable* valuated matroids. Replacing “realizable” by “arbitrary” is a substantial relaxation, and is not permitted by the theorem.

103 Canonical polarization and an exact realization criterion

Definition 103.1 (Canonical polarization). For $q : \Delta_m^d \rightarrow \Gamma \cup \{\infty\}$, its canonical polarization is the function

$$\text{Pol}(q) : \binom{E}{d} \longrightarrow \Gamma \cup \{\infty\}, \quad \text{Pol}(q)(S) = q(c(S)), \quad E = [m] \times [d].$$

We say that $\text{Pol}(q)$ is K -realizable if there exists a rank- d matrix $C \in K^{d \times md}$ with

$$v(\det C_S) = \text{Pol}(q)(S) \quad \text{for all } S \in \binom{E}{d}.$$

This is actual value realization over the specified field, not a closure of a value image.

103.1 From colored minima to a single polarized minor table (source 09)

There is a stronger realization criterion: the minimum in [equation \(102.3\)](#) can be made constant on every color-count class by a finite generic change of generators.

Let $E_i = \{(i, 1), \dots, (i, d)\}$ and $E = \bigsqcup_i E_i$. For a d -subset $I \subseteq E$, let $\alpha(I)_i = |I \cap E_i|$. The *polarization* of a profile h is the function

$$p_h(I) = h(\alpha(I)).$$

It prescribes a valuation for *every* maximal minor of a $d \times dm$ matrix, not just for one minor in each color-count class.

Lemma 103.2 (Simultaneous generic block mixing). *Let $A \in K^{d \times md}$ have rank d , and define q by (102.2). There are invertible real matrices $T_i \in \text{GL}_d(\mathbb{R})$ such that*

$$C = A \text{ diag}(T_1, \dots, T_m) \text{ satisfies } v(\det C_S) = q(c(S)) \text{ for every } S \in \binom{E}{d}.$$

Proof. Temporarily let all entries of the T_i be independent indeterminates. Fix S , set $S_i = S \cap (\{i\} \times [d])$, and write $\alpha = c(S)$. Block Cauchy–Binet gives

$$\det C_S = \sum_{\substack{I \in \binom{E}{d} \\ c(I) = \alpha}} \det A_I \prod_{i=1}^m \det(T_i[I_i, S_i]). \quad (103.1)$$

Empty determinants equal 1. If $q(\alpha) = \infty$, all summands vanish identically, so the required zero determinant is automatic.

Suppose $q(\alpha) < \infty$. Choose I_0 attaining the minimum and put $a = \det A_{I_0}$. After division by a , the polynomial in (103.1) has coefficients in $\mathcal{O}$. Its reduction is not the zero polynomial. To see this without a genericity shortcut, choose one permutation monomial in each $\det(T_i[(I_0)_i, S_i])$. The product monomial records the exact sets of row indices $(I_0)_i$, and therefore cannot occur in a summand indexed by a different I . Its coefficient after division is 1 or -1 . Hence it survives reduction.

There are only finitely many sets S . The product of their nonzero reduced polynomials, together with the polynomials $\det T_i$, is a nonzero polynomial over $\mathbb{R}$. A nonzero polynomial over an infinite field does not vanish at every point: this follows by induction on its number of variables from the univariate root bound. Choose real entries where this product is nonzero. Evaluation of each normalized determinant then has nonzero residue, and therefore valuation zero. Thus its unnormalized valuation is exactly $q(\alpha)$, simultaneously for all S . $\square$

Theorem 103.3 (Exact profile realization; sources 10 and 09). *Let $q : \Delta_m^d \rightarrow \Gamma \cup \{\infty\}$ be not identically ∞ . The following conditions are equivalent:*

- (i) q is the mixed-volume valuation profile of m finite polytopes in K^d .
- (ii) q is the mixed-volume valuation profile of m parallelotopes, each generated by at most d independent vectors.
- (iii) $\text{Pol}(q)$ is K -realizable.

Proof. The implication (i) $\Rightarrow$ (ii) is Theorem 36.2. For (ii) $\Rightarrow$ (iii), pad the independent generators of each parallelotope with zero columns to form a $d \times md$ matrix A . Since the polynomial is nonzero, A has rank d . Proposition 102.1 identifies its colorwise minimum with q , and Lemma 103.2 supplies C .

For (iii) $\Rightarrow$ (i), use the zonotopes formed by the column blocks of C . All determinants of a given color count have the same prescribed valuation, so Proposition 102.1 gives their profile as q . They are finite polytopes. Another application of compression then gives (ii), if desired. $\square$

[write] The equivalence (i) $\Leftrightarrow$ (iii) is Part III's [corollary 40.1](#), re-derived independently by both sources (Part III and this Part come from consecutive batches, and no source cites another). Part III proves necessity from its clone theorem ([theorem 38.1](#): inner segments chosen on a

finite rational grid); the proof above mixes the columns of an arbitrary compressed generator matrix by generic real matrices instead, and source 09's proof below mixes them by nonnegative integer combinations with an explicit bound. Condition (ii) is Part III's [theorem 36.2](#). The hypothesis that q is not identically ∞ excludes the rank-deficient case discussed after the proof; Part III's statement allows it.

Remark 103.4 ([write] Source 09's "Polarized realization criterion", with integer mixing). Source 09 states (i) $\Leftrightarrow$ (iii) in the following form, with $p_h = \text{Pol}(h)$; its last two sentences, the integer mixing clause and the bound $d \binom{dm}{d}$, are not in source 10's statement: A non-identically-infinite function h is a polytope profile if and only if there is a $d \times dm$ matrix A over K such that

$$v(\det A_I) = p_h(I) \quad \text{for every } d\text{-subset } I \subseteq E. \quad (103.2)$$

Starting from the generators in [remark 101.2](#), the columns of A in each color can be chosen as nonnegative integer combinations of those generators. The integer coefficients can be bounded in terms of d, m alone, for example by $d \binom{dm}{d}$.

Second proof (source 09) of (i) $\Leftrightarrow$ (iii), with integer mixing. Start with the compressed generator matrices B_i of sizes $d \times r_i$. Introduce independent variables for an $r_i \times d$ matrix T_i and put $A_i = B_i T_i$. A maximal minor $f_I(T) = \det A_I$ is a polynomial of total degree d in these variables. Its coefficients are signed determinants of original generators. The variables attached to different new columns are distinct. Therefore every assignment of an original generator to each selected column produces its own coefficient monomial; a nonzero original determinant cannot cancel identically in this polynomial.

For $\alpha = \alpha(I)$, all coefficients of f_I have valuation at least $h(\alpha)$. If $h(\alpha)$ is finite, some coefficient has exactly that valuation by [remark 102.3](#). Choose $a_\alpha > 0$ of that valuation. Then $a_\alpha^{-1} f_I$ has finite coefficients and reduces to a nonzero polynomial over $\mathbb{R}$. If $h(\alpha) = +\infty$, f_I is identically zero.

Take the product of the nonzero reduced polynomials for all I having finite prescribed value. It is a nonzero real polynomial of total degree at most $D = d \binom{dm}{d}$. A nonzero polynomial with degree at most D in each variable cannot vanish on the entire grid $\{0, 1, \dots, D\}^q$, where q is its number of variables: apply the univariate root bound successively. Choose a grid point where the product is nonzero. At that integer specialization, every finite minor has exactly the prescribed valuation, and the identically zero ones remain zero. This proves [equation \(103.2\)](#) and the coefficient bound.

Conversely, group the columns of a matrix satisfying [equation \(103.2\)](#) into symmetric segment sums. Every exponent $\alpha \in \Delta_m^d$ occurs as the color count of a d -subset of E . All corresponding determinant valuations equal $h(\alpha)$, so [equation \(102.4\)](#) and positivity give the desired profile. $\square$

This is a concrete realization statement about the polarized table. In tropical terminology, it asks for membership in the rank- d Grassmannian valuation image over the *ordered* field K , rather than merely satisfaction of abstract valuated-matroid exchange. The generic-integer bound is intentionally crude; it makes the finite nature of the selection explicit.

When a block of C has rank less than d , its d generators may produce a zonotope that is not a parallelotope in its affine span. This is why the last proof first concludes (i), rather than calling every degenerate cube image a parallelotope. If q is finite everywhere, all block determinants are nonzero and this issue disappears.

The excluded all- ∞ profile is realized by points, or by any family whose combined directions span a space of dimension less than d . It requires a rank-deficient matrix rather than a Grassmannian point.

103.2 Grassmannian interpretation and its limits

The vector of maximal minors of a rank- d matrix is a Plücker vector. Thus Theorem 103.3 identifies the profile locus with those color-symmetric value vectors in the actual K -valued Grassmannian image. The tropical Grassmannian is traditionally studied as a closure of such value images, often over an algebraically closed field [72]. Our exact field-specific formulation avoids silently replacing real realizability by complex realizability, or exact values by limiting values.

Valuated-matroid exchange conditions, equivalently the short tropical Plücker conditions in the finite uniform case, are necessary but not sufficient for this realization. Canonical polarization and its minimizing inverse are standard operations on M-convex functions [66, Lemma 3.26]; the extra condition here is *determinantal realizability*. Section 108 supplies a concrete obstruction with no infinite coordinates at all.

Definition 103.5. A function $h : \Delta_m^d \rightarrow \Gamma \cup \{+\infty\}$ with nonempty finite-valued domain is *M-convex* if, whenever $h(\alpha), h(\beta)$ are finite and $\alpha_i > \beta_i$, there is j with $\alpha_j < \beta_j$ such that

$$h(\alpha) + h(\beta) \geq h(\alpha - e_i + e_j) + h(\beta + e_i - e_j).$$

We use the minimum convention for this definition.

Remark 103.6 ([write] Exchange-convexity of realizable profiles is Part III's). Source 09's Corollary "M-convexity over arbitrary value groups" states that every nonzero polytope profile is M-convex, without a rank restriction on Γ . This is Part III's theorem 38.4, and source 09's proof is Part III's: take a polarized matrix (theorem 103.3; Part III's clone matrix), choose the two column sets as the first α_k and the first β_k copies of each color, so that they are nested within each color, and apply the determinant exchange argument above; every exchanged element then lies in a deficit color. It is printed once, in Part III.

For perspective, M-convexity of h is also equivalent to the abstract valued basis-exchange property of p_h . In the direction not already implicit above, take $e \in I \setminus J$ of color i . If $\alpha(I)_i \leq \alpha(J)_i$, there is a same-color element $f \in J \setminus I$, and exchanging it changes neither color count. Otherwise the M-convex exchange supplies a deficit color and an element f in that color. This proves the polarized exchange property directly. Realizing that abstract table by actual minors is the additional requirement in theorem 103.3.

103.3 The finite size of a witness is not a decision procedure (source 09)

At most $\binom{dm}{d}$ maximal minors remain after compression. Nevertheless, remark 102.3 is an existential matrix criterion, not a list of necessary and sufficient inequalities in the numbers $h(\alpha)$ alone. Solving it requires compatibility of the minors, including their signs and leading coefficients. Nor does an arbitrary surreal expression come with an effective comparison or normal-form algorithm. These distinctions are important for implementation; see section 111.

104 Standard-part degeneration preserves volume polynomials

The determinant reduction also controls leading coefficients, not just supports, provided we work with the original bodies after a common linear normalization.

The compression theorem intentionally discards leading coefficients. This section uses a different construction, retaining the original vertex sets.

Remark 104.1 ([write] Standard part commutes with bounded volume: Part III’s proposition). Source 10’s Lemma “Standard part of a bounded polytope” and source 09’s Lemma “Bounded shadows” are Part III’s [proposition 34.3](#), whose hull clause is Part I’s [proposition 13.1](#). For $Q = \text{conv}(v_1, \dots, v_N)$ with all $v_j \in \mathcal{O}^d$ and $\bar{Q} = \text{conv}(\text{st } v_1, \dots, \text{st } v_N) \subset \mathbb{R}^d$ they state $\text{st}(Q) = \bar{Q}$, $\text{Vol}_d(Q) \in \mathcal{O}$ and

$$\text{st}(\text{Vol}_d(Q)) = \text{Vol}_d(\bar{Q}), \quad (104.1)$$

and, for bounded polytopes $Q_1, \dots, Q_m$, that standard part commutes with every mixed coefficient, $\text{st}(V_\alpha(Q)) = V_\alpha(\bar{Q})$, that is, with the volume polynomial coefficient by coefficient. Dimension may drop. The statement is printed once, in Part III, where it is proved by semialgebraic continuity; the three manuscripts arrived in consecutive batches and none cites another. The two sources’ proofs differ from Part III’s and from each other, and follow. (Source 09 writes P and $\bar{P}$ for Q and $\bar{Q}$.)

Second proof (source 10) of [proposition 34.3](#), by triangulation. Every convex coefficient is in $[0, 1] \subset \mathcal{O}$. Taking standard parts of a convex combination proves one inclusion in the first equality. Conversely, a real convex combination of the $\text{st } v_j$ lifts using the same real coefficients to a point of Q .

If Q is lower-dimensional, its affine-dependence determinants vanish, so the reduced polytope cannot have larger dimension; both d -volumes are zero. Otherwise triangulate Q into finitely many full-dimensional simplices. Their reduced simplices cover $\bar{Q}$ by the first equality, because standard part commutes with a finite union.

The interiors of two nondegenerate reduced simplices cannot overlap. Indeed, a real point in both interiors has strictly positive barycentric coordinates in each reduced simplex. The original simplices’ determinant matrices are units over $\mathcal{O}$, so solving for the barycentric coordinates of the same real point in K^d gives coordinates whose residues are the prescribed strictly positive numbers. All original barycentric coordinates are then positive, contradicting disjointness of the original interiors.

The nondegenerate reduced simplices therefore give a finite nonoverlapping volume decomposition; degenerate ones contribute zero. Standard part commutes with finite determinant expressions and with absolute value on $\mathcal{O}$. Taking standard parts of the simplex-volume sum gives the desired identity. Finiteness follows either from this sum or from containment in a real-sized cube.

Finally, standard part commutes with Minkowski sums of bounded finite polytopes. Apply (100.4) to the repeated bodies occurring in V_α and use the volume identity term by term. $\square$

Third proof (source 09) of [proposition 34.3](#), by thickening. Let $C = [-1, 1]^d$ and let P^0 be the K -convex hull of the real shadow vertices. There is a nonnegative infinitesimal ϵ bounding the absolute values of all coordinate differences $v_j - \text{st } v_j$; take the maximum of this finite list. Using the same convex weights shows

$$P \subseteq P^0 + \epsilon C, \quad P^0 \subseteq P + \epsilon C.$$

Choose a positive real M with $P, P^0 \subseteq MC$. For $A = P$ or P^0 , mixed-volume monotonicity gives the finite polynomial estimate

$$\begin{aligned} 0 &\leq \text{Vol}_d(A + \epsilon C) - \text{Vol}_d(A) \\ &\leq \sum_{k=1}^d \binom{d}{k} \epsilon^k M^{d-k} \text{Vol}_d(C) = ((M + \epsilon)^d - M^d) \text{Vol}_d(C). \end{aligned}$$

The last expression is infinitesimal. The two inclusions therefore imply that $\text{Vol}_d(P) - \text{Vol}_d(P^0)$ is infinitesimal. Since P^0 has real vertices, its determinant volume is the real volume of $\bar{P}$. This proves [equation \(104.1\)](#).

For polytopes P_i with bounded vertices, $F_P(1)$ is bounded by a real cube volume. All its coefficients are nonnegative, so every coefficient is finite. For each $x \in \mathbb{R}_{>0}^m$, the shadow of $\sum_i x_i P_i$ is $\sum_i x_i \bar{P}_i$; the first part of the proof gives

$$\text{st}(F_P)(x) = F_{\bar{P}}(x).$$

These two real polynomials agree on the positive orthant, so their coefficients agree. $\square$

This proof allows dimensions to drop under standard part. A thin rectangle can have a segment as its shadow. It also avoids introducing a Hausdorff limit indexed by a cofinal sequence in K .

For positive scalars $s_1, \dots, s_m \in K$, let

$$w_i = v(s_i), \quad \mu = \min_{\alpha \in \Delta_m^d} \left(q_P(\alpha) + \sum_i \alpha_i w_i \right).$$

Assume $f_P \neq 0$, so $\mu \in \Gamma$. If $a > 0$ has value μ , define

$$\text{in}_{s,a}(f_P)(x) = \sum_{\alpha \in \Delta_m^d} \text{st}(a^{-1} s^\alpha [x^\alpha] f_P) x^\alpha. \quad (104.2)$$

Each argument of st belongs to $\mathcal{O}$. The polynomial is nonzero, and its support is the set of exponents attaining μ . Its coefficients depend on the actual representatives s_i, a , not just on their valuations.

Theorem 104.2 (Real-polytope initial polynomial). *Every polynomial (104.2) is the volume polynomial of a finite family of polytopes in $\mathbb{R}^d$.*

Proof. Apply compression to P_i , retaining its chosen vertex v_i and parallelotope generators. Multiply the i th block of generators by s_i and concatenate the blocks into A . By [\(101.3\)](#) and [\(102.2\)](#), the smallest valuation of any maximal minor of A is μ . Choose a minimizing basis matrix $D = A_{S_0}$.

Every entry of $D^{-1}A$ belongs to $\mathcal{O}$. Cramer's rule expresses such an entry as a replacement minor divided by $\det D$, and the replacement minor has value at least μ . The translated, normalized polytopes

$$Q_i = D^{-1}(s_i(P_i - v_i))$$

have all vertices in $\mathcal{O}^d$: the upper box containment in [\(101.1\)](#) places them in finite real linear combinations of the bounded columns of $D^{-1}A$.

Let $u = |\det D|$. Independent translation invariance and linear covariance of mixed volume give

$$f_Q(x) = u^{-1} f_P(s_1 x_1, \dots, s_m x_m).$$

[Proposition 34.3](#) identifies its coefficientwise standard part with $f_{\bar{Q}}$. Since u/a is a positive unit, $c = \text{st}(u/a) > 0$, and

$$\text{in}_{s,a}(f_P) = c f_{\bar{Q}}.$$

A common real linear transformation with determinant c , for example $\text{diag}(c, 1, \dots, 1)$, multiplies a volume polynomial by c . The displayed initial polynomial is therefore a real-polytope volume polynomial. $\square$

Corollary 104.3 (Leading-coefficient obstruction). *If a nonnegative polynomial over K has, after a positive variable rescaling and common coefficient normalization, an initial polynomial that is not a real-polytope volume polynomial, then it is not a K -polytope volume polynomial.*

Unlike a compactness argument for an unspecified family of real convex bodies, this proof gives a finite normalization: one minimizing minor bounds every transformed generator and every original vertex. It also works when the value group has higher rank.

104.1 One common affine normalization (source 09)

[write] Source 09 proves the initial-polynomial theorem in the following explicit form. Its normalizer is a maximum absolute determinant over the original scaled vertex differences, which bounds every normalized vertex by 1 and yields the minimum formula (104.4) and the support (104.5). Theorem 104.2 is the same result for an arbitrary normalizing representative a (with a compressed configuration and a minimizing basis). Both are printed because each is used below; in Part III the case $m = 1$ is corollary 35.4.

Theorem 104.4 (Real realization of every normalized initial polynomial; source 09). *Let $P_1, \dots, P_m \subseteq K^d$ have full-dimensional total Minkowski sum, and let $\lambda_i > 0$. Choose an anchor vertex $p_i \in P_i$. Among all vectors $\lambda_i(v - p_i)$, with v a vertex of P_i , choose d vectors forming a matrix B of maximum absolute nonzero determinant. Set*

$$D = |\det B|, \quad Q_i = B^{-1}(\lambda_i(P_i - p_i)).$$

Then every vertex of every Q_i lies in $[-1, 1]^d$, and

$$\text{st}(D^{-1}F_P(\lambda_1 x_1, \dots, \lambda_m x_m)) = F_{\bar{Q}}(x) \neq 0, \quad (104.3)$$

where $\bar{Q}_i$ is the real shadow of Q_i . If $w_i = v(\lambda_i)$, then

$$v(D) = \min_{\alpha \in \Delta_m^d} (h_P(\alpha) + \alpha \cdot w). \quad (104.4)$$

In particular, the support of the real polynomial on the right is exactly

$$S_w = \arg \min_{\alpha \in \Delta_m^d} (h_P(\alpha) + \alpha \cdot w). \quad (104.5)$$

Proof. The direction spaces of the P_i span K^d , so the finite vector list contains a basis. Cramer's rule, using maximality of $|\det B|$, bounds all transformed vertex coordinates by 1. Each Q_i contains 0. Every selected basis vector becomes some e_j in one of the Q_i . Consequently $Q_1 + \dots + Q_m$ contains every e_j and 0, and hence contains their simplex. Its shadow contains the same real simplex.

The translation and linear-change rules in lemma 100.3 give the exact identity over K

$$D^{-1}F_P(\lambda_1 x_1, \dots, \lambda_m x_m) = F_Q(x).$$

Apply proposition 34.3. The shadow polynomial is nonzero since

$$\text{Vol}_d(\bar{Q}_1 + \dots + \bar{Q}_m) \geq 1/d!.$$

For completeness, every coefficient of F_Q is finite because

$$1/d! \leq F_Q(\mathbf{1}) \leq (2m)^d$$

and the coefficients are nonnegative. At least one coefficient is a positive unit: a finite sum of infinitesimals would be infinitesimal, contrary to the lower bound. Thus the minimum valuation of the coefficients of F_Q is zero. Their valuations are

$$h_P(\alpha) + \alpha \cdot w - v(D),$$

which proves [equation \(104.4\)](#). A nonnegative finite coefficient has nonzero standard part precisely when its valuation is zero. This proves [equation \(104.5\)](#). $\square$

104.2 Coordinate independence and weight representatives (source 09)

Proposition 104.5 (Independence of a valuation-minimal basis; source 09). *Fix the anchor vertices and the positive scales. In [theorem 104.4](#), one may use any basis whose determinant has minimum valuation among the nonzero determinants of the vector list. The resulting vertices are finite and the shadow sum is full-dimensional. Two such bases give shadow collections related by a common member of $\text{GL}_d(\mathbb{R})$.*

Proof. Cramer's rule now gives entries in $\mathcal{O}$, rather than necessarily in $[-1, 1]$. A finite list of finite entries has a common real bound, and the selected vectors still become the standard basis. The proof of [theorem 104.4](#) applies with that bound.

For two bases B, B' , all entries of $T = B'^{-1}B$ lie in $\mathcal{O}$. The same holds for T^{-1} . Reducing $TT^{-1} = I$ shows that $\text{st}(T) \in \text{GL}_d(\mathbb{R})$. As $Q'_i = TQ_i$, their shadows satisfy $\bar{Q}'_i = \text{st}(T)\bar{Q}_i$ simultaneously for every i . $\square$

Changing anchor vertices changes each model by a separate translation and leads to the same statement up to those translations. Indeed, all new anchor differences are integral linear combinations of old ones, and vice versa. The minimum determinant valuation is consequently unchanged; the translated normalized vertices remain finite.

The support S_w depends only on w . To specify actual initial coefficients one also chooses representatives λ_i and a positive normalizing scalar. Replacing λ_i by a positive unit multiple rescales the corresponding variable by the positive real standard part of that unit. Replacing D by another scalar of the same valuation changes the initial polynomial by a positive real factor. These are the expected ambiguities; no cross-section of the value group has been assumed.

105 Every exposed initial support is real-representable

For $w \in \Gamma^m$, define

$$\mathcal{J}_w(q) = \left\{ \alpha \in \Delta_m^d : q(\alpha) + \alpha \cdot w = \min_{\beta \in \Delta_m^d} (q(\beta) + \beta \cdot w) \right\}. \quad (105.1)$$

Here Γ is the actual value group, so each w_i is attained by a positive scalar. These are the exposed initial sets. For rank-one real-valued data they are the usual lower-face supports of a regular subdivision. Formula [\(105.1\)](#), not an assumed embedding of Γ into $\mathbb{R}$, is the definition at higher rank.

Lemma 105.1 (A minimum basis yields a real residue configuration). *Let $A \in K^{d \times N}$ have rank d , and let $\mu = \min_{|S|=d} v(\det A_S)$. Choose $D = A_{S_0}$ attaining the minimum. Then $C = D^{-1}A$ has entries in $\mathcal{O}$, its reduction $\bar{C}$ has rank d over $\mathbb{R}$, and*

$$\det \bar{C}_S \neq 0 \iff v(\det A_S) = \mu. \quad (105.2)$$

Proof. Cramer's rule proves that every entry of C has nonnegative valuation. The columns indexed by S_0 form the identity, so the reduced matrix has rank d . For every S , $\det C_S = \det A_S / \det D$ lies in $\mathcal{O}$; it has nonzero residue exactly when its value is zero. Standard part commutes with determinants, which gives (105.2). $\square$

For clarity we include the elementary linear-selection theorem used to interpret the residue configuration. It is the vector-space form of Rado's independence theorem [70].

Remark 105.2 ([write] Linear selection is Part III's lemma). Source 10's Lemma "Linear selection" and source 09's Lemma "Linear independent representatives" are Part III's lemma 38.5, with the same inductive proof (a tight proper set of indices is handled by induction and a quotient; otherwise one quotients by a single chosen vector). It is printed once, in Part III. Source 09 states Part III's form: subspaces $W_1, \dots, W_n$ of a finite-dimensional vector space over any field admit independent vectors $w_i \in W_i$ if and only if

$$\dim \sum_{i \in I} W_i \geq |I| \quad (I \subseteq [n]). \quad (105.3)$$

Source 10 states it with multiplicities: for subspaces $W_1, \dots, W_m$ of a vector space over a field and $\alpha \in \mathbb{N}^m$, it is possible to choose α_i vectors from each W_i , all jointly independent, if and only if

$$\sum_{i \in I} \alpha_i \leq \dim \left(\sum_{i \in I} W_i \right) \quad \text{for every } I \subseteq [m]. \quad (105.4)$$

Its proof reduces this form to Part III's:

Reduction (source 10). Necessity is immediate. For sufficiency, replace W_i by α_i labeled copies, making $n = |\alpha|$ slots $U_1, \dots, U_n$. The hypotheses imply $\dim \sum_{j \in J} U_j \geq |J|$ for every set of slots J : use the set of colors occurring in J in (105.4). We prove independent selection by induction on n . $\square$

Definition 105.3 (Real-representable polymatroid base set). Given subspaces $W_i \subseteq \mathbb{R}^d$ with $\sum_i W_i = \mathbb{R}^d$, define $r(I) = \dim \sum_{i \in I} W_i$. The associated base set is

$$\mathcal{B}(r) = \{\alpha \in \Delta_m^d : \alpha(I) \leq r(I) \text{ for every } I \subseteq [m]\}.$$

It is called real-representable because its rank function is realized by real subspaces, rather than merely satisfying abstract rank axioms.

Theorem 105.4 (Initial-support representability). *For every nonzero polytope profile $q = q_P$ and every $w \in \Gamma^m$, there are real subspaces $W_i \subseteq \mathbb{R}^d$ such that*

$$\mathcal{J}_w(q) = \mathcal{B}(r), \quad r(I) = \dim \sum_{i \in I} W_i. \quad (105.5)$$

Proof. Compress the bodies and scale their generator blocks by positive s_i with $v(s_i) = w_i$. Let A be the resulting rank- d matrix. By Proposition 102.1, a color count lies in $\mathcal{J}_w(q)$ precisely when some maximal minor with that count has globally minimum value. Apply Lemma 105.1; this is precisely the condition that the corresponding columns of $\bar{C}$ can form a basis.

Let W_i be the real span of the columns of color i . Lemma 38.5 characterizes independent selection from these spaces by (105.4). Independent vectors from the spaces can be replaced by actual columns of the same colors: expand their nonzero determinant multilinearly in the spanning columns, and at least one determinant in the expansion is nonzero. Thus the color counts of column bases are exactly $\mathcal{B}(r)$. $\square$

Corollary 105.5 (Squarefree initial obstruction). *If $\mathcal{J}_w(q_P)$ consists only of squarefree vectors $\alpha \in \{0, 1\}^m$, then it is the set of bases of a matroid representable over $\mathbb{R}$.*

Proof. Use the spaces W_i from Theorem 105.4. If some W_i has dimension at least two, two independent columns of that color can be extended to a basis using all the columns. Its color count has an entry at least two, a contradiction. Thus every W_i has dimension at most one. Choose a vector u_i spanning W_i , using zero when $W_i = 0$. The bases of this real vector configuration are exactly $\mathcal{J}_w(q_P)$. $\square$

This tests *every* exposed initial set, not merely the support of the original polynomial. A polynomial can have all coefficients strictly positive and still contain a forbidden squarefree matroid at its leading scale.

105.1 Exact support and rank recovery (source 09)

[write] Source 09's definition of the base set, its support theorem for real polytopes and its corollary follow. Its $B(r)$ is the set $\mathcal{B}(r)$ above. Over $\mathbb{R}$, (105.6) is Part III's [corollary 38.6](#); the rank recovery (105.7) and the converse are source 09's own. The first and last sentences of [corollary 105.8](#) are [theorem 105.4](#) and [corollary 105.5](#); its rank-recovery formula is new.

Definition 105.6. A rank function $r : 2^{[m]} \rightarrow \mathbb{N}$ is *real-linearly representable* if there are real subspaces L_i with

$$r(J) = \dim_{\mathbb{R}} \sum_{i \in J} L_i.$$

It is normalized, monotone, and submodular. When $r([m]) = d$, its integral base set is

$$B(r) = \{\alpha \in \mathbb{N}^m : |\alpha| = d, \alpha(J) \leq r(J) \text{ for every } J \subseteq [m]\}.$$

A base set is not a matroid base set unless its exponents are squarefree.

Submodularity follows from the dimension formula and $\sum_{i \in I \cap J} L_i \subseteq (\sum_{i \in I} L_i) \cap (\sum_{i \in J} L_i)$.

Theorem 105.7 (Support of a real volume polynomial; source 09). *Let $Q_i \subseteq \mathbb{R}^d$ be polytopes with full-dimensional total sum. Put $L_i = \text{span}_{\mathbb{R}}(Q_i - Q_i)$ and $r(J) = \dim \sum_{i \in J} L_i$. Then*

$$\text{supp } F_Q = B(r). \tag{105.6}$$

Furthermore the support recovers the entire rank function:

$$r(J) = \max_{\alpha \in \text{supp } F_Q} \alpha(J). \tag{105.7}$$

Conversely, every real-linearly-representable rank function with total rank d is the support rank function of a collection of real parallelotopes.

Proof. Apply the sandwich construction over $\mathbb{R}$. It preserves the zero pattern and replaces each Q_i by a parallelotope with generators forming a basis of L_i . By [lemma 100.4](#), a coefficient is positive exactly when one can select a linearly independent d -tuple of generators with color multiplicities α .

Such a selection implies $\alpha(J) \leq r(J)$ for every J . Conversely, form a list containing α_i copies of the subspace L_i . For any sublist, let J be the colors that occur. Its length is at most $\alpha(J)$, which is at most $r(J)$ by assumption. Thus [lemma 38.5](#) applies and gives independent vectors with those color multiplicities. Expand each vector in the chosen generators of its color. The

determinant of the resulting independent vectors is nonzero; by multilinearity at least one determinant of actual generators is nonzero. That determinant supplies the desired positive coefficient. This proves [equation \(105.6\)](#).

For [equation \(105.7\)](#), choose $r(J)$ independent generators among the colors in J spanning $\sum_{i \in J} L_i$. Extend them to a basis of $\mathbb{R}^d$ using generators from all colors. Every additional vector can be chosen outside J , since all generators in J are already in the span. The corresponding exponent vector belongs to the support and has $\alpha(J) = r(J)$. The reverse inequality follows from [equation \(105.6\)](#).

For the converse, choose any basis of each prescribed subspace L_i and let Q_i be its symmetric parallelotope. The same determinant argument establishes the claimed support and rank function. $\square$

This is a constructive subspace version of the classical independent-segments criterion [71, Theorem 5.1.8]. It is included as a proof ingredient, not as a claim of novelty about ordinary real mixed-volume positivity.

Corollary 105.8 (Real representability at every scale; source 09). *For every polytope profile h_P and every $w \in \Gamma^m$, the minimum set S_w in [equation \(104.5\)](#) is $B(r_w)$ for a real-linearly-representable rank function r_w of total rank d . It determines r_w by*

$$r_w(J) = \max_{\alpha \in S_w} \alpha(J).$$

If S_w is squarefree, it is the set of bases of a matroid representable over $\mathbb{R}$.

Proof. Use the real polytopes from [theorem 104.4](#) and apply [theorem 105.7](#). If every exponent is squarefree, rank recovery gives $r_w(\{i\}) \leq 1$. Thus each nonzero subspace is a line. Choosing one nonzero vector on each line gives the asserted real matroid representation. $\square$

The corollary is stronger than the assertion that the minimum sets are abstract discrete polymatroid base sets. It imposes compatibility with real linear algebra. A separate difficulty remains: even a realizable support need not have the prescribed mixed-volume coefficients.

Proposition 105.9 (Affine changes of profile). *For $\eta \in \Gamma$ and $w \in \Gamma^m$, the function $q'(\alpha) = q(\alpha) + \eta + \alpha \cdot w$ is realizable if and only if q is realizable.*

Proof. Choose positive scalars b, s_i with $v(b) = \eta$ and $v(s_i) = w_i$. Scale the i th body by s_i and apply to all bodies the common linear map $\text{diag}(b, 1, \dots, 1)$. The profile changes by the specified affine function. Use the inverse transformations for the converse. $\square$

106 A full-support profile that encodes matroid representability

Let M be a loopless rank- d matroid on $[m]$, with rank function r_M and base set $\mathcal{B}(M)$. For $\alpha \in \Delta_m^d$, set

$$q_M(\alpha) = d - r_M(\text{supp } \alpha), \quad \text{supp } \alpha = \{i : \alpha_i > 0\}. \quad (106.1)$$

Every value is finite. Moreover,

$$q_M(\alpha) = 0 \iff \alpha = \mathbf{1}_B \text{ for } B \in \mathcal{B}(M). \quad (106.2)$$

Indeed, $|\text{supp } \alpha| \leq d$, so a support of rank d must have exactly d elements and all exponents must be one.

Theorem 106.1 (Matroid realization encoded by a positive profile). *Let $t \in K$ be positive with $\gamma = v(t) > 0$. For a loopless rank- d matroid M , the following are equivalent:*

- (i) M is representable over $\mathbb{R}$.
- (ii) The function γq_M is the mixed-volume valuation profile of m finite polytopes in K^d .
- (iii) The same profile is realized by m full-dimensional parallelotopes in K^d .

Proof. Suppose (ii). The zero-weight initial set is exactly $\{\mathbf{1}_B : B \in \mathcal{B}(M)\}$ by (106.2). Corollary 105.5 proves (i). The equivalence between (ii) and (iii) follows from compression because the profile is finite everywhere.

For (i) $\Rightarrow$ (ii), choose a real representation $u_1, \dots, u_m \in \mathbb{R}^d$ and put

$$Q = [0, 1]^d, \quad P_i = [0, u_i] + tQ. \quad (106.3)$$

These are full-dimensional polytopes. Consider a determinant term in the coefficient of x^α in their zonotope volume polynomial. The unscaled columns are some of the vectors u_i , using at most one from each color. If the term is nonzero, at most $r = r_M(\text{supp } \alpha)$ independent unscaled columns can occur. The other columns are t times coordinate vectors. Thus every nonzero term has value at least $(d - r)\gamma$.

To attain this bound, choose an independent set of r vectors u_i from the supported colors, using one slot of each of these colors. Complete them to a basis of $\mathbb{R}^d$ with $d - r$ standard coordinate vectors. Distribute those coordinate vectors among the remaining color slots prescribed by α . Each color has all d scaled coordinate directions available, so this distribution is possible. The resulting determinant is a nonzero real constant times t^{d-r} . All coefficient contributions enter with absolute value, so there is no cancellation. The coefficient's value is exactly $(d - r)\gamma = \gamma q_M(\alpha)$. $\square$

Remark 106.2. Loops can be included by taking $u_i = 0$ in (106.3); the same proof works with values up to d . The loopless formulation keeps the range in $\{0, \dots, d - 1\}$ and covers the principal examples.

Theorem 106.1 says more than that some matroids obstruct some supports. It encodes a matroid into a *fully finite* profile, so every prospective realizing body must be full-dimensional. The obstruction is hidden in the scale separation, not in a zero coefficient of the polynomial being realized.

107 Full-support Lorentzian lifts

For a rank- d matroid, write its basis polynomial as

$$b_M(x) = \sum_{B \in \mathcal{B}(M)} \prod_{i \in B} x_i, \quad S = x_1 + \dots + x_m.$$

Define the positive deformation

$$F_{M,t}(x) = b_M(x_1 + tS, \dots, x_m + tS). \quad (107.1)$$

We use the standard Lorentzian class of homogeneous polynomials with nonnegative coefficients. For a cubic with every coefficient positive, strict Lorentzianity means that the Hessian of every first partial derivative is nonsingular with exactly one positive eigenvalue; the analogous quadratic condition is on its own Hessian. This is an algebraic signature condition, so the same definition makes sense over a real closed field. See [69] for the definitions and comparison with convex-geometric volume polynomials.

Proposition 107.1. *For every real $t > 0$, the polynomial $F_{M,t}$ is Lorentzian and all its coefficients are positive. Over K , for positive infinitesimal t with $v(t) = \gamma$,*

$$v([x^\alpha]F_{M,t}) = \gamma q_M(\alpha). \quad (107.2)$$

Its integer-valued order function q_M is M-convex.

Proof. Matroid basis polynomials are Lorentzian, and nonnegative linear substitutions preserve the Lorentzian class [66]. The substitution in (107.1) has matrix $I + tJ$, whose entries are positive for $t > 0$. Selecting the tS term in every factor of any basis monomial contributes $t^d S^d$, proving positivity of every coefficient. The Lorentzian assertion also transfers to K : for fixed (m, d) and this full support it is a finite collection of algebraic Hessian-signature conditions.

For a fixed basis B , producing x^α can retain an unaltered x_i from at most $|B \cap \text{supp } \alpha|$ factors. All other factors must use tS . The maximum of this intersection size over bases is $r_M(\text{supp } \alpha)$, because an independent subset of the support extends to a basis. This maximum is attainable, and the remaining factors of S supply all remaining multiplicities in α . Thus the lowest power of t is $d - r_M(\text{supp } \alpha)$, with a positive integer coefficient. This proves (107.2). Finally apply the Lorentzian tropicalization theorem over real Puiseux series [66, Theorem 3.20] to this polynomial family; it says exactly that its order function is M-convex. $\square$

In the min-plus convention, M-convexity is the exchange condition

$$\begin{aligned} \alpha_i > \beta_i &\implies \text{there exists } j \text{ with } \alpha_j < \beta_j \text{ such that} \\ q(\alpha) + q(\beta) &\geq q(\alpha - e_i + e_j) + q(\beta + e_i - e_j). \end{aligned} \quad (107.3)$$

The theorem just used is only needed to establish this general Lorentzian/M-convex background. Nonrealizability of the Fano example below uses the residue obstruction, and its strict Lorentzianity has a separate direct proof.

Corollary 107.2. *If M is not representable over $\mathbb{R}$, then $F_{M,t}$ has full support and is Lorentzian, but no family of finite K -polytopes has its mixed-volume valuation profile.*

Proof. Combine Proposition 107.1 with Theorem 106.1. $\square$

108 An explicit strictly Lorentzian cubic with an impossible profile

108.1 The polynomial and its coefficient table

Label the seven nonzero vectors of $\mathbb{F}_2^3$ as

$$1 = e_1, \quad 2 = e_2, \quad 3 = e_3, \quad 4 = e_1 + e_2, \quad 5 = e_1 + e_3, \quad 6 = e_2 + e_3, \quad 7 = e_1 + e_2 + e_3.$$

The seven lines of the Fano matroid are

$$124, \quad 135, \quad 236, \quad 167, \quad 257, \quad 347, \quad 456. \quad (108.1)$$

All other three-element subsets are bases. If $e_k(x)$ denotes the k th elementary symmetric polynomial in seven variables, put

$$\begin{aligned} H(x) = e_3(x) - x_1x_2x_4 - x_1x_3x_5 - x_2x_3x_6 \\ - x_1x_6x_7 - x_2x_5x_7 - x_3x_4x_7 - x_4x_5x_6, \end{aligned} \quad (108.2)$$

$$F_t(x) = H(x_1 + tS, \dots, x_7 + tS), \quad S = \sum_{i=1}^7 x_i. \quad (108.3)$$

Each pair belongs to four bases, each point belongs to twelve bases, and there are twenty-eight bases. Expanding by the number of tS factors gives

$$F_t = H + 4tSe_2 + (12t^2 + 28t^3)S^3. \quad (108.4)$$

For $t > 0$, all degree-three monomials occur with positive coefficients. Their complete classification is in Table 12.

Source 09's labelling and the dictionary. [write] Source 09 constructs the same cubic from a different labelling of the Fano plane. In its words:

Label the seven nonzero binary vectors by $1, \dots, 7$, with

$$\begin{aligned} 1 &= (1, 0, 0), & 2 &= (0, 1, 0), & 3 &= (1, 1, 0), & 4 &= (0, 0, 1), \\ 5 &= (1, 0, 1), & 6 &= (0, 1, 1), & 7 &= (1, 1, 1). \end{aligned}$$

The seven dependent triples are

$$\mathcal{L} = \{123, 145, 167, 246, 257, 347, 356\}. \quad (108.5)$$

All other triples are bases; there are $\binom{7}{3} - 7 = 28$ of them. Define

$$B_F(x) = \sum_{\substack{I \subseteq [7], |I|=3 \\ I \notin \mathcal{L}}} \prod_{i \in I} x_i. \quad (108.6)$$

[write] Both line lists are exactly the Fano lines of their labellings (checked in the batch-59 audit). A vertex label of source 09 and a vertex label of source 10 that denote the same binary vector are matched; the matching is the identity except that it exchanges 3 and 4 (source 09's $3 = (1, 1, 0)$ is source 10's $4 = e_1 + e_2$, and source 09's $4 = (0, 0, 1)$ is source 10's $3 = e_3$). It carries source 09's lines

$$123, 145, 167, 246, 257, 347, 356 \quad \text{to} \quad 124, 135, 167, 236, 257, 347, 456,$$

which is source 10's list (108.1). Hence $B_F(x) = H(x_1, x_2, x_4, x_3, x_5, x_6, x_7)$: source 09's basis polynomial B_F is source 10's H after exchanging x_3 and x_4 , its s is S , and its F_t below is source 10's F_t after the same exchange. The coefficient table is invariant under this exchange (it is organized by monomial type), and the two sources' tables coincide; it is printed once, as table 12. Source 09's regularization and coefficient lemma read:

Put $s = x_1 + \dots + x_7$ and $e_2 = \sum_{i < j} x_i x_j$. For $t > 0$, define

$$F_t(x) = B_F(x_1 + ts, \dots, x_7 + ts). \quad (108.7)$$

Lemma 108.1 (Exact coefficient formula; source 09). *For every t in a characteristic-zero field,*

$$F_t = B_F + 4tse_2 + (12t^2 + 28t^3)s^3. \quad (108.8)$$

Its coefficients are given in table 12.

Proof. Let $D_0 = \sum_{i=1}^7 \partial_i$. Each pair belongs to four Fano bases, and each point belongs to twelve. Consequently

$$D_0 B_F = 4e_2, \quad D_0^2 B_F = 24s, \quad D_0^3 B_F = 168.$$

Taylor expansion in the scalar displacement $ts\mathbf{1}$, with s fixed during that expansion, gives equation (108.8). A squarefree monomial of degree three has coefficient 3 in se_2 and 6 in s^3 ; a monomial $x_i^2 x_j$ has coefficients 1 and 3; a cube has coefficients 0 and 1. This yields the table. $\square$

All coefficients are positive for every $t > 0$. In particular, the ordinary support is the entire degree-three simplex, not the Fano base set.

Table 12: All 84 coefficients of F_t . The last column assumes $\nu(t) = \gamma > 0$.

Monomial type	Number	Coefficient	Value
$x_i x_j x_k$, a Fano basis	28	$1 + 12t + 72t^2 + 168t^3$	0
$x_i x_j x_k$, a Fano line	7	$12t + 72t^2 + 168t^3$	γ
$x_i^2 x_j$, $i \neq j$	42	$4t + 36t^2 + 84t^3$	γ
x_i^3	7	$12t^2 + 28t^3$	2γ

108.2 A direct proof of strict Lorentzianity

For a homogeneous cubic with all coefficients positive, the strict Lorentzian condition is that the Hessian of each first partial derivative has one positive and all remaining eigenvalues negative. This is the recursive definition of strict Lorentzianity in degree three [66]. Over a real closed field we can equivalently use the inertia of the associated symmetric bilinear forms.

Proposition 108.2 (Strict Lorentzianity over every real closed field; sources 10 and 09). *For every $t > 0$ in a real closed ordered field, F_t is strictly Lorentzian.*

Proof. Write J for the 7×7 all-ones matrix, I for the identity, and $L = I + tJ$. Let

$$M_k = \text{Hess}(\partial_k H).$$

The k th row and column of M_k are zero. On the other six indices, M_k is the adjacency matrix of the complete tripartite graph $K_{2,2,2}$: the three missing pairs are the pairs on the Fano lines through k . Counting the four bases through each pair gives

$$\sum_{k=1}^7 M_k = 4(J - I). \quad (108.9)$$

The chain rule therefore gives the exact congruence

$$\text{Hess}(\partial_k F_t) = L^T (M_k + 4t(J - I))L. \quad (108.10)$$

Since $\det L = 1 + 7t > 0$, it suffices to calculate the inertia of $N_k = M_k + 4t(J - I)$.

On the three independent directions antisymmetric within one of the three missing pairs, N_k has eigenvalue $-4t$. On the two-dimensional space of vectors constant on each missing pair, with total sum zero and zero k th coordinate, it has eigenvalue $-2 - 4t$. These account for five strictly negative directions.

The remaining plane has orthonormal basis

$$e_k, \quad \frac{1}{\sqrt{6}} \sum_{i \neq k} e_i.$$

The matrix on this plane is

$$\begin{pmatrix} 0 & 4\sqrt{6}t \\ 4\sqrt{6}t & 4 + 20t \end{pmatrix}, \quad \det = -96t^2 < 0. \quad (108.11)$$

It has one positive and one negative eigenvalue. Thus N_k has signature $(1, 6)$ and is nonsingular. Congruence in (108.10) preserves that signature, for every k . Together with the coefficient positivity in Table 12, this is strict Lorentzianity. $\square$

The characteristic polynomial of N_k , independently checked by the accompanying script, is

$$(z + 4t)^3(z + 2 + 4t)^2(z^2 - (4 + 20t)z - 96t^2). \quad (108.12)$$

No numerical eigenvalue approximation enters the proof.

The accompanying program checks both [equation \(108.10\)](#) and [equation \(108.12\)](#) symbolically for all seven indices. No numerical eigenvalue computation is used.

108.3 Why the leading matroid cannot be real

Lemma 108.3 (The Fano configuration is not real-representable; sources 10 and 09). *There are no seven real vectors whose independent triples are exactly the bases complementary to (108.1).*

Proof. Suppose a representation exists. The triple 123 is a basis, so after a linear change and nonzero rescalings of columns we may take $u_1 = e_1, u_2 = e_2, u_3 = e_3$. The lines 124 and 135, together with simplicity, allow further diagonal coordinate rescalings to arrange

$$u_4 = (1, 1, 0), \quad u_5 = (1, 0, 1).$$

The line 236 similarly allows $u_6 = (0, 1, \lambda)$ with $\lambda \neq 0$.

The lines 167, 257, and 347 force u_7 to lie simultaneously in

$$\{(a, b, \lambda b) : a, b \in \mathbb{R}\}, \quad \{(c, d, c) : c, d \in \mathbb{R}\}, \quad \{(e, e, f) : e, f \in \mathbb{R}\}.$$

It follows that its first and second coordinates agree, as do its first and third. Its second coordinate is nonzero, since otherwise u_7 would be zero or parallel to u_1 , contrary to the represented matroid. Hence $\lambda = 1$. But the remaining line 456 would require

$$0 = \det(u_4, u_5, u_6) = \det \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix} = -2,$$

a contradiction. □

Second proof (source 09), in source 09's labelling. Every pair is contained in a prescribed basis, so the vectors must be nonzero and pairwise nonparallel. Since 124 is a basis, use its vectors as coordinate axes. The lines 123 and 145, with separate nonzero rescalings of vectors and axes, allow the normalization

$$v_1 = (1, 0, 0), \quad v_2 = (0, 1, 0), \quad v_4 = (0, 0, 1), \quad v_3 = (1, 1, 0), \quad v_5 = (1, 0, 1).$$

The line 246 gives $v_6 = (0, 1, a)$ with $a \neq 0$ after rescaling. The line 257 forces the first and third coordinates of v_7 to agree; the line 347 forces its first and second coordinates to agree. Pairwise nonparallelness makes this common coordinate nonzero. Hence rescale to $v_7 = (1, 1, 1)$. The line 167 now gives $a = 1$. But then

$$\det(v_3, v_5, v_6) = \det \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix} = -2 \neq 0,$$

contradicting the line 356. □

This characteristic-two obstruction is classical. Its role here is to supply a transparent certificate for a forbidden *initial support*.

Theorem 108.4 (Full-support cubic obstruction; sources 10 and 09). *Let K satisfy (100.1), and let $t > 0$ have $v(t) = \gamma > 0$. The cubic F_t is strictly Lorentzian and every coefficient is positive. Nevertheless, no seven finite polytopes in K^3 have its mixed-volume valuation profile; [source 09] that is, there do not exist finite polytopes $P_1, \dots, P_7 \subseteq K^3$ satisfying*

$$v(c_\alpha(P)) = v([x^\alpha]F_t) \quad (\alpha \in \Delta_7^3). \quad (108.13)$$

More generally, no polynomial whose positive coefficients have the values in the last column of Table 12 is a K -polytope volume polynomial. [source 09] The obstruction persists after adding any affine function $\kappa + \alpha \cdot u$ to the displayed profile, with $\kappa \in \Gamma$ and $u \in \Gamma^7$.

Proof. Strict Lorentzianity is Proposition 108.2. The zero-weight initial support of the tabulated profile is precisely the set of twenty-eight Fano bases. Corollary 105.5 would force that matroid to be real-representable, contradicting Lemma 108.3. The argument depends only on the valuation table, not on any leading coefficient. $\square$

Second proof (source 09), with the affine-shift clause. Strict Lorentzianity and positivity follow from proposition 108.2 and lemma 108.1. By table 12, the least coefficient valuation is zero, attained exactly on the 28 Fano bases. If equation (108.13) held, corollary 105.8 with $w = 0$ would give a real representation of that squarefree minimum set. This contradicts lemma 108.3.

For the affine-shifted profile, choose weight $w = -u$. The minimum set is again exactly the Fano bases; the constant κ does not affect it. The same contradiction applies. $\square$

The last clause covers the natural valuation effects of independently rescaling the polytopes and applying a common invertible linear map. Thus the obstruction is not removed by selecting different units or by renormalizing the overall volume.

Taking $t = \omega^{-1}$ makes this an explicit surreal example. More generally, $t = \omega^{-\gamma}$ works for any positive surreal γ . The obstruction survives multiplication of each coefficient by an arbitrary positive valuation-zero element. By Proposition 105.9, it also survives an arbitrary common valuation shift and arbitrary positive rescaling of the variables.

108.4 All elementary tropical tests still pass

The order function in Table 12, divided by γ , is $3 - r_M(\text{supp } \alpha)$ and hence is M -convex by Proposition 107.1. In particular it obeys every tropical Alexandrov–Fenchel comparison

$$2q(\beta + e_i + e_j) \leq q(\beta + 2e_i) + q(\beta + 2e_j), \quad |\beta| = 1. \quad (108.14)$$

Its canonical polarization has $\binom{21}{3} = 1330$ entries, all finite. For a common index s and distinct indices $a, b, c, d \neq s$, the minimum of

$$p_{sab} + p_{scd}, \quad p_{sac} + p_{sbd}, \quad p_{sad} + p_{sbc}$$

is attained at least twice. The script verifies all $21 \binom{20}{4} = 101745$ such short tropical Plücker relations. Theorem 108.4 nevertheless rules out a determinant realization over K . These short relations describe exchange consistency, not all field-specific realizability constraints.

Corollary 108.5 (Discrete convexity is insufficient; source 09). *The function $h_t(\alpha) = v([x^\alpha]F_t)$ is a finite-valued Γ -valued M -convex function, but it is not a mixed-volume profile. In particular it satisfies the symmetric exchange inequalities*

$$h_t(\alpha) + h_t(\beta) \geq h_t(\alpha - e_i + e_j) + h_t(\beta + e_i - e_j) \quad (108.15)$$

for some j with $\alpha_j < \beta_j$, whenever $\alpha_i > \beta_i$.

Proof. First take t to be the standard positive infinitesimal of the real Puiseux field. [Proposition 108.2](#) and the Lorentzian tropicalization theorem [66, Theorem 3.20] imply M-convexity of the integer-valued function $h_t/v(t)$. [Table 12](#) shows that the same integer function occurs in every field in the statement, multiplied by $\gamma > 0$. Multiplication by a positive element of an ordered abelian group preserves all comparisons of integer multiples, so [equation \(108.15\)](#) follows in general. Non-liftability is [theorem 108.4](#). $\square$

There is also a compact expression for this integer function. If r_F is the binary Fano rank, then

$$h_t(\alpha) = \gamma(3 - r_F(\text{supp } \alpha)), \quad \alpha \in \Delta_7^3. \quad (108.16)$$

The 84 values give 7,056 ordered pairs and 12,047 excess-index exchange obligations. All are checked exactly by `code/09-finite-mixed-volume-models-verify.py`. This check is independent evidence for the finite exchange table, not a replacement for the preceding argument from the classical theorem.

Remark 108.6 (The role of the parameter). The strict Lorentzian statement holds for every ordinary real $t > 0$. The valuation obstruction proved here assumes that t is genuinely infinitesimal. We do not claim from this argument alone that every positive real specialization F_t is nonrealizable, and we do not supply a numerical real-parameter cutoff.

109 Valuations do not determine realizability of leading coefficients

There is a different obstruction even when the profile passes the matroid representability test. Let $x = (x_1, x_2, x_3, x_4)$ and $S = x_1 + x_2 + x_3 + x_4$. Define

$$G_t = e_2(x_1 + tS, \dots, x_4 + tS) = e_2(x) + (3t + 6t^2)S^2. \quad (109.1)$$

Its pure-square coefficients are $3t + 6t^2$, and its six mixed coefficients are $1 + 6t + 12t^2$. It is strictly Lorentzian for $t > 0$: its Hessian is congruent, by the invertible matrix $I + tJ$, to the Hessian $J - I$ of e_2 , which has signature $(1, 3)$.

The initial-support condition does not retain enough information to solve the full coefficient problem. A smaller example makes this distinction explicit.

[write] Source 09 writes $E = e_2(x_1, x_2, x_3, x_4)$, s and E_t for source 10's e_2 , S and G_t ; the two families coincide, $E_t = G_t$. Source 09's lemma is the stronger (it excludes every positive real multiple of e_2), so it is printed with both labels, and source 10's proof of the case $c = 1$ follows it. Source 09's theorem is printed as the merged result, because it adds "even up to a positive common scalar"; source 10's proof, with its different profile witness, is printed as a second proof.

109.1 The classical four-segment obstruction (source 09)

In four variables let

$$E(x) = e_2(x_1, x_2, x_3, x_4) = \sum_{1 \leq i < j \leq 4} x_i x_j.$$

Lemma 109.1 (The equal-area Plücker obstruction; sources 09 and 10). *No positive real multiple of E is the area polynomial of four real polytopes in $\mathbb{R}^2$.*

Proof. If $F_Q = cE$ for $c > 0$, the squarefree support and [theorem 105.7](#) imply that each Q_i has dimension one; each is a nondegenerate segment, since all cross coefficients are positive.

Translate them to $[0, v_i]$. By [equation \(100.6\)](#),

$$|\det(v_i, v_j)| = c \quad (i < j).$$

Let $p_{ij} = \det(v_i, v_j)$. The 2×4 minors satisfy

$$p_{12}p_{34} - p_{13}p_{24} + p_{14}p_{23} = 0.$$

After division by c^2 , each of the three summands is $+1$ or -1 . An odd sum of such numbers cannot be zero over $\mathbb{R}$, a contradiction. $\square$

This is the real example in [66, Remark 4.3]; its elementary proof is reproduced to isolate the coefficient obstruction from representability of the uniform rank-two support.

Second proof (source 10) of the case $c = 1$. Suppose it were. Each pure-square coefficient is zero, so every body has area zero and lies on a line. No body is a point, since all six mixed coefficients are positive. Up to translation the bodies are therefore nonzero segments $[0, u_i]$. [Proposition 102.1](#) gives $|\det(u_i, u_j)| = 1$ for all $i < j$.

Set $p_{ij} = \det(u_i, u_j)$. The two-row Plücker identity is

$$p_{12}p_{34} - p_{13}p_{24} + p_{14}p_{23} = 0.$$

Every product is 1 or -1 , so the left side is a sum of three numbers from $\{1, -1\}$ and cannot be zero. This contradiction proves the claim. $\square$

109.2 A positive infinitesimal family and a genuine profile witness (source 09)

Let now $s = x_1 + \dots + x_4$ and define

$$E_t(x) = E(x_1 + ts, \dots, x_4 + ts) = E(x) + (3t + 6t^2)s^2. \quad (109.2)$$

The square coefficient is $3t + 6t^2$, and every cross coefficient is $1 + 6t + 12t^2$.

Theorem 109.2 (Valuation realizability does not determine initial coefficients; sources 09 and 10). *For a positive infinitesimal $t \in K$, the polynomial E_t is strictly Lorentzian and has positive coefficients. It is not the volume polynomial of four finite K -polytopes in K^2 , even up to a positive common scalar. Nevertheless its valuation profile is realized by four explicitly given full-dimensional parallelograms.*

Proof. The Hessian of E is $J_4 - I_4$, of signature $(1, 3)$. With $A = I_4 + tJ_4$,

$$\text{Hess}(E_t) = A^\top (J_4 - I_4) A.$$

The matrix A is invertible for $t > 0$, so strict Lorentzianity follows. All coefficients in [equation \(109.2\)](#) are positive.

Suppose first that $E_t = F_P$. The least coefficient valuation is zero. By [theorem 104.4](#) with all $\lambda_i = 1$, the positive normalizing scalar D has valuation zero, hence positive real standard part. Reducing the exact normalized polynomial gives

$$F_{\bar{Q}} = \frac{1}{\text{st}(D)} E,$$

contradicting [lemma 109.1](#). If a positive scalar multiple of E_t were a volume polynomial, a common homothety, using a positive square root in K , would make E_t itself one. Thus scalar multiplication does not help.

To realize just the profile, take, for $i = 1, 2, 3, 4$,

$$u_i = (1, i - 1), \quad v = (0, 1), \quad P_i = [0, u_i] + [0, tv].$$

These parallelograms are full-dimensional, and the zonotope formula gives

$$F_P(x) = t \sum_{i=1}^4 x_i^2 + \sum_{1 \leq i < j \leq 4} (j - i + 2t)x_i x_j. \quad (109.3)$$

Both equations (109.2) and (109.3) have square valuations $v(t)$ and cross valuations zero. Their profiles agree, while their cross leading coefficients differ. The positive real numbers $j - i$ in the witness satisfy the signed-minor compatibility that the requested equal coefficients do not. $\square$

Second proof of the non-realizability, with a second profile witness (source 10). The profile is zero at $e_i + e_j$ for $i \neq j$ and γ at $2e_i$. This is $\gamma q_{U_{2,4}}$ for the real-representable uniform matroid of rank two on four elements. For example, it is represented by the columns of

$$U = \begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 2 \end{pmatrix},$$

whose six two-by-two minors are all nonzero. Theorem 106.1 realizes the profile, explicitly using $P_i = [0, u_i] + t[0, 1]^2$.

On the other hand, all coefficients of G_t belong to $\mathcal{O}$ and its coefficientwise standard part is $e_2(x)$. If G_t were a polytope volume polynomial, Theorem 104.2 with $s_i = a = 1$ would realize e_2 by real planar polytopes, contrary to Lemma 109.1. $\square$

Thus three levels must be kept separate: a profile satisfying tropical exchange, a profile admitting actual determinant realization, and a particular positive coefficient vector admitting polytope realization. The Fano cubic separates the first two. The quadratic G_t separates the last two. The underlying undeformed obstructions are classical [66, Remark 4.3]; the finite normalization theorem explains their persistence as infinitesimal, full-support phenomena.

Because this profile really is a polytope profile, all of its weighted initial supports satisfy corollary 105.8. Thus even passing every initial-support test is not enough to realize prescribed initial coefficients. This observation does not decide whether every profile whose initial supports are representable has some coefficient lift; that separate question is left open in section 114.

110 The rank-two boundary: a finite realization algorithm

The obstruction in degree three should be contrasted with degree two. The underlying rank-two tropical Grassmannian fact is classical [72]. We give a direct construction because it works in an arbitrary ordered value group and uses only finitely many field operations and real integer coefficients.

In degree two, the additional realization requirement can always be met. The key construction is the familiar rank-two tree or ultrametric realization behind the tropical Grassmannian [72]. We give a finite version that works over the given field K and an arbitrary value group, without a series cross-section.

Lemma 110.1 (Realizing finite rank-two determinant values; sources 10 and 09). *Let $n \geq 2$, and let $p_{ij} \in \Gamma$ be specified for every $0 \leq i < j < n$. There exist columns $u_i \in K^2$ with $v \det(u_i, u_j) = p_{ij}$ if and only if, for every four distinct indices, the minimum of the three pair-partition sums*

$$p_{ij} + p_{kl}, \quad p_{ik} + p_{jl}, \quad p_{il} + p_{jk} \quad (110.1)$$

is attained at least twice.

Proof. Necessity follows by applying the valuation to the Plücker identity: a sum equal to zero cannot have a unique term of smallest valuation. For sufficiency, choose positive $b_i \in K$ with $v(b_i) = p_{0i}$, for $1 \leq i < n$, and put

$$\lambda_{ij} = p_{ij} - p_{0i} - p_{0j} \quad (1 \leq i < j < n).$$

Using index 0 in (110.1) shows that the minimum of $\lambda_{ij}, \lambda_{ik}, \lambda_{jk}$ is attained at least twice. In particular, $\lambda_{ik} \geq \min(\lambda_{ij}, \lambda_{jk})$.

List the finitely many distinct values as $\ell_1 < \dots < \ell_s$. For each ℓ_r , the relation

$$i \sim_r j \iff i = j \text{ or } \lambda_{ij} > \ell_r$$

is an equivalence relation. Transitivity is exactly the last inequality. Assign distinct integer digits to its different classes, writing $c_r(i)$ for the digit of the class of i . Choose positive $a_r \in K$ with $v(a_r) = \ell_r$ and define the finite sums

$$z_i = \sum_{r=1}^s c_r(i) a_r.$$

If $\lambda_{ij} = \ell_r$, all earlier digits agree and the digits at level r differ. Their difference is a nonzero integer, so

$$v(z_j - z_i) = \ell_r = \lambda_{ij}.$$

Terms from later levels have strictly greater value and cannot cancel the leading term.

Take $u_0 = (0, 1)$ and $u_i = (b_i, b_i z_i)$ for $i \geq 1$. Then

$$v \det(u_0, u_i) = v(-b_i) = p_{0i}, \quad v \det(u_i, u_j) = v(b_i b_j (z_j - z_i)) = p_{ij}.$$

For $n = 2$ there are no λ_{ij} ; use $z_1 = 0$ and the same formula. □

[write] Source 09 proves the sufficiency half (its Lemma “Rank-two realization by finite sums”, with indices $1, \dots, n$ and values w_{ij}) by the same ultrametric construction, and adds:

The use of different integer labels is legitimate because the natural valuation is trivial on nonzero real integers. No assertion about a multiplicative section $\Gamma \rightarrow K^\times$ enters the construction.

Theorem 110.2 (Full-support planar profile criterion; sources 10 and 09). *Let $q : \Delta_m^2 \rightarrow \Gamma$ be finite everywhere. It is the mixed-area valuation profile of m full-dimensional parallelograms if and only if $\text{Pol}(q)$ satisfies the short tropical Plücker conditions (110.1). Equivalently, every full-support M -convex degree-two profile is realizable. [source 09] For any finite-valued function $h : \Delta_m^2 \rightarrow \Gamma$, the following are equivalent: h is M -convex; h is the area-valuation profile of m K -polytopes; and h is the area-valuation profile of m full-dimensional parallelograms. Each parallelogram needs only two segment generators, and no extension of K is necessary.*

Proof. Apply Lemma 110.1 with $n = 2m$ and then Theorem 103.3. Each two-column color block has nonzero determinant because $q(2e_i)$ is finite, so it gives a genuine parallelogram.

For completeness, the exchange equivalence is valid in an arbitrary ordered value group. Let S, T be polarized two-element sets and $a \in S \setminus T$ have color i . If $T \setminus S$ contains a label of the same color, exchanging those labels preserves both color counts. Otherwise color i is in excess in $c(S)$, and M-convex exchange chooses a deficit color j . Some label of color j belongs to $T \setminus S$, giving the required polarized exchange inequality.

Conversely, to compare color counts α, β , realize them by taking the first α_i and first β_i labels of each color. The two label sets are nested within each color. Any exchange of an excess-color label with a label in $T \setminus S$ therefore uses a deficit color and yields the M-convex inequality. For two-element sets, polarized exchange is equivalent to (110.1): the only nontrivial case has four distinct labels, and each of the three pair-partition sums must be at least one of the other two. This is exactly the condition that the minimum is not unique. $\square$

Second proof (source 09), by the four-point cases. A polytope profile is M-convex by theorem 38.4. It remains to construct the parallelograms from an M-convex h . Give each color i two copies, and prescribe the value $h(e_i + e_j)$ on any pair of different copies of colors i, j ; for the two copies of the same color the value is $h(2e_i)$. We verify the four-point condition of lemma 110.1.

For four different colors, the three required pair sums are the three possible partitions of four indices into pairs. M-convexity applied to each partition says that its sum is at least the minimum of the other two; hence the minimum cannot be unique. For colors of multiplicities $(2, 1, 1)$, the condition reduces to

$$h(2e_i) + h(e_j + e_k) \geq h(e_i + e_j) + h(e_i + e_k),$$

which is exactly an M-convex exchange. For multiplicities $(2, 2)$, it reduces to

$$h(2e_i) + h(2e_j) \geq 2h(e_i + e_j),$$

another such exchange. There are no other cases, since each color has only two copies.

Lemma 110.1 therefore realizes all these pair values by a $2 \times 2m$ matrix. The two columns of every color are independent, because their prescribed determinant value is finite. Their symmetric segment sum is a full-dimensional parallelogram. By the determinant formula, its colored area coefficients have exactly the valuations h . $\square$

The finiteness hypothesis on all values is part of the theorem as proved here. A treatment of zeros would require a separate handling of loops and parallel classes in the rank-two table; it is not silently included in the ultrametric construction.

Only representatives of input values are needed: if $r_{ij} > 0$ has value p_{ij} , one may use $b_i = r_{0i}$ and, for a pair attaining a level, $a_r = r_{ij}/(b_i b_j)$. Thus no globally defined cross-section of the value group and no infinite Hahn expansion is required.

110.1 The degree boundary (source 09)

In degree one, every finite profile is realized by intervals: choose positive lengths with the prescribed valuations. Theorem 110.2 settles degree two for every number of colors. The cubic construction in theorem 108.4 will show that, over every non-Archimedean K under consideration, degree three is the first degree at which a finite-valued M-convex profile can fail polytope realization. This is a degree boundary, not a claim that seven is the smallest possible number of cubic variables.

The proof also explains why the quadratic coefficient counterexample in [section 109](#) must fail through its coefficients rather than through its valuation profile. The classical real coefficient classification and the valuation-profile classification have different answers.

Together with the immediate interval construction in dimension one, Theorems [110.2](#) and [108.4](#) show that dimension three is the first dimension in which full-support M-convex profiles can fail mixed-volume realization. This is a statement about dimensions, not about the smallest possible number of bodies in dimension three. The amplitude obstruction of [Section 109](#) already occurs in dimension two and does not contradict this result.

[write] Both sources' "first degree" and "first dimension" statements (source 09's abstract, introduction and the paragraph above; source 10's abstract and the paragraph above) concern finite-valued (equivalently full-support) profiles: full support means that every coefficient is nonzero, that is, that every value is finite. Profiles with zero coefficients are not covered, as source 09 states after [theorem 110.2](#). The one-dimensional case is the interval construction; the two-dimensional case is [theorem 110.2](#); the three-dimensional failure is [theorem 108.4](#).

111 Examples, finite algorithms, and information loss (source 09)

111.1 Two rectangles at unrelated infinitesimal scales

Let $u, v > 0$ be infinitesimals; their values need not be rationally or Archimedean comparable. Put

$$P_1 = [0, 1] \times [0, u], \quad P_2 = [0, v] \times [0, 1].$$

Then

$$F_P(x, y) = (x + vy)(ux + y) = ux^2 + (1 + uv)xy + vy^2. \quad (111.1)$$

Its profile is

$$h(2, 0) = v(u), \quad h(1, 1) = 0, \quad h(0, 2) = v(v).$$

At weight zero the real shadows are a horizontal and a vertical segment, with area polynomial xy . At a weight (a, b) , compare the three values

$$v(u) + 2a, \quad a + b, \quad v(v) + 2b.$$

Theorems [104.4](#) and [105.7](#) explain every tie pattern by a real subspace arrangement, without selecting a real logarithmic scale for u and v .

111.2 Coordinate valuations do not determine determinant valuations

Let $0 < \eta \in \mathfrak{m}$ and consider

$$a = (1, 1), \quad b = (1, 1 + \eta).$$

Every displayed coordinate has valuation zero, while

$$\det(a, b) = \eta.$$

Replacing η by another infinitesimal of a different valuation leaves the coordinate valuation matrix unchanged but alters the area of the parallelotope. For example, in **No** one may compare $\eta = \omega^{-1}$ and $\eta = \omega^{-\omega}$; the latter is smaller than every positive integral power of ω^{-1} . The determinant minimum in [equation \(99.4\)](#) therefore requires exact determinant valuations, including cancellation within a determinant.

There is no contradiction with [lemma 33.2](#). That lemma is applied *after* the determinant has been computed and its absolute value taken, so cancellation between different minors is absent.

111.3 A certificate-oriented algorithm

For a finite representation admitting exact arithmetic, order tests, and valuation tests, the proofs give the following workflow.

Stage 1: per-polytope compression. For each P_i , select an anchor and a row projection of rank r_i . Enumerate the r_i -column minors of the projected difference matrix and choose a maximum absolute nonzero minor. Record the chosen columns and all Cramer coordinates. The inequalities $|a_{kj}| \leq 1$ together with the explicit simplex centroid certify the two inclusions in [equation \(101.4\)](#).

For profiles alone one may instead minimize the *valuation* of the minor. The Cramer coordinates are then finite, although not necessarily bounded by one. A finite common real bound replaces the outer factor 2; the lower simplex inclusion is unchanged. This still proves equality of valuations. Thus an implementation concerned only with profiles can use valuation comparisons for basis selection.

Stage 2: colored determinant table. Assemble at most dm selected columns. Enumerate its maximal minors, compute the color count of each, and retain the minimum determinant valuation in each count class. This produces every $h_P(\alpha)$ by [remark 102.3](#). A straightforward arithmetic-operation bound is the sum of the per-polytope minor enumerations and $O(\binom{dm}{d}d^3)$ operations for the final determinants, before the costs of field arithmetic and sign or valuation decisions are included. No polynomial-time claim in variable d is made.

Stage 3: a chosen leading model. For scales λ_i , return to the *original* vertices. Choose one valuation-minimal global determinant basis, apply its inverse, and reduce the finite coordinates to $\mathbb{R}$. This produces actual initial coefficients, not just a support. Using the compressed parallelotopes here would in general change those coefficients and would be incorrect for this task.

Stage 4: obstruction extraction. Given proposed valuation data, find a weight at which the minimum set is small enough to analyze. Recover the forced rank function by [equation \(105.7\)](#). A nonrepresentable squarefree minimum set certifies non-liftability. With prescribed initial coefficients, test the stronger real volume-polynomial realization problem instead. The Fano and equal-area examples give exact certificates of these two types.

111.4 What remains invisible to the profile

The map $P \mapsto h_P$ deliberately forgets positive real factors, lower terms within coefficients, and most of the ordinary combinatorics. Even when h_P determines a useful family of initial subspace arrangements, it does not reconstruct the original polytopes. Conversely, the face lattices of all P_i do not determine h_P : [equation \(111.1\)](#) varies its profile while each P_i remains a quadrilateral. The determinant model is an exact finite model of this one invariant, not a universal compressed encoding of surreal convex geometry.

112 Certificates, computation, and a formalization route

112.1 A finite profile-extraction procedure

Given exact vertex coordinates and exact access to field order and valuation, the proofs yield the following finite procedure.

First compute each affine dimension and select a maximal vertex simplex in an affine chart. Enumeration is sufficient; no numerical optimization is needed. Record the barycentric coordinates proving the two containments in (101.1). This is the compression certificate.

Next concatenate the at most md generators and evaluate its maximal minors. For every color count, record the least determinant valuation. There are at most $\binom{md}{d}$ determinants, independently of the original vertex counts. The reduction in description size is genuine, but this bound is exponential when dimensions vary and is not a polynomial-time claim in arbitrary dimension.

For an exposed initial set, rescale the color blocks, select a minimum-valued basis D , and form $D^{-1}A$. Its entries have nonnegative value by Cramer's rule. Their residues, with a rank table for the color spans, certify the representable initial polymatroid. Keeping the original vertices under the same normalization gives the stronger initial-polynomial certificate.

To produce a symmetric determinant realization, generic block mixing requires finitely many nonvanishing residue conditions. An explicit matrix of real mixing parameters together with its nonzero reduced polynomial values is a certificate. Searching for one is finite in principle after the relevant residue polynomials are known; the existence proof is not an algorithm for computing standard parts of arbitrary symbolic surreal expressions.

112.2 What can actually be checked

A valuation comparison can be certified by leading terms or, for comparison with a fixed reference magnitude $a > 0$, by explicit positive real constants c, C satisfying $ca \leq |b| \leq Ca$. A vanishing determinant needs an exact algebraic certificate, not a floating-point tolerance. Finite formulas do not make equality of unrestricted surreal expressions decidable.

For nonrealizability, a small initial obstruction can be much shorter than the complete positive data. In the Fano example there are 84 coefficients and 1330 polarized values, but the contradiction reduces to seven prescribed dependent triples, enough independence to normalize three columns, and the final equation $-2 = 0$. Passing 101745 tropical relations does not weaken that contradiction.

112.3 Exact checks supplied with this manuscript

The companion code/10-tropical-mixed-volumes-verify.py was run with SymPy 1.14.0 and performs only exact symbolic or integer computations. It checks the Fano incidences and basis count, expands (108.3), verifies every coefficient in Table 12, checks every exchange obligation (107.3), all cubic comparisons (108.14), all short polarized Plücker relations, the seven Hessian congruences, the characteristic polynomial (108.12), the contradiction determinant, and the quadratic identity (109.1).

The script also applies the rank-two construction to the polarized $U_{2,4}$ profile. Its resulting 2×8 matrix is saved in data/10-tropical-mixed-volumes-rank_two_example.txt. All 28 determinant orders agree with the input profile. The complete 84-row coefficient table is saved in data/10-tropical-mixed-volumes-fano_coefficients.csv, and the exact

Table 13: Executed exact verification. The general proofs do not depend on enumeration of test cases.

Check	Number
Fano coefficient identities	84
M-convex exchange obligations	12047
Cubic tropical Alexandrov–Fenchel comparisons	147
Polarized maximal-minor values	1330
Short tropical Plücker relations	101745
Symbolic first-derivative Hessian identities	7
Rank-two reconstruction determinant orders	28

verification summary in `data/10-tropical-mixed-volumes-verification.json`. These are reproducibility checks, not a substitute for the proofs for arbitrary K, m, d .

112.4 How this could be formalized in ProveIt

[write] This subsection is a proposal. The declarations it names were re-verified in the current ProveIt tree: `valuation`, `valuation_mul`, `valuation_antitone_nonneg`, `tMonomial` and `valuation_tMonomial` in `Algebra/SurrealNumbers/Surreal/Foundations/SignSequenceValuation.lean`; `standardPartHom`, `standardPartHom_surjective` (its surjectivity), `mem_ker_standardPartHom` (its kernel) and `standardPartQuotientEquiv` in `SignSequenceStandardPart.lean` of the same directory. They are inputs that the source cites. No statement of this Part is formalized (nor is any statement of this report; [section 83](#)).

A useful implementation should separate scalar algebra from polytope infrastructure. The inspected scalar sources provide `valuation`, `valuation_mul`, `valuation_antitone_nonneg`, `tMonomial`, and `valuation_tMonomial` [73]. The companion standard-part file defines `standardPartHom`, proves its surjectivity, identifies its kernel, and constructs `standardPartQuotientEquiv` [74]. Finiteness hypotheses on standard-part algebra must be retained: the total standard-part function on infinite inputs is not a field homomorphism.

The least costly first target is generic valued linear algebra: minimum-minor normalization, Cramer’s-rule bounds, reduction of maximal minors, and the finite rank-two reconstruction. These results can be formulated over a valued ordered field with a specified real residue map and only later instantiated on surreal scalars.

A second target is the finite maximal-simplex lemma and the parallelotope containments. A third is the determinant coefficient formula and block-mixing argument. Those three layers isolate most of the new reasoning from the larger task of constructing finite mixed-volume theory over general real closed fields.

The initial-polynomial theorem requires the most geometric work: finite triangulations, volume invariance under triangulation, and the standard-part argument in [Proposition 34.3](#). Merely assuming a mixed-volume interface as axioms would yield a conditional theorem; it would not by itself constitute a complete formalization of the surreal-polytope result. No such complete formalization is claimed in this package.

113 Repository connection and reproducible verification (source 09)

113.1 Pinned context and dependency boundary

The repository snapshot used for the final source check is

b2b626d8537f7dea8f20393ee4714c7201a7001e.

The following sources were read as mathematical context:

Repository source	Relevant content inspected
Algebra/SurrealNumbers/docs/README.md	Collection status, separation of manuscript arguments, finite checks, and Lean coverage.
Algebra/SurrealNumbers/docs/surreal/euclidean-three-space/article.tex	Abstract and opening scope discussion: set-sized real closed fields, finite coordinate geometry, natural valuation, and standard part.

The README was read through its introductory and reading-route sections, and its opening status statement was reread at the pinned commit. The Euclidean report's lines 80–245 were read at that commit. These are *limited source reads*, not an exhaustive audit of the repository's large research collection [76, 77].

The present proofs do not assume any of the Euclidean report's unreviewed claims about rotation groups, analytic trigonometry, or class-sized quotients. They build on its scalar discipline and geometric theme, but the required ordered-field, polytope, and linear-algebra lemmas are stated here with their own arguments or classical references. This independence matters because research prose, executable tests, and checked formal proofs are different forms of evidence.

[write] Re-verified: both files named above are unchanged between b2b626d85 and the current ProveIt tree (same blob identifiers). The Euclidean report is *Euclidean Three-Space over the Surreal Numbers*, a neighbouring report of this collection; nothing of it is imported here.

113.2 Contents and limits of the computational companion

The file `code/09-finite-mixed-volume-models-verify.py` uses exact SymPy arithmetic and writes `data/09-finite-mixed-volume-models-verification_results.json`. The recorded run verifies:

Finite check	Size
Fano coefficient table, including the rank-deficiency formula	84 monomials
Ordered pairs in the exchange enumeration	7,056
Excess-index M-exchange obligations	12,047
Symbolic Hessian congruences and characteristic polynomials	7 indices
Rational maximum-minor sandwich examples	23 configurations
Quadratic reconstruction tests	15 profiles
Reconstructed rank-two maximal-minor valuations	285 minors
Quadratic profile comparison	10 coefficients

The rational tests include point, line, and plane degeneracies as well as full-dimensional random integer configurations, with fixed seed 20260930. The program verifies the inner

inclusion at all cube corners and checks every Cramer coordinate for the outer inclusion. The reconstruction tests cover one through five colors, starting with a Laurent-polynomial profile, forgetting its original vectors, and rebuilding the entire polarized minor table by the ultrametric formula. They check all 285 prescribed pair-minor valuations exactly. It also checks the determinant -2 in the Fano contradiction and the possible signed sums in the quadratic Plücker obstruction.

These computations can expose transcription mistakes in the examples. They do not establish mixed-volume monotonicity, real-closed-field transfer, or the arbitrary-rank standard-part theorem. No symbolic check should be described as a Lean proof, and this package contains no claimed Lean formalization.

[write] The program and its recorded run are named above by their shipped paths; source 09 calls them `verify.py` and `verification_results.json`.

113.3 A feasible formalization decomposition

[write] This subsection is a proposal; it names no Lean declaration. No statement of this Part is formalized.

A repository implementation can avoid formalizing the full surreal class at the outset. The natural interfaces are finite polytopes over a linearly ordered field, a finite-dimensional determinant API, and a convex valuation ring with a residue map. One first proves Cramer's coordinate bound and the explicit centroid sandwich. Mixed-volume monotonicity and the segment determinant formula then yield the profile theorem. The initial-model proof needs the bounded-volume residue lemma and a common determinant basis. The final obstruction is a finite rank argument and an explicit 3×3 determinant computation.

The two most substantial dependencies are therefore a suitable ordered-field mixed-volume library and a residue compatibility theorem. The finite Fano arithmetic is not the difficult part. Module names or repository declarations are not proposed as if they had already been implemented or typechecked.

114 Further research questions

[write] Source 10 proposes ten questions and source 09 ten ("Further research questions and priorities"). Seven pairs ask the same question and are printed once, under source 10's title, with source 09's formulation after source 10's discussion; source 09's three remaining questions follow. The two introductions read:

The following are proposed research directions arising from the proved statements. They are not all asserted to be previously published open conjectures, and their literature status requires separate investigation.

[source 09] The following questions are proposed directions, not claims that the corresponding statements are absent from the literature. Each is distinct from a theorem proved above.

Question 114.1 (Effective equations for the realization locus; sources 10 and 09). For fixed (m, d) , describe the color-symmetric, real-valued Grassmannian locus in Theorem 103.3 by usable finite conditions on q .

The short Plücker conditions suffice for $d = 2$ but fail for $d = 3$. A concrete next case is to determine additional conditions for small m in dimension three, distinguishing equations from real-field constraints. A useful output would be a finite obstruction library or a certified

decision procedure for rational-valued inputs, rather than an unqualified appeal to tropical exchange.

[source 09] *An intrinsic characterization of profiles.* For fixed d, m , eliminate the matrix witness in [remark 102.3](#). Which finite relations on $h : \Delta_m^d \rightarrow \Gamma \cup \{+\infty\}$ characterize those functions whose polarized tables are valuations of realizable maximal minors? [Theorem 110.2](#) settles the finite-valued case in degree two. Targets beyond that result include sparse quadratic profiles and cubic profiles with few colors. Any proposed general criterion must reject [table 12](#) in dimension three, even though its exchange inequalities hold.

Question 114.2 (Compatibility between individually realizable cells; sources 10 and 09). What compatibility conditions between exposed initial configurations are missing from the requirement that each $\mathcal{J}_w(q)$ separately be real-representable?

[Theorem 105.4](#) is only necessary; no sufficiency has been proved. One should either construct a small color-symmetric profile whose cells are individually realizable but do not lift together, or prove sufficiency in a carefully delimited family. The matrices on neighboring cells must come from one valued configuration, which is stronger than unrelated real realizations.

[source 09] *Are all representable initial supports sufficient?* Suppose a finite-valued M-convex profile has a real-linearly-representable minimum set at every weight. Must it have a polytope realization over an appropriate real closed valued extension of $\mathbb{R}$? Or can the real representations at different weights fail to glue to a common valued matrix? The quadratic example does not refute this statement: its *profile* is realizable, although one prescribed coefficient lift is not.

Question 114.3 (The smallest number of bodies for a valuation gap). What is the least m for which a finite M-convex profile on Δ_m^3 fails realization by surreal three-dimensional polytopes?

The present article gives $m \leq 7$. It does not prove that seven is optimal. Since the polarization has $3m$ elements, obstructions might arise from representable-polymatroid constraints even before a seven-color squarefree Fano cell appears. An exact search should test polymatroid cells as well as matroid cells.

Question 114.4 (Leading coefficients above a realizable profile; sources 10 and 09). For a fixed realizable q , characterize the positive leading coefficient data that lift to a polytope volume polynomial.

The quadratic example shows that the profile alone is insufficient. [Theorem 104.2](#) supplies real volume-polynomial constraints on every exposed initial form. Determining how these constraints fit together may produce a finer realization object that retains both scale and amplitude. Even the uniform planar profile with four colors gives a nontrivial test case.

[source 09] *Leading coefficients and compatibility across scales.* Given a realizable profile together with proposed initial real volume polynomials at finitely many weights, determine when they arise from one collection over K . Positive scalar and diagonal-variable ambiguities must be quotiented out. The Plücker obstruction suggests that keeping only initial supports is too weak, while keeping every individual real model without transition data may also be insufficient.

Question 114.5 (Quantitative real specializations; sources 10 and 09). For the explicit families F_t and G_t , determine concrete intervals of ordinary real $t > 0$ on which they do or do not admit polytope realizations.

The infinitesimal proofs do not provide a numerical threshold. One possible route is to make the determinant normalization quantitative and derive a stable inequality excluding

configurations whose leading incidences approximate the forbidden ones. Another is semialgebraic elimination after a suitable bounded representation theorem for *coefficients*; valuation compression alone does not preserve those coefficients.

[source 09] *Quantitative real separation*. Turn the infinitesimal Fano and quadratic obstructions into explicit inequalities with rational constants separating a small real neighborhood of the forbidden leading polynomial from all real convex-body volume polynomials. Classical compactness already gives qualitative separation in the examples [66]; the target here is an explicit, verifiable bound or algebraic certificate, not a new qualitative existence claim.

Question 114.6 (Effective representatives and degree bounds). For rational-valued realizable profiles, what bounds can be placed on the algebraic or Puiseux complexity of a realizing matrix?

The rank-two proof uses finite sums of representatives and is explicit. In higher rank, the exact criterion leaves a genuine lifting problem. Useful bounds would control the number of terms, ramification, algebraic degree, and coefficient heights. Such bounds must be stated for an explicit input representation and must not assume that arbitrary surreal arithmetic is computable.

Question 114.7 (Optimal simultaneous compression; sources 10 and 09). Can the constants in (101.2) be improved while retaining the same finite, field-uniform proof, or can fewer directions suffice when only selected mixed coefficients are required?

A full-dimensional parallelotope already needs d independent directions, so this number is unavoidable for that output class. The interesting refinements concern the comparison constants, shared directions between bodies, selected coefficients, and intrinsic-volume variants. Improvements to constants do not change valuations, but may matter for amplitude estimates and real-specialization thresholds.

[source 09] *Optimal certificates beyond the generator bound*. The dm generator bound is sharp, but many profiles may admit smaller certificates once zero coefficients, shared subspaces, symmetries, or repeated columns are exploited. Determine the minimal colored witness size and the complexity of deciding it. Distinguish the number of columns from the description complexity of their valued entries.

Question 114.8 (Higher-rank iterated residue geometry; sources 10 and 09). Can one organize successive coarsenings of the natural valuation into compatible flags of real-representable initial polymatroids?

Our arguments work at arbitrary rank, but expose one initial scale at a time. A finite collection of values may encode several ordered scales whose successive residue fields are not all $\mathbb{R}$. A careful flag construction should specify those fields and compatibility maps, and avoid replacing higher-rank data by an unjustified single real parameter.

[source 09] *Dependence on the residue field*. Generalize the constructive initial model to real closed residue fields other than $\mathbb{R}$ and to convex coarsenings of the natural valuation. A proof should state explicitly whether it assumes a coefficient-field section. The present argument uses the distinguished embedded copy of $\mathbb{R}$ and its standard-part map; it does not silently supply such a section in every valued field.

Question 114.9 (Relation to the absolute Grassmannian). How much of the numerical mixed-volume information lost under valuation can be recovered from residues of absolute Plücker coordinates and their positive colorwise sums?

Theorem 103.3 gives an exact order-level bridge. A more refined theory could retain leading coefficients of the positive sums and distinguish ordinary, absolute, positive, and real tropical Grassmannian constraints. This would connect directly to the absolute-Grassmannian program of [65], without confusing valuation equality with equality of mixed volumes.

Question 114.10 (A verified obstruction checker; sources 10 and 09). Can ProveIt verify a complete chain from vertex data or a candidate profile to a residue-polymatroid certificate and a small obstruction?

A realistic first milestone is a verified Fano contradiction and a verified rank-two constructor over a generic valued-field interface. The more substantial milestone is instantiating the finite mixed-volume and standard-part layers on the repository’s actual surreal scalars. The executable checks supplied here can guide test cases, but should not be mistaken for that formal proof.

[source 09] *A formal proof and certificate checker.* Implement the maximum-minor sandwich and the finite Fano obstruction first, then formalize the mixed-volume and standard-part interfaces needed by the main theorems. A small checker accepting selected bases, Cramer coordinates, and determinant valuations could be useful even before a general search procedure exists. Its trust boundary should include the field representation and valuation oracle explicitly.

Question 114.11 (Exact algorithms for Hahn inputs; source 09). For finitely presented Hahn or generalized Puiseux entries, design certified algorithms for the determinant valuations required in the three stages of section 111. Give bounds on how much support must be inspected when cancellation occurs. An oracle for leading terms of individual entries alone cannot suffice, as the two-vector example shows.

Question 114.12 (Simultaneous geometry and valuation data; source 09). Which pairs consisting of prescribed face lattices and a prescribed mixed-volume profile admit a common realization? The parallelotope compression answers the valuation problem by changing face lattices. A simultaneous realization problem could detect information not visible to either invariant separately, including interactions with realization spaces and extension complexity.

Question 114.13 (Finite versus general convex objects; source 09). Which conclusions survive for definable convex sets, or for suitably controlled non-polyhedral convex bodies over a real closed valued field? A legitimate extension must define the relevant volume and prove its residue compatibility. Classical integration over $\mathbb{R}$ or a finite polytope proof does not automatically give integration over arbitrary surreal convex classes.

The highest-priority mathematical target is the gluing problem in question 114.2 [[write] source 09’s second question]; the highest-priority computational target is a certified finite-support determinant-valuation engine. The projective volume-polynomial questions discussed in [68] remain separate. A failure of convex polytope realization, even over surreal scalars, supplies no automatic obstruction for arbitrary projective intersection polynomials.

115 Conclusions

115.1 Conclusion of source 10

Mixed-volume valuations of finite surreal polytopes admit a finite linear-algebraic model of uniformly bounded size: d directions per body suffice. Canonical polarization converts their realization into an exact maximal-minor valuation problem, while minimum-basis normalization exposes real polytope geometry in every initial form.

This gives a precise negative answer to a natural sufficiency question: full support and tropical exchange do not characterize realizable mixed-volume profiles. The failure begins in dimension three and can be witnessed by a strictly Lorentzian cubic with only the three valuation levels $0, \gamma, 2\gamma$. In dimension two, those tropical conditions do suffice for the profile, but leading coefficients can still obstruct realization.

The substantive role of surreal numbers here is not a new finite face lattice. It is the exact encoding of multiple scales and the ability to expose algebraic representability constraints at those scales. The next step is to turn the determinant criterion and its residue obstructions into effective, formally verified tools, while keeping valuation, amplitude, and full geometric realization distinct.

115.2 Conclusion of source 09

Finite surreal polytope geometry has a strong algebraic compression at the level of mixed-volume scales. A dimension- d object with arbitrarily many vertices can be replaced, for this invariant, by a parallelotope with at most d generators. Across m polytopes the complete profile is a colored maximal-minor minimum on at most dm columns, and the bound cannot be improved in general within segment-sum witnesses.

Generic integer mixing strengthens this to an exact polarized minor-table criterion. In degree two, an explicit finite ultrametric construction realizes every finite-valued M-convex profile by parallelograms. The positive Fano cubic therefore marks the first degree of failure in that setting.

Leading coefficients require more information. A single common affine normalization of the original scaled polytopes turns their leading volume polynomial into an actual real one. This forces real-linear representability of every initial support and compatibility of its coefficients with real mixed-volume geometry. The full-support Fano cubic violates the first requirement; the four-variable quadratic violates the second while possessing a genuine polytope valuation profile.

These statements are valid at arbitrary natural valuation rank. Their proofs use finite determinants, finite convexity, and standard part rather than analytic completeness of the surreal field. They give a concrete boundary between Lorentzian discrete convexity and realizable multiscale polytope data, together with finite certificates on both sides of that boundary.

116 Source and proof audit (Appendix A of source 10)

116.1 Repository files actually inspected

The following paths are relative to ProveIt. They were inspected at commit

6996fee43cc97b6c16351def7507d59a95bf62f0.

Natural valuation.

Algebra/SurrealNumbers/Surreal/Foundations/SignSequenceValuation.lean

Inspected source lines 1–150; blob identifier 222abd29b3a8f1352b473633383e31e6088d4315. Relevant statements include the natural additive valuation, nonnegative-order comparison, the finite/infinitesimal valuation criteria, and positive monomials with specified values.

Real residue field.

`Algebra/SurrealNumbers/Surreal/Foundations/SignSequenceStandardPart.lean`

Inspected source ranges beginning at lines 1 and 80 through the file’s end; blob identifier `78e7c0a646da5f6b55774f5eebe6e411c4dca273`. Relevant definitions include the finite-element domain, `standardPartHom`, its surjectivity, its infinitesimal kernel, and `standardPartQuotientEquiv`.

The import facade `Algebra/SurrealNumbers/Surreal.lean` and repository search results were also consulted for navigation. They are not evidence that a mixed-volume development has been formally completed. No repository-wide correctness audit or Lean build was attempted. These exact files, rather than an uninspected informal article, are the repository dependencies discussed here.

[write] Re-verified: the two blob identifiers are those of the two files in the current `ProveIt` tree (152 and 222 lines), and the facade `Algebra/SurrealNumbers/Surreal.lean` imports both modules.

116.2 Proof dependencies and limitations

Result	Essential dependencies
Compression	Finite maximal simplex; mixed-volume monotonicity, translation invariance, and homogeneity.
Determinant criterion	Compression; positive determinant-sum formula; block Cauchy–Binet; finite avoidance of nonzero real polynomials.
Initial polynomial	Minimum-minor normalization; finite triangulation; standard part on bounded vertices and determinants.
Initial support	Cramer’s rule; reduction of minors; linear selection.
Matroid profile theorem	Initial squarefree obstruction; independent sets extended to bases; explicit thickened segments.
Fano cubic obstruction	The valuation table; initial support; the explicit characteristic-zero Fano contradiction.
Strict Lorentzian cubic	The seven constant Hessians; an exact congruence; five negative directions and a two-plane determinant.
Planar sufficiency	The classical rank-two tropical relation, proved constructively here over the specified valued field.

The proofs are conventional mathematical proofs, not proof-assistant certificates. The exact script checks finite identities and test families. It does not prove the general compression, normalization, or realization theorems by computation.

116.3 Literature search boundary

The comparison focused on mixed volumes, Lorentzian polynomials, matroid support obstructions, real and tropical Grassmannians, and the repository’s scalar foundations. The 2026 version of [65], the 2026 journal appearance of [69], and the 2026 survey [68] were checked so that older results were not presented as current open questions. Classical results used in the proofs are cited at their point of use. This targeted comparison is not an exhaustive novelty search in valuation theory, polymatroid representation, or non-Archimedean convexity.

117 Reproducing the package (Appendix B of source 10)

The main source is standalone and has an internal bibliography. On a standard TeX Live installation, compile it with

```
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
```

The exact checks require Python 3.10 or newer and SymPy 1.12 or newer:

```
python verify.py
```

This command regenerates the JSON verification report, the CSV coefficient table, and the rank-two example matrix beside the script. The included `data/10-tropical-mixed-volumes-verification.txt` is the saved output of the executed run. No internet access, random sampling, floating-point threshold, or external dataset is used by the verification script.

[write] In `ProveIt` the delivered `article.tex` is not shipped: it is printed in this Part. The script is shipped as `code/10-tropical-mixed-volumes-verify.py` and its build helper as `code/10-tropical-mixed-volumes-build.sh`; the saved output is `data/10-tropical-mixed-volumes-verification.txt`, written by the build helper's tee, not by the script. The script writes its three outputs beside itself under the delivery names (`verification.json`, `fano_coefficients.csv`, `rank_two_example.txt`), so rerun it on a copy, never in `code/`, and without `python -O`, which would disable its assertions; the report's README gives the command.

118 Why the transferred volume facts are finite statements (Appendix A of source 09)

[write] This is the finite transfer argument of Part III ([proposition 33.3](#) and the text before it) and Part I ([theorem 18.1](#)), in source 09's more detailed form; it is printed because source 09's main text refers to it.

Here is a precise route for the use of real-closed-field transfer in the main text. Fix d and a finite bound n on the input vertex lists. For a candidate list of d -simplices whose vertices come from that list, nondegeneracy and the formula for each simplex volume are expressed using polynomial equalities and inequalities in the vertex coordinates. Containment of a point in a simplex is expressed by the existence of nonnegative barycentric coordinates summing to one. Equality between the union of the candidate simplices and the convex hull of the original vertices is therefore a first-order assertion. Intersection conditions for a simplicial complex are also first-order, using the same barycentric descriptions.

There are finitely many candidate vertex triangulations at these fixed sizes. The existence of a triangulation can be expressed as their finite disjunction. Over $\mathbb{R}$ the usual polytope triangulation theorem makes this sentence true; real-closed-field transfer makes it true over K . For any two valid triangulations, equality of their determinant sums is likewise an ordered-field sentence and transfers from the real volume identity. This defines a single-valued semialgebraic determinant-volume function at fixed sizes.

A Minkowski sum of vertex polytopes is the convex hull of all sums of one vertex from each input list. A scaled Minkowski sum has the same finite list format, with coordinates polynomial in the scalar parameters. Consequently its volume is described by a finite semialgebraic formula. The assertion that it equals a homogeneous polynomial of degree d in the nonnegative scalar parameters, with nonnegative coefficients, is a first-order sentence with finitely many unknown coefficients. It holds over $\mathbb{R}$ by Minkowski's theorem, hence over K .

Uniqueness of these coefficients follows from the elementary fact that a polynomial over an infinite ordered field vanishing on its positive orthant is zero. Mixed volumes are then defined from the coefficients by [equation \(100.5\)](#). Translation invariance, homogeneity, symmetry, Minkowski additivity, linear covariance, and monotonicity for finite polytope lists all become finite first-order implications. Inclusion of one polytope in another is itself first-order, so mixed-volume monotonicity transfers as well. One may enlarge the finite vertex bounds for an intermediate Minkowski sum or a refinement; every particular instance still uses only finitely many vertices.

This argument does not assert a first-order definition of Lebesgue integration over arbitrary sets. It transfers a family of finite polytope identities, one finite complexity bound at a time. Nor does it supply an efficient algorithm: quantifier elimination can be expensive, whereas the determinant proofs in the main text are designed to expose smaller certificates for the specific claims under study.

119 Reproducing the exact checks and the document (Appendix B of source 09)

The package contains the TeX source and its compiled PDF. It also includes `code/09-finite-mixed-volume-models-verify.py`, the full results in `data/09-finite-mixed-volume-models-verification_results.json`, a `README.md` [[write] not shipped], and a source audit [[write] 09-finite-mixed-volume-models-SOURCE_AUDIT.md]. The bibliography is embedded in the TeX source; no separate database or external figure files are required.

From the package directory, run

```
python verify.py --output verification_results.json
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
```

The Python program requires SymPy. Its JSON records the Python and SymPy versions from the executed run. It uses integer and rational arithmetic, symbolic polynomial operations, and exact finite enumeration; no external service or numerical optimizer is involved.

The exchange checker enumerates all weak compositions of three into seven parts. For every ordered pair (α, β) and every coordinate with $\alpha_i > \beta_i$, it tests all deficit coordinates j and requires at least one witness for [equation \(108.15\)](#). It stores all 84 coefficient polynomials, their type labels, and their valuations. The Hessian checker computes the seven partial Hessians directly and compares them with the right side of [equation \(108.10\)](#); it also verifies [equation \(108.12\)](#) for each middle matrix. The sandwich checker selects an exact maximum projected minor, reconstructs all differences from its Cramer coordinates, and checks the outer and inner cube conditions. The rank-two checker realizes polarized valuation tables by finite Laurent sums and verifies each of the resulting maximal minors exactly.

All tests passed in the included run. A passing run establishes exactly these finite assertions. It does not certify originality, peer review, the correctness of the underlying computer algebra implementation, or a formal proof of the general theorems.

[[write] In ProveIt the TeX source, the PDF and `README.md` of the delivery are not shipped (the manuscript is printed in this Part); the program, its recorded results and the source audit are shipped under the names given above, and the build file as `code/09-finite-mixed-volume-models-Makefile`. The first command above writes `verification_results.json` into the working directory; run it with an explicit `-output` path outside `data/`, as the report's README shows.

Part VIII

Moving vertices and exponential contact depth

Source and scope. [write] This Part prints, whole, batch 59 manuscript 04, *Exponential Contact Depth in Surreal Polytopes: Face-lattice-forced arithmetic, rigid scale towers, and lower bounds for rational shadow realizations* (archive `surreal_polytopes_contact_depth.zip`, main file `surreal_polytopes_contact_depth/surreal_polytopes_contact_depth.tex`; “Research manuscript prepared for Vladimir Reshetnikov. An AI-assisted contribution to the ProveIt research program”, 30 September 2026; the delivered PDF has 24 pages, A4). In this report it is called *source 11*, after the prefix of its shipped files. It inspected ProveIt at the pin 6996fee.

The “earlier manuscript in this research program” that source 11 cites ([85]) is the base source of Part I, batch 58 manuscript 05. Source 11 read it as a separate manuscript, not as part of this report, and its account of it is accurate: the degree-one realization it contrasts with is Part I’s [theorem 10.4](#). The Part continues Part I. Its [theorem 120.3](#) answers negatively, for rational arcs, Part I’s question “Moving-base determinant degenerations” ([section 22](#)): no bound polynomial in the number of vertices and the dimension exists, even when the shadow keeps every vertex and the dimension. Algebraic and Puiseux arcs are covered only through the normalized ramification statement ([theorem 127.2](#)), and the bounds are not sharp (a factor $16k + 48$ apart). The family is also a lower-bound instance for Part II’s question “Effective controlled specialization” and Part IV’s “Effective algebraic descent”; no upper bound of the kind asked there is proved. Its lemma on the shadow of a finite convex hull ([lemma 121.1](#)) is Part I’s [proposition 13.1](#), and its transfer paragraph restates [theorem 18.1](#); both are kept, with pointers.

Notation in this Part. [write] Source 11’s letters are kept; three macros were renamed without changing the printed output (its two-argument bracket $[a, b]$, its ord_0 for the order at 0, and its plain span).

- *Three degrees.* Part I’s degree ([theorem 9.1](#)) is the least degree of a polynomial *height vector* over a fixed real base. The degree here, δ_{rat} ([definition 120.2](#)), is the homogeneous degree of a rational arc along which *every vertex* moves. Part X’s [theorem 159.2](#) is a third notion, the degree of non-exposedness. The tempting false reading is that Part I’s degree bound $m - 1 \leq n - d - 1$ and the exponential bound here conflict; they measure different deformations ([section 129](#)).
- $m = 4k + 8$ is the number of configuration points, $N = 2m$ the number of vertices and $d = m + 2$ the dimension; in Part I m is the comparison rank.
- $\mathcal{L}, \mathcal{L}_k$ are labeled face lattices; $L(B)$ is the Lawrence polytope of a vector configuration B , not Part III’s scale lattice $L(P)$.
- t is a real parameter in $(0, 1/6)$ or a positive infinitesimal, and s the parameter of a competing arc; $\varepsilon = \omega^{-1}$ where it occurs; τ is the variable of the Puiseux field $\mathbb{R}\langle\tau\rangle$ in [section 127](#), with $v(\tau) = 1$, not the surreal $\omega^{-\omega}$.
- v is an additive valuation with positive value meaning infinitesimal, as in Part III; ord_0 is the order at $s = 0$ of a rational function.
- $\mathcal{A}_k(t)$ and $\mathcal{B}_k(t)$ are the projective and affine configurations; x_i and y_j are recovered cross-ratios; W is a fixed point of the frame, not a matrix.

Abstract of source 11. Finite surreal point configurations have ordinary realizable combinatorics, but this does not imply that their infinitesimal geometry can be represented by low-degree real deformations. We give an explicit quantitative obstruction. For each $k \geq 1$ we construct a labeled polytope with $8k + 16$ vertices in dimension $4k + 10$, a rational one-parameter realization, and a prescribed vertexwise real shadow. Every vertex remains a distinct vertex at the shadow, and the dimension is unchanged. Nevertheless, every rational vertex arc with the same labeled face lattice and that shadow has homogeneous coordinate degree at least

$$\left\lceil \frac{2^k}{16k + 48} \right\rceil.$$

An explicit arc has degree at most 2^k .

The mechanism is a planar projective construction encoding repeated squaring, followed by a Lawrence polytope. From the labeled face lattice of *any* realization we recover a projective configuration and intrinsic cross-ratios $x_0, \dots, x_k$ satisfying $x_{i+1} = x_i^2$. We give the reconstruction by facet cofactors and bound its degree. At the prescribed shadow x_0 vanishes, so the order of x_k is at least 2^k . This proves the lower bound without restricting competing arcs to the displayed Lawrence coordinates.

For a surreal infinitesimal, the recovered valuations satisfy $v(x_i) = 2^i v(x_0)$. Normalizing the deepest observable as the parameter forces Puiseux ramification divisible by 2^k . More generally, an addition chain of length ℓ for an integer q gives the bounds $q/(16\ell + 48)$ and q . All determinant signs in the explicit constructions stabilize already on $0 < t < 1/6$. We explain why these results do not contradict degree-one realizations for height liftings over a fixed real base, and distinguish contact order from valuation rank and from arithmetic-circuit size.

Projective arithmetic and Lawrence reconstruction are classical. The proposed contribution is the explicit shadow-sensitive degree theorem and its quantitative synthesis; literature-wide priority has not been established.

Verification status. The general results have mathematical proofs below. Accompanying exact-arithmetic code checks finite instances and one projective reconstruction. No new Lean formalization or independent refereeing is claimed.

[write] The delivered README (not shipped) adds: the manuscript is AI-assisted and not independently refereed; the invariant concerns rational parametrization degree with a prescribed labeled limit, not arithmetic-circuit length, valuation-group rank, static bit complexity or the size of a complete face-list input; the checks are not a full face-lattice enumeration, a general proof in a theorem prover, a test of all rational arcs, or an independent review of the proof; the repository comparison is limited to the finite-geometry source and the surreal project guide, with no repository-wide audit or new build; the lower-bound proofs do not depend on the cited liftings manuscript; literature-wide priority of the formulation has not been independently established, and no solution of a named published open conjecture is claimed.

120 The question and the main result

120.1 From finite realization to controlled degeneration

A finite list of surreal coordinates can contain extremely separated infinitesimal scales. Its ordinary finite incidence information, however, is specified by finitely many polynomial signs. Real-closed-field transfer therefore does not create new finite realizable oriented matroids merely by replacing $\mathbb{R}$ by **No**. The interesting issue is not whether a real realization exists,

but what information an approaching real realization can preserve and how complicated that approach must be.

An earlier manuscript in this research program proves a particularly strong positive statement for *height liftings over a fixed real base*: a linear height deformation preserves the finite lifted combinatorics and the standard part [85]. Determinants in that setting are linear in the moving heights. [[write] That manuscript is the base source of [Part I](#) (batch 58, manuscript 05); the statement meant is [theorem 10.4](#).] General moving coordinates satisfy nonlinear determinantal constraints. This leads to a concrete question.

Question 120.1. Is there a polynomial bound in the number of vertices and the dimension for the degree of a rational deformation that preserves a finite polytope’s labeled face lattice and approaches a prescribed realizable labeled shadow, whenever such a rational deformation exists?

The answer is negative, even when the shadow loses neither vertices nor dimension. The restriction “whenever such a rational deformation exists” matters: a general semialgebraic curve-selection theorem produces semialgebraic arcs, not necessarily rationally parametrized ones [81]. Our examples already have explicit rational arcs, and the obstruction concerns the minimum degree of all competing rational arcs.

120.2 What is measured

A labeled d -polytope has vertices $q_1, \dots, q_N$ with fixed labels. Its labeled face lattice specifies exactly which subsets of these labels are vertex sets of faces, including the empty face and the whole polytope. A *rational vertex arc* consists of rational functions $q_j(s) \in \mathbb{R}(s)^d$, defined for sufficiently small $s > 0$, whose convex hull has the required labeled face lattice throughout that interval.

Definition 120.2 (Homogeneous degree and rational shadow degree). The homogeneous degree of such an arc is the least D for which every projective vertex $[q_j(s) : 1]$ admits a nonzero vector of polynomials in s of degree at most D . A different common multiplier is allowed for every vertex. Thus an affine polynomial arc of coordinate degree D has homogeneous degree at most D , while rational coordinates are measured after clearing a common denominator separately at each vertex.

For a labeled face lattice $\mathcal{L}$ and a prescribed labeled limiting tuple $Q^0 = (q_1^0, \dots, q_N^0)$, call a rational arc admissible if it realizes $\mathcal{L}$ and tends labelwise to Q^0 . Set

$$\delta_{\text{rat}}(\mathcal{L}, Q^0) = \min\{D : \text{an admissible arc has degree } D\}.$$

Set this value to infinity if there is no admissible arc.

This notion does not charge for writing a repeated power in compressed form. It measures polynomial degree, not straight-line program length, coefficient bit size, or the number of faces. A constant projective change of ambient coordinates preserves the degree of each projective vertex map: it and its inverse are linear on homogeneous coordinates.

Theorem 120.3 (Exponential rational shadow degree). *For every $k \geq 1$, put*

$$m = 4k + 8, \quad N = 2m = 8k + 16, \quad d = m + 2 = 4k + 10.$$

There are an explicitly constructed labeled face lattice $\mathcal{L}_k$ and a rational labeled tuple Q_k^0 of N points in $\mathbb{R}^d$ with the following properties.

- (i) The convex hull of Q_k^0 is d -dimensional and all N points are distinct vertices.
- (ii) A rational arc $P_k(t)$ realizes $\mathcal{L}_k$ for every $0 < t < 1/6$, has vertexwise limit Q_k^0 , and has homogeneous degree at most 2^k .
- (iii) Every rational vertex arc realizing $\mathcal{L}_k$ and tending labelwise to Q_k^0 has degree at least $\lceil 2^k / (16k + 48) \rceil$. Consequently,

$$\left\lceil \frac{2^k}{16k + 48} \right\rceil \leq \delta_{\text{rat}}(\mathcal{L}_k, Q_k^0) \leq 2^k. \quad (120.1)$$

- (iv) The labeled face lattice reconstructs projective observables $x_0, \dots, x_k$ in every realization, with

$$x_{i+1} = x_i^2, \quad x_i = x_0^{2^i}. \quad (120.2)$$

In the displayed arc, $x_0 = t$. In every arc approaching Q_k^0 , $x_0 \rightarrow 0$, while x_0 is nonzero on the punctured arc.

The same construction at a positive surreal infinitesimal realizes $\mathcal{L}_k$ and has standard-part vertex tuple Q_k^0 .

The proof occupies Sections 122–126. The main logical point is the quantifier in part (iii): it applies to *all* rational realizations of the labeled face lattice approaching the shadow, not just to the formulas used to produce one example. Recovering the arithmetic from an arbitrary realization is indispensable.

Corollary 120.4. *Question 120.1 has a negative answer. This remains true with the additional requirement that all vertices and the full dimension survive at the labeled shadow.*

Proof. Both N and d in Theorem 120.3 are linear in k , whereas the lower bound in (120.1) eventually exceeds every polynomial in k . The upper bound proves that the invariant is finite in these examples. $\square$

[[write] Part I’s degree (theorem 9.1) is the least degree of a polynomial height vector over a fixed real base; the degree here is the homogeneous degree of a rational arc along which every vertex moves. Part X’s theorem 159.2 is a third notion, the degree of non-exposedness. Corollary 120.4 answers negatively, for rational arcs, Part I’s question “Moving-base determinant degenerations” (section 22); the note there records the scope.]

120.3 Relation to the repository and to classical work

The inspected repository source `Algebra/SurrealNumbers/Surreal/Algebra/Geometry.1e` at commit `6996fee43cc9` develops field-valued finite coordinate geometry, including dot and cross products, Gram’s identity, and specified Heron and Ptolemy statements [83]. The surrounding project separates the constructed surreal field and mapped Lean results from its unrefereed research reports [84]. The present construction extends the finite determinant viewpoint to higher-dimensional polytope realizations. It does not depend on speculative claims in other reports, and repository membership is not used as a substitute for checking a theorem.

Two central methods are established. Von Staudt constructions encode arithmetic in projective incidences; [79] gives a modern treatment. Lawrence polytopes encode configurations in their combinatorics; Bayer–Sturmfels [78] is the primary source used here. Earlier work on intrinsic spread [80] is an important antecedent to the general phenomenon that combinatorics

can force large coordinate complexity. We do not claim to introduce any of these methods or to solve a previously identified published universality problem.

What is developed here is a specific quantitative statement about a *prescribed labeled degeneration*: an explicit configuration, a noncollapsing polytopal shadow, and a degree estimate for the reconstruction that turns contact order into a lower bound for every rational arc. The result answers the low-degree extension question raised above. Whether an equivalent quantitative formulation already occurs elsewhere in the literature remains a priority question, separate from the proofs.

121 Ordered fields, shadows, and projective notation

121.1 Finite convexity without analytic compactness

Let K be an ordered field. All polytopes below are convex hulls of finitely many points using coefficients in K :

$$\text{conv}_K\{p_1, \dots, p_N\} = \left\{ \sum_j \lambda_j p_j : \lambda_j \geq 0, \sum_j \lambda_j = 1 \right\}.$$

A face is the set on which an affine functional attains its maximum over this hull. The maximum is the largest of its finitely many values at the generators, so no topological compactness assumption is being made. For finite polytopes this description agrees with the usual face lattice. The algebraic arguments below use only ordered-field linear algebra.

When working in $\mathbf{No}$, a finite construction can be carried out in a set-sized ordered subfield containing the coefficients. If real closedness is needed for a transfer statement, close a set-sized coefficient field under the necessary algebraic roots inside $\mathbf{No}$. The explicit construction itself only uses rational functions of one parameter and does not require real closedness. In particular, it is already meaningful in the ordered field $\mathbb{Q}(t)$ with t positive infinitesimal.

121.2 Standard part

For a finite surreal a , write $\text{st}(a)$ for its real standard part. The standard-part map is a ring homomorphism on the finite elements and preserves weak inequalities. Coordinates of all vertices constructed below are finite.

Lemma 121.1 (Shadow of a finite convex hull). *For finite surreal vectors $p_1, \dots, p_N$,*

$$\text{st}(\text{conv}_{\mathbf{No}}\{p_1, \dots, p_N\}) = \text{conv}_{\mathbb{R}}\{\text{st}(p_1), \dots, \text{st}(p_N)\}.$$

Proof. Every convex coefficient satisfies $0 \leq \lambda_j \leq 1$ and is finite. Taking standard parts in a convex combination yields a real convex combination. Conversely, use the same real coefficients on the original vertices to lift any real convex combination of their standard parts. $\square$

[[write] This is Part I's [proposition 13.1](#), proved there for an ordered subfield $\mathbb{R} \subseteq K \subseteq \mathbf{No}$; it is repeated here because the proofs below cite it.]

This lemma concerns sets of points, not face lattices. A standard-part image can acquire new incidences and lose old faces. Later we make that failure explicit even though neither the dimension nor the vertex set collapses.

121.3 Projective points, determinants, and cross-ratios

A nonzero vector $a \in K^r$ determines a projective point $[a] \in \mathbb{P}^{r-1}(K)$. In the projective plane we write points as $(X : Y : Z)$. Joins of distinct points are lines; intersections of distinct lines are points. Collinearity of three points is the vanishing of their homogeneous 3×3 determinant. We keep all vanishing determinants as exact equalities.

A vector configuration $B = (b_1, \dots, b_m)$ has covectors

$$\mathcal{V}(B) = \{(\text{sgn}(c(b_1)), \dots, \text{sgn}(c(b_m))) : c \in (K^r)^*\}.$$

The signs of the maximal minors of a full-rank configuration determine these covectors. This is the standard determinant description of a realizable oriented matroid. One elementary explanation is to ask for a linear functional that vanishes on specified columns and is positive on specified signed remaining columns. The finite linear theorem of the alternative expresses failure by a nonnegative dependence. Minimal such dependences have coefficient signs given by cofactors of independent columns. Thus the feasibility of each sign pattern is determined by the minor signs, over any ordered field. We use this classical linear-algebraic fact, not a new oriented-matroid representation theorem.

For points on a projective line, choose any linear projection of its underlying two-dimensional vector space onto K^2 that is an isomorphism. Write $[a, b]$ for the determinant of the resulting two coordinates. For three distinct reference points O, I, U and $X \neq U$, set

$$\chi(X; O, I, U) = \frac{[X, O][I, U]}{[X, U][I, O]}. \quad (121.1)$$

Changing the projection multiplies every bracket by the same nonzero factor, and rescaling any point cancels between numerator and denominator. Hence (121.1) is intrinsic. With

$$O = (0 : 1), \quad I = (1 : 1), \quad U = (1 : 0), \quad X = (x : 1),$$

it equals x .

122 An explicit projective repeated-squaring configuration

122.1 The frame and the coordinates

Fix six points

$$\begin{aligned} O &= (0 : 0 : 1), & I &= (1 : 0 : 1), & J &= (0 : 1 : 1), \\ U &= (1 : 0 : 0), & V &= (0 : 1 : 0), & W &= (1 : -1 : 0). \end{aligned} \quad (122.1)$$

The four points I, J, U, V form a projective frame: no three are collinear. They determine the remaining two points by

$$O = IU \cap JV, \quad W = IJ \cap UV.$$

The line OU is the number line, with origin O , unit I , and point at infinity U . The point V gives the vertical direction, and W gives slope -1 in the affine chart $Z = 1$.

Let $k \geq 1$, let $0 < t < 1/6$, and put $x_i = t^{2^i}$. Start with

$$X_0 = (t : 0 : 1), \quad A_0 = (0 : t : 1).$$

For each $i = 0, \dots, k-1$, add four points

$$\begin{aligned} B_i &= (1 : x_i : 1), & C_i &= (x_i : x_i^2 : 1), \\ A_{i+1} &= (0 : x_i^2 : 1), & X_{i+1} &= (x_i^2 : 0 : 1). \end{aligned} \quad (122.2)$$

The resulting labeled configuration $\mathcal{A}_k(t)$ has

$$6 + 2 + 4k = 4k + 8 = m \quad (122.3)$$

points. For this repeated-squaring construction they are all projectively distinct when $0 < t < 1/6$. Indeed the nonconstant powers are strictly ordered; points on the two axes, points with first coordinate 1, and interior points C_i have the displayed distinct coordinate patterns.

122.2 Why the incidences force the arithmetic

The construction can be specified without mentioning multiplication:

$$\begin{aligned} A_0 &= X_0W \cap OV, \\ B_i &= A_iU \cap IV, \\ C_i &= OB_i \cap X_iV, \\ A_{i+1} &= C_iU \cap OV, \\ X_{i+1} &= A_{i+1}W \cap OU. \end{aligned} \quad (122.4)$$

All joins and intersections in (122.4) are between distinct points or lines on the specified parameter interval.

Lemma 122.1 (Arithmetic gate). *Suppose a projective configuration has the frame incidences in (122.1), the incidences and nondegeneracies in (122.4), and the same labels. After the unique projective normalization of I, J, U, V to (122.1), it has precisely the coordinates (122.2), with*

$$x_0 = \chi(X_0; O, I, U), \quad x_{i+1} = x_i^2.$$

In particular the equations are forced by incidences, not imposed as additional metric data.

Proof. The frame normalization determines O and W by their displayed intersections. Since $X_0 \in OU$ and $X_0 \neq U$, write $X_0 = (x_0 : 0 : 1)$. The line X_0W has affine equation $X + Y = x_0$, so its intersection with the vertical axis is $A_0 = (0 : x_0 : 1)$.

Assume $X_i = (x_i : 0 : 1)$ and $A_i = (0 : x_i : 1)$. The horizontal line through A_i and the vertical line through I meet at $B_i = (1 : x_i : 1)$. The line OB_i has equation $Y = x_iX$. Intersecting it with the vertical line $X = x_i$ gives $C_i = (x_i : x_i^2 : 1)$. Horizontal projection to the vertical axis gives $A_{i+1} = (0 : x_i^2 : 1)$. The slope -1 projection from there to the horizontal axis gives $X_{i+1} = (x_i^2 : 0 : 1)$. Induction proves the coordinate formulas. Formula (121.1) identifies each number-line coordinate with its intrinsic cross-ratio. $\square$

Only the listed incidences and nondegeneracies are needed for this lemma. The full configuration has additional collinearities, some arising from relations between stages. We do not declare these absent or perturb them away: the oriented matroid used below is the actual one of the displayed configuration, including every additional zero determinant.

Corollary 122.2 (Projective distinction). *For $0 < t, u < 1/6$, a label-preserving projectivity between $\mathcal{A}_k(t)$ and $\mathcal{A}_k(u)$ exists only when $t = u$.*

Proof. The intrinsic cross-ratio of the labels X_0, O, I, U is t in the first configuration and u in the second. $\square$

122.3 A uniform determinant-sign threshold

The highest powers in (122.2) grow exponentially. Yet the interval on which every determinant sign stabilizes can be chosen independently of k .

Lemma 122.3 (Six-term sign certificate). *Let $f(t)$ be a nonzero 3×3 determinant of homogeneous vectors in $\mathcal{A}_k(t)$. Its sign on $0 < t < 1/6$ is the sign of its lowest-degree nonzero coefficient. The same sign is obtained at a positive infinitesimal in an ordered extension of $\mathbb{R}$.*

Proof. Every matrix entry is zero or a signed monomial with coefficient 1 or -1 . The determinant expansion has at most six signed monomials, so after collecting powers its integer coefficients have total absolute value at most six. Write

$$f(t) = t^e \left(a + \sum_{j>e} a_j t^{j-e} \right), \quad a \neq 0.$$

Then $|a| \geq 1$ and $\sum_{j>e} |a_j| \leq 5$. Since the exponents are integers, $0 < t < 1$ gives

$$\left| \sum_{j>e} a_j t^{j-e} \right| \leq 5t < 1 \leq |a|.$$

The leading coefficient therefore controls the sign. A positive infinitesimal satisfies the same inequalities. Identically zero determinants remain zero. $\square$

Thus the entire projective incidence and sign pattern is certified by finite integer computations, despite its large exponents. A convenient ordinary rational sample is $t = 1/7$. The proof does not rely on numerical approximations to near-zero determinants.

123 A bounded affine chart and its degeneration

To form a bounded surreal polytope we first put all configuration points in one affine chart. Set

$$\ell(X, Y, Z) = X + 2Y + Z, \quad b(X : Y : Z) = \left(\frac{X}{\ell}, \frac{Y}{\ell}, 1 \right).$$

This is the projective transformation $(X, Y, Z) \mapsto (X, Y, \ell)$, followed by normalization of its last coordinate. The transformation has determinant one. The chart denominators for O, I, J, U, V, W are respectively $1, 2, 3, 1, 2, -1$; all variable-point denominators are positive. There are no zero denominators.

Write $\mathcal{B}_k(t) = (b_1(t), \dots, b_m(t))$ for the resulting ordered vector configuration in K^3 . In particular,

$$\begin{aligned} b_{X_i} &= \left(\frac{x_i}{1+x_i}, 0, 1 \right), & b_{A_i} &= \left(0, \frac{x_i}{1+2x_i}, 1 \right), \\ b_{B_i} &= \left(\frac{1}{2+2x_i}, \frac{x_i}{2+2x_i}, 1 \right), & b_{C_i} &= \left(\frac{x_i}{1+x_i+2x_i^2}, \frac{x_i^2}{1+x_i+2x_i^2}, 1 \right). \end{aligned} \tag{123.1}$$

Only the normalization of W reverses the sign of its representing vector. Consequently the signs of all minors of $\mathcal{B}_k(t)$ are fixed on $0 < t < 1/6$ by Lemma 122.3, with a single sign reversal for triples containing W .

The fixed frame I, J, U, V remains a projective frame in this chart. In particular, the configuration has rank three after any one column is deleted: if a frame column is deleted, the other three frame columns are independent; otherwise all four remain. This is the no-coloop condition needed below.

Proposition 123.1 (The configuration shadow). *All coordinates of $\mathcal{B}_k(t)$ are bounded near 0. The fixed six points have constant limits, while*

$$b_{X_i}^0 = b_{A_i}^0 = b_{C_i}^0 = b_O = (0, 0, 1), \quad b_{B_i}^0 = b_I = (1/2, 0, 1).$$

The limit configuration still has rank three after deletion of any one column, and no column is zero.

Proof. Take limits in (123.1). The fixed frame survives, and the third coordinate of every column is one. $\square$

Thus many projective points coalesce at the limit. The next construction converts this apparently severe collapse into a polytope degeneration in which *all* vertices remain distinct.

124 Lawrence polytopes and recovery from their faces

We give the needed Lawrence construction and its recovery explicitly. The method is classical [78]; the degree estimate for this recovery is isolated later because it is the quantitative step used in the main theorem.

124.1 The polytope and its distinguished facets

Let $B = (b_1, \dots, b_m)$ be a rank-three vector configuration with no zero columns and with rank three after any one column is removed. Let $e_1, \dots, e_m$ be the standard basis of K^m . In $K^3 \times K^m$ put

$$p_i^- = (\mathbf{0}, e_i), \quad p_i^+ = (b_i, e_i), \quad L(B) = \text{conv}\{p_i^-, p_i^+ : 1 \leq i \leq m\}. \quad (124.1)$$

All vertices lie in the affine hyperplane $\sum_i z_i = 1$.

Lemma 124.1 (Dimension, vertices, and pair-complement facets). *The polytope $L(B)$ has dimension $d = m + 2$ and exactly $2m$ vertices, with labels i^- , i^+ . For every i ,*

$$F_i = \text{conv}\{p_j^-, p_j^+ : j \neq i\}$$

is a facet, supported by $z_i = 0$.

Proof. The lower vertices span an affine space of dimension $m - 1$ in the z coordinates. The differences $p_i^+ - p_i^- = (b_i, 0)$ span a three-dimensional space independent of those directions. Thus the dimension is $m + 2$.

The functional z_i attains its maximum one exactly on the segment with endpoints p_i^-, p_i^+ . These endpoints are distinct because $b_i \neq 0$. A linear functional on K^3 nonzero on b_i distinguishes the endpoints; adding it with a sufficiently small positive coefficient to z_i exposes either desired endpoint. The small coefficient can be chosen by finitely many ordered-field comparisons. Hence all displayed points are vertices.

Deleting pair i leaves $m - 1$ lower vertices of affine dimension $m - 2$ and three independent b directions. Its convex hull has dimension $m + 1 = d - 1$. It is exactly the intersection with $z_i = 0$, so it is a facet. $\square$

The ambient vectors in (124.1) also serve as homogeneous coordinates of these affine points: their length is $m + 3 = d + 1$, and the homogeneous affine-chart functional is $h = \sum_i z_i$. No additional homogeneous coordinate should be appended to this representation. To work in ordinary K^d , simply omit z_m and recover it from $\sum_i z_i = 1$.

124.2 Reading covectors from transversal faces

Call a face *transversal* if it meets every labeled pair. Encode such a face by a sign vector: $+$ when it contains only i^+ , $-$ when it contains only i^- , and 0 when it contains both.

Lemma 124.2 (Face dictionary). *The encoded transversal faces of $L(B)$ are exactly the covectors of B , with the zero covector corresponding to the whole polytope. Moreover, the covectors determine the entire labeled face lattice of $L(B)$.*

Proof. An affine functional on the chart can be represented by a linear functional (c, α) on $K^3 \times K^m$. Its values on pair i are

$$\alpha_i, \quad \alpha_i + c(b_i).$$

For any c , set $\alpha_i = -\max(0, c(b_i))$. Both values are then at most zero, and the maximizing endpoints in pair i are exactly those encoded by $\text{sgn } c(b_i)$. This gives a transversal face with the required sign vector. Conversely, for a face meeting every pair, the difference of the two values forces precisely that choice of endpoints. If $c = 0$, meeting every pair gives the whole polytope.

For completeness, a face that misses some pairs is determined in the same way on the pairs it meets. Its choices there must be restrictions of a covector $\text{sgn } c(B)$. Conversely, given such a covector and a nonempty subset J of pairs to be met, choose

$$\alpha_i = -\max(0, c(b_i)) \quad (i \in J), \quad \alpha_i = -1 - \max(0, c(b_i)) \quad (i \notin J).$$

The maximizing face has exactly those choices on J and misses every other pair. Include the empty face separately. Thus the covector data determine all faces. $\square$

Since maximal-minor signs determine covectors, the fixed minor signs of $\mathcal{B}_k(t)$ give one labeled face lattice on $0 < t < 1/6$. Denote it by $\mathcal{L}_k$. This definition retains every accidental incidence of the actual configuration. It does not require an enumeration of all faces.

124.3 Reconstructing the configuration in an arbitrary realization

Now let Q be any d -dimensional realization of the same labeled face lattice as $L(B)$, where $d = m + 2$. Let $V_i^-, V_i^+ \in K^{d+1}$ be homogeneous representatives of its vertices, chosen in a common positive affine chart. The pair-complement facet F_i has a covector H_i such that

$$H_i(V_j^-) = H_i(V_j^+) = 0 \quad (j \neq i), \quad H_i(V_i^-) > 0, \quad H_i(V_i^+) > 0. \quad (124.2)$$

The signs can be chosen this way because the remaining two vertices lie strictly on the same side of the supporting hyperplane.

Theorem 124.3 (Explicit Lawrence recovery). *The covectors $H_1, \dots, H_m$ are linearly independent, and*

$$S = \bigcap_{i=1}^m \ker H_i$$

is a three-dimensional vector space. The nonzero vectors

$$R_i = H_i(V_i^-)V_i^+ - H_i(V_i^+)V_i^- \quad (124.3)$$

lie in S . Their projective points $[R_i] \in \mathbb{P}(S)$ are independent of the choices of representatives and facet equations. They form a configuration projectively equivalent to a vector configuration having the same covectors as B .

Proof. A linear relation $\sum_i \lambda_i H_i = 0$, evaluated at V_j^- , yields $\lambda_j H_j(V_j^-) = 0$, so every λ_j is zero. Hence

$$\dim S = (d + 1) - m = 3.$$

For a direct normalization, put

$$W_i^\pm = \frac{V_i^\pm}{H_i(V_i^\pm)}.$$

The linear map $H = (H_1, \dots, H_m)$ sends both W_i^- and W_i^+ to e_i . The vectors W_i^- therefore define a splitting of H . Subtracting that splitting identifies K^{d+1} with $S \oplus K^m$ and sends the normalized vertices to

$$(0, e_i), \quad (a_i, e_i), \quad a_i = W_i^+ - W_i^- \in S.$$

The a_i are nonzero because the two vertices are distinct. They span S , since the homogeneous vertices of a full-dimensional realization span K^{d+1} .

The change of affine chart to $\sum_i H_i = 1$ is legitimate on the polytope: this functional is positive on every original vertex, hence throughout the polytope. A projective map whose denominator has one positive sign there preserves convex combinations after renormalizing their positive weights. It therefore preserves the labeled face lattice. The normalized realization is exactly a Lawrence polytope $L(a_1, \dots, a_m)$.

Lemma 124.2 reads the covectors of the a_i from its transversal faces. These faces are the same as those of $L(B)$, so the covectors agree. Finally, (124.3) is a nonzero scalar multiple of a_i . Rescaling H_i or either vertex representative only rescales R_i , establishing the asserted projective independence. $\square$

The proof does not assume that an arbitrary realization was presented in Lawrence coordinates. It constructs such coordinates from the labeled facets. This distinction is essential for a lower bound on *all* realizations.

Corollary 124.4 (Arithmetic read from a labeled face lattice). *Every ordered-field realization of $\mathcal{L}_k$ determines a recovered projective frame and number-line points $X_0, \dots, X_k$. The intrinsic cross-ratios*

$$x_i = \chi(X_i; O, I, U)$$

are defined, are nonzero, and satisfy $x_i = x_0^{2^i}$.

Proof. The recovered configuration has the same covectors, hence the same projective incidences and nonincidences, as $\mathcal{B}_k(t)$. Undoing its affine chart is a projective normalization, so the frame and all nondegenerate intersections needed in Lemma 122.1 are present. That lemma gives the equations. The original number-line points are distinct from O and U ; the same nonincidences hold in the recovered configuration, giving nonzero cross-ratios and nonzero denominators. $\square$

125 A shadow that keeps every vertex and the full dimension

Define

$$P_k(t) = L(\mathcal{B}_k(t)), \quad Q_k^0 = (p_1^{-,0}, p_1^{+,0}, \dots, p_m^{-,0}, p_m^{+,0}),$$

where the limiting vertices are

$$p_i^{-,0} = (0, e_i), \quad p_i^{+,0} = (b_i^0, e_i).$$

As before, the tuple is regarded as an ordinary d -dimensional affine one by omitting the redundant last z coordinate.

Proposition 125.1 (Noncollapsing shadow). *The polytope $P_k^0 = \text{conv}(Q_k^0)$ has dimension $d = m + 2$ and exactly $2m$ distinct vertices. Every pair-complement F_i^0 is still a facet. However, its labeled face lattice is not $\mathcal{L}_k$.*

Proof. Apply Lemma 124.1 to the limiting vector configuration, using Proposition 123.1. Coincident base columns do not produce coincident Lawrence vertices: different pairs have different e_i coordinates, and the two vertices in any pair differ by the nonzero vector b_i^0 .

To see a lost face explicitly, consider the covector obtained by applying $c(x, y, z) = x$ to $\mathcal{B}_k(t)$. Its entry at O is zero and its entry at X_0 is positive. The associated transversal face contains both vertices of the O pair and only the upper vertex of the X_0 pair. At the shadow, $b_{X_0}^0 = b_O$, so no functional can have zero value on the latter and positive value on the former. The same labeled transversal face cannot exist at the shadow, by Lemma 124.2. $\square$

This is a useful separation. The configuration degeneration is visible in the polytope's incidences, but not in its dimension or its number of vertices. [[write] Compare Part II's lost face (14.2), where a vertex is lost at the shadow; here no vertex is lost and the face lattice still changes.] Requiring those two elementary invariants to be preserved does not make the degree problem easy.

Lemma 125.2 (Regularity of recovery at the shadow). *Let $Q(s)$ be any continuous vertexwise deformation tending to Q_k^0 that realizes $\mathcal{L}_k$ for all sufficiently small $s > 0$. Its recovered observable $x_0(s)$ tends to zero. This conclusion does not require rationality of the deformation.*

Proof. At the shadow, each F_i^0 has affine dimension $d - 1$. Choose d of its homogeneous vertex representatives that are linearly independent there. Their cofactor normal remains nonzero for nearby tuples and varies continuously. It gives the facet normal of $F_i(s)$ whenever the tuple realizes $\mathcal{L}_k$.

The values of this normal on the two omitted vertices are nonzero at the shadow: in the displayed Lawrence coordinates the facet is $z_i = 0$ and the omitted values are both one, up to a nonzero normalizing scalar. Consequently (124.3) varies continuously and remains nonzero, including at the shadow. The recovered fixed frame is nondegenerate there. On the recovered number line, the labels O, I, U are distinct at the limit, while X_0 tends to O . Choose two linear coordinates independent on that limiting number line. The denominator in (121.1) is then nonzero at the limit, while its numerator vanishes. Thus $x_0(s) \rightarrow 0$. $\square$

Continuity here is checked in a fixed affine chart containing the limiting vertices. Homogeneous representatives that happen to vanish at $s = 0$ may be replaced locally by normalized representatives; the recovered projective points and cross-ratio do not depend on that choice. The survival of the pair-complement facets and of the projective frame is what makes this argument regular. A singular recovery map at the boundary would not justify the same conclusion.

126 From contact order to a degree lower bound

126.1 Degree of the recovery map

The arithmetic relation alone does not prove a lower bound for coordinates of a polytope. One must estimate how much complexity can be introduced when recovering the arithmetic from those coordinates. The following calculation supplies that missing step.

Lemma 126.1 (Cofactor degree bound). *Suppose a rational vertex arc realizes a Lawrence face lattice as in Theorem 124.3, and every vertex has a homogeneous polynomial representative of degree at most D . Then every recovered point has a homogeneous polynomial representative of degree at most $(d + 2)D$. Every cross-ratio of four recovered collinear points can be written as $p(s)/q(s)$ with*

$$\deg p, \deg q \leq 4(d + 2)D, \quad (126.1)$$

provided that the cross-ratio is defined.

Proof. Work over the ordered field of rational germs $\mathbb{R}(s)$, ordered by their values for sufficiently small positive s . A pair-complement facet has homogeneous rank d . Select d independent homogeneous vertex rows in it, considered over $\mathbb{R}(s)$. Each row has polynomial coordinates of degree at most D . A normal H_i is given by the signed $d \times d$ minors of this $d \times (d + 1)$ matrix. Every coordinate of H_i has degree at most dD . The vector is nonzero over $\mathbb{R}(s)$, and its sign can be oriented correctly on a sufficiently small positive interval.

Thus $H_i(V_i^\pm)$ has degree at most $(d + 1)D$. The explicit recovery formula

$$R_i = H_i(V_i^-)V_i^+ - H_i(V_i^+)V_i^-$$

has coordinate degree at most $(d + 2)D$. We do not divide by facet values or solve a normalized frame here; avoiding such divisions keeps the estimate transparent.

For four recovered points on one line, select two ambient coordinate positions whose projection is independent on that line over $\mathbb{R}(s)$. Such positions exist because the line has vector dimension two. In those two coordinates every bracket has degree at most $2(d + 2)D$. Formula (121.1) is the ratio of two products of two brackets, which proves (126.1). A projection may fail at finitely many real parameter values, but shrinking the punctured interval removes them. Its generic formula is the same intrinsic rational cross-ratio. $\square$

The choices made in this proof need not be nondegenerate at $s = 0$. They need only give polynomial representatives of bounded degree over $\mathbb{R}(s)$. The separate boundary argument in Lemma 125.2 determines the limiting value of the intrinsic cross-ratio. Confusing these two purposes would hide a possible boundary singularity.

Lemma 126.2 (Order of a rational function). *If a nonzero rational function $f = p/q$ has $\deg p, \deg q \leq B$, then*

$$-B \leq \text{ord}_0 f \leq B.$$

If $f(s) \rightarrow 0$ as $s \downarrow 0$, then $\text{ord}_0 f$ is a positive integer.

Proof. For nonzero polynomials, $0 \leq \text{ord}_0 p \leq \deg p$ and $0 \leq \text{ord}_0 q \leq \deg q$. Subtract these orders. Any common polynomial factor cancels from both orders and cannot invalidate the estimate. A rational function tends to zero precisely when the difference of the two orders is positive. $\square$

Theorem 126.3 (A local bound independent of the chosen coordinates). *Let $Q(s)$ be a rational realization arc of $\mathcal{L}_k$ of homogeneous degree D . If its recovered cross-ratio $x_0(s)$ tends to zero, then*

$$D \geq \left\lceil \frac{2^k}{4(d+2)} \right\rceil.$$

More precisely, if $a = \text{ord}_0 x_0$, then

$$4(d+2)D \geq 2^k a. \quad (126.2)$$

Proof. Corollary 124.4 gives the exact identity $x_k = x_0^{2^k}$ in $\mathbb{R}(s)$. Since x_0 is nonzero and tends to zero, Lemma 126.2 gives $a = \text{ord}_0 x_0 \geq 1$. Additivity of orders yields

$$\text{ord}_0 x_k = 2^k a \geq 2^k.$$

On the other hand, Lemma 126.1 represents x_k as a quotient of polynomials of degrees at most $4(d+2)D$. Lemma 126.2 then gives $\text{ord}_0 x_k \leq 4(d+2)D$. This proves both statements. $\square$

126.2 Completion of the main theorem

Proof of Theorem 120.3. The counts are (122.3) and Lemma 124.1. Lemma 122.3, the chart normalization, and Lemma 124.2 prove constancy of the labeled face lattice on $0 < t < 1/6$. Proposition 125.1 proves that the limit keeps all vertices and the dimension.

To prove the upper degree bound, let (X, Y, Z) be a displayed homogeneous vector of $\mathcal{A}_k(t)$ and let $\ell = X + 2Y + Z$. A homogeneous representative of its upper Lawrence vertex is

$$(X, Y, \ell, \ell e_i) \in K^{m+3}. \quad (126.3)$$

The lower vertex is $(0, e_i)$. Every entry of (126.3) has degree at most 2^k . For the fixed point W , the chart value is negative, so one can multiply that one representative by -1 to use a common positive chart; its degree is unchanged. All denominators in the affine formulas have nonzero limits, so this is a rational vertex arc with the asserted limit and degree at most 2^k .

For any competing rational arc with that labeled limit, Lemma 125.2 gives $x_0 \rightarrow 0$. Theorem 126.3 applies, and

$$4(d+2) = 4(4k+12) = 16k+48.$$

This gives the lower bound in (120.1) and proves that it applies to all admissible rational arcs.

Finally, substituting any positive surreal infinitesimal for t preserves every determinant sign by Lemma 122.3. The rational coordinate formulas are finite and their standard parts equal the limits at $t = 0$. Lemma 121.1 identifies the real convex-hull shadow. $\square$

126.3 What the estimate rules out

For this family, a method that always replaces a bounded surreal polytope by a rational arc of degree polynomial in the number of vertices and the dimension cannot preserve both the labeled face lattice and the specified labeled shadow. This conclusion already applies to polynomial-coordinate arcs, since these are a subclass of rational arcs.

The estimate is not limited to paths keeping the lower Lawrence vertices fixed, nor to paths preserving the displayed affine chart or projective frame. A competing path may move every vertex. Its facet equations still reconstruct the arithmetic, and their degrees are included in the bound. Likewise, a rational reparametrization cannot bypass the order argument: its

nonconstant vanishing rational observable still has positive integer order. The refined bound (126.2) shows that making x_0 vanish to higher order only increases the obstruction.

Here are representative bounds. The table gives proven bounds, not the exact minima.

k	Vertices N	Dimension d	Lower bound for δ_{rat}	Upper bound
4	48	26	1	16
8	80	42	2	256
12	112	58	18	4 096
16	144	74	216	65 536
20	176	90	2 850	1 048 576
24	208	106	38 837	16 777 216

The factor linear in k between the two bounds comes from a uniform cofactor estimate. No sharpness claim for that factor is made.

Remark 126.4 (Configuration-level comparison). If one works directly with rational arcs of the planar point configuration, rather than polytopes, the same cross-ratio estimate uses brackets of the original homogeneous points. Degree D then gives numerator and denominator degrees at most $4D$, and the lower bound is $D \geq \lceil 2^k/4 \rceil$. The factor $d + 2$ in the polytope theorem is the cost of recovering points from arbitrary polytope coordinates. This is why merely displaying high-degree coordinates would not have proved the polytopal statement.

127 Surreal valuations and exact ramification

127.1 Intrinsic scale equations

Let v be an additive valuation into an ordered abelian group, with $v(ab) = v(a) + v(b)$. For a field of finite and infinite surreal quantities, we may use the natural valuation relative to the real constants, restricted to a set-sized field containing the construction. Positive valuation means infinitesimal size. No choice of numeric norm is needed.

Theorem 127.1 (Forced contact ratios). *In any ordered-field realization of $\mathcal{L}_k$ equipped with such a valuation, with the recovered observables defined,*

$$v(x_i) = 2^i v(x_0).$$

For a rational arc approaching Q_k^0 , the corresponding integer orders satisfy

$$(\text{ord}_0 x_0, \dots, \text{ord}_0 x_k) = a(1, 2, 4, \dots, 2^k) \quad \text{for some integer } a \geq 1.$$

Proof. Apply the valuation, or the order at zero, to the exact equations $x_i = x_0^{2^i}$. The positivity of a for the specified shadow was proved in Theorem 126.3. $\square$

The vector $(1, 2, 4, \dots, 2^k)$ is independent of the rate at which a rational arc approaches the boundary, up to its common positive integer factor. It records *contact depth*: how rapidly each recovered observable vanishes compared with the initial one. It does not count the rank of a valuation group.

For example, take $\varepsilon = \omega^{-1}$ and substitute $t = \varepsilon$. Then

$$x_i = \omega^{-2^i}.$$

These $k + 1$ observables have exponentially increasing contact orders, but all their valuations lie in one cyclic ordered subgroup. The phenomenon therefore already occurs at valuation rank one. No tower of independent Archimedean classes is required. Nor are there 2^k separate observables: there are $k + 1$, with largest relative order 2^k .

127.2 Normalizing the deepest observable

Let

$$\mathbb{R}\langle\langle\tau\rangle\rangle = \bigcup_{e \geq 1} \mathbb{R}((\tau^{1/e}))$$

be the field of real formal Puiseux series, with $v(\tau) = 1$. Here the ramification denominator of a series means a positive integer e for which all its exponents lie in $(1/e)\mathbb{Z}$.

Theorem 127.2 (Exact ramification after output normalization). *Suppose the deepest observable is used as the parameter, so that $x_k = \tau$, and the initial observable is positive. Then*

$$x_0 = \tau^{1/2^k}.$$

Any Puiseux field $\mathbb{R}((\tau^{1/e}))$ containing the coordinates of such a realization has $2^k \mid e$. Conversely, the explicit construction is realized over $\mathbb{R}((\tau^{1/2^k}))$ after substituting $t = \tau^{1/2^k}$.

Proof. The forced identity is $x_0^{2^k} = \tau$. The ordered positive root is unique, hence is the displayed monomial. Alternatively, its valuation is $1/2^k$. If all vertex coordinates lie in $\mathbb{R}((\tau^{1/e}))$, the rational recovery formulas put x_0 in that same field. Its valuation belongs to $(1/e)\mathbb{Z}$, so $1/2^k = n/e$ for an integer n , forcing $2^k \mid e$. The converse follows by substituting the monomial root in the explicit rational formulas, whose denominators have nonzero constant terms. $\square$

This is a normalized statement, not an absolute obstruction to using an unramified parameter. With x_0 itself as parameter, the construction is rational and unramified; its high powers carry the complexity. With x_k as parameter, those same powers become fractional exponents and the complexity appears as ramification. The two descriptions are compatible, not competing invariants of an unparametrized curve.

127.3 A simple moduli direction can have high contact depth

The displayed arithmetic configuration is generated from its frame and one scalar t by successive unique intersections. Once the projective frame is normalized, there is no additional freedom in these generated points. Yet the output observable is t^{2^k} . Thus a high contact order does not require a complicated parameter-space topology: it is already present on this explicit one-parameter arithmetic branch.

For the displayed Lawrence polytopes, a label-preserving projective equivalence $P_k(t) \cong P_k(u)$ would induce an equivalence of their recovered configurations. Corollary 122.2 therefore gives $t = u$. The varying scale is not an artifact of applying projective coordinate changes to one fixed labeled polytope. This does not classify all extra realization freedoms of the full Lawrence polytope.

128 Addition chains and programmable contact depth

Repeated squaring is one particularly efficient arithmetic circuit. The same construction encodes arbitrary powers with a size controlled by an addition chain.

Definition 128.1. An addition chain for $q \geq 2$ is a strictly increasing sequence

$$1 = a_0 < a_1 < \cdots < a_\ell = q$$

such that each a_j is a sum $a_{p_j} + a_{r_j}$ with $p_j, r_j < j$. Its length is ℓ .

The terminology refers to additions of exponents: the geometric circuit multiplies the represented quantities. Let $y_j = t^{a_j}$. Keep the same six fixed points and the initial pair X_0, A_0 . For gate j introduce

$$\begin{aligned} B_j &= (1 : y_{r_j} : 1), & C_j &= (y_{p_j} : y_{p_j} y_{r_j} : 1), \\ A_j &= (0 : y_j : 1), & X_j &= (y_j : 0 : 1). \end{aligned} \tag{128.1}$$

These are forced by

$$B_j = A_{r_j} U \cap IV, \quad C_j = OB_j \cap X_{p_j} V, \quad A_j = C_j U \cap OV, \quad X_j = A_j W \cap OU.$$

The coordinate proof is the same as in Lemma 122.1, with two possibly different inputs. It gives $y_j = y_{p_j} y_{r_j}$.

Some B_j labels may represent the same projective point when an input is reused. We may retain all these labels. A labeled vector configuration is allowed to have parallel columns, and Lawrence vertices from different labels are still distinct because their e_i coordinates differ. The fixed frame preserves the no-coloop condition. Therefore retaining four new labels per gate gives exact counts without requiring a generic perturbation that would destroy the arithmetic.

Theorem 128.2 (Addition-chain shadow bounds). *For every addition chain of length ℓ ending at q , the above construction gives a labeled polytope family with*

$$N = 8\ell + 16, \quad d = 4\ell + 10$$

and a rational labeled shadow retaining all vertices and the full dimension. Its recovered observables satisfy $y_j = y_0^{a_j}$, and

$$\left\lceil \frac{q}{16\ell + 48} \right\rceil \leq \delta_{\text{rat}}(\mathcal{L}, Q^0) \leq q. \tag{128.2}$$

The entire explicit face lattice is constant on $0 < t < 1/6$. Under the normalization $y_\ell = \tau$, every Puiseux ramification denominator of a coordinate field is divisible by q , and denominator q suffices.

Proof. All coordinates before the affine chart are still zero or signed monomials. Lemma 122.3 therefore applies unchanged, including to identically zero determinants from coincident labels. The largest coordinate degree is q .

The unique-intersection construction gives $y_j = y_0^{a_j}$ by induction on the chain. Lawrence recovery reads those same incidences from any realization of the labeled face lattice. The limiting columns are again O for axis and interior points and I for the B_j points, while the fixed frame survives. Hence the recovery is regular at the labeled shadow and $y_0 \rightarrow 0$ there.

For a competing rational arc of degree D , write $a = \text{ord}_0 y_0 \geq 1$. Then $\text{ord}_0 y_\ell = qa$, whereas Lemma 126.1 bounds this order by $4(d+2)D$. This proves the lower bound. Formula (126.3) gives the upper bound q . The ramification statement follows from $y_0^q = \tau$ exactly as in Theorem 127.2. $\square$

Example 128.3 (A non-power-of-two output). The chain

$$1, 2, 3, 5, 10, 13$$

has length five. It yields a 30-dimensional polytope with 56 vertices whose final recovered observable is y_0^{13} . When that output is the parameter, ramification denominator 13 is forced. The general lower bound in (128.2) is weak for this small example; the example illustrates programmability, not a numerically large degree obstruction.

Within multiplication circuits using at most ℓ gates and starting from the exponent one, no exponent exceeds 2^ℓ : at each gate the largest previous exponent can at most double. Repeated squaring reaches this maximum. This is an optimality statement *within this circuit model*, not an optimal bound for all projective incidence constructions. A binary expansion of any q gives an addition chain of length at most $2\lfloor \log_2 q \rfloor$, so arbitrary integer contact depths can be encoded with $O(\log q)$ vertices and dimension.

There is also an immediate valuation interpretation of each gate:

$$v(y_j) = v(y_{p_j}) + v(y_{r_j}).$$

Thus multiplication circuits give explicit linear constraints on the valuations of recovered observables. The results here concern this multiplicative class; they do not assert a universality theorem for all tropical or valuated-matroid realization spaces.

129 Why the fixed-base positive result does not extend

129.1 Linear signs versus incidence equations

For heights $h_1, \dots, h_n$ on a fixed real base configuration, a lifted minor containing the height row is linear in h . The finite collection of height sign constraints therefore cuts out a relatively open polyhedral region inside a linear subspace. If h^0 belongs to its closure and h^1 belongs to the region, then

$$h(s) = (1-s)h^0 + sh^1, \quad 0 < s < 1,$$

stays in the region: an equality holds along the segment, and a required strict positive linear form is nonnegative at h^0 and positive at h^1 . This is the elementary reason a degree-one finite-data deformation is possible in the fixed-base setting discussed in [85].

Moving projective points are different. Their determinant equalities can express nonlinear arithmetic. Even the graph $y = x^2$ does not admit the same segment argument. The segment from $(0,0)$ to (a,a^2) has coordinates (sa, sa^2) , whereas the square of its first coordinate is s^2a^2 . The defining equality fails for $0 < s < 1$ when $a \neq 0$. Our configuration iterates this obstruction and puts it into a face lattice that can be read from arbitrary polytope coordinates.

[[write] The fixed-base result discussed is Part I's [theorem 10.4](#); its segment proof is the argument just given.]

The prior degree-one result concerns a very special linear slice of configuration space. The present theorem concerns arbitrary moving vertices and exact incidence constraints. There is no contradiction. The comparison identifies the real-base hypothesis as a structural boundary, not merely a convenient assumption.

129.2 Transfer and curve selection remain valid

Without a prescribed shadow, a real realization of the finite combinatorics is easy here: substitute $t = 1/7$. More generally, finite polynomial sign conditions realized over a real closed extension are realizable over $\mathbb{R}$, and over the real algebraic numbers when no nonalgebraic parameters are prescribed. This classical transfer principle is compatible with every lower bound above. [[write] In this report it is Part I's [theorem 18.1](#).]

Similarly, semialgebraic curve selection produces an arc into a semialgebraic realization locus from a point of its closure; quantitative versions control suitable descriptions of such paths [81]. They do not promise a linear rational parametrization. The present examples are unusually explicit because rational arcs exist, but their minimum degree is exponentially large

in a linear-size configuration. A bound for the algebraic description of a semialgebraic arc and a bound for the degree of a rational parametrization are different assertions.

129.3 Three kinds of complexity that must not be conflated

Number of observables versus order. The repeated-squaring example has $k+1$ distinguished observables, not exponentially many. Its largest relative contact order is 2^k .

Degree versus expression length. The expression t^{2^k} has a straight-line program of length k . All points and vertices can be generated by a short arithmetic circuit. The degree lower bound is therefore not a lower bound on the time needed to evaluate a compressed description. Expanding powers into ordinary polynomial coordinate data is precisely the operation that can become expensive.

Vertices versus a complete face-list input. The bound is exponential in the number of vertices along the constructed family. It is not asserted to be exponential in the length of an input that explicitly lists every face. The number of faces of a high-dimensional polytope can itself be large. No general computational-complexity lower bound for algorithms reading such a list is claimed.

129.4 Essential hypotheses and unproved strengthenings

The lower bound uses a *labeled* face lattice and a *labeled* shadow. If the shadow is discarded, a constant rational realization has degree zero. If only an unlabeled limiting polytope is prescribed, the frame and the arithmetic labels may need separate reconstruction, which is not established here. A constant permutation of labels can be handled once its effect on the marked frame is known, but that is not the same as a label-free theorem.

The incidences are exact. Generic perturbations may remove the collinearities that force multiplication. We do not claim that these examples are simplicial, simple, or uniformly realizable with the same obstruction. Their dimension grows with k . We also do not claim a fixed-dimensional version, an exact value of δ_{rat} , or a universal upper bound for arbitrary realization spaces. These are specific further questions rather than consequences silently folded into the main theorem.

130 Exact checks and a route to formal verification

130.1 What the accompanying program verifies

The accompanying `verify.py` [write] (shipped as `code/11-contact-depth-verify.py`) uses sparse integer polynomials to build the homogeneous projective coordinates. It expands every 3×3 minor, checks the listed incidence identities, verifies the six-term coefficient bound, and compares the leading-coefficient sign with the exact sign at $t = 1/7$. It also checks that a fixed rank-three witness survives every single-column deletion.

For $k = 1, \dots, 6$, the run supplied in `verification.json` [write] (shipped as `data/11-contact-depth-verification.json`) checked 12 180 minors in total. The signs in the next table are those after the affine-chart normalization, in the label order used by the program. Zeros are exact polynomial identities, not values that happen to be small at a numeric sample.

k	Points m	Minors	Zero	Positive	Negative	Max. coefficient ℓ^1
1	12	220	29	113	78	3
2	16	560	59	272	229	4
3	20	1 140	102	534	504	6
4	24	2 024	161	925	938	6
5	28	3 276	239	1 471	1 566	6
6	32	4 960	339	2 198	2 423	6

A separate exact rational test treats $k = 1$. It mixes all homogeneous ambient coordinates by an invertible projective transformation, recovers the twelve pair-complement facet normals by nullspaces, reconstructs the projective points using (124.3), and obtains the cross-ratios

$$x_0 = 1/7, \quad x_1 = 1/49.$$

This specifically tests the coordinate-independent recovery rather than only the original arithmetic formulas. The test uses SymPy; the sparse minor checks use Python's integer arithmetic and standard library.

These checks are not an enumeration of the full face lattice, not a test of every possible rational path, and not a proof of the infinite family. The proofs above establish those general claims. The code is a finite consistency check of formulas, signs, counts, and a sample recovery map.

130.2 A dependency-ordered Lean development

[[write] A proposal; nothing of this Part is formalized. The repository file named below, `Algebra/SurrealNumbers/Surreal/Algebra/Geometry.lean`, is unchanged between the pin and the current tree and does contain the finite coordinate geometry described (`Surreal.Complexify.dot`, `cross`, `ptolemy`, `heron_factorization` and their relatives); it contains no polytope statement of this Part.]

The repository already organizes finite coordinate geometry over an ordered base field rather than silently substituting real-valued metrics [83]. The present results admit a similar layered design. A practical formalization should begin with the generic finite algebra, not with class-sized surreal objects.

Layer 1: determinants and gates. Represent plane points by nonzero triples and lines by nonzero covectors, with incidence given by their pairing. Prove the coordinate identities for the frame and each multiplication gate. Polynomial normalization and nonzero hypotheses are the main obligations; all arithmetic equalities reduce to finite ring identities.

Layer 2: finite Lawrence faces. Define the two vertices over each label. Prove their affine rank, the vertex property, and the pair-complement facets. Establish the transversal face dictionary directly from the two functional values. A formal bridge between maximal-minor signs and covectors can be imported from a suitable verified linear-feasibility development or proved separately. This bridge is a real prerequisite, not something to replace with numerical tests.

Layer 3: recovery. Prove independence of the facet normals by evaluation at omitted vertices. Construct the splitting of their combined linear map. Establish that the recovered projective vectors are nonzero and that rescalings do not change the cross-ratios. Treat the common positive affine chart explicitly.

Layer 4: polynomial degrees and orders. Formalize the degree bounds for minors, evaluations, recovered vectors, and brackets. Define the order at zero of a rational function and prove its degree bound and multiplicativity. These lemmas yield Theorem 126.3 without requiring analysis or surreal normal forms.

Layer 5: the boundary and surreal specialization. Prove regularity at the shadow by fixed nonvanishing minors and a nonvanishing cross-ratio denominator. For the explicit surreal specialization, prove the finite standard-part formulas and apply the convex-hull lemma. The valuation and ramification corollaries can then be added with their normalization hypotheses visible.

Formal work on polyhedral face lattices over ordered fields exists in Coq [82], but it is not a Lean implementation of the present construction. The article supplies a dependency plan, not untested Lean files advertised as checked. No new statement in this package is claimed to have passed a theorem prover.

131 Further research questions and concrete directions

Question 131.1 (Close the degree gap). What is the exact value of $\delta_{\text{rat}}(\mathcal{L}_k, Q_k^0)$, or its asymptotic value up to an absolute constant? The present bounds differ by a factor $16k + 48$. A sharper recovery expression may improve the lower bound; conversely, moving the frame and the paired vertices may reduce the degree of an optimal path. Any proposed compression must still reproduce the intrinsic order 2^k after recovery.

Question 131.2 (Keep the ambient dimension fixed). Can one obtain an explicit exponential rational shadow-degree lower bound in one fixed dimension, while retaining all vertices and that dimension at the shadow? The Lawrence construction used here increases dimension linearly with the arithmetic circuit. A fixed-dimensional incidence encoding would need a quantitative reconstruction map that remains regular at the prescribed boundary, not merely a qualitative realization-space simulation.

Question 131.3 (Remove labels or recover them combinatorially). Can the frame, the paired vertices, and the number-line labels be made intrinsically recognizable in the unlabeled face lattice with only a linear increase in size? One approach is to attach asymmetrical combinatorial markers, then check that these markers do not destroy the regularity of the shadow or hide large coordinate-degree costs. The result would replace the marked degeneration invariant by an unmarked one.

Question 131.4 (Replace exact incidences by robust inequalities). Is there an analogue for uniform oriented matroids, or for simple or simplicial polytopes, in which contact depth is forced by strict inequalities rather than exact arithmetic equations? A direct generic perturbation is insufficient: it erases the multiplication identities. A viable construction would have to control errors relative to the output scale, not merely make them small in absolute value.

Question 131.5 (Compare rational degree with semialgebraic description degree). For a labeled realization locus and a boundary tuple, compare the minimum degree of a rational vertex arc, when finite, with the minimum algebraic description degree of a semialgebraic arc. Curves of positive genus show why rational parametrization should not be assumed automatically. The quantitative framework of [81] suggests a route to upper bounds for the second invariant, while the present construction gives explicit lower bounds for the first.

Question 131.6 (Classify regular recoverable shadows). The proof uses two boundary conditions: every pair-complement facet keeps its rank, and the recovered frame stays nondegenerate. Can these be replaced by a necessary-and-sufficient combinatorial or determinantal criterion for regularity of Lawrence recovery at a labeled shadow? Such a criterion would let one transfer many other arithmetic circuit obstructions to polytopal degenerations with controlled constants.

Question 131.7 (Several inputs and valuation constraints). What systems of linear valuation relations and inequalities can be realized by finite multiplication-and-addition incidence circuits with controlled polytopal size? Multiplication gives exact addition of valuations. Ordinary addition introduces cancellation and a minimum rule, so signed leading coefficients must be tracked as well. A useful next target is a theorem with an explicit circuit-to-polytope map and an explicit description of the admissible valuation region.

Question 131.8 (Optimize geometric circuit size). Let $c(q)$ be the least number of additional labeled projective points needed, in a specified gate model and with a fixed frame, to force a recovered output x^q . How does $c(q)$ compare with ordinary addition-chain length? The four-label multiplication gate gives a concrete upper bound. Lower bounds must distinguish genuinely generated points from preassigned high-complexity constants and must specify whether coincident labels may be reused.

Question 131.9 (Build a verified compiler with honest output guarantees). Can one implement a compiler from an arithmetic incidence circuit to a labeled Lawrence polytope together with checked certificates for rank, facets, recovery, and contact-order lower bounds? The output should retain both a compressed arithmetic representation and a degree budget for expanded homogeneous coordinates. The distinction prevents a small circuit from being mistaken for a low-degree deformation. The repeated-squaring family is a useful regression test.

Question 131.10 (Beyond rational functions of one infinitesimal). For a fixed finite configuration of general surreal coordinates, when can its relevant incidence data and prescribed shadow be represented in a one-parameter rational field, and when is an algebraic or higher-rank valued field genuinely needed? The present examples need only one infinitesimal. They isolate a degree obstruction from genuinely higher-rank obstructions, which should be studied separately.

132 Conclusion

Finite surreal combinatorics can be real-realizable and still resist low-degree realization *toward a specified shadow*. The explicit family here keeps all vertices and the dimension, has a rational model, and uses only one infinitesimal. What forces its complexity is an exact projective arithmetic relation recovered from the labeled face lattice:

$$x_k = x_0^{2^k}.$$

The cofactor recovery bound converts that relation into an exponential lower bound for every rational vertex arc approaching the given boundary. With another choice of parameter, the same relation becomes an exact ramification obstruction.

The result is best understood as a quantitative use of classical projective arithmetic and Lawrence geometry. Its significance for the surreal-polytope program is a precise boundary: finite sign preservation and real realizability do not imply low-degree shadow realizability once the coordinates themselves move. The special degree-one theorem for height liftings survives because its constraints are linear; it does not extend to general finite polytopes.

133 Proof, provenance, and verification ledger (Appendix A of source 11)

Component	Role and dependencies	Status in this package
Finite real realization	Real-closed-field transfer of finitely many polynomial signs; background only	Classical, not a novelty claim
Projective arithmetic	Frame and unique intersections force multiplication; Lemma 122.1	Classical method; explicit formulas proved here
Lawrence face dictionary	Functional values on each pair; Lemma 124.2	Classical method, with direct proof
Lawrence recovery	Pair-complement facets, independent normals, positive chart, splitting; Theorem 124.3	Classical construction, reproved in the form used
Uniform 1/6 threshold	Six signed monomials in each determinant; Lemma 122.3	Direct bound proved here; finite checks supplied
Noncollapsing shadow	Fixed frame and nonzero limiting columns; Proposition 125.1	Explicit family proved here
Rational degree bound	Recovery degrees plus order of cross-ratios; Theorems 126.3 and 120.3	Main quantitative result proved here; priority unverified
Valuation and ramification	Exact power relation and specified parameter normalization	Corollaries proved here
Addition-chain version	Same recovery and a two-input gate	Generalization proved here
Finite program	12 180 minor checks and one exact transformed recovery	Executed successfully; not general formal verification
Lean extension	Dependency plan in Section 130	Not implemented or checked in this package

The repository inspection used the snapshot

6996fee43cc97b6c16351def7507d59a95bf62f0

and focused on the surreal project guide and the finite geometry source identified in the bibliography. It was not a repository-wide proof audit or a new build of that snapshot. The present proofs are independent of unmapped research claims elsewhere in the repository.

134 Reproduction notes (Appendix B of source 11)

[[write] Shipped names: the program `verify.py` is `code/11-contact-depth-verify.py`, `verification.json` is `data/11-contact-depth-verification.json`, the requirements file is `data/11-contact-depth-requirements.txt` and the build script `code/11-contact-depth-build.sh`. The LaTeX source, PDF and README are not shipped; this Part is the article. The default output of the program is `verification.json` in the current directory, so a rerun must pass an explicit `-output` path outside the report, as its README shows.]

The archive contains the LaTeX source, the compiled PDF, the exact verification program, its JSON output, and a README with build commands. No external paper or font file is bundled. The bibliography is embedded in the LaTeX source so that BibTeX or a separate bibliography database is not required.

For the supplied finite checks, run

```
python verify.py --max-k 6 --output verification.json
```

SymPy 1.14.0 was used for the projective reconstruction test. The option `--skip-reconstruction` omits that dependency and runs only the standard-library sparse-polynomial checks. The program deliberately limits its finite-check parameter to $1 \leq k \leq 12$: expanding and evaluating large exponents can be expensive, although the formulas have short circuit descriptions. That program limit is unrelated to the mathematical theorem, which holds for every integer $k \geq 1$.

Part IX

Omnific integer hulls and the rational-normal dichotomy

Source and scope. [write] This Part prints, whole, manuscript 05 of batch 59, printed here as source 12: *A Rational-Normal Dichotomy for Omnific Integer Hulls: Finite Graver Certificates and Irrational Obstructions in Surreal Polyhedra*, headed “Surreal geometry and discrete optimization”, author line “Research draft prepared for Vladimir Reshetnikov” (the PDF metadata of the delivered file reads “ChatGPT”), dated 30 September 2026 (archive `Omnific_Integer_Hulls_Research.zip`, main file `omnific-integer-hulls/article.tex`; delivered PDF of 28 pages on Letter paper, not shipped). It was written against ProveIt commit 6996fee43; it arrived in commit ed80eba06 and was placed in commit f2cb2034e. Its program, recorded run, recorded console output, build record, build file and two audit notes are shipped as `code/12-omnific-hulls-verify.py`, `data/12-omnific-hulls-verification_results.json`, `data/12-omnific-hulls-verification_output.txt`, `data/12-omnific-hulls-build_report.json`, `code/12-omnific-hulls-Makefile`, `12-omnific-hulls-SOURCES_AND_SCOPE.md` and `12-omnific-hulls-PROOF_AUDIT.md`. Its README and checksum list `SHA256SUMS.txt` are not shipped; they survive in the delivery archive of arrival commit ed80eba06, and the README’s status paragraph is printed below. Every section, result, proof, example, remark, figure, table and research question of the manuscript is printed here, in the manuscript’s order, with two structural changes: its three appendices are printed as the ordinary [sections 149 to 151](#) at the end of this Part, and its bibliography is printed as the list “References of Part IX”. Its formalization section stays in this Part ([section 146](#)). In this Part “this article”, “this draft” and “the manuscript” mean source 12, and “the repository” means ProveIt at the inspected commit.

Relation to the other Parts. Source 12 treats integer *hulls* where Part IV ([Part IV](#)) counts lattice points, and it uses Part IV’s omnific interface: the first sentence of [lemma 137.1](#) is Part IV’s bounded-element property ([47.1](#)), which is the Lean theorem `Surreal.Foundations.SignSequence.omnific_bounded_iff` ([section 146](#) lists the other Lean declarations of the integer-part interface). Its graph-Graver argument ([lemma 139.1](#)) is the ordinary-integer case of Part I’s finite Graver certificate ([lemma 17.1](#)): the same Dickson finiteness, applied to the graph lattice of $[A - I_m]$ rather than to Part I’s lattice of affine dependences. Its same-matrix descent theorem ([theorem 142.4](#)) is the integer-hull instance of the report’s “no new finite face lattices” spine, which Parts I, IV and V state for finite vertex lists ([theorem 18.1](#), [proposition 48.1](#),

theorem 74.3): the omnific integer hull of a bounded rational-normal polyhedron is a finite polytope whose face lattice already occurs for an ordinary integer right-hand side with the same matrix. For integer hulls, source 12 partly answers Part IV’s question “Other integer parts and unbounded geometry” (**section 58**): arbitrary integer parts are treated in **corollary 141.4**, infinite coordinates and right-hand sides in **theorems 141.2** and **142.4**; the counting half remains open (its question Q10 in **section 147**). Its nearest relatives outside this report are listed after its own statement of prior work in **section 136**. The positive theorems hold for an arbitrary integer part of an arbitrary ordered field (**corollary 141.4**); the obstruction theorems of **section 143** use the surreal splitting $\mathbf{Oz} = \Pi \oplus \mathbb{Z}$, and source 12 warns that they should not be extended to every integer part.

Notation in this Part. [write] The following symbols mean something different elsewhere in the report (**section 1.3**); the Part-local meaning is the one in force here.

- $\mathbf{Oz}$ is the ring of omnific integers, as in Part IV; Π is the real vector space of purely infinite normal forms (not a product sign), ct the constant term and $\lfloor \cdot \rfloor_{\mathbf{Oz}}$ the omnific floor.
- $A \in \mathbb{Z}^{m \times d}$ is a fixed *constraint matrix*: its m rows are the normals of the inequalities $Ax \leq b$. It is not Part I’s base configuration A (whose columns are points, homogenized to $\tilde{A}$), and m is not Part I’s comparison rank or Part III’s number of bodies.
- $G_A \subset \mathbb{Z}^d$ is the first-coordinate projection of the Graver basis of $[A - I_m]$, the conformally minimal vectors (g, Ag) of the graph lattice (**139.1**). The tempting false reading is that it is Part I’s G_A , the Graver basis of $\Lambda = \ker_{\mathbb{Z}} \tilde{A}$: the two are Graver bases of different lattices, finite by the same Dickson argument. Part X’s G_A is a face of a polar moment hull, a third object.
- $h \in \Pi^m$ is the purely infinite part of the floored right-hand side $\beta = h + z$, and $h(A)$ in **corollary 140.4** is a number of lines; neither is Part I’s height vector h .
- ε is an arbitrary positive infinitesimal, except in **proposition 143.6**, where $\varepsilon = \omega^{-2}$; it is never the report’s fixed ω^{-1} in this Part.
- $L_a = a \cdot \mathbb{Z}^d$ is an additive subgroup of $\mathbb{R}$; it is not Part III’s scale lattice $L(P)$ or Part IV’s count $L_P(n)$. T is the set of small-slack rows in **lemma 141.1** and an omnific quotient in $B = 6T + r$; r is a slack offset, a real right-hand side or a residue in $\{0, \dots, 5\}$; H is a set-sized $\mathbb{Q}$ -subspace of Π (with $D_H = H \oplus \mathbb{Z}$), a tail slack H_{jI} , or a purely infinite parameter; $D_I = \det A_I$ and D is an integer part. None of them is Part III’s or Part IV’s T , H or S .
- “Bounded” means bounded by surreal (omnific) bounds, not by a real box as in Part IV’s counting statements. No valuation is used in this Part.
- **Theorem 142.4** (same-matrix ordinary realization of a face lattice) is not Part IV’s trace-preserving algebraic descent (**theorem 50.1**); both source labels are `thm:descent`, and the sub-prefixes keep them apart.
- Macro change only: source 12’s `\argmax` (defined but not used in its text) is the report’s `\Argmax`; the printed output is unchanged.

Abstract of source 12. A bounded polyhedron over the surreal numbers may contain a proper class of omnific integer points. Nevertheless, when its facet inequalities have fixed ordinary rational normals, its omnific integer hull is generated by a uniformly bounded finite collection of points. We prove an explicit formula: every vertex is an intersection of a basis of

the original hyperplanes after subtracting a bounded ordinary integer slack vector. The slack bounds are computed from one finite ordinary Graver test set. This also gives a universal rational normal-fan refinement, finite tail-and-residue tests for the complete face lattice, and a same-matrix descent theorem: every such omnific face lattice already occurs for an ordinary integer right-hand side.

The rationality restriction is sharp in a universal sense. For a fixed real matrix, finite generation and attainment for all boxed surreal right-hand sides hold exactly when each nonzero row is proportional to a rational vector. A single irrational normal produces a bounded example whose linear objective has neither a maximum nor a supremum in the surreal order. We give full proofs, an exact six-residue knapsack calculation, executable finite checks, a formalization dependency plan, and twelve further research questions.

Status and provenance (title page of source 12). This is an AI-assisted, unrefereed research manuscript. The arguments are proved here at the mathematical level; no Lean verification or independent priority certification is claimed. The questions settled are formulated in this manuscript, not advertised as resolutions of a named long-standing conjecture. Classical Presburger and Graver machinery is credited explicitly. Repository statements are used only in the scopes identified in Section 136.

Status (README of source 12). This is an AI-assisted, unrefereed mathematical research draft. Proofs are provided in the manuscript, but no independent correctness or priority certification is claimed. It does not purport to settle a named long-standing conjecture. Classical Presburger and Graver theory is explicitly credited.

No new Lean formalization is included or claimed. Repository scope was inspected, not independently rebuilt. The executable checks are exact finite ordinary-integer tests, not a general implementation of surreal arithmetic and not a proof-assistant verification of the theorems.

135 The problem and the main conclusions

There are two very different ways to ask whether a surreal polytope has new combinatorics. One can study the convex hull of a prescribed finite point configuration. Alternatively, one can intersect a bounded polyhedron with the omnific integers and then take its convex hull. The second operation is not a finite construction at the outset. In fact, a bounded interval of infinite length already contains a proper class of omnific integers. Classical arguments that begin by enumerating all lattice points therefore do not apply.

This article studies the second question. Throughout, $d, m \geq 1$ are ordinary finite integers. We write $\mathbf{No}$ for Conway's ordered field and $\mathbf{Oz}$ for its omnific integer subring. If $A \in \mathbb{Z}^{m \times d}$ and $b \in \mathbf{No}^m$, put

$$P_A(b) = \{x \in \mathbf{No}^d : Ax \leq b\}, \quad S_A(b) = P_A(b) \cap \mathbf{Oz}^d, \quad \mathcal{H}_A(b) = \text{conv}_{\mathbf{No}} S_A(b). \quad (135.1)$$

The convex hull means *finite* convex combinations with surreal coefficients. Neither a topological closure nor an infinite summation is built into this definition. Empty hulls are allowed.

The important asymmetry is this: the matrix A is ordinary and fixed, whereas b and every optimization objective may be arbitrary surreals. The right-hand side can involve many incomparable magnitudes and normal forms with infinite supports. No one-parameter asymptotic representation is assumed.

135.1 A finite formula, rather than only an existence theorem

The finite set $G_A \subset \mathbb{Z}^d \setminus \{0\}$ defined in Section 139 depends only on A . Set

$$M_i = \max_{g \in G_A} |A_i g| \in \mathbb{N}, \quad \beta_i = \lfloor b_i \rfloor_{\mathbf{Oz}}. \quad (135.2)$$

For an index set $I \subseteq \{1, \dots, m\}$ of size d , A_I denotes its square row submatrix, with rows in increasing order. Define

$$C_A(\beta) = \left\{ v = A_I^{-1}(\beta_I - r) : |I| = d, \det A_I \neq 0, \right. \\ \left. r \in \mathbb{Z}^I, 0 \leq r_i < M_i, v \in \mathbf{Oz}^d, Av \leq \beta \right\}. \quad (135.3)$$

This is a finite list with repetitions removed, even when $S_A(b)$ is a proper class. Its construction does not enumerate lattice points in a large box.

Theorem 135.1 (Finite candidate formula). *If $P_A(b)$ is bounded, then*

$$\mathcal{H}_A(b) = \text{conv}_{\mathbf{No}} C_A(\beta).$$

In particular, every nonempty $\mathcal{H}_A(b)$ is a finite polytope, all its vertices belong to $\mathbf{Oz}^d$, and

$$f_0(\mathcal{H}_A(b)) \leq N(A) := \sum_{\substack{|I|=d \\ \det A_I \neq 0}} \prod_{i \in I} M_i. \quad (135.4)$$

Every surreal linear objective attains its maximum on $S_A(b)$ when this class is nonempty.

A factor $M_i = 0$ contributes no allowed slack for a basis containing that row. Such a row is necessarily zero, so it cannot occur in a nonsingular basis. The theorem includes lower-dimensional and degenerate polyhedra. The proof of finiteness comes first, in Section 140; the exact candidate formula is proved in Section 141. This ordering prevents an implicit assumption that the unknown hull already has finitely many edges.

135.2 What survives, and what fails, beyond rational normals

Say that a nonzero real vector is *rational in direction* if it is a nonzero real scalar multiple of a vector in $\mathbb{Q}^d$. We prove in Section 143 that a fixed real matrix has finite omnific integer hulls for *all* boxed surreal right-hand sides exactly when every nonzero row is rational in direction. The equivalent optimization property allows *all surreal objectives*; it is not restricted to rational objectives.

For example, consider the four-sided bounded surreal polytope

$$P = \{(x, y) \in [-\omega, \omega]^2 : y - \sqrt{2}x \leq \tfrac{1}{2}\}. \quad (135.5)$$

The objective $y - \sqrt{2}x$ has no largest value on $P \cap \mathbf{Oz}^2$. More strongly, its values have no least upper bound in $\mathbf{No}$. Thus its omnific integer hull cannot be generated by finitely many points. This is not a contradiction with boundedness: order completeness and ordinary compactness cannot be imported into the surreal field.

On the positive side, the entire face lattice of $\mathcal{H}_A(b)$ is determined by a finite collection of ordinary congruences and signs of rational linear expressions in the right-hand side. Consequently, for each nonempty bounded instance there exists $\beta' \in \mathbb{Z}^m$ such that

$$\mathcal{H}_A(b) \quad \text{and} \quad \text{conv}_{\mathbb{R}}\{x \in \mathbb{Z}^d : Ax \leq \beta'\}$$

have isomorphic face lattices. The same matrix A works on both sides. The surreal hull may have infinite edge lengths, but it acquires no new finite face-lattice type in this fixed rational-normal family.

136 Repository context and classical antecedents

136.1 The precise connection to ProveIt

The repository was inspected on 30 September 2026. Its recursively listed main tree had revision

6996fee43cc97b6c16351def7507d59a95bf62f0.

The relevant foundations are in `Algebra/SurrealNumbers/`. Its README documents the constructed surreal field, the omnific subring, constant extraction, finite-element rigidity, the omnific floor, and congruence reduction modulo ordinary integers [86]. These are exactly the local algebraic features used below. We do not infer unrelated mathematical conclusions from the repository’s existence or from its broad project descriptions.

The independent bridge is `Logic/PresburgerArithmetic/`. Its README describes an executable Lean Cooper eliminator for ordinary integer arithmetic and an independent Rocq/Coq proof of the normalized one-variable elimination step [87]. The model-theoretic passage from ordinary integers to the additive omnific structure is proved mathematically in Section 138. It is *not* already a kernel-checked consequence merely because these two repository developments are present.

There is also a close prior theme in the repository’s report *Omnific Groups and Lattices*. Its Part III, assembled from a manuscript entitled *Shortest Vectors and Missing Infima over the Omnific Integers*, studies rational flags and failures of lattice minima [88]. Accordingly, we do not claim to originate the general observation that irrationality can obstruct omnific lattice optimization. The objects and conclusions here are different: bounded linear-inequality slices, a universal row-direction characterization, finite vertex certificates, and same-matrix combinatorial descent. The comparison is based on the report’s documented scope, not an independent audit of every proof in its long merged manuscript.

[write] Other reports of this collection treat nearby phenomena, and none of them contains the theorems of this Part: the closest-point theory of *Omnific Groups and Lattices*, Part III (`ogl:lat:thm:closestflag`, `ogl:lat:thm:alltargets`: a quadratic analogue of Theorems 141.2 and 143.5, with a rational-flag criterion); the flag criterion and Pell example of *Hahn–Tate Uniformization at Arbitrary Valuation Rank* (`tate:theta:thm:flag`, `tate:theta:ex:irrational`); the failure of inequational transfer in *Set-Sized Quotients of the Omnific Integers* (`osq:prop:conservative`, witness $x = \sqrt{2}\omega$, $y = \omega$); the elementary embedding $(\mathbb{Z}, L_{\text{rat}}) \leq (\mathbf{Oz}, L_{\text{rat}})$ of *Exponential Relations over Omnific Integers* (`exr:thm:rational`), which is the model-theoretic base of Lemma 138.1; the linear systems with ordinary right-hand sides of *Omnific Integers and Omnific–Diophantine Geometry* (`odg:thm:linear`, equations only); and Part IV of this report (Theorem 55.2, Corollary 55.5).

136.2 Which ingredients are classical

The additive theory used here is the theory of $\mathbb{Z}$ -groups: discretely ordered abelian groups with distinguished least positive element 1 and divisible quotient by $\mathbb{Z}$. Quantifier elimination in the language with ordinary congruences, as recalled by Jeřábek [89], is the reason a fixed ordinary arithmetic certificate continues to work with non-Archimedean parameters. Cluckers’ work [90] places finite piecewise-linear and congruence descriptions of parameter families in the wider theory of Presburger sets.

The ordinary integer-optimization ingredients are Graver test sets and the normal-fan refinement supplied by a zonotope of edge directions. Onn’s monograph [91] develops the

algorithmic theory. Onn and Rozenblit [92], in particular their zonotope lemmas and their Graver edge-direction lemma, provide direct classical antecedents. Those classical facts are not presented as new results. We reconstruct the needed finite arguments in the present setting and identify the additional issues caused by omnific multiplicities and proper-class feasible collections.

The contribution of this draft is the resulting theorem package and its explicit proofs: the passage through fixed finite conformal decompositions, the bounded-slack basis formula, the tail/residue compiler, the exact universal rational-normal boundary, and their combination into a surreal integer-hull theory. A focused source comparison cannot certify that every statement is previously unpublished. Publication-level priority and correctness still call for independent review.

137 Omnific arithmetic and the meaning of boundedness

137.1 The normal-form splitting

Write Π for the real vector space of purely infinite normal forms: its nonzero elements have only strictly positive growth exponents in Conway's normal form. The symbol ω^γ here denotes a Conway monomial. Then

$$\mathbf{Oz} = \Pi \oplus \mathbb{Z} \quad \text{as additive groups.} \quad (137.1)$$

Every $z \in \mathbf{Oz}$ has a unique expression $z = h + n$ with $h \in \Pi$ and $n \in \mathbb{Z}$. Its constant term is $\text{ct}(z) = n$. The crucial order property is

$$h \in \Pi \setminus \{0\} \implies |h| > r \quad \text{for every } r \in \mathbb{R}_{>0}. \quad (137.2)$$

Indeed, the leading nonzero term has positive growth exponent and dominates all ordinary constants. Cancellation in a finite real linear combination can remove leading terms, but the result is either zero or another purely infinite normal form. No negative growth exponent is created.

Lemma 137.1 (Finite omnific elements and ordinary remainders). *An omnific integer bounded in absolute value by an ordinary real is an ordinary integer. In particular, 1 is the least positive omnific integer. For every ordinary integer $n > 0$ and every $z \in \mathbf{Oz}$, there are unique $q \in \mathbf{Oz}$ and $r \in \{0, \dots, n-1\}$ such that $z = nq + r$.*

Proof. The first claim follows immediately from (137.1) and (137.2). For the division assertion, write $z = h + k$ and use ordinary integer division $k = nq_0 + r$. Then $q = h/n + q_0 \in \mathbf{Oz}$. If two remainders occur, their difference is a multiple of n lying strictly between $-n$ and n , which forces equality. $\square$

[write] The first sentence of lemma 137.1 is Part IV's bounded-element property (47.1); this lemma and the omnific floor below are the "integer-part interface" whose Lean counterparts are listed after table 18.

The omnific floor is characterized by

$$\lfloor x \rfloor_{\mathbf{Oz}} \leq x < \lfloor x \rfloor_{\mathbf{Oz}} + 1.$$

For completeness, if $x = h + a + \varepsilon$ is its normal-form separation into a purely infinite part, a real constant, and an infinitesimal part, its floor is $h + \lfloor a \rfloor$, except that one subtracts 1 when $a \in \mathbb{Z}$ and $\varepsilon < 0$. This explains the otherwise easy-to-miss negative infinitesimal boundary. Uniqueness follows from discreteness.

Lemma 137.2 (Flooring the right-hand side). *For an ordinary integer matrix A , a surreal vector b , and $x \in \mathbf{Oz}^d$,*

$$Ax \leq b \iff Ax \leq \beta, \quad \beta_i = \lfloor b_i \rfloor_{\mathbf{Oz}}.$$

Proof. Each coordinate of Ax is omnific. An omnific integer strictly above β_i is at least $\beta_i + 1 > b_i$. $\square$

We henceforth use $S(\beta) = \{x \in \mathbf{Oz}^d : Ax \leq \beta\}$ when the matrix is understood. The linear relaxation with right-hand side β is contained in $P_A(b)$ and is therefore bounded whenever $P_A(b)$ is bounded.

137.2 A bounded lattice slice need not be a set

Proposition 137.3 (Proper-class intervals). *In the full class field $\mathbf{No}$, if $\Omega > 0$ is infinite, then $[0, \Omega] \cap \mathbf{Oz}$ is a proper class.*

Proof. Let $c\omega^\gamma$ be the leading term of Ω , with $c > 0$ and $\gamma > 0$. For every ordinal $\alpha \geq 1$ define

$$u_\alpha = \omega^{\gamma/(\alpha+1)}.$$

Its growth exponent lies strictly between 0 and γ , so u_α is an omnific integer and $0 < u_\alpha < \Omega$. Distinct ordinals give distinct exponents and monomials. A set containing all these values would admit an injection from the ordinals, which is impossible. $\square$

Thus even the interval hull $\text{conv}_{\mathbf{No}}([0, \Omega] \cap \mathbf{Oz})$, when $\Omega \in \mathbf{Oz}$, is generated by two points while the underlying collection of lattice points is proper-class sized. The word “finite” in the main theorem refers to the number of generators, not the number of feasible points.

We may work in a standard class foundation, or in a universe-relative interpretation with a larger ambient universe. Nothing below requires applying set-sized compactness to a proper-class structure. Every model-theoretic step is localized to set-sized additive substructures in the next section.

138 The Presburger bridge, with its size boundary made explicit

138.1 Set-sized $\mathbb{Z}$ -groups inside the omnific integers

For a set-sized $\mathbb{Q}$ -linear subspace $H \subseteq \Pi$, put

$$D_H = H \oplus \mathbb{Z} \subseteq \mathbf{Oz}.$$

It is an ordered abelian group with distinguished least positive element 1, and $D_H/\mathbb{Z} \cong H$ is divisible. Hence D_H is a $\mathbb{Z}$ -group. We use the classical completeness and quantifier elimination theorem for these structures in the language

$$0, 1, +, -, \leq, \quad x \equiv y \pmod{n} \quad (n = 2, 3, \dots).$$

All coefficients and moduli in this language are ordinary integers. Multiplication of two variables is not present.

Lemma 138.1 (Elementary localization). *If $H \subseteq H'$ are set-sized $\mathbb{Q}$ -subspaces of Π , then $D_H \subseteq D_{H'}$ is an elementary embedding in the Presburger language. Every finite tuple of omnific integers belongs to some such D_H .*

Proof. A finite tuple lies in the group obtained by taking the $\mathbb{Q}$ -span of its purely infinite parts. Addition and order are preserved by inclusion. Congruences are preserved and reflected: for $h + k \in D_H$,

$$h + k \equiv 0 \pmod{n} \iff k \equiv 0 \pmod{n},$$

and h/n is still in H . Quantifier elimination then makes every formula absolute between the two groups. $\square$

This supplies a precise meaning to applying a fixed Presburger formula to omnific parameters. One evaluates its quantifier-free equivalent; all finite witnesses and counterwitnesses lie in suitable larger groups $D_{H'}$. There is no appeal to a proper-class compactness theorem.

138.2 Attainment for ordinary objectives

Lemma 138.2 (Transferred bounded optimization). *Let $A \in \mathbb{Z}^{m \times d}$ and $q \in \mathbb{Z}^d$ be fixed. Suppose $S(\beta) \neq \emptyset$ and every $x \in S(\beta)$ satisfies $-U \leq x_j \leq U$, where $U \in \mathbf{Oz}_{\geq 0}$. Then some $v \in S(\beta)$ maximizes $q \cdot x$ on all of $S(\beta)$.*

Proof. Over the ordinary integers, the following statement is true: for every $\beta \in \mathbb{Z}^m$ and every integer $U \geq 0$, if

$$T(\beta, U) = \{x : Ax \leq \beta, -U \leq x_j \leq U\}$$

is nonempty, there is a $v \in T(\beta, U)$ such that $q \cdot y \leq q \cdot v$ for all $y \in T(\beta, U)$. It is true there because $T(\beta, U)$ is finite. For these fixed A, d, m, q the displayed assertion is a single Presburger sentence, so it holds in every D_H .

Choose D_H containing β, U and a feasible witness. Obtain $v \in D_H^d$ from that sentence. Any potential better omnific point y belongs, together with these parameters, to a larger $D_{H'}$. By lemma 138.1, the assertion that v is optimal remains true in $D_{H'}$. Therefore no such y exists. The box is redundant by hypothesis, so v is optimal on $S(\beta)$ itself. $\square$

A bounded surreal polyhedron supplies an ordinary finite tuple of surreal coordinate bounds; applying the omnific ceiling to their maximum supplies U . Multiplying an ordinary rational objective by a positive common denominator extends the lemma to $q \in \mathbb{Q}^d$.

It is essential not to replace the proof by “the boxed omnific set is finite.” The ordinary finiteness is used only to establish one first-order sentence before transfer. The omnific conclusion is an attainment statement, not a finiteness statement.

139 One finite test set controls all surreal objectives

139.1 Conformal order and the graph lattice

For real or surreal vectors u, v of the same length, write $u \sqsubseteq v$ when each pair of coordinates has compatible signs and $|u_j| \leq |v_j|$. For ordinary integer vectors define

$$E_A(u) = (u, Au) \in \mathbb{Z}^{d+m}.$$

The graph lattice consists of the integer kernel vectors of the matrix $[A \ -I_m]$, written as (u, Au) . Define

$$G_A = \{g \in \mathbb{Z}^d \setminus \{0\} : E_A(g) \text{ is conformally minimal among nonzero graph-lattice vectors}\}. \quad (139.1)$$

Thus G_A is the first-coordinate projection of the Graver basis of $[A - I_m]$. In particular, the signs of both u and Au are part of the definition. Omitting either block can invalidate the feasibility argument.

Lemma 139.1 (Ordinary finite conformal decomposition). *The set G_A is finite and symmetric. Every $u \in \mathbb{Z}^d$ admits*

$$u = \sum_{g \in G_A} n_g g, \quad n_g \in \mathbb{N}, \quad (139.2)$$

where every used $E_A(g)$ is conformal with $E_A(u)$.

Proof. In each of the finitely many coordinate orthants, take absolute values. Dickson's lemma gives finitely many minimal elements of the resulting subset of $\mathbb{N}^{d+m}$. This proves finiteness; negation proves symmetry. One elementary proof of Dickson's lemma is by induction on the number of coordinates: an infinite sequence of natural-number vectors has an infinite subsequence nondecreasing in its first coordinate, and iteration produces two vectors ordered coordinatewise. An infinite antichain of minimal elements is therefore impossible.

For a nonzero u , choose a nonzero conformally minimal graph vector $E_A(g) \sqsubseteq E_A(u)$. If $u \neq g$, both g and $u - g$ lie in the same closed graph orthant as u , and the integer ℓ^1 -norm of the graph vector strictly decreases on passing to $u - g$. Induction gives a conformal sum. Collecting repeated directions yields the fixed-index expression (139.2). The zero vector has all coefficients zero. $\square$

[write] The finiteness argument is the one of Part I's finite Graver certificate (lemma 17.1), there applied to the lattice of integer affine dependences $\Lambda = \ker_{\mathbb{Z}} \tilde{A}$ of an integral point configuration and here to the graph lattice of $[A - I_m]$; the two sets called G_A are different ("Notation in this Part").

139.2 Infinite multiplicities, but still only finitely many directions

Lemma 139.2 (Omnific conformal decomposition). *For every $u \in \mathbf{Oz}^d$ there are $\lambda_g \in \mathbf{Oz}_{\geq 0}$, indexed by the ordinary finite set G_A , such that*

$$u = \sum_{g \in G_A} \lambda_g g, \quad (139.3)$$

and, whenever $\lambda_g > 0$, every coordinate of $E_A(g)$ has the same weak sign as the corresponding coordinate of $E_A(u)$, with zero coordinates required to remain zero. For every used g one moreover has $E_A(g) \sqsubseteq E_A(u)$.

Proof. For fixed A and its finite set G_A , the existence of the coefficients in (139.2), together with the stated orthant conditions, is a Presburger statement about u . Each product is multiplication by a fixed ordinary integer coordinate of g ; no variable multiplication is hidden in this assertion. Lemma 139.1 makes this statement true over $\mathbb{Z}$. Transfer in a D_H containing u gives the required omnific coefficients.

A positive λ_g is at least 1. In any coordinate there is no cancellation among the summands. Thus $|(E_A u)_j| \geq \lambda_g |(E_A g)_j| \geq |(E_A g)_j|$ for a used g , which proves the final assertion. $\square$

One cannot instead perform an induction of " $|u|$ many steps" for infinite u . The fixed finite index set in (139.3) is the mechanism that makes the transfer legitimate. Infinite multiplicities are coefficients, not an illicit number of summands.

Lemma 139.3 (Every used direction is a feasible unit move). *If $x, y \in S(\beta)$ and g occurs with positive coefficient in a conformal decomposition of $y - x$, then $x + g \in S(\beta)$.*

Proof. Fix row i . If $A_i(y - x) \leq 0$, conformity gives $A_i g \leq 0$, so the inequality remains satisfied. If $A_i(y - x) > 0$, conformity and the unit bound give

$$0 \leq A_i g \leq A_i(y - x) \leq \beta_i - A_i x.$$

Again $A_i(x + g) \leq \beta_i$. The coordinates of $x + g$ remain omnific. $\square$

Theorem 139.4 (Finite augmentation criterion). *For $c \in \mathbf{No}^d$ and $x \in S(\beta)$, the following are equivalent:*

- (i) x maximizes $c \cdot y$ on $S(\beta)$;
- (ii) there is no $g \in G_A$ with $x + g \in S(\beta)$ and $c \cdot g > 0$.

Proof. The first implication is immediate. Conversely, if y is better than x , use lemma 139.2 to write

$$0 < c \cdot (y - x) = \sum_{g \in G_A} \lambda_g(c \cdot g).$$

A finite sum of nonpositive terms cannot be positive in an ordered field. Hence some used g has $c \cdot g > 0$, and lemma 139.3 makes it a feasible improving move. $\square$

This theorem does not assert that repeated unit augmentation terminates after an ordinary finite number of steps from an arbitrary omnific starting point. It asserts that optimality itself has a finite local certificate. Existence of an optimizer must still be supplied; the next section does so without a transfinite algorithm.

140 Finite hulls and a universal rational normal fan

140.1 A small ordered-field linear-algebra lemma

We use two finite linear facts over ordered fields. First, a finite system of rational linear equalities and strict inequalities has a solution over an ordered extension of $\mathbb{Q}$ if and only if it has a rational solution. Second, a point outside the convex hull of finitely many points can be strictly separated from that hull by a linear functional over the same ordered field. Both follow from finite Fourier–Motzkin elimination; a certificate-level explanation is given in Section 149. Neither fact uses order completeness.

Consider the central hyperplane arrangement

$$\mathcal{A}_A = \{c \in \mathbb{R}^d : c \cdot g = 0\}, \quad g \in G_A, \quad (140.1)$$

with repeated hyperplanes removed. A *chamber* is a full-dimensional sign region of this arrangement. Choose one rational point in every chamber and clear denominators positively, obtaining a finite set $Q_A \subset \mathbb{Z}^d$. Every $q \in Q_A$ satisfies $q \cdot g \neq 0$ for all $g \in G_A$.

Lemma 140.1 (Finite objective representatives). *For every $c \in \mathbf{No}^d$ there is a $q \in Q_A$ such that*

$$c \cdot g > 0 \implies q \cdot g > 0, \quad c \cdot g < 0 \implies q \cdot g < 0 \quad (g \in G_A).$$

More generally, there is a rational objective with exactly the same three-way sign pattern as c on all of G_A , including all zero entries.

Proof. For the first assertion, impose just the strict inequalities prescribed by the nonzero signs of c . The rational system is feasible in an ordered extension, hence feasible over $\mathbb{Q}$. Its real solution set is nonempty and open. Avoiding the finitely many hyperplanes gives a chamber, whose representative has the required signs. For the second assertion, also impose the rational linear equalities corresponding to zero signs and apply the same feasibility fact inside their rational solution subspace. $\square$

140.2 Construction of the hull from finitely many optima

Theorem 140.2 (Finite generation and attainment). *Suppose $S(\beta)$ is nonempty and contained in a box with omnific bounds. For each $q \in Q_A$ there is a unique maximizer $v_q \in S(\beta)$. With $V(\beta) = \{v_q : q \in Q_A\}$, one has*

$$\text{conv}_{\mathbf{No}} S(\beta) = \text{conv}_{\mathbf{No}} V(\beta). \quad (140.2)$$

Every $c \in \mathbf{No}^d$ is maximized on $S(\beta)$ by at least one point of $V(\beta)$.

Proof. Existence of v_q follows from lemma 138.2. To prove uniqueness, let x, y be distinct optimal points for a generic q . A conormal decomposition of $y - x$ has at least one positive coefficient. Every used unit move from x is feasible. Optimality gives $q \cdot g \leq 0$, and genericity makes each such value strictly negative. Hence

$$q \cdot (y - x) = \sum_g \lambda_g (q \cdot g) < 0,$$

a contradiction.

Now fix an arbitrary surreal c . Choose $q \in Q_A$ as in lemma 140.1. If a g were a feasible improving move for c at v_q , it would also improve q , which is impossible. The augmentation criterion shows that v_q maximizes c on all of $S(\beta)$.

Let $H = \text{conv}_{\mathbf{No}} V(\beta)$. Certainly $H \subseteq \text{conv}_{\mathbf{No}} S(\beta)$. If some $s \in S(\beta)$ were outside H , finite strict separation would give $c \in \mathbf{No}^d$ with

$$c \cdot s > \max_{v \in V(\beta)} c \cdot v.$$

This contradicts the just-proved supporting property of $V(\beta)$. Therefore $S(\beta) \subseteq H$, proving (140.2). $\square$

Every member of $V(\beta)$ is a vertex because it is the unique maximizer of an objective on the whole hull; conversely every vertex of a finite hull belongs to its finite generating set. Distinct chambers may select the same vertex. The construction works without assuming the hull is full-dimensional.

140.3 Edge directions and face selection

For a finite polytope we use its algebraic exposed faces, including the whole polytope and the empty face. Finite ordered-field polyhedral theory identifies these with the usual face lattice; no appeal to order-topological compactness is needed.

Theorem 140.3 (Uniform face control). *For fixed A , the following hold for every nonempty bounded $S(\beta)$.*

- (i) Objectives with the same three-way sign pattern on G_A have exactly the same maximizing lattice points and expose the same hull face.
- (ii) Every face is exposed by an ordinary rational objective.
- (iii) Every edge of the hull is parallel to a member of G_A .
- (iv) The normal fan of the ordinary rational zonotope $Z_A = \sum_{g \in G_A^+} [-g, g]$, where G_A^+ selects one direction from each distinct unoriented line in G_A , refines the normal fan of the hull, interpreted over **No**.

Proof. The augmentation criterion characterizes the maximizing points entirely through the feasible moves and the signs of $c \cdot g$. This proves (i). A hull point is a finite convex combination of lattice points; it is optimal exactly when all points given positive weight are optimal. Thus maximizing lattice sets indeed determine the exposed face. Part (ii) follows from the exact-sign assertion of lemma 140.1; the zero objective exposes the whole hull.

For (iii), let v, w be the endpoints of an edge exposed by c . In a conformal decomposition of $w - v$, every used $v + g$ is feasible and has $c \cdot g \leq 0$. The equality $c \cdot (w - v) = 0$ forces $c \cdot g = 0$ for every used g . Thus $v + g$ is also on the edge. At least one such g exists, and it is a nonzero ordinary integer vector parallel to $w - v$.

The normal cones of Z_A are exactly the closed sign cones of the hyperplane arrangement. Part (i), including its boundary sign patterns, says that each such cone selects a compatible face of the hull. This is the claimed fan refinement. Equivalently, one may first use (iii) and the elementary zonotope edge-direction refinement argument. $\square$

Corollary 140.4 (A chamber bound). *Let $h(A)$ be the number of distinct unoriented lines represented by G_A . Then*

$$f_0(\text{conv}_{\mathbf{No}} S(\beta)) \leq 2 \sum_{k=0}^{d-1} \binom{h(A)-1}{k}, \quad (140.3)$$

where binomial coefficients outside their ordinary range are zero.

Proof. There is at most one chosen vertex per chamber. The maximum number of chambers of h central hyperplanes in dimension d is at most $2 \sum_{k=0}^{d-1} \binom{h-1}{k}$. To see the bound, add one hyperplane: new chambers are bounded by the number of chambers of the induced central arrangement in dimension $d - 1$. Induction and Pascal's identity give the formula, with the one-hyperplane and one-dimensional cases as bases. $\square$

The bound and the zonotope method are classical in origin [92]. What required proof here was that they can be applied without assuming a finite lattice set or a pre-existing finite integer hull. The two bounds (135.4) and (140.3) are generally incomparable; their minimum is available.

141 The bounded-slack basis formula

Recall the ordinary integer thresholds $M_i = \max_{g \in G_A} |A_i g|$. They depend only on the matrix, not the right-hand side, the number of normal-form terms, or an objective.

Lemma 141.1 (Small-slack rows span at a generic optimum). *Let v be a maximizer for an objective q satisfying $q \cdot g \neq 0$ for all $g \in G_A$. Set $s = \beta - Av \in \mathbf{Oz}_{\geq 0}^m$ and*

$$T = \{i : s_i < M_i\}.$$

Then $\text{rank } A_T = d$. Moreover each s_i with $i \in T$ is an ordinary integer in $\{0, \dots, M_i - 1\}$.

Proof. The last assertion is [lemma 137.1](#). Suppose the rank assertion fails. The rational nullspace of A_T contains a nonzero ordinary integer vector u . Take an ordinary conformal decomposition of u and choose any used $g \in G_A$. For every $i \in T$, the equality $A_i u = 0$ and conformity force $A_i g = 0$. For $i \notin T$,

$$s_i \geq M_i \geq |A_i g|.$$

It follows that both $v + g$ and $v - g$ satisfy every inequality and lie in $\mathbf{Oz}^d$. Since $q \cdot g \neq 0$, one of these two points improves the objective. This contradiction proves full rank. $\square$

Theorem 141.2 (Explicit vertex candidates). *For a bounded $P_A(b)$, let $\beta = \lfloor b \rfloor_{\mathbf{Oz}}$ coordinatewise and define $C_A(\beta)$ by (135.3). Then*

$$\mathcal{H}_A(b) = \text{conv}_{\mathbf{No}} C_A(\beta).$$

The set $S_A(b)$ is nonempty exactly when $C_A(\beta)$ is nonempty. Every vertex has a representation

$$v = A_I^{-1}(\beta_I - r), \quad 0 \leq r_i < M_i, \quad r_i \in \mathbb{Z}. \quad (141.1)$$

There are at most $N(A)$ labels (I, r) to test.

Proof. Flooring does not change the lattice slice. If that slice is empty, so is the candidate set, and the assertion is immediate. Otherwise, [theorem 140.2](#) expresses its hull as the convex hull of the generic optimizers v_q . For each such optimizer, [lemma 141.1](#) supplies a nonsingular row basis $I \subseteq T$. Taking $r = s_I$ gives (141.1), and all these v_q therefore belong to $C_A(\beta)$. Conversely every accepted candidate is itself a feasible omnific lattice point. The two convex-hull inclusions follow. Counting the allowed ordinary slack vectors for each basis gives $N(A)$. $\square$

Remark 141.3 (What the proximity assertion does and does not say). The point $A_I^{-1}\beta_I$ is an intersection of d original supporting hyperplanes, but it need not satisfy all the other inequalities. The theorem places v at a bounded ordinary displacement $-A_I^{-1}r$ from such a *basis intersection*. It does not, without an additional argument, place v near a feasible vertex of the original linear relaxation. Keeping this distinction avoids an unjustified stronger proximity claim.

141.1 An extension beyond the surreal field

Corollary 141.4 (Arbitrary integer parts). *Let K be an ordered field and $D \subset K$ an integer part: a discretely ordered subring with least positive element 1 such that every $x \in K$ has a $a \in D$ with $a \leq x < a + 1$. For fixed $A \in \mathbb{Z}^{m \times d}$ and bounded $P_A(b) \subset K^d$, the conclusions of [theorems 140.2, 140.3](#) and [141.2](#) hold with K, D in place of $\mathbf{No}, \mathbf{Oz}$.*

Proof. For $z \in D$ and ordinary $n > 0$, take the integer-part floor of z/n . The remainder lies in $D \cap [0, n)$ and is one of $0, \dots, n - 1$, by discreteness. Thus the additive group of D is a $\mathbb{Z}$ -group. Its ordinary bounded elements are ordinary integers, and the integer-part property supplies D -valued bounds for bounded subsets of K^d . Presburger transfer, conformal decomposition, rational objective representatives, and finite ordered-field separation now prove the same results word for word. In a set-sized field the localization discussion is unnecessary. $\square$

The purely infinite/constant decomposition is not needed for the positive finite-hull theorem itself. It becomes essential in the particularly explicit compiler below and in the sharp irrational obstruction. The negative theorem should not be indiscriminately extended to every integer part of every field.

142 A finite tail-and-residue compiler

142.1 Integrality and feasibility without enumerating supports

Write the floored right-hand side uniquely as

$$\beta = h + z, \quad h \in \Pi^m, \quad z \in \mathbb{Z}^m.$$

For a candidate label (I, r) , its unfiltered point splits as

$$v_{I,r} = A_I^{-1}h_I + A_I^{-1}(z_I - r). \quad (142.1)$$

The first term is purely infinite coordinatewise; the second is rational. This separates exactly what the large scales can affect from what the ordinary residue arithmetic can affect.

Theorem 142.1 (Exact finite tests). *Let $D_I = \det A_I \neq 0$. The candidate $v_{I,r}$ belongs to $\mathbf{Oz}^d$ if and only if the ordinary vector congruence*

$$\text{adj}(A_I)(z_I - r) \equiv 0 \pmod{|D_I|} \quad (142.2)$$

holds componentwise. For every row j , set

$$H_{jI} = h_j - A_j A_I^{-1} h_I \in \Pi, \quad K_{jIr} = z_j - A_j A_I^{-1} (z_I - r) \in \mathbb{Q}. \quad (142.3)$$

The inequality $A_j v_{I,r} \leq \beta_j$ holds exactly when

$$H_{jI} > 0 \quad \text{or} \quad (H_{jI} = 0 \text{ and } K_{jIr} \geq 0). \quad (142.4)$$

In particular, $H_{jI} < 0$ always rejects the candidate at row j .

Proof. A point of $\Pi \oplus \mathbb{Q}$ is omnific precisely when its constant term is an integer. Applying the adjugate formula to the second term in (142.1) gives (142.2). Its first term is already in Π^d and introduces no residue obstruction. The slack at row j is $H_{jI} + K_{jIr}$. If $H_{jI} \neq 0$, its sign dominates every ordinary rational constant. If $H_{jI} = 0$, the sign is exactly that of K_{jIr} . These alternatives give (142.4). $\square$

All tail observables in this theorem are from the fixed finite family $h_j - A_j A_I^{-1} h_I$; they do not depend on r . Arbitrarily deep cancellation inside one of them must be resolved exactly, but there are only finitely many such expressions. Leading exponents alone need not determine their signs. An infinite normal-form support is not truncated in the argument.

Corollary 142.2 (No new vertex tail span). *Every vertex coordinate has purely infinite part in $\text{span}_{\mathbb{Q}}\{h_1, \dots, h_m\}$. In particular, construction of the omnific integer hull introduces no new rationally independent tail parameter.*

Proof. Use (142.1) for a vertex and note that A_I^{-1} has ordinary rational entries. $\square$

This is a statement about an additive $\mathbb{Q}$ -span, not a claim that the normal forms become finitely supported or that their exponent groups acquire finite rank.

142.2 All face incidences from finitely many signs

Choose one rational objective in each nonempty sign cell of $\mathcal{A}_A$, including cells on hyperplanes and the zero cell. Call the resulting finite set $\widehat{Q}_A$. Unlike Q_A , it contains objectives exposing non-vertex faces.

Theorem 142.3 (Finite combinatorial parameter data). *For fixed A , there is a finite collection of Presburger predicates in $\beta \in \mathbf{Oz}^m$ that determines, for every nonempty bounded instance, the complete labeled face lattice of $\text{conv}_{\mathbf{No}} S(\beta)$. These predicates can be taken to record:*

- (i) *which candidate labels (I, r) are integral and feasible;*
- (ii) *which pairs of accepted candidate labels denote the same point;*
- (iii) *for each $q \in \widehat{Q}_A$, all signs of $q \cdot (v_{I,r} - v_{J,s})$ between accepted candidates.*

After $\beta = h + z$ is split, these predicates reduce to finitely many signs of rational linear expressions in h , ordinary linear comparisons in z , and ordinary congruences in z .

Proof. There are at most $N(A)$ labels. Integrality can be written directly as the congruences $\text{adj}(A_I)(\beta_I - r) \equiv 0 \pmod{|D_I|}$ in the additive $\mathbb{Z}$ -group; feasibility is a rational linear inequality in β . Coincidences and the objective differences in (iii) are also rational linear expressions with rational constant terms. Multiplying by positive ordinary common denominators puts all comparisons and equalities in Presburger form.

By [theorem 141.2](#), accepted candidates generate the hull. For each q , the signs in (iii) select exactly the candidate points on its maximizing face. By [theorem 140.3](#), $\widehat{Q}_A$ exposes every nonempty face. After duplicate points and duplicate face sets are identified, inclusion among these finite candidate sets is precisely face inclusion. Indeed, each face is the convex hull of its maximizing candidates. The empty face is adjoined separately.

Finally, every rational linear expression in β plus an ordinary constant splits into a purely infinite expression in h plus an ordinary rational expression in z . Its sign and its equality to zero are decided by the same alternatives as [\(142.4\)](#); its ordinary congruences depend only on the constant part. This proves the last assertion. $\square$

142.3 Same-matrix descent of the face lattice

Theorem 142.4 (Ordinary realization with the same matrix). *Let $A \in \mathbb{Z}^{m \times d}$ be fixed. For every nonempty bounded omnific slice $S(\beta)$, there exists an ordinary $\beta' \in \mathbb{Z}^m$ such that the ordinary integer slice $S_{\mathbb{Z}}(\beta') = \{x \in \mathbb{Z}^d : Ax \leq \beta'\}$ is nonempty and bounded and*

$$\text{conv}_{\mathbf{No}} S(\beta) \quad \text{and} \quad \text{conv}_{\mathbb{R}} S_{\mathbb{Z}}(\beta')$$

have isomorphic face lattices. The isomorphism may preserve the candidate labels and all maximizing-face selections of $\widehat{Q}_A$.

Proof. Record the finite truth pattern of the predicates in [theorem 142.3](#) at β , and let $\Phi(t)$ be their conjunction, negating those that are false. Choose an omnific bound U for $S(\beta)$. The following parameter-free Presburger sentence is true in a set-sized D_H containing the data:

$$\exists t \exists U \geq 0 \left[\Phi(t) \wedge \exists x (Ax \leq t) \wedge \forall x (Ax \leq t \Rightarrow \bigwedge_j (-U \leq x_j \leq U)) \right]. \quad (142.5)$$

Completeness of the theory of $\mathbb{Z}$ -groups makes this sentence true over the ordinary integers. Let β', U' be ordinary witnesses. Their integer slice is nonempty and finite. Its finite candidate

data agree with the original truth pattern, so [theorem 142.3](#) identifies the face lattices and the stated labels. $\square$

The theorem preserves finite combinatorics, not edge lengths, support values, normal-form coefficients, or an infinite family of external inequalities. It does not replace a surreal parameter by a real number in its complete ordered-field type. It realizes a specifically bounded finite list of arithmetic predicates.

142.4 What is effective and what remains an oracle

For a fixed ordinary A , a finite exact workflow is now available. Precompute G_A , the thresholds M_i , the nonsingular bases, and the rational objective representatives. Floor the right-hand side and split $\beta = h + z$. Enumerate at most $N(A)$ labels, apply [theorem 142.1](#), and use the objective comparisons of [theorem 142.3](#) to assemble the faces.

This is an exact finite reduction *relative to* operations on the input surreal parameters: omnific flooring, constant extraction, rational linear combination, equality, and order comparison. It is not a Turing algorithm for arbitrary unnamed surreal numbers. A representation with undecidable equality cannot be turned into a computable one merely by this finite reduction. For symbolic inputs supporting those operations, however, the same precomputed ordinary certificate works at every scale.

There is also no polynomial-time claim when A varies. Graver bases and the bound $N(A)$ can be large. The point is independence from the magnitude, birthday, support length, and number of asymptotic scales of b , not a hidden small complexity bound in the ordinary matrix data.

143 The sharp rational-normal boundary

143.1 Real linear images have no hidden infinitesimal remainder

Lemma 143.1 (Finite real-linear images). *Let $a \in \mathbb{R}^d$ and $x \in \mathbf{Oz}^d$. If $a \cdot x$ is bounded in absolute value by an ordinary real, then*

$$a \cdot x \in L_a := a \cdot \mathbb{Z}^d.$$

More generally, every $a \cdot x$ is either an ordinary member of L_a or has nonzero purely infinite part and is infinite in magnitude.

Proof. Write $x = h + n$ coordinatewise. Since Π is an $\mathbb{R}$ -vector space, $a \cdot h \in \Pi$. If it is nonzero it dominates the ordinary real $a \cdot n$ in magnitude. If it is zero, $a \cdot x = a \cdot n \in L_a$. $\square$

This is not true for an arbitrary real vector in $\mathbf{No}^d$: its coordinates may have infinitesimal normal-form tails. The assertion depends on the omnific support restriction.

Lemma 143.2 (The real subgroup dichotomy). *For $a \in \mathbb{R}^d \setminus \{0\}$, either a is rational in direction and L_a is discrete, or L_a is dense in $\mathbb{R}$.*

Proof. If all ratios to one nonzero coordinate are rational, clearing denominators shows that L_a is contained in a cyclic discrete subgroup of $\mathbb{R}$; as a nonzero finitely generated subgroup of that cyclic group it is itself cyclic. Conversely, if one ratio is irrational, two coordinates generate a subgroup containing nonzero elements arbitrarily close to zero. For a direct proof, apply the pigeonhole principle to the fractional parts of $0, \theta, \dots, N\theta$, where θ is the irrational ratio. The difference of two fractional parts gives a nonzero integer combination of 1 and θ of

absolute value at most $1/N$. Multiplying back by the chosen coordinate gives arbitrarily small positive elements of L_a . For any real interval, choose such a positive element shorter than its length; an ordinary integer multiple lies inside the interval. Hence L_a is dense. $\square$

143.2 A boxed objective with no supremum

Theorem 143.3 (Dense-image obstruction). *Let $a \in \mathbb{R}^d$ be not rational in direction, let $\Omega > 0$ be an infinite surreal, and choose $r \in \mathbb{R} \setminus L_a$. Then*

$$S = \{x \in \mathbf{Oz}^d : |x_j| \leq \Omega, a \cdot x \leq r\}$$

is nonempty, the objective $a \cdot x$ has neither a maximum nor a supremum in $\mathbf{No}$, and $\text{conv}_{\mathbf{No}} S$ is not finitely generated.

Proof. Every ordinary integer vector lies in the box. Density of L_a gives ordinary vectors whose objective values lie arbitrarily close to r from below. In particular, the slice is nonempty. For an arbitrary feasible x , lemma 143.1 says that its objective is either negative infinite or an ordinary real in L_a . It cannot equal r . In the first case an ordinary feasible value improves it; in the second, density gives an ordinary feasible value strictly between $a \cdot x$ and r . Thus there is no maximum.

To prove the stronger order-theoretic statement, every positive infinitesimal ε makes $r - \varepsilon$ an upper bound: all finite feasible values have a positive *real* gap from r , and the remaining values are negative infinite. Suppose U is any upper bound. If $U \geq r$, the upper bound $r - \varepsilon$ is smaller. If $U < r$, put $\delta = r - U > 0$. Density of the ordinary feasible values forces δ to be infinitesimal; otherwise a positive real gap below r would contain a value larger than U . Then $r - 2\delta < U$ is again an upper bound. There is no least one.

Finally, suppose the convex hull were generated by finitely many points $v_1, \dots, v_k$. A linear objective has a largest value on that finite list and hence on its convex hull. Each v_i is itself a finite convex combination of points of S , so its objective value is at most one of those lattice values. A maximum on the hull would therefore be a maximum on S , contrary to the first part. $\square$

Remark 143.4 (The exact real threshold for this objective). For the same dense L_a and infinite box, the principal objective $a \cdot x$ attains its maximum under $a \cdot x \leq r \in \mathbb{R}$ precisely when $r \in L_a$. In that case an ordinary integer vector with value r is feasible and optimal. This statement concerns that one objective; it does not assert finite generation of the entire integer hull.

In (135.5), $a = (-\sqrt{2}, 1)$ and $r = 1/2$ work because $1/2 \notin \mathbb{Z} + \sqrt{2}\mathbb{Z}$. The finite objective values approach $1/2$ in the ordinary real sense, but $1/2$ is *not* their surreal supremum. Also, ordinary real convergence does not imply convergence in the surreal order topology: a positive infinitesimal neighborhood can exclude every term. Accordingly, no topological nonclosedness claim is being deduced from this example.

143.3 A universal characterization for fixed real matrices

Theorem 143.5 (Rational-normal dichotomy). *Fix $R \in \mathbb{R}^{m \times d}$. For $b \in \mathbf{No}^m$ and $\Omega \in \mathbf{No}_{>0}$ set*

$$S_R(b, \Omega) = \{x \in \mathbf{Oz}^d : Rx \leq b, |x_j| \leq \Omega\}.$$

The following conditions are equivalent:

- (i) *Every nonzero row of R is rational in direction.*

- (ii) For every b, Ω with $S_R(b, \Omega) \neq \emptyset$, its **No**-convex hull is generated by finitely many points.
- (iii) For every such b, Ω and every $c \in \mathbf{No}^d$, the objective $c \cdot x$ attains a maximum on $S_R(b, \Omega)$.

When these conditions hold, the number of vertices has a bound depending only on R and d after a once-for-all row normalization, not on b or Ω .

Proof. Assume (i). Write each nonzero row as $R_i = \lambda_i A_i$, where $\lambda_i > 0$ is real and $A_i \in \mathbb{Z}^d$. A negative scalar can be absorbed into A_i . Divide its right-hand side by λ_i and append the box normals $\pm e_j$. Zero rows either make the slice empty or can be omitted. This produces a fixed ordinary integer matrix and a bounded surreal polyhedron. [Theorems 140.2](#) and [141.2](#) give (ii), (iii), and the uniform bound.

Condition (ii) implies (iii) even if the finite generators are not originally chosen from the lattice: maximizing on the finite hull and then expanding a maximal generator as a finite convex combination gives a lattice maximizer, as in the proof of [theorem 143.3](#).

Finally suppose (i) fails, and let $a = R_i$ be an irrational-direction row. Take $\Omega = \omega$ and choose $r \in (0, 1) \setminus L_a$; this is possible since L_a is countable. Put $b_i = r$ and $b_j = \omega^2$ for every other row. In the box, $|R_j x| \leq C_j \omega < \omega^2$ for an ordinary real C_j , so those other inequalities are inactive. The remaining slice is the one in [theorem 143.3](#), and its principal objective has no maximum. Thus (iii) fails, proving the converse. $\square$

The theorem quantifies over *every* right-hand side. It does not say that one irrational normal spoils every instance. For example,

$$\{(x, y) \in [-\omega, \omega]^2 : y - \sqrt{2}x \leq 0\}$$

has four vertices

$$(-\omega/\sqrt{2}, -\omega), \quad (\omega, -\omega), \quad (\omega, \omega), \quad (\omega/\sqrt{2}, \omega),$$

all of which are omnific. Its omnific integer hull is the polytope itself. This also illustrates that irrational edge directions are perfectly possible outside the rational-normal class.

143.4 An integer right-hand side and an infinitesimal witness

The choice $r = 1/2$ makes the density obstruction especially transparent, but a nonintegral right-hand side is not essential to failure of finite generation.

Proposition 143.6 (Failure with omnific-integer right-hand sides). *Let*

$$S_1 = \{(x, y) \in \mathbf{Oz}^2 : |x|, |y| \leq \omega, y - \sqrt{2}x \leq 1\}, \quad \varepsilon = \omega^{-2}.$$

The real objective $a(x, y) = y - \sqrt{2}x$ has maximum 1, but the surreal objective

$$c(x, y) = y - \sqrt{2}x + \varepsilon y \tag{143.1}$$

has no maximum. Consequently $\text{conv}_{\mathbf{No}} S_1$ is not finitely generated, even though every displayed right-hand side is omnific.

Proof. The point $(0, 1)$ attains $a = 1$. Write $x = h_x + n_x$ and $y = h_y + n_y$. On the face $a = 1$, real-linear splitting and irrationality force

$$n_x = 0, \quad n_y = 1, \quad h_y = \sqrt{2}h_x.$$

Consequently all its lattice points have the form

$$(x, y) = (u/\sqrt{2}, u + 1), \quad u \in \Pi, \quad -\omega \leq u < \omega. \quad (143.2)$$

The upper inequality is strict because $u = \omega$ would give $y = \omega + 1$. Conversely, if $u < \omega$ is purely infinite, the positive purely infinite quantity $\omega - u$ exceeds 1, so the upper box inequality holds. The lower bound and the x -bounds follow directly. This proves the exact parameterization.

There is no largest u in (143.2): replace it by $u' = (u + \omega)/2 \in \Pi$, which is strictly larger and still below ω . On this face $c = 1 + \varepsilon(u + 1)$, so c has no maximum there.

Every point off the face has a either negative infinite or an ordinary real strictly below 1. Since $\varepsilon y \leq \omega^{-1}$, its value of c is strictly below 1. The point $(0, 1)$ has value $1 + \varepsilon$. Thus a global maximizer, were one to exist, would have to lie on the face, where none exists. Finite generation is excluded as before. $\square$

This example separates two logically different issues. The principal supporting inequality may attain equality at lattice points, yet the resulting face may have no extremal point for a secondary objective. The infinitesimal term in (143.1) detects that missing extremum without allowing the secondary objective to compensate for a positive real loss in the principal one.

144 Totally unimodular systems and a complete knapsack example

144.1 The totally unimodular specialization

Theorem 144.1 (Flooring suffices for a totally unimodular matrix). *Suppose $A \in \mathbb{Z}^{m \times d}$ is totally unimodular and $P_A(b)$ is bounded. Then*

$$\mathcal{H}_A(b) = \{x \in \mathbf{No}^d : Ax \leq \lfloor b \rfloor_{\mathbf{Oz}}\}. \quad (144.1)$$

Proof. Let Q be the polyhedron on the right. If it is empty there is nothing to prove. A nonempty bounded polyhedron over an ordered field is the convex hull of its finitely many basis vertices, including the lower-dimensional case; this is the finite algebraic version of the vertex description recalled in Section 149. At a vertex, choose a nonsingular active basis A_I . Total unimodularity gives $\det A_I = \pm 1$, so A_I^{-1} has integer entries. Its vertex $A_I^{-1}\beta_I$ therefore belongs to $\mathbf{Oz}^d$. All vertices of Q lie in its omnific slice, proving (144.1). $\square$

This is the classical total-unimodularity integrality argument, now with omnific right-hand sides. Without total unimodularity, simply flooring the right-hand side is not sufficient; the following example gives the exact additional cuts.

144.2 The matrix, its test set, and its finite bound

Consider

$$S_B = \{(x, y) \in \mathbf{Oz}_{\geq 0}^2 : 2x + 3y \leq B\}, \quad B \in \mathbf{Oz}_{\geq 0}, \quad A = \begin{pmatrix} -1 & 0 \\ 0 & -1 \\ 2 & 3 \end{pmatrix}. \quad (144.2)$$

Its graph Graver set is

$$G_A = \pm\{(1, 0), (0, 1), (1, -1), (2, -1), (3, -2)\}. \quad (144.3)$$

Here is a complete proof rather than a conclusion inferred from a bounded search. In quadrants with x, y of the same sign, the coordinate directions suffice. By overall negation, the remaining case is $(x, -t)$ with $x, t \geq 0$. If $2x \geq 3t$, write $t = 2k + r$ with $r \in \{0, 1\}$. Then

$$(x, t) = k(3, 2) + r(2, 1) + (x - 3k - 2r)(1, 0), \quad (144.4)$$

and every coefficient is nonnegative. If $2x \leq 3t$, write $x = 3k + r$ with $r \in \{0, 1, 2\}$. Then

$$(x, t) = k(3, 2) + r(1, 1) + (t - 2k - r)(0, 1). \quad (144.5)$$

These are conformal in both coordinates and in $2x - 3t$. Returning to $(x, -t)$ gives exactly the directions in (144.3). Each listed direction is irreducible: its coordinate bounds leave only the smaller integer vectors just covered by these two cones, and none has the requisite third-coordinate sign and bound. For $(3, -2)$ that bound is zero, forcing $2x - 3t = 0$ and hence the primitive pair $(3, 2)$; the other four cases are immediate from their coordinate bounds. Thus the set is precisely G_A .

The thresholds are

$$(M_1, M_2, M_3) = (3, 2, 3), \quad N(A) = 3 \cdot 2 + 3 \cdot 3 + 2 \cdot 3 = 21.$$

There are five unoriented direction lines, giving the independent chamber bound 10. Both bounds apply to every nonnegative omnific B , regardless of its magnitude. The actual family has at most five vertices.

144.3 Three cuts give the exact hull

Theorem 144.2 (Exact 2–3 knapsack hull). *Write $B = 6T + r$ with $T \in \mathbf{Oz}$, $T \geq 2$, and $r \in \{0, 1, 2, 3, 4, 5\}$. Then $\text{conv}_{\mathbf{No}} S_B$ is exactly the polytope*

$$\left\{ (x, y) \in \mathbf{No}_{\geq 0}^2 : \begin{array}{l} 2x + 3y \leq B, \\ x + y \leq \lfloor B/2 \rfloor_{\mathbf{Oz}}, \\ x + 2y \leq \lfloor 2B/3 \rfloor_{\mathbf{Oz}}, \\ y \leq \lfloor B/3 \rfloor_{\mathbf{Oz}} \end{array} \right\}. \quad (144.6)$$

Its counterclockwise vertices, beginning at the origin, are the six rows of Table 16.

Proof. All three added cuts are valid for lattice points. The first follows from $2(x + y) \leq 2x + 3y \leq B$. The second follows from $3(x + 2y) = 3x + 6y \leq 4x + 6y \leq 2B$, and the third from $3y \leq B$. The left-hand sides are omnific, so their rational upper bounds can be floored.

Ordinary remainder arithmetic gives

$$\lfloor B/2 \rfloor_{\mathbf{Oz}} = 3T + \lfloor r/2 \rfloor, \quad \lfloor 2B/3 \rfloor_{\mathbf{Oz}} = 4T + \lfloor 2r/3 \rfloor, \quad \lfloor B/3 \rfloor_{\mathbf{Oz}} = 2T + \lfloor r/3 \rfloor.$$

Intersecting successive active lines of (144.6) gives Table 16. Each listed point satisfies all inequalities, and successive segments account for the boundary: the inward coordinate axes bound the lower edges, and the upper edges have respectively the active slopes among -1 , $-2/3$, $-1/2$, and 0 . Their intercepts are the displayed floored values. For $T \geq 2$ the listed consecutive vertices are distinct and occur in the stated order; intermediate cuts can be redundant. Thus the intersection is the convex hull of the listed points. Every coordinate in the table is omnific, and every listed point satisfies $2x + 3y \leq B$. The cut polytope is therefore contained in $\text{conv}_{\mathbf{No}} S_B$, while cut validity gives the opposite inclusion. $\square$

r	Vertices of the hull for $B = 6T + r, T \geq 2$
0	$(0, 0), (3T, 0), (0, 2T)$
1	$(0, 0), (3T, 0), (3T - 1, 1), (2, 2T - 1), (0, 2T)$
2	$(0, 0), (3T + 1, 0), (1, 2T), (0, 2T)$
3	$(0, 0), (3T + 1, 0), (3T, 1), (0, 2T + 1)$
4	$(0, 0), (3T + 2, 0), (2, 2T), (0, 2T + 1)$
5	$(0, 0), (3T + 2, 0), (3T + 1, 1), (1, 2T + 1), (0, 2T + 1)$

Table 16: A complete periodic vertex formula. The parameter T is omnific, not restricted to an ordinary integer.

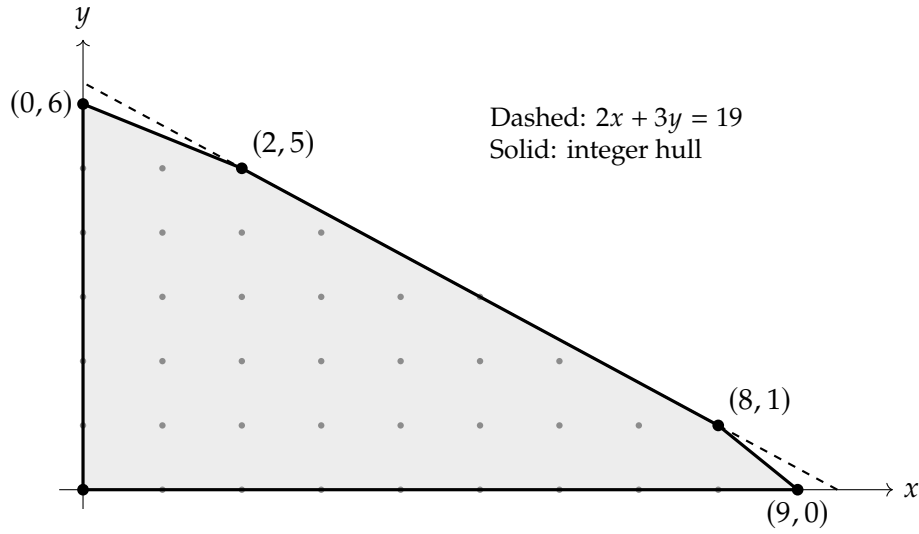


Figure 7: The finite specialization $T = 3, r = 1$. This is an exact ordinary model illustrating the five-vertex pattern, not a scale drawing of infinite surreal lengths.

144.4 An explicit multiscale instance

Take the purely infinite parameter

$$H = \omega^2 + \sqrt{3}\omega + \omega^{1/2}, \quad B = H + 13.$$

Then $B = 6T + 1$ with $T = H/6 + 2$. The omnific integer hull has precisely the five vertices

$$(0, 0), \quad (H/2 + 6, 0), \quad (H/2 + 5, 1), \quad (2, H/3 + 3), \quad (0, H/3 + 4). \quad (144.7)$$

This example carries several scales exactly while the ordinary residue $13 \bmod 6$ selects the combinatorial row. The same formulas remain valid if H is replaced by any positive purely infinite normal form, including one with infinite support. The example does not rely on the displayed support having only three terms.

145 Executable checks and their exact scope

The accompanying code/12-omnific-hulls-verify.py uses only Python's standard library. All geometric decisions are made with integers and `fractions.Fraction`; no floating-point

tolerance appears. It is a finite arithmetic verification program, not a general implementation of surreal arithmetic and not a proof assistant certificate.

The supplied run produced the results in Table 17. For every $B = 0, \dots, 245$, the program independently enumerates ordinary lattice points, computes their monotone-chain convex hull, constructs the 21-label candidate list, and enumerates the vertices of the cut system. All three hulls agree. For $B = 6T + r$ with $T = 2, \dots, 40$, it also verifies the exact six-residue table. The finite computations support the formulas; the proof for infinite T is the algebraic argument above, not numerical extrapolation.

Exact check family	Count
Conformal decompositions for nonzero vectors in $[-80, 80]^2$	25,920
Finite tests excluding smaller conformal vectors for listed generators	110
Independent integer-hull comparisons, $0 \leq B \leq 245$	246
Lattice points enumerated across those instances	426,236
Checks of the six-residue vertex table, $2 \leq T \leq 40$	234
Local-versus-global augmentation checks, $0 \leq B \leq 35$	77,469
Exact rational-square certificates for the irrational example	4

Table 17: Actual results of the included verification run. The largest vertex count observed in the knapsack family was five.

For the last check family, the program certifies

$$\frac{1}{2} - \frac{1}{D} < m - \sqrt{2}n < \frac{1}{2}$$

for

$$(D, n, m) = (10, 11, 16), (100, 64, 91), (1000, 781, 1105), (10000, 12671, 17920).$$

It does so by comparing the rational squares of $m - 1/2$ and $m - 1/2 + 1/D$ with $2n^2$. These are exact witnesses of ordinary approach to $1/2$; the nonexistence of a surreal supremum is proved separately in [theorem 143.3](#).

Run the checks from the package directory with

```
python code/12-omnific-hulls-verify.py \
    -output /path/to/scratch/verification_results.json
```

The JSON file and the recorded console output are both included. ([write] Source 12 printed `python verify.py -output verification_results.json`, run from its package directory. Without `-output` the program writes `verification_results.json` into the current directory; pass a path outside the report directory, as above, so that the shipped record is never overwritten. The recorded JSON file and console output are shipped as `data/12-omnific-hulls-verification_results.json` and `data/12-omnific-hulls-verification_output.txt`. The shipped `code/12-omnific-hulls-Makefile` still names the delivery paths `verify.py`, `verification_results.json`, `verification_output.txt` and `article.tex`. The README of source 12 adds: “Do not use Python’s `-0` flag: the comparisons are implemented as assertions.”) The program has assertions for every claimed comparison and exits with an error if any check fails. Completeness of the general Graver construction for arbitrary matrices is not tested by this one example; its mathematical proof is [lemma 139.1](#).

146 Formalization architecture and proof-audit boundaries

[write] This section is a proposal: nothing of this Part is formalized. The Lean declarations that already cover its first layer are listed after [table 18](#).

146.1 A field-generic core and a surreal-specific interface

The positive theorem should be formalized for an integer part of an ordered field before specializing to surreals. This keeps ordinal and normal-form machinery out of the finite combinatorial proof. The surreal-specific interface then supplies flooring, constant extraction, finite-element rigidity, and closure of purely infinite elements under real scalar multiplication.

A useful implementation split is shown in [Table 18](#). These are proposed mathematical contracts, not names of declarations asserted to exist in the repository. The inspected repository does not by itself certify any of the new hull theorems in this article.

Proposed layer	Contract to establish
Integer-part interface	Least positive element, ordinary division and remainders, floors, and bounded-element rigidity.
Presburger bridge	Interpret the fixed additive language in the integer part; prove transfer of sentences and the bounded-optimization instance.
Finite conformal test set	A finite graph-Graver certificate, its ordinary conformal decomposition theorem, and the fixed-formula transferred version.
Optimization and geometry	Feasible unit moves, local optimality, rational objective cells, finite strict separation, and finite hull generation.
Candidate compiler	Full rank of small-slack rows, adjudicate candidate formulas, congruence tests, and extraction of a finite face lattice.
Surreal obstruction	The real-vector-space tail splitting, dense real images, missing suprema, and the integer-right-hand-side infinitesimal example.

Table 18: A proposed proof-assistant dependency decomposition.

[write] At the write phase (ProveIt HEAD 4e128356d) the first row (“Integer-part interface”) is already proved in Lean for the actual omnific ring, with no sorry: `Surreal.Foundations.SignSequence.omnific_bounded_iff` (an omnific integer bounded by an ordinary real is an ordinary integer) and `Surreal.Foundations.SignSequence.omnific_least_positive` (`Surreal/Foundations/OmnificUnits.lean`, lines 65 and 104); `Surreal.Foundations.SignSequence.omnificFloor_spec` and `Surreal.Foundations.SignSequence.existsUnique_omnific_integerPart` (`Surreal/Foundations/OmnificFloor.lean`, lines 112 and 138); `Surreal.Foundations.SignSequence.existsUnique_omnific_division` (ordered division by any positive omnific integer, `Surreal/Foundations/OmnificDivision.lean`, line 62); and `Surreal.Foundations.SignSequence.omnificQuotientZModEquiv` ($\mathbf{Oz}/n\mathbf{Oz} \cong \mathbb{Z}/n\mathbb{Z}$, `Surreal/Foundations/OmnificResidues.lean`, line 81). Paths are relative to `Algebra/SurrealNumbers/`. The Presburger bridge and every later row are not formalized, and nothing in this Part is.

For a universe-relative surreal type, the field-generic theorems avoid proper-class quantification completely. The full-class interpretation of this article is recovered by the finite-data localization already proved. These are two compatible presentations, not a request to introduce a type whose elements form the collection of all sets.

The repository’s ordinary Cooper implementation is especially relevant to the finite predicates in [theorem 142.3](#). It does not automatically prove their soundness under omnific

interpretation. That soundness is a separate semantic theorem, after which the same finite syntax can be reused. Likewise, an executable list of candidate points is not a proof that all vertices occur in it until the small-slack theorem has been connected to the checker.

146.2 The five delicate steps

The following are the principal places where an apparently shorter argument would be invalid.

Bounded is not finite. The boxed lattice slice may be a proper class. Ordinary finite optimization is used only to prove a Presburger sentence, then transferred with parameters.

Infinite coefficients are not infinitely many steps. The decomposition $u = \sum_{g \in G_A} \lambda_g g$ has an ordinary finite index set. Its coefficients may be infinite. No induction on an omnific norm or termination of unit augmentation is asserted.

A polytope cannot be assumed before it is constructed. Finite optimal representatives and strict separation prove finite generation before normal-fan or edge arguments are applied to the integer hull.

Small slacks are ordinary. The rank argument depends on $s_i < M_i$ with M_i ordinary. Replacing this by an unspecified surreal slack bound would not give finitely many offsets.

A real limiting value need not be a surreal supremum. The irrational example has upper bounds $r - \varepsilon$ for every positive infinitesimal ε . Its real limiting value is not the least upper bound in **No**, and real sequences cannot settle order-topological questions.

146.3 What has actually been checked

The mathematical dependencies are explicit: normal-form arithmetic at the interface, the classical $\mathbb{Z}$ -group completeness theorem, finite conformal integer decompositions, and finite ordered-field linear separation. The manuscript supplies the remaining arguments. The code checks finite examples and exact algebraic comparisons. The LaTeX source compiles independently of the repository. No repository build, new Lean proof, independent peer review, or exhaustive literature-priority certification is claimed.

147 Further research questions and topics

These are proposed next questions for this research program. They are not all asserted to be previously published open problems; several ask for an explicit omnific or proof-certified version of a classical construction.

- Q1. Smaller universal test sets.** Which proper subsets of the graph Graver set still certify all linear optima for the fixed matrix A and all omnific right-hand sides? Distinguish sets sufficient for generic objectives, all face objectives, and just the small-slack rank argument. A reduction here improves both the fan and the candidate bound.

- Q2. Efficient finite atlases.** The finite-atlas consequence in Section 150 makes the complete collection of face lattices decidable from a certified test set. Can the atlas be computed without enumerating all sign and congruence patterns? Determine which adjacent objective chambers must select the same vertex uniformly in the parameters and exploit that coalescence.
- Q3. Sharp matrix-dependent bounds.** Find useful sharp or asymptotically sharp bounds for the maximum vertex count and the minimum necessary offset ranges as functions of the arithmetic of A . The explicit bound $N(A)$ is deliberately robust, not expected to be optimal. Total unimodularity provides a benchmark where all offset corrections can be eliminated.
- Q4. Unbounded rational-normal polyhedra.** Extend the finite-certificate presentation to unbounded omnific slices using finite vertices, lineality data, and recession directions. The target should specify exactly which sums of an integer hull and a rational recession cone are meant and handle slices with no vertices. A bare invocation of a classical Minkowski decomposition does not explain the class-sized arithmetic interface.
- Q5. Mixed integer and continuous coordinates.** For variables in $\mathbf{Oz}^p \times \mathbf{No}^q$, determine a finite certificate theory that combines real elimination with discrete residues. Establish the precise conditions under which projection commutes with the mixed integer hull and whether a fixed rational normal arrangement still suffices.
- Q6. Instancewise irrational criteria.** The universal characterization does not classify individual irrational-normal instances. The right-hand sides 0, $1/2$, and 1 in the two-dimensional examples already exhibit three different behaviors of the principal objective. Characterize finite generation using the interaction between real constant fibers and purely infinite feasible faces, rather than row rationality alone.
- Q7. Secondary objectives and ordered hierarchies.** Generalize proposition 143.6 to a finite hierarchy of real supporting functionals perturbed by successively smaller surreal coefficients. When does attainment at every preceding level coexist with failure at the next? Identify finite conditions guaranteeing that the resulting lexicographic face has an omnific extremal point.
- Q8. Comparison of integer parts.** The positive theorem holds for arbitrary integer parts, but the obstruction uses the split real-vector-space tail of $\mathbf{Oz}$. Compare integer hulls for other integer parts of the same ordered field. Determine which structural properties of the integer part, beyond its additive Presburger theory, govern real irrational coefficient systems.
- Q9. Effective symbolic surreal domains.** Implement the candidate compiler for a specified representation supporting exact floor and equality/order tests on rational linear tail expressions. Track certificate size as a function of expression size, rather than birthday. An exact treatment of cancellation is necessary; inspecting only leading valuation data is insufficient.
- Q10. Counting without confusing cardinality and length.** A literal ordinary cardinality cannot count all points of a proper-class omnific interval. Seek explicitly axiomatized valuations or specialization invariants that extend ordinary Ehrhart data on finite rational polytopes. Any proposed invariant must state its additivity, domain, and relation to ordinary finite specializations; substituting ω into a polynomial is not by itself a cardinality theorem.
- Q11. Certified integer optimization in the repository.** Formalize the semantic Presburger

bridge and the small-slack rank theorem, then connect a candidate-list checker to actual omnific convex hulls. An early milestone is the full six-residue knapsack table. The next is the field-generic theorem for arbitrary integer parts, separating code execution from the trusted completeness argument.

Q12. Functorial and geometric interactions. Study how these finite integer hulls behave under rational affine maps, Minkowski sums, and products, and compare their normal-fan atlas with the repository’s other multiscale polytope constructions. Even over ordinary integers, taking an integer hull does not generally commute with projection; seek useful sufficient hypotheses, especially integral or totally unimodular fiber descriptions.

[write] Q10 is the counting half of Part IV’s question “Other integer parts and unbounded geometry” (section 58), which this Part answers for integer hulls only (corollary 141.4 and theorems 141.2 and 142.4). The “repository’s other multiscale polytope constructions” of Q12 now include the other Parts of this report.

148 Conclusion

The decisive finiteness is not finiteness of the feasible lattice collection. It is finiteness of an ordinary arithmetic test set. A fixed rational matrix admits one finite conformal certificate; Presburger transfer allows its multiplicities to become omnific without allowing its list of directions to become infinite. Rational objective cells then give finitely many supporting points, and a small-slack rank argument makes those points explicit.

The resulting geometry has a clear boundary. Rational-normal families have uniformly finite integer hulls, ordinary edge directions, an exact parameter-to-face-lattice compiler, and same-matrix ordinary realizations of all their finite combinatorial types. Real irrational normals can instead produce unattained objectives and missing surreal suprema inside bounded polytopes. Even an attained principal objective and omnific-integer right-hand sides do not remove every obstruction.

Thus the non-Archimedean extension is neither an automatic source of new finite polytope types nor a harmless change of scalars for lattice geometry. It has an exact rational regime in which finite arithmetic controls all scales, and an irrational regime in which the arithmetic of constant fibers and the order structure of purely infinite faces must both be retained.

149 Finite linear certificates over an ordered field (Appendix A of source 12)

For clarity, this appendix spells out the linear-algebra facts used in the proofs. Let K be any ordered field; it need not be Archimedean or real closed. All systems and matrices in this appendix are finite.

149.1 Elimination and rational feasibility

To eliminate one variable t from a system of linear inequalities, divide rows with nonzero t -coefficient by its absolute value and collect lower bounds $t \geq L_i(x)$ and upper bounds $t \leq U_j(x)$. Existence of t is equivalent to $L_i(x) \leq U_j(x)$ for every pair, together with the rows not involving t . Strict inequalities remain strict whenever one of the paired bounds is strict. Equalities may first be solved or expressed as two weak inequalities. If only lower or only

upper bounds remain, a witness is obtained by adding or subtracting 1 from an extremal bound as necessary. If both sides remain, a midpoint works when a strict gap is required; equal endpoints are allowed only when both bounds permit equality.

These operations are valid over every ordered field. When all starting coefficients and constants are rational, back-substitution chooses rational witnesses. Therefore feasibility of such a system over K implies feasibility over $\mathbb{Q}$. An arbitrary sign condition $c \cdot g > 0$ is an ordinary strict linear inequality. Thus both versions of [lemma 140.1](#) follow from this finite procedure, including exact zero signs.

149.2 Strict separation of a point from a finite hull

Membership of s in the convex hull of $v_1, \dots, v_k$ is the feasibility of

$$\sum_i \lambda_i v_i = s, \quad \sum_i \lambda_i = 1, \quad \lambda_i \geq 0.$$

Finite elimination gives the usual alternative for this system: either it is feasible, or there are $a \in K^d$ and $t \in K$ such that

$$a \cdot v_i + t \geq 0 \quad \text{for all } i, \quad a \cdot s + t < 0.$$

One way to track the alternative is to retain the nonnegative row combinations produced during elimination; an infeasible final constant inequality supplies the displayed multiplier certificate. Equality rows have unrestricted multipliers because they were entered with both signs. This is the finite Farkas alternative and uses only field operations and order. Taking $c = -a$ gives

$$c \cdot s > t \geq c \cdot v_i \quad \text{for every } i,$$

which is exactly the strict separator required in [theorem 140.2](#).

149.3 Vertices of a bounded polyhedron

A short constructive argument gives the finite vertex description used for total unimodularity. At a point x of a nonempty bounded polyhedron, let the active-row matrix have rank $r < d$. Choose a nonzero direction u annihilated by its rows. The feasible scalars t for $x + tu$ form a closed interval in K , obtained from finitely many upper and lower linear bounds. Boundedness ensures both endpoints are finite. Since every currently inactive row has positive slack, 0 is strictly between the two endpoints. At each endpoint a new active row is independent of the old active rows: a dependent row would also annihilate u and could not become newly tight along the line.

Thus x is a convex combination of two points of strictly larger active rank. Repeat at most $d - r$ times along each branch. This is an ordinary finite binary decomposition ending at points with active rank d , namely basis vertices. There are only finitely many nonsingular row bases. Hence the polyhedron is the convex hull of finitely many such vertices. The argument also covers a polyhedron lying in a proper affine subspace, because its active rank simply starts higher. The same finite elimination and supporting-hyperplane arguments give the usual exposed-face interpretation for finite polytopes.

150 An effective finite atlas of all realizable face lattices (Appendix B of source 12)

Corollary 150.1 (Finite atlas from an ordinary certificate). *Given an ordinary matrix A together with its complete finite conformal test set, there is an ordinary terminating procedure that lists all face lattices of nonempty bounded omnific slices $S(\beta)$, up to isomorphism, and supplies an ordinary integer right-hand side realizing each listed type. It also computes the exact maximum vertex count among these slices when any such slice exists.*

Proof. Construct the finite predicate list of [theorem 142.3](#). There are only finitely many possible truth patterns. For each pattern, form its formula $\Phi(t)$ and ask whether the Presburger sentence (142.5) is true over $\mathbb{Z}$. Presburger decidability answers each question. When the answer is yes, ordinary enumeration of integer tuples (t, U, x) , using the corresponding decidable boundedness and nonemptiness conditions, eventually finds witnesses; equivalently one can use witness extraction during elimination. The resulting finite bounded integer slice has an explicitly computable finite hull and face lattice.

Every surviving pattern is realized over $\mathbb{Z}$, hence over $\mathbf{Oz}$ with the same boundedness statement by elementary localization. Conversely every nonempty bounded omnific slice contributes a surviving pattern by [theorem 142.4](#). Remove isomorphic duplicates by finite face-poset comparison. Counting vertices in the finite list gives the asserted maximum. If no pattern survives, there is no nonempty bounded slice. $\square$

For an ordinary right-hand side, why can no new infinite omnific feasible point invalidate the ordinary boundedness assertion? The boundedness property is itself the Presburger sentence with those ordinary parameters. Its truth over $\mathbb{Z}$ transfers to every relevant D_H , so such an omnific point cannot exist. In this special case all feasible omnific coordinates are bounded by an ordinary integer and are therefore ordinary integers.

This corollary is a decidability and existence-of-algorithm statement, not a practical running-time claim. It records a consequence that would otherwise be easy to leave incorrectly listed as an unresolved question. The research challenge in Q2 is efficient construction and compression of the atlas, not its bare finiteness or computability from the stated certificate.

151 Notation and dependency ledger (Appendix C of source 12)

Notation	Meaning
$\mathbf{No}, \mathbf{Oz}$	Conway's surreal field and omnific integer subring.
Π	Purely infinite additive part; a real vector space, with no nonzero ordinary-bounded elements.
$\text{ct}, \lfloor \cdot \rfloor_{\mathbf{Oz}}$	Ordinary integer constant term on $\mathbf{Oz}$ and omnific floor on $\mathbf{No}$.
$A, \beta = h + z$	Fixed ordinary integer constraint matrix and floored omnific right-hand side, split into tail and integer constant.
$E_A(u)$	Graph vector (u, Au) , used to require conformity in variables and row values simultaneously.
G_A	Finite conformally minimal graph directions in $\mathbb{Z}^d$.
$M_i, N(A)$	Ordinary slack thresholds and the total bound on candidate labels.
$C_A(\beta)$	Integral feasible basis intersections with bounded ordinary slack offsets. Not every accepted candidate is necessarily a vertex.

Notation	Meaning
$Q_A, \widehat{Q}_A$	Finite rational objective representatives for chambers, respectively all nonempty arrangement sign cells.
$L_a = a \cdot \mathbb{Z}^d$	Real additive image controlling the finite values of a real objective on omnific points.

The proof dependencies of the central conclusions can be read without following the order of exposition:

omnific splitting and ordinary division $\implies$ $\mathbb{Z}$ -group localization,
 ordinary graph Graver decomposition + Presburger transfer
 $\implies$ finite augmentation criterion,
 ordinary-objective attainment + finite rational sign cells + strict separation
 $\implies$ finite hull,
 finite hull + small-slack rank $\implies$ explicit candidate formula,
 candidate formula + finite objective comparisons
 $\implies$ tail/residue compiler and same-matrix descent,
 real-linear tail splitting + dense real subgroup
 $\implies$ irrational obstruction and universal dichotomy.

The dependence on classical Presburger completeness is explicit, not removed by the finite numerical checks. The dependence on ordinal normal forms is confined to the surreal interface and the negative examples, not hidden in the ordinary Graver certificate.

Part X

Beyond finite presentations

Source and scope. [write] This Part prints, whole, manuscript 06 of batch 59, printed here as source 13: *Beyond Finite Surreal Polytopes: Exposed faces, sharp infinitesimal depth, and finite-description rigidity*, headed “Surreal geometry / finite and infinite presentations”, author line “Prepared for Vladimir Reshetnikov with ChatGPT” (the PDF metadata of the delivered file reads the same), dated 30 September 2026 (archive `Beyond_Finite_Surreal_Polytopes.zip`, main file `Beyond_Finite_Surreal_Polytopes/article.tex`; delivered PDF of 26 pages on Letter paper, not shipped). It was written against ProveIt commit 6996fee43; it arrived in commit ed80eba06 and was placed in commit f2cb2034e. Its program, recorded run, recorded console output and build file are shipped as `code/13-beyond-finite-verify.py`, `data/13-beyond-finite-verification_results.json`, `data/13-beyond-finite-verification_stdout.txt` and `code/13-beyond-finite-Makefile`; its three notes, delivered in a directory `notes/`, are shipped as `13-beyond-finite-SOURCES_AND_SCOPE.md`, `13-beyond-finite-PROOF_AUDIT.md` and `13-beyond-finite-BUILD_REPORT.md` (whose SHA-256 lines describe the delivered `article.tex` and `article.pdf`, neither of which is shipped). Its README and checksum list `SHA256SUMS.txt` are not shipped; they survive in the delivery archive of arrival commit ed80eba06, and the README’s status section is printed below. Every section, result, proof, example, remark, figure, table and research question of the manuscript is printed here, in the manuscript’s order, with three structural changes: its two appendices are printed as the ordinary [sections 167](#) and [168](#) at the end of this Part; its bibliography is printed

as the list “References of Part X”; and its theorem environments are the report’s (source 13’s aliascnt setup is dropped; as in the source, all numbered statements share one counter). Its formalization route (the subsection “A practical Lean decomposition” of [section 164](#)) stays in this Part. In this Part “this manuscript” and “the package” mean source 13 and its delivery.

Relation to the other Parts. Source 13 continues Part III’s question “Beyond finite vertex sets” ([section 43](#)) and partly answers it: with set-sized presentations, a class that is both a small hull and a small halfspace intersection has finite subpresentations of both ([theorem 157.2](#)), so definable small hulls and definable small H-classes are finite polytopes ([corollaries 157.3](#) and [157.4](#)), while the rational moment hull and its polar are explicit non-polytopal small classes ([theorems 161.1](#) and [161.2](#)); the volume half of that question is untouched. It also continues Part I’s scalar-extension correction ([section 20](#), subsection “A real face and its scalar extension are different sets”): [corollary 154.4](#) extends the trace correspondence of faces to arbitrary *sets* of generators, and the paragraph after it keeps the same warning that a single exposing functional need not descend. Its exposing normals are surreal, so it is consistent with Part V’s vertex invisible to all real objectives ([example 67.5](#)); its polarity theorem ([theorem 158.1](#)) concerns infinite presentations and is complementary to Part I’s finite polar residue ([proposition 14.1](#)). Its opening observation that a finite list of surreal coordinates produces no new finite face lattice is the report’s “no new finite face lattices” spine ([theorem 18.1](#)). Source 13 credits the lexicographic exposure theorem, the degree of non-exposedness and its codimension bound to Martínez-Legaz (1997); its [theorem 159.2](#) is a translation of that invariant, as it says in [section 152](#). Its positive theorems are proved for **No** in a fixed universe and, with “set” replaced by “of cardinality less than κ ”, for every κ -saturated real closed field ([theorem 163.1](#)).

Notation in this Part. [write] The following symbols and words mean something different elsewhere in the report ([section 1.3](#)); the Part-local meaning is the one in force here. No symbol of source 13 has been renamed.

- *Small* means indexed by a set in one fixed universe of NBG set theory with Global Choice ([section 153](#)); in [theorem 163.1](#) it means of cardinality less than κ . The size condition concerns generators, rows, parameters and formulas. The tempting false reading is that a small hull $V(S)$ or a small H-class is itself a set: it is in general a proper class, since the hull of two distinct points already contains a copy of the surreal interval $[0, 1]$.
- $V(S) = \text{conv}_{\mathbf{No}} S$ (finite convex combinations, S a set) and $H(A, b)$ (a set-indexed intersection of closed halfspaces). The sans-serif V is neither Part III’s mixed volume V nor Part I’s dependence space V .
- *Closed* and *bounded* are pointwise order-box notions; “bounded” means contained in a box of some *surreal* radius, not a real one (contrast Part IV’s real box). $\mathcal{O}$ is the ring of finite surreals, the report’s $\mathcal{O}$.
- ε is any positive element infinitesimal relative to $\mathbb{R}$, not necessarily ω^{-1} ; the η of [section 162](#) ranges over positive infinitesimals and is not $\omega^{-\omega}$.
- *Degree*: $m_\varepsilon(C, F)$ and $\delta_C(F)$ in [theorem 159.2](#) are the degree of non-exposedness of a real face, a third notion of degree in this report (see the paragraph after that theorem); $r_C(F)$ is an exposed-chain length, and δ is also a radius in [corollary 155.2](#).
- $m_d(t)$ is the moment curve, M_d its rational hull, H_d the polar, and F_A, G_A their faces, with $k = \lfloor d/2 \rfloor$; this G_A is not the Graver set G_A of Part IX or Part I’s Graver basis G_A , and m_d, m_ε are not Part I’s rank m . K_d is the compact real “paraboloid tower” of [section 160](#); K elsewhere is an ordered field.

- $C^\uparrow = \text{conv}_{\mathbf{No}}(C(\mathbb{R}))$ is the generator extension of a real convex set; it is generally not the definable extension $C(\mathbf{No})$.

Abstract of source 13. Finite polytope theory over the surreal numbers is governed by ordered-field algebra. A different phenomenon appears when one permits a set of generators or a set of inequalities while retaining all surreal scalars. We prove a finite-description rigidity theorem: if a finite-dimensional convex class is both the finite-convex-combination hull of a set of surreal points and the intersection of a set of surreal halfspaces, then both original presentations reduce to finite subpresentations. More generally, a small definable intersection contained in a small convex hull admits a finite sandwich.

The same saturation principle gives strong separation of disjoint set-generated hulls, exposes every nonempty proper face, and forces every nonempty face of a set-presented halfspace intersection to be cut out by at most d active input inequalities. Polarity turns nonfinite generation into a genuine set-versus-class obstruction. We also identify polynomial infinitesimal exposure degree with the classical degree of non-exposedness, construct explicit compact examples attaining every codimension bound, and classify all faces of the hull of the rational moment curve and of its polar. The hull has only simplex faces on at most $\lfloor d/2 \rfloor$ rational parameters; the polar has no faces below dimension $\lceil d/2 \rceil$.

All claims are proved in a fixed class-theoretic universe. The crucial restriction is that the presentations are sets, not proper classes. This is an unrefereed research manuscript, not a Lean-certified development; priority of the new formulations and consequences has not been established.

Title page of source 13. A companion research direction for ProveIt, with explicit provenance, proof dependencies, counterexamples, and reproducible finite checks.

Status and important distinctions (README of source 13). This is an AI-assisted, unrefereed proof manuscript. Its main conclusions have written proofs, but they are not Lean-certified or independently refereed. Priority of the proposed geometric formulations and consequences has not been established. No named longstanding conjecture is claimed solved.

Martínez-Legaz’s 1997 lexicographical exposure theorem, classical degree of non-exposedness, and codimension bound are credited as prior results. The new polynomial notation does not create a new real invariant. Saturation, real closedness, finite Farkas theory, and the compactness mechanism are also prior mathematical tools.

“Small” means externally set-indexed in one fixed universe. The convex classes themselves are generally proper classes. Convex combinations have ordinary finite support. A hull of the real points of a convex body is NOT its full semialgebraic extension to all surreal points.

152 The question beyond finite polytopes

A finite list of surreal coordinates does not, by itself, produce an unprecedented finite face lattice. Finite linear feasibility, affine dependence, and the vertex–halfspace correspondence are algebraic facts about ordered fields. Their validity is not tied to Archimedean scalars. The ordered-field Farkas theory has both a classical literature and recent formal verification in Lean [97, 98].

The productive change of question is therefore not merely to replace a finite real vertex matrix by a finite surreal matrix. It is to ask what happens when the *presentation* is allowed

to be an arbitrary set, while the ambient scalars range over the entire proper class **No**. Two apparently natural generalizations then diverge:

$$V(S) = \text{conv}_{\mathbf{No}} S, \quad S \subseteq \mathbf{No}^d \text{ a set,} \quad (152.1)$$

$$H(A, b) = \bigcap_{i \in I} \{x \in \mathbf{No}^d : a_i \cdot x \leq b_i\}, \quad I \text{ a set.} \quad (152.2)$$

Every convex combination in (152.1) has ordinary finite support. No infinite sum or completion is intended.

The central answer

For nonempty classes in ordinary finite dimension,

$$\text{small V-classes} \cap \text{small H-classes} = \text{finite polytopes.}$$

Moreover, equality of the two presentations forces finite sublists of *those very presentations* to define the same polytope.

This is not the assertion that every closed bounded surreal convex class is a polytope. Quite the opposite: we construct closed bounded, countably generated convex classes with infinitely many vertices and very sparse face dimensions. They have no set-sized halfspace presentation whatsoever. On the dual side, a countable intersection of closed bounded intervals can have no extreme points at all.

152.1 Relation to the repository

The consulted ProveIt material includes the surreal-number project overview and its report on surreal fields across universes [93, 94]. The overview distinguishes its constructed, formally checked field foundations from research reports whose statements are not automatically machine-checked. The across-universes report studies omitted cuts and the saturation of old surreal fields; its own README explicitly does not claim Lean verification. That is the relevant bridge: our geometric mechanism is *small-cut realization in the full current-universe field*, not a new assumption of metric compactness.

The proofs below do not import any unrefereed theorem from the repository. They use the classical real-closedness and cut properties of **No**, finite ordered-field convexity, and first-order compactness. They can therefore be reviewed independently of the repository's other research claims. The source notes record the consulted paths and the indexed commit ([write] shipped as 13-beyond-finite-SOURCES_AND_SCOPE.md). No repository build or axiom audit was performed for this manuscript.

152.2 What is established, and what is not claimed

The main results and their logical roles are as follows.

Result	Precise content
Separation and exposure	Disjoint small hulls admit a uniform surreal separator; every nonempty proper face of a small hull is exposed (theorems 155.1 and 155.3).
Finite active faces	Every nonempty face of a small halfspace intersection is an intersection with at most d active input hyperplanes (theorem 156.2).
Double compression	A small V/H equality reduces simultaneously to finite subpresentations (theorem 157.2).
Polar obstruction	A nonpolyhedral small hull with the origin interior has a polar not generated by any set of points (theorem 158.1).
Exact exposure degree	Polynomial degree in one infinitesimal equals Martínez-Legaz's classical degree of non-exposedness; an explicit compact family attains the bound (theorems 159.2 and 160.2).
Moment-curve faces	The proper face lattice of the rational moment hull is exactly its $\lfloor d/2 \rfloor$ -point simplex skeleton; the polar has a complementary dimension gap (theorems 161.1 and 161.2).

The lexicographical exposure theorem, its degree invariant, and its codimension bound were already proved by Martínez-Legaz in 1997 [99]. We explicitly credit them. Our infinitesimal degree statement is a translation and application of that theory, not a claim to have discovered the invariant. The finite-compression arguments are applications of familiar model-theoretic compactness. Their geometric combination, sharp examples, and class-size consequences are the research contribution proposed here; an exhaustive priority review remains to be done. No named longstanding conjecture is asserted to have been settled.

153 Size conventions and the saturation engine

153.1 One universe, ordinary finite dimension

We work in a fixed universe of NBG set theory with Global Choice. The field $\mathbf{No}$ is a proper class, and each individual surreal is a set-coded number. The integer $d \geq 1$ is always an ordinary finite natural number. A *small* family means a family indexed by a set in this universe. A small family can be uncountable, and its parameters can have very large birthdays. A family of definable classes is understood through a set of formula-and-parameter codes, not as a set whose members are proper classes. Likewise, a face lattice described below is represented by the set of facial traces or finite affine-span codes, with its inclusion relation.

A subclass of $\mathbf{No}^d$ is convex when it is closed under all two-point convex combinations with coefficients in $\mathbf{No}$. A face is a convex subclass F of a convex class C such that

$$x, y \in C, \quad 0 < t < 1, \quad (1-t)x + ty \in F \quad \implies \quad x, y \in F.$$

The empty face and C itself are included. An exposed proper face has the form $\{x \in C : u \cdot x = c\}$, where $u \neq 0$ and $u \cdot C \leq c$.

For $x \in \mathbf{No}^d$, let $\|x\|_\infty = \max_j |x_j|$. We call a class *closed* when every point outside it has a disjoint positive-radius open box, and *bounded* when it is contained in a box of some surreal radius. These are pointwise descriptions; we do not treat the collection of all open subclasses as an ordinary set topology. Relative interior is defined in the same way inside the affine hull.

A small presentation is not a small convex class. A hull of two distinct points already contains a copy of the entire surreal interval $[0, 1]$, a proper class. For example, that interval contains distinct $\omega^{-\alpha}$ for all positive ordinals α . All size restrictions below concern generators, rows, parameters, or formulas—not the elements of the convex hull.

153.2 Cuts and relative embeddings

The classical surreal cut property says that for sets $L, R \subseteq \mathbf{No}$ with $L < R$, there is $z \in \mathbf{No}$ such that $L < z < R$, including when either side is empty. We also use that $\mathbf{No}$ is real closed. The relationship between these facts, universality, and absolute saturation is developed in Ehrlich's work [95]; fields of surreal numbers are studied further in [96].

Lemma 153.1 (Relative embedding). *Let $K \subseteq \mathbf{No}$ be a set-sized real closed subfield, and let E/K be a set-sized ordered real closed field extension. There is an ordered-field embedding $E \hookrightarrow \mathbf{No}$ that is the identity on K .*

Proof. Well-order the elements of E . Construct embeddings of real closed intermediate subfields by recursion through this set-sized well-order. At a successor stage, an element not in the present real closed field F is transcendental over F . Its position determines a cut of F : the two sets of elements below and above it partition F . Transport this cut to the image of F and realize it in $\mathbf{No}$.

The cut-realizing number is not in the image of F , since every element of that image lies on one side of the cut. It is therefore transcendental over that real closed image. The resulting map on $F(a)$ preserves order: a polynomial over a real closed field factors into real linear factors and quadratic factors of fixed sign, so the cut determines the signs of all its nonzero values at a transcendental element. Extend this embedding to the ordered real closure of $F(a)$ inside E and $\mathbf{No}$. Ordered real closures are unique up to ordered isomorphism over the base. At a limit stage take the union; a union of a chain of real closed fields is real closed. The recursion has set length, so all cuts used are sets. $\square$

Theorem 153.2 (Small-type realization). *Let $p(x)$ be a set of ordered-field formulas in a fixed finite tuple of variables, with a set of parameters from $\mathbf{No}$. If every finite subset of p has a solution in $\mathbf{No}$, then all of p has a solution in $\mathbf{No}$.*

Proof. Put all parameters in a set-sized real closed subfield $K \subseteq \mathbf{No}$. For each finite conjunction from p , its existential closure is true in $\mathbf{No}$. Quantifier elimination for real closed fields makes that sentence equivalent to a quantifier-free statement over K , and hence true in K . This use of quantifier elimination is pointwise; no set-sized model with underlying domain $\mathbf{No}$ is being postulated.

Apply ordinary first-order compactness to the elementary diagram of K and the formulas p with new constants for x . A set-sized elementary extension E realizes them. It is a real closed ordered extension of K . Embed E in $\mathbf{No}$ over K by lemma 153.1. Quantifier elimination again makes this embedding elementary for ordered-field formulas, so the image of the realizing tuple realizes p . $\square$

We will repeatedly use two special cases. A small collection of positive surreals has a positive strict lower bound, and a small collection of surreals has a strict upper bound. These are immediate cuts. Less obviously, infinitely many simultaneous linear equations, inequalities, and nonmembership conditions can be solved whenever all finite subcollections can.

Remark 153.3 (What saturation does not say). The theorem gives realization for a set of formulas. Replacing that set by a proper class is invalid. Nor may one silently replace the full current-universe $\mathbf{No}$ by an arbitrary birthday cutoff or by an old field $\mathbf{No}^M$ viewed from a larger universe. Section 163 isolates the precise hypothesis needed in those settings.

154 Finite convexity and facial traces

154.1 The finite algebraic backbone

We recall the finite results used below. Over an ordered field, a point in a convex hull in dimension d is a combination of at most $d + 1$ generators. Indeed, if a positive-support representation has more terms, choose an affine dependence (μ_j) , so that $\sum \mu_j = 0$ and $\sum \mu_j s_j = 0$. After changing sign, some $\mu_j > 0$. Subtract $\min_{\mu_j > 0} \lambda_j / \mu_j$ times that dependence from the coefficients. At least one coefficient vanishes and none becomes negative. Iterate. This is the usual algebraic proof of Carathéodory's theorem.

Finite systems of linear inequalities admit Fourier–Motzkin elimination: a variable with lower and upper bounds is eliminated by all pairwise comparisons of those bounds. Strict bounds are retained when appropriate. Only field arithmetic and order are involved. Its consequences include finite Farkas alternatives, invariance of finite linear feasibility under ordered-field extensions, and finite polyhedral descriptions of finite convex hulls. The finite vertex–halfspace theorem for bounded polyhedra and exposure of all finite-polytope faces are used in their standard ordered-field form [97, 98].

For separation alone it would also suffice to use real-closed-field transfer: for fixed finite numbers of vertices and a fixed dimension, the assertion that two disjoint finite hulls have a strict separator is a first-order statement true over $\mathbb{R}$.

Lemma 154.1 (Finite ingredients). *In finite dimension over an ordered field, the following hold.*

- (i) *Disjoint nonempty finite convex hulls admit a strict affine separator. The separator can be rescaled to have any specified positive finite-list margin.*
- (ii) *Every nonempty proper face of a finite polytope is exposed.*
- (iii) *If $P = \{x : a_i \cdot x \leq b_i, i \in J\} \neq \emptyset$ with J finite, then $v \cdot x \leq \beta$ is valid on P exactly when*

$$v = \sum_{i \in J} \lambda_i a_i, \quad \sum_{i \in J} \lambda_i b_i \leq \beta, \quad \lambda_i \geq 0.$$

154.2 A face is determined by its generators

Lemma 154.2 (Trace and affine hull). *Let $C = \text{conv}_K S$ over an ordered field K , and let F be a nonempty face. Then*

$$F = \text{conv}_K(S \cap F), \quad F = C \cap \text{aff}_K F.$$

Every nonempty finite-dimensional convex class has a relative interior point. In particular, a face has one.

Proof. In a finite convex combination belonging to a face, every generator with positive coefficient belongs to that face. This follows by isolating one coefficient and applying the segment definition; the single-coefficient case is immediate. It proves the first equality.

Choose an affinely independent finite set spanning the affine hull of a nonempty convex class. Its simplex lies in the class, and its barycenter contains a relative open box in the affine span, as is seen from its strictly positive barycentric coordinates. This proves the relative interior assertion using only finite inequalities.

Choose $p \in \text{ri } F$ and $x \in C \cap \text{aff } F$. For sufficiently small $t > 0$, $y = p - t(x - p)$ lies in F . The point p is an interior point of the segment $[x, y]$ unless $x = p$. The face property gives $x \in F$, proving the second equality. $\square$

Proposition 154.3 (Finite character of facial traces). *For $I \subseteq S$, the following are equivalent:*

- (i) $I = S \cap F$ for a face F of $\text{conv}_K S$.
- (ii) For every finite $T \subseteq S$, there is an affine function over K that is zero on $T \cap I$ and strictly negative on $T \setminus I$.

Under these conditions, $F = \text{conv}_K I$. The empty and full traces are allowed.

Proof. If F exists, its intersection with $\text{conv}_K T$ is a face of that finite polytope, generated by $T \cap I$. Finite exposure gives the affine function when this face is nonempty and proper. Use the constant functions -1 and 0 in the empty and full cases.

Conversely, a point of $S \setminus I$ cannot belong to $\text{conv}_K I$: a hypothetical expression would involve only finitely many generators, and the corresponding affine function would be both negative and zero there. Thus the trace of $\text{conv}_K I$ is exactly I .

Suppose an interior point of a segment $[x, y]$ in $\text{conv}_K S$ belongs to $\text{conv}_K I$. Gather the finite generators in expressions for x, y , and that point into one finite set T . The affine function for T is nonpositive at x, y and zero at their strict convex combination. Both values are zero. In the chosen representations of x, y , all generators with positive weight must therefore belong to I . Thus $x, y \in \text{conv}_K I$. $\square$

Corollary 154.4 (Scalar extension preserves faces). *For an ordered-field extension $K \subseteq L$ and a set $S \subseteq K^d$, the face lattices of $\text{conv}_K S$ and $\text{conv}_L S$ correspond canonically by their traces on S . Face dimensions are preserved. In particular,*

$$F \longmapsto \text{conv}_L(S \cap F)$$

is an inclusion-preserving bijection of faces.

Proof. The finite systems in [proposition 154.3](#) are feasible over K if and only if they are feasible over L , by finite linear elimination. The trace criterion therefore gives the same collection of subsets of S in both fields. Affine ranks of finite matrices over K do not change after scalar extension. $\square$

Although the face lattice is preserved, the existence of a *single* exposing functional is an infinite system and need not descend to K . That distinction is essential later.

[write] Part I makes the same distinction for a finite list of real vertices ([section 20](#), subsection “A real face and its scalar extension are different sets”); [corollary 154.4](#) extends the face correspondence to an arbitrary set of generators.

155 Separation, closedness, and exposure of small hulls

Theorem 155.1 (Uniform strong separation). *Let $S, T \subseteq \mathbf{No}^d$ be nonempty sets such that $V(S) \cap V(T) = \emptyset$. There exist $u \in \mathbf{No}^d \setminus \{0\}$ and $c \in \mathbf{No}$ with*

$$u \cdot s \leq c - 1 \quad (s \in S), \quad u \cdot t \geq c + 1 \quad (t \in T).$$

The same inequalities hold throughout the two hulls.

Proof. View the displayed inequalities as a type in the $d + 1$ unknowns (u, c) . Every finite subsystem concerns disjoint finite hulls. Add one fixed generator from each side if a side is absent from that subsystem. Apply [lemma 154.1\(i\)](#) and rescale a strict separator around the

midpoint of its finite gap so that both margins are at least 1. Thus every finite subsystem is feasible. By [theorem 153.2](#), the whole type is realized. Since both S and T are nonempty, its solution cannot have $u = 0$. Finite convex combinations preserve the inequalities. $\square$

Corollary 155.2 (Bounded and closed). *Every small hull $V(S)$ is order bounded and closed. For each exterior point there is a strictly separating affine halfspace, and consequently*

$$V(S) = \bigcap \{H : H \text{ is a closed affine halfspace containing } V(S)\}.$$

The intersection on the right is generally proper-class indexed.

Proof. A surreal M strictly greater than all $|s_j|$ exists because this is a set of coordinates. Every finite convex combination is contained in $[-M, M]^d$.

For $x \notin V(S)$ with S nonempty, apply [theorem 155.1](#) to S and $\{x\}$. If $\|y - x\|_\infty < \delta$, where

$$\delta = \frac{1}{1 + \sum_j |u_j|},$$

then $|u \cdot (y - x)| < 1$. Hence this open box around x misses $V(S)$. The separating halfspace also excludes x , proving the intersection formula. Empty hulls satisfy the assertions directly. $\square$

Boundedness here means bounded by a surreal, not by an ordinary real. Closedness says nothing about attainment of infima or about finite subcovers of arbitrary open covers.

Theorem 155.3 (All small-hull faces are exposed). *Let $C = V(S)$ for a set $S \subseteq \mathbf{No}^d$, and let F be a nonempty proper face of C . Put $I = S \cap F$. There are $u \neq 0$ and c such that*

$$\begin{aligned} u \cdot i &= c & (i \in I), \\ u \cdot s &\leq c - 1 & (s \in S \setminus I). \end{aligned}$$

In particular, $F = \{x \in C : u \cdot x = c\}$.

Proof. By [lemma 154.2](#), $F = V(I)$, so I is nonempty; $S \setminus I$ is also nonempty because F is proper. A finite subsystem of the displayed equations and inequalities concerns a finite set $T \subseteq S$. Add fixed elements of I and $S \setminus I$ to T . The nonempty proper face $F \cap V(T)$ of the finite polytope $V(T)$ is exposed. Its finitely many outside generators have strictly positive slack. Rescale by the reciprocal of their minimum slack. This satisfies the finite subsystem.

Realize the entire type by [theorem 153.2](#). For $x = \sum_j \lambda_j s_j \in C$, its slack satisfies

$$c - u \cdot x \geq \sum_{s_j \notin I} \lambda_j.$$

It is zero for $x \in V(I)$, and zero slack in any representation forces all positively weighted generators to belong to I . This proves the claimed exposed-face identity. The outside inequalities force $u \neq 0$. $\square$

Remark 155.4 (The margin is a generator margin). There is no positive uniform gap on all of $C \setminus F$. If $p \in F$ and $s \notin F$, points $(1 - t)p + ts$ approach F as $t > 0$ decreases. The unit margin in [theorem 155.3](#) concerns the specified outside generators. Confusing these two assertions would make the theorem false even for an ordinary segment.

Corollary 155.5 (Number of faces). *Every extreme point of $V(S)$ belongs to S . Every nonempty face is determined by the affine hull of at most $d + 1$ elements of S . Thus its face collection is set-indexed, and if S is infinite it has at most $|S|$ faces.*

Proof. A singleton face is generated by its trace, so its point belongs to S . For any nonempty face choose at most $d + 1$ generators spanning its affine hull; lemma 154.2 recovers the face by intersection with that affine hull. There are only $|S|$ finite subsets of an infinite set S . $\square$

156 Small halfspace systems: certificates and active faces

Fix a set I and a system

$$H = \{x \in \mathbf{No}^d : a_i \cdot x \leq b_i \text{ for every } i \in I\}.$$

The rows may be redundant, and zero rows are permitted.

Theorem 156.1 (Finite certificates from small systems). *The following statements hold.*

- (i) *If $H = \emptyset$, some finite subsystem is infeasible.*
- (ii) *If $H \subseteq \{x : v \cdot x \leq \beta\}$, the same implication already holds for a finite subsystem of the original rows.*
- (iii) *If $H \neq \emptyset$, every valid inequality has a finite Farkas certificate:*

$$v = \sum_{i \in J} \lambda_i a_i, \quad \sum_{i \in J} \lambda_i b_i \leq \beta, \quad J \subseteq I \text{ finite}, \quad \lambda_i \geq 0.$$

Proof. For (i), finite feasibility of all rows would imply feasibility by theorem 153.2. For (ii), if no finite subsystem implied the desired inequality, the type consisting of every input row together with $v \cdot x > \beta$ would be finitely satisfiable. Its realization would contradict validity. For (iii), the finite subsystem supplied by (ii) is nonempty because it contains H . Apply finite Farkas theory to it. $\square$

This is a finiteness theorem for *each consequence*, not a finiteness theorem for the entire row system. Different consequences may require different finite sublists.

156.1 A uniform neighborhood in the active affine space

At $x \in H$, define

$$A(x) = \{i \in I : a_i \cdot x = b_i\}, \quad W_x = \bigcap_{i \in A(x)} \ker a_i.$$

The inactive slacks form a set of positive numbers. A single positive surreal can be chosen smaller than all their suitably normalized values. This eliminates the familiar real phenomenon of infinitely many nearly active inequalities creating a new local boundary without an active row.

Theorem 156.2 (Finite active-face theorem). *For each $x \in H$, choose $J \subseteq A(x)$ whose normals span all the active normals. It can be chosen with $|J| \leq d$. Then*

$$F_x = H \cap \bigcap_{i \in J} \{y : a_i \cdot y = b_i\}$$

is the minimal face of H containing x , and $x \in \text{ri } F_x$. Every nonempty proper face of H is such a slice. It is exposed by the sum of at most d of the original row normals.

Proof. For each inactive row let $s_i = b_i - a_i \cdot x > 0$. Choose $\delta > 0$ with

$$\delta < \frac{s_i}{1 + \sum_{j=1}^d |a_{ij}|} \quad (i \notin A(x)).$$

The cut property gives such a δ ; if there are no inactive rows, take any positive δ . If $w \in W_x$ and $\|w\|_\infty < \delta$, all active rows remain equalities and all inactive rows remain satisfied. Thus $x + (W_x \cap (-\delta, \delta)^d) \subseteq H$.

Finite-dimensional linear algebra gives the stated J . Its common kernel is W_x , so the displayed slice has affine hull $x + W_x$ and contains a relative neighborhood of x . Each active equality cuts out a face, and therefore F_x is a face. If another face G contains x , then for every $y \in F_x$ the small relative neighborhood permits a point of F_x just beyond x on the line from y to x . The face property at x forces $y \in G$. Hence F_x is minimal.

For an arbitrary nonempty face F , choose $x \in \text{ri } F$ using [lemma 154.2](#). The same line argument shows that every face containing x contains F , so $F = F_x$. Finally, all the slacks of the rows in J are nonnegative on H . Their sum vanishes exactly when each vanishes. Thus the sum of their normals exposes the slice. If the slice is proper, that sum cannot be zero: evaluated at x , a zero summed normal would give zero summed bound and force the summed slack to vanish everywhere on H . $\square$

Corollary 156.3 (Vertices come from active bases). *A point $x \in H$ is extreme if and only if its active row normals span $\mathbf{No}^d$. Equivalently, x is the unique solution of d independent active input equations. Consequently $\text{ext } H$ is a set.*

Proof. The minimal face at x has direction space W_x and a relative neighborhood in that space. It is a singleton precisely when $W_x = 0$. Independent d -tuples of rows are set-indexed, and each has at most one solution, given by a finite matrix inverse. Their feasible solutions form a set containing every extreme point. $\square$

Remark 156.4. No boundedness hypothesis was needed in [theorem 156.2](#). The system can have no vertices, and it can require genuinely infinitely many rows even though every one of its faces has a finite active description.

157 Finite sandwiches and V/H rigidity

A class is *ordered-field definable* if it is defined by one ordered-field formula with finitely many surreal parameters. By real closed quantifier elimination, these are the semialgebraic classes. A small intersection of such classes need not itself be definable.

Theorem 157.1 (Finite sandwich). *Let $(D_i)_{i \in I}$ be a set-indexed family of ordered-field definable subclasses of $\mathbf{No}^d$, with a set of parameters in total. Let $S \subseteq \mathbf{No}^d$ be a set. If*

$$D := \bigcap_{i \in I} D_i \subseteq V(S),$$

then there are finite $I_0 \subseteq I$ and $S_0 \subseteq S$ such that

$$\bigcap_{i \in I_0} D_i \subseteq V(S_0).$$

Empty intersections and empty hulls have their usual meanings.

Proof. Suppose no such finite pair exists. In a finite tuple of variables x , consider the type

$$\{x \in D_i : i \in I\} \cup \{x \notin V(T) : T \subseteq S \text{ finite}\}.$$

Membership in a finite hull is an ordered-field formula: quantify its finitely many nonnegative coefficients with sum 1. The finite subsets of S form a set, so this is a small type.

Take any finite part of it. Let I_0 be the rows occurring in that part, and let S_0 be the union of the finitely many sets T occurring in its nonmembership conditions. By the assumed failure of every finite sandwich, a point lies in all D_i , $i \in I_0$, but not in $V(S_0)$. It then lies in none of the smaller hulls $V(T)$. Thus the type is finitely satisfiable. A realization lies in D but in no finite hull from S , contradicting $D \subseteq V(S)$. $\square$

Theorem 157.2 (Double finite compression). *Suppose*

$$C = V(S) = \bigcap_{i \in I} \{x : a_i \cdot x \leq b_i\},$$

where S and I are sets. There are finite sublists $S_0 \subseteq S$ and $I_0 \subseteq I$ with

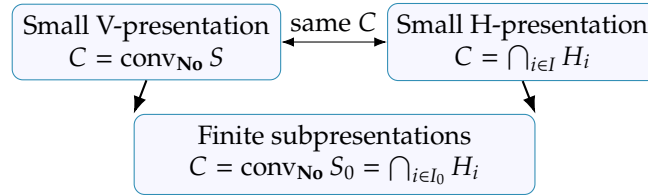
$$C = V(S_0) = \bigcap_{i \in I_0} \{x : a_i \cdot x \leq b_i\}.$$

Conversely, every finite polytope admits both sorts of small presentation.

Proof. Apply [theorem 157.1](#) to the halfspaces. It gives

$$\bigcap_{i \in I_0} H_i \subseteq V(S_0) \subseteq C \subseteq \bigcap_{i \in I} H_i.$$

All inclusions are equalities. The converse is the finite vertex–halfspace theorem; lower-dimensional affine constraints can be written as pairs of inequalities. $\square$



157.1 Definability is already a finiteness condition

Corollary 157.3 (Small hulls and definability). *For a small hull $C = V(S)$, the following are equivalent:*

- (i) C is a finite polytope.
- (ii) C is ordered-field definable.
- (iii) C is an intersection of a set of ordered-field definable classes.

In (iii), finite generators from the original S already generate C .

Proof. Finite hulls are definable, and a definable class is a singleton-indexed intersection. For (iii), the finite sandwich yields $C \subseteq V(S_0)$ for a finite $S_0 \subseteq S$, while the reverse inclusion is automatic. $\square$

Corollary 157.4 (Definable small H-classes). *If a small halfspace intersection H is ordered-field definable, then a finite subsystem of its original rows defines H . Hence it is a polyhedron; if nonempty and bounded, it is a finite polytope.*

Proof. If no finite subsystem sufficed, the type consisting of every row and the single formula $x \notin H$ would be finitely satisfiable. Saturation would contradict the definition of H . The remaining assertions are finite polyhedral theory. $\square$

These results have a useful interpretation. A small hull is a small *union* of definable finite hulls. If it is also a small *intersection* of definable classes, saturation forces finite compression. This is a standard compactness mechanism with unusually strong geometric consequences because all externally set-sized data are small relative to the full surreal field.

158 Polarity and the obstruction to vertex generation

For a convex class P containing 0, write

$$P^\circ = \{y \in \mathbf{No}^d : y \cdot x \leq 1 \text{ for every } x \in P\}.$$

If $P = V(S)$, testing the generators suffices, so P° has the small H-presentation $y \cdot s \leq 1$ for $s \in S$.

Theorem 158.1 (Small-hull polarity). *Let $P = V(S)$ be full-dimensional and let $0 \in \text{ri } P$. Then P° is closed, bounded, and has 0 in its interior. Moreover,*

$$P^{\circ\circ} = P, \quad P^\circ \text{ is a small hull} \iff P \text{ is a finite polytope.}$$

Proof. Choose $M > 0$ with $P \subseteq [-M, M]^d$. Then a sufficiently small box around 0, for example of radius $1/(2dM)$, lies in P° . Choose $r > 0$ so small that each $\pm re_j$ lies in P . Testing these points gives $|y_j| \leq 1/r$ for $y \in P^\circ$, proving boundedness. Closedness follows directly from the defining halfspaces.

The inclusion $P \subseteq P^{\circ\circ}$ is immediate. If $x \notin P$, strong separation gives $u \cdot P \leq c - 1$ and $u \cdot x \geq c + 1$. Since $0 \in P$, $c \geq 1$. Hence $u/c \in P^\circ$, but $(u/c) \cdot x > 1$. This proves the reverse inclusion.

If P° is a small hull, it has both a small V- and a small H-presentation. By [theorem 157.2](#), it is a finite polytope. Its finite vertex list then gives a finite H-presentation of $P^{\circ\circ} = P$; the original small V-presentation and rigidity make P finite. Conversely, the polar of a finite polytope containing 0 in its interior is a bounded finite polyhedron, hence a finite polytope. $\square$

Corollary 158.2 (A bounded small H-class is generated by its extreme points only in the finite case). *Let H be a nonempty bounded small halfspace intersection. Then*

$$H = \text{conv}_{\mathbf{No}}(\text{ext } H) \iff H \text{ is a finite polytope.}$$

The same equivalence holds when the hull on the left is replaced by its order-box closure.

Proof. By [corollary 156.3](#), $\text{ext } H$ is a set. Equality would give a small V-presentation and therefore a finite polytope. The converse is finite polytope theory. By [corollary 155.2](#), the hull of the set $\text{ext } H$ is already closed, so taking its closure does not change it. $\square$

This is not a contradiction to a real compact-convex generation theorem: order bound-ness and pointwise closedness in $\mathbf{No}^d$ supply neither its compactness hypothesis nor its Archimedean setting. The failure is sharper than merely requiring more vertices. In [theorem 158.1](#), a nonfinite small hull has a polar that cannot be generated by *any set* of points, even points that are not extreme.

159 Real faces and one-infinitesimal exposure

For a real convex set $C \subseteq \mathbb{R}^d$, define its *generator extension*

$$C^\uparrow = \text{conv}_{\mathbf{No}} C.$$

The generators here are the actual real points of C . By [corollary 154.4](#), its faces are the classes $F^\uparrow$ for real faces F of C ; its affine dimension is unchanged. By [theorem 155.3](#), all these faces are exposed over $\mathbf{No}$.

159.1 The classical invariant

For a nonempty proper real face F of C , let $r_C(F)$ be the minimum length of a chain

$$C = F_0 \supsetneq F_1 \supsetneq \cdots \supsetneq F_r = F$$

in which F_j is an exposed face of F_{j-1} , using a real affine functional on the latter's affine hull. Martínez-Legaz proved that every face admits such a chain, defined its degree of non-exposedness, and established the bound [\[99\]](#)

$$\delta_C(F) = r_C(F) - 1 \leq \dim C - \dim F - 1.$$

Recent work studies the extension of face-characterization questions to infinite-dimensional vector spaces [\[100\]](#); that is a different change of scope from the finite-dimensional, proper-class scalar setting used here.

For completeness, the existence and dimension bound also follow by an ordinary supporting-hyperplane argument. Choose $p \in \text{ri} F$. Unless the current convex set is F , the point p lies on its relative boundary: a face containing a relative interior point would be the entire set. A real supporting hyperplane at p cuts a proper exposed face containing F . Continue in that face's affine hull. At every step the dimension strictly decreases. A proper inclusion of nonempty faces cannot preserve affine dimension, by [lemma 154.2](#); consequently the process reaches F after at most $\dim C - \dim F$ steps.

159.2 An exact polynomial translation

Fix a positive $\varepsilon \in \mathbf{No}$ infinitesimal relative to $\mathbb{R}$: $0 < \varepsilon < r$ for every real $r > 0$. A finite polynomial with real coefficients has the sign of its first nonzero coefficient when evaluated at ε . In particular, ε is transcendental over $\mathbb{R}$.

Definition 159.1. Let $m_\varepsilon(C, F)$ be the least $m \geq 0$ for which there are real linear functionals $u_0, \dots, u_m$ such that

$$U_\varepsilon = u_0 + \varepsilon u_1 + \cdots + \varepsilon^m u_m$$

exposes $F^\uparrow$ in $C^\uparrow$. Functionals are considered on $\text{aff } C$, and may be extended to $\mathbb{R}^d$. A constant term in an affine functional is absorbed by evaluating its difference from a fixed $p \in F$.

Theorem 159.2 (Polynomial degree equals classical non-exposedness). *For every nonempty proper face F of a real finite-dimensional convex set C ,*

$$m_\varepsilon(C, F) = r_C(F) - 1 = \delta_C(F).$$

It is independent of the chosen positive real-infinitesimal ε , and

$$0 \leq m_\varepsilon(C, F) \leq \dim C - \dim F - 1.$$

Proof. Choose a shortest exposed chain of length r , a point $p \in F$, and real linear functionals $u_0, \dots, u_{r-1}$ for its successive steps, normalized only by subtracting their values at p . For a real point $x \in C \setminus F$, let $j \in \{0, \dots, r-1\}$ be the index with $x \in F_j \setminus F_{j+1}$. Then $u_i(x - p) = 0$ for $i < j$ and $u_j(x - p) < 0$. The subsequent values can have either sign, but the first nonzero real coefficient determines

$$U_\varepsilon(x - p) = \sum_{i=0}^{r-1} \varepsilon^i u_i(x - p) < 0.$$

It is zero for real $x \in F$. Finite surreal convex combinations preserve nonpositivity. A combination has value zero only if every generator with positive coefficient belongs to F . Thus U_ε exposes $F^\uparrow$, and $m_\varepsilon(C, F) \leq r - 1$.

Conversely, suppose a polynomial of degree m exposes $F^\uparrow$. For real $x \in C$, the coefficient vector

$$(u_0(x - p), u_1(x - p), \dots, u_m(x - p))$$

is lexicographically nonpositive, and it is zero precisely on F . Indeed, a nonzero coefficient vector cannot evaluate to zero at the transcendental ε . Restrict successively to the zero sets of u_0 , then u_1 , and so on. At each stage, the next functional is nonpositive on the current face. Its zero set is therefore exposed there. Discard stages at which the functional is constant on the current face. The remaining chain reaches F in at most $m + 1$ strict steps, giving $r_C(F) \leq m + 1$. The two bounds prove the formula. $\square$

The new notation m_ε does not introduce a different real invariant. The theorem identifies exactly what a one-infinitesimal polynomial encoding measures. Its proof also explains why the coefficients must be real, or belong to a specified field over which ε is infinitesimal. Allowing arbitrary surreal coefficients would make the “degree” vacuous: any surreal exposing normal could be placed in degree zero.

[write] This is the report’s third notion of degree. Part I’s degree (theorem 9.1) is the least degree of a polynomial height vector over a fixed real base; Part VIII’s (definition 120.2) is the homogeneous degree of a rational arc along which every vertex moves; the degree here, $m_\varepsilon(C, F) = \delta_C(F)$, is the degree of non-exposedness of a real face.

160 An explicit sharp family in every codimension

We now construct compact real convex bodies for which the bound in theorem 159.2 is attained. This supplies concrete test objects rather than merely an abstract dimension estimate.

Set $K_1 = [0, 1]$. For $d \geq 2$, write $x = (z, t) \in \mathbb{R}^{d-1} \times \mathbb{R}$ and set

$$G_d = \{(z, \|z\|_2^2) : z \in \mathbb{R}^{d-1}, \|z\|_2 \leq 1\}, \quad (160.1)$$

$$K_d = \text{conv}_{\mathbb{R}}((K_{d-1} \times \{0\}) \cup G_d). \quad (160.2)$$

Each K_d is compact: the union inside the hull is compact, and its convex hull is the continuous image of a compact collection of at most $d + 1$ points and their simplex of coefficients. It is full-dimensional because $K_{d-1} \times \{0\}$ spans its base hyperplane and G_d contains points above that hyperplane. These real sets are also semialgebraic, by their finite quantified descriptions and real quantifier elimination.

Lemma 160.1 (A forced first exposure). *The origin is an extreme point of K_d . Its only proper exposed face containing it, for $d \geq 2$, is $K_{d-1} \times \{0\}$.*

Proof. Every last coordinate in (160.2) is nonnegative, and a point of G_d has last coordinate zero only at the origin. Thus

$$K_d \cap \{t = 0\} = K_{d-1} \times \{0\}.$$

Extremality of the origin follows inductively from extremality in this base face.

Suppose a nonzero real linear functional $(a, b) \cdot (z, t)$ has its maximum at 0 on K_d . For any fixed $v \in \mathbb{R}^{d-1}$ and all sufficiently small positive real s , both

$$s a \cdot v + b s^2 \|v\|_2^2 \leq 0, \quad -s a \cdot v + b s^2 \|v\|_2^2 \leq 0$$

hold. Dividing by s and letting s decrease through ordinary reals forces $a \cdot v = 0$. Hence $a = 0$, and the nonzero supporting functional has $b < 0$. Its maximizing face is exactly the displayed base face. $\square$

Theorem 160.2 (Sharp degree and sharp depth). *For every $d \geq 1$,*

$$r_{K_d}(\{0\}) = d, \quad m_\varepsilon(K_d, \{0\}) = d - 1.$$

An exposing functional on $K_d^\uparrow$ is

$$L_\varepsilon(x_1, \dots, x_d) = -x_d - \varepsilon x_{d-1} - \dots - \varepsilon^{d-1} x_1.$$

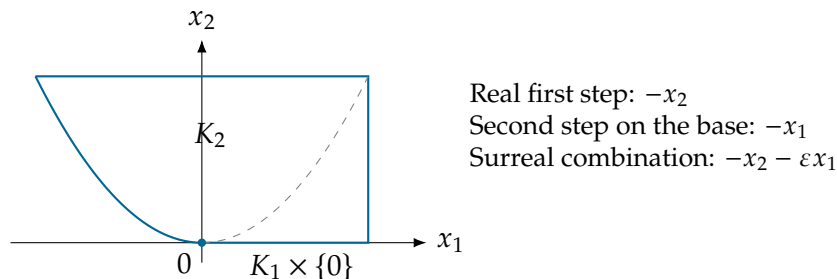
For every pair $d > f \geq 0$, the product

$$C = K_{d-f} \times [-1, 1]^f, \quad F = \{0\} \times [-1, 1]^f$$

has $\dim C = d$, $\dim F = f$, and $m_\varepsilon(C, F) = d - f - 1$.

Proof. For $d = 1$, the origin is exposed by $-x_1$, so the depth is 1. For $d \geq 2$, lemma 160.1 forces the first step in every exposed chain to be the base $K_{d-1} \times \{0\}$. Consequently the minimum depth is one more than the depth in K_{d-1} , giving d by induction. The degree formula follows from theorem 159.2. The displayed functional is obtained by using the successive negative coordinate functionals along the forced chain.

For the product, that same polynomial functional, ignoring the last f coordinates, gives the upper bound. Any exposing polynomial for the product face restricts on $K_{d-f} \times \{0\}$ to one exposing its origin. It therefore has degree at least $d - f - 1$, proving the lower bound. $\square$



160.1 The necessary semantic distinction

Generator extension is not definable extension

For a semialgebraic real set C , the two classes

$$C^\uparrow = \text{conv}_{\mathbf{No}}(C(\mathbb{R})) \quad \text{and} \quad C(\mathbf{No})$$

are generally different. The second means evaluating a defining semialgebraic formula with arbitrary surreal variables. The first retains only the real points as generators. The exposure theorems in Sections 159–160 concern the first.

The sharp family gives an exact witness. For $d \geq 2$, the full semialgebraic extension of its paraboloid contains the point with

$$x_{d-1} = -\varepsilon/2, \quad x_d = \varepsilon^2/4, \quad x_j = 0 \quad (j < d-1).$$

At this point,

$$L_\varepsilon(x) = -\varepsilon^2/4 + \varepsilon^2/2 = \varepsilon^2/4 > 0.$$

Thus L_ε does *not* support $K_d(\mathbf{No})$ at the origin. Since it is nonpositive on $K_d^\uparrow$, the displayed point is not in that set-generated hull. This explicitly prevents the erroneous conclusion that all faces of every semialgebraic surreal convex body are exposed.

161 The rational moment curve: a complete face classification

The next example exhibits genuinely infinite combinatorics. Fix $d \geq 2$ and define

$$m_d(t) = (t, t^2, \dots, t^d), \quad M_d = \text{conv}_{\mathbf{No}}\{m_d(t) : t \in \mathbb{Q}\}, \quad k = \lfloor d/2 \rfloor.$$

Although the rational curve is unbounded in real Euclidean space, all its coordinates are finite reals. Hence $M_d \subseteq [-\omega, \omega]^d$. It is closed by [corollary 155.2](#) and has full affine dimension by the Vandermonde determinant.

Theorem 161.1 (All faces of the rational moment hull). *The nonempty proper faces of M_d are exactly*

$$F_A = \text{conv}_{\mathbf{No}}\{m_d(a) : a \in A\}, \quad A \subseteq \mathbb{Q}, \quad 1 \leq |A| \leq \lfloor d/2 \rfloor.$$

Each F_A is an $(|A| - 1)$ -simplex. Every rational curve point is an extreme point. There is no set-sized halfspace presentation of M_d .

Proof. First, let A have at most k elements. The real polynomial

$$q_A(t) = \prod_{a \in A} (t - a)^2$$

has degree at most d , is nonnegative on $\mathbb{R}$, and vanishes exactly at A . Express it as $q_A(t) = c - u \cdot m_d(t)$, allowing zero coefficients through degree d . This affine slack is nonnegative at all generators, and zero exactly at the generators indexed by A . It exposes F_A . Vandermonde determinants show that any at most $d + 1$ distinct moment points are affinely independent, so F_A is the indicated simplex.

Conversely, let F be a nonempty proper face. By [theorem 155.3](#), it is exposed by a surreal affine functional, with nonzero slack polynomial

$$q(t) = c - u \cdot m_d(t) \in \mathbf{No}[t], \quad \deg q \leq d, \quad q(t) \geq 0 \quad (t \in \mathbb{Q}).$$

The polynomial is nonzero because a zero polynomial would give zero slack on all generators, hence the entire hull as the face. Put $B = \max_j |q_j| > 0$, where q_j are its finitely many coefficients, including the constant coefficient. All coefficients of q/B are finite surreals, and at least one is exactly 1 or -1 . Take their real standard parts to obtain a nonzero polynomial $\bar{q} \in \mathbb{R}[t]$ of degree at most d . Standard part exists for a finite surreal by the real cut it determines; it preserves sums, products, and weak inequalities among finite elements.

For rational t , $\bar{q}(t) = \text{st}(q(t)/B) \geq 0$. Real continuity and the density of $\mathbb{Q}$ imply $\bar{q} \geq 0$ throughout $\mathbb{R}$. Every rational zero of q is a zero of $\bar{q}$. A nonzero real polynomial nonnegative on all of $\mathbb{R}$ has even multiplicity at each distinct real root. Therefore the set of rational zeros of q has at most $\lfloor d/2 \rfloor$ elements. By [lemma 154.2](#), these are exactly the parameters generating F .

Singleton choices of A prove that all rational curve points are extreme. There are infinitely many of them. A set-sized H-presentation would make M_d a finite polytope by [theorem 157.2](#), contradicting its infinitely many extreme points. $\square$

The last argument can also be read as a classification of the real finite-convex-combination hull of the rational moment curve, transported to **No** by [corollary 154.4](#). Its special surreal feature is that this hull becomes order bounded and closed without gaining the extra face strata that a usual real compactification would introduce.

161.1 The resulting face poset

Including the empty face and the top element, its abstract face poset is

$$\{\emptyset\} \cup \{A \subseteq \mathbb{Q} : 1 \leq |A| \leq k\} \cup \{\top\},$$

ordered by inclusion among the finite subsets, with $\top$ above all. Thus every collection of at most k vertices spans a face, but there are no other proper faces.

Dimension	Nonempty proper face dimensions	Example of missing faces
2, 3	0 only	No edges
4, 5	0, 1	No two-dimensional faces
6, 7	0, 1, 2	No three-dimensional faces
d	$0, \dots, \lfloor d/2 \rfloor - 1$	No facets

This is not a finite-polytope face lattice, and ordinary gradedness by geometric dimension fails at its top element. Every finite subhull is an ordinary polytope, but a face of a finite subhull need not remain a face when more rational generators are admitted. The passage to the directed union of finite hulls is exactly where finite polytope intuition ceases to determine the global face structure.

161.2 The polar has a complementary face-dimension gap

The polar can be classified just as explicitly. Choose

$$b = \frac{1}{d+1} \sum_{r=0}^d m_d(r), \quad P_d = M_d - b,$$

and set

$$H_d = P_d^\circ = \{y \in \mathbf{No}^d : y \cdot (m_d(t) - b) \leq 1 \text{ for every } t \in \mathbb{Q}\}.$$

The barycenter b lies in the interior of the full-dimensional simplex on $m_d(0), \dots, m_d(d)$, so $0 \in \text{int } P_d$. Both P_d and H_d are closed, bounded, and full-dimensional.

Theorem 161.2 (Complete dual classification; no vertices in the polar). *The nonempty proper faces of H_d are exactly*

$$G_A = \{y \in H_d : y \cdot (m_d(a) - b) = 1 \text{ for every } a \in A\}, \quad 1 \leq |A| \leq k,$$

where A ranges over finite subsets of $\mathbb{Q}$ and $k = \lfloor d/2 \rfloor$. They satisfy

$$\dim G_A = d - |A|, \quad G_A \subseteq G_B \iff B \subseteq A.$$

In particular, H_d has no nonempty faces of dimension less than $\lceil d/2 \rceil$, and has no extreme points. The correspondence $F_A - b \leftrightarrow G_A$, together with the interchange of the empty and full faces, is an order-reversing bijection of the two face posets. For corresponding proper faces the dimensions sum to $d - 1$.

Proof. At $y \in H_d$, its active parameters are those $t \in \mathbb{Q}$ with $y \cdot (m_d(t) - b) = 1$. If this set is nonempty, it is the trace of a nonempty proper exposed face of P_d . The classification in [theorem 161.1](#) makes it a set A of at most k parameters.

Its active row vectors are linearly independent. Indeed, if $\sum_{a \in A} \lambda_a (m_d(a) - b) = 0$, taking the scalar product with y gives $\sum_a \lambda_a = 0$. Hence $\sum_a \lambda_a m_d(a) = 0$ is an affine dependence among at most k distinct moment points. Vandermonde independence forces every $\lambda_a = 0$. The active-face theorem therefore makes the minimal face at y equal to G_A , of dimension $d - |A|$. Every nonempty proper face has a relative interior point, so this accounts for all possibilities.

It remains to prove that every listed A actually occurs, without additional active parameters. Write $q_A(t) = c_A - u_A \cdot m_d(t)$ for the square-product polynomial from the previous theorem, and put

$$\alpha_A = c_A - u_A \cdot b = \frac{1}{d+1} \sum_{r=0}^d q_A(r) > 0.$$

The strict inequality holds because $q_A \geq 0$ and A cannot contain all $d+1$ distinct parameters $0, \dots, d$. For $y_A = u_A/\alpha_A$,

$$1 - y_A \cdot (m_d(t) - b) = q_A(t)/\alpha_A \geq 0$$

for all rational t , with equality exactly on A . Thus $y_A \in H_d$ has precisely the required active set. The preceding rank calculation and active-face theorem prove its face dimension.

If $B \subseteq A$, the equations give $G_A \subseteq G_B$. Conversely, such an inclusion puts y_A in G_B , so its exact active set implies $B \subseteq A$. This proves both distinctness and the asserted order relation. Since $d - |A| \geq d - k = \lceil d/2 \rceil \geq 1$, there are no zero-dimensional faces. The full dual correspondence and the dimension identity follow from the two classifications. $\square$

Dimension	Proper face dimensions of P_d	Proper face dimensions of H_d
2	0	1
3	0	2
4	0, 1	2, 3
5	0, 1	3, 4
6	0, 1, 2	3, 4, 5

Thus this is an explicit mutually polar pair with dual face posets and all proper faces exposed, but with no facets on one side and no vertices on the other. Each minimal nonempty face on the H-side is still positive-dimensional. It has no nonempty proper face of its own, because faces of faces are faces of the ambient convex class. By [theorem 158.1](#), H_d is not the convex hull of any set of points.

162 One-dimensional examples and failure modes

162.1 A closed bounded hull with no minimum

Let

$$C = \text{conv}_{\mathbf{No}}\{1/n : n \geq 1\} = \bigcup_{n \geq 1} [1/n, 1].$$

The equality follows because every finite convex combination lies between its smallest generator and 1, and every interval on the right is already a two-generator hull. This class is closed and bounded by [corollary 155.2](#). It has maximum 1 but no minimum: if $x \geq 1/n$, then $1/(n+1) < x$ is another element. Its only nonempty proper face is $\{1\}$. It is not a finite polytope, hence has no small H-presentation.

There is a direct cut proof of the last assertion. A nontrivial halfspace in one dimension is either a lower bound $x \geq a$ or an upper bound $x \leq b$. Validity on C forces $b \geq 1$, and it forces $a < 1/n$ for every n ; equality for some n would fail at $1/(n+1)$. Given any set of such lower bounds, the cut property supplies

$$a < z < 1/n \quad \text{for every listed } a \text{ and every } n, \quad z > 0.$$

This z satisfies all listed halfspaces but is not in C .

There is nevertheless a proper-class H-presentation:

$$C = \{x : x \leq 1, x \geq \eta \text{ for every positive real-infinitesimal } \eta \in \mathbf{No}\}.$$

Every member of C exceeds all positive infinitesimals. Conversely, a positive infinitesimal x violates the inequality indexed by $2x$; a nonpositive x violates any positive one. This is a transparent example of why the word “set” cannot be omitted from the rigidity theorem.

162.2 A countable bounded H-system with no vertices

Consider

$$H = \bigcap_{n \geq 1} [-1 - 1/n, 1 + 1/n].$$

It contains $[-1, 1]$ and is bounded by $[-2, 2]$. Every defining inequality is strict at every feasible point: equality in a row indexed by n would violate the tighter row indexed by $n+1$. Thus no point has an active nonzero row. The active-face theorem shows that every point is an interior point and that H has no extreme points. It is closed as an intersection of closed intervals. Hence this nonempty bounded class is simultaneously open and closed in the order-box sense.

One can see the local neighborhood directly. At a point $x \in H$, all row slacks are positive, and a positive surreal smaller than every slack allows both $x + \delta$ and $x - \delta$ to stay feasible. No finite subsystem defines H : a finite list allows one of its outer endpoints, whereas the full list excludes that endpoint. It follows either directly or by [theorem 157.2](#) that H is not a small hull.

162.3 The finite surreals as a hull

Another simple example is

$$\text{conv}_{\mathbf{No}} \mathbb{R} = \mathcal{O} := \{x \in \mathbf{No} : |x| < r \text{ for some real } r > 0\}.$$

Every finite combination of real endpoints is finite, and every finite surreal lies between two reals. The class $\mathcal{O}$ is a small hull, closed and bounded by ω , but has no extreme points. It

cannot be a small halfspace intersection. This example shows that small V-classes need not be generated by their extreme points either; the general face exposure theorem asserts exposure of *existing* faces, not their existence or sufficiency for generation.

162.4 Why finite real approximation is not enough

A positive real number substituted for ε in the sharp examples need not expose the intended face: real points can have first-stage slack smaller than that fixed perturbation scale. Likewise, no finite list of rows in the countable interval example captures all its infinitesimal boundary information. Exact finite checks are useful for finding algebraic errors, but the small-cut or saturation step is logically indispensable.

163 Cardinal versions and the precise role of the universe

The proofs use no multiplication peculiar to Conway normal forms. They apply to a more general saturated real closed field whenever the presentations fit below its saturation threshold.

Theorem 163.1 (Set-sized saturated-field version). *Let K be a κ -saturated real closed ordered field, where κ is uncountable. Replace “set” in theorems 155.1, 155.3, 156.1, 156.2, 157.1 and 157.2 by “of cardinality less than κ ,” including all parameter families. The resulting statements hold over K .*

Proof. Every type used in the proofs has fewer than κ formulas and parameters. Taking all finite subsets of a family of size less than κ still yields fewer than κ many subsets when the family is infinite; finite families cause no issue. Only finite unions of parameter families are needed. Use κ -saturation in place of theorem 153.2. Uniform local radii are obtained from the same small positive-bound types. $\square$

In particular, no regularity assumption on κ is needed for these finite-union cardinal estimates. The full surreal case is the external version in which every set-sized family is admissible.

Proposition 163.2 (One-dimensional geometry detects cuts). *Let K be an ordered field and κ an uncountable cardinal. The following are equivalent:*

- (i) *Every cut $L < R$ with $|L \cup R| < \kappa$ has a strict interpolant, including cuts with an empty side.*
- (ii) *Every subset of K of size less than κ is order bounded, and any two disjoint hulls of nonempty such subsets in dimension one have a strict affine separator.*

If K is real closed, these conditions also imply realization of every finitely satisfiable partial type with fewer than κ formulas and parameters in finitely many variables. Thus they give all the geometric conclusions of theorem 163.1.

Proof. For (i) $\Rightarrow$ (ii), use cuts with an empty side to obtain bounds. Two disjoint nonempty convex subsets of a line are ordered one wholly before the other. After interchanging the generators if necessary, the corresponding cut $S < T$ therefore has an interpolant, which is a strict separator.

For (ii) $\Rightarrow$ (i), when both L, R are nonempty their convex hulls are disjoint, and a strict affine separator gives a point strictly between all elements of L and all elements of R . If a side is empty, use the assumed bound, enlarging a weak bound by 1 when necessary.

For the real closed assertion, put the parameters in a real closed subfield $K_0 \subseteq K$ of size less than κ . Quantifier elimination and compactness, followed by downward Löwenheim–Skolem,

give a realizing real closed extension E/K_0 of size less than κ . The relative embedding proof of lemma 153.1 embeds E into K over K_0 : all intermediate fields and cuts have size bounded by $\max(|K_0|, |E|, \aleph_0) < \kappa$. This realizes the type in K . $\square$

Thus an omitted small cut in an old field is already an obstruction to one-dimensional uniform separation. This is the elementary geometric link to the repository's across-universes program [94]. No particular forcing or gap-spectrum assertion from that report is required for this implication.

164 Proof review, exact checks, and a formalization route

164.1 Dependency structure

There are three independent proof streams.

Input	Consequences
Cuts + real closedness + compactness	Small-type realization; strong separation; all small-hull faces exposed; finite implication certificates; finite sandwiches and V/H rigidity.
Positive small lower bounds + finite linear algebra	Uniform neighborhoods in active affine spaces; finite active-face descriptions; the set of vertices of a small H-class.
Real facial chains + infinitesimal sign rule	Exact polynomial degree; recursive sharp compact bodies; explicit distinction between generator extension and definable extension.

The moment-curve classification combines exposure, a coefficientwise standard-part argument, elementary polynomial root multiplicity, and Vandermonde independence. Polarity uses separation and finite-description rigidity, not any unproved compactness principle.

The most sensitive checks are the set-sized nature of each type, the restriction of the unit exposure margin to generators, the fact that the hulls themselves can be proper classes, and the distinction between $C^\uparrow$ and $C(\text{No})$. The proofs explicitly address all four.

164.2 What the supplied code checks

The package contains a standard-library Python program using exact rational arithmetic ([write] shipped as `code/13-beyond-finite-verify.py`). It checks square-product moment polynomials, affine Vandermonde determinants, the leading coefficient signs of sampled recursive sharp-family generators, the identity $\varepsilon^2/4$ in the definable-extension counterexample, the exact active normals and ranks in the polar classification, and finite nonnegative row-combination certificates. The recorded results can be reproduced with

```
python3 code/13-beyond-finite-verify.py \
  --output /path/to/scratch/verification_results.json
```

([write] Source 13 printed `python3 code/verify.py -output data/verification_results.json`. The program writes a file only when `-output` is given; pass a path outside the report directory, as above, so that the shipped record `data/13-beyond-finite-verification_results.json` (console output `data/13-beyond-finite-verification_stdout.txt`) is never overwritten. The shipped `code/13-beyond-finite-Makefile` still names `code/verify.py`, `data/verification_results.json` and `article.tex`.) These are finite regression checks,

not evidence of model-theoretic compactness, not a numerical implementation of the entire surreal field, and not a replacement for any proof. In particular, a loop over rational samples does not establish the moment-curve face classification; its upper bound is the standard-part and even-multiplicity argument.

164.3 A practical Lean decomposition

[write] This subsection is a proposal: nothing of this Part is formalized. At the write phase no repository-authored Lean file of ProveIt mentions Farkas duality, so its first layer has no counterpart in the repository either.

A formalization should separate abstract finite-dimensional convexity from the implementation of the surreal field. The first layer needs affine faces, finite hull membership, finite exposure, and ordered-field Farkas certificates. Existing ordered-field duality work provides a relevant reference point [98], but this manuscript has not verified that its exact declarations are already present in the repository.

The second layer should take a *saturation interface* as a hypothesis: every suitably small finitely satisfiable family of formulas in finite coordinate types has a realization. Under this interface, the sandwich proof is short and reusable. The active-face theorem can be formalized under the weaker interface providing a positive number below a small family of positive numbers.

Only the third layer should instantiate these interfaces for a universe-indexed surreal construction. In Lean, a proper class is not an ordinary type inside the same universe; the smallness relationship must be expressed with universes and lifting. A theorem about a fixed small surreal type must not be silently advertised as saturation for every set of that same universe. That foundational bookkeeping is a substantive part of the proposed implementation, not a notation change.

Finally, the sharp-family and moment-curve arguments are separable algebraic targets. Finite polynomial sign-at-infinitesimal lemmas, coefficientwise standard part, and Vandermonde determinants can be formalized before the full saturation theorem. No Lean file is supplied here because no corresponding compilation was performed; a plausible uncompiled script would not constitute verification.

165 Further research questions

The following are proposed research directions, not claims that each is a previously published open problem. The theorems above supply constraints that a successful answer must respect.

Question 165.1 (Birthdays and explicit separators). Given a set of generators with a bound on their birthdays and a description of a facial trace, what bounds can be proved for the birthday of an exposing normal? Can a finite-complexity trace yield a normal with a specified normal-form support or a controlled number of scales?

The general saturation proof is existential. It does not imply effective extraction from an arbitrary presentation. A useful first target is a countable family of generators given by polynomial or rational maps with real coefficients and a semialgebraically described trace.

Question 165.2 (Exact presentation thresholds in set-sized fields). For a κ -saturated real closed field K , how small can an H-presentation of a nonpolyhedral hull of fewer than κ generators be? Can the lower bound κ forced by rigidity be attained, and which cut cofinalities determine the exact minimum?

The one-dimensional cut examples suggest that different ordered fields with the same elementary theory may have different exact presentation spectra. Any proposed answer should separate the number of generators, the number of rows, and the size of the parameter field.

Question 165.3 (Polars beyond small generation). For a bounded small H-class, characterize the points missing from the hull of its extreme points. Can one describe them intrinsically by types, active-row limits, or cuts in normal directions? What additional class-sized data recover the full polar of a small V-class?

The rational moment-curve polar gives a concrete nonfinite family for which any set of proposed generators must fail. A satisfactory description must do more than enumerate additional vertices.

Question 165.4 (Computation of exposure depth). For real semialgebraic C and a specified face F , what is the complexity of computing the classical degree $\delta_C(F)$ and a shortest exposing chain? Can this be turned into a symbolic certificate of minimal infinitesimal polynomial degree?

The family K_d gives a test with a provably forced first step at every stage. It would be useful to find similarly transparent families with prescribed branching in their shortest exposed chains, or with much smaller semialgebraic descriptions at the same depth.

Question 165.5 (Infinite face-poset realization). Which set-sized posets occur as face lattices of small hulls in fixed finite-dimensional surreal space? How can one characterize the additional constraints beyond finite facial compatibility and an affine-dimension bound?

The rational moment hull realizes a full finite-subset skeleton followed by a dimension jump to the top. It shows that the gradedness and facet structure of finite polytopes are not automatic. A systematic theory should encode how faces of finite subhulls disappear under enlargement.

Question 165.6 (Ground fields for exposing coefficients). For $S \subseteq K^d$ in a proper ordered subfield $K \subseteq \mathbf{No}$, determine when every facial trace can be exposed using finitely many infinitesimal scales with coefficients in K . What field extension is minimally needed when this fails?

For $K = \mathbb{R}$, the exact answer for generator extensions is controlled by classical exposure depth. For general K , missing cuts can obstruct the supporting-hyperplane constructions used in the real proof, even though the full surreal exposing normal still exists by saturation.

Question 165.7 (Completion choices). Which explicitly defined enlargements of finite-combination hulls admit a useful V/H duality beyond finite polytopes? Possibilities include specified summable combinations or intersections indexed by controlled proper classes. Which choices preserve a bipolar theorem and a tractable face notion?

The rigidity theorem rules out obtaining a genuinely broader duality while simultaneously retaining both set-sized presentations and only finite convex combinations. At least one of those conditions has to change.

Question 165.8 (A checked abstract compactness library). Can the finite-sandwich, finite-consequence, and active-face arguments be formalized once for saturated ordered fields and then instantiated for surreal models, nonstandard real fields, and relevant universe-relative subfields with explicit size hypotheses?

This is a focused integration target for ProveIt. It would connect the repository's field foundations and across-universe questions to a geometric API without requiring a general theory of proper-class topological compactness.

166 Conclusion

The most important distinction in this geometry is not finite versus infinite coordinates. It is the distinction between a finite presentation, a set-sized presentation, and a proper-class presentation. In the full surreal field, saturation makes a set-sized family of generators behave as if every needed strict gap could be chosen simultaneously. That gives closedness and strong exposure. On the inequality side it makes local boundaries finitely active and every valid linear consequence finitely certified. When the two sorts of presentation describe the same class, the finite-sandwich theorem forces them both to collapse to finite ones.

The examples show that this is a structural boundary, not a vacuous rephrasing of finite polytope theory. Countably generated hulls can have infinitely many exposed vertices and no facets. Countably presented bounded H-classes can have no vertices. Polarity can take a hull with a countable list of generators to a class that no set can generate. Infinitesimal functionals can expose every old real face while failing on the full semialgebraic extension of the very same real body.

The proofs establish these statements under explicit hypotheses. Their formal verification, effective refinements, and full priority assessment remain separate tasks. They provide a concrete research program with sharp obstructions and exact examples, rather than an unsupported claim that changing the scalar field resolves a classical open conjecture.

167 Notation and scope reference (Appendix A of source 13)

Notation or phrase	Meaning
No	The full surreal ordered field of the fixed current universe.
Small	Indexed by an external set; in theorem 163.1 , of size less than κ .
d	Ordinary finite dimension, never a surreal integer.
$V(S)$	Finite-convex-combination hull with arbitrary surreal coefficients. The generator family S is a set.
Small H-class	An intersection of a set of closed affine surreal halfspaces. The feasible class need not be a set.
Polytope	Convex hull of an ordinary finite list of points.
Closed	Every exterior point has a disjoint positive-radius order box.
Bounded	Contained in a box of some surreal radius; not necessarily bounded by a real radius.
$C^\uparrow$	Generator extension $\text{conv}_{\mathbf{No}}(C(\mathbb{R}))$ of a real convex set.
$C(\mathbf{No})$	Extension by a semialgebraic formula, when one is specified; generally different from $C^\uparrow$.
$\delta_C(F)$	Martínez-Legaz's degree of non-exposedness of a real face.
$m_\varepsilon(C, F)$	Minimum degree of an exposing polynomial in ε with real linear-functional coefficients.
M_d	Hull of the moment curve at all rational parameters; not a finite cyclic polytope.

168 Provenance and proof-status statement (Appendix B of source 13)

The repository was consulted through its connected GitHub interface on 30 September 2026. Search results were pinned to commit

6996fee43cc97b6c16351def7507d59a95bf62f0.

The two principal README files were fetched from the default branch on that date. Their returned blob hashes were, respectively,

cf63fc5ff9ca9d8877e90f8e69ec34e6ded19d23,
bfb819fc0cff4c42961a2048d21845af5a9522fd.

The package's source notes ([write] 13-beyond-finite-SOURCES_AND_SCOPE.md) distinguish that indexed commit from the individual returned blobs; they do not assert that all fetched files were read at a simultaneously pinned checkout.

[write] Re-verified at the write phase: these are the blob hashes of Algebra/SurrealNumbers/README.md and Algebra/SurrealNumbers/docs/foundations-and-computation/surreal-fields-across-universes/README.md, both at commit 6996fee43 and at HEAD 4e128356d; the second README still states that nothing in that report is formalized or Lean-verified.

The manuscript is AI-assisted and unrefereed. It contains complete written proofs of its principal claims. It does not contain a proof-assistant certificate, and its Python checks are explicitly finite algebraic regressions. The original 1997 paper was consulted directly to establish that lexicographical exposure, degree of non-exposedness, and the codimension bound are prior results. No source or quotation is used as a substitute for the proofs of the finite-description and surreal consequences developed here.

Provenance and references

A Provenance of the report

[write] This appendix was written in the merge.

A.1 Sources

The report was built in the batch-58 write phase from six manuscripts, all dated 30 September 2026 and delivered on that day in six archives (arrival commit `b2b626d`); they were placed in commit `b30441a`, which staged manuscript 05’s `article.tex` and `README.md` as delivered and every other staged file under its manuscript’s prefix. Manuscripts are numbered by arrival within the batch (manuscript 01, on a Keller map, went to another report).

No.	Title (archive; main file)	Delivered PDF, author line	Pin	Printed as
05	<i>Finite-Scale Geometry of Surreal Liftings</i> (<code>surreal_liftings_research.zip</code> ; <code>surreal_liftings/article.tex</code>)	25 pp. A4; research manuscript prepared for Vladimir Reshetnikov, an AI-assisted contribution to the ProveIt research program	<code>5e0f4e0</code>	Part I , base (whole)
02	<i>Surreal Polytopes at Finite and Transfinite Scales</i> (<code>surreal_polytopes_research.zip</code> ; <code>surreal_polytopes/article.tex</code>)	28 pp. Letter; research article prepared for Vladimir Reshetnikov	<code>6996fee</code>	Part I (merged) and Part II
06	<i>Surreal Polytopes: Affine Residues, Mixed-Volume Scales, and Simultaneous Segment Models</i> (<code>Surreal_Polytopes_Research_Package.zip</code> ; <code>Surreal_Polytopes/article.tex</code>)	27 pp. A4; research article prepared for Vladimir Reshetnikov	<code>6996fee</code>	Part III (whole)
03	<i>Infinitesimal Boundary Geometry of Surreal Polytopes: Lexicographic traces, one-parameter descent, and Ehrhart universality</i> (<code>surreal_polytope_research.zip</code> ; <code>surreal_polytopes/surreal_polytopes.tex</code>)	28 pp. Letter; research manuscript prepared for Vladimir Reshetnikov	<code>6996fee</code>	Part IV (whole)
07	<i>Invisible Complexity in Surreal Polytopes: Explicit polygons with identical all-order jets and identical initial slack matrices</i> (<code>surreal_polytopes.zip</code> ; <code>surreal_polytopes/surreal_polytopes.tex</code>)	26 pp. A4; research manuscript prepared for Vladimir Reshetnikov	<code>6996fee</code>	Part V , base (whole)

No.	Title (archive; main file)	Delivered PDF, author line	Pin	Printed as
04	<i>Beyond-All-Orders Extension Complexity in Surreal Polygon Geometry: Flat deformations, classical face lattices, and a sharp-order complexity gap</i> (surreal_polytopes_beyond_a11_orders.zip; surreal_polytopes_beyond_a11_orders/article.tex)	26 pp. A4; research manuscript prepared for Vladimir Reshetnikov, developed with ChatGPT; independent review and formalization pending	6996fee	Part V (merged)

The pin 6996fee is commit 6996fee43cc97b6c16351def7507d59a95bf62f0 (30 September 2026, 16:07:39 UTC); 5e0f4e0 is 5e0f4e05046bacda3c6ba2643f19b91aa632436b, its second parent. Archives 02, 03 and 07 all wrap an inner directory named surreal_polytopes/; they are different packages.

A.2 What each manuscript contributed

Manuscript 05 (base; Part I). The frame of Part I: the affine gauge and dependence space, the simultaneous leading-term compression ([theorem 6.1](#) and [corollary 6.3](#)), the flag classification ([theorem 8.2](#)), the exact germ degrees ([theorems 9.1](#) and [9.2](#)) and the failure of a single real parameter ([proposition 9.4](#)); determinant and barycentric certificates, the explicit threshold ([lemma 10.3](#)), the linear realization with the exact shadow and the separation of the two degrees ([theorems 10.4](#) and [10.6](#)); shadows and the canonical refinement chain; the higher-rank tropical bridge; the boundary examples; toric initial ideals through Graver certificates ([theorem 17.2](#)); transfer and shadow arcs ([theorem 18.1](#)); the corrections to the private draft ([section 20](#)); the verifier and the halting-problem obstruction ([proposition 21.1](#)); eight research questions, its dependency ledger and a Lean plan.

Manuscript 02 (Parts I and II). In Part I: its introduction (three operations that change scale), its working field and common initial normal-form cuts, the cut-compatible compression ([theorem 7.1](#)) and the transfinite-delay example, the fixed-base comparison-rank theorem with its cut clause ([theorem 11.2](#)), the stage threshold ([theorem 11.4](#)), positive sums and the classification of support-initial histories ([lemma 12.1](#) and [theorem 12.2](#)), the sharp parabolic family and the hexagon ([theorem 12.3](#)), the finite polar identity, the lost face and the optimality band ([propositions 14.1](#) and [14.2](#)), the finite Puiseux model ([theorem 19.1](#)), the certificate workflow and the query-model bound ([proposition 21.2](#)), and a second, covector-level proof of transfer. In Part II: the four-point test, the rational alternation, the universal reciprocal-series histories, the independent transfinite endpoint, finite determinacy for uniform ordinary series and the tail-truncation caveat ([lemma 26.1](#), [theorems 26.2](#), [26.3](#) and [26.6](#), and [proposition 26.7](#)), its recorded tests and proof-dependency audit, eight research directions, the exact recursion and its reproducibility notes; and a Lean plan.

Manuscript 06 (Part III). Moving coordinates over any set-sized real closed $K \supseteq \mathbb{R}$ with its natural valuation: the scale lattice $L(P)$ and the affine residue $\text{Res}(P)$ with its normalization theorem, determinant certificates (at most d^d determinants), the simultaneous inner-segment models that preserve every pure mixed-volume valuation on a finite rational grid, the Grassmannian and clone results, exchange-convexity of valuation profiles over any ordered group, the classification of probes and finite probe sets, the distortion bound, the full

characterization corollary, a one-parameter specialization of finite-support Hahn certificates, its examples, ten research questions, its audit appendix and a Lean plan.

Manuscript 03 (Part IV). Shadows and real traces $\text{Tr}_{\mathbb{R}}(P) = P \cap \mathbb{R}^d$ of a surreal polytope: at most $d + 1$ lexicographic levels per inequality, one-parameter descent to the real closure of $\mathbb{R}(t)$, face implantation, the counting functions $(n + 1)^2 - \lceil \alpha n \rceil$ for every irrational $0 < \alpha < 1$ (with an algebraic instance), the second asymptotic coefficient and its attainable interval, rational flags and the dichotomy for counting functions, twelve research questions, a dependency map, its two appendices and a Lean plan. It uses `omni fic_bounded_iff` as an input.

Manuscript 07 (base; Part V). The frame of Part V: for every $n \geq 4$, two strictly convex n -gons with the same shadow, oriented matroid, all-order jets and leading slack data, one of logarithmic linear and semidefinite extension complexity and one of complexity at least of order $\sqrt{n}$; generic linear and semidefinite lower bounds; one fixed field of valuation rank three and the order-unit criterion with the sharpness of rank two; residue descent for every convex valuation ring; precision and conservativity theorems; the bounds table; ten research questions and a Lean plan.

Manuscript 04 (Part V). A second construction on one circle: the regular 2^m -gon R_n and a cyclic n -gon P_n° agreeing below every power of ε , with $m \leq \text{xc}(R_n) \leq 2m$ and $\lceil \sqrt{n} \rceil \leq \text{xc}(P_n^\circ) \leq \min(n, \lceil 24\sqrt{n} \rceil)$ ([theorem 70.1](#)); the field- and scale-general criterion that gaps in all-order jet fibres exist exactly when the flat ideal is nonzero, exactly when $v(\varepsilon)$ is not an order unit ([theorem 72.7](#)); the rank-two fields K_n , a finite-order rank-one analogue ([theorem 74.1](#)), the failure of lifting coherent finite-order factorizations ([theorem 71.2](#)), what asymptotic data cannot recover, algebraic finite-accuracy approximants ([corollary 74.6](#)), a rational-chart generalization ([theorem 76.1](#)), an in-principle decision procedure ([proposition 77.1](#)), nine research questions and its repository and notation appendices; second proofs of the flat-ideal lemma, the order-unit theorem, the scalar-extension corollary, the normalization theorem and algebraic realization (with its point case).

A.3 Where the merge had to choose

- *Bases.* Manuscript 05 is the base of Part I and of the report: on the lifting theorem it shares with 02 it assumes no more (an ordered subfield $\mathbb{R} \subseteq K \subseteq \mathbf{No}$, real closure only where needed; 02 assumes a real closed K) and concludes more (exact leading terms, the flag classification, exact germ degrees). Manuscript 07 is the base of Part V: it covers every $n \geq 4$ (04: $n = 2^m$), linear and semidefinite lifts, residue descent for every convex valuation ring (04: the standard part and nonnegative rank only), every fixed dimension, and a sharper parameter count ($dN \leq r(r - 1)$ against 04's $s \leq r^2$).
- *Printed once (Part I).* `stconv = conv st` (05 and 02; one proof, both labels); the slack certificate (05's [proposition 10.2](#), carrying 02's label, with 02's statement and proof in [remark 11.1](#)); the failure of a single real parameter (05's [proposition 9.4](#); 02's remark kept and marked as the same result); real-closed-field transfer (05's [theorem 18.1](#), with 02's statement appended in its covector form and 02's proof as a second proof).
- *Printed beside each other (Part I).* 02's [theorems 7.1](#) and [11.2](#) beside 05's [theorem 6.1](#) and [corollary 6.3](#), because 02's cut clauses are new and proved by the argument that also proves the shared clauses; 02's [theorem 11.4](#) beside 05's [lemma 10.3](#), because its constants differ (both valid; both give $1/34$ and $1/10$ on the examples); 02's [theorem 19.1](#)

beside the bounded half of [theorem 18.1](#); 02’s query-model [proposition 21.2](#) beside 05’s halting-problem [proposition 21.1](#) (different obstructions).

- *Printed once (Part V), with both labels and 04’s proof as a second proof:* the flat-ideal lemma, the order-unit theorem ([theorem 72.2](#)), the scalar-extension corollary ([corollary 74.7](#)), the normalization theorem ([theorem 73.2](#), 04’s statement in a remark), algebraic realization ([theorem 74.3](#), 04’s point case in a remark), and precision with 04’s tolerances as part (b) of [theorem 74.2](#). Printed beside each other: 04’s transcendence-degree proposition beside [theorem 67.1](#) (07’s statement does not cover the circle), 04’s proposition on Yannakakis’s factorization beside 07’s displayed factorization, and the two different reflection lemmas; 04’s $\text{st conv} = \text{conv st}$ is printed in its conventions with a pointer to [proposition 13.1](#).
- *Moved.* Manuscript 02’s Section 6, recorded tests, audit table, research directions, conclusion and appendices form Part II; its workflow and query bound sit in 05’s computation section. Every manuscript’s Lean roadmap is in [section 83](#). Source appendices became ordinary sections at the end of their Parts. In Part V, 04’s Sections 1–4 are distributed into 07’s first sections, 04’s construction and its Section 7 are sections of their own after 07’s Section 8, 04’s criterion joins 07’s field section, and 04’s charts, computation and appendices are sections before the merged questions.
- *Questions.* Each manuscript’s research questions stay at the end of its material, with [write] cross-references where two manuscripts ask the same thing (05’s questions 1, 2, 4 and 5 and four of 02’s directions; in Part V three of 07’s questions are merged with three of 04’s, and four more are re-scoped by notes; Parts III and IV point to 02’s directions and to each other); a question that another manuscript answers in part is re-scoped by a note rather than printed as open.
- *Paper and layout.* A4 and the base’s fonts; one theorem counter numbered within sections; source numbers are not kept.

A.4 Renamed symbols

Only letters that would otherwise clash inside a Part, or that denote the fixed surreals ω^{-1} and $\omega^{-\omega}$, were renamed; no normalization changed.

Part	Source letter	Printed	Where and why
I	02: $t = \omega^{-1}$	ε	in 02’s material of Part I (working field, example 7.3 , theorems 12.2 and 12.3 , (14.2) , proposition 21.2); t is 05’s real parameter there. 02’s introduction and Part II keep t , the power-series variable evaluated at $t = \varepsilon$
I	05: $\varepsilon_1, \varepsilon_2$	ε, τ	section 16.5 : $\varepsilon_1 = \omega^{-1}$, $\varepsilon_2 = \omega^{-\omega}$
III	06: macro <code>\Span</code> (plain span)	<code>\spanplain</code>	a macro clash with 05’s $\text{span}_{\mathbb{R}}$; printed output unchanged. 06’s $\eta = \omega^{-\omega}$ is kept
V	07: $\rho = \omega^{-\omega}$	τ	main construction, $V_{\rho} \rightarrow V_{\tau}$; 07’s re-chosen scale in corollary 72.4 and theorem 74.2 keeps ρ
V	07: auxiliary τ	σ	proof of theorem 74.2
V	04: $\mathcal{I}_{\varepsilon}, v, \Gamma$	$\mathfrak{I}_{\varepsilon}, v, G$	one flat-ideal symbol, one valuation letter and one value-group letter in Part V
V	04: v_i	μ'_i	monomials in the proof of theorem 74.1
V	04: $P_n, p_i, P_{n,M}$	$P_n^{\circ}, p_i^{\circ}, P_{n,M}^{\circ}$	07’s P_n is a different polygon
V	04: $\mathcal{T} \ni \tau, c_{\tau}, b_{\tau}$	$\mathcal{D} \ni \delta, c_{\delta}, b_{\delta}$	tolerance set of theorem 74.2

Part	Source letter	Printed	Where and why
V	04: “algebraic finite-accuracy shadows”	“approximants”	they are real-algebraic approximants, not standard parts (corollary 74.6)

Labels and citation keys received the prefixes of [section 1](#); a few labels were added on unlabelled items that the merged text cites. The report’s `README.md` lists the label counts, the added labels, and the correspondence between the delivered files and the shipped ones.

A.5 Batch 59: the sources of Parts VI–X

[write] Parts VI–X were built in the batch-59 write phase from six further manuscripts, all dated 30 September 2026 and delivered on that day in six archives (arrival commit `ed80eba`). They were placed in commit `f2cb2034e` as additions to this report: each manuscript’s code, data and audit files were staged under a prefix continuing the report’s local sequence (`08-` to `13-`), and the manuscripts, their READMEs, PDFs and checksum lists were not staged; they survive in the arrival commit. Manuscripts are numbered by arrival within batch 59 (archive modification time); the report calls each by its file prefix, *source 08* to *source 13*, because the batch-59 numbers 01–06 would collide with the batch-58 numbers used in Parts I–V.

Source (b59)	Title (archive; main file)	Delivered PDF, author line	Pin	Printed as
08 (01)	<i>Mixed-Volume Tomography of Surreal Polytopes: Determinant ideals, complete scale spectra, and an adaptive–nonadaptive dichotomy</i> (surreal_mixed_volume_tomography.zip; surreal_mixed_volume_tomography/article.tex)	24 pp. Letter; research manuscript prepared for Vladimir Reshetnikov	6996fee	Part VI (whole)
09 (02)	<i>Finite Models and Real Degenerations of Surreal Mixed Volumes: Parallelotope compression, representable initial supports, and explicit Lorentzian non-liftability</i> (surreal_mixed_volumes.zip; surreal_mixed_volumes/article.tex)	28 pp. A4; research manuscript, prepared for Vladimir Reshetnikov	b2b626d	Part VII (merged)
10 (03)	<i>Tropical Mixed Volumes of Surreal Polytopes: Parallelotope compression, Grassmannian realization, and full-support matroid obstructions</i> (surreal_mixed_volumes_research.zip; surreal_mixed_volumes/article.tex)	25 pp. A4; research manuscript prepared for the ProveIt research program	6996fee	Part VII , base (whole)
11 (04)	<i>Exponential Contact Depth in Surreal Polytopes: Face-lattice-forced arithmetic, rigid scale towers, and lower bounds for rational shadow realizations</i> (surreal_polytopes_contact_depth.zip; surreal_polytopes_contact_depth/surreal_polytopes_contact_depth.tex)	24 pp. A4; research manuscript prepared for Vladimir Reshetnikov, an AI-assisted contribution to the ProveIt research program	6996fee	Part VIII (whole)

Source (b59)	Title (archive; main file)	Delivered PDF, author line	Pin	Printed as
12 (05)	<i>A Rational-Normal Dichotomy for Omnific Integer Hulls: Finite Graver Certificates and Irrational Obstructions in Surreal Polyhedra</i> (Omnific_Integer_Hulls_Research.zip; omnific-integer-hulls/article.tex)	28 pp. Letter; research draft prepared for Vladimir Reshetnikov	6996fee	Part IX (whole)
13 (06)	<i>Beyond Finite Surreal Polytopes: Exposed faces, sharp infinitesimal depth, and finite-description rigidity</i> (Beyond_Finite_Surreal_Polytopes.zip; Beyond_Finite_Surreal_Polytopes/article.tex)	26 pp. Letter; prepared for Vladimir Reshetnikov with ChatGPT	6996fee	Part X (whole)

The pin b2b626d is commit b2b626d8537f7dea8f20393ee4714c7201a7001e (30 September 2026, 17:24:44 UTC), the batch-58 arrival commit, a child of 6996fee. Archives 09 and 10 both wrap an inner directory named `surreal_mixed_volumes/`; they are different packages. No batch-59 manuscript saw this report: it was placed at b30441a after all six pins. Source 11 cites batch-58 manuscript 05 by title; no other batch-59 source cites another manuscript.

A.6 What each batch-59 manuscript contributed

Source 08 (Part VI). The displacement module (Part III’s scale lattice) as the exact object seen by mixed-volume valuations; the colorful determinant formula for arbitrary finite generating lists ([theorem 87.2](#)); the classification and box realization of one-body scale spectra, orthogonal invariance, visible dimension and the residue flag; singular lengths equal scale exponents; partial and full stability with a sharp threshold; the reference-dependent $d + 1$ test and the classification by all first-slot probes (both printed once, in Part III); the nonadaptive obstruction in the plane ([theorem 91.2](#)); the exact two-body interval ([theorem 92.1](#)); the complementary-minor formula; polarity as module duality ([theorem 93.3](#)); algorithms, a certificate boundary, worked scale data, scalar extension; nine questions, its verification record, dependency ledger and two appendices.

Source 11 (Part VIII). Labeled Lawrence polytopes with $8k + 16$ vertices in dimension $4k + 10$ whose shadow keeps every vertex and the dimension, and whose rational vertex arcs need homogeneous degree between $\lceil 2^k / (16k + 48) \rceil$ and 2^k ([theorem 120.3](#)); the projective repeated-squaring gate, the uniform $1/6$ sign threshold, explicit Lawrence recovery from any realization ([theorem 124.3](#)), the cofactor degree bound, valuation and ramification identities, the addition-chain generalization ([theorem 128.2](#)), the comparison with Part I’s degree-one realization, ten further questions, a Lean plan and its ledger.

Source 10 (base; Part VII). Over any set-sized real closed $K \supseteq \mathbb{R}$ with its natural valuation: simultaneous parallelotope compression with Part III’s constants (printed there); an exact three-way realization criterion by canonical polarization and real block mixing ([theorem 103.3](#)); every exposed coefficientwise initial polynomial is a real-polytope volume polynomial ([theorem 104.2](#)), and every exposed initial support a real-representable polymatroid base set ([theorem 105.4](#)); the matroid-profile theorem ([theorem 106.1](#)): for a loopless rank- d matroid M the profile $\gamma(d - r_M(\text{supp } \alpha))$ is realizable exactly when M is representable over $\mathbb{R}$, with

a Lorentzian-lift corollary; the explicit strictly Lorentzian Fano cubic with an unrealizable full-support profile ([theorem 108.4](#); 84 coefficients at three levels, 101,745 short Plücker checks); the $U_{2,4}$ amplitude example, the rank-two lemma and planar sufficiency ([theorem 110.2](#)); a certificate procedure, a formalization proposal citing nine Lean declarations of the repository, and ten questions.

Source 09 (Part VII). An independent answer to the same prompt under the same hypotheses. It adds the optimality of the dm generator bound ([corollary 101.3](#)), the colored-minimum and integer-mixing forms of the realization criterion (bound $d\binom{dm}{d}$), an explicit maximum-determinant form of the initial-polynomial theorem with its independence of coordinates up to one element of $GL_d(\mathbb{R})$ ([proposition 104.5](#)), rank recovery and its converse for real supports, the affine-shift clause of the Fano theorem, the coefficient obstruction “even up to a positive scalar” with a parallelogram witness ([theorem 109.2](#)), two-rectangle and two-vector examples, a certificate algorithm, a literature discussion (Brändén–Huh’s Remark 4.3, Menges, Huh’s Example 3.4 not answered), a finite transfer appendix and ten questions, seven of them shared with source 10.

Source 12 (Part IX). Integer-hull theory over the omnific integers: for a fixed ordinary integer constraint matrix and arbitrary surreal right-hand sides, the omnific integer hull of a bounded polyhedron, although its lattice slice can be a proper class, is a finite polytope generated by an explicit candidate set of basis intersections with bounded ordinary slack offsets, computed from one finite graph-Graver test set by Presburger transfer ([theorem 141.2](#)); a universal zonotope normal-fan refinement; a finite tail-and-residue compiler for the whole face lattice; same-matrix descent of every such face lattice to an ordinary right-hand side ([theorem 142.4](#)); the extension to any integer part of any ordered field ([corollary 141.4](#)); the rational-normal dichotomy ([theorem 143.5](#)), with an irrational row giving no maximum and no surreal supremum, even for omnific right-hand sides; a total-unimodularity corollary, the six-residue 2–3 knapsack hull, exact checks, a formalization plan and twelve questions.

Source 13 (Part X). Set-sized presentations over the full surreal field: realization of small types, uniform strong separation of disjoint small hulls, exposure of every face of a small hull, finite Farkas certificates and at most d active rows for faces of small halfspace intersections, and double finite compression ([theorem 157.2](#)): a class that is both a small hull and a small halfspace class has finite subpresentations of both, so definable small hulls are polytopes; polarity as a set-versus-class obstruction; polynomial infinitesimal exposure degree equals Martínez-Legaz’s degree of non-exposedness (credited) with a sharp family ([theorem 159.2](#), [theorem 160.2](#)); the faces of the rational moment-curve hull and of its vertex-free polar; one-dimensional failure modes, a κ -saturated version, a cut criterion, a Lean decomposition proposal and eight questions.

A.7 Where the batch-59 merge had to choose

- *Position.* Parts VI–X were appended after the roadmap chapter of Parts I–V, so that no existing number changed; the report’s structure otherwise groups the formalization roadmaps before the provenance.
- *Duplicates of Part III.* Source 08’s adaptive test and its all-probes corollary have Part III’s statements and proofs and are printed once, in Part III, carrying source 08’s labels; source 08’s formulations are kept as remarks in Part VI. Its colorful determinant formula

and module factorization are printed in Part VI beside Part III’s (different constants and enumeration), as is its difference-body form of Part III’s distortion proposition.

- *Part VII.* Source 10 is the base: the hypotheses are the same, its compression constants are sharper (source 09’s symmetric version is kept as a marked remark), and its matroid criterion is more general. A result both prove is printed once with both labels and, where the proofs differ, source 09’s as a second proof: the witness-size corollary, the Fano theorem, strict Lorentzianity and non-representability, the rank-two lemma and planar sufficiency. The coefficient obstruction is printed in source 09’s stronger form (“even up to a positive scalar”) with source 10’s proofs and witness as second routes. The two Fano cubics are one polynomial; one table is printed, with the dictionary between the labellings (they differ by exchanging 3 and 4). Results already in Part III (positive sums, parallelotope compression, standard part of volume, the Rado lemma, exchange-convexity) are printed there once; Part VII keeps marked pointers carrying the sources’ labels and their different proofs. Sources 10 and 09 have twenty questions; seven pairs were merged, giving thirteen. Four equation labels of source 09 whose displays are printed once under source 10’s labels were dropped, after every reference to them was redirected (the report’s README lists them).
- *Parts IX and X.* Neither source repeats a Part III theorem. Source 12’s [lemma 137.1](#) and [lemma 139.1](#) overlap Part IV’s bounded-element property and Part I’s finite Graver certificate but say more, and are printed in full with pointers. The paragraph listing the collection’s other treatments of irrational obstructions to omnific extrema, written after source 12’s own statement of prior work, uses the titles those reports carry. Source 13’s aliascnt theorem setup is replaced by the report’s.
- *Appendices and questions.* As in Parts I–V, each source’s appendices are ordinary sections at the end of its Part, its research questions end its material, and each source’s formalization proposal stays in its Part.
- *Red flags fixed in the write.* Source 08’s closing claim that “the planar relative-position problem is completely solved” is printed with its scope (two bodies, prescribed spectra) and a note; source 11’s bibliography entry names a file, `surreal_liftings_article.tex`, that does not exist, and a note gives the report’s Part I instead; source 12’s statement that its formalization contracts are not repository declarations under-claims, and a note names the omnific Lean declarations that already prove its first row.

A.8 Symbols renamed in Parts VI–X

Part	Source letter	Printed	Where and why
VI	08: $\mathcal{L}(P)$, the displacement module	$L(P)$	it is Part III’s scale lattice
VI	08: $\mathfrak{d}$, the determinant ideal	$\mathfrak{D}$	Part III’s ideal
VI	08: macro <code>\val</code> ($= v$)	v	printed output unchanged; the host’s <code>\val</code> is v
VIII	11: macros <code>\br</code> , <code>\ord</code> , <code>\Span</code>	<code>\brr</code> , <code>\ordz</code> , <code>\spanplain</code>	macro clashes with Part I; printed output unchanged
VII	09: macros <code>\vol</code> ($= \text{Vol}$), <code>\Span</code> ; 10: <code>\Span</code>	<code>\Vol</code> , <code>\spanplain</code>	macro clashes; printed output unchanged

Part	Source letter	Printed	Where and why
VII	09: “Research question” (own counter)	question	one theorem counter; 09’s one textual reference to its question 2 is redirected
IX	12: macro <code>\argmax</code>	<code>\Argmax</code>	defined but unused in 12’s text; the host’s <code>\argmax</code> differs

References

[write] The twelve manuscripts' reference lists, grouped by Part and numbered consecutively; an entry cited by two manuscripts of one Part is printed once.

Parts I and II (manuscripts 05 and 02)

- [1] Harry Gonshor, *An Introduction to the Theory of Surreal Numbers*, London Mathematical Society Lecture Note Series 110, Cambridge University Press, 1986. Chapter 5, “Normal Form,” pp. 52–94 [write] (the page range is manuscript 02's; both manuscripts cite this book). doi:10.1017/CBO9780511629143.
- [2] Jesús A. De Loera, Jörg Rambau, and Francisco Santos, *Triangulations: Structures for Algorithms and Applications*, Algorithms and Computation in Mathematics 25, Springer, 2010. doi:10.1007/978-3-642-12971-1.
- [3] Michael Joswig and Ben Smith, “Convergent Hahn series and tropical geometry of higher rank,” *Journal of the London Mathematical Society* **107** (2023), 1450–1481. doi:10.1112/jlms.12716; arXiv:1809.01457. The rank-two decomposition and the concluding questions are relevant comparisons, not new claims of this manuscript.
- [4] Rocco Chirivì, Martina Costa Cesari, Xin Fang, and Peter Littelmann, “Higher rank Gelfand–Kapranov–Zelevinsky fans,” arXiv:2604.20250v2, 13 September 2026. Sections 2–3 concern higher-rank cones and subdivisions; Sections 4–5 concern quasi-valuations and degenerations.
- [5] Saugata Basu, Richard Pollack, and Marie-Françoise Roy, *Algorithms in Real Algebraic Geometry*, 1st edition, Algorithms and Computation in Mathematics 10, Springer, 2003. doi:10.1007/978-3-662-05355-3.
- [6] Saugata Basu and Marie-Françoise Roy, “Quantitative Curve Selection Lemma,” arXiv:1803.00505v3, 19 July 2021; first version 2018. [write] Manuscript 02 cites the 2018 version and uses only its classical qualitative curve-selection and algebraic-Puiseux framework, not a quantitative complexity claim.
- [7] Sonja Petrović, “On the universal Gröbner bases of varieties of minimal degree,” arXiv:0711.2714, 2007. Cited for the classical Graver/universal-Gröbner background; the needed ordered-weight argument is proved in Lemma 17.1.
- [8] Vladimir Reshetnikov, ProveIt repository, Algebra/SurrealNumbers/Surreal/Algebra/Geometry.lean, inspected at commit 5e0f4e0, 30 September 2026. Fixed source link.
- [9] ProveIt, Algebra/SurrealNumbers/README.md and Algebra/SurrealNumbers/docs/README.md, consulted 30 September 2026. Project and report guide. Descriptions of formalization status are repository reports, not a fresh build certification by this manuscript.
- [10] *Finite Convexity and Exact Lexicographic Optimization over the Surreals: A finite-dimensional theorem package and a one-objective compiler for multiscale linear programs*, research draft prepared for the Surreal repository, 22 September 2026, 8 pages. Research-library copy: article(20260923-000126).pdf. Proposition 7.1 and Theorem 9.1, printed pages 5–6, are examined in Section 20. No public repository location for this copy is assumed. [write] The draft is not in ProveIt; see the note at the head of section 20.
- [11] L. van den Dries, Alfred Tarski's elimination theory for real closed fields, *Journal of Symbolic Logic* **53**(1) (1988), 7–19. doi:10.2307/2274424.
- [12] M. Joswig, G. Loho, B. Lorenz, and B. Schröter, Linear programs and convex hulls over fields of Puiseux fractions, in *Mathematical Aspects of Computer and Information Sciences (MACIS 2015)*, Lecture Notes in Computer Science 9582, Springer, 2016, pp. 429–445. doi:10.1007/978-3-319-32859-1_37. arXiv:1507.08092.

- [13] M. Wortel, Lexicographic cones and the ordered projective tensor product, arXiv preprint, 2018. Proposition 4.4. [arXiv:1812.04830](https://arxiv.org/abs/1812.04830).
- [14] C. W. Lee and F. Santos, Subdivisions and triangulations of polytopes, in J. E. Goodman, J. O'Rourke, and C. D. Tóth (eds.), *Handbook of Discrete and Computational Geometry*, 3rd ed., CRC Press, 2017, Chapter 16, pp. 415–447. In particular, Theorem 16.4.1. [Author-hosted chapter](#).
- [15] J. Bochnak, M. Coste, and M.-F. Roy, *Real Algebraic Geometry*, Ergebnisse der Mathematik und ihrer Grenzgebiete, 3rd series, vol. 36, Springer, 1998. [doi:10.1007/978-3-662-03718-8](https://doi.org/10.1007/978-3-662-03718-8).
- [16] V. Reshetnikov and contributors, *ProveIt*, GitHub repository, inspected 30 September 2026. Pinned commit `6996fee43cc97b6c16351def7507d59a95bf62f0`. Within `Algebra/SurrealNumbers/`, consulted `README.md`, `docs/README.md`, and the pinned source `Surreal/Algebra/Geometry.lean`. [Pinned subproject](#). See the accompanying source notes for the distinction between repository claims, inspected source, and formal builds not performed.

Part III (manuscript 06)

- [17] Vladimir Reshetnikov and contributors, *ProveIt: Algebra/SurrealNumbers*, repository README (which links a formalization ledger), inspected at commit `6996fee43cc97b6c16351def7507d59a95bf62f0`, 30 September 2026. <https://github.com/VladimirReshetnikov/ProveIt/blob/6996fee43cc97b6c16351def7507d59a95bf62f0/Algebra/SurrealNumbers/README.md>. Repository assertions are distinguished from the new proofs in this article.
- [18] ProveIt research collection, *Reversion Beyond Archimedean Valuations: Strong Hahn convergence, sharp Newton depth, and minimal finite coefficient certificates*, 29 September 2026, with repository editorial notes; inspected at the same commit as [17]. https://github.com/VladimirReshetnikov/ProveIt/blob/6996fee43cc97b6c16351def7507d59a95bf62f0/Analysis/Transseries/docs/series-and-transseries/Reversion_Beyond_Archimedean_Valuations/article.tex.
- [19] Philip Ehrlich, *The absolute arithmetic continuum and the unification of all numbers great and small*, Bulletin of Symbolic Logic **18** (2012), no. 1, 1–45. DOI: [10.2178/bsl/1327328438](https://doi.org/10.2178/bsl/1327328438).
- [20] Michel Coste, *An Introduction to Semialgebraic Geometry*, lecture notes, Institut de Recherche Mathématique de Rennes, 2002. Author-institution archive: <https://univ-rennes.hal.science/hal-05407113>.
- [21] Xavier Allamigeon, Ricardo D. Katz, and Pierre-Yves Strub, *Formalizing the Face Lattice of Polyhedra*, in Automated Reasoning—IJCAR 2020, Lecture Notes in Computer Science **12167**, Springer, 2020, 185–203. DOI: [10.1007/978-3-030-51054-1_11](https://doi.org/10.1007/978-3-030-51054-1_11). Open text: <https://pmc.ncbi.nlm.nih.gov/articles/PMC7324146/>.
- [22] Rolf Schneider, *Convex Bodies: The Brunn–Minkowski Theory*, second expanded edition, Encyclopedia of Mathematics and its Applications **151**, Cambridge University Press, 2014. ISBN 9781107601017. Especially the chapters on Minkowski addition, mixed volumes, and inequalities for mixed volumes.
- [23] Gennadiy Averkov, Katherina von Dichter, Simon Richard, and Ivan Soprunov, *Mixed volumes of zonoids and the absolute value of the Grassmannian*, arXiv:2404.02842v2, 1 November 2024. <https://arxiv.org/html/2404.02842v2>. The cited version supplies the motivating connection and proposed tropical direction; no assertion about the complete status of its open questions after that version is required here.
- [24] Petter Brändén and June Huh, *Lorentzian polynomials*, Annals of Mathematics (2) **192** (2020), no. 3, 821–891. Preprint: <https://arxiv.org/abs/1902.03719>.
- [25] Luca Battistella, Kevin Kühn, Arne Kuhrs, Martin Ulirsch, and Alejandro Vargas, *Buildings, valuated matroids, and tropical linear spaces*, Journal of the London Mathematical Society **109** (2024), no. 1, e12850. DOI: [10.1112/jlms.12850](https://doi.org/10.1112/jlms.12850). Preprint: <https://arxiv.org/abs/2304.09146>.

Part IV (manuscript 03)

- [26] V. Reshetnikov, *ProveIt: Surreal numbers project README*, repository documentation, consulted September 30, 2026. <https://github.com/VladimirReshetnikov/ProveIt/tree/main/Algebra/SurrealNumbers>. The live README was used for orientation; pinned substantive sources follow.
- [27] V. Reshetnikov, *The surreal and surcomplex reports*, `Algebra/SurrealNumbers/docs/README.md`, ProveIt commit 6996fee43cc97b6c16351def7507d59a95bf62f0. [Commit-pinned report guide on GitHub](#).
- [28] V. Reshetnikov, *Units and finite elements of the actual omnific ring*, `OmniFicUnits.lean`, ProveIt commit 6996fee43cc97b6c16351def7507d59a95bf62f0. [Commit-pinned Lean source on GitHub](#).
- [29] H. Gonshor, *An Introduction to the Theory of Surreal Numbers*, London Mathematical Society Lecture Note Series 110, Cambridge University Press, 1986. <https://doi.org/10.1017/CB09780511629143>.
- [30] S. Basu, R. Pollack, and M.-F. Roy, *Algorithms in Real Algebraic Geometry*, Algorithms and Computation in Mathematics 10, Springer, 2003. <https://doi.org/10.1007/978-3-662-05355-3>.
- [31] S. Basu and M.-F. Roy, *Quantitative Curve Selection Lemma*, arXiv:1803.00505, 2018. <https://arxiv.org/abs/1803.00505>.
- [32] J. F. Fernando, *On a Nash curve selection lemma through finitely many points*, arXiv:2504.03348, 2025. <https://arxiv.org/abs/2504.03348>. *Revista Matemática Iberoamericana*, <https://doi.org/10.4171/rmi/1554>. Cited here for its explicit account of classical Nash curve selection, not for a claim that this manuscript requires its stronger multi-point result.
- [33] M. Conforti, M. De Santis, M. Di Summa, and F. Rinaldi, *Scanning integer points with lex-inequalities: a finite cutting plane algorithm for integer programming with linear objective*, *4OR* 19 (2021), 531–548. <https://doi.org/10.1007/s10288-020-00459-6>.
- [34] S. V. Sam and K. M. Woods, *A finite calculus approach to Ehrhart polynomials*, *Electronic Journal of Combinatorics* 17 (2010), R68. <https://doi.org/10.37236/340>. <https://arxiv.org/abs/0904.0679>.
- [35] D. Cristofaro-Gardiner, T. X. Li, and R. P. Stanley, *New examples of period collapse*, arXiv:1509.01887, 2015. <https://arxiv.org/abs/1509.01887>. See also the authors' *Irrational triangles with polynomial Ehrhart functions*, *Discrete & Computational Geometry* 61 (2019), 227–246, <https://doi.org/10.1007/s00454-018-0036-7>.
- [36] J. P. Bell, K. Gunn, K. D. Nguyen, and J. C. Saunders, *A general criterion for the Pólya–Carlson dichotomy and application*, arXiv:2206.00862, 2022. <https://arxiv.org/abs/2206.00862>. The classical Pólya–Carlson theorem is stated as Theorem 1.1.
- [37] T. Yu, *An analytic proof of multivariate Pólya–Carlson theorem*, *Acta Mathematica Sinica, English Series* 41 (2025), 2707–2712. <https://doi.org/10.1007/s10114-025-4213-3>. Preprint: *A multivariate version of Pólya–Carlson theorem*, arXiv:2312.00428, <https://arxiv.org/abs/2312.00428>.
- [38] R. P. Stanley, *Differentiably finite power series*, *European Journal of Combinatorics* 1 (1980), 175–188. [https://doi.org/10.1016/S0195-6698\(80\)80051-5](https://doi.org/10.1016/S0195-6698(80)80051-5).

Part V (manuscripts 07 and 04)

- [39] Vladimir Reshetnikov, *ProveIt: machine-checked mathematics and research artifacts*, repository documentation, accessed 30 September 2026. <https://github.com/VladimirReshetnikov/ProveIt>. Snapshot metadata: 6996fee43cc97b6c16351def7507d59a95bf62f0. [source 07]

- [40] Vladimir Reshetnikov, *ProveIt, Algebra/SurrealNumbers: README and report catalogue*, accessed 30 September 2026. <https://github.com/VladimirReshetnikov/ProveIt/tree/main/Algebra/SurrealNumbers>. Documentation reports formalization status; no independent build was performed for this manuscript. [source 07]
- [41] *Finite Convexity and Exact Lexicographic Optimization over the Surreals: A finite-dimensional theorem package and a one-objective compiler for multiscale linear programs*, research draft prepared for the Surreal repository, 22 September 2026. Prior research-library artifact titled `article(20260923-000126).pdf`; consulted for scope and overlap. [source 07] [write] A private draft, not part of ProveIt; it could not be checked.
- [42] Harry Gonshor, *An Introduction to the Theory of Surreal Numbers*, London Mathematical Society Lecture Note Series 110, Cambridge University Press, 1986. <https://doi.org/10.1017/CB09780511629143>. [source 07]
- [43] Samuel Fiorini, Thomas Rothvoß and Hans Raj Tiwary, *Extended formulations for polygons*, *Discrete & Computational Geometry* **48** (2012), 658–668. <https://arxiv.org/abs/1107.0371>. <https://doi.org/10.1007/s00454-012-9421-9>. [sources 07 and 04]
- [44] Volker Kaibel and Kanstantsin Pashkovich, *Constructing Extended Formulations from Reflection Relations*, arXiv:1011.3597, 2010. <https://arxiv.org/abs/1011.3597>. [sources 07 and 04] [write] Source 04 cites the conference version: in *Integer Programming and Combinatorial Optimization (IPCO 2011)*, Lecture Notes in Computer Science **6655**, Springer, 2011, 287–300, https://doi.org/10.1007/978-3-642-20807-2_23.
- [45] Mihalis Yannakakis, *Expressing combinatorial optimization problems by linear programs*, *Journal of Computer and System Sciences* **43** (1991), 441–466. [https://doi.org/10.1016/0022-0000\(91\)90024-Y](https://doi.org/10.1016/0022-0000(91)90024-Y). [sources 07 and 04]
- [46] João Gouveia, Pablo A. Parrilo and Rekha R. Thomas, *Lifts of convex sets and cone factorizations*, *Mathematics of Operations Research* **38** (2013), 248–264. <https://arxiv.org/abs/1111.3164>. <https://doi.org/10.1287/moor.1120.0575>. [source 07]
- [47] João Gouveia, Richard Z. Robinson and Rekha R. Thomas, *Worst-Case Results for Positive Semidefinite Rank*, arXiv:1305.4600v2, 30 May 2014. <https://arxiv.org/abs/1305.4600>. In particular, Theorem 2.1 and its real-field parameter-count argument. [source 07]
- [48] Hamza Fawzi, João Gouveia, Pablo A. Parrilo, Richard Z. Robinson and Rekha R. Thomas, *Positive semidefinite rank*, *Mathematical Programming* **153** (2015), 133–177. <https://arxiv.org/abs/1407.4095>. <https://doi.org/10.1007/s10107-015-0922-1>. [source 07]
- [49] Assia Mahboubi and Cyril Cohen, *Formal proofs in real algebraic geometry: from ordered fields to quantifier elimination*, arXiv:1201.3731, 2012. <https://arxiv.org/abs/1201.3731>. [source 07]
- [50] A. Tarski, *A Decision Method for Elementary Algebra and Geometry*, prepared with the assistance of J. C. C. McKinsey, University of California Press, 1951; RAND report R-109. <https://www.rand.org/pubs/reports/R109.html>. [source 04]
- [51] G. E. Collins, Quantifier elimination for real closed fields by cylindrical algebraic decomposition, in *Automata Theory and Formal Languages*, Lecture Notes in Computer Science **33**, Springer, 1975, 134–183. https://doi.org/10.1007/3-540-07407-4_17. [source 04]
- [52] L. van den Dries and P. Ehrlich, Fields of surreal numbers and exponentiation, *Fundamenta Mathematicae* **167** (2001), 173–188. <https://doi.org/10.4064/fm167-2-3>. [source 04]
- [53] M. Kwan, L. Saueremann, and Y. Zhao, Extension complexity of low-dimensional polytopes, *Transactions of the American Mathematical Society* **375** (2022), 4209–4250. <https://doi.org/10.1090/tran/8614>. Preprint, version 3: <https://arxiv.org/abs/2006.08836>. [source 04]

- [54] V. Reshetnikov et al., *ProveIt: Surreal-number library and research collection*, repository snapshot 6996fee43cc97b6c16351def7507d59a95bf62f0, inspected September 30, 2026. <https://github.com/VladimirReshetnikov/ProveIt/blob/6996fee43cc97b6c16351def7507d59a95bf62f0/Algebra/SurrealNumbers/README.md>. [source 04]
- [55] *The surreal and surcomplex reports*, ProveIt reader's guide, same snapshot and access date. <https://github.com/VladimirReshetnikov/ProveIt/blob/6996fee43cc97b6c16351def7507d59a95bf62f0/Algebra/SurrealNumbers/docs/README.md>. [source 04]
- [56] *Surreal and Surcomplex Numbers in Symbolic Computer Algebra*, report README in ProveIt, same snapshot and access date. <https://github.com/VladimirReshetnikov/ProveIt/blob/6996fee43cc97b6c16351def7507d59a95bf62f0/Algebra/SurrealNumbers/docs/foundations-and-computation/computer-algebra/README.md>. [source 04]

Part VI (source 08)

- [57] V. Reshetnikov and contributors, *ProveIt*, repository and root README, snapshot 6996fee43cc97b6c16351def7507d59a95bf62f0, inspected 30 September 2026. <https://github.com/VladimirReshetnikov/ProveIt/tree/6996fee43cc97b6c16351def7507d59a95bf62f0>.
- [58] V. Reshetnikov and contributors, *Surreal-number project and research-report guide*, Algebra/SurrealNumbers/README.md and Algebra/SurrealNumbers/docs/README.md, in [57], same pinned snapshot. <https://github.com/VladimirReshetnikov/ProveIt/blob/6996fee43cc97b6c16351def7507d59a95bf62f0/Algebra/SurrealNumbers/README.md>.
- [59] *Euclidean Three-Space over the Surreal Numbers*, research-report README, prepared for Vladimir Reshetnikov, Algebra/SurrealNumbers/docs/surreal/euclidean-three-space/README.md, in [57], same pinned snapshot. The README distinguishes report claims from mapped formal proofs. <https://github.com/VladimirReshetnikov/ProveIt/blob/6996fee43cc97b6c16351def7507d59a95bf62f0/Algebra/SurrealNumbers/docs/surreal/euclidean-three-space/README.md>.
- [60] M. Joswig, G. Loho, B. Lorenz, and B. Schröter, *Linear programs and convex hulls over fields of Puiseux fractions*, in *Mathematical Aspects of Computer and Information Sciences (MACIS 2015)*, Lecture Notes in Computer Science **9582**, Springer, 2016, pp. 429–445. https://doi.org/10.1007/978-3-319-32859-1_37. Preprint: <https://arxiv.org/abs/1507.08092>, version 2.
- [61] R. Schneider, *Convex Bodies: The Brunn–Minkowski Theory*, second expanded edition, Encyclopedia of Mathematics and its Applications **151**, Cambridge University Press, 2014. <https://doi.org/10.1017/CB09781139003858>. See the chapters on mixed volumes and inequalities for mixed volumes.
- [62] A. Chernikov and A. Mennen, *Combinatorial properties of non-archimedean convex sets*, Pacific Journal of Mathematics **323** (2023), 1–30. <https://doi.org/10.2140/pjm.2023.323.1>. Preprint: <https://arxiv.org/abs/2109.04591>, version 2.
- [63] K. Kaveh and P. Makhnatch, *Invariant factors as limit of singular values of a matrix*, Arnold Mathematical Journal **8** (2022), 561–571. <https://arxiv.org/abs/1811.07706>, version 2.
- [64] P. Brändén and J. Huh, *Lorentzian polynomials*, Annals of Mathematics **192** (2020), 821–891. <https://doi.org/10.4007/annals.2020.192.3.4>. Preprint: <https://arxiv.org/abs/1902.03719>, version 8.

Part VII (sources 10 and 09)

- [65] Gennadiy Averkov, Katherina von Dichter, Simon Richard, and Ivan Soprunov, *Mixed volumes of zonoids and the absolute value of the Grassmannian*, arXiv:2404.02842, version 3, 13 July 2026. [arXiv:2404.02842](https://arxiv.org/abs/2404.02842).

- [write] Part III cites version 2 of this preprint; version 3 could not be checked offline in the write phase, and no [write] text relies on it.
- [66] Petter Brändén and June Huh, *Lorentzian polynomials*, *Annals of Mathematics* **192** (2020), no. 3, 821–891. doi:10.4007/annals.2020.192.3.4. Author version consulted: [arXiv:1902.03719](https://arxiv.org/abs/1902.03719), version 8. [write] Source 09 cites this work too; its entry reads: P. Brändén and J. Huh, *Lorentzian polynomials*, *Annals of Mathematics* **192** (2020), 821–891. Updated author manuscript, arXiv:1902.03719v8 (17 July 2024). <https://arxiv.org/abs/1902.03719v8>. See especially Theorem 3.20, Theorem 4.1, and Remark 4.3.
- [67] Harry Gonshor, *An Introduction to the Theory of Surreal Numbers*, London Mathematical Society Lecture Note Series, vol. 110, Cambridge University Press, 1986. doi:10.1017/CBO9780511629143. [write] Source 09 cites this work too; its entry reads: H. Gonshor, *An Introduction to the Theory of Surreal Numbers*, London Mathematical Society Lecture Note Series, vol. 110, Cambridge University Press, 1986. [Publisher’s book page](#).
- [68] June Huh, *Volume polynomials*, arXiv:2601.13249, 2026. arXiv:2601.13249. [write] Source 09 cites this work too; its entry reads: J. Huh, *Volume polynomials*, arXiv:2601.13249v4 (4 June 2026). <https://arxiv.org/html/2601.13249v4>. Sections 2–3 distinguish convex-body polynomials from projective volume polynomials; Example 3.4 states the separate projective Fano question. [write] Version 4 and its Example 3.4, cited by both sources, could not be checked offline in the write phase; no [write] text relies on them.
- [69] Amelie Menges, *Comparing the sets of volume polynomials and Lorentzian polynomials*, *Combinatorics, Probability and Computing* **35** (2026), no. 1, 26–39; published online 19 September 2025. doi:10.1017/S0963548325100151. [write] Source 09 cites this work too; its entry reads: A. Menges, *Comparing the sets of volume polynomials and Lorentzian polynomials*, *Combinatorics, Probability and Computing* **35** (2026), 26–39; published online 19 September 2025. <https://doi.org/10.1017/S0963548325100151>. Preprint: <https://arxiv.org/abs/2310.07020>.
- [70] Richard Rado, *A theorem on independence relations*, *Quarterly Journal of Mathematics, Oxford Series* **13** (1942), 83–89. doi:10.1093/qmath/os-13.1.83.
- [71] Rolf Schneider, *Convex Bodies: The Brunn–Minkowski Theory*, second expanded edition, *Encyclopedia of Mathematics and its Applications*, vol. 151, Cambridge University Press, 2014. [Publisher record](#). [write] Source 09 cites this work too; its entry reads: R. Schneider, *Convex Bodies: The Brunn–Minkowski Theory*, 2nd expanded edition, *Encyclopedia of Mathematics and its Applications*, vol. 151, Cambridge University Press, 2014. <https://doi.org/10.1017/CBO9781139003858>. Chapter 5 contains the mixed-volume facts used here.
- [72] David Speyer and Bernd Sturmfels, *The tropical Grassmannian*, *Advances in Geometry* **4** (2004), no. 3, 389–411. arXiv:math/0304218. [write] Source 09 cites this work too; its entry reads: D. Speyer and B. Sturmfels, *The tropical Grassmannian*, *Advances in Geometry* **4** (2004), 389–411. <https://doi.org/10.1515/adv.2004.023>. Preprint: <https://arxiv.org/abs/math/0304218>. The rank-two case is the tree-space realization used as context here.
- [73] Vladimir Reshetnikov et al., ProveIt repository, *SignSequenceValuation.lean*, source at commit 6996fee43cc97b6c16351def7507d59a95bf62f0, inspected 30 September 2026. [Pinned source file](#).
- [74] Vladimir Reshetnikov et al., ProveIt repository, *SignSequenceStandardPart.lean*, source at the same pinned commit, inspected 30 September 2026. [Pinned source file](#).
- [75] A. Tarski, *A Decision Method for Elementary Algebra and Geometry*, prepared for publication with the assistance of J. C. C. McKinsey, revised edition, University of California Press, 1951. RAND report R-109 edition: <https://www.rand.org/pubs/reports/R109.html>.

- [76] V. Reshetnikov and contributors, *ProveIt: the surreal and surcomplex reports*, repository documentation at commit `b2b626d8537f7dea8f20393ee4714c7201a7001e`, read 30 September 2026. [Pinned repository README](#).
- [77] *Euclidean Three-Space over the Surreal Numbers*, AI-assisted research report in `ProveIt, Algebra/SurrealNumbers/docs/surreal/euclidean-three-space/article.tex`, at the same pinned commit. Abstract and opening scope discussion inspected; no full-report verification is claimed. [Pinned article source](#).

Part VIII (source 11)

- [78] M. Bayer and B. Sturmfels, *Lawrence polytopes*, *Canadian Journal of Mathematics* **42** (1990), no. 1, 62–79. [doi:10.4153/CJM-1990-004-4](#). The classical Lawrence construction and its reconstruction properties are the main geometric antecedents used here.
- [79] L. Kühne, R. Pendavingh, and G. Yashfe, *Von Staudt constructions for skew-linear and multilinear matroids*, *Combinatorial Theory* **3** (2023), no. 1, paper 16. [doi:10.5070/C63160429](#); [arXiv:2012.07361](#). Cited for the projective-arithmetic method, not for the shadow-degree bound of this article.
- [80] J. E. Goodman, R. Pollack, and B. Sturmfels, *The intrinsic spread of a configuration in $\mathbb{R}^d$* , *Journal of the American Mathematical Society* **3** (1990), no. 3, 639–651. [doi:10.2307/1990931](#). A classical antecedent on configuration complexity; no detailed theorem from this paper is required in the proofs above.
- [81] S. Basu and M.-F. Roy, *Quantitative curve selection lemma*, [arXiv:1803.00505v3](#), 19 July 2021; first version 2018. Cited for the quantitative semialgebraic curve-selection framework.
- [82] X. Allamigeon, R. D. Katz, and P.-Y. Strub, *Formalizing the face lattice of polyhedra*, in *Automated Reasoning—IJCAR 2020*, Lecture Notes in Computer Science **12167**, Springer, 2020, 185–203. [doi:10.1007/978-3-030-51054-1_11](#).
- [83] V. Reshetnikov, *ProveIt, Algebra/SurrealNumbers/Surreal/Algebra/Geometry.lean*, commit `6996fee43cc9`, inspected 30 September 2026. <https://github.com/VladimirReshetnikov/ProveIt/blob/6996fee43cc97b6c16351def7507d59a95bf62f0/Algebra/SurrealNumbers/Surreal/Algebra/Geometry.lean>. Only the finite-geometry scope described in the source is used in the repository comparison.
- [84] V. Reshetnikov, *ProveIt: Surreal numbers project guide*, `Algebra/SurrealNumbers/README.md`, inspected 30 September 2026. <https://github.com/VladimirReshetnikov/ProveIt/tree/6996fee43cc97b6c16351def7507d59a95bf62f0/Algebra/SurrealNumbers>. The guide distinguishes research manuscripts from mapped formal results.
- [85] *Finite-Scale Geometry of Surreal Liftings: Sharp dependence compression, exact polynomial degree bounds, and linear realization of finite combinatorics*, research manuscript prepared for Vladimir Reshetnikov, 30 September 2026, `surreal_liftings_article.tex` and companion PDF. User-library manuscript, AI-assisted and not independently refereed. Cited for the fixed-real-base comparison and its distinction between finite signs and all real dependence evaluations. None of the lower-bound proofs in the present article depends on that manuscript. [write] In *ProveIt* this manuscript is Part I of this report (batch 58, manuscript 05, delivered as `surreal_liftings/article.tex`); the file name `surreal_liftings_article.tex` above does not exist, and the companion PDF is not shipped.

Part IX (source 12)

- [86] Vladimir Reshetnikov and contributors, *ProveIt: Algebra/SurrealNumbers*, repository README and documented formalization scope. Inspected 30 September 2026; tree revision `6996fee43cc97b6c16351def7507d59a95bf62f0`. <https://github.com/VladimirReshetnikov/ProveIt/blob/6996fee43cc97b6c16351def7507d59a95bf62f0/Algebra/SurrealNumbers/README.md>.

- [87] Vladimir Reshetnikov and contributors, *ProveIt: Presburger arithmetic is decidable*, Logic/PresburgerArithmetic/README.md. Same inspected revision and date as [86]. <https://github.com/VladimirReshetnikov/ProveIt/blob/6996fee43cc97b6c16351def7507d59a95bf62f0/Logic/PresburgerArithmetic/README.md>.
- [88] Vladimir Reshetnikov and contributors, *Omnific Groups and Lattices*, merged research-report README, particularly Part III, *Shortest Vectors and Missing Infima over the Omnific Integers*. Same inspected revision and date as [86]. <https://github.com/VladimirReshetnikov/ProveIt/blob/6996fee43cc97b6c16351def7507d59a95bf62f0/Algebra/SurrealNumbers/docs/surreal/omnific-groups-and-lattices/README.md>.
- [89] Emil Jeřábek, *Rigid models of Presburger arithmetic*, Mathematical Logic Quarterly **65** (2019), no. 1, 108–115. Consulted version: arXiv:1803.05797v2, 8 November 2018. <https://arxiv.org/abs/1803.05797v2>. <https://doi.org/10.1002/malq.201800019>.
- [90] Raf Cluckers, *Presburger sets and p -minimal fields*, Journal of Symbolic Logic **68** (2003), 153–162. <https://arxiv.org/abs/math/0206197>.
- [91] Shmuel Onn, *Nonlinear Discrete Optimization: An Algorithmic Theory*, Zurich Lectures in Advanced Mathematics, European Mathematical Society, 2010. <https://doi.org/10.4171/093>.
- [92] Shmuel Onn and Michal Rozenblit, *Convex Integer Optimization by Constantly Many Linear Counterparts*, arXiv:1208.5639v1, 28 August 2012; journal reference recorded by arXiv: Linear Algebra and its Applications **447** (2014), 88–109. In particular, the zonotope refinement and counting lemmas and the graph Graver edge-direction argument. <https://arxiv.org/abs/1208.5639v1>.

Part X (source 13)

- [93] V. Reshetnikov and contributors, *ProveIt: Surreal project overview*, Algebra/SurrealNumbers/README.md, consulted 30 September 2026. <https://github.com/VladimirReshetnikov/ProveIt>. Individual path and returned blob recorded in Appendix B and the source notes. ([write] Appendix B is section 168; the source notes are 13-beyond-finite-SOURCES_AND_SCOPE.md.)
- [94] V. Reshetnikov and contributors, *Surreal Fields Across Set-Theoretic Universes*, research-report README, Algebra/SurrealNumbers/docs/foundations-and-computation/surreal-fields-across-universes/README.md, consulted 30 September 2026. Unreferenced repository report; used for motivation and scope, not as a proof dependency.
- [95] P. Ehrlich, *The absolute arithmetic continuum and the unification of all numbers great and small*, Bulletin of Symbolic Logic **18** (2012), no. 1, 1–45. doi:10.2178/bsl/1327328438. Official issue record: <https://www.math.ucla.edu/~asl/bsl/1801-toc.htm>.
- [96] L. van den Dries and P. Ehrlich, *Fields of surreal numbers and exponentiation*, Fundamenta Mathematicae **167** (2001), 173–188. doi:10.4064/fm167-2-3. Erratum, **168** (2001), 295–297.
- [97] A. Charnes and W. W. Cooper, *The strong Minkowski–Farkas–Weyl theorem for vector spaces over ordered fields*, Proceedings of the National Academy of Sciences **44** (1958), no. 9, 914–916. doi:10.1073/pnas.44.9.914.
- [98] M. Dvorak and V. Kolmogorov, *Duality theory in linear optimization and its extensions: formally verified*, Annals of Formalized Mathematics **2** (2026), no. 1. doi:10.46298/afm.14253. Consulted preprint version 3: <https://arxiv.org/html/2409.08119v3>. The ordered-field finite duality statements are relevant here; the paper’s separately adjoined infinite values are not surreal scalars.
- [99] J. E. Martínez-Legaz, *Lexicographical characterization of the faces of convex sets*, Acta Mathematica Vietnamica **22** (1997), no. 1, 207–211. <https://math.ac.vn/uploads/files/9701207.pdf>. See Theorem 2, Definition 3, Proposition 4, and Proposition 6.

- [100] V. V. Gorokhovik, *Characterizations of Faces of Convex Sets in Infinite-dimensional Vector Spaces*, arXiv:2506.08742, version 1 (2025). <https://arxiv.org/html/2506.08742v1>. Used to distinguish the prior face-characterization literature from the finite-dimensional scalar-extension questions considered here.